

BSAVA Small Animal Formulary

**11th edition • Part B:
Exotic Pets**

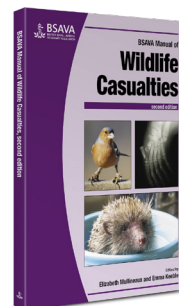
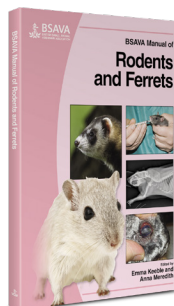
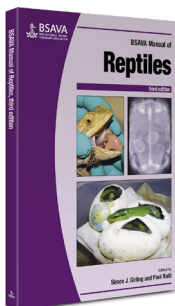
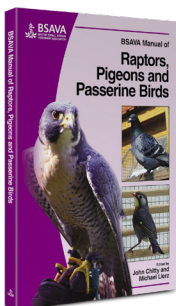
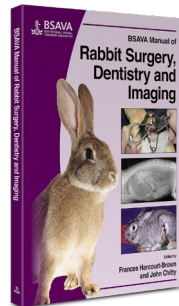
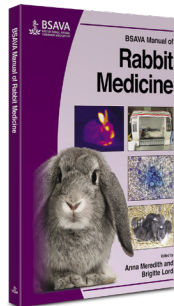
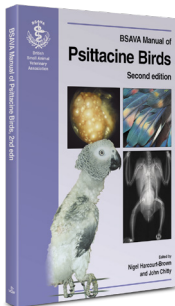
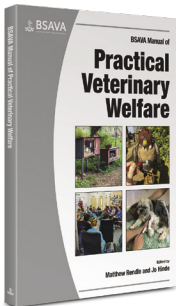
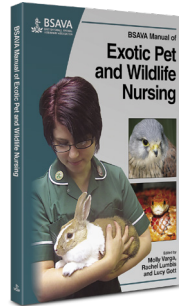
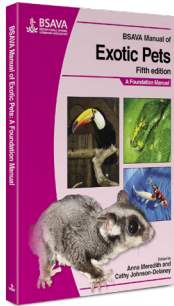
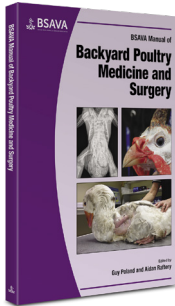
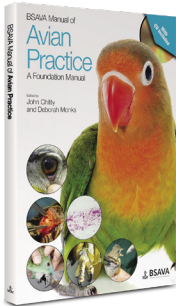
Editor:
Joanna Hedley



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BRITISH SMALL ANIMAL VETERINARY ASSOCIATION

Small Animal Formulary

11th edition – Part B: Exotic Pets

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Small Animal Formulary – Part B: Exotic Pets

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The publishers, editors and contributors cannot take responsibility for information provided on dosages and methods of application of drugs mentioned or referred to in this publication. Details of this kind must be verified in each case by individual users from up to date literature published by the manufacturers or suppliers of those drugs. Veterinary surgeons are reminded that in each case they must follow all appropriate national legislation and regulations (for example, in the United Kingdom, the prescribing cascade) from time to time in force.

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Booksvets.blogspot.com

V E T E R I N A R Y L I B R A R Y

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Foreword

The *BSAVA Small Animal Formulary: Part B Exotic Pets* has continued to develop into a unique compilation of information on authorized products, expert opinion and published evidence on an extensive number of medications for the increasing array of pet species presented in small animal practice. This invaluable publication provides both students and clinicians with an instantly accessible, current and reviewed resource when prescribing for species that may be seen less frequently in practice with very contrasting differences in drug doses, safety and metabolism factors. It is a hugely popular membership benefit and held in high regard in both the UK and around the world.

The 11th edition has been fully updated and sees an additional 7 new drug monographs added, detailing drug doses for an extensive variety of species including a wide variety of small mammals, birds, reptiles, amphibians and fish. Each drug entry provides available data for formulations, action, uses, contraindications, drug interactions and adverse reaction risks alongside safety handling information in a simple alphabetical user-friendly format. Additionally, the 11th edition contains expanded sedation protocols for fish.

As per Part A: Canine and Feline, there is the important addition of a clear visual indicator on every antibiotic to show its European Medicines Agency (EMA) category with expanded guidelines explaining the EMA categories and promoting responsible use of antibiotics to protect public and animal health. In tandem with this there is a new appendix with guidance on the responsible use of parasiticides in companion animal practice.

Joanna Hedley and her team are to be congratulated on the valuable additions and updates to this must-have 11th edition of the *BSAVA Small Animal Formulary for Exotic Pets*.

Alison Speakman BVMS Hons PhD CertAVP GPCert (VPS) MRCVS
BSAVA President 2022–2023

Preface

Welcome to Part B of the 11th edition of the *BSAVA Small Animal Formulary*, covering those drugs used in exotic pets. Treatment of exotic pets can be challenging due to the wide variety of species and conditions that may be encountered, in addition to the many anatomical and physiological differences between the different species groups. Although our knowledge is progressing all the time, unfortunately there is still a lack of pharmacological studies and clinical efficacy trials for many commonly used drugs in these species. Consequently, many drug dosages are extrapolated from those used in other species or have been derived from anecdotal recommendations based on apparent clinical efficacy. As in the previous edition, primary references indicate where information is available, but for most drugs it is important to be aware that evidence is limited. Additional references are provided in the online version of the Formulary and we look forward to expanding this resource in the future.

Since the last edition, various new drugs have become available and information on other drugs has increased with some modifications to recommended dosages or treatment protocols. Some drugs have been deleted if they are no longer available or their use is no longer recommended. The majority of drugs listed are still not authorized for use in these species, so need to be prescribed via the cascade with informed client consent.

I would like to thank all the Editorial Panel members for their hard work reviewing each monograph and ensuring that dosages and evidence are as up to date as possible. I would also like to thank the BSAVA Publishing team at Woodrow House for all their editorial and administrative assistance, without which this Formulary would not have been possible.

Joanna Hedley

January 2023

Introduction

Notes on the monographs

- **Name.** The rINN generic name is used where this has been agreed. When a choice of names is available the more commonly used in the UK has been provided. The list of trade names is not necessarily comprehensive, and the mention or exclusion of any particular commercial product is not a recommendation or otherwise as to its value. Any omission of a product that is authorized for a particular small animal indication is purely accidental. All monographs were updated in the period March–September 2022. Products that are not authorized for veterinary use by the Veterinary Medicines Directorate (VMD) are marked with an asterisk. Products that are authorized for use in one or more of the exotic species listed in a monograph are marked by bold text. Note that an indication that a product is authorized does not necessarily mean that it is authorized for all species and indications listed in the monograph; users should check individual data sheets. You may also wish to refer to the VMD's Product Information Database (www.vmd.defra.gov.uk/ProductInformationDatabase/).
- **Formulations.** Only medicines and formulations that are available in the UK have been included – many others are available outside the UK and some medicines in different formulations. Common trade names of human medicines are provided. In many cases they are available as generic formulations and may be cheaper. However, be careful of assuming that the bioavailability of one brand is the same as that of another. Avoid switching between brands unnecessarily.
- **Action and Use.** Veterinary surgeons using this publication are warned that **many of the drugs and doses listed are not currently authorized** by the VMD or the European Medicines Agency (EMA) (either at all or for a particular species), or manufacturers' recommendations may be limited to particular indications. **The decision, and therefore the responsibility, for prescribing any drug for an animal lies solely with the veterinary surgeon.** Expert assistance should be obtained when necessary. The 'cascade' and its implications are discussed below. For information on combination drugs, it is important to refer to all relevant monographs.
- **Safety and handling.** This section only outlines specific risks and precautions for a particular drug that are in addition to the general advice given below in the 'Health and safety in dispensing' section. A separate Appendix deals with chemotherapeutic drugs.
- **Contraindications and Adverse reactions.** The list of adverse reactions is not intended to be comprehensive and is limited to those effects that may be of clinical significance. The information for both of these sections is taken from published veterinary and human references and not just from product literature.
- **Drug interactions.** A listing of those interactions which may be of clinical significance.
- **Doses.** These are based on those recommended by the manufacturers in their data sheets and package inserts or are based on those given in published articles or textbooks or are based on clinical experience. **These recommendations should be used only as guidelines and should not be considered appropriate for every case.** Clinical judgement must take precedence. Where possible, doses have been given for individual species; however, sometimes

generalizations are used. 'Mammals' includes ferrets, lagomorphs and rodents. Doses for small primates and other exotic mammals, such as African pygmy hedgehogs and sugar gliders, are given on a species basis only. 'Birds' includes psittacines, raptors, pigeons and others. 'Reptiles' includes chelonians, lizards and snakes. Amphibians and pet fish are also included. Except where indicated, all doses given for reptiles assume that the animal is kept within its Selected Temperature Range (T_s). Animals that are maintained at different temperatures may have different rates of metabolism and therefore the dose (and especially the frequency) that is required may require alteration. The colour banding on the antibiotic drug monographs denotes their category as classified by the EMA ([see Appendix for Guidelines on responsible antibiotic use](#)).

- **References.** Primary references are given to support doses where available and are denoted by a letter. Such primary references describe a pharmacokinetic or clinical efficacy trial in that species or species group (not case reports). Primary references are generally not given where doses are considered very well established in the scientific literature, particularly for rabbits and rodents (e.g. from studies on laboratory mammals), or where the product is authorized in that species or species group. For some drugs, non-primary references are also given and are denoted by a number. These generally refer to sources of information such as textbooks where no other primary information can be found and the reference is deemed a useful resource in itself. All other doses can be assumed to be taken from other non-primary sources such as other formularies, book chapters or reviews, or are anecdotal.

Distribution categories

Authorized small animal medicines within Great Britain now fall within the first four categories below and all packaging supplied by drug manufacturers and distributors was changed in 2008. Medical products not authorized for veterinary use retain their former classification (e.g. GSL, P, POM). Other laws apply in other jurisdictions. Some nutritional supplements (nutraceuticals) are not considered medicinal products and therefore are not classified. Where a product does not have a marketing authorization it is designated 'general sale'.

AVM-GSL: Authorized veterinary medicine – general sales list. This may be sold by anyone.

NFA-VPS: Non-food animal medicine – veterinarian, pharmacist, Suitably Qualified Person (SQP). These medicines for companion animals must be supplied by a veterinary surgeon, pharmacist or SQP. An SQP must be registered with the Animal Medicines Training Regulatory Authority (AMTRA). Veterinary nurses can become SQPs, but it is not automatic.

POM-VPS: Prescription-only medicine – veterinarian, pharmacist, SQP (formerly PML livestock products, MFSX products and a few P products). These medicines for food-producing animals (including horses) can only be supplied on an oral or written veterinary prescription from a veterinary surgeon, pharmacist or SQP and can only be supplied by one of those groups of people in accordance with the prescription.

POM-V: Prescription-only medicine – veterinarian. These medicines can only be supplied against a veterinary prescription that has been prepared

(either orally or in writing) by a veterinary surgeon to animals under their care following a clinical assessment and can only be supplied by a veterinary surgeon or pharmacist in accordance with the prescription.

Exemptions for Small Pet Animals (ESPA): These products, formerly known as Small Animal Exemption Scheme (SAES) products, are veterinary medicinal products (VMPs) that can be administered to minor species of small animals kept as pets (e.g. caged birds, fish, companion rabbits and small rodents), the active ingredient of which has been declared by the Secretary of State as not requiring veterinary control. They do not have a marketing authorization and are therefore not required to prove safety, quality or efficacy. They may be sold retail by anyone but must be manufactured to the same standards as authorized medicines and are subject to pharmacovigilance reporting. A wholesale dealer's authorization (WDA) is required to sell these products wholesale.

CD: Controlled Drug. A substance controlled by the Misuse of Drugs Act 1971 and Regulations. The CD is followed by (Schedule 1), (Schedule 2), (Schedule 3), (Schedule 4) or (Schedule 5) depending on the Schedule to The Misuse of Drugs Regulations 2001 (as amended) in which the preparation is included. You could be prosecuted for failure to comply with this act. Prescribers are reminded that there are additional requirements relating to the prescribing of Controlled Drugs. For more information see the *BSAVA Guide to the Use of Veterinary Medicines* at www.bsavalibrary.com/medicinesguide.

Schedule 1: Includes LSD, cannabis, lysergide and other drugs that are not used medicinally. Possession and supply are prohibited except in accordance with Home Office Authority.

Schedule 2: Includes etorphine, fentanyl, ketamine, morphine, methadone, papaveretum, pethidine, secobarbital (quinalbarbitone), diamorphine (heroin), cocaine and amphetamine. Record all purchases and each individual supply (within 24 hours). Registers must be kept for 2 calendar years after the last entry. Drugs must be kept under safe custody (locked secure cabinet), except secobarbital. There are specific requirements regarding the destruction of Schedule 2 Controlled Drugs, which may require an independent veterinary surgeon or person authorized by the Secretary of State to witness.

Schedule 3: Includes buprenorphine, pentazocine, the barbiturates (e.g. pentobarbital and phenobarbital but not secobarbital – which is Schedule 2), midazolam, tramadol, gabapentin and others. Buprenorphine, with some others (e.g. diethylpropion and temazepam), must be kept under safe custody (locked secure cabinet); it is advisable that all Schedule 3 drugs are locked away. Tramadol, gabapentin and pregabalin are exempt from safe custody requirements but must follow CD prescription writing requirements. Retention of invoices for 2 years is necessary.

Schedule 4: Includes most of the benzodiazepines except midazolam and temazepam (which are now Schedule 3), and androgenic and anabolic steroids (e.g. clenbuterol, nandrolone). Exempted from control when used in normal veterinary practice.

Schedule 5: Includes preparations (such as several codeine products) which, because of their strength, are exempt from virtually all Controlled Drug requirements other than the retention of invoices for 2 years.

Extemporaneous preparations

Extemporaneous preparations (also known as ‘veterinary specials’ and denoted as **VSP** in the Formulations section of the monographs) are products that do not hold a marketing authorization. These products have not been assessed against the same standards of quality, safety (for the target animal, user, consumer and environment) and efficacy as authorized veterinary medicines. They can legally be prescribed, supplied and used under the last step of the cascade (see below). Extemporaneous preparations carry a higher risk than authorized medicines and this should be taken into consideration when prescribed. Manufacturers of extemporaneous products must be authorized (i.e. hold a ManSA) and comply with the general principles of Good Manufacturing Practice (GMP). Their facilities, equipment and procedures are regularly inspected to ensure they manufacture extemporaneous products to a set quality standard ([see Appendix for Useful websites](#)).

The prescribing cascade

Veterinary medicinal products (VMPs) must be administered in accordance with the prescribing cascade, as set out in The Veterinary Medicines Regulations (VMR) 2013. The current law starts from the principle that all VMPs must be authorized, and that use of an unauthorized medicine, or use of an authorized medicine in an unauthorized way, is an offence. However, the law recognizes that there are circumstances where the benefits of treatment of animals with unauthorized medicines outweigh the risks, particularly where there are not veterinary authorized medicines for a condition or for a species. As a result, the legislators have given veterinary surgeons an exemption from the legal requirement to use an authorized VMP for the species and condition that they are treating. This exemption is known as the prescribing cascade.

‘Off-label’ use of medicines

‘Off-label’ use is the use of medicines outside the terms of their marketing authorization (MA). It may include medicines authorized outside the UK or NI that are used in accordance with an import certificate issued by the VMD. A veterinary surgeon, with detailed knowledge of the medical history and clinical status of a patient, may reasonably prescribe a medicine ‘off-label’ in accordance with the prescribing cascade. Authorized medicines have been scientifically assessed against statutory criteria of safety, quality and efficacy when used in accordance with the authorized recommendations on the product literature. Use of an unauthorized medicine provides none of these safeguards and may, therefore, pose potential risks that the authorization process seeks to minimize. Note that the use of an ESPA medicine not in accordance with the product’s labelling would be classed as use under the cascade and considered an extemporaneous preparation.

Medicines may be used ‘off-label’ for a variety of reasons, including:

- No authorized product is suitable for the condition or specific subpopulation being treated
- Need to alter the duration of therapy, dosage, route of administration, etc., to treat the specific condition presented
- An authorized product has proved ineffective in the circumstances of a particular case (all cases of suspected lack of efficacy of authorized veterinary medicines should be reported to the VMD).

Responsibility for the use of a medicine 'off-label' lies solely with the prescribing veterinary surgeon. They should inform the owner of the reason why a medicine is to be used 'off-label' and record this reason in the patient's clinical notes. When electing to use a medicine 'off-label' always:

- Discuss all therapeutic options with the owner
- Use the cascade to determine your choice of medicine
- Obtain signed informed consent if an unauthorized product is to be used, ensuring that all potential problems are explained to the client
- Administer unauthorized medicines against a patient-specific prescription. Do not administer to a group of animals if at all possible.

An 'off-label' medicine must show a comparative clinical advantage to the authorized product in the specific circumstances presented (where applicable). Medicines may be used 'off-label' in the following ways (this is not an exhaustive list):

- Authorized product at an unauthorized dose
- Authorized product for an unauthorized indication
- Authorized product used outwith the authorized age range
- Authorized product administered by an unauthorized route
- Authorized product used to treat an animal in an unauthorized physiological state, e.g. pregnancy (i.e. an unauthorized indication)
- Product authorized for use in humans or a different animal species to that being treated.

Adverse effects may or may not be specific for a species, and idiosyncratic reactions are always a possibility. If no adverse effects are listed, consider data from different species. When using novel or unfamiliar drugs, consider pharmaceutical and pharmacological interactions. In some species, and with some diseases, the ability to metabolize/excrete a drug may be impaired/enhanced. Use the lowest dose that might be effective and the safest route of administration. Ensure that you are aware of the clinical signs that may suggest toxicity. Information on 'off-label' use may be available from a wide variety of sources (**see Appendix for Further reading and Useful websites**).

Marketing authorizations

Following the exit of the United Kingdom (UK) from the European Union (EU) there are now three MAs for veterinary medicines in the UK. The options that veterinary surgeons must consider under the prescribing cascade depend on whether they are working in Great Britain (GB) or Northern Ireland (NI). The Northern Ireland Protocol requires that the use of veterinary medicines in Northern Ireland be compliant with the regulations of the European veterinary medicines regulatory system. In practice, for the cascade this means that where no suitable NI-authorized veterinary medicine is available, veterinary surgeons in NI should consider using an EU authorized veterinary medicine before using an alternative medicine authorized elsewhere in the world.

The three types of authorization are:

- UK MA – a product with a UK MA has been approved for marketing in England, Wales, Scotland and Northern Ireland. These MAs were issued pre-2021 and retain a UK-wide authorization
- GB MA – a product with a GB MA has been approved for marketing in England, Wales and Scotland. These authorizations have been

assessed and issued by the VMD against the requirements set out within the UK VMR 2013, as amended

- NI MA – a product with an NI MA has been approved for marketing in Northern Ireland. There are four different routes to obtain an NI MA:
 - A national authorization (valid in NI only)
 - A centralized authorization issued by the European Commission (valid in all EU member states, including NI)
 - A mutual recognition procedure (valid in select EU member states where the product is already authorized on a national basis in one of them)
 - A decentralized procedure (valid in select EU member states).

Compliance with the cascade

The cascade differs depending on whether the prescribing veterinary surgeon is based in Great Britain or Northern Ireland.

Veterinary surgeons in Great Britain: The steps, in descending order of suitability, are:

- Veterinary medicine with an MA valid in GB or UK wide for the indicated species and condition
- Veterinary medicine with an MA valid in NI for the indicated species and condition
- Veterinary medicine with an MA valid in GB, NI or UK wide for a different species or condition
- Human medicine with an MA valid in GB, NI or UK wide OR an authorized veterinary medicine from outside the UK. In the case of a food-producing animal, the medicine must be authorized in a food-producing species
- Extemporaneous preparation prepared by a veterinary surgeon, pharmacist or person holding an appropriate Manufacturer's Authorization, located in the UK
- In exceptional circumstances, a human medicine may be imported from outside of the UK.

NOTE: For products not authorized in GB or UK wide, a Special Import Certificate from the VMD is required.

Veterinary surgeons in Northern Ireland: The steps, in descending order of suitability, are:

- Veterinary medicine with an MA valid in NI or UK wide for the indicated species and condition
- Veterinary medicine with an MA valid in NI or UK wide for a different species or condition
- Human medicine with an MA valid in NI or UK wide or a veterinary medicine with an MA valid in an EU member state. In the case of a food-producing animal, the medicine must be authorized in a food-producing species
- Extemporaneous preparation prepared by a veterinary surgeon, pharmacist or person holding an appropriate Manufacturer's Authorization, located in the UK
- In exceptional circumstances, a veterinary medicine with an MA in GB or outside the EU may be imported, or a human medicine from outside NI may be imported.

NOTE: For products not authorized in NI or UK wide, a Special Import Certificate from the VMD is required.

Drug storage and dispensing

For further information on the storage and dispensing of medicines see the *BSAVA Guide to the Use of Veterinary Medicines* available at www.bsavalibrary.com/medicinesguide. Note that veterinary surgeons may only supply a veterinary medicine from practice premises that are registered with the RCVS and that these premises must be inspected. It is recommended that, in general, medications are kept in and dispensed in the manufacturer's original packaging. Medicines can be adversely affected by adverse temperatures, excessive light, humidity and rough handling. Loose tablets or capsules that are repackaged from bulk containers should be dispensed in child-resistant containers and supplied with a package insert (if one exists). Tablets and capsules in foil strips should be sold in their original packaging or in a similar cardboard box for smaller quantities. Preparations for external application should be dispensed in coloured fluted bottles. Oral liquids should be dispensed in plain glass bottles with child-resistant closures.

All medicines should be labelled. The label should include:

- The owner's name and address
- Identification of the animal
- Date of supply (and, if applicable, the expiry date)
- Product name (and strength)
- Total quantity of the product supplied in the container
- Instructions for dosage
- Practice name and address
- The name of the veterinary surgeon who prescribed the medication (if not an authorized use)
- Any specific pharmacy precautions (including storage, disposal, handling)
- The wording 'Keep out of reach of children' and 'For animal treatment only'
- Withdrawal period, if relevant
- Any other necessary warnings.

The words 'For external use only' should be included on labels for products for topical use. All labels should be typed. If this information will not fit on a single label then it is permissible to include the information on a separate sheet.

For medicines that are not authorized for veterinary use, and even for some that are, it is useful to add to the label or on a separate information sheet the likely adverse effects, drug interactions and the action to be taken in the event of inadvertent mis-dosing or incorrect administration written in plain English.

In order to comply with the current Veterinary Medicines Regulations, records of all products supplied on prescription must be kept for 5 years. When a batch is brought into use in a practice, the batch number and date should be recorded. It is not necessary to record the batch number of each medication used for a given animal.

Owners should be informed that any unused or left over medication should be returned to the practice for safe disposal.

Health and safety in dispensing

All drugs are potentially poisonous to humans as well as animals. Toxicity may be mild or severe and includes carcinogenic and teratogenic

effects. Warnings are given in the monographs. However, risks to humans dispensing medicines are not always well characterized and idiosyncratic reactions may occur. **It is good practice for everyone to wear protective clothing (including gloves) when directly handling medicines**, not to eat or drink (or store food or drink) near medicines, and to wash their hands frequently when working with medicines. Gloves, masks and safety glasses should be worn if handling potentially toxic liquids, powders or broken tablets. Do not break tablets of antineoplastic cytotoxic drugs, and use laminar flow cabinets for the preparation and dispensing of these medications (**see Appendix for Safety and handling of chemotherapeutic agents**).

Many prescribers and users of medicines are not aware of the carcinogenic potential of the drugs they are handling. Below are lists of medicines included in the BSAVA Formulary that are known or potential carcinogens or teratogens. The lists are not all-inclusive: they include only those substances that have been evaluated. Most of the drugs are connected only with certain kinds of cancer. The relative carcinogenicity of the agents varies considerably and some do not cause cancer at all times or under all circumstances. Some may only be carcinogenic or teratogenic if a person is exposed in a certain way (for example, ingesting as opposed to touching the drug). For more detailed information refer to the International Agency for Research on Cancer (IARC) or the National Toxicology Program (NTP) (information is available on their respective websites).

Examples of drugs known or suspected to be human carcinogens (c) or teratogens (t):

- ACE inhibitors (t), e.g. benazepril, enalapril
- Androgenic (anabolic) steroids (t, c)
- Antibiotics (c), e.g. metronidazole, chloramphenicol
- Antibiotics (t) e.g. aminoglycosides, doxycycline, trimethoprim, sulphonamides
- Antifungals (c), e.g. fluconazole, itraconazole
- Antineoplastic drugs (c, t) – all
- Antithyroid drugs (t), e.g. carbimazole/methimazole
- Beta-blockers (t)
- Deferoxamine (t)
- Diltiazem (t)
- Finasteride (t)
- Immunosuppressives (c), e.g. azathioprine, ciclosporin
- Methotrexate (t)
- Misoprostol (t)
- NSAIDs (t)
- Penicillamine (t)
- Phenoxybenzamine (c)
- Progestagens (c) and some oestrogens (c)
- Vitamin A (t)

Note that most carcinogens are also likely to be teratogens.

Acepromazine (ACP)

(Acecare, Acesedate, ACP) **POM-V**

Formulations: Injectable: 2 mg/ml solution. Oral: 10 mg, 25 mg tablets.

Action: Phenothiazine with depressant effect on the CNS, thereby causing sedation and a reduction in spontaneous activity.

Use: Sedation or pre-anaesthetic medication in small mammals and other exotic species. ACP raises the threshold for cardiac arrhythmias and has antiemetic properties. Sedation is unreliable when ACP is used alone; combining ACP with an opioid drug improves sedation (neuroleptanalgesia) and the opioid provides analgesia. The depth of sedation is dose-dependent up to a plateau (0.1–5 mg/kg dependent on species and size). Increasing the dose above this does little to improve the predictability of achieving adequate sedation but increases the risk of incurring adverse effects, the severity of adverse effects and the duration of action of any effects (desirable or adverse). Onset of sedation is 20–30 min after i.m. administration; clinical doses cause sedation for up to 6 hours. The oral dose of ACP tablets required to produce sedation varies between individual animals and high doses can lead to very prolonged sedation.

Safety and handling: Normal precautions should be observed.

Contraindications: Hypotension due to shock, trauma or cardiovascular disease. Avoid in animals <3 months old and animals with liver disease. Avoid in gerbils as lowers the seizure threshold in susceptible strains. Not recommended in birds.

Adverse reactions: Rarely, healthy animals may develop profound hypotension following administration of phenothiazines. Supportive therapy to maintain body temperature and fluid balance is indicated until the animal is fully recovered. Can lead to seizures in gerbils.

Drug interactions: Other CNS depressant agents (e.g. barbiturates, propofol, alfaxalone, volatile anaesthetics) will cause additive CNS depression if used with ACP. Doses of other anaesthetic drugs should be reduced when ACP has been used for premedication. Quinidine given with phenothiazines may cause additional cardiac depression. Increased levels of both drugs may result if propranolol is administered with phenothiazines. As phenothiazines block alpha-adrenergic receptors, concomitant use with adrenaline may lead to unopposed beta activity, thereby causing vasodilation and tachycardia. Antidiarrhoeal mixtures (e.g. kaolin/pectin, bismuth salicylate) and antacids may cause reduced GI absorption of oral phenothiazines.

DOSES

When used for sedation is generally given as part of a combination. See Appendix for sedation protocols in all species.

Mammals: Ferrets: 0.2–0.5 mg/kg i.m., s.c., p.o.; Rabbits: 0.1–1.0 mg/kg i.m., s.c.; Guinea pigs: 0.5–5 mg/kg i.m., s.c., p.o.; Rats: 0.5–2.5 mg/kg i.m., s.c., p.o.; Mice: 0.5–5 mg/kg i.m., s.c., p.o.; Hamsters: 0.5–5 mg/kg i.m., s.c., p.o.; Gerbils: Not recommended; Primates (small): 0.5–1 mg/kg p.o., s.c., i.m.; Sugar gliders: 1–2 mg/kg i.m., s.c., p.o.; Hedgehogs: 0.1–1 mg/kg p.o., s.c., i.m. The higher doses should only be given orally.

Birds: Not recommended.

Reptiles: 0.1–0.5 mg/kg i.m.^a.

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A**Amphibians, Fish:** No information available.**B****References**

- * Alves-Junior JR, Bosso AC, Andrade MB *et al.* (2012) Association of acepromazine with propofol in giant Amazon turtles (*Podocnemis expansa*) reared in captivity. *Acta Cirurgica Brasileira* **27**(8), 552–556

C**D****Acetaminophen** see **Paracetamol****E****F****Acetazolamide**(Diamox*, Diamox SR*) **POM****G****Formulations:** Injectable: 500 mg vial (powder for reconstitution).

Oral: 250 mg tablets, capsules.

H**Action:** Systemic carbonic anhydrase inhibitor.**I**

Use: Treatment of gas bubble disease in seahorses but also requires the correction of contributing causes such as bacterial dermatitis, faulty air pumps and algal overgrowth. Has also been used to treat ocular gas bubbles (both intraocular and retrobulbar) in fish caused by supersaturation of the water or suspected disorders of the pseudobranch or choroidal rete.

J**K**

Safety and handling: Normal precautions should be observed. The use of 100 micron (or smaller) sock filters are advised to help remove tableting compounds when crushed tablets are used as an immersion treatment.

L**Contraindications:** No information available.**M**

Adverse reactions: In fish, the drug may interfere with normal carbonic anhydrase activity in the vascular rete of other organs such as the swim bladder and result in disturbance of normal buoyancy. Neurological signs and mortality have been associated with high doses.

N**O****Drug interactions:** No information available.**P****DOSES****Q**

Fish: Seahorses: 6 mg/l by immersion, change daily for 4–8 days; 2.5–5.0 mg/kg i.m. q3–7days for up to 3 doses, intracoelomic for gas bubble disease; **Other species:** 6 mg/kg by peribulbar injection.

R**S****Mammals, Birds, Reptiles, Amphibians:** No information available.**T****U****Acetic acid****V**

Formulations: Available in a variety of concentrations including glacial acetic acid (100%) and vinegar formulations (5–20%).

Action: Not known.**W**

Use: Treatment of ectoparasites (e.g. protozoans, monogeneans, hirudinea, crustaceans) in fish. Other treatments are considered safer to use. Management of intestinal dysbiosis in birds, especially macrorhabdiosis. While evidence for efficacy is low (most data is taken from use in broiler poultry), this is much safer than any of the alternative therapies for macrorhabdiosis and may be suitable for use in affected flocks or for in-contact birds. For sick birds, amphotericin-B is a more proven therapy to be given on an individual basis.

X**Y****Z**

Safety and handling: Corrosive to skin and irritant to mucous membranes.

Contraindications: No information available.

Adverse reactions: May be toxic to small tropical fish.

Drug interactions: No information available.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Birds: 15 ml apple cider vinegar/l drinking water. Alternatively quarter to half a teaspoon daily for adult birds in fresh food; 1–2 drops daily in handrearing formula for chicks.

Fish: 2 ml/l by immersion (glacial acetic acid) for 30–60 seconds.

Mammals, Reptiles, Amphibians: No information available.

Acetylcysteine

(StromEase, Ilube*, Parvolex*) **POM-V, POM**

Formulations: **Injectable:** 200 mg/ml solution. **Topical:** 25 mg/ml eye drops (StromEase), 5% ophthalmic solution in combination with 0.35% hypromellose ophthalmic drops in 10 ml bottle.

Action: Decreases the viscosity of bronchial secretions, maintains glutathione levels in the liver and has some anticollagenase activity.

Use: As a mucolytic in respiratory disease. Reduces the extent of liver injury in cases of paracetamol poisoning but no reports on this use in exotic species. In rabbits direct application into the ear has been reported as beneficial in cases of secretory otitis media, reducing inflammation and preventing long-term fibrotic changes. In birds, used as a mucolytic agent via nebulization for respiratory disease and to improve penetration of nebulized drugs through mucus exudates. In reptiles, it may be applied to retained spectacles to aid removal.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Acetylcysteine has caused hypersensitivity and bronchospasm when used in the pulmonary tree. When given orally for paracetamol poisoning it may cause GI effects (nausea, vomiting) and, rarely, urticaria.

Drug interactions: In cases of paracetamol poisoning the concurrent administration of activated charcoal is controversial as it may also reduce acetylcysteine absorption.

DOSES

Mammals: Ferrets, Rabbits, Rodents, Primates: Mucolytic: 20 mg/ml solution (dilute with saline) nebulized for 30–60 min as required (e.g. q6–12h); Otic lavage: 1–2 ml of a 20% solution.

Birds: Mucolytic: 22 mg/ml sterile water for nebulization. May be combined with other drugs (e.g. amikacin at 6 mg/ml, gentamicin at 10 mg/ml or kanamycin at 3–15 mg/ml).

Reptiles: Mucolytic: 20 mg/ml solution (dilute with saline) nebulized for 15–30 min as required.

Amphibians, Fish: No information available.

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A**Acetylsalicylic acid** see **Aspirin****B****C****Aciclovir**(Aciclovir*, Zovirax*) **POM****D**

Formulations: Ophthalmic: 3% ointment in 4.5 g tubes. Oral: 200 mg, 800 mg tablets; 200 mg/5 ml, 400 mg/5 ml suspension. **Injectable:** 250 mg, 500 mg vials for reconstitution. **Topical:** 5% ointment in 10 g, 25 g tubes.

E**F**

Action: Inhibits viral replication (viral DNA polymerase); depends on viral thymidine kinase for phosphorylation.

G**H**

Use: Management of ocular herpesvirus lesions in mammals although unable to eradicate latent viral infection. Used to treat systemic herpesvirus infections in various avian species (e.g. Pacheco's disease in psittacine birds) and also for prophylaxis of psittacine herpesvirus in outbreaks. In reptiles, it is used to treat chelonian herpesvirus infections.

I**J**

Safety and handling: Normal precautions should be observed.

K

Contraindications: No information available.

L

Adverse reactions: Ocular irritation may occur and the frequency of application should be reduced if this develops. Treatment should not be continued for >3 weeks. May cause vomiting in birds.

M

Drug interactions: Avoid concurrent administration of benzylpenicillin as associated with reduced elimination of acyclovir in red-eared sliders^{b,1}.

N**DOSES****O**

Mammals: Rabbits: Topically, 1–5 times daily^a.

P

Birds: Raptors: 330 mg/kg p.o. q12h for 7–14 days; **Psittacids:** 20–40 mg/kg i.m. q12h or 80 mg/kg p.o. q8h for 7 days^{b,c} or 400 mg/kg in food in aviaries^c; **Canaries:** 80 mg/kg p.o. q6h for 10 days for canary pox¹.

Q

Reptiles: Testudo species: 80 mg/kg p.o. q8–24h; topically to oral lesions q8–24h^d; **Terrapene species:** 40 mg/kg p.o. q24h^e.

R

Amphibians, Fish: No information available.

S**References****T**

^a Trousdale MD, Dunkel EC and Nesburn AB (1980) Effect of acyclovir on acute and latent herpes simplex virus infections in the rabbit. *Investigative Ophthalmology & Visual Science* **19**(11), 1336–1341

U

^b Norton TM, Gaskin J, Kollias GV et al. (1991) Efficacy of acyclovir against herpesvirus infection in Quaker parakeets. *American Journal of Veterinary Research* **52**(12), 2007–2009

V

^c Norton TM, Kollias GV, Clark CH et al. (1992) Acyclovir (Zovirax®) Pharmacokinetics in Quaker parakeets, (*Myiopsitta monachus*). *Journal of Veterinary Pharmacology and Therapeutics* **15**(3), 252–258

W

^d Gaio C, Rossi T, Villa R et al (2007) Pharmacokinetics of acyclovir after a single oral administration in marginated tortoises (*Testudo marginata*). *Journal of Herpetological Medicine and Surgery* **17**, 8–11

X

^e Allender MC, Mitchell MA, Yarborough J and Cox S (2013) Pharmacokinetics of a single oral dose of acyclovir and valacyclovir in North American box turtles (*Terrapene species*). *Journal of Veterinary Pharmacology and Therapeutics* **36**(2), 205–208

Y

¹ Mayahi M and Talazade F (2010) Case report: treatment of canary pox with orally administration of Acyclovir. *Journal of US-China Medical Science* **7**(4), 45–46

Z

ACP see **Acepromazine****Acriflavine (Acriflavinium chloride)**(Acriflavin, Acriflavine) **ESPA****Formulations:** Immersion: dry powder. Liquid: proprietary liquid formulations, occasionally mixed with proflavine.**Action:** No information available.**Use:** Treatment of external fungal, bacterial and parasitic diseases in fish. Some bacterial resistance exists and acriflavine is considered to be less effective than other medications. It is strongly recommended that a proprietary formulation is used initially to avoid problems related to purity and enable accurate dosing.**Safety and handling:** Powdered formulation is irritant and potentially mutagenic.**Contraindications:** No information available.**Adverse reactions:** May cause reproductive failure in some species (e.g. guppies). Will kill aquatic plants.**Drug interactions:** No information available.**DOSES****See Appendix for guidelines on responsible parasiticide use.****Fish:** 5 mg/l by immersion for 3–5 days, 10 mg/l by immersion for 2 h or 500 mg/l for 30 min; use proprietary formulations at the manufacturer's recommended dose rate.**Mammals, Birds, Reptiles, Amphibians:** No information available.**Acriflavinium chloride** see **Acriflavine****ACTH** see **Tetracosactide****Activated charcoal** see **Charcoal****Adrenaline (Epinephrine)**(Adrenaline*, Epinephrine*) **POM****Formulations:** **Injectable:** Range of concentrations for injection: 0.1–10 mg/ml, equivalent to 1:10,000 to 1:100.**Action:** Adrenaline exerts its effects via alpha-1 and -2, and beta-1 and -2 adrenoreceptors.**Use:** Cardiac resuscitation, status asthmaticus and to offset the effects of histamine release in severe anaphylactoid reactions. The effects of adrenaline vary according to dose. Infusions of low doses mainly result in beta-adrenergic effects (increases in cardiac output, myocardial oxygen consumption, and a reduced threshold for arrhythmias with peripheral vasodilation and a fall in diastolic blood pressure). At high doses alpha-1 effects predominate, causing a rise in systemic vascular resistance, diverting blood to the central organs; however, this may improve cardiac output and blood flow. Adrenaline is not a substitute for**A****B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A fluid replacement therapy. Respiratory effects include bronchodilation and an increase in pulmonary vascular resistance. Renal blood flow is moderately decreased. The duration of action of adrenaline is short (2–5 min). Beware of using in animals with diabetes mellitus (monitor blood glucose concentration), hypertension or hyperthyroidism. Use with caution in hypovolaemic animals. Overdosage can be fatal so check dose, particularly in small patients. Intracardiac injection is not recommended. Used for the treatment of cardiac arrest in fish.

E **Safety and handling:** Do not confuse adrenaline vials of different concentrations. Adrenaline is sensitive to light and air: do not use if it is pink, brown or contains a precipitate. It is unstable in 5% dextrose.

F **Contraindications:** The use of human adrenaline pen injections is not recommended for the treatment of suspected anaphylaxis. The doses in such pens are usually too small to be effective in most normal animals and by the time the animal has collapsed would be unlikely to have any effect on outcome. If such pen injections are administered by owners, then, in common with medical practice, patients must be carefully monitored for at least 6 hours.

J **Adverse reactions:** Increases myocardial oxygen demand and produces arrhythmias including ventricular fibrillation. These may be ameliorated by administering oxygen and slowing the heart rate with beta-2 antagonists. Other adverse effects include tachycardia, arrhythmias, dry mouth and cold extremities. Repeated injections can cause necrosis at the injection site.

M **Drug interactions:** Toxicity may occur if used with other sympathomimetic amines because of additive effects. The effects of adrenaline may be potentiated by antihistamines and thyroxine. Propranolol may block the beta effects of adrenaline, thus facilitating an increase in blood pressure. Alpha blocking agents or diuretics may negate or diminish the pressor effects. When adrenaline is used with drugs that sensitize the myocardium (e.g. high doses of digoxin) monitor for signs of arrhythmias. Hypertension may result if adrenaline is used with oxytocic agents.

DOSES

Mammals: Ferrets: 20 µg (micrograms)/kg s.c., i.m., i.v., intratracheal;

Rabbits: Cardiac resuscitation: 100 µg/kg i.v., repeated and/or higher doses (up to 200 µg/kg) may be required at intervals of 2–5 min;

Guinea pigs: 3 µg/kg i.v.; **Other rodents:** 10 µg/kg i.v. as required.

Birds: 0.1–1.0 mg/kg i.m., i.v., intraosseous, intracardiac, intratracheal.

Reptiles: Resuscitation: 0.5 mg/kg i.v., intraosseous; 1 mg/kg intratracheal diluted in 1 ml saline/100 g bodyweight; To speed recovery from anaesthesia: 0.1 mg/kg i.m.^a.

Fish: 0.01–0.5 mg/kg i.m., i.v., i.p., intracardiac.

Amphibians: No information available.

References

^a Goe A, Shmalberg J, Gatson B et al. (2016) Epinephrine or GV-26 electrical stimulation reduces inhalant anesthetic recovery time in common snapping turtles (*Chelydra serpentina*). *Journal of Zoo and Wildlife Medicine* **47**(2), 501–507

Aglepristone

(Alizin) **POM-V**

Formulations: Injectable: 30 mg/ml solution.

Action: Progesterone receptor blockage leads to reduced progesterone support for pregnancy.

Use: Termination of pregnancy in mammals. No specific information is available for exotic mammal species. In bitches confirmed as pregnant, a partial abortion may occur in up to 5% and a similar situation may occur in other mammals; owners should be warned. A clinical examination (uterine palpation) is always recommended after treatment in order to confirm termination. After induced abortion an early return to oestrus is frequently observed (the oestrus-to-oestrus interval may be significantly shortened). Can also be used for the treatment of pyometra in some mammals, although recurrence is fairly common.

Safety and handling: Use with care. People who are or may become pregnant should not handle this drug.

Contraindications: Consider avoiding in animals with diagnosed or suspected hypoadrenocorticism or diabetes mellitus.

Adverse reactions: Transient pain at the injection site; any local inflammation produced resolves uneventfully. In animals treated in the later stages of gestation, abortion may be accompanied by the physiological signs of parturition, i.e. fetal expulsion, anorexia, mammary congestion.

Drug interactions: Aglepristone binds to glucocorticoid receptors and may therefore interfere with the actions of glucocorticoids; however, the clinical significance of this is unclear.

DOSES

Mammals: Rabbits: Pregnancy termination: 10 mg/kg on days 6 and 7 post implantation; Guinea pigs: Pyometra/metritis: 10 mg/kg on days 1, 2 and 8; Rats: 10 mg/kg s.c. 24 hours apart on days 11 and 12 after mating (i.e. at mid-pregnancy)¹.

Birds, Reptiles, Amphibians, Fish: No information available.

References

- ¹ Gogny A and Fiéni F (2016) Aglepristone: A review on its clinical use in animals. *Theriogenology* **85**(4), 555–566

Alfaxalone

(Alfaxan) **POM-V**

Formulations: Injectable: 10 mg/ml solution; the alfaxalone is solubilized in a cyclodextrin. Single or multidose bottles are available. Multidose bottles contain ethanol, chlorocresol and benzethonium chloride as preservative.

Action: Anaesthesia induced by the CNS depressant effect of the alfaxalone.

Use: Induction agent used before inhalational anaesthesia, or as a sole anaesthetic agent for examination or surgical procedures. Authorized in pet rabbits as an induction agent prior to inhalation anaesthesia. As with all i.v. anaesthetic drugs, premedication will reduce the dose required for induction and maintenance of anaesthesia. Pre-oxygenation is

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A recommended before i.v. induction as transient post-induction apnoea can occur. The drug should be given slowly and to effect in order to prevent inadvertent overdose. Analgesia is insufficient for surgery; other analgesic drugs such as opioids should be incorporated into the anaesthetic protocol. Alfaxalone is shorter acting and causes less excitement during recovery than the alfaxalone/alfadalone combination previously available. Alfaxalone can be given i.m. or s.c. to provide sedation. The drug should be administered in the cranial half in reptiles to avoid shunting to the liver and first-pass metabolism. Do not use in combination with other i.v. anaesthetic agents. Used for the induction and maintenance of anaesthesia in fish; the alfaxalone/alphadalone combination previously available has been used parenterally in large fish.

Safety and handling: The single-dose vial does not contain an antimicrobial preservative; thus it is recommended that the remainder of an opened bottle is discarded after single use within 24 hours.

Contraindications: The use of alfaxalone by i.m. injection in koi carp is not recommended since it can produce prolonged anaesthesia lasting several hours, with prolonged apnoea and high mortality.

Adverse reactions: In mammals a slight increase in heart rate can occur immediately after i.v. injection as a compensatory response to maintain blood pressure in the face of mild hypotension. This effect can be minimized by slow i.v. injection. Transient post-induction apnoea can also occur. As with all anaesthetic drugs, respiratory depression can occur with overdoses. In Quaker parrots, alfaxalone alone at 10mg/kg i.m. induced muscle tremors. These were improved if midazolam was given as well, though recoveries were longer.

Drug interactions: No information available.

DOSES

See Appendix for sedation protocols in all species.

Mammals: Unpremedicated: **Ferrets:** 5–10 mg/kg i.v., i.m. (provides variable sedation); **Guinea pigs:** 20 mg/kg i.m., i.p.; **Rats:** 2–5 mg/kg i.p., i.v.; **Other rodents:** 20 mg/kg i.m.; **Marmosets:** 12 mg/kg i.m.;

Other primates: 2 mg/kg s.c. with medetomidine and midazolam. Premedicated: **Rabbits:** Authorized for use following premedication with opioid/alpha-2: 1–3 mg/kg i.v. or 3–6 mg/kg i.m.^{a,b}; **Rats:** 2–3 mg/kg i.m. following premedication with opioid and midazolam; **Primates:** Induction with 0.5 mg/kg i.v. after sedation with dexmedetomidine, butorphanol and midazolam or 1–4 mg/kg i.m. with ketamine or butorphanol/medetomidine; **Hedgehogs:** 3–5 mg/kg s.c. with midazolam.

Birds: Significant species differences have been reported. For large birds and those with a dive response: 2–4 mg/kg i.v. to effect; **Budgerigars:** 10–15 mg/kg s.c.^c (lower dose suitable for shorter non-invasive investigations); **Bengalese finches:** 10–50 mg/kg s.c. with or without midazolam and/or butorphanol. Increasing dose linked with increasing length of sedation rather than depth^d.

Reptiles: 5–12 mg/kg i.v., intraosseous for induction (light sedation) up to 30 mg/kg i.m. for surgical anaesthesia. Effects are dependent on species, injection site and temperature; **Chelonians:** 5 mg/kg i.v. or 10 mg/kg i.m. (light sedation) up to 20 mg/kg i.m. for surgical anaesthesia; **Macquarie river turtles:** 9 mg/kg i.v.; **Lizards, Snakes:** <9 mg/kg i.v.; **Green iguanas:** 10–20 mg/kg i.m. or 5 mg/kg i.v. (for sufficient sedation to allow intubation); **Veiled chameleons:** 5 mg/kg i.v.

(for sufficient sedation to allow intubation); **Bearded dragons**: 5–12 mg/kg i.v.; **Chinese water dragons**, **Leopard geckos**: 5 mg/kg i.v.; **Galliwasp**: 15 mg/kg i.m.; **Ball pythons**: 10–30 mg/kg i.m.; **Garter snakes**: 30 mg/kg intracoelomic; **Corn snakes**: 15 mg/kg s.c.^e.

Amphibians: Only provides light sedation when administered i.m. or by immersion bath. **Bullfrogs**: 10–17.5 mg/kg i.m.; **Australian green tree frogs**: 30 mg/kg i.m.; **Green and golden bell frogs**, **Booroolong frogs**: 20 mg/kg i.m.^f; **Oriental fire-bellied toads**: 200 mg/l immersion bath; **Blue poison dart frogs**: 20–30 mg/kg s.c.; **Spanish ribbed newt**: 15–30 mg/kg i.m.

Fish: 5–10 mg/l by immersion for induction of anaesthesia; 1–2.5 mg/l by immersion for maintenance of anaesthesia^g.

References

- ^a Grint NJ, Smith HE and Senior JM (2008) Clinical evaluation of alfaxalone in cyclohexadrin for the induction of anaesthesia in rabbits. *Veterinary Record* **163**(13), 395–396
- ^b Huynh M, Poumeyrol S, Pignon C, Le Teuff G and Zilberstein L (2014) Intramuscular administration of alfaxalone for sedation in rabbits. *Veterinary Record* **176**, 255
- ^c Escalante GC, Balko JA and Chinnadurai SK (2018) Comparison of the sedative effects of alfaxalone and butorphanol-midazolam administered intramuscularly in budgerigars (*Melopsittacus undulatus*). *Journal of Avian Medicine and Surgery* **32**(4), 279–285
- ^d Perrin KL, Nielsen JB, Thomsen AF and Bertelsen MF (2017) Alfaxalone anaesthesia in the Bengalese finch (*Lonchura domestica*). *Journal of Zoo and Wildlife Medicine* **48**(4), 1146–1153
- ^e Rockwell K, Boykin K, Padlo J *et al.* (2021) Evaluating the efficacy of alfaxalone in corn snakes (*Pantherophis guttatus*). *Veterinary Anaesthesia and Analgesia* **48**(3), 364–371
- ^f Sladakovics I, Johnson RS and Vogelneist L (2014) Evaluation of intramuscular alfaxalone in three Australian frog species (*Litoria caerulea*, *Litoria aurea*, *Litoria booroolongensis*). *Journal of Herpetological Medicine and Surgery* **24**(1), 36–42
- ^g Minter LJ, Bailey KM, Harms CA, Lewbart GA and Posner LP (2014) The efficacy of alfaxalone for immersion anaesthesia in koi carp (*Cyprinus carpio*). *Veterinary Anaesthesia and Analgesia* **41**(4), 398–405

Allopurinol

(Zyloric*) **POM**

Formulations: Oral: 100 mg, 300 mg tablets.

Action: Xanthine oxidase inhibition decreases formation of uric acid by blocking the conversion of hypoxanthine to xanthine, and of xanthine to uric acid.

Use: Treatment of hyperuricaemia and articular gout in birds and reptiles. It should be remembered that, unlike mammals, uric acid is a natural end-product of protein metabolism in birds and reptiles. Therefore use of allopurinol is symptomatic only and the primary cause should be addressed. Ensure patients are adequately hydrated. Clinical response is variable in birds (e.g. studies in red-tailed hawks showed no significant effect on plasma uric acid levels). The dosage should be reduced as plasma uric acid levels decrease.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: In red-tailed hawks, doses of 50 mg/kg and 100 mg/kg have been reported as toxic and causing renal failure.

Drug interactions: In humans, allopurinol may enhance the effects of azathioprine and theophylline.

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A**DOSES**

Birds: 10–30 mg/kg p.o. q12h (all species)^{a,b}; **Pigeons:** 830 mg/l drinking water; **Psittacids:** 300 mg/l drinking water.

Reptiles: 10–20 mg/kg p.o. q24h (most species); **Chelonians:** 50 mg/kg p.o. q24h for 30 days then q72h^c; **Green iguanas:** 25 mg/kg p.o. q24h^d.

Mammals, Amphibians, Fish: No information available.

B**References****C**

^a Lumeij JT, Sprang EPM and Redig PT (1998) Further studies on allopurinol-induced hyperuricaemia and visceral gout in Red-tailed hawks (*Buteo jamaicensis*). *Avian Pathology* **24**(4), 390–393

D

^b Poffers J, Lumeij JT, Timmermans-Sprang EP and Redig PT (2002) Further studies on the use of allopurinol to reduce plasma uric acid concentrations in the Red-tailed hawk (*Buteo jamaicensis*) hyperuricaemic model. *Avian Pathology* **31**(6), 567–572

E

^c Kolle P (2001) Efficacy of allopurinol in European tortoises with hyperuricaemia. *Proceedings of the Association of Reptilian and Amphibian Veterinarians*, p. 186

F

^d Hernandez-Divers SJ, Martinez-Jimenez D, Bush S *et al.* (2008) Effects of allopurinol on plasma uric acid levels in normouricaemic and hyperuricaemic green iguanas (*Iguana iguana*). *Veterinary Record* **162**(4), 112–115

G

Alphaxalone see **Alfaxalone**

H**I**

Aluminium antacids (Aluminium hydroxide)

J

(Alucap*. With alginate: Acidex*, Gastrocote*, Gaviscon Advance*, Peptac*. With magnesium salt: Asilone*, Maalox*, Mucogel*) **P, GSL**

K**L**

Formulations: Oral: Aluminium hydroxide is available as a dried gel. Other products are composite preparations containing a variety of other compounds including magnesium oxide, hydroxide or trisilicate, potassium bicarbonate, sodium carbonate and bicarbonate, alginates and dimeticone. Aluminium hydroxide content varies.

M

Action: Neutralizes gastric hydrochloric acid. May also bind bile acids and pepsin and stimulate local prostaglandin (PGE-1) production. Also binds inorganic phosphate (PO_4^{3-}) in the GI tract, making it unavailable for absorption.

N

Use: Management of gastritis and gastric ulceration. In renal failure, to lower serum phosphate levels in animals with hyperphosphataemia. Frequent administration is necessary to prevent rebound acid secretion. Phosphate-binding agents are usually only used if low-phosphate diets are unsuccessful. Monitor serum phosphate levels at 10–14 day intervals and adjust dosage accordingly if trying to normalize serum concentrations. Thoroughly mix the drug with food to disperse it throughout the GI tract and to increase its palatability.

O

Safety and handling: Long-term use (many years) of oral aluminium products in humans has been associated with aluminium toxicity and possible neurotoxicity. This is unlikely to be a problem in veterinary medicine.

P

Contraindications: No information available.

Q

Adverse reactions: Constipation may occur. This is an effect of the aluminium compound and is counteracted by inclusion of a magnesium

R**S****T****U****V****W****X****Y****Z**

salt. Reduced food intake and bodyweight may result at the higher doses used in rats.

Drug interactions: Do not administer digoxin, tetracycline or fluoroquinolone products orally within 2 hours of aluminium salts as their absorption may be impaired.

DOSES

Mammals: Rabbits: 30–60 mg/kg p.o. q8–12h; Guinea pigs: 20–40 mg/animal p.o. prn for hyperphosphataemia due to renal failure; Rats: 2.4–9.6 g/kg/day for chronic renal failure^a.

Birds: 30–90 mg/kg p.o. q12h.

Reptiles: 100 mg/kg p.o. q24h.

Amphibians, Fish: No information available.

References

- ^a Sanai T, Okuda S, Onoyama K *et al.* (1991) Effect of different doses of aluminium hydroxide on renal deterioration and nutritional state in experimental chronic renal failure. *Mineral and Electrolyte Metabolism* **17**(3), 160–165

Aluminium hydroxide see Aluminium antacids

Amantadine

(Lysovir*, Symmetrel*) **POM**

Formulations: Oral: 100 mg capsule; 10 mg/ml syrup.

Action: Provides analgesia through NMDA antagonist action which may potentiate the effects of other analgesics. Anti-viral activity proposed for avian influenza and bornavirus infections. In birds, pharmacokinetics have been studied in orange-winged Amazons, but there have been no studies on analgesic efficacy; for this reason it is not recommended as a first-line analgesic.

Use: Adjunctive analgesic in animals that are unresponsive to opioids, or that require chronic pain relief in a home environment (e.g. osteoarthritis or cancer pain). Proventricular dilatation disease (PDD) in parrots.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: In humans, minor GI and CNS effects have been reported, although these have not been reported in animals.

Drug interactions: No information available.

DOSES

Mammals: Ferrets: 3–5 mg/kg. May have a role in the treatment of neuropathic pain in rabbits and rodents at canine doses (3–5 mg/kg p.o. q24h).

Birds: 25 mg/kg p.o. q12h; Orange-winged Amazon parrots: 5 mg/kg p.o. q24h^a.

Reptiles, Amphibians, Fish: No information available.

References

- ^a Berg KJ, Guzman DSM, Knych HK, Drzenovich TL and Paul-Murphy JR (2020) Pharmacokinetics of amantadine after oral administration of single and multiple doses to orange-winged Amazon parrots (*Amazona amazonica*). *American Journal of Veterinary Research* **81**(8), 651–655

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A**Amethocaine** see **Tetracaine****B****C****Amikacin**(Amikacin*, Amikin*) **POM****D****Formulations:** Injectable: 50 mg/ml, 250 mg/ml solutions.**E****Action:** Aminoglycosides inhibit bacterial protein synthesis. They are bactericidal and their mechanism of killing is concentration-dependent, leading to a marked post-antibiotic effect, allowing prolonged dosing intervals (which may reduce toxicity).**F****G****Use:** Active against many Gram-negative bacteria, *Staphylococcus aureus* and *Nocardia* spp., including some that may be resistant to gentamicin. *Streptococci* and anaerobes are usually resistant. Its use is only indicated after sensitivity testing has been performed and the organism shown to be resistant to other aminoglycosides such as gentamicin. Activity at low oxygen sites may be limited. Movement across biological membranes may also be limited, hence systemic levels require parenteral administration, and access to sites such as the CNS and ocular fluids is very limited. Monitoring serum amikacin levels should be considered to ensure therapeutic levels and minimize toxicity, particularly in neonates, geriatric patients and those with reduced renal function. Monitoring renal function is also advisable during treatment of any animal. Intravenous doses should be given slowly, generally over 30–60 min. Concurrent fluid therapy is advised.**L****M****Safety and handling:** Normal precautions should be observed.**N****Contraindications:** If possible avoid use in animals with reduced renal function.**O****Adverse reactions:** Nephrotoxic and ototoxic. Due to potential nephrotoxicity in birds it should be used with care, but it has been used for treating resistant *Pseudomonas* infections**P****Q****Drug interactions:** Synergism may occur *in vivo* when aminoglycosides are combined with beta-lactam antimicrobials. Avoid the concurrent use of other nephrotoxic, ototoxic or neurotoxic agents (e.g. amphotericin B, furosemide). Aminoglycosides may be inactivated *in vitro* by beta-lactam antibiotics (e.g. penicillins, cephalosporins) or heparin; do not give these drugs in the same syringe. Can potentiate neuromuscular blockade so avoid use in combination with neuromuscular blocking agents.**R****S****T****DOSES****U****Classified as category C (Caution) by the EMA.****See Appendix for guidelines on responsible antibiotic use.****V****Mammals:** Ferrets: 8–16 mg/kg i.v., i.m., s.c. q8–24h; Rabbits: 2–10 mg/kg i.v., i.m., s.c. q8–12h; **Rodents:** 5–15 mg/kg i.v., i.m., s.c. q8–12h. Concurrent fluid therapy advised, especially if hydration status poor or uncertain. Nebulization: 10 mg in 10 ml of water for injection over 15–30 minutes q6–12h.**W****X****Birds:** Red-tailed hawks: 15–20 mg/kg i.m. q24h^a; Cockatiels, Amazon parrots: 10–20 mg/kg i.m., s.c., i.v. q8–12h^b. Nebulization: 6 mg/ml (may be given in combination with acetylcysteine).**Y****Z****Reptiles:** Gopher tortoises: 5 mg/kg i.m. q48h^c; Gopher snakes: 5 mg/kg i.m. once, then 2.5 mg/kg i.m. q72h at 25°C^d; concurrent fluid

therapy is advised due to potential nephrotoxicity; **Ball pythons:** 3.48 mg/kg i.m. once^e. Nebulization: 5 mg/ml for 15–30 minutes q12h.

Amphibians: 5–10 mg/kg s.c., i.m. q24–48h¹.

Fish: 5 mg/kg i.m., intracoelomic q72h for 3 treatments.

References

- ^a Bloomfield RB, Brooks D and Vulliet R (1997) The pharmacokinetics of a single intramuscular dose of amikacin in red-tailed hawks (*Buteo jamaicensis*). *Journal of Zoo and Wildlife Medicine* **28**(1), 55–61
- ^b Ramsay EC and Vulliet R (1993) Pharmacokinetic properties of gentamicin and amikacin in the cockatiel. *Avian Diseases* **37**(2), 628–634
- ^c Caligiuri R, Kollias GV, Jacobson E et al. (1990) The effects of ambient temperature on amikacin pharmacokinetics in gopher tortoises. *Journal of Veterinary Pharmacology and Therapeutics* **13**, 287–291
- ^d Mader DR, Conzelman GM Jr and Baggot JD (1985) Effects of ambient temperature on the half-life and dosage regimen of amikacin in the gopher snake. *Journal of the American Veterinary Medical Association* **187**(11), 1134–1136
- ^e Johnson JH, Jensen JM, Brumbaugh GW and Booth DM (1997) Amikacin pharmacokinetics and the effects of ambient temperature on the dosing regimen in ball pythons (*Python regius*). *Journal of Zoo Wildlife and Medicine* **28**(1), 80–88
- ¹ Letcher J and Papich M (1994) Pharmacokinetics of intramuscular administration of three antibiotics in bullfrogs (*Rana catesbeiana*). *Proceedings of the Association of Reptilian and Amphibian Veterinarians/American Association of Zoo Veterinarians Conference*, pp. 79–93

Amino acid solutions

(Duphalyte, Aminoplasmal*, Aminoven*, Clinimix*, Glamin*, Hyperamine*, Intrafusin*, Kabiven*, Kabiven Peripheral*, Nutriflex*) **POM, POM-V**

Formulations: **Injectable:** Synthetic crystalline L-amino acid solutions for i.v. use only. Numerous human products are available, varying in concentrations of amino acids. Most products also contain electrolytes. Some products contain varying concentrations of glucose.

Action: Support protein anabolism, arrest protein and muscle wasting, and maintain intermediary metabolism.

Use: Amino acid solutions supply essential and non-essential amino acids for protein production. They are used parenterally in patients requiring nutritional support but unable to receive enteral support. The authorized veterinary preparation (Duphalyte) contains insufficient amino acids to meet basal requirements for protein production and is intended as an aid for i.v. fluid support. All products are hyperosmolar. The use of concentrated amino acid solutions for parenteral nutrition support should not be undertaken without specific training and requires central venous access and intensive care monitoring. Parenteral nutrition may also be able to meet the patient's requirements for fluids, essential electrolytes (sodium, potassium, magnesium) and phosphate. Additionally, if treatment is prolonged, vitamins and trace elements may need to be given. Intravenous lines for parenteral nutrition should be dedicated for that use alone and not used for other medications. As many of the available amino acids solutions contain potassium, the maximal acceptable rates of infusion will depend on the potassium content of the amino acid preparation.

Safety and handling: Normal precautions should be observed.

Contraindications: Dehydration, hepatic encephalopathy, severe azotaemia, shock, congestive heart failure and electrolyte imbalances.

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A

Adverse reactions: The main complications of parenteral nutrition are metabolic, including hyperglycaemia, hyperlipidaemia, hypercapnia, acid–base disturbances and electrolyte disturbances. Other complications include catheter-associated thrombophlebitis, bacterial colonization of the catheter and resulting bacteraemia and septicaemia. Potentially life-threatening electrolyte imbalances including hypophosphataemia may also be seen (also referred to as refeeding syndrome). As with other hyperosmolar solutions, severe tissue damage could occur if extravasated, though this has not been reported.

B**C****D****E**

Drug interactions: Consult specific product data sheet(s).

F

DOSES

G

Mammals: Rabbits, Rodents: Anecdotally, these products have been used either alone or diluted with lactated Ringer's solution at a 1:5 ratio and given at a total volume of approximately 100 ml/kg/day.

H

Birds: Up to 10 ml/kg/day.

I

Reptiles, Amphibians, Fish: No information available.

J

Aminophylline

K

(Aminophylline*) **POM**

L

Formulations: Injectable: 25 mg/ml solution. **Oral:** 100 mg tablet. For modified-release preparations **see Theophylline** (100 mg of aminophylline is equivalent to 79 mg of theophylline).

M**N**

Action: Aminophylline is a stable mixture of theophylline (q.v.) and ethylenediamine. Causes inhibition of phosphodiesterase, alteration of intracellular calcium, release of catecholamine, and antagonism of adenosine and prostaglandin, leading to bronchodilation and other effects.

O**P**

Use: Spasmolytic agent and has a mild diuretic action. It is used in the treatment of small airway disease. Beneficial effects include bronchodilation, enhanced mucociliary clearance, stimulation of the respiratory centre, increased sensitivity to $P_a\text{CO}_2$, increased diaphragmatic contractility, stabilization of mast cells and a mild inotropic effect. Aminophylline has a low therapeutic index and should be dosed on a lean bodyweight basis. Administer with caution in patients with severe cardiac disease, gastric ulcers, hyperthyroidism, renal or hepatic disease, severe hypoxia or severe hypertension. Therapeutic plasma aminophylline values are 5–20 µg(micrograms)/ml.

Q**R****S****T**

Safety and handling: Do not mix aminophylline in a syringe with other drugs.

U**V**

Contraindications: Patients with known history of arrhythmias or seizures.

W**X****Y****Z**

Adverse reactions: Vomiting, diarrhoea, polydipsia, polyuria, reduced appetite, tachycardia, arrhythmias, nausea, twitching, restlessness, agitation, excitement and convulsions. Most adverse effects are related to the serum level and may be symptomatic of toxic serum concentrations. The severity of these effects may be decreased by the use of modified-release preparations. They are more likely to be seen with more frequent administration. Aminophylline causes intense local pain when given i.m. and is very rarely used or recommended via this route.

Drug interactions: Agents that may increase the serum levels of aminophylline include cimetidine, diltiazem, erythromycin, fluoroquinolones and allopurinol. Phenobarbital may decrease the serum concentration of aminophylline. Aminophylline may decrease the effects of pancuronium. Aminophylline and beta-adrenergic blockers (e.g. propranolol) may antagonize each other's effects. Aminophylline administration with halothane may cause an increased incidence of cardiac dysrhythmias, and with ketamine an increased incidence of seizures.

DOSES

Mammals: Ferrets: 4.4–6.6 mg/kg p.o., i.m. q12h; Rabbits: 2.0 mg/kg i.v. q12h (slowly for emergency bronchodilation); Guinea pigs: 50 mg/kg p.o. q12h.

Birds: 4–5 mg/kg i.m., i.v. q6–12h; **Raptors:** 8–10 mg/kg p.o., i.m., i.v. q6–8h.

Reptiles: 2–4 mg/kg i.m. once.

Amphibians, Fish: No information available.

Amitriptyline

(Amitriptyline*) **POM**

Formulations: Oral: 10 mg, 25 mg, 50 mg tablets; 5 mg/ml, 10 mg/ml solutions.

Action: Blocks noradrenaline and serotonin re-uptake in the brain, resulting in antidepressive activity.

Use: Management of chronic anxiety problems, including 'compulsive disorders', separation anxiety and psychogenic alopecia. Used for a minimum of 30 days for feather plucking in birds. Amitriptyline is bitter and can be distasteful. Some caution and careful monitoring is warranted in patients with cardiac or renal disease.

Safety and handling: Normal precautions should be observed.

Contraindications: Hypersensitivity to tricyclic antidepressants, glaucoma, history of seizures or urinary retention, severe liver disease.

Adverse reactions: Sedation, dry mouth, vomiting, excitability, arrhythmias, hypotension, syncope, increased appetite, weight gain and, less commonly, seizures and bone marrow disorders have been reported in humans. The bitter taste may cause ptialism. Extrapyrimal side effects have been seen in a macaw.

Drug interactions: Should not be used with monoamine oxidase inhibitors or drugs metabolized by cytochrome P450 2D6, e.g. chlorphenamine, cimetidine. If changing medication from one of these compounds, a minimal washout period of 2 weeks is recommended (the washout period may be longer if the drug has been used for a prolonged period of time).

DOSES

Mammals: Rats: 5–20 mg/kg p.o. q24h.

Birds: 1–9 mg/kg p.o. q12h. Studies suggest higher dose rates generally needed. However, dosing should be started low and titrated upwards given the occurrence of severe extrapyramidal signs seen at 5 mg/kg^{a,b}.

Reptiles, Amphibians, Fish: No information available.

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A**References**

- ^a Barboza T and Beaufreire H (2017) Extrapryramidal side effects in a blue and gold macaw (*Ara ararauna*) treated with amitriptyline. *Journal of Veterinary Behaviour* **22**, 19–23
- ^b Visser M, Ragsdale MM and Boothe DM (2015) Pharmacokinetics of amitriptyline HCl and its metabolites in healthy African grey parrots (*Psittacus erithacus*) and cockatoos (*Cacatua species*). *Journal of Avian Medicine and Surgery* **29**(4), 275–281

B**C****D****E****Amlodipine**(Amodip, Amlodipine*, Istin*) **POM-V, POM****F**

Formulations: Oral: 1.25 mg (cats), 5 mg, 10 mg tablets; 1 mg/ml, 2 mg/ml sugar-free solutions. Special reformulations available: 0.3125 mg tablets; 12.5 mg/ml transdermal gel. **VSP**

G**H**

Action: Calcium-channel antagonist with predominant action affecting peripheral arteriolar vasculature, causing vasodilation and reducing afterload. Has mild negative inotropic and chronotropic effects, which are negligible at therapeutic doses.

I**J**

Use: Hypertension. Amlodipine is metabolized in the liver and dosage should be reduced when there is hepatic dysfunction.

K

Safety and handling: Normal precautions should be observed.

L

Contraindications: Avoid in cardiogenic shock, severe hepatic failure and pregnancy.

M

Adverse reactions: Lethargy, hypotension or inappetence are rare side effects.

N

Drug interactions: Little is known in birds. Hepatic metabolism may be impaired by drugs such as cimetidine, ciclosporin, ketoconazole and itraconazole and therefore increase circulating doses. CYP3A4 inducers (such as rifampin) may reduce circulating amlodipine levels. Hypotension is a risk if combined with other antihypertensives (e.g. ACE inhibitors, diuretics, beta-blockers).

O**P****DOSES**

Birds: Psittacids: 0.1–0.4 mg/kg p.o. q24h^a, 1 mg/kg p.o. q12h¹.

Q

Mammals, Reptiles, Amphibians, Fish: No information available.

R**References**

- ^a Fink DM and Mans C (2021) Use of amlodipine in psittacine birds: 5 cases (2010–2018). *Journal of Avian Medicine and Surgery* **35**(2), 155–160
- ¹ Montesinos Barcelo A and Ardiaca Garcia M (2017) *Guía de terapeutica en Animales Exoticos*. Multimedia Ediciones Veterinarias, Barcelona

S**T****U****V****Amoxicillin (Amoxycillin)**(Amoxinsol, Amoxycare, Amoxypen, Bimoxyl, Clamoxyl, Duphamox, Huvamox, Rhemox, Solamocta) **POM-V****W**

Formulations: Injectable: 150 mg/ml suspension. Oral: 40 mg, 200 mg, 250 mg tablets; suspension which when reconstituted provides 50 mg/ml.

X**Y**

Action: Binds to penicillin-binding proteins involved in bacterial cell wall synthesis, thereby decreasing cell wall strength and rigidity, affecting cell division, growth and septum formation. These antimicrobials act in a time-dependent bactericidal fashion.

Z

Use: Active against certain Gram-positive and Gram-negative aerobic organisms and many obligate anaerobes but not against those that produce penicillinases (beta-lactamases), e.g. *Escherichia coli*, *Staphylococcus aureus*. The more difficult Gram-negative organisms (*Pseudomonas*, *Klebsiella*) are usually resistant. Amoxicillin is excreted well in bile and urine. Oral amoxicillin may be given with or without food. Since amoxicillin works in a time-dependent fashion, it is important to maintain levels above the MIC for a high percentage of the time. In practical terms this means that the dosing interval is critical and missing doses can seriously compromise efficacy. In ferrets it is used in combination with bismuth subsalicylate, ranitidine or omeprazole and metronidazole ('triple therapy') for treatment of *Helicobacter mustelae* infection. Used to treat bacterial diseases in fish.

Safety and handling: Refrigerate oral suspension after reconstitution; discard if solution becomes dark or after 7 days.

Contraindications: Avoid oral antibiotics in critically ill patients, as absorption from the GI tract may be unreliable. Avoid use in animals with reported sensitivity to penicillins. Do not administer penicillins by any route to guinea pigs, chinchillas, hamsters, gerbils or degus. Do not administer oral penicillins to rabbits.

Adverse reactions: Nausea, diarrhoea and skin rashes are the commonest adverse effects. Oral doses can cause fatal enterotoxaemia in small herbivores (e.g. rabbits), hamsters and gerbils.

Drug interactions: Avoid concurrent use with bacteriostatic antibiotics (e.g. tetracycline, erythromycin, chloramphenicol). Do not mix in the same syringe as aminoglycosides. A synergistic effect is seen when beta-lactam and aminoglycoside antimicrobials are used concurrently.

DOSES

Classified as category D (Prudence) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Ferrets: 10–30 mg/kg s.c., p.o. q12h; Rabbits: 7 mg/kg s.c. q24h; Rats, Mice: 100–150 mg/kg i.m., s.c. q12h; Primates: 11 mg/kg p.o. q12h or 11 mg/kg s.c., i.m. q24h; Prosimians: 10–20 mg/kg p.o. q 12h; Sugar gliders: 30 mg/kg p.o., s.c. q12–24h for 14 days; Hedgehogs: 15 mg/kg p.o., s.c., i.m. q8–12h. Guinea pigs, Chinchillas, Hamsters, Gerbils, Degus: Do not use.

Birds: 150–175 mg/kg i.m., s.c. q8–12h (q24h for long-acting preparations); Raptors, Parrots: 175 mg/kg p.o. q12h; Pigeons: 1–1.5 g/l drinking water q24h for 3–5 days or 100–200 mg/kg p.o. q6–8h; Waterfowl: 1 g/l drinking water alternate days for 3–5 days or 300–500 mg/kg p.o. (in soft food) for 3–5 days; Passerines: 1.5 g/l drinking water for 3–5 days.

Reptiles: 10–20 mg/kg i.m., p.o. q12–24h.

Fish: 40–80 mg/kg in feed q24h for 10 days or 12.5 mg/kg i.m. once (long-acting preparation)^a.

Amphibians: No information available.

References

^a Brown AG and Grant AN (1992) Use of amoxycillin by injection in Atlantic salmon broodstock. *The Veterinary Record* **131**(11), p.237

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A **Amoxicillin/Clavulanate** see **Co-amoxiclav**
Amoxycillin see **Amoxicillin**

B

C

Amphotericin B

D

(Abelcet*, AmBisome*, Amphocil*, Fungizone*) **POM**

E

Formulations: Injectable: 50 mg/vial powder for reconstitution.

F

Action: Binds to sterols in fungal cell membrane creating pores and allowing leakage of contents.

G

Use: Management of systemic fungal infections and leishmaniasis.

H

Given the risk of severe toxicity it is advisable to reserve use for severe/potentially fatal fungal infections only. Abelcet, AmBisome and Amphocil are lipid formulations that are less toxic. Physically incompatible with electrolyte solutions. Lipid formulations are far less toxic than

I

conventional formulations for i.v. use because the drug is targeted to macrophages, but these preparations are far more expensive. Solutions are usually given i.v. but if regular venous catheterization is problematic then an s.c. alternative has been used for cryptococcosis and could potentially be used for other systemic mycoses. Renal values and electrolytes should be monitored pre- and post- each treatment;

J

urinalysis and liver function tests weekly. If considering use in patients with pre-existing renal insufficiency (where other treatment options

K

have failed and benefits outweigh risks), consider lipid formulations, concurrent saline administration and dose reduction. Has been used

L

topically in reptiles to treat pulmonary candidiasis. In amphibians, amphotericin B has been used as a water treatment for the management of chytridiomycosis in adult animals.

O

Safety and handling: Keep in the dark, although loss of drug activity is negligible for at least 8 hours in room light. After initial reconstitution (but not further dilution), the drug is stable for 1 week if refrigerated and stored in the dark. Do not dilute in saline. Pre-treatment heating of the reconstituted concentrated solution to 70°C for 20 min produces superaggregates which are less nephrotoxic. To produce a lipid-formulated product if not commercially available mix 40 ml sterile saline, 10 ml of lipid infusion (q.v.) and 50 mg of the reconstituted concentrated solution.

P

Contraindications: Do not use in renal or hepatic failure.

T

Adverse reactions: Include hypokalaemia, leading to cardiac arrhythmias, phlebitis, hepatic failure, renal failure, vomiting, diarrhoea, pyrexia, muscle and joint pain, anorexia and anaphylactoid reactions. Nephrotoxicity is a major concern; do not use other nephrotoxic drugs concurrently.

U

Nephrotoxicity may be reduced by saline infusion (20 ml/kg over 60 minutes) prior to administration of amphotericin B. Fever and vomiting may be decreased by pre-treating with aspirin, diphenhydramine or an antiemetic. Amphotericin B is toxic to birds when administered systemically; administer fluids and monitor carefully if giving i.v. Care is needed if amphotericin B is nebulized as birds may ingest it when preening. Can cause bronchospasm in humans if nebulized. Amphotericin B is acutely toxic to *Alytes muletensis* tadpoles at 8 µg (micrograms)/ml.

V

X

Y

Z

Drug interactions: Amphotericin may increase the toxic effects of fluorouracil, doxorubicin and methotrexate. Flucytosine is synergistic with amphotericin B *in vitro* against *Candida*, *Cryptococcus* and *Aspergillus*.

DOSES

Mammals: Ferrets: 0.4–0.8 mg/kg i.v. q7d for treatment of blastomycosis; Rabbits: 1 mg/kg i.v. q24h (desoxycholate form) or 5 mg/kg i.v. q24h (liposomal form); Guinea pigs: 1.25–2.5 mg/kg s.c. q24h for cryptococcosis; Mice: 0.11 mg/kg s.c. q24h or 0.43 mg/kg p.o. q24h.

Birds: Systemic fungal infections: 1–1.5 mg/kg i.v. q8–12h for 3–5 days (give with 10–15 ml/kg saline) or 1 mg/kg in 2 ml sterile water intratracheal q8–12h for 12 days then q48h for 5 weeks; Parrots: 100–300 mg/kg p.o. q12–24h for *Macrorhabdus* infection, although resistance has been reported^a; Passerines: 100 mg/kg p.o. q8–12h or 1–5 g/l drinking water or 1 mg/ml in saline nebulized for 15 min q12h.

Reptiles: 0.5–1 mg/kg i.v., intracoelomic q24–72h for 2–4 weeks, concurrent fluid therapy is advised due to potential nephrotoxicity; Respiratory infections: Nebulize 5 mg in 150 ml saline for 30–60 min q12h; may also use topically on lesions q12h.

Amphibians: Internal mycoses: 1 mg/kg intracoelomic q24h; *Batrachochytrium dendrobatidis* sporangia and zoospores: 8–15 µg (micrograms)/ml in water for 48 h^{b,c}.

Fish: No information available.

References

- ^a Baron HR, Leung KC, Stevenson BC, Gonzalez MS and Phalen DN (2018) Evidence of amphotericin B resistance in *Macrorhabdus ornithogaster* in Australian cage-birds. *Medical Mycology* **57**(4), 421–428
- ^b Holden WM, Ebert AR, Canning PF and Rollins-Smith LA (2014) Evaluation of amphotericin B and chloramphenicol as alternative drugs for treatment of chytridiomycosis and their impacts on innate skin defences. *Applied Environmental Microbiology* **80**(13), 4034–4041
- ^c Martel A, Van Rooij P, Vercauteren G *et al.* (2011) Developing a safe antifungal treatment protocol to eliminate *Batrachochytrium dendrobatidis* from amphibians. *Medical Mycology* **49**(2), 143–149

Ampicillin

(Amfipen, Ampicaps, Ampicare, Duphacillin) **POM-V**

Formulations: **Injectable:** Ampicillin sodium 250 mg, 500 mg powders for reconstitution (human licensed product only); 150 mg/ml suspension, 100 mg/ml long-acting preparation. **Oral:** 500 mg tablets; 250 mg capsule.

Action: Binds to penicillin-binding proteins involved in bacterial cell wall synthesis, thereby decreasing cell wall strength and rigidity, affecting cell division, growth and septum formation. It acts in a time-dependent bactericidal fashion.

Use: Active against many Gram-positive and Gram-negative aerobic organisms and obligate anaerobes, but not against those that produce penicillinases (beta-lactamases), e.g. *Escherichia coli*, *Staphylococcus aureus*. The difficult Gram-negative organisms such as *Pseudomonas aeruginosa* and *Klebsiella* are usually resistant. Ampicillin is excreted well in bile and urine. Maintaining levels above the MIC is critical for efficacy and thereby prolonged dosage intervals or missed doses an compromise therapeutic response. Dose and dosing interval is determined by infection site, severity and organism. Oral bioavailability is reduced in the presence of food. Used to treat bacterial disease in fish.

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

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Safety and handling: After reconstitution the sodium salt will retain adequate potency for up to 8 hours if refrigerated, but use within 2 hours if kept at room temperature.

B

Contraindications: Avoid the use of oral antibiotic agents in critically ill patients, as absorption from the GI tract may be unreliable. Avoid use in animals with reported sensitivity to penicillins. Do not administer penicillins by any route to guinea pigs, chinchillas, hamsters, gerbils or degus. Do not administer oral penicillins to rabbits.

C**D**

Adverse reactions: Nausea, diarrhoea and skin rashes are the commonest adverse effects. Oral doses can cause fatal enterotoxaemia in small herbivores (e.g. rabbits), hamsters and gerbils.

E**F****G****H**

Drug interactions: Avoid the concurrent use of ampicillin with bacteriostatic antibiotics (e.g. tetracycline, erythromycin, chloramphenicol). Do not mix in the same syringe as aminoglycosides. A synergistic effect is seen when beta-lactam and aminoglycoside antimicrobials are used concurrently.

I

DOSES

J

Classified as category D (Prudence) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

K

Mammals: Ferrets: 5–30 mg/kg i.m., s.c. q12h; Rats, Mice: 25 mg/kg i.m., s.c. q12h or 50–200 mg/kg p.o. q12h; Primates: 20 mg/kg p.o., i.m., i.v. q8h; Rabbits, Guinea pigs, Chinchillas, Hamsters, Gerbils, Degus: Do not use p.o., caution is advised with its use parenterally.

L**M**

Birds: 50–100 mg/kg i.v., i.m. q8–12h, 150–200 mg/kg p.o. q8–12h, 1–2 g/l drinking water, 2–3 g/kg in soft feed^a.

N

Reptiles: 10–20 mg/kg s.c., i.m. q12–24h (most species); Hermann's tortoises: 50 mg/kg i.m. q12h.

O

Fish: 50–80 mg/kg in feed q24h for 10 days.

P

Amphibians: No information available.

Q

References

- ^a Ensley PK and Janssen DL (1981) A preliminary study comparing the pharmacokinetics of ampicillin given orally and intramuscularly to psittacines: Amazon parrots (*Amazona* spp.) and Blue-naped parrots (*Tanygnathus lucionensis*). *Journal of Zoo Animal Medicine* **12**(2), 42–47

R**S**

Amprolium

T

(Amproline, Coccibal, Coxoid, Eimeryl, Harkers Pigeon Coccidiosis Treatment, Surricoxx) **AVM-GSL, ESPA**

U

Formulations: Oral: 3.84% solution for dilution in water.

V

Action: Thiamine analogue that disrupts protozoal metabolism.

W

Use: Coccidiosis in homing/racing pigeons and small mammals. Limit duration of therapy to 2 weeks.

X

Safety and handling: Normal precautions should be observed.

Y

Contraindications: No information available.

Adverse reactions: Prolonged high doses can cause thiamine deficiency.

Z

Drug interactions: No information available.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Mammals: Ferrets: 19 mg/kg p.o. q72h; Rabbits: 20 mg/kg p.o. q24h for 2–4 weeks; Chinchillas: 10–15 mg/kg p.o. total daily dose divided q8–24h; Gerbils, Hamsters, Rats, Mice: 10–20 mg/kg p.o. total daily dose divided q8–24h.

Birds: Pigeons: 28 ml of the concentrated solution in 4.5 l of drinking water for 7 days; in severe outbreaks continue with half-strength solution (14 ml per 4.5 l) for a further 7 days; medicated water should be discarded after 24 hours; **Passerines:** 50–100 mg/l drinking water for 5 days or longer.

Reptiles, Amphibians, Fish: No information available.

Aniline green see **Malachite green**

Apomorphine

(Apovomin, Emedog) **POM-V**

Formulations: Injectable: 1 mg/ml solution (Emedog), 3 mg/ml solution (Apovomin).

Action: Stimulates emesis through D2 dopamine receptors in the chemoreceptor trigger zone.

Use: Induction of self-limiting emesis within a few minutes of administration in animals where vomiting is desirable, e.g. following the ingestion of toxic, non-caustic foreign material. Emesis generally occurs rapidly and within a maximum of 10 min. Further doses depress the vomiting centre and may not result in any further vomiting.

Safety and handling: Normal precautions should be observed.

Contraindications: Induction of emesis is contraindicated if strong acid or alkali has been ingested, due to the risk of further damage to the oesophagus. Induction of vomiting is contraindicated if the animal is unconscious, fitting, or has a reduced cough reflex, or if the poison has been ingested for >2 hours, or if the ingesta contains paraffin, petroleum products or other oily or volatile organic products, due to the risk of inhalation. Contraindicated in rabbits as they are unable to vomit. Contraindicated in rodents as their stomach walls lack the strength to tolerate forced emesis.

Adverse reactions: Apomorphine may induce excessive vomiting, respiratory depression and sedation.

Drug interactions: In the absence of compatibility studies, apomorphine must not be mixed with other products. Antiemetic drugs, particularly antidopaminergics (e.g. phenothiazines) may reduce the emetic effects of apomorphine. Additive CNS or respiratory depression may occur when apomorphine is used with opiates or other CNS or respiratory depressants.

DOSES

Mammals: Ferrets: 70 µg (micrograms)/kg s.c.

Birds, Reptiles, Amphibians, Fish: No information available.

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A

Ara-C see **Cytarabine**
Ascorbic acid see **Vitamin C**

B**C**

Asparaginase (L-Asparaginase, Crisantaspase)

D

(Asparaginase*, Elspar*, Erwinase*) **POM**

E

Formulations: Injectable: vials of 5,000 or 10,000 IU powder for reconstitution.

F

Action: Lymphoid tumour cells are not able to synthesize asparagine and are dependent upon supply from the extracellular fluid.

G

Asparaginase deprives malignant cells of this amino acid, which results in cessation of protein synthesis and cell death.

H

Use: Main indication is treatment of lymphoid malignancies.

I

Safety and handling: **Cytotoxic drug; see specialist texts for further advice on chemotherapeutic agents.** Store in a refrigerator.

J

Contraindications: Patients with active pancreatitis or a history of pancreatitis. History of anaphylaxis associated with previous administration.

K

Adverse reactions: Anaphylaxis may follow administration, especially if repeated. Premedication with an antihistamine is recommended before administration. GI disturbances, hepatotoxicity and coagulation deficits may also be observed.

L**M**

Drug interactions: Administration with or before vincristine may reduce clearance of vincristine and increase toxicity; thus, if used in combination the vincristine should be given 12–24 hours before the enzyme.

N**O**

DOSES

P

See Appendix for chemotherapy protocols in ferrets.

Q

Mammals: Ferrets: 10,000 IU/m² s.c. weekly for the first 3 weeks as part of a chemotherapeutic protocol.

R

Reptiles: Bearded dragons: 400 IU/kg s.c. at 3 week intervals as part of a chemotherapeutic protocol^a.

S

Birds, Amphibians, Fish: No information available.

T

References

- ^a Keeney CMH, Intile JL, Sims CS and Harrison TM (2021) Lymphoid leukemia in five bearded dragons (*Pogona vitticeps*). *Journal of the American Veterinary Medical Association* **258**(7), 748–757

U

Aspirin (Acetylsalicylic acid)

V

(Aspirin BP* and component of many others) **P**

W

Formulations: Oral: 75 mg, 300 mg tablets.

X

Action: Produces irreversible inhibition of COX by acetylation, thereby preventing the production of both prostaglandins and thromboxanes from membrane phospholipids.

Y

Use: Prevention of arterial thromboembolism. Also can be used to control mild to moderate pain, although NSAIDs that are more selective for the COX-2 enzyme have a better safety profile; not an NSAID of choice for analgesia in small mammals. Administration of aspirin to

Z

animals with renal disease must be carefully evaluated. It is advisable to stop aspirin before surgery (at least 2 weeks) to allow recovery of normal platelet function and prevent excessive bleeding.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not give aspirin to dehydrated, hypovolaemic or hypotensive patients or those with GI disease. Do not give to pregnant animals or animals <6 weeks old.

Adverse reactions: GI ulceration and irritation are common side effects of all NSAIDs. It is advisable to stop therapy if diarrhoea or nausea persists beyond 1–2 days. Stop therapy immediately if GI bleeding is suspected and begin symptomatic treatment. There is a small risk that NSAIDs may precipitate cardiac failure in humans and this risk in animals is unknown. All NSAIDs carry a risk of renal papillary necrosis due to reduced renal perfusion caused by a reduction in the production of renal prostaglandins. This risk is greatest when NSAIDs are given to animals that are hypotensive or animals with pre-existing renal disease.

Drug interactions: Do not administer concurrently or within 24 hours of other NSAIDs and glucocorticoids. Do not administer with other potentially nephrotoxic agents, e.g. aminoglycosides.

DOSES

Mammals: Ferrets: 10–20 mg/kg p.o. q24h; Rabbits: 100 mg/kg p.o. q12–24h; Rodents: 50–150 mg/kg p.o. q4–8h; Primates: 5–10 mg/kg p.o. q4–6h.

Birds: Parrots: 5 mg/kg p.o. q8h.

Reptiles, Amphibians, Fish: No information available.

Atenolol

(Atenolol*, Tenormin*) **POM**

Formulations: Oral: 25 mg, 50 mg, 100 mg tablets; 5 mg/ml syrup. Injectable: 0.5 mg/ml.

Action: Cardioselective beta-adrenergic blocker. It is relatively specific for beta-1 adrenergic receptors but can antagonize beta-2 receptors at high doses. Blocks the chronotropic and inotropic effects of beta-1 adrenergic stimulation on the heart, thereby reducing myocardial oxygen demand. Bronchoconstrictor, vasodilatory and hypoglycaemic effects are less marked due to its cardioselective nature.

Use: Cardiac tachyarrhythmias, hyperthyroidism, hypertrophic cardiomyopathy, obstructive cardiac disease (severe aortic or pulmonic stenosis) and systemic hypertension. Less effective when used alone for ventricular arrhythmias unless the arrhythmia is mediated by sympathetic tone. It is recommended to withdraw therapy gradually in patients who have been receiving the drug chronically. Alternatively, sotalol has been used in birds.

Safety and handling: Normal precautions should be observed.

Contraindications: Patients with bradyarrhythmias, acute or decompensated congestive heart failure. Relatively contraindicated in animals with medically controlled congestive heart failure as is poorly tolerated.

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A

Adverse reactions: Most frequently seen in geriatric patients with chronic heart disease or in patients with acute or decompensated heart failure. Include bradycardia, AV block, myocardial depression, heart failure, syncope, hypotension, hypoglycaemia, bronchospasm and diarrhoea. Depression and lethargy may occur as a result of atenolol's high lipid solubility and its penetration into the CNS.

B**C****D****E****F****G****H****I****J**

Drug interactions: Do not administer concurrently with alpha-adrenergic agonists (e.g. phenylpropanolamine). The hypotensive effect of atenolol is enhanced by many agents that depress myocardial activity including anaesthetic agents, phenothiazines, antihypertensive drugs, diuretics and diazepam. There is an increased risk of bradycardia, severe hypotension, heart failure and AV block if atenolol is used concurrently with calcium-channel blockers. Concurrent digoxin administration potentiates bradycardia. The metabolism of atenolol is accelerated by thyroid hormones; thus the dose of atenolol may need to be decreased when initiating carbimazole therapy. Atenolol enhances the effects of muscle relaxants (e.g. suxamethonium, tubocurarine). Hepatic enzyme induction by phenobarbital may increase the rate of metabolism of atenolol. The bronchodilatory effects of theophylline may be blocked by atenolol. Atenolol may enhance the hypoglycaemic effect of insulin.

DOSES

K

Mammals: Ferrets: 3.125–6.25 mg/animal p.o. q24h; Rabbits: 0.5–2 mg/kg p.o. q24h; Rats: 0.2–2 mg/kg p.o. q24h; Mice: 2–10 mg/kg i.v., i.p. q24h.

L**M**

Birds: 5–10 mg/kg p.o. q12–24h¹ or alternatively sotalol may be used at 1 mg/kg p.o. q12h².

N

References

O

¹ Beaufre H, Schilliger L and Pauriat R (2016) Cardiovascular System. *Current Therapy in Exotic Pet Practice*, ed. MA Mitchell and TN Tully. Elsevier, St Louis

P

² Oster SC, Jung S and Moon R (2019) Resolution of supraventricular arrhythmia using sotalol in an adult golden eagle (*Aquila chrysaetos*) with presumed atherosclerosis. *Journal of Exotic Pet Medicine* **29**, 136–141

Q**R**

Atipamezole

S

(Alzane, Antisedan, Atipam, Revertor, Sedastop) **POM-V**

T

Formulations: Injectable: 5 mg/ml solution.

U

Action: Selective alpha-2 adrenoreceptor antagonist.

V

Use: Reverses the sedative effects of medetomidine or dexmedetomidine; will also reverse other alpha-2 agonists to provide a quick recovery from anaesthesia and sedation. It also reverses other effects such as the analgesic, cardiovascular and respiratory effects of alpha-2 agonists. Atipamezole does not alter the metabolism of medetomidine or dexmedetomidine but occupies the alpha-2 receptor, preventing binding of the drug. The duration of action of atipamezole and medetomidine or dexmedetomidine are similar, so re-sedation is uncommon. Routine administration of atipamezole i.v. is not recommended because the rapid recovery from sedation is usually associated with excitation, though i.v. administration may be indicated in an emergency (e.g. excessive sedation from medetomidine or dexmedetomidine, or cardiovascular complications).

W**X****Y****Z**

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Transient over-alertness and tachycardia may be observed after overdosage. This is best handled by minimizing external stimuli and allowing the animal to recover quietly.

Drug interactions: No information available.

DOSES

Mammals: Ferrets, Rodents, Primates, Hedgehogs: 5 times the previous medetomidine dose s.c., i.m.^a or 10 times the previous dexmedetomidine dose (i.e. equal volume of solution to medetomidine or dexmedetomidine if using 0.5 mg/ml solution); **Rabbits, Marsupials:** 2.5 times the previous medetomidine dose or 5 times the previous dexmedetomidine dose (i.e. half the volume of medetomidine or dexmedetomidine if using 0.5 mg/ml solution). When medetomidine or dexmedetomidine has been administered at least an hour before, dose of atipamezole can be reduced by half and repeated if recovery is slow^b.

Birds: 2.5–5 times the previous medetomidine or dexmedetomidine dose i.m., i.v.^c.

Reptiles, Amphibians: 5 times the previous medetomidine dose s.c., i.m. or 5–10 times the previous dexmedetomidine dose (i.e. equal volume of solution to medetomidine, half to equal the volume of solution of dexmedetomidine if using 0.5 mg/ml solution)^{d,e}.

Fish: 5 times the previous medetomidine or dexmedetomidine dose i.m., i.v.^f.

References

- ^a Jang HS, Choi HS, Lee SH, Jang KH and Lee MG (2009) Evaluation of the anaesthetic effects of medetomidine and ketamine in rats and their reversal with atipamezole. *Veterinary Anaesthesia and Analgesia* **36**(4), 319–327
- ^b Williams AM and Wyatt JD (2007) Comparison of subcutaneous and intramuscular ketamine-medetomidine with and without reversal by atipamezole in Dutch belted rabbits (*Oryctolagus cuniculus*). *Journal of the American Association of Laboratory Animal Science* **46**(6), 16–20
- ^c Sandmeier P (2000) Evaluation of medetomidine for short-term immobilization of domestic pigeons (*Columba livia*) and Amazon parrots (*Amazona* spp.). *Journal of Avian Medicine and Surgery* **14**(1), 8–14
- ^d Sleeman JM and Gaynor J (2000) Sedative and cardiopulmonary effects of medetomidine and reversal with atipamezole in Desert tortoises (*Gopherus agassizi*). *Journal of Zoo and Wildlife Medicine* **31**(1), 28–35
- ^e Yaw TJ, Mans C, Martinelli L and Sladky KK (2020) Comparison of subcutaneous administration of alfaxalone–midazolam–dexmedetomidine with ketamine–midazolam–dexmedetomidine for chemical restraint in juvenile blue poison dart frogs (*Dendrobates tinctorius azureus*). *Journal of Zoo and Wildlife Medicine* **50**(4), 868–873
- ^f Williams TD, Rollins M and Block BA (2004) Intramuscular anesthesia of bonito and Pacific mackerel with ketamine and medetomidine and reversal of anesthesia with atipamezole. *Journal of the American Veterinary Medical Association* **225**(3), 417–421

Atracurium

(Tracrium*) **POM**

Formulations: Injectable: 10 mg/ml solution.

Action: Inhibits the actions of acetylcholine at the neuromuscular junction by binding competitively to the nicotinic acetylcholine receptor on the post-junctional membrane.

Use: Neuromuscular blockade during anaesthesia. This may be to improve surgical access through muscle relaxation, to facilitate positive pressure ventilation or for intraocular surgery in rabbits. May also be

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A used for mydriasis to allow ocular examination and/or surgery in birds. Atracurium has an intermediate duration of action (15–35 min) and is non-cumulative due to non-enzymatic (Hofmann) elimination. It is therefore suitable for administration to animals with renal or hepatic disease. Monitoring (using a nerve stimulator) and reversal of the neuromuscular blockade is recommended to ensure complete recovery before the end of anaesthesia. Hypothermia, acidosis and hypokalaemia will prolong the duration of action of neuromuscular blockade. Use the low end of the dose range in patients with myasthenia gravis and ensure that neuromuscular function is monitored during the period of the blockade and recovery using standard techniques.

Safety and handling: Store in refrigerator.

Contraindications: Do not administer unless the animal is adequately anaesthetized and facilities to provide positive pressure ventilation, monitor and support blood pressure are available. Avoid in patients where underlying conditions may exacerbate the cardiovascular effects of atracurium, for example those with hypovolaemia, compromised cardiac function or a high risk of intraoperative haemorrhage.

Adverse reactions: Can precipitate the release of histamine after rapid i.v. administration, resulting in bronchospasm and hypotension. Diluting the drug in normal saline and giving the drug slowly i.v. minimizes these effects but hypotension and also hypothermia are still considered likely intraoperative complications when clinical doses of atracurium are administered to pet rabbits. Effects are not thought to be dose-dependent.

Drug interactions: Neuromuscular blockade is more prolonged when atracurium is given in combination with volatile anaesthetics, aminoglycosides, clindamycin or lincomycin.

DOSES

Mammals: Rabbits: 0.44 ± 0.4 mg/kg i.v.^a.

Birds: 0.25 mg/kg i.v.; 1 mg/kg i.v.^{b,c}.

Reptiles, Amphibians, Fish: No information available.

References

- ^a Adami C, Sanchez RF and Monticelli P (2019) Use of atracurium and its reversal with neostigmine in 14 pet rabbits undergoing ophthalmic surgery: a retrospective study. *Veterinary Record* **184**(14), 443–443
- ^b Nicholson A and Ilkiw JE (1992) Neuromuscular and cardiovascular effects of atracurium in isoflurane-anesthetized chickens. *American Journal of Veterinary Research* **53**, 2337–2342
- ^c Pascoe P and Hawkins M (2017) The effect of systemic atracurium on pupillary area in chickens. *Veterinary Anaesthesia and Analgesia* **44**(5), 1262.e10

Atropine

(Atropine Sulfate) **POM**

Formulations: Injectable: 0.6 mg/ml, 1 mg/ml. **Ophthalmic:** 0.5%, 1% solution in single-use vials, 5 ml bottle; 1% ointment.

Action: Blocks the action of acetylcholine at muscarinic receptors at the terminal ends of the parasympathetic nervous system, reversing parasympathetic effects and producing mydriasis, tachycardia, bronchodilation and general inhibition of GI function.

Use: Prevent or correct bradycardia and bradyarrhythmias, to dilate pupils, in the management of organophosphate and carbamate

toxicities, and in conjunction with anticholinesterase drugs during antagonism of neuromuscular block. Routine administration prior to anaesthesia as part of premedication is no longer recommended; it is better to monitor heart rate and give atropine to manage a low heart rate if necessary. Some rabbits and rodents have endogenous plasma atropinesterase so efficacy may be reduced if used at standard doses. Ophthalmic use is ineffective in birds and reptiles because of the complex arrangement of musculature in the iris and ciliary body. Atropine has a slow onset of action (10 min i.m., 2–3 min i.v.), therefore it is important to wait for an adequate period of time for the desired effect before redosing. The ophthalmic solution tastes very bitter and can cause hypersalivation; therefore the ophthalmic ointment preparation is preferred. Used for the treatment of organophosphate poisoning in fish.

Safety and handling: The solution does not contain any antimicrobial preservative, therefore any remaining solution in the vial should be discarded after use. The solution should be protected from light.

Contraindications: Glaucoma, lens luxation, keratoconjunctivitis sicca.

Adverse reactions: Include sinus tachycardia (usually of short duration after i.v. administration), blurred vision from mydriasis, which may worsen recovery from anaesthesia, and drying of bronchial secretions. Atropine increases intraocular pressure and reduces tear production. Ventricular arrhythmias may be treated with lidocaine if severe. Other GI side effects such as ileus and vomiting are rare in small animals. May be associated with prolonged ileus in reptiles.

Drug interactions: Atropine is compatible (for at least 15 min) mixed with various medications but not with bromides, iodides, sodium bicarbonate, other alkalis or noradrenaline. The following may enhance the activity of atropine: antihistamines, quinidine, pethidine, benzodiazepines, phenothiazines, thiazide diuretics and sympathomimetics. Combining atropine and alpha-2 agonists is not recommended. Atropine may aggravate some signs seen with amitraz toxicity, leading to hypertension and gut stasis.

DOSES

Mammals: Ferrets: 0.02–0.04 mg/kg s.c., i.m., i.v.; **Rabbits:** 0.1–0.5 mg/kg i.m., i.v., endogenous atropinase levels may make repeat injections q10–15min necessary; **Chinchillas, Guinea pigs:** 0.1–0.2 mg/kg s.c., i.m.; **Rodents:** Organophosphate poisoning may require up to 10 mg/kg i.v., i.m. q20min; Bradyarrhythmias: 0.01–0.03 mg/kg i.v.; low doses may exacerbate bradycardia; repetition of the dose will usually promote an increase in heart rate; 0.03–0.04 mg/kg i.m. can be given to prevent development of bradycardia during administration of potent opioids such as fentanyl; **Primates:** 0.02–0.05 mg/kg s.c., i.m., i.v.; **Sugar gliders:** 0.01–0.02 mg/kg s.c., i.m.; **Hedgehogs:** 0.01–0.05 mg/kg s.c., i.m.; **Others:** 0.04–0.1 mg/kg i.m., s.c.

Birds: Organophosphate poisoning: 0.04–0.5 mg/kg i.v., i.m. q4h; Supraventricular bradycardia: 0.01–0.02 mg/kg i.v. once.

Reptiles: 0.01–0.04 mg/kg i.m., i.v.; May be ineffective in green iguanas^a.

Amphibians: 0.1 mg/kg s.c., i.m., i.v., intracoelomic as required (organophosphate toxicity).

Fish: 0.1–0.5 mg/kg s.c., i.m., i.v., intracoelomic as required (organophosphate toxicity).

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A**References**

- ^a Pace L and Mader DR (2002) Atropine and glycopyrrolate, route of administration and response in the green iguana (*Iguana iguana*). *Proceedings of the Association of Reptilian and Amphibian Veterinarians*, pp. 79

B**C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z****Azathioprine**(Azathioprine*, Imuran*) **POM****Formulations:** Oral: 25 mg, 50 mg tablets.

Action: Inhibits purine synthesis, which is necessary for cell proliferation especially of leucocytes and lymphocytes. It suppresses cell-mediated immunity, alters antibody production and inhibits cell growth.

Use: Management of immune-mediated diseases. Often used in conjunction with corticosteroids. Routine haematology (including platelets) should be monitored closely: initially every 1–2 weeks; and every 1–2 months when on maintenance therapy. Use with caution in patients with hepatic disease. In animals with renal impairment, dosing interval should be extended. Clinical responses can take up to 6 weeks. Mycophenolic acid may be preferred if a more rapid response is required.

Safety and handling: **Cytotoxic drug; see specialist texts for further advice on chemotherapeutic agents.** Azathioprine tablets should be stored at room temperature in well closed containers and protected from light.

Contraindications: Do not use in patients with bone marrow suppression or those at high risk of infection.

Adverse reactions: Bone marrow suppression is the most serious adverse effect. This may be influenced by the activity of thiopurine s-methyltransferase, which is involved in the metabolism of the drug and which can vary between individuals due to genetic polymorphism. Avoid rapid withdrawal.

Drug interactions: Enhanced effects and increased azathioprine toxicity when used with allopurinol. Increased risk of azathioprine toxicity with aminosaliclates and corticosteroids.

DOSES

Mammals: Ferrets: 0.5 mg/kg p.o. q48h to 5 mg/kg p.o. q24h for eosinophilic gastroenteritis (limited evidence); 0.9 mg/kg p.o. q24–72h with prednisolone and dietary management for inflammatory bowel disease.

Birds, Reptiles, Amphibians, Fish: No information available.

Azithromycin(Zithromax*) **POM**

Formulations: Oral: 250 mg capsule; 200 mg/5 ml suspension (reconstitute with water).

Action: Binds to the 50S bacterial ribosome (like erythromycin), inhibiting peptide bond formation and has bactericidal or bacteriostatic activity depending on the susceptibility of the organism. Azithromycin has a longer tissue half-life than erythromycin, shows better oral absorption and is better tolerated in humans.

Use: Alternative to penicillin in allergic individuals as it has a similar, although not identical, antibacterial spectrum. It is active against Gram-positive cocci (some *Staphylococcus* species are resistant), Gram-positive bacilli, some Gram-negative bacilli (*Haemophilus*, *Pasteurella*), mycobacteria, obligate anaerobes, *Chlamydia*, *Mycoplasma* and *Toxoplasma*. Some strains of *Actinomyces*, *Nocardia* and *Rickettsia* are also inhibited. Most strains of the Enterobacteriaceae (*Pseudomonas*, *Escherichia coli*, *Klebsiella*) are resistant. Useful in the management of respiratory tract, mild to moderate skin and soft tissue, and non-tubercular mycobacterial infections. Used to treat chlamydiosis in birds. Has been used in primates (macaques) as an anti-malarial. Doses are empirical and subject to change as experience with the drug is gained. More work is needed to optimize the clinically effective dose rate. Azithromycin activity is enhanced in an alkaline pH; administer on an empty stomach where possible.

Safety and handling: Normal precautions should be observed.

Contraindications: Avoid in renal and hepatic failure in all species.

Adverse reactions: In humans similar adverse effects to those of erythromycin are seen, i.e. vomiting, cholestatic hepatitis, stomatitis and glossitis, but the effects are generally less marked than with erythromycin.

Drug interactions: Azithromycin may increase the serum levels of methylprednisolone, theophylline and terfenadine. The absorption of digoxin may be enhanced.

DOSES

Classified as category C (Caution) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Ferrets: 5 mg/kg p.o. q24h for 5 days as part of a protocol for treatment of *Helicobacter*; Rabbits: 4–5 mg/kg i.m. q48h for 7 days is effective against syphilis; 15–30 mg/kg p.o. q24h for respiratory infections; 50 mg/kg p.o. q24h azithromycin with 40 mg/kg p.o. q12h rifampin for staphylococcal osteomyelitis^a; **Rats, Mice:** 50 mg/kg p.o. q12h for 14 days; **Prairie dogs:** 15–30 mg/kg p.o. q24h for 15 days; **Primates:** 25 mg/kg s.c. q24h^b, 40 mg/kg p.o. q24h.

Birds: Chlamydiosis: 40 mg/kg p.o. q24–48h for up to 45 days^c; *Mycoplasma*: 50–80 mg/kg p.o. q24h for 3 days, then 4 days off; repeat for up to 3 weeks; **Blue-and-gold macaws:** 10 mg/kg p.o. q48h for susceptible infections^d; **Pigeons:** 20 mg/kg p.o. q12h for 12 days appeared effective as part of treatment for cutaneous pox virus infection^e.

Reptiles: 10 mg/kg p.o. q3d for skin infections; q5d for respiratory tract infections; q7d for liver and kidney infections^f.

Amphibians, Fish: No information available.

References

- ^a Shirtliff ME, Mader JT and Cathoun J (1999) Oral rifampin plus azithromycin or clarithromycin to treat osteomyelitis in rabbits. *Clinical Orthopaedics and Related Research* **359**, 229–236
- ^b Puri SK and Singh N (2000) Azithromycin: antimalarial profile against blood- and sporozoite-induced infections in mice and monkeys. *Experimental Parasitology* **94**(1), 8–14
- ^c Sanchez-Migallon Guzman D, Diaz-Figueroa O, Tully T Jr et al. (2010) Evaluating 21-day doxycycline and azithromycin treatments for experimental *Chlamydophila psittaci* infection in cockatiels (*Nymphicus hollandicus*). *Journal of Avian Medicine and Surgery* **24**(1), 35–45
- ^d Carpenter JW, Olsen JH, Randle-Port M, Koch DE, Isaza R and Hunter RP (2005) Pharmacokinetics of azithromycin in the blue-and-gold macaw (*Ara ararauna*) after intravenous and oral administration. *Journal of Zoo and Wildlife Medicine* **36**(4), 606–609

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A

^e Sudhakara RB, and Sivajothi S (2018) Therapeutic management of cutaneous form of pox in pigeons with azithromycin. *International Journal of Avian and Wildlife Biology* **3**(2), 136–137

B

^f Coke RL, Hunter RP, Isaza R et al. (2003) Pharmacokinetics and tissue concentrations of azithromycin in ball pythons (*Python regius*). *American Journal of Veterinary Research* **64**(2), 225–228

C

D

Aztreonam

E

(Azactam*) **POM**

F

Formulations: Injectable: 1 g, 2 g powder for solution.

G

Action: Bactericidal by interfering with cell wall synthesis.

H

Use: Treatment of Gram-negative bacterial infections in fish.

I

Safety and handling: Normal precautions should be observed.

J

Contraindications: No information available.

K

Adverse reactions: No information available.

L

Drug interactions: No information available.

M

DOSES

N

Classified as category A (Avoid) by the EMA.

O

See Appendix for guidelines on responsible antibiotic use.

P

Fish: 100 mg/kg i.m., intracoelomic q48h for 7 treatments.

Q

Mammals, Birds, Reptiles, Amphibians: No information available.

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Benazepril

(Benazecare Flavour, Benefortin, Cardalis, Fortekor, Fortekor-Plus, Kelapril, Nelio, Prilben) **POM-V**

Formulations: Oral: 2.5 mg, 5 mg, 20 mg tablets. Available in compound preparations with spironolactone (2.5 mg benazepril/20 mg spironolactone; 5 mg benazepril/40 mg spironolactone; 10 mg benazepril/80 mg spironolactone) (Cardalis) and pimobendan (1.25 mg pimobendan/2.5 mg benazepril; 5 mg pimobendan/10 mg benazepril) (Fortekor-Plus).

Action: ACE inhibitor. It inhibits conversion of angiotensin I to angiotensin II and inhibits the breakdown of bradykinin. Overall effect is a reduction in preload and afterload via venodilation and arteriodilation, decreased salt and water retention via reduced aldosterone production and inhibition of the angiotensin-aldosterone-mediated cardiac and vascular remodelling. Efferent arteriolar dilation in the kidney can reduce intraglomerular pressure and therefore glomerular filtration. This may decrease proteinuria.

Use: Treatment of congestive heart failure in most species and chronic renal insufficiency (anecdotal only). Often used in conjunction with diuretics when heart failure is present as most effective when used in these cases. Can be used in combination with other drugs to treat heart failure (e.g. pimobendan, spironolactone, digoxin). May reduce blood pressure in hypertension. Benazepril undergoes significant hepatic metabolism and may not need dose adjustment in renal failure. ACE inhibitors are more likely to cause or exacerbate prerenal azotaemia in hypotensive animals and those with poor renal perfusion (e.g. acute, oliguric renal failure). Use cautiously if hypotension, hyponatraemia or outflow tract obstruction are present. Regular monitoring of blood pressure, serum creatinine, urea and electrolytes is strongly recommended with ACE inhibitor treatment. Hypotension, azotaemia and hyperkalaemia are all indications to stop or reduce ACE inhibitor treatment in rabbits. The use of ACE inhibitors in animals with cardiac disease stems mainly from extrapolation from theoretical benefits and studies showing a benefit in other species with heart failure and different cardiac diseases (mainly dogs and humans).

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use in cases of cardiac output failure.

Adverse reactions: Potential adverse effects include hypotension, hyperkalaemia and azotaemia. Monitor blood pressure, serum creatinine and electrolytes when used in cases of heart failure. Dosage should be reduced if there are signs of hypotension (weakness, disorientation). Anorexia, vomiting and diarrhoea are rare. It is not recommended for breeding, pregnant or lactating animals, as safety has not been established. In rabbits, treatment with ACE inhibitors can be associated with an increase in azotaemia.

Drug interactions: Concomitant usage with potassium-sparing diuretics (e.g. spironolactone) or potassium supplements could result in hyperkalaemia. However, in practice, spironolactone and ACE inhibitors appear safe to use concurrently. There may be an increased risk of nephrotoxicity and decreased clinical efficacy when used with NSAIDs. There is a risk of hypotension with concomitant administration of

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A diuretics, vasodilators (e.g. anaesthetic agents, antihypertensive agents) or negative inotropes (e.g. beta-blockers).

B DOSES

Mammals: Ferrets: 0.25–0.5 mg/kg p.o. q24h; Rabbits, Guinea pigs:

Starting dose 0.05 mg/kg p.o. q24h. Dose may be increased to a maximum of 0.1 mg/kg; **Rats:** ACE inhibitors have been used to mitigate protein losing nephropathy in rats at 0.5–1.0 mg/kg p.o. q24h^a.

Birds: 0.5 mg/kg p.o. q24h.

Reptiles, Amphibians, Fish: No information available.

E References

- ^a Mudagal M, Patel J, Nagalakshmi NC and Asif Ansari M (2011) Renoprotective effects of combining ACE inhibitors and statins in experimental diabetic rats. *DARU Journal of Pharmaceutical Sciences* **19**(5), 322–325

H Benzalkonium chloride (Quaternary ammonium compound)

(Ark-Klens, **F10 Antiseptic Solution**) **ESPA**

Formulations: Various formulations including 12.5% liquid (Ark-Klens), benzalkonium chloride 5.4 g and polyhexanide 0.4 g liquid (F10 Antiseptic Solution Concentrate), benzalkonium chloride 0.4 g, polyhexanide 0.03 g and cypermethrin 0.25 g spray (F10® Germicidal Wound Spray with Insecticide). Also available in pre-diluted ready-to-use solution (1:250), barrier ointment and shampoo formulations.

Action: Biocidal action thought to cause damage to cell membranes and leakage of cell contents.

Use: Disinfection of equipment. May be used for nebulizing, nasal or sinus flushing and wound flushing at appropriate dilutions. Treatment of external bacterial infections of the skin and gills in fish. Quaternary ammonium compounds are more toxic at high temperatures and in water with low hardness, therefore dose rates should be reduced by 50%. Benzalkonium chloride has a low therapeutic index.

Safety and handling: Irritant to skin and eyes.

Contraindications: No information available.

Adverse reactions: No information available.

Drug interactions: No information available.

T DOSES

Mammals, Birds, Reptiles, Amphibians: Nebulization, nasal, sinus, wound flushing; use 1:250 ml F10 Antiseptic solution^a.

Fish: 10 mg/l by immersion for 5–10 min; 0.5 mg/l by prolonged immersion.

W References

- ^a Drake GJ, Koepfel K and Barrows M (2010) Disinfectant (F10SC) nebulisation in the treatment of 'red leg syndrome' in amphibians. *Veterinary Record* **166**, 593–594

Benzocaine

(Optomease Vet) **POM-V**

Formulations: 100% powder for dissolution in acetone or ethanol.
Benzocaine 200 mg/ml liquid (Optomease Vet) concentrate solution.

Action: Benzocaine is lipid soluble and is rapidly absorbed across the gills and skin, resulting in anaesthesia by impeding peripheral nerve signal transmission to the CNS.

Use: Sedation, immobilization, anaesthesia and euthanasia of fish. Ideally, the drug should be used in water from the tank or pond of origin to minimize problems due to changes in water chemistry. The dry powder is poorly soluble in water and must be made into a stock solution with acetone or ethanol (e.g. 100 g/l) for more accurate dosing. Before use, the pH of the anaesthetic solution should be buffered with sodium bicarbonate to the same pH as the water of origin. The anaesthetic solution should be used on the day of preparation and be well aerated during use. Food should be withheld from fish for 12–24 h prior to anaesthesia to reduce the risk of regurgitation. The stage of anaesthesia reached is determined by the concentration used and the duration of exposure, since absorption continues throughout the period of immersion. Potency decreases with higher temperatures. Different species vary in their response and may require different concentrations. It is recommended to use the lower dose rates to test the selected drug concentration and exposure time with a small group before medicating large numbers. Fish may retain some movement during anaesthesia, making it less desirable to use during surgery. Anaesthetized fish should be returned to clean water from their normal environment to allow recovery. For euthanasia, use 5–10 times the normal anaesthetic dose and keep the fish in the solution for at least 60 minutes after respiration ceases. Has previously been used in amphibians but can result in apnoea and the use of alternatives such as tricaine are generally preferred.

Safety and handling: The powder should be stored dry and stock solutions should be stored in the dark.

Contraindications: No information available.

Adverse reactions: Can induce apnoea in amphibians.

Drug interactions: No information available.

DOSES

Amphibians: Has previously been used for anaesthesia but alternative anaesthetic agents now preferred.

Fish: Induction of anaesthesia: 25–200 mg/l by immersion; Maintenance of anaesthesia: 15–40 mg/l by immersion¹.

Mammals, Birds, Reptiles: No information available.

References

¹ Sneddon LU (2012) Clinical anesthesia and analgesia in fish. *Journal of Exotic Pet Medicine* **21**, 32–43

Benzyl penicillin see **Penicillin G**

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A

Betamethasone

(Fuciderm, Norbet, Otomax, Betnesol*, Maxidex*) **POM-V, POM**

Formulations: **Injectable:** 4 mg/ml solution for i.v. or i.m. use.

Oral: 0.25 mg tablet. **Topical:** 0.1% cream with 0.5% fusidic acid.

Ophthalmic/Otic: 0.1% solution; 0.88 mg/ml suspension with clotrimazole and gentamicin. Betamethasone is also present in varying concentrations in several topical preparations with or without antibacterials.

Action: Alters the transcription of DNA, leading to alterations in cellular metabolism which cause reduction in inflammatory responses. Has high glucocorticoid but low mineralocorticoid activity. Betamethasone also antagonizes insulin and ADH.

Use: Short-term relief of many inflammatory but non-infectious conditions. Long duration of activity and therefore not suitable for long-term daily or alternate-day use. On a dose basis, 0.12 mg betamethasone is equivalent to 1 mg prednisolone. Prolonged use of glucocorticoids suppresses the hypothalamic–pituitary axis, resulting in adrenal atrophy. Animals on chronic corticosteroid therapy should be given tapered decreasing doses when discontinuing the drug. The use of long-acting steroids in most cases of shock is of no benefit and may be detrimental.

Safety and handling: Wear gloves when applying cream.

Contraindications: Do not use in pregnant animals. Systemic corticosteroids are generally contraindicated in patients with renal disease and diabetes mellitus. Topical corticosteroids are contraindicated in ulcerative keratitis.

Adverse reactions: Catabolic effects of glucocorticoids lead to weight loss and cutaneous atrophy. Iatrogenic hyperadrenocorticism may develop. Vomiting, diarrhoea and GI ulceration may develop. Glucocorticoids may increase glucose levels and decrease serum T3 and T4 values. Impaired wound healing and delayed recovery from infections may be seen. Corticosteroids should be used with care in birds as there is a high risk of immunosuppression and side effects, such as hepatopathy and a diabetes mellitus-like syndrome. In rabbits, even small single doses can potentially cause severe adverse reactions. Ferrets are particularly susceptible to GI ulceration, and concurrent gastric protectants may be advisable, especially in stressed animals.

Drug interactions: There is an increased risk of GI ulceration if used concurrently with NSAIDs. Glucocorticoids antagonize the effect of insulin. Phenobarbital may accelerate the metabolism of corticosteroids and antifungals (e.g. itraconazole) may decrease it. There is an increased risk of hypokalaemia when used concurrently with acetazolamide, amphotericin and potassium-depleting diuretics (furosemide, thiazides).

DOSES

Mammals: 0.1 mg/kg s.c. q24h; Otic: 4 drops of polypharmaceutical to affected ear q12h; Ocular: 1 drop of ophthalmic solution to affected eye q6–8h; Skin: Apply cream to affected area q8–12h.

Birds: Not recommended.

Reptiles, Amphibians, Fish: No information available.

Bethanecol

(Myotonine*) **POM**

Formulations: Oral: 10 mg tablets.

Action: A muscarinic agonist (cholinergic or parasympathomimetic) that increases urinary bladder detrusor muscle tone and contraction.

Use: Management of urinary retention with reduced detrusor tone. It does not initiate a detrusor reflex and is ineffective if the bladder is areflexic. Best given on an empty stomach to avoid GI distress.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use when urethral resistance is increased unless in combination with agents that reduce urethral outflow pressure (e.g. phenoxybenzamine).

Adverse reactions: Vomiting, diarrhoea, GI cramping, anorexia, salivation and bradycardia (with overdosage). Treat overdoses with atropine.

Drug interactions: No information available.

DOSES

Mammals: Rabbits: 2.5–5 mg/kg p.o. q12h. Titrate doses upwards to avoid adverse effects.

Birds, Reptiles, Amphibians, Fish: Not indicated.

Bismuth salts (Bismuth carbonate, subnitrate and subsalicylate: tri-potassium di-citrate bismuthate (bismuth chelate))

(De-Noltab*, Pepto-Bismol*) **AVM-GSL, P**

Formulations: Oral: De-Noltab: tablets containing the equivalent of 120 mg bismuth oxide. Pepto-Bismol: bismuth subsalicylate suspension.

Action: Bismuth is a gastric cytoprotectant with activity against spiral bacteria. Bismuth chelate is effective in healing gastric and duodenal ulcers in humans, due to its direct toxic effects on gastric *Helicobacter pylori* and by stimulating mucosal prostaglandin and bicarbonate secretion. It is often used in conjunction with an H₂ receptor antagonist. Bismuth subsalicylate has a mild anti-inflammatory effect.

Use: Acute oral poisoning, gastric ulceration and flatulent diarrhoea. Doses are empirical.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use where specific oral antidotes are being administered in cases of poisoning. Do not use if the patient is unconscious, fitting, or has a reduced cough reflex, or in cases of intestinal obstruction, or where enterotomy or enterectomy is to be performed.

Adverse reactions: Avoid long-term use (except chelates) as absorbed bismuth is neurotoxic. Bismuth chelate is contraindicated in renal impairment. Nausea and vomiting reported in humans.

Drug interactions: Absorption of tetracyclines is reduced by bismuth and specific antidotes may also be affected.

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A**DOSES**

Mammals: Ferrets: Pepto-Bismol: 17.6 mg/kg or 0.25–1.0 ml/kg p.o. q4–8h; Rabbits: Pepto-Bismol: 0.3–0.6 ml/kg p.o. q4–6h.

Birds, Reptiles, Amphibians, Fish: No information available.

B**C****D**

Bright green see **Malachite green**
British anti-lewisite see **Dimercaprol**

E**F****Bromhexine**

(Bisolvon, Bromhex-Air Forte, Exflow, Tudomax) **POM-V**

G

Formulations: Injectable: 3 mg/ml solution. Oral: 10 mg/g powder.

H

Action: A bronchial secretolytic that disrupts the structure of acid mucopolysaccharide fibres in mucoid sputum and produces a less viscous mucus, which is easier to expectorate.

I

Use: To aid the management of respiratory diseases.

J

Safety and handling: Normal precautions should be observed.

K

Contraindications: No information available.

L

Adverse reactions: No information available.

M

Drug interactions: No information available.

N**DOSES****O**

Mammals: 0.3 mg/animal p.o. q24h or via nebulizer as 0.15 mg/ml for 20–30 minutes, 1–3 times daily. Do not use the powdered version in a nebulizer, it will foam and block the line.

P

Birds: 1.5 mg/kg i.m., p.o. q12–24h^a.

R

Reptiles: 0.1–0.2 mg/kg p.o. q24h.

S

Amphibians, Fish: No information available.

T**References****U**

^a Sumano H, Gracia I, Capistrán A *et al.* (1995) Use of ambroxol and bromhexine as mucolytics for enhanced diffusion of furaltadone into tracheobronchial secretions in broilers. *British Poultry Science* **36**(3), 503–507

V**W****Bronopol****X**

(Pyceze) **POM-V**

Y

Formulations: Immersion: 500 mg/l liquid.

Z

Action: Inhibition of dehydrogenase activity, causing membrane damage.

Use: Treat and control external fungal infection in fish and fish eggs.

Safety and handling: Irritating to eyes, lungs and skin.

Contraindications: No information available.

Adverse reactions: No information available.

Drug interactions: No information available.

DOSES

Fish: 20 mg/l by immersion for 30 min q24h for 14 treatments;

Eggs: 50 mg/l by immersion for 30 min^a.

Mammals, Birds, Reptiles, Amphibians: No information available.

References

- ^a Branson E (2002) Efficacy of bronopol against infection of rainbow trout (*Oncorhynchus mykiss*) with the fungus *Saprolegnia* species. *Veterinary Record* **151**(18), 539–541

Budesonide

(Benacort*, Budelin*, Budenofalk*, Cortiment*, Entocort*, Pulmicort*) **POM**

Formulations: Oral: 3 mg gastroresistant capsule, 3 mg capsule containing gastroresistant slow-release granules, 9 mg sustained-release gastrointestinal tablet. **Rectal:** 2 mg (total dose) rectal foam, 0.02 mg/ml enema. **Inhaled powder:** 100–400 µg (micrograms) per dose. **Nasal spray:** 64–100 µg per dose.

Action: Anti-inflammatory and immunosuppressive steroid.

Use: A novel steroid that is metabolized on its first pass through the liver in humans and therefore might be expected to have reduced systemic side effects. It has been suggested that it may be effective as a monotherapy in ferrets with inflammatory bowel disease when compared with other steroids (such as prednisolone). Systemic side effects were still seen in some patients. The dosing of this drug is unclear as it comes in a capsule of 0.3 mg and the dose is extrapolated from humans (no real pharmacokinetic data/hepatic metabolism data available in small animals). The uncoated powder for inhalant use in people (for which no information is available in small animals) should not be used for oral administration because of hydrolysis by gastric acid.

Safety and handling: Normal precautions should be observed.

Contraindications: Intestinal perforation; severe hepatic impairment.

Adverse reactions: In theory, the rapid metabolism should give minimal systemic adverse effects. However, signs of iatrogenic hyperadrenocorticism (hair loss, muscle wastage, elevation in liver enzymes, hepatomegaly, lethargy, polyphagia and polyuria/polydipsia) may develop. In theory, sudden transfer from other steroid therapy might result in signs related to reductions in steroid levels. Corticosteroids can potentially cause severe immunosuppression in rabbits; use with care.

Drug interactions: Additive effect if given with other corticosteroids. The metabolism of corticosteroids may be decreased by antifungals. Avoid using with antacids, erythromycin, cimetidine, itraconazole and other drugs that inhibit the liver enzymes that metabolize budesonide.

DOSES

Mammals: Ferrets: Doses of up to 1 mg/ferret p.o. q24h have been suggested for the management of IBD.

Birds, Reptiles, Amphibians, Fish: No information available.

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Bupivacaine

(Marcain*, Sensorcaine*) **POM**

B

Formulations: Injectable: 2.5, 5.0, 7.5 mg/ml solutions, 2.5, 5.0 mg/ml solution with 1:200,000 adrenaline.

C

Action: Reversible blockade of the sodium channel in nerve fibres produces local anaesthesia.

D**E**

Use: Provision of analgesia by perineural nerve blocks, regional and epidural techniques. Onset of action is significantly slower than lidocaine (20–30 min for epidural analgesia) but duration of action is relatively prolonged (6–8h). Lower doses should be used when systemic absorption is likely to be high (e.g. intrapleural analgesia). Small volumes of bupivacaine can be diluted with normal saline to enable wider distribution of the drug for perineural blockade. Doses of bupivacaine up to 2 mg/kg q8h are unlikely to be associated with systemic side effects if injected perineurally, epidurally or intrapleurally. Combining bupivacaine with lidocaine can prolong the duration of the sensory block whilst limiting the duration of the motor block compared with administration of bupivacaine alone.

F**G****H****I****J**

Safety and handling: Normal precautions should be observed.

K

Contraindications: Do not give i.v. or use for i.v. regional anaesthesia. Use of bupivacaine with adrenaline is not recommended when local vasoconstriction is undesirable (e.g. end arterial sites) or when a significant degree of systemic absorption is likely. In birds there are concerns over the longevity of bupivacaine and the development of high plasma levels several hours following subcutaneous use. It is therefore important not to exceed the maximum dose and lidocaine may be a safer option.

L**M****N**

Adverse reactions: Inadvertent intravascular injection may precipitate severe cardiac arrhythmias that are refractory to treatment.

O**P**

Drug interactions: All local anaesthetics share similar side effects, therefore the dose of bupivacaine should be reduced when used in combination with other local anaesthetics.

Q

DOSES

R

Mammals:

S**T****U****V****W****X****Y****Z**

- Perineural: Volume of injection depends on the site of placement and size of the animal. As a guide: 0.1 ml/kg for femoral, radial and sciatic nerve block; 0.15 ml/kg for combined ulnar, musculocutaneous, median and ulnar nerve blocks; 0.3 ml/kg for brachial plexus nerve block; 0.25–1 ml total volume for blockade of the infraorbital, mental, maxillary and mandibular nerves. Choose an appropriate concentration of bupivacaine to achieve a 1–2 mg/kg dose within these volume guidelines.
- Epidural: 1.6 mg/kg (analgesia to level of L4), 2.3 mg/kg (analgesia to level of T11–T13); 1 mg/kg bupivacaine combined with preservative-free morphine 0.1 mg/kg. Limit the total volume of solution injected into the epidural space to 0.33 ml/kg in order to limit the cranial distribution of drugs in the epidural space and prevent adverse pressure effects.
- Interpleural: 1 mg/kg diluted with normal saline to a total volume of 1–5 ml depending on the size of the animal. The solution can be instilled via a thoracotomy tube. Dilution reduces pain on injection due to the acidity of bupivacaine.

- **Ferrets:** 1–2 mg/kg local infusion; **Rabbits:** 1 mg/kg local infusion, do not exceed 2 mg/kg; **Guinea pigs:** 1–2 mg/kg for specific nerve blocks (may require dilution with saline for local infusion); **Rats:** 1–2 mg/kg local infusion or intradermally once.

Birds: <2 mg/kg local infusion^{a,b}; may mix with dimethylsulfoxide (DMSO) for topical application preoperatively in bumblefoot.

Reptiles: 1 mg/kg local infusion or used in combination with lidocaine for neuraxial anaesthesia in chelonians¹ and bearded dragons^c.

Amphibians, Fish: No information available.

References

- Glatz PC, Murphy LB and Reston AP (1992) Analgesic therapy of beak-trimmed chickens. *Australian Veterinary Journal* **69**(1), 18
- Machin KL and Livingston A (2001) Plasma bupivacaine levels in mallard ducks (*Anas platyrhynchos*) following a single subcutaneous dose. *Proceedings of the Annual Conference of the AAZV/AAVW/ARAV/NAZWV*, pp. 159–163
- Ferreira TH, Fink DM and Mans C (2021) Evaluation of neuraxial administration of bupivacaine in bearded dragons (*Pogona vitticeps*). *Veterinary Anaesthesia and Analgesia* **48**(5), 798–803
- Mans C (2014) Clinical technique: Intrathecal drug administration in turtles and tortoises. *Journal of Exotic Pet Medicine* **23**(1), 67–70

Buprenorphine

(Bupaq, Buprecare, Bupredine, Buprevet, Vetergesic)

POM-V CD SCHEDULE 3

Formulations: **Injectable:** 0.3 mg/ml solution; available in 1 ml vials that do not contain a preservative, or in 10 ml multidose bottle that contains chlorocresol as a preservative.

Action: Analgesia through high affinity, low intrinsic activity and slow dissociation with the mu receptor.

Use: Relief of mild to moderate perioperative pain. As a partial agonist it antagonizes the effects of full opioid agonists (e.g. methadone, fentanyl), although the clinical relevance of interactions between full mu agonists and buprenorphine has recently been questioned. However, in practice it is not recommended to administer buprenorphine when the subsequent administration of full mu agonists is likely. If analgesia is inadequate after buprenorphine, a full mu agonist may be administered without delay. Buprenorphine may be mixed with acepromazine or dexmedetomidine to provide sedation for minor procedures or pre-anaesthetic medication. Response to all opioids is variable between individuals; therefore assessment of pain after administration is imperative. Onset of action of buprenorphine may be slower than methadone (>15 min). Duration of effect is approximately 6 hours in rabbits and rodents. Buprenorphine is metabolized in the liver; some prolongation of effect may be seen with impaired liver function. The multidose preparation is unpalatable given sublingually due to the preservative. Be careful of species differences in effect in birds. Appears active for 2–5 hours in birds. Used for the relief of pain in fish, although it is considered to have poor analgesic properties in rainbow trout. In reptiles, studies show no evidence of analgesic efficacy.

Safety and handling: Normal precautions should be observed.

Contraindications: Combination with full mu agonists is not

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A recommended for analgesia; therefore, do not use for premedication when administration of potent opioids during surgery is anticipated.

B **Adverse reactions:** As a partial mu agonist, side effects are rare after clinical doses. Pain on i.m. injection of the multidose preparation has been anecdotally reported. The taste of the multidose preparation is aversive for some species.

C **Drug interactions:** In common with other opioids, buprenorphine will reduce the doses of other drugs required for induction and maintenance of anaesthesia.

D **DOSES**

F **When used for sedation is generally given as part of a combination. See Appendix for sedation protocols in all species.**

G **Mammals:** Analgesia: **Ferrets:** 0.01–0.05 mg/kg s.c., i.m., i.v. q6h^a (high end of dose range may have some sedative effects); **Rabbits:** 0.03–0.06 mg/kg s.c., i.m., i.v. q6–12h (doses <0.03 mg/kg have very limited analgesic effects but still have some sedative effects); **Guinea pigs, Chinchillas:** 0.05–0.2 mg/kg s.c., i.v. q4–8h^{b,c}. The high end of the dose range may be needed for full analgesic effect, but use with caution as individual response is variable and sedation may be seen at higher doses; Oral transmucosal delivery appears effective in rabbits, guinea pigs and chinchillas but only the preservative-free version should be used by this route. **Mice:** 0.05–0.1 mg/kg i.m., s.c. q6–12h; **Primates:** 0.01–0.03 mg/kg i.m, i.v q6–12h. **Hedgehogs:** 0.01–0.05 mg/kg s.c. q24h^d.

Birds: Analgesia: 0.01–0.05 mg/kg i.v., i.m. q8–12h. **Grey parrots:** 0.25 mg/kg i.m. q7h (extrapolated from pharmacokinetic study but efficacy unproven); **Cockatiels:** dose rates of 1.8 mg/kg or less had no effect on foot withdrawal following a thermal stimulus; **American kestrels:** A dose of 1.8 mg/kg s.c. using a sustained-release formulation provided apparent analgesic effects for up to 24h^e; **Red-tailed hawks:** Doses of a concentrated formulation provided adequate plasma levels for 24h and 48h at doses of 0.3 mg/kg and 1.8 mg/kg, respectively.

Reptiles: Analgesia: Doses of 0.01–0.1 mg/kg i.m. q24–48h have been suggested but analgesic efficacy is not established. If used, administration into the front limbs is recommended over the hind limbs for optimal systemic drug concentrations.

Amphibians: **Leopard frogs:** 38 mg/kg s.c.; **Newts:** 50 mg/kg intracoelomic q24h^f.

Fish: Analgesia: 0.01–0.1 mg/kg i.m.

T **References**

- ^a Katzenbach JE, Wittenburg LA, Allweiler SI, Gustafson DL and Johnston MS (2018) Pharmacokinetics of single-dose buprenorphine, butorphanol, and hydromorphone in the domestic ferret (*Mustela putorius furo*). *Journal of Exotic Pet Medicine* **27**(2), 95–102
- ^b Sadar MJ, Knych HK, Drazenovich TL and Paul-Murphy JR (2018) Pharmacokinetics of buprenorphine after intravenous and oral transmucosal administration in guinea pigs (*Cavia porcellus*). *American Journal of Veterinary Research* **79**(3), 260–266
- ^c Fox L and Mans C (2018) Analgesic efficacy and safety of buprenorphine in chinchillas (*Chinchilla lanigera*). *Journal of the American Association for Laboratory Animal Science* **57**(3), 286–290
- ^d Doss GA and Mans C (2020) Antinociceptive efficacy and safety of subcutaneous buprenorphine hydrochloride administration in African pygmy hedgehogs (*Atelerix albiventris*). *Journal of the American Veterinary Medical Association* **257**(6), 618–623
- ^e Guzman DSM, Ceulemans SM, Beaufrière H, Olsen GH and Paul-Murphy JR (2018) Evaluation of the thermal antinociceptive effects of a sustained-release buprenorphine formulation after intramuscular administration to American kestrels (*Falco sparverius*). *Journal of Avian Medicine and Surgery* **32**(1), 1–7

- ^f Koeller CA (2009) Comparison of buprenorphine and butorphanol analgesia in the eastern red-spotted newt (*Notophthalmus viridescens*). *Journal of the American Association of Laboratory Animal Science* **48**(2), 171–175

Buserelin

(Busol, Fertigest, **Receptal**, Veterelin) **POM-V**

Formulations: **Injectable:** 4 µg (micrograms)/ml solution. Authorized for use in rabbits, trout and certain large animal species.

Action: Synthetic GnRH (gonadotrophin releasing hormone) analogue that stimulates LH (luteinizing hormone) and FSH (follicle stimulating hormone) production, thus causing oestrus to develop and progress.

Use: Authorized in rabbits for the improvement of conception rate and induction of ovulation (Busol, Fertigest, Receptal, Veterelin). To supplement natural LH in cases of ovulation failure or delay. Will also induce lactation postpartum in mammals. In males, it may stimulate testosterone secretion and is indicated in the management of genital hypoplasia and reduced libido. In ferrets it may be used in the management of signs of oestrus. In rabbits it is used to induce ovulation postpartum for insemination and to improve conception rates. In guinea pigs it can be used to resolve ovarian cysts if hormone-responsive. Used in birds for chronic egg laying (must be combined with husbandry changes). Used in trout to facilitate stripping in both male and female fish in spawning condition, and reduce mortality due to egg binding.

Safety and handling: People who are or may become pregnant should not handle this drug.

Contraindications: No information available.

Adverse reactions: Anaphylactic reactions may occasionally occur.

Drug interactions: No information available.

DOSES

Mammals: **Ferrets:** 1.5 µg (micrograms)/animal i.m. q24h for 2 days; **Rabbits:** 0.8 µg/animal s.c. at time of insemination or mating; **Guinea pigs:** 25 µg/guinea pig, repeat in 2 weeks.

Birds: 0.5–1.0 µg (micrograms)/kg i.m. q48h, up to 3 times^a.

Fish: 3–4 µg (micrograms)/kg i.m. once, strip fish 2–3 days later^b.

Reptiles, Amphibians: No information available.

References

- ^a Lovas EM, Johnston SD and Filippich LJ (2010) Using a GnRH agonist to obtain an index of testosterone secretory capacity in the cockatiel (*Nymphicus hollandicus*) and sulphur-crested cockatoo (*Cacatua galerita*). *Australian Veterinary Journal* **88**, 52–56
- ^b Arabacı M, Diler İ and Sarı M (2004) Induction and synchronisation of ovulation in rainbow trout, *Oncorhynchus mykiss*, by administration of emulsified buserelin (GnRH_a) and its effects on egg quality. *Aquaculture* **237** (1–4), 475–484

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A Butorphanol**B** (Butador, Dolorex, Torbugesic, Torbutrol, Torphadine, Torphasol) **POM-V****C Formulations:** Injectable: 10 mg/ml solution. **Oral:** 5 mg, 10 mg tablets.**D Action:** Analgesia resulting from affinity for the kappa opioid receptor. Also has mu receptor antagonist properties and an antitussive action resulting from central depression of the cough mechanism.**E Use:** Management of mild perioperative pain. Provision of sedation through combination with acepromazine or alpha-2 agonists. Potent antitussive agent indicated for the relief of acute or chronic non-productive cough associated with tracheobronchitis, tracheitis, tonsillitis or laryngitis resulting from inflammatory conditions of the upper respiratory tract. Butorphanol has a very rapid onset and relatively short duration of action; in different models analgesia has been shown to last between 45 minutes and 4 hours. Butorphanol is metabolized in the liver and some prolongation of effect may be seen with impaired liver function. Butorphanol is unlikely to be adequate for the management of severe pain. Higher doses of full mu agonists may be needed to provide additional analgesia after butorphanol but it is not necessary to wait 4h after butorphanol administration to give other opioids. Response to all opioids appears to be very variable between individuals; therefore assessment of pain after administration is imperative. Be careful of species differences in effect in birds. There is limited evidence of analgesic efficacy in reptiles but does have sedative effects. Butorphanol has shown beneficial effects in koi following abdominal surgery but had no clear benefit in chain dogfish or zebrafish.**K Safety and handling:** Protect from light.**O Contraindications:** Animals with diseases of the lower respiratory tract associated with copious mucous production. Premedication when administration of potent opioids during surgery is anticipated. Decreased ventilation rate and buoyancy problems have been noted in koi.**Q Adverse reactions:** As a kappa agonist/mu antagonist, side effects such as respiratory depression, bradycardia and vomiting are rare after clinical doses. Cough suppression following torbugesic tablets may be associated with mild sedation.**S Drug interactions:** In common with other opioids, butorphanol will reduce the doses of other drugs required for induction and maintenance of anaesthesia. Combination with full mu agonists is not recommended for analgesia, addition of butorphanol will reduce analgesia produced from the full mu agonist.**T DOSES****U When used for sedation is generally given as part of a combination. See Appendix for sedation protocols in all species.****V Mammals:** Analgesia: Ferrets: 0.1–0.5 mg/kg s.c. q4–6h; Rabbits: 0.1–0.5 mg/kg s.c. q4h; Guinea pigs: 0.2–2 mg/kg s.c. q4h; Chinchillas: 0.5–2 mg/kg s.c. q4h; Gerbils, Hamsters, Rats, Mice: 1–5 mg/kg s.c. q4h; Primates: 0.01–0.02 mg/kg i.v., s.c., p.o. q6–12h; Sugar gliders: 0.5 mg/kg i.m. q8h; Hedgehogs: 0.2–0.5 mg/kg i.m.**Z Birds:** Analgesia: 0.3–4 mg/kg i.m., i.v. q2–12h^{a,b}; Use lower end doses for raptors.

Reptiles: Doses of 0.5–2 mg/kg i.m. q24h have been suggested for sedation but are not proven to provide analgesia. **Bearded dragons, Green iguanas:** 1.5 mg/kg i.m. q24h^c.

Amphibians: **Leopard frogs:** 0.2–0.4 mg/kg i.m.; **Newts:** 0.5 mg/l of water^{d,e}.

Fish: Analgesia: 0.25–5 mg i.m.; **Koi:** 10 mg/kg i.m.^f.

References

- ^a Guzman DS, Flammer K, Paul-Murphy JR, Barker SA and Tully TN Jr (2011) Pharmacokinetics of butorphanol after intravenous, intramuscular and oral administration in Hispaniolan Amazon parrots (*Amazona ventralis*). *Journal of Avian Medicine and Surgery* **25**(3), 185–191
- ^b Paul-Murphy JR, Brunson DB and Miletic V (1999) Analgesic effects of butorphanol and buprenorphine in conscious African grey parrots (*Psittacus erithacus erithacus* and *Psittacus erithacus timneh*). *American Journal of Veterinary Research* **60**(10), 1218–1221
- ^c Greenacre CB, Takle G, Schmacher JP et al. (2006) Comparative anti-nociception of morphine, butorphanol and buprenorphine versus saline in the green iguana (*Iguana iguana*) using electrostimulation. *Journal of Herpetological Medicine and Surgery* **16**, 88–92
- ^d Koeller CA (2009) Comparison of buprenorphine and butorphanol analgesia in the eastern red-spotted newt (*Notophthalmus viridescens*). *Journal of the American Association of Laboratory Animal Science* **48**(2), 171–175
- ^e Terril-Robb L, Suckow M and Gridgesby C (1996) Evaluation of the analgesic effects of butorphanol tartrate, xylazine hydrochloride and flunixin meglumine in leopard frogs (*Rana pipiens*). *Contemporary Topics in Laboratory Animal Science* **35**, 54–56
- ^f Baker TR, Baker BB, Johnson SM and Sladky KK (2013) Comparative analgesic efficacy of morphine sulfate and butorphanol tartrate in koi (*Cyprinus carpio*) undergoing unilateral gonadectomy. *Journal of the American Veterinary Medical Association* **243**(6), 882–890

Butylscopolamine (Hyoscine)

(Buscopan) **POM-V, P**

Formulations: **Injectable:** 4 mg/ml butylscopolamine + 500 mg/ml metamizole in 100 ml multidose bottle (Buscopan Compositum); 20 mg/ml butylscopolamine only, in 2 ml ampoules. **Oral:** 10 mg tablet containing butylscopolamine only.

Action: Inhibits M1 muscarinic acetylcholine receptors in the GI and urinary tracts causing smooth muscle relaxation but does not cross blood–brain barrier.

Use: Control of GI pain in rabbits. Control of pain associated with urinary obstruction in rabbits. Must only be used in combination with investigations into the cause of abdominal pain or definitive relief of urinary obstruction.

Safety and handling: Avoid self-injection: metamizole can cause reversible but potentially serious agranulocytosis and skin allergies. Protect solution from light.

Contraindications: Intestinal obstruction. Contraindicated for the treatment of GI ileus in rabbits.

Adverse reactions: Dry mouth, blurred vision, hesitant micturition and constipation at doses acting as gut neuromuscular relaxants. The i.m. route may cause a local reaction.

Drug interactions: Metamizole should not be given to animals that have been treated with a phenothiazine, as hypothermia may result. Effects may be potentiated by concurrent use of other anticholinergic or analgesic drugs.

DOSES

Mammals: **Rabbits:** 0.1 ml/kg i.v., i.m. q12h (Buscopan Compositum) as analgesic/antispasmodic.

Birds, Reptiles, Amphibians, Fish: No information available.

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A

Cabergoline

(Galastop) **POM-V**

B

Formulations: Oral: 50 µg (micrograms)/ml solution.

C

Action: Potent selective inhibition of prolactin.

D

Use: Has been used for treatment of pituitary adenomas in rats. Cabergoline does not seem reliably effective in inducing abortion in the rat, mouse or rabbit. Generally ineffective if given for short periods in rabbits due to lack of clearly defined oestrus. May have beneficial effect in birds with chronic egg laying and other hormonal disorders related to high prolactin levels when combined with management changes. In birds it is also conjectured that its action could be mediated via its effect as a dopamine agonist. There do appear differences in both role of prolactin in different species' breeding cycles and also in whether or not dopamine is involved in prolactin release in different species so consequently, responses can be variable^a. For chronic laying problems, alteration of light cycles may give more consistent results.

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Safety and handling: Normal precautions should be observed.

K

Contraindications: Should not be used in combination with hypotensive drugs or in animals in a hypotensive state.

L

Adverse reactions: Not reported in exotic species. However, in dogs and cats vomiting or anorexia may occur after the first one or two doses in a small proportion of cases; there is no need to discontinue treatment unless vomiting is severe or it persists beyond the second dose. In some animals a degree of drowsiness may be seen in the first 2 days of dosing. May induce transient hypotension. Shown not to impair fertility in the male rat, non-teratogenic in mice and rabbits, and does not affect the latter phase of gestation or parturition in the female rat^b.

M

N

O

Drug interactions: Metoclopramide antagonizes the effects on prolactin.

P

DOSES

Q

Mammals: Rats: 10–50 µg (micrograms)/kg p.o. q12–24h or 600 µg/kg p.o. q72h for pituitary adenoma and associated mammary pathology^{c,1}.

R

Birds: 10–50 µg (micrograms)/kg p.o. q24h².

S

Reptiles, Amphibians, Fish: No information available.

T

References

U

^a Smiley KO and Adkins-Regan E (2016) Prolactin is related to individual differences in parental behavior and reproductive success in a biparental passerine, the zebra finch (*Taeniopygia guttata*). *General and Comparative Endocrinology* **234**, 88–94

V

^b Beltrame D, Longo M and Mazué G (1996) Reproductive toxicity of cabergoline in mice, rats and rabbits. *Reproductive Toxicology* **10**(6), 471–483

W

X

^c Eguchi K, Kawamoto K, Uozumi T *et al.* (1995) Effect of cabergoline, a dopamine agonist, on estrogen-induced rat pituitary tumors: *In Vitro Culture Studies. Endocrine Journal* **42**(3), 413–420

Y

¹ Mayer J, Sato A, Kiupel M, DeCubellis J and Donnelly T (2011) Extralabel use of cabergoline in the treatment of a pituitary adenoma in a rat. *Journal of the American Veterinary Medical Association* **239**(5), 656–660

Z

² Chitty J, Raftery A and Lawrie A (2006) Use of cabergoline in companion psittacine birds. *Proceedings of the Association of Avian Veterinarians* pp. 65–67

CaEDTA see **Edetate calcium disodium**

Calcium salts (Calcium borogluconate, Calcium carbonate, Calcium chloride, Calcium gluconate, Calcium lactate)

(Zolcal D, Many cattle preparations, e.g. Calcibor. Minor component of **Aqupharm No. 11**. Several generic formulations) **POM-V, ESPA**

Formulations: There are many formulations available; a selection is given here:

- Injectable: 200 mg/ml calcium borogluconate solution equivalent to 15 mg/ml calcium formed from 168 mg/ml of calcium gluconate and 34 mg/ml boric acid (Calcibor 20); 100 mg/ml (10%) calcium chloride solution containing 27.3 mg/ml elemental calcium (= 1.36 mEq calcium/ml = 680 µmol (micromoles)/ml); 100 mg/ml calcium gluconate solution 10 ml ampoules containing 9 mg elemental calcium/ml (= mEq calcium/ml)
- Oral solution: 35 mg/ml calcium gluconate, 25 IU/ml cholecalciferol, 2 mg/ml magnesium (Zolcal D)
- Note on other formulations: 11.2 mg calcium gluconate, 13.3 mg calcium borogluconate, 7.7 mg calcium lactate, 3.6 mg calcium chloride; each contains 1 mg elemental calcium = 0.5 mEq calcium.

Action: Calcium is an essential element involved in maintenance of numerous homeostatic roles and key reactions including activation of key enzymes, cell membrane potentials and nerve and musculoskeletal function.

Use: Management of hypocalcaemia and hyperkalaemic cardiotoxicity associated with urinary obstruction. Calcium gluconate and borogluconate are preferred for this. Serum calcium levels and renal function tests should be assessed before starting therapy. ECG monitoring during i.v. infusions is advised. Avoid using mixed electrolyte solutions intended for cattle use if possible. Treatment of hyperkalaemic cardiotoxicity with calcium rapidly corrects arrhythmias but effects are short-lived (5–10 min to effect) and i.v. glucose 0.5–1 g/kg ± insulin may be needed to decrease serum potassium. Parenteral calcium should be used very cautiously in dehydrated birds and reptiles and in patients receiving digitalis glycosides or those with cardiac or renal disease. Used before oxytocin in medical treatment of birds and reptiles with egg retention/dystocia. Calcium salts are given to many captive birds (especially Grey parrots) and reptiles as a routine dietary supplement for prevention of nutritional secondary hyperparathyroidism.

Safety and handling: Normal precautions should be observed.

Contraindications: Ventricular fibrillation or hypercalcaemia. Calcium should be avoided in pregnancy unless there is a deficient state. Hyperkalaemia associated with hypoadrenocorticism is often associated with hypercalcaemia and therefore additional calcium is not recommended in those cases.

Adverse reactions: Hypercalcaemia can occur, especially in renal impairment or cardiac disease. Tissue irritation is common and can occur with injectable preparation regardless of route. Rapid injection may cause hypotension, cardiac arrhythmias and cardiac arrest. Perivascular administration is treated by stopping the infusion, infiltrating the tissue with normal saline and topical application of corticosteroids.

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A Be careful using i.v. in dehydrated reptiles and birds as has been suggested to precipitate gout.

B

C **Drug interactions:** Patients on digitalis glycosides are more prone to develop arrhythmias if given i.v. calcium. All calcium salts may antagonize verapamil and other calcium-channel blockers. Calcium borogluconate is compatible with most i.v. fluids except those containing other divalent cations or phosphorus. Calcium borogluconate is reportedly compatible with lidocaine, adrenaline and hydrocortisone. Calcium chloride is incompatible with amphotericin B, cefalotin sodium and chlorphenamine. Calcium gluconate is incompatible with many drugs, including lipid emulsions, propofol, amphotericin B, cefamandole, naftate, cefalotin sodium, dobutamine, methylprednisolone sodium succinate and metoclopramide. Consult manufacturers' data sheets for incompatibilities with other solutions.

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DOSES

I **Mammals:** Guinea pigs, Chinchillas: 100 mg/kg calcium gluconate i.m., i.p. once, repeat as necessary; **Primates:** 200 mg/kg calcium gluconate s.c., i.m., i.v. once, repeat if necessary; 10–20 mg/kg calcium chloride i.v. slow infusion; 1 ml/kg calcium glubionate p.o. q12h; **Sugar gliders:** 100 mg/kg calcium gluconate s.c. q12h for 3–5 days (dilute in saline to 10 mg/ml) for nutritional secondary hyperparathyroidism; 150 mg/kg calcium glubionate p.o. q24h; **Hedgehogs:** Hypocalcaemia: 50 mg/kg calcium gluconate i.m.

M **Birds:** Egg retention, hypocalcaemia: 150–200 mg/kg calcium borogluconate i.m., s.c., slow i.v. once; Hypocalcaemia: 5–10 mg/kg calcium gluconate i.m. q12h.

N **Reptiles:** Dystocia, hypocalcaemia: 100 mg/kg calcium borogluconate i.v., i.m., s.c. or 50–100 mg/kg calcium gluconate i.m., s.c. once, repeat as necessary. 1 ml/kg p.o. q24h (Zolcal D).

O **Amphibians:** 1 mg/kg calcium glubionate p.o. q24h; Hypocalcaemic tetany: 100–200 mg/kg calcium gluconate s.c. once; Nutritional secondary hyperparathyroidism: 2.3% bath of calcium gluconate (with 2–3 IU/ml vitamin D3).

P **Fish:** No information available.

Carbimazole

(Vidalta) **POM-V**

Formulations: Oral: 10 mg, 15 mg tablets in a sustained-release formulation.

Action: Carbimazole is metabolized to the active drug thiamazole, which interferes with the synthesis of thyroid hormones.

Use: Control of thyroid hormone levels in guinea pigs.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Vomiting (in relevant species) and inappetence/anorexia may be seen but are often transient. Jaundice, cytopenias, immune-mediated diseases and dermatological changes (pruritus, alopecia and self-induced trauma) are reported but rarely seen. Treatment of hyperthyroidism can decrease glomerular filtration

rate, thereby raising serum urea and creatinine values, and can occasionally unmask occult renal failure. Animals that have an adverse reaction to thiamazole are likely also to have an adverse reaction to carbimazole.

Drug interactions: Carbimazole should be discontinued before iodine-131 treatment. Do not use with low iodine prescription diets.

DOSES

Mammals: Guinea pigs: 1–2 mg/kg p.o. q24h¹.

Birds, Reptiles, Amphibians, Fish: No information available.

References

- ¹ Künzel F, Hierlmeier B, Christian M and Reifinger M (2013) Hyperthyroidism in four Guinea pigs: clinical manifestations, diagnosis and treatment. *Journal of Small Animal Practice* **54**(12), 667–671

Carbomer 980

(LubriThal) **P, general sale**

Formulations: Ophthalmic: 0.2% (10 g tube, single-use vial), 0.25% (10 g tube) gel. This formulation is marketed specifically for small animals. Other formulations are widely available for general sale.

Action: Mucinomimetic, replacing both aqueous and mucin components of the tear film.

Use: Tear replacement and beneficial for management of quantitative (keratoconjunctivitis sicca (KCS) or dry eye) and qualitative tear film disorders. It has longer corneal contact time than the aqueous tear substitutes.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: It is tolerated well and ocular irritation is unusual.

Drug interactions: No information available.

DOSES

Mammals: 1 drop per eye q4–6h. Use during anaesthesia: apply every 15 minutes to avoid ocular drying, especially if ketamine combinations are used or forced warm air warming is employed.

Birds, Reptiles, Amphibians, Fish: No information available.

Carboplatin

(Carboplatin*, Paraplatin*) **POM**

Formulations: Injectable: 10 mg/ml solution.

Action: Binds to DNA to form intra- and interstrand crosslinks and DNA-protein crosslinks, resulting in inhibition of DNA synthesis and function.

Use: May be of use in a number of neoplastic diseases including anal adenocarcinoma, squamous cell carcinoma, ovarian carcinoma, mediastinal carcinoma, pleural adenocarcinoma, nasal carcinoma and thyroid adenocarcinoma. The drug is highly irritant and must be administered via a preplaced i.v. catheter. Do not use needles or i.v. sets

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containing aluminium as precipitation of the drug may occur. This drug is generally now preferred over cisplatin due to reduced GI and renal toxicity. Use with caution in patients with abnormal renal function, active infections, hearing impairment or pre-existing hepatic disease. Has been used in renal adenocarcinoma in budgerigars and squamous cell carcinoma (mixed with bird's own plasma for intralesional use) and bile duct carcinoma in Amazon parrots. However most reports are in single or small case numbers- as such the literature should always be checked before use especially when looking to start therapy in a species in which this drug has not previously been reported. The health risks to owners of (particularly indoor) birds from such treatments need to be considered carefully before recommending chemotherapy in birds.

Safety and handling: **Potent cytotoxic drug that should only be prepared and administered by trained personnel. See specialist texts for further advice on chemotherapeutic agents.**

Contraindications: No information available.

Adverse reactions: Include myelosuppression, nephrotoxicity, ototoxicity, nausea, vomiting, electrolyte abnormalities, neurotoxicity and anaphylactic reactions. However, produces fewer adverse reactions than cisplatin.

Drug interactions: Concomitant use of aminoglycosides or other nephrotoxic agents may increase risk of nephrotoxicity. May adversely affect the safety and efficacy of vaccinations.

DOSES

Birds: 5 mg/kg i.v., intraosseous over 3 mins; 5 mg/kg intralesional use (mixed with plasma to make a concentration of 10 mg/ml) for squamous cell carcinoma^{a,b}.

Mammals, Reptiles, Amphibians, Fish: No information available.

References

- ^a Filippich LJ, Charles BG, Sutton RH and Bucher AM (2004) Carboplatin pharmacokinetics following a single-dose infusion in sulphur-crested cockatoos (*Cacatua galerita*). *Australian Veterinary Journal* **82**(6), 366–369
- ^b Antonissen G, Devreese M, De Baere S *et al.* (2015) Comparative pharmacokinetics and allometric scaling of carboplatin in different avian species. *PLOS One* **10**(7), e0134177

Carnidazole

(Harkers Carnidazole, Spartrix) **AVM-GSL, ESPA**

Formulations: Oral: 10 mg tablet.

Action: Coccidiocidal; mode of action not known.

Use: Pigeon canker (*Trichomonas columbae*); treat all birds in loft simultaneously. It should be used in conjunction with good loft hygiene, including disinfection of feed and water bowls. Appears safe in a range of raptor species.

Safety and handling: Direct contact with the product must be avoided. Wear impermeable gloves when handling.

Contraindications: Not to be used in birds intended for human consumption.

Adverse reactions: No information available.

Drug interactions: None known.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Birds: Raptors: 25–30 mg/kg p.o., one dose normally sufficient but can be repeated next day if needed; **Psittacids:** 30–50 mg/kg p.o., repeat after 2 weeks; **Pigeons:** 12.5–25 mg p.o. once; **Other birds:** 20–30 mg/kg p.o. once.

Mammals, Reptiles, Amphibians, Fish: No information available.

Carprofen

(Canidryl, Carprieve, Carprodyl, Carprox Vet, Rimadyl, Rimifin, Rycarfa) **POM-V**

Formulations: Injectable: 50 mg/ml. Oral: 20 mg, 50 mg, 100 mg tablets (in plain and palatable formulations).

Action: Preferentially inhibits COX-2 enzyme, thereby limiting the production of prostaglandins involved in inflammation. Other non-COX-mediated mechanisms are suspected to contribute to the anti-inflammatory effect but these have not yet been identified.

Use: Control of postoperative pain and inflammation following surgery and reduction of chronic inflammation, e.g. degenerative joint disease, osteoarthritis. All NSAIDs should be administered cautiously in the perioperative period. Although carprofen preferentially inhibits COX-2, it may still adversely affect renal perfusion during periods of hypotension. If hypotension during anaesthesia is anticipated, delay carprofen administration until the animal is fully recovered from anaesthesia and is normotensive. Liver disease will prolong the metabolism of carprofen, leading to the potential for drug accumulation and overdose with repeated dosing. Prolonged long-term treatment should be under veterinary supervision. Use with caution in birds with dehydration, shock and renal dysfunction. Carprofen has induced some analgesic effects in rainbow trout.

Safety and handling: Formulations that use palatable tablets can be extremely palatable. Animals have been reported to eat tablets spontaneously, resulting in overdose. Ensure that tablets are stored out of animal reach. Store injectable solution in the refrigerator; once broached the product is stable for use at temperatures up to 25°C for 28 days.

Contraindications: Do not give to dehydrated, hypovolaemic or hypotensive patients or those with GI disease or blood clotting abnormalities. Administration of carprofen to animals with renal disease must be carefully evaluated and is not advisable in the perioperative period. Do not give to pregnant animals or animals <6 weeks old. Care may be needed when using in vultures.

Adverse reactions: GI signs may occur in all animals after NSAID administration. Stop therapy if this persists beyond 1–2 days. Some animals develop signs with one NSAID and not another. A 1–2 week wash-out period should be allowed before starting another NSAID after cessation of therapy. Stop therapy immediately if GI bleeding is suspected. There is a small risk that NSAIDs may precipitate cardiac failure in humans and this risk in animals is unknown. Fish may exhibit depressed activity at higher dose rates.

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A **Drug interactions:** Different NSAIDs should not be administered within 24 hours of each other or glucocorticoids as they are more ulcerogenic when used concurrently. The nephrotoxic tendencies of all NSAIDs are significantly increased when administered concurrently with other nephrotoxic agents (e.g. aminoglycosides).

C **DOSES**

D **Mammals:** Ferrets: 1–5 mg/kg total daily dose p.o.; Rabbits: 2–4 mg/kg s.c. q24h or 1.5 mg/kg p.o. q24h^a; Rodents: 2–5 mg/kg total daily dose i.v., i.m., s.c., p.o. in single or two divided doses^{b,c}; Others: 2–4 mg/kg i.v., i.m., s.c. q24h.

F **Birds:** 1–5 mg/kg i.m., s.c., p.o. q12–24h (higher rate appears effective for 24 hours). Note: 3 mg/kg i.m. q12h was not sufficient to provide analgesia in experimentally induced arthritis^d.

G **Reptiles:** 4 mg/kg s.c., i.m., p.o. once, then 2 mg/kg s.c., i.m., p.o. q24h; Bearded dragons: 2 mg/kg i.m. q24h.

H **Fish:** 1–5 mg i.m.

I **Amphibians:** No information available.

J **References**

- K** ^a Hawkins MG, Taylor IT, Craigmill AL and Tell LA (2008) Enantioselective pharmacokinetics of racemic carprofen in New Zealand white rabbits. *Journal of Veterinary Pharmacology and Therapeutics* **31**(5), 423–430
- L** ^b Roughan JV and Flecknell PA (2004) Behaviour-based assessment of the duration of laparotomy-induced abdominal pain and the analgesic effects of carprofen and buprenorphine in rats. *Behavioural Pharmacology* **15**, 461–472
- M** ^c Zegre-Cannon C, Kissling GE, Goulding DR, King-Herbert AP and Blankenship-Paris T (2011) Analgesic effects of tramadol, carprofen or multimodal analgesia in rats undergoing ventral laparotomy. *Laboratory Animal* **40**(3), 85–93
- N** ^d Paul-Murphy JR, Sladky KK, Krugner-Higby LA et al. (2009) Analgesic effects of carprofen and liposome-encapsulated butorphanol tartrate in Hispaniolan parrots (*Amazona ventralis*) with experimentally induced arthritis. *American Journal of Veterinary Research* **70**(10), 1201–1210

P **Carvedilol**

Q (Carvedilol*) **POM**

R **Formulations:** Oral: 3.125 mg, 6.25 mg, 12.5 mg, 25 mg tablets.

S **Action:** Non-selective beta-adrenergic blocker with the afterload reduction properties of an alpha-1 adrenergic blocker. Additional antioxidant properties may decrease the oxidant stress associated with heart failure.

T **Use:** Has been advocated for use as an adjunctive therapy in the management of chronic heart failure due to valvular disease or DCM and as an antihypertensive drug in patients that do not respond to first-line therapy. Veterinary experience is limited and benefit has not been established. Limited data on pharmacokinetics and pharmacodynamics in animals. Treatment should not be started until congestive heart failure has been stabilized for at least 2 weeks initially. Since it undergoes extensive hepatic metabolism, caution should be exercised in patients with hepatic insufficiency.

Y **Safety and handling:** Normal precautions should be observed.

Z **Contraindications:** Patients with bradyarrhythmias, acute or decompensated heart failure and bronchial disease. Do not administer concurrently with alpha-adrenergic agonists (e.g. adrenaline).

Adverse reactions: Potential side effects include lethargy, diarrhoea, bradycardia, AV block, myocardial depression, exacerbation of heart failure, syncope, hypotension and bronchospasm. A reduction in the glomerular filtration rate may exacerbate pre-existing renal impairment.

Drug interactions: The hypotensive effect of carvedilol is enhanced by many agents that depress myocardial activity including anaesthetic agents, phenothiazines, antihypertensive drugs, diuretics and diazepam. There is an increased risk of bradycardia, severe hypotension, heart failure and AV block if carvedilol is used concurrently with calcium-channel blockers. Hypotensive effect may be antagonized by NSAIDs. Concurrent digoxin administration potentiates bradycardia. Carvedilol may enhance the hypoglycaemic effect of insulin. Carvedilol increases plasma concentration of ciclosporin. Rifampin can decrease carvedilol plasma concentrations.

DOSES

Mammals: Rats: 2–30 mg/kg p.o. q24h^a; Hamsters: 1–11 mg/kg p.o. q24h^{b,c,d}. Start at low end of dose range.

Birds: 1–9 mg/kg p.o. q12–24h.

Reptiles, Amphibians, Fish: No information available.

References

- ^a Watanabe K, Ohta Y, Nakazawa M *et al.* (2000) Low dose carvedilol inhibits progression of heart failure in rats with dilated cardiomyopathy. *Pharmacology* **130**(7), 1489–1495
- ^b Cruz N, Arocho L, Rosario L and Crespo MJ (2007) Chronic administration of carvedilol improves cardiac function in 6-month-old Syrian cardiomyopathic hamsters. *Pharmacology* **80**, 144–150
- ^c Nanjo S, Yamazaki J, Yoshikawa K, Ishii T and Togane Y (2006) Carvedilol prevents myocardial fibrosis in hamsters. *International Heart Journal* **47**(4), 607–61
- ^d Zendaoui A, Lachance D, Roussel E, Couet J and Arsenault M (2011) Usefulness of carvedilol in the treatment of chronic aortic valve regurgitation. *Circulation: Heart Failure* **4**(2), 207–213

CCNU see **Lomustine**

Cefalexin (Cephalexin)

(Cefaseptin, Cephacare, Ceporex, Rilexine, Therios)

POM-V

Formulations: Injectable: 180 mg/ml (18%) suspension. Oral: 50 mg, 75 mg, 120 mg, 250 mg, 300 mg, 500 mg, 600 mg, 750 mg tablets; granules which, when reconstituted, provide a 100 mg/ml oral syrup.

Action: A 1st generation cephalosporin that binds to proteins involved in bacterial cell wall synthesis, thereby decreasing cell wall strength and rigidity, and affecting cell division. Resistant to some bacterial beta-lactamases, particularly those produced by *Staphylococcus* spp. As for other beta-lactam antibiotics, works in a time-dependent fashion.

Use: Active against several Gram-positive and Gram-negative organisms (e.g. *Staphylococcus*, *Pasteurella* and *Escherichia coli*). *Pseudomonas* and *Proteus* are often resistant. Maintaining levels above the MIC is critical for efficacy and prolonged dosage intervals or missed doses can compromise therapeutic response. As such it is only really viable as an option in companion birds where owners are genuinely able to dose their birds accurately. Dose and dosing interval is determined by

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infection site, severity and organism. In severe or acute conditions, doses may be doubled or given at more frequent intervals.

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Safety and handling: Reconstituted oral drops should be stored in the refrigerator and discarded after 10 days.

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Contraindications: Patients hypersensitive to penicillins may also be sensitive to cephalosporins (cross-hypersensitivity in <10% of human patients); avoid use in animals with reported sensitivity to other beta-lactam antimicrobials. Avoid oral administration in small herbivores (e.g. rabbits).

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Adverse reactions: Vomiting (where species are susceptible) and diarrhoea most common; administration with food may reduce these. Can cause fatal enterotoxaemia in small herbivores especially if given via the oral route which should be avoided. Injection may be painful.

Drug interactions: Bactericidal activity may be affected by concomitant use of bacteriostatic agents (e.g. erythromycin, oxytetracycline). May be an increased risk of nephrotoxicity if cephalosporins are used with amphotericin or loop diuretics (e.g. furosemide); monitor renal function. Do not mix in the same syringe as aminoglycosides.

DOSES

Classified as category C (Caution) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

K**L****M****N****O**

Mammals: Ferrets: 15–30 mg/kg p.o. q8–12h; Rabbits: 15–20 mg/kg s.c. q12–24h; Guinea pigs: 25 mg/kg i.m. q12–24h; Primates: 20–30 mg/kg p.o. q12h; Sugar gliders: 30 mg/kg p.o., s.c. q24h; Hedgehogs: 25 mg/kg p.o. q8h; Others: 15–30 mg/kg i.m. q8–12h.

Birds: 35–100 mg/kg p.o., i.m. q6–8h^a.

Reptiles: 20–40 mg/kg p.o. q12h.

Amphibians, Fish: No information available.

P**Q****R****S**

References

- ^a Bush M, Locke D, Neal LA and Carpenter JW (1981) Pharmacokinetics of cephalothin and cephalexin in selected avian species. *American Journal of Veterinary Research* **42**, 1014–1017

T**U****V****W****X****Y****Z**

Cefotaxime

(Cefotaxime*) **POM**

Formulations: Injectable: 500 mg, 1 g, 2 g powders for reconstitution.

Action: A 3rd generation cephalosporin that binds to proteins involved in bacterial cell wall synthesis, thereby decreasing cell wall strength and rigidity, and affecting cell division. Resistant to many bacterial beta-lactamases, particularly those produced by *Staphylococcus* spp. As other beta-lactam antibiotics, works in a time-dependent fashion.

Use: Good activity against many Gram-negative organisms, especially Enterobacteriaceae (not *Pseudomonas*) but lower activity against many Gram-positive organisms than 1st and 2nd generation cephalosporins. It is important to maintain tissue concentrations above the MIC. Use should be reserved for: patients with acute sepsis or serious infections; where cultures are pending or culture and sensitivity testing shows sensitivity; or where authorized preparations are not appropriate, and the animal is not a good candidate for intensive aminoglycoside therapy

(e.g. pre-existing renal dysfunction). Use with care in patients with renal disease and consider increasing dose interval. *Escherichia coli* resistance reported in feral birds ^a.

Safety and handling: The reconstituted solution is stable for 10 days when refrigerated.

Contraindications: Patients hypersensitive to penicillins may also be sensitive to cephalosporins (cross-hypersensitivity in <10% of human patients); avoid use in animals with reported sensitivity to other beta-lactam antimicrobials. Avoid oral administration in small herbivores (e.g. rabbits).

Adverse reactions: May produce pain on injection. GI disturbance and superinfection with resistant microorganisms is a potential risk. Can cause fatal enterotoxaemia in small herbivores especially if given via the oral route, which should be avoided.

Drug interactions: Bactericidal activity may be affected by concomitant use of bacteriostatic agents (e.g. oxytetracycline, erythromycin). The cephalosporins are synergistic with the aminoglycosides, but should not be mixed in the same syringe. May be increased risk of nephrotoxicity if cephalosporins are used with amphotericin or loop diuretics (e.g. furosemide); monitor renal function.

DOSES

Classified as category B (Restrict) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Rabbits, Primates: 50 mg/kg i.m., i.v. q8h.

Birds: 50–100 mg/kg i.m. q6–8h.

Reptiles: 20–40 mg/kg i.m. q24h.

Amphibians, Fish: No information available.

References

- ^a Zurfluh K, Nüesch-Inderbilen M, Stephan R and Hächler H (2013) Higher generation cephalosporin-resistant *Escherichia coli* in feral birds in Switzerland. *International Journal of Antimicrobial Agents* **41(30)**, 296–297

Cefovecin

(Convenia) **POM-V**

Formulations: Injectable: Lyophilized powder which when reconstituted contains 80 mg/ml cefovecin.

Action: A 3rd generation cephalosporin that binds to proteins involved in bacterial cell wall synthesis, thereby decreasing cell wall strength and rigidity, and affecting cell division. Resistant to some bacterial beta-lactamases. Assumed to work in a time-dependent fashion as other beta-lactam antibiotics.

Use: In line with rational antimicrobial use, cefovecin should not be considered if a 14 day course of antimicrobial would not ordinarily be required for the infection being treated. Specifically indicated for the prolonged treatment of skin and soft tissue infections and for infections of the urinary tract. Also used as part of the management of severe periodontal disease. Good efficacy against organisms commonly associated with these conditions (e.g. *Staphylococcus*, *Streptococcus*, *Escherichia coli*, *Pasteurella multocida*, *Proteus*).

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Activity against anaerobes such as *Prevotella*, *Fusobacterium*, *Bacteroides* and *Clostridium* also appears to be good. Not active against *Pseudomonas*. Due to unique pharmacokinetic profile, cefovecin has an extremely long half-life in some mammals but not in ferrets, primates, birds or reptiles. Cefovecin concentrations were detectable for 4 days in white bamboo sharks and 7 days in copper rockfish following a single subcutaneous injection but there was some variability in plasma concentrations between individual fish.

Safety and handling: Store in the refrigerator even prior to reconstitution; use reconstituted drug within 28 days.

Contraindications: Patients hypersensitive to penicillins may also be sensitive to cephalosporins (cross-hypersensitivity in <10% of human patients); avoid use in animals with reported sensitivity to other beta-lactam antimicrobials. Do not use in rabbits, guinea pigs and other small herbivores. Avoid use during lactation and in pregnant animals, as safety has not been established.

Adverse reactions: Appears to be relatively safe but has not been assessed in renal disease. Reported adverse reactions include mild GI disturbance and transient swelling at the injection site. Can cause fatal enterotoxaemia in small herbivores especially if given via the oral route, which should be avoided.

Drug interactions: Highly bound to plasma proteins, therefore it would be prudent to exhibit caution when using in conjunction with other highly protein-bound drugs such as furosemide and NSAIDs.

DOSES

Classified as category B (Restrict) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Ferrets: 8 mg/kg s.c. q48–72h.

Birds, Reptiles: Initial data appear to show it is not practicable (half-life <2h in poultry, <4h in hens and green iguanas^a, <7h in red-eared sliders^b, ~20h in Hermann's tortoises^c). Further data show that the half-life in parrots is similar to that in chickens.

Fish: White bamboo sharks: 8 mg/kg s.c. q4d^d; Copper rockfish: 16 mg/kg s.c. q7d^e.

Amphibians: No information available.

References

- ^a Thuesen R, Bertelsen MF, Brimer L and Skaanild MT (2009) Selected pharmacokinetic parameters for cefovecin in hens and green iguanas. *Journal of Veterinary Pharmacology and Therapeutics* **32**(6), 631–617
- ^b Synniewski LA, Maxwell LK, Murray JK, Brandão JL and Papich MG (2017) Cefovecin pharmacokinetics in the red-eared slider. *Journal of Exotic Pet Medicine* **26**(2), 108–113
- ^c Nardini G, Barbarossa A, Dall'Occo A *et al.* (2014) Pharmacokinetics of cefovecin sodium after subcutaneous administration to Hermann's tortoises (*Testudo hermanni*). *American Journal of Veterinary Research* **75**(10), 918–923
- ^d Steeil JC, Schumacher J, George RH *et al.* (2014) Pharmacokinetics of cefovecin (Convenia®) in white bamboo sharks (*Chiloscyllium plagiosum*) and Atlantic horseshoe crabs (*Limulus polyphemus*). *Journal of Zoo and Wildlife Medicine* **45**(2), 389–392
- ^e Seeley KE, Wolf KN, Bishop MA, Turnquist M and KuKanich B (2016) Pharmacokinetics of long-acting cefovecin in copper rockfish (*Sebastes caurinus*). *American Journal of Veterinary Research* **77**(3), 260–264

Ceftazidime

(Fortum*, Kefadim*) **POM**

Formulations: Injectable: 500 mg, 1 g, 2 g, 3 g powders for reconstitution.

Action: A 3rd generation cephalosporin that binds to proteins involved in bacterial cell wall synthesis, thereby decreasing cell wall strength and rigidity, and affecting cell division. Resistant to some bacterial beta-lactamases. As other beta-lactam antibiotics, works in a time-dependent fashion.

Use: Higher activity against many Gram-negative organisms but lower activity against many Gram-positive organisms when compared to 1st and 2nd generation cephalosporins. Very good activity against *Pseudomonas* in humans. Use should be limited to cases with a confirmed susceptibility and acute sepsis or serious infections where authorized preparations are found to be inappropriate. Limited information on clinical pharmacokinetics in animal species and doses given below are empirical. Important to maintain tissue concentrations above the MIC with regular doses. Used for the treatment of bacterial disease in fish.

Safety and handling: Normal precautions should be observed.

Contraindications: Patients hypersensitive to penicillins may also be sensitive to cephalosporins (cross-hypersensitivity in <10% of human patients); avoid use in animals with reported sensitivity to other beta-lactam antimicrobials. Avoid oral administration in small herbivores (e.g. rabbits).

Adverse reactions: GI disturbances associated with drug use in humans. Pain may be noted following injection. Can cause fatal enterotoxaemia in small herbivores especially if given via the oral route, which should be avoided.

Drug interactions: Bactericidal activity may be affected by concomitant use of bacteriostatic agents (e.g. oxytetracycline, erythromycin). May be an increased risk of nephrotoxicity if cephalosporins are used with amphotericin or loop diuretics (e.g. furosemide); monitor renal function. Do not mix in the same syringe as aminoglycosides. Ceftazidime is synergistic with the aminoglycoside antimicrobials *in vivo* (often used in humans for pseudomonal infection in neutropenic patients).

DOSES

Classified as category B (Restrict) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Rabbits: 100 mg/kg i.m. q12h; Primates: 50 mg/kg i.m., i.v. q8h.

Birds: 75–200 mg/kg i.v., i.m. q6–12h.

Reptiles: Most species: 20 mg/kg i.m., s.c., i.v. q72h^a; more frequent dosing (q24–48h) has been suggested in chameleons. Less frequent dosing (q120h) has been suggested in eastern box turtles, yellow-bellied sliders and river cooters^b.

Amphibians: Most species: 20 mg/kg s.c., i.m. q48–72h; Northern leopard frogs: 20 mg/kg s.c. q24h^c; Eastern hellbenders: 20 mg/kg s.c. q5d^d.

Fish: 20 mg/kg i.m. q72h for 3–5 treatments¹.

References

^a Lawrence K, Muggleton PW and Needham J (1984) Preliminary study on the use of ceftazidime, a broad spectrum cephalosporin antibiotic, in snakes. *Research in Veterinary Science* **36**(1), 16–20

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

- ^b Cerreta AJ, Lewbart GA, Dise DR and Papich MG (2018) Population pharmacokinetics of ceftazidime after a single intramuscular injection in wild turtles. *Journal of Veterinary Pharmacology and Therapeutics* **41**(4), 495–501
- ^c Hawkins SJ, Cox SK and Sladky KK (2021) Pharmacokinetics of ceftazidime in Northern leopard frogs (*Lithobates pipiens*) at two different doses and administration routes. *American Journal of Veterinary Research* **82**(7), 560–565
- ^d Musgrave KE, Hanley CS and Papich MG (2021) Population pharmacokinetics of ceftazidime after a single subcutaneous injection and normal oral and cloacal bacterial flora survey in eastern hellbenders (*Cryptobranchus alleganiensis alleganiensis*). *Journal of Zoo and Wildlife Medicine* **52**(1), 90–96
- ¹ Govett PD, Buur J, Krein A, McDowell T and Weber ES (2009) Pharmacokinetics of ceftazidime in koi (*Cyprinus carpio*) after a single dose intramuscular or intracoelomic administration. *Proceedings of the International Association of Aquatic Animal Medicine 40th annual conference*, p.19

Ceftiofur

(Excenel) **POM-V**

Formulations: Injectable: 1 g, 4 g powder for reconstitution; 50 mg/ml suspension.

Action: A 3rd generation cephalosporin that binds to proteins involved in bacterial cell wall synthesis, thereby decreasing cell wall strength and rigidity and affecting cell division. Resistant to many bacterial beta-lactamases, particularly those produced by *Staphylococcus* spp. Uniquely among the cephalosporins, ceftiofur is metabolized to desfuroylceftiofur which is an active metabolite. Action is time-dependent.

Use: Higher activity against many Gram-negative organisms, especially Enterobacteriaceae (not *Pseudomonas*) but lower activity against many Gram-positives than 1st and 2nd generation cephalosporins. Use should be reserved for patients suffering from acute sepsis or serious infections where cultures are pending, other authorized preparations are not appropriate and the animal is not a good candidate for intensive aminoglycoside therapy (pre-existing renal dysfunction). Important to maintain tissue concentrations above the MIC. Use with care in patients with renal disease and consider increasing dose interval.

Safety and handling: Store powder in the refrigerator; once reconstituted store in the refrigerator and discard after 7 days or within 12 hours if stored at room temperature.

Contraindications: Patients hypersensitive to penicillins may also be sensitive to cephalosporins (cross-hypersensitivity in <10% of human patients); avoid use in animals with reported sensitivity to other beta-lactam antimicrobials. Avoid oral administration in small herbivores (e.g. rabbits).

Adverse reactions: May produce pain on injection. GI disturbance and superinfection with resistant microorganisms is a potential risk. May be an increased risk of nephrotoxicity if cephalosporins are used with amphotericin or loop diuretics (e.g. furosemide); monitor renal function. Can cause fatal enterotoxaemia in small herbivores especially if given via the oral route, which should be avoided.

Drug interactions: Bactericidal activity may be affected by concomitant use of bacteriostatic agents (e.g. oxytetracycline, erythromycin). The cephalosporins are synergistic with the aminoglycosides, but should not be mixed in the same syringe.

DOSES

Classified as category B (Restrict) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Primates: 1.1–2.2 mg/kg i.m. q24h; Hedgehogs: 20 mg/kg s.c. q12–24h.

Birds: Red-tailed hawks: 10 mg/kg q36–45h, 20 mg/kg q96–120h (longer dosage intervals for treatment of Gram-positive compared to Gram-negative infections)^a; Bald eagles: 10 mg/kg i.m. q60h or 20 mg/kg i.m. q80h^b; Amazons: 10 mg/kg i.m. q8–12h; Cockatiels: 10 mg/kg i.m. q4h; in some cases, higher doses (up to 100 mg/kg) may be necessary. 2 mg/kg given by i.v. regional limb perfusion achieved greater concentrations in joints so may be an option when treating susceptible infections in localized joints that are refractory to less critical antibiotics^c.

Reptiles: Chelonians: 4 mg/kg i.m. q24h; Green iguanas: 5 mg/kg i.m., s.c. q24h^d; Bearded dragons: 30 mg/kg i.m., s.c. lasts >12 days^e;

Ball pythons: 15 mg/kg i.m. lasts >5 days^f.

Amphibians, Fish: No information available.

References

- ^a Sadar MJ, Hawkins MG, Byrne BA *et al.* (2015) Pharmacokinetics of a single intramuscular injection of ceftiofur crystalline-free acid in red-tailed hawks (*Buteo jamaicensis*). *American Journal of Veterinary Research* **76**(12), 1077–1084
- ^b Sadar MJ, Cox SK, Duvall A and Jones MP (2021) Pharmacokinetics of a single intramuscular injection of ceftiofur crystalline free acid in bald eagles (*Haliaeetus leucocephalus*). *Journal of Avian Medicine and Surgery* **35**(3), 290–294
- ^c Knafo SE, Graham JE and Barton BA (2019) Intravenous and intraosseous regional limb perfusion of ceftiofur sodium in an avian model. *American Journal of Veterinary Research* **80**(6), 539–546
- ^d Benson KG, Tell LA, Young LA, Wetzlich S and Craigmill AL (2003) Pharmacokinetics of ceftiofur sodium after intramuscular or subcutaneous administration in green iguanas (*Iguana iguana*). *American Journal of Veterinary Research* **64**(10), 1278–1282
- ^e Churgin SM, Musgrave KE, Cox SK and Sladky KK (2014) Pharmacokinetics of subcutaneous versus intramuscular administration of ceftiofur crystalline-free acid to bearded dragons (*Pogona vitticeps*). *American Journal of Veterinary Research* **75**(5), 453–459
- ^f Adkesson MJ, Fernandez-Varon E, Cox S and Martin-Jiménez T (2011) Pharmacokinetics of a long-acting ceftiofur formulation (ceftiofur crystalline free acid) in the ball python (*Python regius*). *Journal of Zoo and Wildlife Medicine* **42**(3), 444–450

Cefuroxime

(Zinacef*, Zinnat*) **POM**

Formulations: Injectable: 250 mg, 750 mg, 1.5 g powders for reconstitution (sodium salt). Oral (as cefuroxime axetil): 125 mg, 250 mg tablets; 125 mg/5 ml suspension.

Action: A 2nd generation cephalosporin that binds to proteins involved in bacterial cell wall synthesis, thereby decreasing cell wall strength and rigidity, and affecting cell division. Resistant to some bacterial beta-lactamases. As other beta-lactam antibiotics, works in a time-dependent fashion. Cefuroxime axetil is hydrolysed in intestinal mucosa and liver to yield active drug giving oral bioavailability.

Use: Higher activity against many Gram-negative organisms when compared to 1st generation cephalosporins. Good activity against a wider spectrum of Enterobacteriaceae (not *Pseudomonas*). Many obligate anaerobes also susceptible. It is a time-dependent antimicrobial, so maintaining levels above the MIC is important for efficacy. Limited

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A applications in veterinary species and limited pharmacokinetic data make appropriate dose selection problematical.

B **Safety and handling:** Normal precautions should be observed.

C **Contraindications:** Patients hypersensitive to penicillins may also be sensitive to cephalosporins (cross-hypersensitivity in <10% of human patients); avoid use in animals with reported sensitivity to other beta-lactam antimicrobials. Avoid oral administration in small herbivores (e.g. rabbits).

D **Adverse reactions:** May cause pain on i.m. and s.c. injection. GI disturbance has been reported in humans, particularly associated with the oral axetil formulation. Can cause fatal enterotoxaemia in small herbivores especially if given via the oral route, which should be avoided.

E **Drug interactions:** Bactericidal activity may be affected by concomitant use of bacteriostatic agents (e.g. oxytetracycline, erythromycin). May be an increased risk of nephrotoxicity if cephalosporins are used with amphotericin or loop diuretics (e.g. furosemide); monitor renal function. Synergistic with aminoglycosides, do not mix in the same syringe.

F **DOSES**

G **Classified as category C (Caution) by the EMA.**

H **See Appendix for guidelines on responsible antibiotic use.**

I **Mammals:** Rabbits: 18.75 mg/kg i.v. q8h^a.

J **Reptiles:** 100 mg/kg i.m. q24h for 10 days at 30°C.

K **Birds, Amphibians, Fish:** No information available.

L **References**

M ^a Pützler J, Arens D, Metsemakers W *et al.* (2017) Duration of antibiotic prophylaxis with intravenous cefuroxime affects infection rate with *Staphylococcus aureus* in an open fracture model in rabbits. *Orthopaedic Proceedings* **99–B(S22)**, 63

N **Celecoxib**

O (Celebrex*, Celecoxib*) **POM**

P **Formulations:** Oral: 100 mg, 200 mg capsules.

Q **Action:** Preferentially inhibits COX-2 enzyme, thereby limiting the production of prostaglandins involved in inflammation.

R **Use:** Has been used for analgesia in the management of arthritis in rats and in the management of proventricular dilatation disease (PDD) in parrots. However, recent studies appear to cast doubt on its efficacy in PDDa. It may also be used as a conventional anti-inflammatory in birds.

S **Safety and handling:** Normal precautions should be observed.

T **Contraindications:** Do not give to dehydrated, hypovolaemic or hypotensive patients, or those with GI disease or clotting problems. Administration of celecoxib to animals with renal disease must be carefully evaluated and is not advisable in the perioperative period. Do not give to animals <6 weeks of age.

U **Adverse reactions:** GI signs may occur in all animals after NSAID administration. Stop therapy if this persists beyond 1–2 days. Some animals develop signs with one NSAID and not another. A 1–2 week wash-out period should be allowed before starting another NSAID after cessation of therapy. Stop therapy immediately if GI bleeding is suspected. There is a small risk that NSAIDs may precipitate cardiac failure in humans and this risk in animals is unknown.

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Drug interactions: No information available.

DOSES

Mammals: Rats: 10–20 mg/kg p.o.^b.

Birds: 10–20 mg/kg p.o. q24h^{a,c}.

Reptiles, Amphibians, Fish: No information available.

References

- ^a Escandon P, Heatley JJ, Tizard I *et al.* (2019) Treatment with nonsteroidal anti-inflammatory drugs fails to ameliorate Pathology in cockatiels experimentally infected with parrot Bornavirus-2. *Veterinary Medicine: Research and Reports* **10**, 185
- ^b Millecamps M, Jourdan D, Leger S *et al.* (2005) Circadian pattern of spontaneous behavior in monoarthritic Rats: a novel global approach to evaluation of chronic pain and treatment effectiveness. *Arthritis & Rheumatology* **52(11)**, 3470–3478
- ^c Gasthuys E, Houben R, Haesendonck R *et al.* (2019) Development of an *in vivo* lipopolysaccharide inflammation model to study the pharmacodynamics of COX-2 inhibitors celecoxib, mavacoxib, and meloxicam in cockatiels (*Nymphicus hollandicus*). *Journal of Avian Medicine and Surgery* **33(4)**, 349–360

Cephalexin see Cefalexin

Charcoal (Activated charcoal)

(Actidose-Aqua*, Charcodote*, Liqui-Char*) **AVM-GSL**

Formulations: Oral: 50 g activated charcoal powder or premixed slurry (200 mg/ml).

Action: Absorbs toxins, fluids and gases in the GI tract. Activated charcoal has increased porosity and enhanced absorptive capacity.

Use: In acute poisoning with organophosphates, carbamates, chlorinated hydrocarbons, strychnine, ethylene glycol, inorganic and organic arsenical and mercurial compounds, polycyclic organic compounds (most pesticides) and dermal toxicants that may be ingested following grooming. As a general rule administer at a dose of at least 10 times the volume of intoxicant ingested. Repeat dosing as required if emesis or massive toxin ingestion occurs. Repeated dosing necessary if highly lipid-soluble toxins, which are likely to undergo enterohepatic recirculation, have been ingested. Used in the treatment of oiled birds.

Safety and handling: Activated charcoal powder floats, covering everything in the area; prepare very carefully as it will stain permanently. Paste forms are available which are more controllable.

Contraindications: Activated charcoal should not be used prior to the use of emetics.

Adverse reactions: Charcoal colours stools black, which is medically insignificant but may be alarming to the owner.

Drug interactions: Activated charcoal reduces the absorption and therefore efficacy of orally administered drugs.

DOSES

Mammals: 0.5–5 g/kg p.o.

Birds: 3.75 g/kg p.o.

Reptiles: Chelonians: 2–8 g/kg by stomach tube.

Amphibians, Fish: No information available.

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Chitosan

B(Ipakitine) **GSL****C****Formulations:** Oral: Powder containing 8% chitosan, 10% calcium carbonate and 82% lactose.**D****Action:** Adsorbent for intestinal uraemic toxins, including phosphate.**E****Use:** The combination has been suggested to reduce serum urea and phosphate in chronic renal disease in ferrets, based on its use in cats. Phosphate-binding agents are usually only used if low phosphate diets are unsuccessful. Monitor serum phosphate levels at 10–14 day intervals and adjust dosage accordingly if trying to normalize serum concentrations. As formulation contains lactose, use with care in diabetic and lactose-intolerant animals.**F****G****Safety and handling:** Normal precautions should be observed.**H****Contraindications:** Not advised in rabbits due to the risk of hypercalcaemia.**I****Adverse reactions:** Hypercalcaemia, possibly due to the calcium carbonate component.**J****Drug interactions:** None noted.**K**

DOSES

L**Mammals:** Ferrets: 200 mg/kg p.o. q12h mixed with food (anecdotal);**M**Rats: 200 mg/kg chitosan p.o. q24h has been shown to have hepatic protective effects^a; 165–825 mg/kg chitosan p.o. q24h has been shown to have renal protective effects^b.**N****Birds, Reptiles, Amphibians, Fish:** No information available.**O**

References

P^a Ozcelik E, Uslu S, Erkasap N and Karimi H (2014) Protective effect of chitosan treatment against acetaminophen-induced hepatotoxicity. *The Kaohsiung Journal of Medical Sciences* **30**(6), 286–290**Q**^b Chou CK, Li YC, Chen SM, Shih YM and Lee JA (2015) Chitosan prevents gentamicin-induced nephrotoxicity via a carbonyl stress-dependent pathway. *BioMed Research International* 2015, 675714**R**

Chlorambucil

S(Leukeran*) **POM****T****Formulations:** Oral: 2 mg tablet.**U****Action:** Alkylating agent that inhibits DNA synthesis and function through cross-linking with cellular DNA. Cell cycle non-specific.**V****Use:** Management of some malignancies, lymphoproliferative, myeloproliferative and immune-mediated diseases. Immunosuppressive effect is not well defined and therefore it should only be considered where more established therapies such as prednisolone and azathioprine have failed. Has been used for treatment of lymphosarcoma in cockatoos.**W****X****Safety and handling:** **Cytotoxic drug; see specialist texts for further advice on chemotherapeutic agents.** Tablets should be stored in a closed, light-protected container under refrigeration (2–8°C).**Y****Z****Contraindications:** Bone marrow suppression, factors predisposing to infection.

Adverse reactions: Anorexia, nausea, vomiting, leucopenia, thrombocytopenia, anaemia (rarely), neurotoxicity, alopecia (rarely) and slow regrowth of clipped hair coat.

Drug interactions: Drugs that stimulate hepatic cytochrome P450 system increase cytotoxic effects. Prednisolone has a synergistic effect in the management of lymphoid neoplasia.

DOSES

See Appendix for chemotherapy protocols in ferrets.

Mammals: Ferrets: Used as part of a multi-drug protocol for lymphoma.

Birds: Lymphosarcoma: 1–2 mg/kg p.o. twice weekly¹; 2 mg/kg p.o. twice weekly for thymic lymphoma in a Java Sparrow².

Reptiles, Amphibians, Fish: No information available.

References

- ¹ Rivera S, McClearen JR and Reacill DR (2009) Treatment of non-epitheliotropic cutaneous B-cell lymphoma in an Umbrella cockatoo (*Cacatua alba*). *Journal of Avian Medicine and Surgery* **23**(4), 294–302
- ² Yu PH and Chi CH (2015) Long-term management of thymic lymphoma in a Java sparrow (*Lonchura oryzivora*). *Journal of Avian Medicine and Surgery* **29**(1), 51–55

Chloramine-T (Sodium N-chloro 4-methylbenzenesulfonamide, Sodium N-chloro-para-toluenesulfonylamide)

(Chloramine T, Halamid) **ESPA**

Formulations: 100% white powder.

Action: The chemical slowly decomposes in water to the hypochlorite anion to produce weak hypochlorous acid, which decomposes further to active chlorine and oxygen to provide a disinfectant effect.

Use: For the treatment of external water-borne bacterial infections, particularly bacterial gill disease, and some protozoan and trematode ectoparasites in fish. It should not be mixed in metal containers or ones in which other chemicals have been previously mixed. Do not use in bright sunlight or in water <12°C. Turn off UV and carbon filter systems for 24 hours. More toxic in water with low hardness and low pH. Make up fresh solution immediately prior to use since product degrades in water. Do not dispose of product in drains or waterways.

Safety and handling: Corrosive. Wear gloves and eye protection.

Contraindications: No information available.

Adverse reactions: Hyperplastic thickening and oedema of respiratory epithelia at significantly higher dose rates causing increased respiratory rates.

Drug interactions: The chemical should not be used with other chemicals (other than salt), particularly reducing agents such as formalin.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Fish: 1–2 mg/l by prolonged immersion; repeat daily for 4 days if necessary.

Mammals, Birds, Reptiles, Amphibians: No information available.

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Chloramphenicol

(Chloramphenicol*, Chloromycetin Ophthalmic Ointment*, Chloromycetin Redidrops*, Kemeticine*) **POM**

C Formulations: **Injectable:** 1 g powder for reconstitution. **Ophthalmic:** 1% ointment; 0.5% solution. **Oral:** 250 mg capsules.

D Action: Bacteriostatic antimicrobial that acts by binding to the 50S ribosomal subunit of susceptible bacteria, preventing bacterial protein synthesis.

E Use: Broad spectrum of activity against Gram-positive (e.g. *Streptococcus*, *Staphylococcus*), Gram-negative (e.g. *Brucella*, *Salmonella*, *Haemophilus*) and obligate anaerobic bacteria (e.g. *Clostridium*, *Bacteroides fragilis*). Other sensitive organisms include *Chlamydia*, *Mycoplasma* (unreliable in treatment of ocular mycoplasmosis) and *Rickettsia*. Used in amphibians for chytridiomycosis; safe for larvae, metamorphs and adults. Resistant organisms include *Nocardia* and *Mycobacterium*. Acquired resistance may occur in Enterobacteriaceae. High lipid solubility makes it suitable for the treatment of intraocular infections. It will also access the CNS. However, due to concerns of resistance development and human toxicity, use should be restricted to individual animals where there is a specific indication such as salmonellosis resistant to other antimicrobials or deep infections of the eye. Patients with hepatic or renal dysfunction may need adjustment to dose. Decrease dose or increase dosing interval in neonates. Use with caution or avoid in nursing animals, especially those with neonates, as crosses into milk.

N Safety and handling: Humans exposed to chloramphenicol may have an increased risk of developing a fatal aplastic anaemia. Products should be handled with care; use impervious gloves and avoid skin contact.

P Contraindications: In amphibians, chloramphenicol might dramatically alter the protective natural skin microbiome when used as an antifungal agent.

Q Adverse reactions: Dose-related reversible bone marrow suppression can develop in all species. Unlike humans, the development of irreversible aplastic anaemia in veterinary species does not appear to be a significant problem. Other adverse effects include nausea, vomiting, diarrhoea and anaphylaxis.

T Drug interactions: Irreversible inhibitor of a large number of hepatic cytochrome P450-dependent enzymes and so increases plasma levels of pentobarbital, phenobarbital and oral hypoglycaemic agents. Recovery requires synthesis of new liver enzymes and can take up to 3 weeks. Rifampin accelerates the metabolism of chloramphenicol, thus decreasing serum levels. Chloramphenicol may inhibit activity of bactericidal antimicrobials such as the aminoglycosides and beta-lactams. May also be an inhibitory effect if used in combination with macrolide or lincosamide antimicrobials.

DOSES

Y Classified as category C (Caution) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Z Mammals: Ferrets: 25–50 mg/kg p.o., s.c., i.m., i.v. q12h; Rabbits: 50 mg/kg p.o., s.c. q12–24h; Guinea pigs: 0.83 mg/ml drinking water;

Mice: 50 mg/kg i.m., p.o. q12h, 200 mg/kg p.o. q12h or 0.5 mg/ml drinking water; **Gerbils:** 1 mg/ml drinking water; **Other rodents:** 30–50 mg/kg i.v., i.m., s.c., p.o. q8–12h; **Primates:** 25–50 mg/kg s.c., i.m., i.v. q8h; **Sugar gliders:** 50 mg/kg p.o. q12h; **Hedgehogs:** 30–50 mg/kg p.o., s.c., i.m., i.v. q12h. Ophthalmic: 1 drop q4–8h; ointment q8–12h.

Birds: 50 mg/kg i.v., i.m. q8h or 75 mg/kg p.o. q8h; **Pigeons:** 25 mg/kg p.o. q12h.

Reptiles: Most species: 40–50 mg/kg i.m., s.c., p.o. q24h; **Rat snakes, King snakes, Indigo snakes:** 50 mg/kg i.m. q12h; **Boids:** 50 mg/kg i.m. q12–24h; **Water snakes (Nerodia spp.):** 50 mg/kg i.m. q12–72h^a.

Amphibians: 50 mg/kg s.c., i.m., intracoelomic q12–24h or 10–20 mg/l as a bath^{b,c,d}. Concentrations <200 mg/l as a bath may be required in hellbenders for treatment of chytridiomycosis as lower doses appear ineffective.

Fish: No information available.

References

- ^a Clark CH, Rogers ED and Milton JL (1985) Plasma concentrations of chloramphenicol in snakes. *American Journal of Veterinary Research* **46**(12), 2654–2657
- ^b Bishop PJ, Speare R, Poulter R et al. (2009) Elimination of the amphibian chytrid fungus *Batrachochytrium dendrobatidis* by Archey's frog (*Leiopelma archeyi*). *Diseases of Aquatic Organisms* **84**(1), 9–15
- ^c Holden WM, Ebert AR, Canning PF and Rollins-Smith LA (2014) Evaluation of amphotericin B and chloramphenicol as alternative drugs for treatment of chytridiomycosis and their impacts on innate skin defences. *Applied Environmental Microbiology* **80**(13), 4034–4041
- ^d Young S, Speare R, Berger L and Skerratt LF (2012) Chloramphenicol with fluid and electrolyte therapy cures terminally ill green tree frogs (*Litoria caerulea*) with chytridiomycosis. *Journal of Zoo and Wildlife Medicine* **43**(2), 330–337

Chlorhexidine

(Antisept, Clorexyderm, CLX wipes, Douxo Pyo, Ermidra foam, Hibiscrub, Iryplus eye wipes, Malaseb, Microbex, Otodine, Stomidine F, Savlon*, TrizChlor*)

POM-V, AVM-GSL

Formulations: Topical shampoo: 2% chlorhexidine + 2% miconazole (Malaseb); 31.2 mg/ml chlorhexidine (Microbex); 3% chlorhexidine + 0.5% climbazole + phytosphingosine (Duoxo Pyo shampoo/mousse/pads); 4% chlorhexidine (Clorexyderm 4% shampoo, spray, Ermidra foam); **Cleansing solution:** sodium acetate + chlorhexidine digluconate + acetic acid (Antisept spray); **Spot gel:** chlorhexidine 4% + Triz-EDTA + polyvinylpyrrolidone + lanolin + glycerine (Clorexyderm spot gel); 1.5% chlorhexidine + cetrimide (Savlon); **Surgical scrub solution:** 4% chlorhexidine + isopropyl alcohol (Hibiscrub); **Mouthwash:** 0.12% chlorhexidine (Chlorohex); **Topical skin cleaner:** chlorhexidine + Tris-EDTA + zinc gluconate + glycerine + climbazole + benzyl alcohol + propylene glycol (CLX wipes); **Ear/wound cleaner:** 0.15% chlorhexidine + EDTA (TrizChlor); **Ear cleaner:** chlorhexidine + Tris-EDTA + lactic acid (Otodine); **Toothpaste:** 0.07% chlorhexidine, Tris-EDTA (Stomidine F).

Action: Chemical antiseptic that disrupts bacterial cell membrane.

Use: Topical treatment of bacterial, dermatophyte and *Malassezia* skin infections in animals as a shampoo (Malaseb). Topical treatment of *Malassezia* infections (Microbex). Washing surgical instruments, routine

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antisepsis for surgical operations (Savlon, Hibiscrub) and dental hygiene (Chlorohex). Topical treatment of mild pruritus. Concurrent systemic antibacterial therapy is generally advised when treating bacterial skin infections. Leave in contact with the skin for 5–10 minutes prior to washing off. Ear cleaners for cleansing and removal of cerumen. Chlorhexidine as a single agent is not consistently effective as an antifungal. Use diluted, but very sparingly to reduce gross contamination only in the preparation of surgical sites in fish.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not instil into ears where the integrity of the tympanum is unknown. Do not use on eyes. Has been associated with skin damage in amphibians so use not recommended ^a.

Adverse reactions: Ototoxic. May irritate mucous membranes.

Drug interactions: Not known.

DOSES

Mammals: Apply to affected area q8h at 0.5–2.0% concentrations. 0.05% solution in water can be used as a safe wound flush. When treating dermatophytosis continue treatment for 2 weeks after apparent clinical cure and negative fungal culture results. Otic: Dilute topical products to a 1.0% concentration and apply topically q8–12h.

Birds: Apply 2–3 times weekly. May be used less frequently once infection is controlled.

Reptiles: 0.05% solution in water has been suggested as a safe wound flush q24h.

Amphibians: Not recommended.

Fish: Use 0.025% solution very sparingly to remove gross contamination only¹.

References

^a Philips BH, Crim MJ, Hankenson FC *et al.* (2015) Evaluation of presurgical skin preparation agents in African clawed frogs (*Xenopus laevis*). *Journal of the American Association for Laboratory Animal Science* **54**(6), 788–798

¹ Harms CA and Lewbart GA (2000) Surgery in fish. *Veterinary Clinics: Exotic Animal Practice* **3**(3), 759–774

Chloroquine (Chloroquine diphosphate, Chloroquine phosphate)

(Avloclor*) **POM**

Formulations: Oral: 250 mg tablets.

Action: Antiprotozoal; mode of action not known but may enter the cell of marine protozoan ectoparasites causing cell lysis due to disruption of normal cell membrane function.

Use: Treatment of haemoparasites, most commonly *Plasmodium* in a variety of zoo and wild animal species, although rarely indicated in exotic pets. Treatment of skin and gill disease due to infestation with marine protozoan ectoparasites (e.g. *Cryptocaryon irritans* (marine white spot), *Brooklynella*, *Uronema*) and dinoflagellates (e.g. *Amyloodinium ocellatum* (marine velvet)). When using in aquatic set-ups, do not use activated carbon, UV sterilization, ozone or protein skimmer during treatment. Change 25% of the water before each dose.

Safety and handling: Normal precautions should be observed.

Contraindications: Highly toxic to micro- and macroalgae and various invertebrates, particularly corals, clams and echinoderms. Do not use in reef tanks.

Adverse reactions: May cause reduced appetite.

Drug interactions: No information available.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Birds: Raptors: 10–20 mg/kg p.o., then 5–10 mg/kg at 6, 24, 48h. May be combined with primaquine.

Reptiles: Tortoises: 125 mg/kg p.o. q48h for 3 doses.

Fish: For Amyloodinium: 10 mg/l by immersion, repeat after 21 days prn; For protozoans: 10–15 mg/l by immersion then 10 mg/l q7–10d for 3 doses (more effective if salinity is reduced to 12–13 ppt). Add activated carbon to remove the drug if no relapse apparent after 21 days.

Mammals, Amphibians: No information available.

Chlorphenamine (Chlorpheniramine)

(Piriton*) **POM, GSL**

Formulations: Injectable: 10 mg/ml solution. Oral: 4 mg tablet, 0.4 mg/ml syrup.

Action: Binds to H1 histamine receptors to prevent histamine binding.

Use: Management of allergic disease and prevention and early treatment of anaphylaxis. Commonly used as premedication before transfusions and certain chemotherapeutic agents. Specific doses for animals have not been determined by pharmacokinetic studies. Use with caution in cases with urinary retention, angle-closure glaucoma and pyloroduodenal obstruction.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: May cause mild sedation. May reduce seizure threshold.

Drug interactions: No information available.

DOSES

Mammals: Ferrets: 1–2 mg/kg p.o. q8–12h; Rabbits: 0.2–0.4 mg/kg p.o. q12h; Rodents: 0.6 mg/kg p.o. q24h; Primates: 0.5 mg/kg p.o. q24h.

Birds, Reptiles, Amphibians, Fish: No information available.

Cholestyramine see Colestyramine

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Chorionic gonadotrophin (Human chorionic gonadotrophin, hCG)

B(Chorulon) **POM-V****C****Formulations:** Injectable: 1500 IU powder for reconstitution.**D****Action:** In females, induces follicular maturation, ovulation and development of the corpus luteum. In males, stimulates testosterone secretion.**E****Use:** Used to supplement or replace LH (luteinizing hormone) in cases of ovulation failure or delay, to induce lactation post-partum, or in females who fail to hold to mating. Used in ferrets to treat persistent oestrus. In guinea pigs it can be used to resolve ovarian cysts if hormone-responsive. In male animals it may increase libido; may also assist the treatment of cryptorchidism before surgical castration, provided inguinal canal remains patent and therapy is started early. May have an effect in preventing egg laying in birds. Used in amphibians for mating or release of sperm in males, followed with GnRH (gonadotropin-releasing hormone) in 8–24h, and induction of ovulation in females (can be used with progesterone or PMSG).**F****G****H****I****J****Safety and handling:** Reconstituted vials do not contain any preservative and so should be discarded within 24 hours.**K****Contraindications:** No information available.**L****Adverse reactions:** Anaphylactic reactions may occasionally occur.**M****Drug interactions:** No information available.**N**

DOSES

O**Mammals:** Ferrets: 100 IU/animal i.m. q14d for 2 doses; Rabbits: 20–25 IU/animal i.v. once to induce ovulation; Guinea pigs: 100 IU/kg i.m. weekly for 3 doses; Primates: 250 IU i.m. once.**P****Birds:** 500–1000 IU/kg i.m. on days 1, 3 and 7 q3–6wk to inhibit egg laying^a.**Q****Amphibians:** 300–400 IU s.c., i.m.^b.**R****Reptiles, Fish:** No information available.**S**

References

^a Lightfoot TL (2000) Clinical use and preliminary data of chorionic gonadotropin administration in psittacines. *Proceedings of the Annual Conference of the American Avian Veterinarians*, pp. 17–21

^b Kouba AJ, del Barco-Trillo J, Vance CK, Milam C and Carr M (2012) A comparison of human chorionic gonadotropin and luteinizing hormone releasing hormone on the induction of spermiation and amplexus in the American toad (*Anaxyrus americanus*). *Reproductive Biology and Endocrinology* **10**, 59

T**U****V**

Ciclosporin (Cyclosporin(e))

W(Atopica, Optimimmune, Neoral*, Sandimmun*) **POM-V, POM****X****Formulations:** Ophthalmic: 0.2% ointment (Optimmune). Oral: 10 mg, 25 mg, 50 mg, 100 mg capsules; 100 mg/ml solution. Injectable: 50 mg/ml solution.**Y****Action:** T lymphocyte inhibition.**Z****Use:** Used for disseminated idiopathic myositis and pure red cell aplasia in ferrets, and sebaceous adenitis in rabbits. May be of use in

the treatment of bornavirus in cockatiels. No evidence of systemic or ocular toxicity following ocular administration but systemic absorption has been reported. Recommended that bacterial and fungal infections are treated before use. Whilst the nephrotoxicity seen in human patients does not appear to be common in animals, care should be taken in treating animals with renal impairment, and creatinine levels should be monitored regularly. Use with caution in patients with pre-existing infections and monitor for opportunistic infections.

Safety and handling: Use gloves to prevent cutaneous absorption.

Contraindications: Do not use in progressive malignant disorders. Do not give live vaccines during treatment or within a 2-week interval before or after treatment. The manufacturer does not recommend use in diabetic animals.

Adverse reactions: Hypertrichosis is common. There may be immediate discomfort on topical application (blepharospasm). Transient vomiting and diarrhoea may follow systemic administration; these are usually mild and do not require cessation of treatment. Infrequently observed adverse effects include: anorexia; mild to moderate gingival hyperplasia; papillomatous lesions of the skin; red and swollen pinnae; muscle weakness; and muscle cramps. These effects resolve spontaneously after treatment is stopped. Systemic treatment may be associated with an increased risk of malignancy.

Drug interactions: The metabolism of ciclosporin is reduced, and thus serum levels increased, by various drugs that competitively inhibit or induce enzymes involved in its metabolism, particularly cytochrome P450, including diltiazem, doxycycline and imidazole antifungal drugs. In humans there is an increased risk of nephrotoxicity if ciclosporin is administered with aminoglycosides, NSAIDs, quinolones, or trimethoprim/sulphonamides; concomitant use of ciclosporin not recommended. Increased risk of hyperkalaemia if used with ACE inhibitors. As a substrate and inhibitor of the MDR-1 P-glycoprotein transporter, co-administration of ciclosporin with P-glycoprotein substrates such as macrocyclic lactones (e.g. ivermectin and milbemycin) could decrease the efflux of such drugs from blood–brain barrier cells, potentially resulting in signs of CNS toxicity. Ciclosporin has been shown to affect glucose metabolism and decrease the effects of insulin and to do so as much as low doses of glucocorticoids; therefore, use with caution in diabetic patients.

DOSES

Mammals: Ferrets: 5 mg/kg p.o. q24h; Rabbits: 5 mg/kg p.o. q24h for idiopathic sebaceous adenitis¹; Rats: 10 mg/kg p.o. q24h.

Birds: Cockatiels: 0.2 mg/animal p.o. to stop development of avian bornavirus^a. For anti-inflammatory activity, doses of 30–60 mg/kg p.o. q12–24h have been used in various parrot species, however, in a Harris' hawk these dose rates produced toxic signs – pharmacokinetic study showed that a dose rate of 5 mg/kg was required to maintain plasma levels without toxicity².

Reptiles, Amphibians, Fish: No information available.

References

^a Hameed SS, Guo J, Tizard I, Shivaprasad HL and Payne S (2018) Studies on immunity and immunopathogenesis of parrot bornaviral disease in cockatiels. *Virology* **515**, 81–91

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- ¹ Jassies-van der Lee A, Van Zeeland Y, Kik M and Schoemaker N (2009) Successful treatment of sebaceous adenitis in a rabbit with ciclosporin and triglycerides. *Veterinary Dermatology* **20**(1), 67–71
- ² Chitty J (2017) An update on pyoderma in Harris' hawks. *Proceedings of the International Conference on Avian, Reptile and Exotic Mammals*. Venice pp. 354–355

Cimetidine

(Zitac, Cimetidine*, Dyspamet*, Tagamet*) **POM-V, POM**

Formulations: **Injectable:** 100 mg/ml solution in 2 ml ampoule.

Oral: 100 mg, 200 mg, 400 mg, 800 mg tablets; 40 mg/ml syrup.

Action: Histamine (H₂) receptor antagonist, blocking histamine-induced gastric acid secretion. Rapidly absorbed with high bioavailability; undergoes hepatic metabolism and renal excretion. Plasma half-life is about an hour.

Use: Management of idiopathic, uraemic or drug-related erosive gastritis, gastric and duodenal ulcers, oesophagitis, and hypersecretory conditions secondary to gastrinoma or mast cell neoplasia. Efficacy against NSAID-induced ulcers is controversial. Reduction of vomiting due to gastritis and gastric ulceration is typically achieved in about 2 weeks but animals should be treated for at least 2 weeks after the remission of clinical signs, so minimum of 28 days recommended. If considered successful, medication can then be stopped. Rebound gastric acid secretion may be seen on cessation of cimetidine, so therapy should be tapered. A 2-week medication-free period should be allowed to see if vomiting occurs again. If the animal starts vomiting again after a medication-free period, treatment can be re-initiated, without risk for intolerance. Depending on the response, treatment can be adapted to the individual animal until the response is considered to be adequate and then continued at this level. Concomitant treatment with sucralfate may be helpful, and dietary measures should always be maintained. If used i.v., should be administered over 30 min to prevent cardiac arrhythmias and hypotension. Dosage should be reduced for animals with renal impairment. Less effective at reducing gastric acidity than more modern H₂ blockers and proton pump inhibitors. Cimetidine has minimal prokinetic effects.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Rare, although hepatotoxicity and nephrotoxicity have been reported in humans. Adverse reactions are generally minor even at high doses. In humans cimetidine has been associated with headache, gynaecomastia and decreased libido.

Drug interactions: Retards oxidative hepatic drug metabolism by binding to the microsomal cytochrome P450. May increase plasma levels of beta-blockers (e.g. propranolol), calcium-channel blockers (e.g. verapamil), diazepam, lidocaine, metronidazole, pethidine and theophylline. When used with other agents that cause leucopenia may exacerbate the problem. Sucralfate may decrease bioavailability; although there is little evidence to suggest this is of clinical importance it may be a wise precaution to administer sucralfate at least 2 hours before cimetidine. Stagger oral doses by 2 hours when used with other antacids, digoxin, itraconazole or maropitant.

DOSES

Mammals: Ferrets: 5–10 mg/kg i.m., s.c., p.o. q8h; Rabbits: 5–10 mg/kg p.o. q6–8h; Rodents: 5–10 mg/kg p.o., s.c., i.m., i.v. q6–12h; Primates: 10 mg/kg p.o., s.c., i.m. q8h; Hedgehogs: 10 mg/kg p.o. q8h.

Birds: 5 mg/kg i.m., p.o. q8–12h.

Reptiles: 4 mg/kg i.m., p.o. q8–12h.

Amphibians, Fish: No information available.

Ciprofloxacin

(Ciloxan*, Ciproxin*) **POM**

Formulations: Oral: 100 mg, 250 mg and 500 mg tablets; 50 mg/ml suspension. **Injectable:** 2 mg/ml for i.v. infusion. **Ophthalmic:** 0.3% solution in 5 ml bottle; 0.3% ointment in 3.5 g tube.

Action: Bactericidal through inhibition of bacterial DNA gyrase.

Use: Ideally fluoroquinolone use should be reserved for infections where culture and sensitivity testing predicts a clinical response and where first- and second-line antimicrobials would not be effective. Broad-spectrum activity against wide range of Gram-negative and some Gram-positive aerobes; some activity against *Mycoplasma* and *Chlamydia*. Active against many ocular pathogens, including *Staphylococcus* and *Pseudomonas aeruginosa*, although there is increasing resistance amongst staphylococci and streptococci. The eye drop formulation is also used in reptiles for the topical management of wounds or stomatitis.

Safety and handling: Normal precautions should be observed.

Contraindications: Fluoroquinolones are relatively contraindicated in growing small mammals as cartilage abnormalities have been reported in young mice, rats, guinea pigs and rabbits after administration of other fluoroquinolones. Care may be needed using fluoroquinolones in breeding females.

Adverse reactions: May cause local irritation after application. In humans the following are reported: local burning and itching; lid margin crusting; hyperaemia; taste disturbances; corneal staining, keratitis, lid oedema, lacrimation, photophobia, corneal infiltrates; nausea; and visual disturbances.

Drug interactions: No information available.

DOSES

Classified as category B (Restrict) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Ferrets: 10–30 mg/kg p.o. q24h; Rabbits: 10–20 mg/kg p.o. q24h or 1 drop to affected eye q6h; loading dose can be used 1 drop to affected eye q15min for 4 doses; Guinea pigs, Chinchillas: 5–25 mg/kg p.o. q12h; Hamsters: 10–20 mg/kg p.o. q12h; Other rodents: 7–25 mg/kg p.o. q12h; Primates: 10–20 mg/kg p.o. q12h^a; Sugar gliders: 10 mg/kg p.o. q12h; Hedgehogs: 5–20 mg/kg p.o. q12h.

Birds: 5–20 mg/kg i.v., i.m., p.o. q12h^b; Raptors: 50 mg/kg p.o. q12h^c.

Reptiles: Topical: 1 drop to affected eyes or in wounds or topical application for stomatitis; Systemic: 10 mg/kg p.o., s.c., i.m. q24–48h.

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A **Amphibians:** 10 mg/kg p.o. q24–48h for 7 days; 500–750 mg/75 l as a bath for 6–8h q24h.

B **Fish:** 15 mg/kg i.v., i.m. q2–5days^d.

References

- C** ^a Nelson M, Stagg AJ, Stevens DJ *et al.* (2011) Post-exposure therapy of inhalational anthrax in the common marmoset. *International Journal of Antimicrobial Agents* **38**(1), 60–64
- D** ^b Atta AH and Sharif L (1997) Pharmacokinetics of ciprofloxacin following intravenous and oral administration in broiler chickens. *Journal of Veterinary Pharmacology and Therapeutics* **20**(4), 326–329
- E** ^c Isaza R, Budsberg SC, Sundlof SF and Baker B (1993) Disposition of ciprofloxacin in red-tailed hawks (*Buteo jamaicensis*) following a single oral dose. *Journal of Zoo and Wildlife Medicine* **24**(4), 498–502
- F** ^d Nouws JFM, Grondel JL, Schutte AR and Laurensen J (1988) Pharmacokinetics of ciprofloxacin in carp, African catfish and rainbow trout. *Veterinary Quarterly* **10**(3), 211–216

Cisapride

(Cisapride) **POM-V**

Formulations: Various formulations (including tablets and suspensions) available as a veterinary special depending on requirements. **VSP**

Action: GI prokinetic agent related to metoclopramide but has no central antiemetic activity.

Use: Primarily used in GI stasis in rabbits and guinea pigs. Its licence for humans was withdrawn due to potentially fatal cardiac arrhythmias and it is solely available to the veterinary profession on a named patient or named veterinary surgeon 'special' basis. Recent work has cast doubt upon the efficacy of this drug in at least some circumstances, in clinically healthy animals. It should be used with awareness of its potential limitations, and other drugs considered as well as or instead of this^{a,b}.

Safety and handling: Normal precautions should be observed.

Contraindications: In humans, simultaneous administration of cisapride and ranitidine may cause nausea. In rabbits it is difficult to judge whether this is a significant issue, but should be considered if the patient exhibits signs of inappetence or discomfort or suspected nausea, and the doses staggered by at least 2 hours.

Adverse reactions: Abdominal cramps and diarrhoea may develop, especially at higher doses or if used alongside other prokinetic agents. Fatal cardiac arrhythmias have not been reported in rabbits or rodents.

Drug interactions: Opioid analgesics and antimuscarinics (e.g. atropine) may antagonize the effects of cisapride. In humans, it is known that drugs inhibiting the cytochrome P450 3A4 enzymes which metabolize cisapride (clarithromycin, erythromycin, itraconazole) when taken alongside cisapride have led to fatal arrhythmias.

DOSES

Mammals: Rabbits, Guinea pigs: 0.1–1.0 mg/kg (typically 0.5 mg/kg) p.o. q8–12h; **Primates:** 0.2 mg/kg p.o. q12h^c; **Sugar gliders:** 0.25 mg/kg p.o., s.c. q8–24h.

Birds, Reptiles, Amphibians, Fish: No information available.

References

- Z** ^a Feldman ER, Singh B, Mishkin NG *et al.* (2021) Effects of cisapride, buprenorphine, and their combination on gastrointestinal transit in New Zealand white rabbits. *Journal of the American Association for Laboratory Animal Science* **60**(2), 221–228

- ^b Mans C, Fink DM and Ciarrocchi C (2021) Effects of oral cisapride administration in chinchillas (*Chinchilla lanigera*) with experimentally induced fecal output reduction. *Journal of Exotic Pet Medicine* **38**, 21–25
- ^c Yogo K, Onoma M, Ozaki K et al. (2008) Effects of oral mitemincal (GM-611), erythromycin, EM-574 and cisapride on gastric emptying in conscious rhesus monkeys. *Digestive Diseases and Sciences* **53**(4), 912–91

Clarithromycin

(Klaricid*) **POM**

Formulations: Oral: 250 mg, 500 mg tablets; 125 mg/5 ml suspension; 250 mg/5 ml suspension; 250 mg granules sachet (to be dissolved in water). **Injectable:** 500 mg vial for reconstitution.

Action: Derived from erythromycin and with greater activity. Bactericidal (time-dependent) or bacteriostatic properties, depending on concentration and susceptibility. Binds to the 50S ribosome, inhibiting peptide bond formation.

Use: Alternative to penicillin in penicillin-allergic humans as it has a similar, although not identical, antibacterial spectrum. Active against Gram-positive cocci (some *Staphylococcus* spp. resistant), Gram-positive bacilli, some Gram-negative bacilli (e.g. *Pasteurella*) and some spirochaetes (e.g. *Helicobacter*). Some strains of *Actinomyces*, *Nocardia*, *Chlamydia*, *Mycoplasma* and *Rickettsia* also inhibited. Most strains of Enterobacteriaceae (*Pseudomonas*, *Escherichia coli*, *Klebsiella*) are resistant. Highly lipid-soluble and useful against intracellular pathogens. Particularly useful in management of respiratory tract infections, mild to moderate skin and soft tissue infections, and non-tubercular mycobacterial infections. For the latter used in combination with enrofloxacin and rifampin. Activity is enhanced in an alkaline pH; administer on an empty stomach. There is limited information regarding use in animals. Use with caution in animals with hepatic dysfunction. Reduce dose in animals with renal impairment. In ferrets it has been used successfully to treat pneumonia due to *Mycobacterium abscessus* and it can be used in combination with omeprazole or ranitidine and bismuth plus amoxicillin or metronidazole for treatment of *Helicobacter mustelae*. In tortoises it has been used to resolve clinical signs in severe cases of mycoplasmosis.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: In humans similar adverse effects to those of erythromycin are seen, i.e. vomiting, cholestatic hepatitis, stomatitis and glossitis.

Drug interactions: May increase serum levels of several drugs, including methylprednisolone, theophylline, omeprazole and itraconazole. The absorption of digoxin may be enhanced.

DOSES

Classified as category C (Caution) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Ferrets: 8–12.5 mg/kg p.o. q12–24h; Rabbits: 80 mg/kg p.o. q12h with rifampin at 40 mg/kg p.o. q12h for *Staphylococcus osteomyelitis*; Rats: 3.5–10 mg/kg p.o. q8–12h; Primates: 10 mg/kg p.o. q12h^a, 20 mg/kg p.o. q24h^b; Hedgehogs: 5.5 mg/kg p.o. q12h.

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Birds: Most species: 85 mg/kg p.o. q24h (anecdotal). Used at 55 mg/kg p.o. q24h as part of multi-drug mycobacterial therapy in a Bruce's green pigeon¹.

B

Reptiles: Desert tortoises: 15 mg/kg p.o. q48–72h^{c,d}.

C

Amphibians, Fish: No information available.

D

References

E

^a Dubois A, Berg DE, Fiala N *et al.* (1998) Cure of *Helicobacter pylori* infection by omeprazole-clarithromycin-based therapy in non-human primates. *Journal of Gastroenterology* **33**, 18–22

F

^b Badyal DK, Garg SK (2000) Effects of clarithromycin on the pharmacokinetics of carbamazepine in rhesus monkeys. *Methods and Findings in Experimental and Clinical Pharmacology* **22**, 581–584

G

^c Wimsatt JH, Johnson J, Mangone BA *et al.* (1999) Clarithromycin pharmacokinetics in the desert tortoise (*Gopherus agassizii*). *Journal of Zoo and Wildlife Medicine* **30**(1), 36–43

H

^d Wimsatt J, Tothill A, Offermann CF, Sheehy JG and Peloquin CA (2008) Long-term and per rectum disposition of clarithromycin in the desert tortoise (*Gopherus agassizii*). *Journal of the American Association of Laboratory Animal Science* **47**(4), 41–45

I

¹ Elizabeth JB and Geoffrey WP (2018) Resolution of a localized granuloma caused by *Mycobacterium avium-intracellulare* complex on the cere of a Bruce's green pigeon (*Treron waalia*). *Journal of Avian Medicine and Surgery* **32**(4), 322–327

J**K**

Clazuril

L

(Appertex, Harkers CoxiTabs) **AVM-GSL, ESPA**

M

Formulations: Oral: 2.5 mg tablet.

N

Action: Coccidiocidal; mode of action unclear.

O

Use: Treatment and control of coccidiosis (*Eimeria labbeana*, *E. columbarum*) in homing and show pigeons. Birds should be treated following transportation to shows or races where they may have been exposed to coccidia.

P

Safety and handling: Normal precautions should be observed.

Q

Contraindications: Do not use in birds intended for human consumption.

R

Adverse reactions: No information available.

S

Drug interactions: Do not administer with drugs that may cause vomiting.

T

DOSES

U

See Appendix for guidelines on responsible parasiticide use.

V

Birds: Raptors: 30 mg/kg once; Pigeons: 5–10 mg/kg once (treat all birds in loft simultaneously)^{a,b}; **Psittacids:** 7 mg/kg p.o. q2d for 2 doses.

W

Mammals, Reptiles, Amphibians, Fish: No information available.

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References

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^a Maes L, Coussemment W, Desplenter L and Marsboom R (1988) Safety of a new anticoccidial agent, clazuril, during reproduction in carrier pigeons. *Tijdschr Diergeneeskde* **113**(4), 195–204

Z

^b Vercruyse J (1990) Efficacy of toltrazuril and clazuril against experimental infections with *Eimeria labbeana* and *E. columbarum* in racing pigeons. *Avian Diseases* **34**(1), 73–79

Clindamycin

(Antirobe, Clinacin, Clindacyl, Clindaseptin) **POM-V**

Formulations: Oral: 25 mg, 75 mg, 150 mg, 300 mg capsules and tablets; 25 mg/ml solution.

Action: Lincosamide antibiotic that binds to the 50S ribosomal subunit, inhibiting peptide bond formation. May be bactericidal or bacteriostatic depending on susceptibility.

Use: Bone and joint infections associated with Gram-positive bacteria; pyoderma; toxoplasmosis and infections associated with the oral cavity. Active against Gram-positive cocci (including penicillin-resistant staphylococci), many obligate anaerobes, mycoplasmas and *Toxoplasma gondii*. May be used in combination with toltrazuril for treatment of *Choleoeimeria*. Attains high concentrations in bone and bile. Being a weak base, it becomes ion-trapped (and therefore concentrated) in fluids that are more acidic than plasma, such as prostatic fluid, milk and intracellular fluid. There is complete cross resistance between lincomycin and clindamycin, and partial cross resistance with erythromycin. Use with care in individuals with hepatic or renal impairment.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not administer lincosamides to rabbits, other small herbivores or hamsters.

Adverse reactions: Colitis, vomiting and diarrhoea are reported. Oral lincosamides can cause fatal enterotoxaemia in small herbivores and hamsters. Clindamycin should not be administered orally or into abscesses which connect with the oral cavity.

Drug interactions: May enhance the effect of non-depolarizing muscle relaxants (e.g. tubocurarine) and may antagonize the effects of neostigmine and pyridostigmine. Do not administer with macrolide, chloramphenicol or other lincosamide antimicrobials as these combinations are antagonistic.

DOSES

Classified as category C (Caution) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Ferrets: 5.5–11 mg/kg p.o. q12h (toxoplasmosis: 12.5–25 mg/kg p.o. q12h); Rats: 7.5–25 mg/kg p.o. q24h^a; Primates: 10 mg/kg p.o. q12h, 20 mg/kg i.m. q8h^b; Sugar gliders, Hedgehogs: 5.5–10 mg/kg p.o. q12h.

Birds: 25 mg/kg p.o. q8h, 50 mg/kg p.o. q12h or 100 mg/kg p.o. q24h. Pigeons: 100 mg/kg p.o. q6h^c.

Reptiles: 2.5–5 mg/kg p.o. q24h. Used for treatment of *Choleoeimeria* in Rankin's dragons at 5 mg/kg p.o. q24h for 7 days in combination with toltrazuril^d.

Amphibians, Fish: No information available.

References

- ^a Yang SH and Lee MG (2007) Dose-independent pharmacokinetics of clindamycin after intravenous and oral administration to rats: contribution of gastric first-pass effect to low bioavailability. *International Journal of Pharmaceutics* **332**(1–2), 17–23
- ^b Picardi JL, Lewis HP, Tan JS and Phair JP (1975) Clindamycin concentrations in the central nervous system of primates before and after head trauma. *Journal of Neurosurgery* **43**(6), 717–720

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- ^c Lenarduzzi T, Langston C and Ross MK (2011) Pharmacokinetics of clindamycin administered orally to pigeons. *Journal of Avian Medicine and Surgery* **25**(4), 259–265
- ^d Stöhr AC, Globokar-Vrhovec M and Pantchev N (2020) *Choleoecimeria* spp. prevalence in captive reptiles in Germany and a new treatment option in a Lawson's Dragon (*Pogona henrylawsoni*). *Journal of Herpetological Medicine and Surgery* **30**(4), 261–269

Clomipramine

(Clomicalm) **POM-V**

Formulations: Oral: 5 mg, 20 mg, 80 mg tablets.

Action: Both clomipramine and its primary metabolite desmethyl-clomipramine are active in blocking serotonin and noradrenaline re-uptake in the brain, with resultant anxiolytic, antidepressant and anticomulsive effects.

Use: Used in association with a behaviour modification plan for the management of anxiety-related disorders in animals, including 'compulsive behaviours', noise fears and urine spraying. Care required before use in animals with a history of constipation, epilepsy, glaucoma, urinary retention or arrhythmias. Used in birds for feather plucking, especially if due to separation anxiety; should be used in association with a behaviour modification plan. Can be used with benzodiazepines.

Safety and handling: Normal precautions should be observed.

Contraindications: Patients sensitive to tricyclic antidepressants. Do not give with, or within 2 weeks of, monoamine oxidase inhibitors (e.g. selegiline). Not recommended for use in male breeding animals, as testicular hypoplasia may occur.

Adverse reactions: May cause sporadic vomiting, changes in appetite or lethargy. Vomiting may be reduced by co-administration with a small quantity of food.

Drug interactions: May potentiate the effects of the antiarrhythmic drug quinidine, anticholinergic agents (e.g. atropine), other CNS active drugs (e.g. barbiturates, benzodiazepines, general anaesthetics, neuroleptics), sympathomimetics (e.g. adrenaline) and coumarin derivatives. Simultaneous administration with cimetidine may lead to increased plasma levels of clomipramine. Plasma levels of certain antiepileptic drugs, (e.g. phenytoin and carbamazepine), may be increased by co-administration with clomipramine.

DOSES

Mammals: Rats: 16–32 mg/kg p.o. q12h.

Birds: 0.5–1 mg/kg p.o. q6–8h or 3 mg/kg p.o. q12h^a–24h^b.

Reptiles, Amphibians, Fish: No information available.

References

- ^a Seibert LM, Crowell-Davis SL, Wilson GH and Ritchie BW (2004) Placebo-controlled clomipramine trial for the treatment of feather picking disorder in cockatoos. *Journal of the American Animal Hospital Association* **40**(4), 261–269
- ^b Welle KR (2019) Rectricectomy with pygostylectomy for resolution of feather trauma in an Umbrella Cockatoo (*Cacatua alba*). *Journal of Avian Medicine and Surgery* **33**(2), 193–197

Clotrimazole

(Canesten*, Clotrimazole*, Lotriderm*) **POM**

Formulations: Topical: 1% cream; 1% solution. Many other products available; some contain corticosteroids.

Action: Topical imidazole with an inhibitory action on the growth of pathogenic dermatophytes, *Aspergillus* and yeasts by inhibiting cytochrome P450-dependent ergosterol synthesis.

Use: Superficial fungal infections. Naso-sinal infections including aspergillosis, particularly in birds.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: No information available.

Drug interactions: No information available.

DOSES

Mammals: Rabbits: Otic: Instil 3–5 drops in ear q12h; Topical: Apply to affected area and massage in gently q12h; if no improvement in 4 weeks re-evaluate therapy or diagnosis; **Guinea pigs, Chinchillas, Rats, Hamsters, Other small mammals:** Topical application to defined lesions q12h for 3–6 weeks.

Birds: Endoscopic: 10 mg/kg applied directly to fungal lesions. Nasal flush: 10 mg/ml (flush volume 20 ml/kg). Intratracheal: 2–3 ml of a 1% solution nebulized for periods of 1 hour q24h or topically.

Reptiles: Apply topically to lesion q12h.

Amphibians, Fish: No information available.

Cloxacillin

(Opticlox, Orbenin) **POM-V**

Formulations: Ophthalmic: Cloxacillin benzathine ester 16.7% suspension.

Action: Beta-lactamase-resistant penicillin which is bactericidal and works in a time-dependent fashion. Binds to penicillin-binding proteins involved in cell wall synthesis, thereby decreasing bacterial cell wall strength and rigidity, and affecting cell division, growth and septum formation.

Use: Narrow spectrum antimicrobial. Less active than penicillin G or V against *Streptococcus*. Specifically indicated for ocular infections with beta-lactamase-producing *Staphylococcus*.

Safety and handling: Normal precautions should be observed.

Contraindications: Avoid use in animals which have displayed hypersensitivity reactions to other antimicrobials within the beta-lactam family (which includes cephalosporins). Do not administer penicillins by any route to guinea pigs, chinchillas, hamsters, gerbils or degus. Do not administer oral penicillins to rabbits.

Adverse reactions: Avoid use in small herbivores (e.g. rabbits), hamsters and gerbils as ingestion of penicillins can cause fatal enterotoxaemia.

Drug interactions: Avoid the concomitant use of bacteriostatic antibiotics (chloramphenicol, erythromycin, tetracycline).

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A**DOSES****Classified as category D (Prudence) by the EMA.****B****See Appendix for guidelines on responsible antibiotic use.****C**

Mammals: While topical cloxacillin is effective against *Treponema* localized to the eyelids, its use can treat local symptoms but leave the rabbit in a carrier state, and is not advised.

D

Birds: Apply 1/10th of a tube (0.3 g) q24h.

E

Reptiles, Amphibians, Fish: No information available.

F

Co-amoxiclav (Amoxicillin/Clavulanate, Amoxycillin/Clavulanic acid)

G

(Clavabactin, Clavaseptin, Clavucil, Clavudale, Combisyn, Kesium, Nisamox, Noroclav, Synulox, Augmentin*) **POM-V, POM**

H**I**

Formulations: **Injectable:** 175 mg/ml suspension (140 mg amoxicillin, 35 mg clavulanate); 600 mg powder (500 mg amoxicillin, 100 mg clavulanate); 1.2 g powder (1 g amoxicillin, 200 mg clavulanate) for reconstitution (Augmentin). **Oral:** 50 mg, 250 mg, 500 mg tablets each containing amoxicillin and clavulanate in a ratio of 4:1; Palatable drops which when reconstituted with water provide 40 mg amoxicillin and 10 mg clavulanic acid per ml.

J**K****L****M**

Action: Amoxicillin binds to penicillin-binding proteins involved in bacterial cell wall synthesis, thereby decreasing cell wall strength and rigidity, affecting cell division, growth and septum formation. The addition of the beta-lactamase inhibitor clavulanate increases the antimicrobial spectrum against those organisms that produce beta-lactamase, such as *Staphylococcus* and *Escherichia coli*.

N**O****P**

Use: Active against Gram-positive and Gram-negative aerobic organisms and many obligate anaerobes. Penicillinase-producing *Escherichia coli* and *Staphylococcus* are susceptible, but difficult Gram-negative organisms such as *Pseudomonas aeruginosa* and *Klebsiella* are often resistant. Dose and dosing interval will be determined by infection site, severity and organism. The predominant bacterial infections in reptiles are Gram-negative and many are resistant to penicillins.

Q**R****S****T****U**

Safety and handling: Tablets are wrapped in foil moisture-resistant packaging; do not remove until to be administered. Refrigerate oral suspension and i.v. solution after reconstitution. Discard oral suspension and i.v. formulation if they become dark or after 10 days. A small amount of discoloration of the i.v. solution is acceptable.

V**W****X****Y**

Contraindications: Avoid oral antibiotic agents in critically ill patients, as absorption from the GI tract may be unreliable; such patients may require i.v. formulation. Avoid use in animals which have displayed hypersensitivity reactions to other antimicrobials within the beta-lactam family (which includes cephalosporins). Use of penicillins should be avoided in rabbits, other small herbivores, hamsters and gerbils, especially via the oral route.

Z

Adverse reactions: Nausea, diarrhoea and skin rashes are the commonest adverse effects. Oral doses of penicillins can cause fatal enterotoxaemia in small herbivores, hamsters and potentially gerbils.

Drug interactions: Avoid the concurrent use of amoxicillin with bacteriostatic antibiotics (e.g. tetracycline, erythromycin). Do not mix in the same syringe as aminoglycosides. Do not use with allopurinol in birds. Synergism may occur between the beta-lactam and aminoglycoside antimicrobials *in vivo*.

DOSES

Classified as category C (Caution) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Ferrets: 12.5–20 mg/kg i.m., s.c. q12h; Rats, Mice: 100 mg/kg p.o., s.c. q12h; Primates: 15 mg/kg p.o. q12h; Sugar gliders, Hedgehogs: 12.5 mg/kg p.o., s.c. q12h.

Birds: 125–150 mg/kg p.o., i.v. q12h; 125–150 mg/kg i.m. q24h^a.

Reptiles: Asian house geckos: 10 mg/kg p.o. q24h^b.

Amphibians, Fish: No information available.

References

^a Orosz SE, Jones MO, Cox SK, Zagaya NK and Frazier DL (2000) Pharmacokinetics of amoxicillin plus clavulanic acid in blue-fronted Amazon parrots (*Amazona aestival aestival*). *Journal of Avian Medicine and Surgery* **14**(2), 107–112

^b Agius JE, Rose K, Emery JP and Phalen DN (2021) Experimental infection of Asian house geckos with *Enterococcus lacertideformis* demonstrates multiple disease transmission routes and the *in vivo* efficacy of antibiotics. *Scientific Reports* **11**(1), 1–15

Colchicine

(Colchicine*) **POM**

Formulations: Oral: 0.5 mg tablet.

Action: Colchicine inhibits collagen synthesis, may enhance collagenase activity and blocks the synthesis and secretion of serum amyloid A.

Use: Management of fibrotic hepatic and pulmonary diseases, oesophageal stricture and renal amyloidosis. In birds, used for gout and hepatic cirrhosis/fibrosis. Due to the relatively high incidence of adverse reactions, this drug should be used with caution.

Safety and handling: Protect from light.

Contraindications: Pregnancy.

Adverse reactions: Commoner adverse effects include vomiting, abdominal pain and diarrhoea. Rarely, renal damage, bone marrow suppression, myopathy and peripheral neuropathy may develop. Colchicine may increase serum ALP, decrease platelet counts and cause false-positive results when testing urine for RBCs and haemoglobin. Overdoses can be fatal.

Drug interactions: Possible increased risk of nephrotoxicity and myotoxicity when colchicine given with ciclosporin. NSAIDs, especially phenylbutazone, may increase the risks of thrombocytopenia, leucopenia or bone marrow depression when used concurrently with colchicine. Many anticancer chemotherapeutics may cause additive myelosuppressive effects when used with colchicine.

DOSES

Birds: 0.04 mg/kg p.o. q12h^{1,2}. Rainbow lorikeets: 0.1 mg/kg p.o. q12h³.

Mammals, Reptiles, Amphibians, Fish: No information available.

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References

- ¹ Hoefer H (1991) Hepatic fibrosis and colchicine therapy. *Journal of the Association of Avian Veterinarians* **5**, 193
- ² Romano J (2013) Therapeutic review: colchicine. *Journal of Exotic Pet Medicine* **22(4)**, 405–408
- ³ Mastakov A, Kelly-Bosma M, Allavena R and Doneley R (2022) Polycystic kidney disease in 3 juvenile Rainbow Lorikeets (*Trichoglossus moluccanus*). *Journal of Avian Medicine and Surgery* **35(4)**, 464–474

Colestyramine (Cholestyramine)

(Questran*) **POM**

Formulations: Oral: 4 g powder/sachet.

Action: Ion exchange resin.

Use: In rabbits and susceptible rodents for absorbing toxins produced in the GI tract following the development of overgrowth of *Clostridium*.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Constipation may develop.

Drug interactions: Colestyramine reduces the absorption of digoxin, anticoagulants, diuretics and thyroxine.

DOSES

Mammals: Rabbits: 2–4 g/animal p.o. syringed gently with 20 ml water q24h; Guinea pigs: 1 g/animal mixed with water p.o. q24h.

Birds, Reptiles, Amphibians, Fish: No information available.

Copper sulphate (Cupric sulphate)

(Proprietary formulations are available) **ESPA**

Formulations: Liquid for immersion.

Action: No information available.

Use: For the treatment of protozoan and monogenean ectoparasites, some fungi and bacteria in fish. It is strongly recommended that a proprietary formulation is used initially to avoid problems related to purity and to enable accurate dosing. Copper has a low therapeutic index and is toxic to gill tissue. It is commonly used to treat diseases in marine aquaria by prolonged immersion but is toxic to invertebrates, elasmobranchs, algae and many plants. Although copper can be used to treat ectoparasites in freshwater fish, other remedies are safer. The solubility of copper is reduced at higher pH and it is absorbed or inactivated by high levels of calcareous materials and organic matter. Free copper ion levels must be maintained between 0.15–0.20 mg/l for the chemical to be effective – this requires monitoring using a commercial test kit then adjusting the dose initially twice daily. A stock solution is made by dissolving 1 g of copper sulphate pentahydrate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) in 250 ml distilled water to produce 1 mg copper/ml and enables accurate dosing of the chemical. Copper can be removed using inactivated carbon and water changes.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use in the presence of invertebrates.

Adverse reactions: No information available.

Drug interactions: No information available.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Fish: 100 mg/l by immersion for 1–5 minutes; 0.15 mg/l by prolonged immersion until therapeutic effect (at least 10 days); follow the manufacturer's recommendations for proprietary formulations.

Mammals, Birds, Reptiles, Amphibians: No information available.

Cordycepic acid see **Mannitol**,
Crisantaspase see **Asparaginase**
Cupric sulphate see **Copper sulphate**

Cyclophosphamide

(Cyclophosphamide*, Endoxana*) **POM**

Formulations: Injectable: 100 mg, 200 mg, 500 mg, 1000 mg powder for reconstitution. Oral: 50 mg tablets.

Action: Metabolites crosslink DNA resulting in inhibition of DNA synthesis and function.

Use: Treatment of lymphoproliferative diseases, myeloproliferative disease and immune-mediated diseases. The use of cyclophosphamide in immune-mediated haemolytic anaemia is controversial and therefore it is not recommended as an immunosuppressant in the management of this disease. May have a role in management of certain sarcomas and carcinomas. Use with caution in patients with renal failure; dose reduction may be required. Has been used in treatment of lymphosarcoma in cockatoos and lymphoma in the green iguana.

Safety and handling: **Cytotoxic drug; see specialist texts for further advice on chemotherapeutic agents.**

Contraindications: No information available.

Adverse reactions: Myelosuppression, with the nadir usually occurring 7–14 days after the start of therapy depending on species; regular monitoring of WBCs recommended. A metabolite of cyclophosphamide (acrolein) may cause a sterile haemorrhagic cystitis. The cystitis may be persistent. This risk may be reduced by increasing water consumption or by giving furosemide to ensure adequate urine production. Other effects include vomiting, diarrhoea, hepatotoxicity, nephrotoxicity and a reduction in hair growth rate.

Drug interactions: Increased risk of myelosuppression if thiazide diuretics given concomitantly. Absorption of orally administered digoxin may be decreased, may occur several days after dosing. Barbiturates increase cyclophosphamide toxicity due to increased rate of conversion to metabolites. Phenothiazines and chloramphenicol reduce cyclophosphamide efficacy. If administered with doxorubicin there is an increased risk of cardiotoxicity. Insulin requirements are altered by concurrent cyclophosphamide.

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A**DOSES****See Appendix for chemotherapy protocols in ferrets.****B****Mammals:** Ferrets: As part of a multi-drug protocol for lymphoma.**Birds:** Lymphosarcoma: 200 mg/m² intraosseous q7d.**C****Reptiles:** 3 mg/kg i.v. q2wk as part of a chemotherapy protocol in one case report in the green iguana¹.**D****Amphibians, Fish:** No information available.**E****References**¹ Folland DW, Johnston MS, Thamm DH and Reavill D (2011) Diagnosis and management of lymphoma in a green iguana (*Iguana iguana*). *Journal of the American Veterinary Medical Association* **239**(7), 985–991**F****G****Cyclosporin(e)** see **Ciclosporin****H****I****Cypermethrin****(F10 Germicidal Wound Spray with Insecticide) ESPA****J****Formulations:** Topical: various formulations including benzalkonium chloride 0.4 g polyhexanide 0.03 g and cypermethrin 0.25 g spray; barrier ointment and shampoo formulations.**K****L****Actions:** Acts as a sodium open-channel blocker, resulting in muscular convulsions and death in arthropods.**M****Use:** To repel insects and eliminate fly strike infestations.**N****Safety and handling:** Normal precautions should be observed. May be harmful to aquatic organisms. Do not contaminate aquaria, ponds or waterways.**O****Contraindications:** Do not use in fish or other aquatic organisms. Do not use on cats.**P****Adverse reactions:** No adverse reactions reported in small mammals if used at dosage regime advised by manufacturer.**Q****Drug interactions:** No information available.**R****DOSES****See Appendix for guidelines on responsible parasiticide use.****S****Mammals:** Apply ointment or spray to wound site and repeat if necessary once a week. May need to be re-applied if the animal is exposed to rain.**T****Fish:** Do not use.**U****Birds, Reptiles, Amphibians:** No information available.**V****Cyproheptadine****(Periactin*) POM****W****Formulations:** Oral: 4 mg tablet.**X****Action:** Binds to and blocks the activation of H1 histamine and serotonin receptors.**Y****Use:** Management of allergic disease and appetite stimulation. Use with caution in cases with urinary retention, angle-closure glaucoma and pyloroduodenal obstruction. Specific doses for animals have not been**Z**

determined by pharmacokinetic studies and clinical effectiveness has not been established.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: May cause mild sedation, polyphagia, weight gain. May reduce seizure threshold.

Drug interactions: No information available.

DOSES

Mammals: Ferrets, Chinchillas, Guinea pigs: 0.5 mg p.o. q12h.

Birds, Reptiles, Amphibians, Fish: No information available.

Cyromazine

(Rearguard) **NFA-VPS**

Formulations: Topical: 6% solution in ready-to-use bottle with built-in applicator sponge.

Action: Inhibits larval development by inhibiting the deposition of chitin into the cuticle.

Use: Authorized in domestic rabbits for the prevention of blowfly strike for up to 10 weeks after application (Rearguard). To be applied in early summer before any flies are seen. Will not kill adult flies or larger maggots already present, but prevents eggs developing into maggots. Bathing the rabbit after application may reduce efficacy so is not recommended.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use in rabbits less than 10 weeks of age, or pregnant or breeding does. Do not apply to broken skin.

Adverse reactions: Transient inappetence may occur for 24–48h after treatment.

Drug interactions: No information available.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Mammals: Apply topically to fur from middle of the back to tip of tail and between back legs q6–10wks.

Birds, Reptiles, Amphibians, Fish: No information available.

Cytarabine (Cytosine arabinoside, Ara-C)

(Cytarabine*, Cytosar-U*) **POM**

Formulations: Injectable: 100 mg, 500 mg powders for reconstitution.

Action: The active nucleotide metabolite ara-CTP is incorporated into DNA and inhibits pyrimidine and DNA synthesis. Cytarabine is therefore S phase-specific.

Use: Management of lymphoma in the ferret.

Safety and handling: **Cytotoxic drug; see specialist texts for further advice on chemotherapeutic agents.** After reconstitution, store at room temperature and discard after 48 hours or if a slight haze develops.

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Contraindications: Do not use if there is evidence of bone marrow suppression or substantial hepatic impairment.

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Adverse reactions: Vomiting, diarrhoea, leucopenia. As it is a myelosuppressant, careful haematological monitoring is required.

C

Conjunctivitis, oral ulceration, hepatotoxicity and fever have also been seen.

D

Drug interactions: Oral absorption of digoxin is decreased. Activity of gentamicin may be antagonized. Simultaneous administration of methotrexate increases the effect of cytarabine.

E**F**

DOSES

See Appendix for chemotherapy protocols in ferrets.

G

Mammals: Ferrets: 1 mg/animal p.o., s.c. q24h for 2 days.

H

Birds, Reptiles, Amphibians, Fish: No information available.

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Deferoxamine (Desferrioxamine)

(Desferal*) **POM**

Formulations: Injectable: 500 mg vial for reconstitution.

Action: Deferoxamine chelates iron, and the complex is excreted in the urine.

Use: To remove iron from the body following poisoning. Also used for haemochromatosis (common in toucans, hornbills and softbills, though unusual in other birds).

Safety and handling: Normal precautions should be observed.

Contraindications: Avoid in severe renal disease.

Adverse reactions: Administration i.m. is painful. Anaphylactic reactions and hypotension may develop if administered rapidly i.v.

Drug interactions: No information available.

DOSES

Birds: Various doses proposed, ranging from 20 mg/kg p.o. q4h to 40 mg/kg i.m. q24h (mynahs) to 100 mg/kg p.o., s.c., i.m. q24h for up to 14 weeks^{a,1}. 100 mg/kg p.o. q24h given to common mynahs for 1 month was effective in reducing liver iron levels – effects equivalent to regular phlebotomy^b.

Mammals, Reptiles, Amphibians, Fish: No information available.

References

- ^a Dierenfeld E and Phalen DN (2006) A comparison of four regimens for treatment of iron storage disease using the European Starling (*Sturnus vulgaris*) as a model. *Journal of Avian Medicine and Surgery* **20**(2), 74–79
- ^b Jalili N, Zaeemi M, Razmyar J and Azzizadeh M (2021) Comparison of iron chelators used alone or in combination with phlebotomy in common mynahs (*Acridotheres tristis*) with experimental iron storage disease. *Avian Pathology* **50**(4), 350–356
- ¹ Sheppard C and Dierenfeld E (2002) Iron storage disease in birds: speculation on etiology and implications for captive husbandry. *Journal of Avian Medicine and Surgery* **16**(3), 192–197

Desferrioxamine see Deferoxamine

Deslorelin

(Suprelorin) **POM-V**

Formulations: Implant containing 4.7 mg or 9.4 mg of active product.

Action: Desensitizes GnRH receptors, thereby decreasing release of LH and FSH. This leads to reduction in testosterone and sperm production.

Use: Authorized for the induction of temporary infertility in healthy, entire, sexually mature male ferrets for up to 4 years. (Suprelorin 9.4 mg implant). Can also be used for chemical contraception in other susceptible species. In males, there is a 14-day surge in testosterone followed by lowering of levels. Infertility is achieved from 6 weeks up to at least 6 months after initial treatment. Treated male animals should therefore still be kept away from females within the first 6 weeks following initial treatment. Any mating that occurs more than 6 months after the administration of the product may result in pregnancy. Disinfection of the site should be undertaken prior to implantation to avoid introduction of infection. If the

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A hair is long, a small area should be clipped. The product should be implanted subcutaneously in the loose skin on the back between the lower neck and the lumbar area. Avoid injection of the implant into fat, as release of the active substance might be impaired in areas of low vascularization. The biocompatible implant does not require removal.

B However, should it be necessary to end treatment, implants may be surgically removed. Implants can be located using ultrasonography.

C Deslorelin implants effectively prevent reproduction and the musky odour of intact male and female ferrets, and are therefore considered a suitable alternative for surgical neutering in these animals. Surgical neutering of ferrets has been implicated as an aetiological factor in the development of hyperadrenocorticism in this species. Deslorelin implants can be given to neutered ferrets to decrease the development or progression of adrenocortical disease. In rabbits, the effects on reproductive behaviour and fertility are considered negligible in males^a, although implants appear to reduce ovarian activity if placed prior to puberty in females^b. However, the possibility of masking future uterine pathology makes use in rabbits difficult to advise without careful monitoring. The efficacy in preventing ovarian cysts in guinea pigs is poor. The use of the implant in birds and reptiles has been reported for a range of behavioural and reproductive (e.g. excessive egg laying) disorders but with variable effect and duration of action between species. In Quaker parrots and cockatiels, one study showed a 6 month prevention of egg laying using a 4.7 mg implant^c. The variability in responses and relatively high treatment failure rate in suppressing egg laying may be related to differing uses of GnRH in different bird species. In male birds, has been used to counter aggressive behaviour with variable response. Has also been used for management of a Sertoli cell tumour in budgerigars and ovarian neoplasia in cockatiels^{d,e}.

D In reptiles, reproductive activity was suppressed in female iguanas but efficacy has not been established in other reptiles.

E **Safety and handling:** People who are or may become pregnant should not handle this drug.

F **Contraindications:** No information available.

G **Adverse reactions:** Moderate swelling at the implant site may be observed for 14 days. A significant decrease in testicle size will be seen during the treatment period. In very rare cases, a testicle may be able to ascend the inguinal ring. In guinea pigs, risk of metritis due to persistent closure of the vaginal membrane.

H **Drug interactions:** No information available.

I **DOSES**

J **Mammals:** Ferrets: 1 implant per animal. The 9.4 mg implant (authorized in males only) lasts approximately 3–4 years. The 4.7 mg implant is not authorized in either sex and lasts approximately 1–2 years. Females will come into oestrus for approximately 2 weeks after implantation; **K** **Guinea pigs:** 1 implant per animal for contraception in females, although appears ineffective in males^f; **L** **Rats:** 1 implant per animal provides contraceptive and reduced reproductive behaviour effects for >1 year in males and females; **M** **Other species:** Effects are variable and depend on the exact pathology, species and sex.

N **Birds, Reptiles, Amphibians:** 1 implant per animal regardless of size, placed subcutaneously. The 9.4 mg implant appears to last longer than the 4.7 mg but duration of either is variable.

O **Fish:** No information available.

References

- ^a Goericke-Pesch S, Groeger G and Wehrend A (2015) The effects of a slow release GnRH agonist implant on male rabbits. *Animal Reproduction Science* **152**, 83–89
- ^b Geyer A, Daub L, Otzdorff C *et al.* (2016) Reversible estrous cycle suppression in prepubertal female rabbits treated with slow-release deslorelin implants. *Theriogenology* **85(2)**, 282–287
- ^c Petritz OA, Sanchez-Migallon Guzman D, Paul-Murphy J *et al.* (2013) Evaluation of the efficacy and safety of single administration of 4.7 mg deslorelin acetate implants on egg production and plasma sex hormones in Japanese quail (*Coturnix coturnix japonica*). *American Journal of Veterinary Research* **74(2)**, 316–323
- ^d Keller KA, Beaufrière H, Brandão J *et al.* (2013) Long-term management of ovarian neoplasia in two cockatiels (*Nymphicus hollandicus*). *Journal of Avian Medicine and Surgery* **27(1)**, 44–52
- ^e Straub J and Zenker I (2013) First experience in hormonal treatment of sertoli cell tumors in budgerigars (*M. undulatus*) with absorbable extended release GnRH chips (Suprelorin®). *1st International Conference on Avian, Herpetological and Exotic Mammal Medicine*. Wiesbaden, pp. 299–301
- ^f Forman C, Wehrend A and Goericke-Pesch S (2016) Deslorelin implants are suitable for contraception in female, but not male guinea pigs. *Proceedings of the 8th International Symposium on Canine and Feline Reproduction, Paris*

Dexamethasone

(Aurizon, Dexadreson, Dexafort, Dexa-ject, Rapidexon, Voren, Dexamethasone*, Maxidex*, Maxitrol*) **POM-V, POM**

Formulations: Ophthalmic: 0.1% solution (Maxidex, Maxitrol). Maxitrol also contains polymyxin B and neomycin. **Injectable:** 2 mg/ml solution; 1 mg/ml, 3 mg/ml suspension; 2.5 mg/ml suspension with 7.5 mg/ml prednisolone. **Oral:** 0.5 mg tablet. (1 mg of dexamethasone is equivalent to 1.1 mg of dexamethasone acetate, 1.3 mg of dexamethasone isonicotinate or dexamethasone sodium phosphate, or 1.4 mg of dexamethasone trioxa-undecanoate).

Action: Alters the transcription of DNA, leading to alterations in cellular metabolism which cause a reduction in inflammatory response.

Use: Anti-inflammatory drug. Also used to prevent and treat anaphylaxis associated with transfusion or chemotherapeutic agents. Anti-inflammatory potency is 7.5 times greater than prednisolone. On a dose basis 0.15 mg dexamethasone is equivalent to 1 mg prednisolone. Dexamethasone has a long duration of action and low mineralocorticoid activity and is particularly suitable for short-term high-dose therapy in conditions where water retention would be a disadvantage. Unsuitable for long-term daily or alternate-day use. Animals on chronic therapy should be tapered off steroids when discontinuing the drug. Use shorter-acting preparations wherever possible in birds and rabbits. In birds, corticosteroids should not be used where immunosuppression is to be avoided (e.g. dyspnoea where aspergillosis has not been ruled out). The use of long-acting steroids in most cases of shock and spinal injury is of no benefit and may be detrimental. Used for the treatment of shock, trauma and chronic stress in fish.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use in pregnant animals. Systemic corticosteroids are generally contraindicated in patients with renal disease and diabetes mellitus. Impaired wound healing and delayed recovery from infections may be seen. Topical corticosteroids are contraindicated in ulcerative keratitis.

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Adverse reactions: A single dose of dexamethasone or dexamethasone sodium phosphate suppresses adrenal gland function for up to 32 hours. Prolonged use of glucocorticoids suppresses the hypothalamic-pituitary axis (HPA), causing adrenal atrophy, elevated liver enzymes, cutaneous atrophy, weight loss, PU/PD, vomiting and diarrhoea. GI ulceration may develop. Hyperglycaemia and decreased serum T4 values may be seen in patients receiving dexamethasone. Corticosteroids should be used with care in birds as there is a high risk of immunosuppression and side effects, such as hepatopathy and a diabetes mellitus-like syndrome. In rabbits, even small single doses can potentially cause severe adverse reactions. Ferrets are particularly susceptible to GI ulceration, and concurrent gastric protectants may be advisable, especially in stressed animals.

Drug interactions: There is an increased risk of GI ulceration if used concurrently with NSAIDs. The risk of developing hypokalaemia is increased if corticosteroids are administered concomitantly with amphotericin B or potassium-depleting diuretics (furosemide, thiazides). Dexamethasone antagonizes the effect of insulin. Phenobarbital or phenytoin may accelerate the metabolism of glucocorticoids and antifungals (e.g. itraconazole) may decrease it.

DOSES

Mammals: Ophthalmic: Apply small amount of ointment to affected eye(s) q6–24h or 1 drop of solution in affected eye(s) q6–12h. **Ferrets:** 0.5–2.0 mg/kg s.c., i.m., i.v. q24h; 1 mg/kg i.m., i.v. once followed with prednisolone (post adrenalectomy); **Rabbits:** 0.2–0.6 mg/kg s.c., i.m., i.v. q24h; **Guinea pigs:** 0.6 mg/kg i.v., i.m., s.c. q24h (pregnancy toxemia); **Primates:** 0.25–1 mg/kg i.v., i.m., s.c. q24h (inflammation); **Sugar gliders:** 0.1–0.6 mg/kg s.c., i.m., i.v. (anti-inflammatory); 0.5–2 mg/kg s.c., i.m., i.v. (shock); **Hedgehogs:** 0.1–1.5 mg/kg i.m.; **Others:** anti-inflammatory: 0.055–0.2 mg/kg i.m., s.c. q12–24h tapering dose over 3–14 days.

Birds: 2–6 mg/kg i.v., i.m. q12–24h.

Reptiles: Inflammation, non-infectious respiratory disease: 0.1–4 mg/kg i.m., i.v. q24h (lower end of dose range recommended).

Amphibians: 1.5 mg/kg s.c., i.m. q24h.

Fish: 1–2 mg/kg i.m., intracoelomic or 10 mg/l for 60 minute bath q12–24h.

Dexmedetomidine

(Dexdomitor, Sedadex, Sileo) **POM-V**

Formulations: Injectable: 0.1 mg/ml and 0.5 mg/ml solution.

Oral: 0.1 mg/ml gel.

Action: Agonist at peripheral and central alpha-2 adrenoreceptors producing dose-dependent sedation, muscle relaxation and analgesia.

Use: To provide sedation and premedication when used alone or in combination with opioid analgesics and may also provide additional analgesia. Dexmedetomidine combined with ketamine can be used to provide a short duration (20–30 min) of surgical anaesthesia. Dexmedetomidine is the pure dextroenantiomer of medetomidine. As the levomedetomidine enantiomer is largely inactive, dexmedetomidine is twice as potent as the racemic mixture (medetomidine). Administration of dexmedetomidine reduces the biological load presented to the animal, resulting in quicker metabolism of concurrently administered anaesthetic

drugs and a potentially faster recovery from anaesthesia.

Dexmedetomidine is a potent drug that causes marked changes in the cardiovascular system, including an initial peripheral vasoconstriction that results in an increase in blood pressure and a compensatory bradycardia. After 20–30 min vasoconstriction wanes, while blood pressure returns to normal values. Heart rate remains low due to the central sympatholytic effect of alpha-2 agonists. These cardiovascular changes result in a fall in cardiac output; central organ perfusion is well maintained at the expense of redistribution of blood flow away from the peripheral tissues. Respiratory system function is well maintained; respiration rate may fall but is accompanied by an increased depth of respiration. Oxygen supplementation is advisable in all animals that have received dexmedetomidine for sedation. Combining dexmedetomidine with an opioid provides improved analgesia and sedation. Lower doses of dexmedetomidine should be used in combination with other drugs. Reversal of dexmedetomidine sedation or premedication with atipamezole at the end of the procedure shortens the recovery period, which is advantageous. Analgesia should be provided with other classes of drugs before atipamezole. High doses ($>10 \mu\text{g}$ (micrograms)/kg) are associated with greater physiological disturbances than doses of $1\text{--}10 \mu\text{g/kg}$. Using dexmedetomidine in combination with opioids in the lower dose range can provide good sedation and analgesia with minimal side effects. The lower concentration of dexmedetomidine is designed to increase the accuracy of dosing in smaller species. In birds alpha-2 agents can be effective when used intranasally or intramuscularly. Lack of predictable effect and slowing of respiratory and heart rates make them less desirable when used as sole agents. They are used occasionally in combination but other combinations may be more predictable and safer (e.g. midazolam/butorphanol).

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use in animals with cardiovascular or other systemic disease. Use of dexmedetomidine in geriatric patients is not advisable. It should not be used in pregnant animals, nor in animals likely to require or receiving sympathomimetic amines.

Adverse reactions: Causes diuresis by suppressing ADH secretion, a transient increase in blood glucose by decreasing endogenous insulin secretion, and mydriasis and decreased intraocular pressure. Vomiting after i.m. administration is common, so dexmedetomidine should be avoided when vomiting is contraindicated (e.g. foreign body, raised intraocular pressure). Due to effects on blood glucose, use in diabetic animals is not recommended. Spontaneous arousal from deep sedation following stimulation can occur with all alpha-2 agonists, aggressive animals sedated with dexmedetomidine must still be managed with caution.

Drug interactions: No information available.

DOSES

When used for sedation is generally given as part of a combination. See Appendix for sedation protocols in all species.

Mammals: Ferrets: $0.04\text{--}0.1 \text{ mg/kg s.c., i.m.}$; Rabbits: $0.025\text{--}0.05 \text{ mg/kg i.v. or } 0.05\text{--}0.15 \text{ mg/kg s.c., i.m.}$

Birds: Buzzards: $0.1 \text{ mg atropine} + 25 \mu\text{g}$ (micrograms)/kg dexmedetomidine i.m.; Kestrels: $0.05 \text{ mg atropine} + 75 \mu\text{g/kg}$ dexmedetomidine i.m. as induction for isoflurane general anaesthesia^a.

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Reptiles: Most species 0.05–0.1 mg/kg usually in combination with ketamine and/or opioids/midazolam/alfaxalone to provide light anaesthesia (**see Appendix**). **Tegus:** 0.2 mg/kg (analgesia)^b; **Ball pythons:** 0.1–0.2 mg/kg (sedation or possible analgesia)^c.

B**C**

Amphibians: **Blue poison dart frogs:** 5mg/kg s.c. in combination with midazolam and ketamine/alfaxalone for sedation^d.

D

Fish: No information available.

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References

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^a Santangelo B, Ferrari D, Di Martino I *et al.* (2009) Dexmedetomidine chemical restraint of two raptor species undergoing inhalation anaesthesia. *Veterinary Research Communications* **33**(1), 209–211

G

^b Bisetto SP, Melo CF and Carregaro AB (2018) Evaluation of sedative and antinociceptive effects of dexmedetomidine, midazolam and dexmedetomidine–midazolam in tegus (*Salvator merianae*). *Veterinary Anaesthesia and Analgesia* **45**(3), 320–328

H

^c Bunke LG, Sladky KK and Johnson SM (2018) Antinociceptive efficacy and respiratory effects of dexmedetomidine in ball pythons (*Python regius*). *American Journal of Veterinary Research* **79**(7), 718–726

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^d Yaw TJ, Mans C, Martinelli L and Sladky KK (2020) Comparison of subcutaneous administration of alfaxalone–midazolam–dexmedetomidine with ketamine–midazolam–dexmedetomidine for chemical restraint in juvenile blue poison dart frogs (*Dendrobates tinctorius azureus*). *Journal of Zoo and Wildlife Medicine* **50**(4), 868–873

J**K**

Dextrose **see Glucose**

L**M**

Diazepam

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(Diazedor, Diazemuls*, Diazepam Rectubes*, Stesolid*, Valium*, Ziapam and several others) **POM-V, POM**

O

Formulations: **Injectable:** 5 mg/ml solution. **Oral:** 2 mg, 5 mg, 10 mg tablets; 2 mg/5 ml solution. **Rectal:** 2 mg/ml (1.25, 2.5 ml tubes), 4 mg/ml (2.5 ml tubes) solutions; 10 mg suppositories.

P**Q**

Action: Enhances activity of the major inhibitory central nervous system neurotransmitter, gamma-aminobutyric acid (GABA), through binding to the benzodiazepine site of the GABA_A receptor.

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Use: Anticonvulsant, anxiolytic and skeletal muscle relaxant (e.g. urethral muscle spasm and tetanus). Diazepam is the drug of choice for the short-term emergency control of severe epileptic seizures and status epilepticus. In guinea pigs it may be used to reduce the excitability associated with extreme pruritus (e.g. with ectoparasitic infestations). It may also be used in combination with ketamine to offset muscle hypertonicity associated with ketamine, and with opioids and/or acepromazine for pre-anaesthetic medication in the critically ill. It provides very poor sedation or even excitation when used alone in healthy animals. Used in birds for the short-term management of feather plucking. Diazepam has a high lipid solubility, which facilitates its oral absorption and rapid central effects. It is rarely used for sedation even in combination with midazolam being preferred. Liver disease will prolong duration of action. In the short term, repeated doses of diazepam or a CRI will lead to drug accumulation and prolonged recovery. Flumazenil (a benzodiazepine antagonist) will reverse the effects of diazepam. The development of dependence on benzodiazepines may occur after regular use, even with therapy of only a few weeks, and the dose should be gradually reduced in these cases if the benzodiazepine is being withdrawn.

Safety and handling: Substantial adsorption of diazepam may occur on to some plastics and this may cause a problem when administering diazepam by continuous i.v. infusion. The use of diazepam in PVC infusion bags should be avoided; giving sets should be kept as short as possible and should not contain a cellulose propionate volume-control chamber. If diazepam is given by continuous i.v. infusion the compatible materials include glass, polyolefin, polypropylene and polyethylene.

Contraindications: Benzodiazepines should be avoided in patients with CNS depression, respiratory depression, severe muscle weakness or hepatic impairment (as may worsen hepatic encephalopathy). They are also contraindicated in the long-term treatment of behavioural disorders due to the risks of disinhibition and interference with memory and learning.

Adverse reactions: Sedation, muscle weakness and ataxia are common. Rapid i.v. injection or oral overdose may cause marked paradoxical excitation (including aggression) and elicit signs of pain; i.v. injections should be made slowly (over at least 1 minute for each 5 mg). Intramuscular injection is painful and results in erratic drug absorption. Rectal administration is effective for emergency control of seizures if i.v. access is not possible, but the time to onset is delayed to 5–10 min. The duration of action may be prolonged after repeated doses in rapid succession, in older animals, those with liver dysfunction and those receiving beta-1 antagonists. Chronic dosing leads to a shortened half-life due to activation of the hepatic microsomal enzyme system and tolerance to the drug may develop. The propylene glycol formulation of injectable diazepam can cause thrombophlebitis, therefore the emulsion formulation is preferred for i.v. injection.

Drug interactions: Do not dilute or mix with other agents. Due to extensive metabolism by the hepatic microsomal enzyme system, interactions with other drugs metabolized in this way are common. Cimetidine and omeprazole inhibit metabolism of diazepam and may prolong clearance. Concurrent use of phenobarbital may lead to a decrease in the half-life of diazepam. An enhanced sedative effect may be seen if antihistamines or opioid analgesics are administered with diazepam, and diazepam will reduce the dose requirement of other anaesthetic agents. When given with diazepam the effects of digoxin may be increased. Diazepam may be used in combination with tricyclic antidepressant therapy for the management of more severe behavioural responses.

DOSES

When used for sedation is generally given as part of a combination. See Appendix for sedation protocols in all species.

Mammals: Ferrets: Seizures: 2–5 mg/kg i.m. once; urethral sphincter muscle relaxation post urinary catheterization or obstruction: 0.5 mg/kg p.o., i.m., i.v. q6–8h; **Rabbits:** Epileptic seizures, pre-anaesthetic sedation, muscle relaxation: 1–5 mg/kg i.v.; **Guinea pigs:** 0.5–5.0 mg/kg i.m. as required; **Chinchillas, Rats, Mice, Hamsters, Gerbils:** 2–5 mg/kg i.m., i.p. once; **Primates:** Sedation and seizures: 0.25–0.5 mg/kg p.o., i.v. (sedation); 0.5–1 mg/kg i.m., i.v. (seizures); **Sugar gliders, Hedgehogs:** Sedation and seizures: 0.5–2 mg/kg p.o., s.c., i.m.

Birds: Epileptic seizures: 0.1–1 mg/kg i.v., i.m. once; **Raptors:** 0.2 mg/kg p.o. q24h as appetite stimulant; **Doves:** 0.2 mg/kg combined with ketamine for sedation^a.

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A**Reptiles:** Treatment of seizures: 0.5–2.5 mg/kg i.m., i.v.**Amphibians, Fish:** No information available.**B****References**

- ^a Abbas SW, Ali MN, Abbas G *et al.* (2018). Comparative effectiveness of general anesthesia in doves using a combination of ketamine and diazepam. *Advances in Zoology and Botany* **6**(4), 95–100

C**D****Diazoxide**(Eudemine*) **POM****E****F****Formulations:** Injectable: 15 mg/ml solution. Oral: 50 mg tablet.**G****Action:** A diuretic that causes vasodilation and inhibits insulin secretion by blocking calcium mobilization.**H****Use:** Used to manage hypoglycaemia caused by hyperinsulinism due to insulinoma in ferrets. In humans it is also used in the short-term management of acute hypertension.**I****Safety and handling:** Normal precautions should be observed.**J****Contraindications:** No information available.**K****Adverse reactions:** The commonest adverse effects are anorexia, vomiting and diarrhoea. Hypotension, tachycardia, bone marrow suppression, pancreatitis, cataracts and electrolyte and fluid retention may occur. Drug efficacy may diminish over a period of months.**L****M****Drug interactions:** Phenothiazines and thiazide diuretics may increase the hyperglycaemic activity of diazoxide, whilst alpha-adrenergic blocking agents (e.g. phenoxybenzamine) may antagonize the effects of diazoxide.**N****DOSES****O****Mammals:** Ferrets: 5–30 mg/kg p.o. q12h.**P****Birds, Reptiles, Amphibians, Fish:** No information available.**Q****Diclofenac**(Voltarol Ophtha*, Voltarol Ophtha Multidose*) **POM****R****S****Formulations:** Ophthalmic: 0.1% solution in 5 ml bottle and in single-use vial.**T****Action:** COX inhibitor that produces local anti-inflammatory effects.**U****Use:** Used in cataract surgery to prevent intraoperative miosis and reflex (axonal) miosis caused by ulcerative keratitis. Used to control pain and inflammation associated with corneal surgery and in ulcerative keratitis when topical corticosteroid use is contraindicated.**V****Safety and handling:** Normal precautions should be observed.**W****Contraindications:** No information available.**X****Adverse reactions:** As with other topical NSAIDs, diclofenac may cause local irritation. Topical NSAIDs should be used with caution in ulcerative keratitis as they can delay epithelial healing. Topical NSAIDs, and most specifically diclofenac, have been associated with an increased risk of corneal 'melting' (keratomalacia) in humans, although this has not been reported in the veterinary literature. Topical NSAIDs have the potential to increase intraocular pressure and should be used with caution in animals**Y****Z**

predisposed to glaucoma. Regular monitoring is advised. Use of systemic formulations has been associated with death in some species of birds.

Drug interactions: Ophthalmic NSAIDs may be used safely with other ophthalmic pharmaceuticals, although concurrent use of drugs which adversely affect the corneal epithelium (e.g. gentamicin) may lead to increased corneal penetration of the NSAID. The concurrent use of topical NSAIDs with topical corticosteroids has been identified as a risk factor in humans for precipitating corneal problems.

DOSES

Mammals: Topically as required (from 1 drop q30m for 2 h prior to cataract surgery, to 1 drop q4h for general anti-inflammatory effects)^a.

Birds: Not recommended (see note above on adverse reactions).

Reptiles, Amphibians, Fish: No information available.

References

- ^a Waterbury L and Flach A (2006) Comparison of ketorolac tromethamine, diclofenac sodium and loteprednol etabonate in an animal model of ocular inflammation. *Journal of Ocular Pharmacology and Therapeutics* **22**, 155–159

Diflubenzuron (Dimilin)

(Aradol, CestoNemEx, Parazin) **ESPA**

Formulations: Immersion: 1 g/l (CestoNemEx), 1.64 g/l (Aradol); 4 g tablet for dissolution in water (Parazin).

Action: Inhibits chitin synthesis during ecdysis (moulting) of exoskeleton in crustacean parasites.

Use: Treatment of crustacean ectoparasites in fish (e.g. *Argulus*, *Ergasilus*, *Lernaea*). Only effective during moulting of immature life stages of the parasites, including eggs, but not adults. It is highly toxic to non-target species and must not be used in the presence of invertebrates. It is persistent in the environment. Some formulations may require approval for use from relevant authorities in some countries.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use in the presence of aquatic invertebrates.

Adverse reactions: No information available.

Drug interactions: No information available.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Fish: 0.03 mg/l by prolonged immersion or 0.01 mg/l by prolonged immersion q6d for 3 treatments^{1,2}; follow the manufacturer's recommendations for proprietary formulations.

Mammals, Birds, Reptiles, Amphibians: No information available.

References

- ¹ Noga EJ (2010) *Fish Disease Diagnosis and Treatment*, 2nd edn. Wiley-Blackwell, Oxford
- ² Stoskopf MK (1993) *Fish Medicine*. Saunders, Philadelphia

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Digoxin

(Digoxin*, Lanoxin*, Lanoxin PG*) **POM**

Formulations: Oral: 62.5 µg (micrograms), 125 µg, 250 µg tablets; 50 µg/ml elixir. **Injectable:** 100 µg/ml, 250 µg/ml.

Action: Inhibits Na⁺/K⁺ ATPase, leading to an increase in intracellular sodium. Sodium is exchanged for calcium, resulting in an increase in intracellular calcium and hence a mild positive inotropic effect. Digoxin slows the heart rate by decreasing the rate of sinoatrial node firing and inhibiting AV nodal conduction. These effects result primarily from parasympathetic activation and sympathetic inhibition, although it may also produce a modest direct depression of nodal tissue. The combination of a slower heart rate and increased force of contraction increases cardiac output in patients with supraventricular tachyarrhythmias. Digoxin improves baroreceptor reflexes that are impaired in heart failure.

Use: Management of heart failure and supraventricular tachyarrhythmias. It is primarily used to control the ventricular rate in cases of heart failure with concurrent atrial fibrillation. Digoxin/diltiazem combination therapy results in more effective rate control than monotherapy. In ferrets and rabbits it is used in the treatment of dilated cardiomyopathy. Serum levels should be checked after 5–7 days, with a sample taken at least 8 hours post-pill. The bioavailability of digoxin varies between the different formulations: i.v. = 100%; tablets = 60%; and elixir = 75%. If toxic effects are seen or the drug is ineffective, serum levels of digoxin should be assessed; the ideal therapeutic level is a trough serum concentration in the region of 1 ng (nanogram)/ml to optimize beneficial effects and minimize toxic side effects, with a suggested range of 0.6–1.2 ng/ml. Decreased dosages or an increase in dosing intervals may be required in geriatric patients, obese animals or those with significant renal dysfunction. The intravenous route is rarely indicated and, if used, should be done very slowly and with extreme care.

Safety and handling: Normal precautions should be observed.

Contraindications: Frequent ventricular arrhythmias or atrioventricular block.

Adverse reactions: Hypokalaemia predisposes to toxicity in all species. Signs of toxicity include anorexia, vomiting, diarrhoea, depression or arrhythmias (e.g. AV block, bigeminy, paroxysmal ventricular or atrial tachycardias with block, and multiform ventricular premature contractions). Lidocaine and phenytoin may be used to control digoxin-associated arrhythmias. Intravenous administration may cause vasoconstriction.

Drug interactions: Antacids, chemotherapy agents (e.g. cyclophosphamide, cytarabine, doxorubicin, vincristine), cimetidine and metoclopramide may decrease digoxin absorption from the GI tract. The following may increase the serum level, decrease the elimination rate or enhance the toxic effects of digoxin: amiodarone, antimuscarinics, diazepam, erythromycin, loop and thiazide diuretics (hypokalaemia), oxytetracycline, quinidine and verapamil. Spironolactone may enhance or decrease the toxic effects of digoxin.

DOSES

Mammals: Ferrets: 5–10 µg (micrograms)/kg p.o. q12–24h; Rabbits: 3–30 µg/kg p.o. q24–48h; **Hamsters:** 50–100 µg/kg p.o. q12–24h;

Primates: 20–120 µg/kg p.o., i.m., i.v. q12–24h, 10 µg/kg p.o. q24h (CHF marmosets); **Hedgehogs:** 10 µg/kg p.o. q12–24h.

Birds: **Raptors:** 20–50 µg (micrograms)/kg p.o. q12h for 2–3 days then reduce to 10 µg/kg p.o. q12–24h; **Pigeons, Parrots:** 20–50 µg/kg p.o. q24h^{a,1}.

Reptiles, Amphibians, Fish: No information available.

References

- ^a Wilson RC, Zenoble RD, Horton Jr CR and Ramsey DT (1989) Single dose digoxin pharmacokinetics in the Quaker Conure (*Myiopsitta monachus*). *Journal of Zoo and Wildlife Medicine* **20**(4), 432–434
- ¹ Fitzgerald BC, Dias S and Martorell J (2018) Cardiovascular drugs in avian, small mammal, and reptile medicine. *Veterinary Clinics of North America: Exotic Animal Practice* **21**(2), 399–442

Diltiazem

(Hypercard, Adizem*, Angitil*, Diltiazem*, Dilzem*, Slozem*, Tildiem*, Viazem*) **POM-V, POM**

Formulations: Oral: 10 mg (Hypercard), 60 mg (generic) tablets and variety of capsule sizes. Long-acting preparations authorized for humans are available but their pharmacokinetics have been little studied in animals to date.

Action: Inhibits inward movement of calcium ions through slow (L-type) calcium channels in myocardial cells, cardiac conduction tissue and vascular smooth muscle; vascular smooth muscle is more sensitive to diltiazem than myocardial tissues (relative activity of 7:1). Diltiazem causes a reduction in myocardial contractility (negative inotrope, although less effective than verapamil), depressed electrical activity (retarded atrioventricular conduction) and decreases vascular resistance (vasodilation of cardiac vessels and peripheral arteries and arterioles).

Use: Used in ferrets and rabbits as in other species to control supra-ventricular tachyarrhythmias and for hypertrophic cardiomyopathy. Used to decrease the ventricular rate in atrial fibrillation either as monotherapy or in combination with digoxin. Digoxin/diltiazem combination therapy results in more effective rate control than monotherapy. Diltiazem is preferred to verapamil by many because it has effective antiarrhythmic properties with minimal negative inotropy. Diltiazem is less effective than amlodipine in the management of hypertension. Reduce the dose in patients with hepatic or renal impairment.

Safety and handling: Normal precautions should be observed.

Contraindications: Diltiazem is contraindicated in patients with second- or third-degree AV block, marked hypotension or sick sinus syndrome, and should be used cautiously in patients with systolic dysfunction or acute or decompensated congestive heart failure.

Adverse reactions: No information available.

Drug interactions: If diltiazem is administered concurrently with beta-adrenergic blockers (e.g. propranolol), there may be additive negative inotropic and chronotropic effects. The co-administration of diltiazem and beta-blockers is not recommended. The activity of diltiazem may be adversely affected by calcium salts or vitamin D. There are conflicting data regarding the effect of diltiazem on serum digoxin levels and monitoring of these levels is recommended if the drugs are

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

used concurrently. Cimetidine inhibits the metabolism of diltiazem, thereby increasing plasma concentrations. Diltiazem enhances the effect of theophylline, which may lead to toxicity. It may affect quinidine and ciclosporin concentrations. Diltiazem may displace highly protein-bound agents from plasma proteins. Diltiazem may increase intracellular vincristine levels by inhibiting outflow of the drug from the cell.

DOSES

Mammals: Ferrets: 1.5–7.5 mg/kg p.o. q12h; Rabbits: 0.5–1.0 mg/kg p.o. q12–24h.

Birds, Reptiles, Amphibians, Fish: No information available.

Dimercaprol (British anti-lewisite)

(Dimercaprol*) **POM**

Formulations: Injectable: 50 mg/ml solution in peanut oil.

Action: Chelates heavy metals.

Use: Treatment of acute toxicity caused by arsenic, gold, bismuth and mercury, and used as an adjunct (with edetate calcium disodium) in lead poisoning.

Safety and handling: Normal precautions should be observed.

Contraindications: Severe hepatic failure.

Adverse reactions: Intramuscular injections are painful. Dimercaprol-metal complexes are nephrotoxic. This is particularly so with iron, selenium or cadmium; do not use for these metals. Alkalinization of urine during therapy may have protective effects for the kidney.

Drug interactions: Iron salts should not be administered during therapy.

DOSES

Birds: 2.5 mg/kg i.m. q4h for 2 days then q12h until signs resolve^{1,2}.

Mammals, Reptiles, Amphibians, Fish: No information available.

References

- ¹ De Francisco N, Ruiz Troya JD and Agüera EI (2003) Lead and lead toxicity in domestic and free living birds. *Avian Pathology* **32**(1), 3–13
- ² Richardson JA, Murphy LA, Khan SA and Means C (2001) Managing pet bird toxicoses. *Exotic DVM* **3**(1), 23–27

Dimethylsulfoxide (DMSO)

(Rimso-50*) **POM**

Formulations: Injectable: 50%, 90% liquid; medical grade only available as a 50% solution, other formulations are available as an industrial solvent. Topical: 70%, 90% gel; 70% cream.

Action: The mechanism of action is not well understood. Antioxidant activity has been demonstrated in certain biological settings and is thought to account for the anti-inflammatory activity.

Use: Management of otitis externa and haemorrhagic cystitis induced by cyclophosphamide. Although efficacy is unproven it has been used in the treatment of renal amyloidosis and prior to surgical treatment of pododermatitis (bumblefoot) in birds. In humans, DMSO is authorized

for the treatment of interstitial cystitis. DMSO is very rapidly absorbed through the skin following administration by all routes and is distributed throughout the body. Metabolites of DMSO are excreted in the urine and faeces. DMSO is also excreted through the lungs and skin, producing a characteristic sulphuric odour. Humans given DMSO experience a garlic-like taste sensation after administration.

Safety and handling: Should be kept in a tightly closed container because it is very hygroscopic. Gloves should be worn during topical application and the product should be handled with care.

Contraindications: No information available.

Adverse reactions: Adverse effects include local irritation and erythema caused by local histamine release. Intravenous administration of solutions with concentrations >20% may cause haemolysis and diuresis.

Drug interactions: DMSO should not be mixed with other potentially toxic ingredients when applied to the skin because of profound enhancement of systemic absorption.

DOSES

Birds: May be applied topically to lesions (e.g. bumblefoot) as an anti-inflammatory prior to surgery. May be combined with other drugs as a carrying agent.

Mammals, Reptiles, Amphibians, Fish: No information available.

Dimilin see **Diflubenzuron**

Dinoprost tromethamine (Prostaglandin F₂)

(Enzaprost, Lutalyse) **POM-V**

Formulations: Injectable: 5 mg/ml solution.

Action: Stimulates uterine contraction, causes cervical relaxation and inhibits progesterone production by the corpus luteum.

Use: Used in the termination of pregnancy at any stage of gestation and to stimulate uterine contractions in the treatment of open pyometra. Possesses prokinetic effects on the caecum of the rabbit. In birds used topically in cases of egg retention.

Safety and handling: People who are or may become pregnant and asthmatics should avoid handling this drug.

Contraindications: Do not use for the treatment of closed pyometra as there is a risk of uterine rupture. Do not use in birds if the ureterovaginal sphincter is not dilated.

Adverse reactions: Hypersalivation, panting, tachycardia, vomiting, urination, defecation, transient hyperthermia, locomotor incoordination and mild CNS signs have been reported. Such effects usually diminish within 30 min of drug administration. There is no adverse effect on future fertility. Severe adverse effects have been reported in birds.

Drug interactions: The effect of oxytocin would be potentiated by prostaglandins and inhibited by progestogens.

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A**DOSES**

Mammals: Rabbits: 0.2 mg/kg single i.m. injection following 3 days of oral liquid paraffin to assist in emptying impacted caecal contents.

Birds: 0.02–0.1 mg/kg i.m., intracloacal once.

Reptiles, Amphibians, Fish: No information available.

B**C****D****Dinoprostone (Prostaglandin E2)**

(Prostin E2*) **POM**

E**F**

Formulations: Topical: 0.4 mg/ml gel.

G

Action: Stimulates uterine contraction, causes cervical relaxation and inhibits progesterone production by the corpus luteum.

H

Use: Used to relax the vagina and induce uterine contractions in egg-bound birds. Used in preference to Prostaglandin F as better effects in sphincter relaxation.

I

Safety and handling: People who are or may become pregnant and asthmatics should avoid handling this drug.

J

Contraindications: No information available.

K

Adverse reactions: Uterine rupture may occur.

L

Drug interactions: No information available.

L**DOSES****M**

Birds: Apply a thin layer of gel to the cloacal mucosa once.

N

Mammals, Reptiles, Amphibians, Fish: No information available.

O**Diphenhydramine**

(Nytol*) **P**

P

Formulations: Oral: 25 mg tablet; 2 mg/ml solution. Other products are available of various concentrations and most contain other active ingredients.

Q**R**

Action: The antihistaminergic (H1) effects are used to reduce pruritus and prevent motion sickness. It is also a mild anxiolytic and sedative.

S

Use: In birds it is used in the management of allergic rhinitis, hypersensitivity reactions and allergic dermatopathies. In ferrets it is used before vaccination if a previous vaccine reaction has been encountered, or for prevention of sneezing or coughing when these interrupt eating or sleeping. In rabbits it may also be used to reduce possible nausea associated with torticollis. Liquid is very unpalatable.

U**V**

Safety and handling: Normal precautions should be observed.

W

Contraindications: Urine retention, glaucoma and hyperthyroidism.

X

Adverse reactions: May cause hypotension in birds.

Y

Drug interactions: An increased sedative effect may occur if used with benzodiazepines or other anxiolytics/hypnotics. Avoid the concomitant use of other sedative agents. Diphenhydramine may enhance the effect of adrenaline and partially counteract anticoagulant effects of heparin.

Z

DOSES

Mammals: Ferrets: 0.5–2 mg/kg p.o., s.c. q8–12h for allergic reactions; for preventative management of vaccination reactions, give the high end of the dose range by i.m. injection prior to vaccination; **Rabbits:** 2 mg/kg p.o., s.c., i.m. q12h for torticollis; **Guinea pigs:** 1–5 mg/kg s.c. prn; **Chinchillas, Hamsters, Rats, Mice:** 1–2 mg/kg p.o., s.c. q12h; **Primates:** 5 mg/kg/day p.o., i.m., i.v. total dose divided q6–8h.

Birds: 1–4 mg/kg p.o. q8–12h.

Reptiles, Amphibians, Fish: No information available.

Domperidone

(Domperidone*, Motilium*) **POM**

Formulations: Oral: 10 mg tablet, 1 mg/ml suspension.

Action: A potent antiemetic with a similar mechanism of action to metoclopramide, but with fewer adverse CNS effects as it cannot penetrate the blood–brain barrier. It is gastrokinetic in humans and rabbits.

Use: Treatment of vomiting and reduced GI motility. In rabbits, it has a significant prokinetic action on gastric emptying.

Safety and handling: Normal precautions should be observed.

Contraindications: GI obstruction or perforation.

Adverse reactions: No information available.

Drug interactions: No information available.

DOSES

Mammals: Rabbits: 0.5 mg/kg p.o. q12h.

Birds, Reptiles, Amphibians, Fish: No information available.

Dopamine

(Dopamine*) **POM**

Formulations: Injectable: 200 mg in 5 ml vial (40 mg/ml solution), 800 mg in a 5 ml vial (160 mg/ml solution).

Action: Dopamine is an endogenous catecholamine and precursor of noradrenaline, with direct and indirect (via release of noradrenaline) agonist effects on dopaminergic and beta-1 and alpha-1 adrenergic receptors.

Use: Improvement of haemodynamic status. Main indications are treatment of shock following correction of fluid deficiencies, acute heart failure, and support of blood pressure during anaesthesia. Dobutamine is preferred for support of systolic function in patients with heart failure. Dopamine is a potent and short-acting drug, therefore it should be given in low doses by continuous rate infusion, and accurate dosing is important. Dopamine should be diluted in normal saline to an appropriate concentration. At low doses (<10 µg (micrograms)/kg/min) dopamine acts on dopaminergic and beta-1 adrenergic receptors, causing vasodilation, increased force of contraction and heart rate, and resulting in an increase in cardiac output and organ perfusion; systemic

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A vascular resistance remains largely unchanged. At higher doses
B (>10 µg/kg/min) dopaminergic effects are overridden by the alpha effects,
C resulting in an increase in systemic vascular resistance and reduced
D peripheral blood flow. Dopamine has been shown to vasodilate mes-
E enteric blood vessels via DA1 receptors. There may be an improvement in
F urine output but this may be entirely due to inhibition of proximal tubule
G sodium ion reabsorption and an improved cardiac output and blood
H pressure rather than directly improving renal blood flow. The dose of
I dopamine should be adjusted according to clinical effect, therefore
J monitoring of arterial blood pressure during administration is advisable.
K All sympathomimetic drugs have pro-arrhythmic properties, therefore
L ECG monitoring is advised.

Safety and handling: Solution should be discarded if it becomes discoloured.

Contraindications: Discontinue or reduce the dose of dopamine should cardiac arrhythmias arise.

Adverse reactions: Extravasation of dopamine causes necrosis and sloughing of surrounding tissue due to tissue ischaemia. Should extravasation occur, infiltrate the site with a solution of 5–10 mg phentolamine in 10–15 ml of normal saline using a syringe with a fine needle. Nausea, vomiting, tachyarrhythmias and changes in blood pressure are the most common adverse effects. Hypotension may develop with low doses, and hypertension may occur with high doses. Sudden increases in blood pressure may cause a severe bradycardia. All dopamine-induced arrhythmias are most effectively treated by stopping the infusion.

Drug interactions: Risk of severe hypertension when monoamine oxidase inhibitors, doxapram and oxytocin are used with dopamine. Halothane may increase myocardial sensitivity to catecholamines. The effects of beta-blockers and dopamine are antagonistic.

DOSES

Mammals: Guinea pigs: 0.08 mg/kg i.v. prn.

Birds: 5–10 µg (micrograms)/kg/min to counter isoflurane-induced hypotension. More effective than dobutamine^a.

Reptiles, Amphibians, Fish: No information available.

References

- ^a Schnellbacher RW, Da Cunha AF, Beaufrère H *et al.* (2012) Effects of dopamine and dobutamine on isoflurane-induced hypotension in Hispaniolan Amazon parrots (*Amazona ventralis*). *American Journal of Veterinary Research* **73**(7), 952–958

Dorzolamide

(CoSopt*, Dorzolamide*, Dorzolamide with Timolol*, Trusopt*) **POM**

Formulations: Ophthalmic drops: 20 mg/ml (2%) (Dorzolamide, Trusopt), 2% dorzolamide + 0.5% timolol (CoSopt, Dorzolamide with Timolol); 5 ml bottle, single-use vials (CoSopt, Trusopt).

Action: Carbonic anhydrase inhibitor. Reduces intraocular pressure by reducing the rate of aqueous humour production by inhibiting the formation of bicarbonate ions within the ciliary body epithelium.

Use: In the control of all types of glaucoma, either alone or in combination with other topical drugs. It may be less tolerated than

brinzolamide because of its less physiological pH of 5.6. The concurrent use of a topical and a systemic carbonic anhydrase inhibitor is not beneficial as there is no additional decrease in intraocular pressure compared with either route alone.

Safety and handling: Normal precautions should be observed.

Contraindications: Severe hepatic or renal impairment. Dorzolamide/timolol is not the drug of choice in uveitis or anterior lens luxation.

Adverse reactions: Local irritation and blepharitis. Dorzolamide may cause more local irritation than brinzolamide. Dorzolamide/timolol causes miosis.

Drug interactions: No information available.

DOSES

Mammals: Rabbits: 1 drop per eye q8–12h; Rats: 1 drop 1% solution (dilute standard formulation with sterile water) q12h.

Birds: 1 drop per eye q12h^{1,2}. Appeared effective in combination with timolol maleate in a white-bellied caique³.

Reptiles, Amphibians, Fish: No information available.

References

- ¹ Fordham M, Rosenthal K, Durham A, Duda L and Komáromy AM (2010) Intraocular osteosarcoma in an Umbrella cockatoo (*Cacatua alba*). *Veterinary Ophthalmology* **13**, 103–108
- ² Jayson S, Guzman DSM, Petritz O, Freeman K and Maggs DJ (2014) Medical management of acute ocular hypertension in a western screech owl (*Megascops kennicottii*). *Journal of Avian Medicine and Surgery* **28(1)**, 38–45
- ³ Cococcetta C, Collarile T, Masi M *et al.* (2021) Antemortem diagnosis of congenital glaucoma in a white-bellied caique (*Pionites leucogaster*). *Veterinary Ophthalmology* **24(1)**, 93–96

Doxapram

(Doxapram hydrochloride) **POM-VPS, POM-V, POM**

Formulations: Injectable: 20 mg/ml solution.

Action: Stimulates respiration by increasing the sensitivity of aortic and carotid body chemoreceptors to arterial gas tensions.

Use: Stimulates respiration during and after general anaesthesia. In neonatal animals, used to stimulate or initiate respiration after birth, particularly in those delivered by caesarean section. The dose should be adjusted according to the requirements of the situation; adequate but not excessive doses should be used. A patent airway is essential. Must not be used indiscriminately to support respiratory function. Severe respiratory depression should be controlled by tracheal intubation, followed by IPPV and then resolution of the initiating cause. Duration of effect in mammals is 15–20 mins. Has also been used to aid assessment of laryngeal function under light anaesthesia. Not effective at stimulating respiration in the face of hypoxaemia (pre-oxygenation of hypoventilating neonates is recommended, along with removal of obstructive secretions). Used for the treatment of respiratory depression in fish.

Safety and handling: Protect from light.

Contraindications: Do not use in animals without a patent airway.

Adverse reactions: Overdose can cause excessive hyperventilation,

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which may be followed by reduced carbon dioxide tension in the blood leading to cerebral vasoconstriction. This could result in cerebral hypoxia in some animals. Doxapram is irritant and may cause a thrombophlebitis, avoid extravasation or repeated i.v. injection into the same vein. Use doxapram injection with caution in neonates because it contains benzyl alcohol which is toxic. Overdosage signs include hypertension, skeletal muscle hyperactivity, tachycardia and generalized CNS excitation including seizures; treatment is supportive using short-acting i.v. barbiturates or propofol and oxygen. Effects in pregnant/lactating animals are not known.

Drug interactions: Hypertension may occur with sympathomimetics. The use of theophylline concurrently with doxapram may cause increased CNS stimulation. As doxapram may stimulate the release of adrenaline, its use within 10 min of the administration of anaesthetic agents that sensitize the myocardium to catecholamines (e.g. halothane) should be avoided. Doxapram is compatible with 5% dextrose or normal saline, but is incompatible with sodium bicarbonate or thiopental. High doses administered during or after anaesthesia with halogenated hydrocarbon anaesthetics, such as halothane, may precipitate cardiac arrhythmias. Doxapram injection should be used with extreme caution in animals that have been sedated with morphine. Administration of doxapram at 10 mg/kg to such animals may be followed by convulsions.

DOSES

Mammals: Ferrets, Rabbits, Chinchillas, Rats, Mice, Hamsters, Gerbils, Hedgehogs: 2–10 mg/kg i.v., i.m., i.p., sublingual once; Guinea pigs: 2–5 mg/kg i.v., s.c., i.p. once; Primates: 2 mg/kg i.v.; Sugar gliders: 2 mg/kg s.c., i.m., i.v. once³.

Birds: 5–20 mg/kg i.m., i.v., intratracheal, intraosseous once¹.

Reptiles: 4–12 mg/kg i.m., i.v., p.o. once; Ball pythons: 10 mg/kg s.c. once^a.

Fish: 5 mg/kg i.v., intracoelomic², topically to the gills prn.

Amphibians: No information available.

References

- ^a Karklus AA, Sladky KK and Johnson SM (2021) Respiratory and antinociceptive effects of dexmedetomidine and doxapram in ball pythons (*Python regius*). *American Journal of Veterinary Research* **82**(1), 11–21
- ¹ Lierz M and Korbel R (2012) Anesthesia and analgesia in birds. *Journal of Exotic Pet Medicine* **21**(1), 44–58
- ² Hadfield C, Whitaker BR and Clayton LA (2007) Emergency and critical care of fish. *Veterinary Clinics of North America: Exotic Animal Practice* **10**, 647–675
- ³ Lennox AM (2007) Emergency and critical care procedures in sugar gliders (*Petaurus breviceps*), African hedgehogs (*Atelerix albiventris*), and prairie dogs (*Cynomys* spp). *Veterinary Clinics of North America: Exotic Animal Practice* **10**(2), 533–555

Doxepin

(Sinepin*, Sinequan*, Zonalon*) **POM**

Formulations: Oral: 25 mg, 50 mg capsules.

Action: Doxepin blocks noradrenaline and serotonin re-uptake in the brain, resulting in antidepressive activity, while the H₁ and H₂ blockage result in antipruritic effects. Its metabolite, desmethyldoxepin, is also psychoactive.

Use: Management of pruritus and psychogenic dermatoses where there is a component of anxiety, including compulsive disorders. Data are lacking as to its efficacy at the suggested doses. Used in birds for organophosphate toxicity. When used in birds it is important to monitor for cardiac arrhythmias.

Safety and handling: Normal precautions should be observed.

Contraindications: Hypersensitivity to tricyclic antidepressants, glaucoma, history of seizures or urinary retention and severe liver disease.

Adverse reactions: Sedation, dry mouth, diarrhoea, vomiting, excitability, arrhythmias, hypotension, syncope, increased appetite, weight gain and, less commonly, seizures and bone marrow disorders have been reported in humans.

Drug interactions: Should not be used with monoamine oxidase inhibitors or drugs which are metabolized by cytochrome P450 2D6 (e.g. chlorphenamine and cimetidine).

DOSES

Birds: Feather plucking: 1–2 mg/kg p.o. q12h¹; organophosphate toxicity: 0.2 mg/kg i.m. q4h.

Mammals, Reptiles, Amphibians, Fish: No information available.

References

- ¹ van Zeeland YRA, Spruit BM, Rodenburg TA et al. (2009) Feather damaging behaviour in parrots: a review with consideration of comparative aspects. *Applied Animal Behaviour Science* **121**(2), 75–95

Doxorubicin (Adriamycin)

(Doxorubicin*) **POM**

Formulations: Injectable: 10 mg, 50 mg powders for reconstitution; 10 mg, 50 mg/vial solutions.

Action: Inhibits DNA synthesis and function.

Use: Treatment of lymphoma in the ferret. Has been used to treat lymphoma in a green iguana. It may be used alone or in combination with other antineoplastic therapies. Premedication with i.v. chlorphenamine or dexamethasone is recommended. Doxorubicin is highly irritant and must be administered via a preplaced i.v. catheter. The reconstituted drug should be administered over a minimum period of 10 min into the side port of a freely running i.v. infusion of 0.9% NaCl. Do not use heparin flush. May need to reduce dose in patients with liver disease. Use with caution in patients previously treated with radiation as can cause radiation recall.

Safety and handling: **Potent cytotoxic drug that should only be prepared and administered by trained personnel. See specialist texts for further advice on chemotherapeutic agents.** After reconstitution the drug is stable for at least 48h at 4°C. A 1.5% loss of potency may occur after 1 month at 4°C but there is no loss of potency when frozen at –20°C. Filtering through a 0.22 µm (micrometre) filter will ensure adequate sterility of the thawed solution. Store unopened vials under refrigeration.

Contraindications: Do not use in patients with existing cardiac or renal disease.

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A **Adverse reactions:** Allergic reactions have been reported; acute anaphylactic reactions should be treated with adrenaline, steroids and fluids. Doxorubicin causes a dose-dependent cumulative cardiotoxicity in dogs (leading to cardiomyopathy and congestive heart failure), and this may be of concern in ferrets. The risk of cardiotoxicity is greatly increased when the cumulative dose is $>240 \text{ mg/m}^2$. It may also cause tachycardia and arrhythmias on administration; monitor with ECG and/or echocardiograms. Anorexia, vomiting, severe leucopenia, thrombocytopenia, haemorrhagic gastroenteritis and nephrotoxicity are the major adverse effects. A CBC and platelet count should be monitored whenever therapy is given. If the neutrophil count drops below $3 \times 10^9/\text{l}$ or the platelet count drops below $50 \times 10^9/\text{l}$, treatment should be suspended. Once the counts have stabilized, doxorubicin can be restarted at the same dose. If haematological toxicity occurs again, or if GI toxicity is recurrent, the dose should be reduced by 10–25%.

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D Extravasation injuries secondary to perivascular administration may be serious, with severe tissue ulceration and necrosis possible. Dexrazoxane can be used to treat extravasation if it occurs. Ice compresses may also be beneficial (applied for 15 min q6h).

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J **Drug interactions:** Barbiturates increase plasma clearance of doxorubicin. The agent causes a reduction in serum digoxin levels. Do not mix doxorubicin with other drugs. Doxorubicin is incompatible with dexamethasone, 5-fluorouracil and heparin; concurrent use will lead to precipitate formation.

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DOSES

M **See Appendix for chemotherapy protocols in ferrets.**

N **Mammals:** Ferrets: 20 mg/m^2 or $1\text{--}2 \text{ mg/kg}$ i.v. every 3 weeks for a maximum of 5 doses. Also used as part of a multi-drug protocol for lymphoma.

O **Birds:** 2 mg/kg i.v. infused in normal saline over 20 minutes^a.

P **Reptiles:** $0.26\text{--}0.75 \text{ mg/kg}$ i.v. reported as part of a successful chemotherapy protocol in a green iguana¹.

Q **Amphibians, Fish:** No information available.

References

R ^a Gilbert CM, Filippich LJ and Charles BG (2004) Doxorubicin pharmacokinetics following a single-dose infusion to sulphur-crested cockatoos (*Cacatua galerita*). *Australian Veterinary Journal* **82**(12), 769–772

S ¹ Folland DW, Johnston MS, Thamm DH and Reavill D (2011) Diagnosis and management of lymphoma in a green iguana (*Iguana iguana*). *Journal of the American Veterinary Medical Association* **239**(7), 985–991

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Doxycycline

V (Doxatib, Doxx-Sol, Doxyseptin 300, HydroDoxx, Karidox, Ornicure, Powdox, Pulmodox, Ronaxan, Soludox, Vibramycin*, Vibravenos*) **POM-V**

W

X **Formulations:** Oral: 20 mg, 100 mg (Ronaxan), 300 mg (Doxyseptin) tablets; 260 mg/sachet powder (Ornicure); 100 mg/ml oral solution (Karidox, Powdox). Oral solution also available as a veterinary special if lower concentration required for accurate dosing. **Injectable:** 20 mg/ml long-acting injection (Vibravenos; import on an STC), 500 mg/g powder (Doxatib, Doxx-Sol, Hydrodox, Soludox). **VSP**

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Action: Bacteriostatic agent inhibiting protein synthesis at the initiation step by interacting with the 30S ribosomal subunit.

Use: Authorized for use in pigeons (racing pigeons) and ornamental birds, particularly Psittaciformes for treatment of infections, specifically *Chlamydia psittaci*, *Pasteurella multocida* and *Mycoplasma* spp. (Orniculture). Antibacterial (including spirochaetes such as *Helicobacter* and *Campylobacter*), antirickettsial, antimycoplasmal and antichlamydial activity. It is the drug of choice to treat avian chlamydiosis; treatment may be required for 6 weeks in birds. It is not affected by, and does not affect, renal function as it is excreted in faeces, and is therefore recommended when tetracyclines are indicated in animals with renal impairment. It is preferred by some authors to oxytetracycline for use in birds. Being extremely lipid-soluble, it penetrates well into prostatic fluid and bronchial secretions. Administer with food. Injection is very irritant in birds: must alternate injection sites or divide dose if large volume to inject.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not administer to pregnant animals. Do not administer if there is evidence of oesophagitis or dysphagia.

Adverse reactions: Nausea, vomiting and diarrhoea. Oesophagitis and oesophageal ulceration may develop; administer with food or a water bolus to reduce this risk. Administration during tooth development may lead to discoloration of the teeth, although the risk is less than with other tetracyclines.

Drug interactions: Absorption of doxycycline is reduced by antacids, calcium, magnesium and iron salts, although the effect is less marked than that seen with water-soluble tetracyclines. Phenobarbital and phenytoin may increase its metabolism, thus decreasing plasma levels. Do not use in combination with bactericidal antimicrobials.

DOSES

Classified as category D (Prudence) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Rabbits: 2.5–4 mg/kg p.o. q24h; Rats, Mice: 5 mg/kg p.o. q12h; **Other rodents:** 2.5 mg/kg p.o. q12h or 70–100 mg/kg s.c., i.m. of the long-acting preparation (Vibravenos); **Primates:** 2.5 mg/kg p.o. q24h; 3–4 mg/kg p.o. q12h; 5 mg/kg p.o. q12h for 1 day then 2.5 mg/kg p.o. q24h (for *Balantidium coli*); **Hedgehogs:** 2.5–10 mg/kg p.o., s.c., i.m. q12h.

Birds: **Raptors:** 50 mg/kg p.o. q12h, 100 mg/kg i.m. q7d (Vibravenos); **Parrots:** 15–50 mg/kg p.o. q24h, 1000 mg/kg in soft food/dehulled seed, 75–100 mg/kg i.m. q7d (Vibravenos; lowest dose rate for macaws)^{a,b,c,d}; course of treatment with doxycycline for chlamydiosis = 45 days; **Passerines/Pigeons:** 40 mg/kg p.o. q12–24h, 200–500 mg/l in water (soft or deionized water only).

Reptiles: 50 mg/kg i.m. once, then 25 mg/kg i.m. q72h; 5–10 mg/kg p.o. q24h.

Amphibians: 50 mg/kg i.m. q7d.

Fish: No information available.

References

- ^a Flammer K and Papich M (2005) Assessment of plasma concentrations and effects of injectable doxycycline in three psittacine species. *Journal of Avian Medicine and Surgery* **19**(3), 216–224

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^b Flammer K and Whitt-Smith D (2001) Plasma concentrations of doxycycline in selected psittacine birds when administered in water for potential treatment of *Chlamydophila psittaci* infection. *Journal of Avian Medicine and Surgery* **15**(4), 276–282

^c Powers LV, Flammer K and Papich M (2000) Preliminary investigation of doxycycline plasma concentrations in Cockatiels (*Nymphicus hollandicus*) after administration by injection or in water or feed. *Journal of Avian Medicine and Surgery* **14**(1), 23–30

^d Saturation T (2008) Administration of doxycycline in drinking water for treatment of spiral bacterial infection in Cockatiels. *Journal of Zoo and Wildlife Medicine* **39**(3), 499–501

Edetate calcium disodium (CaEDTA)

(Ledclair*) **POM**

Formulations: Injectable: 200 mg/ml solution. Not available in the UK currently.

Action: Heavy metal chelating agent.

Use: Lead and zinc poisoning. Dilute strong solution to a concentration of 10 mg/ml in 5% dextrose before use. Blood lead levels may be confusing, therefore monitor clinical signs during therapy. Measure blood lead levels 2–3 weeks after completion of treatment in order to determine whether a second course is required or if the animal is still being exposed to lead. Ensure there is no lead in the GI tract before administering (e.g. use laxatives) nor any residual heavy metals elsewhere (e.g. powdered zinc in coat or an environmental source).

Safety and handling: Normal precautions should be observed.

Contraindications: Use with caution in patients with impaired renal function.

Adverse reactions: Reversible nephrotoxicity is usually preceded by other clinical signs of toxicity (e.g. depression, vomiting, diarrhoea). Injections are painful.

Drug interactions: No information available.

DOSES

Mammals: Ferrets: 20–30 mg/kg s.c. q12h; Rabbits: 27.5 mg/kg s.c. q6h; Rodents: 25–30 mg/kg s.c. q6–12h. Repeat treatment for 5 days on, 5 days off until blood lead or zinc levels are normal when measured at the end of the period off treatment.

Birds: 35–50 mg/kg i.m., s.c. q12h for 5 days followed by 2 days of no treatment, then repeat until metal particles are no longer visible on radiographs^a. 100 mg/kg i.m. weekly has been proposed in zinc toxicosis.

Reptiles: 10–40 mg/kg i.m. q12h.

Amphibians, Fish: No information available.

References

^a Sears J, Cooke SW, Cooke ZR and Heron TJ (1989) A method for the treatment of lead poisoning in the mute swan (*Cygnus olor*) and its long-term success. *British Veterinary Journal* **145**(6), 586–595

Emamectin benzoate

(Fluke-Solve Plus, Lice-Solve, Slice) **ESPA, POM-V**

Formulations: Immersion: 10 g, 100 g sachets of powder containing 1.4% emamectin; 10 g, 50 g, 100 g sachets of powder containing praziquantel, nitroscanate and emamectin benzoate. In food: 2.5 kg pouch of powder containing 5 g emamectin benzoate.

Action: Irreversibly binds to and opens GABA and glutamate-gated channels, leading to flaccid paralysis and death of the parasite.

Use: Treatment of crustacean parasites of fish (*Argulus*, *Ergasilus* spp.). For use in closed water systems. Turn off UV and carbon filters for 24 h after adding the product to the water. Switch off protein skimmers during treatment. Ensure good aeration of the water during treatment. Emamectin rapidly biodegrades in sunlight.

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A Safety and handling: Normal precautions should be observed. Do not dispose of down drains or in waterways.

B Contraindications: Do not use at the same time as other fish or pond treatments. Do not use in ponds containing orfe (*Leuciscus* spp.) or terrapins.

C Adverse reactions: No information available.

D Drug interactions: No information available.

E DOSES

See Appendix for guidelines on responsible parasiticide use.

Fish: 56 µg (micrograms)/l by prolonged immersion, repeat after 1 month. **Goldfish:** 0.05 mg/kg bodyweight daily in food for 7 days; **Koi:** 0.005 mg/kg bodyweight daily in food for 7 days^a.

Mammals, Birds, Reptiles, Amphibians: No information available.

H References

^a Hanson SK, Hill JE, Watson CA, Yanong RP and Endris R (2011) Evaluation of emamectin benzoate for the control of experimentally induced infestations of *Argulus* sp. in goldfish and koi carp. *Journal of Aquatic Animal Health* **23**(1), 30–34

Emodepside

(Dronspot, Profender) **POM-V**

L Formulations: Topical: 21.4 mg/ml emodepside with praziquantel solution in spot-on pipettes of various sizes.

M Action: Stimulates presynaptic secretin receptors resulting in paralysis and death of the parasite.

N Use: Treatment of roundworms (adult and immature) and tapeworms (adult) including *Toxocara cati*, *Toxascaris leonina*, *Ancylostoma tubaeforme*, *Aelurostrongylus abstrusus*, *Dipylidium caninum*, *Taenia taeniaeformis*, *Echinococcus multilocularis*. Do not shampoo until substance has dried. Treatment of a variety of reptile endoparasites including oxyurids, ascarids, strongylids, trichostrongylids and capillarids.

Q Safety and handling: People who are or may become pregnant should not handle this drug.

R Contraindications: Do not use in pregnant mammals. Studies in rabbits and rats suggest it may interfere with fetal development *in utero*.

S Adverse reactions: Ingestion may result in salivation or vomiting. Harmful to aquatic animals.

T Drug interactions: Possible interaction with P-glycoprotein substrates/inhibitors.

U DOSES

See Appendix for guidelines on responsible parasiticide use.

W Mammals: Rabbits: 0.14 ml/kg topically once (reported for *Trichostrongylus colubriformis*).

X Reptiles: <1.12 ml/kg applied topically once (corresponding to 24 mg emodepside and 96 mg praziquantel/kg)^{a,b}.

Birds, Amphibians, Fish: No information available.

Y References

^a Schilliger L, Betremieux O, Rochet J, Krebber R and Schaper R (2009) Absorption and efficacy of a spot-on combination containing emodepside plus praziquantel in reptiles. *Revue de Médecine Vétérinaire* **160**(12), 557–561

Z

- ^b Tang PK, Pellett S, Blake D and Hedley J (2017) Efficacy of a topical formulation containing emodepside and praziquantel (Profender®, Bayer) against nematodes in captive tortoises. *Journal of Herpetological Medicine and Surgery* **27**(3), 116–122

Enalapril

(Enacard) **POM-V**

Formulations: Oral: 1 mg, 2.5 mg, 5 mg, 10 mg, 20 mg tablets.

Action: ACE inhibitor. It inhibits conversion of angiotensin I to angiotensin II and inhibits the breakdown of bradykinin. Overall effect is a reduction in preload and afterload via venodilation and arteriodilation, decreased salt and water retention via reduced aldosterone production and inhibition of the angiotensin-aldosterone-mediated cardiac and vascular remodelling. Efferent arteriolar dilation in the kidney can reduce intraglomerular pressure and therefore glomerular filtration. This may decrease proteinuria.

Use: Treatment of congestive heart failure caused by mitral regurgitation or dilated cardiomyopathy. Often used in conjunction with diuretics when heart failure is present as most effective when used in these cases. Can be used in combination with other drugs to treat heart failure (e.g. pimobendan, spironolactone, digoxin). May be beneficial in cases of chronic renal insufficiency, particularly protein-losing nephropathies. May reduce blood pressure in hypertension. ACE inhibitors are more likely to cause or exacerbate prerenal azotaemia in hypotensive animals and those with poor renal perfusion (e.g. acute, oliguric renal failure). Use cautiously if hypotension, hyponatraemia or outflow tract obstruction are present. Regular monitoring of blood pressure, serum creatinine, urea and electrolytes is strongly recommended with ACE inhibitor treatment. Hypotension, azotaemia and hyperkalaemia are all indications to stop or reduce ACE inhibitor treatment in rabbits. Has been used alongside amlodipine in treatment of hypertension in birds

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use in cases of cardiac output failure.

Adverse reactions: Potential adverse effects include hypotension, hyperkalaemia and azotaemia. Monitor blood pressure, serum creatinine and electrolytes when used in cases of heart failure. Dosage should be reduced if there are signs of hypotension (weakness, disorientation). Anorexia, vomiting and diarrhoea are rare. It is not recommended for breeding, pregnant or lactating animals, as safety has not been established. In rabbits, treatment with ACE inhibitors can be associated with an increase in azotaemia and significant hypotension; treatment should start at the lower end of the range.

Drug interactions: Concomitant treatment with potassium-sparing diuretics (e.g. spironolactone) or potassium supplements could result in hyperkalaemia. However, in practice, spironolactone and ACE inhibitors appear safe to use concurrently. There may be an increased risk of nephrotoxicity and decreased clinical efficacy when used with NSAIDs. There is a risk of hypotension with concomitant administration of diuretics, vasodilators (e.g. anaesthetic agents, antihypertensive agents) or negative inotropes (e.g. beta-blockers).

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A**DOSES**

Mammals: Ferrets: 0.25–0.5 mg/kg p.o. q24–48h; **Rabbits:** 0.25–0.5 mg/kg p.o. q24–48h; **Rodents:** 0.5–1.0 mg/kg p.o. q24h; **Primates:** 0.5 mg/kg p.o. q48h; **Sugar gliders:** 0.22–0.44 mg/kg p.o. q24h; **Hedgehogs:** 0.5–1 mg/kg p.o. q24h¹.

Birds: 1.25 mg/kg p.o. q12h².

Reptiles, Amphibians, Fish: No information available.

B**C****D****E****References**

¹ Delk KW, Eshar D, Garcia E and Harkin K (2014) Diagnosis and treatment of congestive heart failure secondary to dilated cardiomyopathy in a hedgehog. *Journal of Small Animal Practice* **55**(3), 174–177

² Fitzgerald BC, Dias S and Martorell J (2018) Cardiovascular drugs in avian, small mammal, and reptile medicine. *Veterinary Clinics: Exotic Animal Practice* **21**(2), 399–442

F**G****H****I****Enilconazole**

(Imaverol) **POM-VPS**

J

Formulations: Topical: 100 mg/ml (10%) liquid.

K

Action: Inhibition of cytochrome P450-dependent synthesis of ergosterol in fungal cells, causing increased cell wall permeability and allowing leakage of cellular contents.

L

Use: Fungal infections of the skin and nasal aspergillosis. Has been shown to inhibit zoospores of *Batrachochytrium dendrobatidis* *in vitro* but there is no information on systemic clinical treatment of amphibians with chytridiomycosis.

M**N**

Safety and handling: Normal precautions should be observed.

O

Contraindications: No information available.

P

Adverse reactions: Hepatotoxic if swallowed. Avoid contact with eyes.

Q**DOSES**

Mammals: **Rabbits:** Dilute 1:50 in water to produce a 2 mg/ml (0.2%) solution. Apply topically to lesions every 1–3 days for 3–4 applications and then check success of therapy with fungal culture; **Hamsters:** 0.2% rinse topically every 7 days until fungal cultures negative.

R**S**

Birds: Dilute 1:10 in water and give 0.5 ml/kg/day intratracheally for 7–14 days^a.

T

Reptiles: Dilute 1:50 in water. Apply topically to lesions every 2–3 days.

U

Amphibians, Fish: No information available.

V**References**

^a Perelman B, Smith B, Bronstein D, Gur-Lavie A and Kuttin ES (1992) Use ofazole compounds for the treatment of experimental aspergillosis in turkeys. *Avian Pathology* **21**(4), 591–599

W**X****Y****Z**

Enrofloxacin

(**Baytril**, **Enrobactin**, **Enrocare**, **Enrotab**, **Enrotron**, **Enrox**, **Enroxil**, **Floxabactin**, **Floxibac**, **Lanfloxx**, **Orniflox**, **Powerflox**, **Quinoflox**, **Spectron**, **Xeden**, **Zobuxa**) **POM-V**

Formulations: **Injectable:** 25 mg/ml, 50 mg/ml, 100 mg/ml solutions.

Oral: 15 mg, 50 mg, 150 mg, 250 mg tablets; 25 mg/ml solution, 100 mg/ml solution.

Action: Enrofloxacin is a bactericidal antimicrobial which inhibits bacterial DNA gyrase. The bactericidal action is concentration-dependent, meaning that pulse-dosing regimens may be effective, particularly against Gram-negative bacteria.

Use: Authorized in rabbits, rodents, ornamental birds and reptiles for the treatment of bacterial infections of the alimentary and respiratory tracts and the treatment of skin and wound infections (different formulations may be authorized for different target species and indications). Ideally fluoroquinolone use should be reserved for infections where culture and sensitivity testing predicts a clinical response and where first- and second-line antimicrobials would not be effective. Active against *Mycoplasma* and many Gram-positive and Gram-negative organisms, including *Pasteurella*, *Staphylococcus*, *Pseudomonas aeruginosa*, *Klebsiella*, *Escherichia coli*, *Mycobacterium*, *Proteus* and *Salmonella*. Relatively ineffective against obligate anaerobes. Fluoroquinolones are highly lipophilic drugs that attain high concentrations within cells in many tissues and are particularly effective in the management of soft tissue, urogenital (including prostatic) and skin infections. For the treatment of non-tubercular mycobacterial disease, enrofloxacin can be combined with clarithromycin and rifampin. Administration by i.v. route is not authorized but has been used in cases of severe sepsis. If this route is used, administer slowly as the carrier contains potassium. Dilute solution for injection 1 in 4 with water if dosing small mammals orally. Switch to oral medications in birds as soon as possible. Used for the treatment of bacterial disease in fish, particularly those caused by Gram-negative organisms.

Safety and handling: Normal precautions should be observed.

Contraindications: Fluoroquinolones are relatively contraindicated in growing small mammals as cartilage abnormalities have been reported in young mice, rats, guinea pigs and rabbits after administration of fluoroquinolones similar to enrofloxacin. Care may be needed using fluoroquinolones in breeding females.

Adverse reactions: In birds, joint lesions have been induced in nestling pigeons with high doses of enrofloxacin, and in all species, muscle necrosis may be seen following i.m. administration. Enrofloxacin should be used with caution in epileptics until further information is available, as in humans they potentiate CNS adverse effects when administered concurrently with NSAIDs. Excitation and diarrhoea have been reported in Galapagos tortoises.

Drug interactions: Adsorbents and antacids containing cations (Mg^{2+} , Al^{3+}) may bind to fluoroquinolones and prevent their absorption from the GI tract. The absorption of fluoroquinolones may also be inhibited by sucralfate and zinc salts; separate doses of these drugs by at least 2 hours. Fluoroquinolones increase plasma theophylline concentrations. Cimetidine may reduce the clearance of fluoroquinolones and so

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should be used with caution in combination with these drugs. Efficacy may be reduced due to chelation when used as a bath treatment for fish in hard water with high levels of divalent metal cations (Ca^{2+} , Mg^{2+}), such as in seawater.

DOSES

Classified as category B (Restrict) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Authorized at 5 mg/kg q12h but may not be effective at this dose in every case so higher doses commonly used^a. **Ferrets:** 5–10 mg/kg p.o., s.c., i.m. q12h or 10–20 mg/kg p.o., s.c., i.m. q24h; **Rabbits:** 10–20 mg/kg p.o., s.c., i.v. q24h; **Rodents:** 5–20 mg/kg s.c., p.o. q12–24h; **Primates:** 5 mg/kg p.o., s.c. q12–24h; **Sugar gliders:** 5 mg/kg p.o., s.c., i.m. q12h; **Hedgehogs:** 2.5–10 mg/kg p.o., s.c., i.m. q12h; **Others:** 5–10 mg/kg s.c., p.o. q12h or 20 mg/kg s.c., p.o. q24h.

Birds: 10 mg/kg p.o., i.m. q12h (authorized dose) or 100–200 mg/l drinking water^b.

Reptiles: Variable absorption when given p.o. so i.m. administration may be more appropriate in critically ill animals. **Most species:** 5–10 mg/kg i.m., p.o. q24–48h; **Indian star tortoises:** 5 mg/kg i.m. q12–24h; **Gopher tortoises:** 5 mg/kg i.m. q24–48h; **Hermann's tortoises:** 10 mg/kg intracoelomic; **Red-eared sliders:** 5 mg/kg i.m. or 10 mg/kg p.o., i.m., s.c., intracoelomic q24h or 500 mg/l as a 6–8h bath q24h^c; **Savannah monitors:** 10 mg/kg i.m. q5d; **Green iguanas:** 5 mg/kg p.o., i.m. q24h; **Bearded dragons:** 10 mg/kg i.m.^d; **Asian house geckos:** 10 mg/kg p.o. q24h; **Burmese pythons:** 10 mg/kg i.m., then q48h; **Pit vipers:** 10 mg/kg i.m. q48–72h; **South American Rattlesnakes:** 10 mg/kg i.m.

Amphibians: 5–10 mg/kg p.o., s.c., i.m. topical application q24h^e.

Fish: 5–10 mg/kg i.m., intracoelomic, p.o. q24–48h for 7 doses or 2.5–5 mg/l by immersion for 5 h q24–48h for 5–7 treatments^f.

References

- ^a Mähler M, Stünkel S, Ziegowski C and Kunstýř I (1995) Inefficacy of enrofloxacin in the elimination of *Pasteurella multocida* in rabbits. *Laboratory Animals* **29**(2), 192–199
- ^b Flammer K, Aucoin DP and Whitt DA (1991) Intramuscular and oral disposition of enrofloxacin in African grey parrots following single and multiple doses. *Journal of Veterinary Pharmacology and Therapeutics* **14**(4), 359–366
- ^c James SB, Calle PP, Raphael BL et al. (2003) Comparison of injectable versus oral enrofloxacin pharmacokinetics in red-eared slider turtles (*Trachemys scripta elegans*). *Journal of Herpetological Medicine and Surgery* **13**(1), 5–10
- ^d Salvadori M, Vercelli C, De Vito V et al. (2017) Pharmacokinetic and pharmacodynamic evaluations of a 10 mg/kg enrofloxacin intramuscular administration in bearded dragons (*Pogona vitticeps*): a preliminary assessment. *Journal of Veterinary Pharmacology and Therapeutics* **40**(1), 62–69
- ^e Howard AM, Papich MG, Felt SA et al. (2010) The pharmacokinetics of enrofloxacin in adult African clawed frogs (*Xenopus laevis*). *Journal of the American Association for Laboratory Animal Science* **49**(6), 800–804
- ^f Lewbart G, Vaden S, Deen J et al. (1997) Pharmacokinetics of enrofloxacin in the red pacu (*Colossoma brachypomum*) after intramuscular, oral and bath administration. *Journal of Veterinary Pharmacology and Therapeutics* **20**, 124–128

Epinephrine see **Adrenaline**

Epoetin alfa, Epoetin beta see **Erythropoietin**

Equine chorionic gonadotrophin see **Serum gonadotrophin**

Erythromycin

(Erythrocin, Erythromycin*, Erythroped*) **POM-V, POM**

Formulations: **Injectable:** 200 mg/ml solution; 1 g/vial powder for reconstitution. **Oral:** 250 mg, 500 mg tablets/capsules; 25 mg/ml suspension; powder authorized for chickens.

Action: May be bactericidal (time-dependent) or bacteriostatic, depending upon drug concentration and bacterial susceptibility. It binds to the 50S ribosome, inhibiting peptide bond formation.

Use: Has a similar antibacterial spectrum to penicillins. It is active against Gram-positive cocci (some *Staphylococcus* species are resistant), Gram-positive bacilli and some Gram-negative bacilli (*Pasteurella*). Some strains of *Actinomyces*, *Nocardia*, *Chlamydophila* and *Rickettsia* are also inhibited by erythromycin. Most of the Enterobacteriaceae (*Pseudomonas*, *Escherichia coli*, *Klebsiella*) are resistant. It is used in hamsters to treat proliferative ileitis (*Lawsonia intracellularis*) and in ferrets to control *Campylobacter* infection, although it may not eliminate intestinal carriage of this organism. Being a lipophilic weak base, it is concentrated in fluids that are more acidic than plasma, including milk, prostatic fluid and intracellular fluid. Resistance to erythromycin can be quite high, particularly in staphylococcal organisms. Erythromycin acts as a GI prokinetic by stimulating motilin receptors. Different esters of erythromycin are available. It is likely that the kinetics and possibly the toxicity will differ, depending on the ester used. Erythromycin's activity is enhanced in an alkaline pH. As the base is acid-labile it should be administered on an empty stomach. Used for the treatment of bacterial diseases in fish where practical.

Safety and handling: Normal precautions should be observed.

Contraindications: In humans the erythromycin estolate salt has been implicated in causing cholestatic hepatitis. Although not demonstrated in veterinary medicine, this salt should be avoided in animals with hepatic dysfunction. Avoid oral administration in small herbivores (e.g. rabbits).

Adverse reactions: Care should be taken in cases of hepatic or renal impairment. Avoid oral administration in small herbivores (e.g. rabbits), as can cause fatal enterotoxaemia. Its use as a prokinetic in these species is not advised. Using as an immersion treatment for fish may be harmful to biological filtration systems and is not advised.

Drug interactions: Erythromycin may enhance the absorption of digoxin from the GI tract and increase serum levels of cisapride, methylprednisolone, theophylline and terfenadine. The interactions with cisapride and terfenadine proved particularly significant in human medicine, leading to fatal or near-fatal arrhythmias in some patients receiving both drugs. Erythromycin should not be used in combination with other macrolides, lincosamides or chloramphenicol antimicrobials as antagonism may occur.

DOSES

Classified as category C (Caution) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Ferrets: 10 mg/kg p.o. q6h; Hamsters, Rats, Mice: 20 mg/kg p.o. q12h or 0.13 mg/ml drinking water; Primates: 20–50 mg/kg i.m. q12h; 75 mg/kg p.o. q12h; Hedgehogs: 10 mg/kg p.o., i.m. q12h.

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Birds: 20 mg/kg i.m., s.c. q8h; 60 mg/kg p.o. q12h or 125 mg/l of drinking water; 200 mg/kg soft feed.

B

Fish: 10–20 mg/kg i.m. q24h for 1–3 treatments or 100–200 mg/kg p.o. q24h for 7–21 days.

C

Reptiles, Amphibians: No information available.

D**E**

Erythropoietin (Epoetin alfa, Epoetin beta)

(Eprex*, Neorecormon*) **POM**

F

Formulations: Injectable: 1000 IU, 2000 IU, 5000 IU powders for reconstitution; 2000 IU/ml, 4000 IU/ml, 10,000 IU/ml, 40,000 IU/ml solutions. Eprex is epoetin alfa. Neorecormon is epoetin beta.

G

Action: Stimulates division and differentiation of red blood cells.

H

Use: Recombinant human erythropoietin (r-HuEPO) is predominantly used to treat anaemia associated with chronic renal failure, although it is also used to treat anaemic human patients with cancer and rheumatoid arthritis. Erythropoietin is not indicated in conditions where high serum concentrations of the hormone already exist (e.g. haemolytic anaemia, anaemia due to blood loss), where the anaemia is due to iron deficiency or where systemic hypertension is present. Monitoring and/or supplementation of iron may be necessary, especially if response to treatment is poor. Darbepoetin may be a better choice in many cases.

I**J****K****L****M**

Safety and handling: Normal precautions should be observed.

N

Contraindications: Conditions where high serum concentrations of erythropoietin already exist.

O

Adverse reactions: Local and systemic allergic reactions may rarely develop (skin rash at the injection site, pyrexia, arthralgia and mucocutaneous ulcers).

P

Drug interactions: No information available.

Q

DOSES

R

Mammals: Ferrets, Rabbits: Epoetin alfa: 50–150 IU/kg i.m., s.c. q48–72h; Hedgehogs: 100 IU/kg s.c. q48–72h

S

Birds, Reptiles, Amphibians, Fish: No information available.

T

Ethylene glycol monophenyl ether see **Phenoxyethanol**

U**V**

Eugenol (Isoeugenol, Oil of clove)

(Koi calm) **ESPA**

W

Formulations: Immersion: Oil for dissolution in water.

X

Action: Produces anaesthesia by impeding peripheral nerve signal transmission to the CNS.

Y

Use: For the sedation, immobilization, anaesthesia and euthanasia of fish. Ideally, the drug should be used in water from the tank or pond of origin to minimize problems due to changes in water chemistry.

Z

Over-the-counter preparations contain about 1 g eugenol per ml but are poorly soluble in water and must be made into a stock solution (e.g. 1:9 with 95% ethanol produces 100 mg/ml) for more accurate dosing. The anaesthetic solution should be used on the day of preparation and be well aerated during use. Food should be withheld from fish for 12–24 h before anaesthesia to reduce the risk of regurgitation. The stage of anaesthesia reached is determined by the concentration used and the duration of exposure since absorption continues throughout the period of immersion. Different species vary in their response and may require different concentrations. It is recommended to use the lower dose rates to test the selected drug concentration and exposure time with a small group before medicating large numbers. Fish may retain some movement during anaesthesia, making it less desirable to use during surgery. Anaesthetized fish should be returned to clean water from their normal environment to allow recovery. For euthanasia, use 5–10 times the normal anaesthetic dose and keep the fish in the solution for at least 60 minutes after respiration ceases. Can also be used for sedation and anaesthesia of amphibians but effects variable. Unreliable method of euthanasia for amphibians so not recommended ^a.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Cardiorespiratory depression and death have been noted in some tropical marine fish (e.g. tangs, surgeonfish). Can be associated with reversible gastric prolapse in amphibians.

Drug interactions: No information available.

DOSES

Amphibians: Leopard frogs: 255 mg/l resulted in variable degree of sedation ^b; African clawed frogs: 350 mg/l resulted in anaesthesia after 5–10 minutes immersion ^c; Tiger salamanders: 450 mg/l resulted in anaesthesia in ~10 minutes in 80% of animals ^d.

Fish: Anaesthesia: 40–120 mg/l by immersion; Euthanasia: 500–1000 mg/ml by immersion or undiluted 10 drops/l (whisk to thoroughly mix with water) by immersion ¹. Follow manufacturer's recommendations for proprietary formulations.

Mammals, Birds, Reptiles: No information available.

References

- ^a Galex IA, Gallant CM, D'Avignon N *et al.* (2020) Evaluation of effective and practical euthanasia methods for larval African clawed frogs (*Xenopus laevis*). *Journal of the American Association of Laboratory Animal Science* **59**(3), 269–274
- ^b Lafortune M, Mitchell MA and Smith JA (2001) Evaluation of medetomidine, clove oil and propofol for anesthesia of Leopard Frogs, (*Rana pipiens*). *Journal of Herpetological Medicine and Surgery* **11**(4), 13–18
- ^c Guénette SA, Hélie P, Beaudry F and Vachon P (2007) Eugenol for anesthesia of African clawed frogs (*Xenopus laevis*). *Veterinary Anaesthesia and Analgesia* **34**(3), 164–170
- ^d Mitchell MA, Riggs SM, Singleton CB, Diaz-Figueroa O and Hale LK (2009) Evaluating the clinical and cardiopulmonary effects of clove oil and propofol in tiger salamanders (*Ambystoma tigrinum*). *Journal of Exotic Pet Medicine* **18**(1), 50–56
- ¹ Sneddon LU (2012) Clinical anaesthesia and analgesia in fish. *Journal of Exotic Pet Medicine* **21**, 32–43

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Famciclovir

(Famciclovir*, Famvir*) **POM****B****Formulations:** Oral: 125 mg, 250 mg, 500 mg tablets.**C****Action:** Inhibits viral replication (viral DNA polymerase); depends on viral thymidine kinase for phosphorylation.**D****Use:** Famciclovir is a pro-drug for penciclovir, which is closely related to aciclovir. Famciclovir is virostatic and is unable to eradicate latent viral infection. Has been used experimentally in ducks to control hepatitis B virus infection.**E****F****Safety and handling:** Normal precautions should be observed.**G****Contraindications:** No information available.**H****Adverse reactions:** Little information available. A dose of 90 mg/kg caused no changes in liver enzymes (single dose) but adversely affected the conjunctival goblet cell density and therefore the quality of the tear film in cats.**I****Drug interactions:** No information available.**J**

DOSES

K**Birds:** Ducklings: 10 mg/kg p.o. q24h^a.**L****Reptiles:** 10–30 mg/kg p.o. q24h used to manage ranavirus outbreak in Eastern box turtles^b.**M****Mammals, Amphibians, Fish:** No information available.**N**

References

O^a Lin E, Luscombe C, Colledge D, Wang YY and Locarnini S (1998) Long-term therapy with the guanine nucleoside analog penciclovir controls chronic duck hepatitis B virus infection *in vivo*. *Antimicrobial Agents and Chemotherapy* **42**(8), 2132–2137**P**^b Sim RR, Allender MC, Crawford LK *et al.* (2016) Ranavirus epizootic in captive eastern box turtles (*Terrapene carolina carolina*) with concurrent herpesvirus and mycoplasma infection: management and monitoring. *Journal of Zoo and Wildlife Medicine* **47**(1), 256–270**Q****R**

Famotidine

(Pepcid*) **POM****S****Formulations:** Oral: 20 mg, 40 mg tablets.**T****Action:** Potent histamine (H₂) receptor antagonist blocking histamine-induced gastric acid secretion. It is many times more potent than cimetidine but has poorer oral bioavailability (37%).**U****Use:** Management of gastric and duodenal ulcers, idiopathic, uraemic or drug-related erosive gastritis, oesophagitis, and hypersecretory conditions secondary to gastrinoma, mast cell neoplasia or short bowel syndrome. Reduction of vomiting due to gastric ulceration is typically achieved in about 2 weeks. However, animals should be treated for at least 2 weeks after the remission of clinical signs, so a minimum treatment duration of 28 days is recommended. Currently cimetidine is the only antiulcer drug with a veterinary market authorization. Has little effect on GI motility in humans.**V****X****Safety and handling:** Normal precautions should be observed.**Y****Contraindications:** No information available.**Z****Adverse reactions:** In humans famotidine has fewer side effects than cimetidine.

Drug interactions: Famotidine is devoid of many of the interactions of the H₂ related antagonist cimetidine.

DOSES

Mammals: Ferrets: 0.25–0.5 mg/kg p.o., i.v., s.c. q24h; Rabbits: 0.5 mg/kg p.o., s.c., i.v. q12h; Guinea pigs: 0.4 mg/kg p.o., s.c. q24h; Chinchillas: 0.5 mg/kg s.c. q24h; Primates: 0.5 mg/kg p.o. q24h; Hedgehogs: 0.5 mg/kg p.o. q12h¹, 1 mg/kg s.c. q24h, 0.5 mg/kg p.o. q12h.

Birds, Reptiles, Amphibians, Fish: No information available.

References

- ¹ Hryszyn TM, Malmberg JL and Johnston MS (2019) Diagnosis and clinical management of eosinophilic gastroenteritis in an African pygmy hedgehog (*Atelerix albiventris*). *Journal of Exotic Pet Medicine* **30**, 88–91

Febantel see **Pyrantel**

Fenbendazole

(Granofen, **Lapizole**, **Panacur**, Zerofen, Various authorized proprietary products) **NFA-VPS, ESPA**

Formulations: Oral: 222 mg/g granules (22%); 20 mg/ml oral suspension (2%); 25 mg/ml oral suspension (2.5%); 100 mg/ml oral suspension (10%); 187.5 mg/g oral paste (18.75%).

Action: Inhibits fumarate reductase system of parasites thereby blocking the citric acid cycle and also reduces glucose absorption by the parasite.

Use: Treatment of oxyurids, ascarids (including larval stages), hookworms, whipworms, tapeworms (*Taenia*), *Oslerus osleri*, *Aelurostrongylus abstrusus*, *Angiostrongylus vasorum*, *Capillaria aerophila*, *Ollulanus tricuspis*, *Physaloptera rara* and *Paragonimus kellicotti* infections. Fenbendazole has 100% efficacy in clearing *Giardia* cysts. It is used in rabbits for the treatment of *Encephalitozoon cuniculi*. Unlike some other benzimidazoles, fenbendazole is safe to use in pregnant animals. Used for the treatment of non-encysted gastrointestinal nematodes, some cestodes (*Bothriocephalus*) and external monogenean parasites in fish.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Birds, especially vultures and storks, and some raptors, are more sensitive to adverse reactions affecting bone marrow, intestinal and liver functions. In pigeons and doves, mortality of 50% has occurred at doses of 20 mg/kg p.o. given on 3 consecutive days. Vomiting, depression and death within 96 h are recorded in some raptors. It is very important when using suspension formulations that the suspension is shaken well before each dispensing and each dosing. If not, the contents are likely to settle and accidental overdosage result when using the last doses. Feather damage has also been reported in pigeons. Has been associated with profound leucopenia in *Testudo hermanni* following two courses of 50 mg/kg fenbendazole given daily for 5 days per course. Avoid in reptiles with suspected septicaemia. Some species of fish (e.g. discus) may react adversely to some formulations of

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fenbendazole by immersion resulting in neurological signs and mortality. The sudden death of a large burden of nematodes may cause tissue damage or intestinal blockage.

Drug interactions: No information available.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Mammals: Rabbits: *E. cuniculi*: 20 mg/kg p.o. q24h for 28 days^a; Oxyuriasis: 50 ppm (50 mg/kg) in feed for 5 days^b; **Primates:** 50 mg/kg p.o. q24h for 3–14 days; *Filaroides* and *Strongyloides*: 20 mg/kg p.o. q24h for 14 days; **Sugar gliders:** 20–50 mg/kg p.o. q24h for 3 days, repeat in 14 days; **Hedgehogs:** 10–30 mg/kg p.o. q24h for 5 days; 10–15 mg/kg p.o. q14days for 2–3 treatments; **Other small mammals:** 20–50 mg/kg p.o. q24h for 5 consecutive days; the higher end of the range is suggested for giardiasis only.

Birds:

- Nematodes^c: 20–100 mg/kg p.o., administer 2 doses separated by 10 days; capillariasis: 25 mg/kg p.o. q24h for 5 consecutive days; **Pigeons:** 16 mg/kg p.o. once, repeat after 10 days if necessary or 10–20 mg/kg p.o. q24h for 3 days, repeat after 2 weeks; **Passerines:** 20 mg/kg p.o. q24h for 3 doses. Not advisable to give more than 50 mg/kg in unfamiliar species.
- Giardiasis: 50 mg/kg p.o. q24h for 3 doses^d.

Reptiles:

- Nematodes: 50–100 mg/kg p.o., per cloaca once or 20–25 mg/kg p.o., per cloaca q24h over 3–5 day course^e; Repeated doses after 2 weeks have been advised but may be unnecessary as complete effect may not be seen until 31 days post-treatment^f.
- Giardiasis and flagellates: 50 mg/kg p.o. q24h for 3–5 days.

Amphibians: 100 mg/kg p.o., repeat in 2 weeks.

Fish: External monogenean parasites: 25 mg/l by immersion for 12 h; GI nematodes: 2 mg/l by immersion q7days for 3 doses, 50 mg/kg p.o. q24h for 2 days, repeat in 14 days.

References

- Suter C, Müller-Doblies UU, Hatt JM and Deplazes P (2001) Prevention and treatment of *Encephalitozoon cuniculi* infection in rabbits with fenbendazole. *Veterinary Record* **148**(15), 478–480
- Düwel D and Brech K (1981) Control of oxyuriasis in rabbits by fenbendazole. *Laboratory Animal* **15**(2), 101–105
- Lawrence K (1983) Efficacy of fenbendazole against nematodes of captive birds. *Veterinary Record* **112**, 433–434
- Yazwinski TA, Andrews P, Holtzen H et al. (1986) Dose-titration of fenbendazole in the treatment of poultry nematodiasis. *Avian Diseases* **30**(4), 716–718
- Holt PE (1982) Efficacy of fenbendazole against the nematodes of reptiles. *Veterinary Record* **110**, 302–304
- Giannetto S, Brianti E, Poglayen G et al. (2007) Efficacy of oxfendazole and fenbendazole against tortoise (*Testudo hermanni*) oxyurids. *Parasitology Research* **100**, 1069–1073
- Noga EJ (2010) *Fish Disease – Diagnosis and Treatment, 2nd edn*. Wiley-Blackwell, Oxford

Fentanyl

(Fentadon, Durogesic*, Fentanyl*, Fentora*, Sublimaze*)

POM-V CD SCHEDULE 2, POM CD SCHEDULE 2

Formulations: Oral: 100 µg (micrograms), 200 µg, 400 µg, 600 µg, 800 µg tablets. **Injectable:** 50 µg/ml solution. **Transdermal:** 12.5 µg/h, 25 µg/h, 50 µg/h, 75 µg/h, 100 µg/h patches.

Action: Synthetic pure mu (OP3) receptor agonist.

Use: Very potent opioid analgesic (50 times more potent than morphine) used to provide profound intraoperative analgesia. Can also be used at low dose rates for postoperative analgesia. Use of potent opioids during anaesthesia contributes to a balanced anaesthesia technique, therefore the dose of other concurrently administered anaesthetic agents should be reduced. Fentanyl has a rapid onset of action after i.v. administration and short duration of action (10–20 min depending on dose). After prolonged administration (>4 hours) or high doses its duration of action is significantly prolonged as the tissues become saturated. It can be used intraoperatively to provide analgesia by intermittent bolus doses or by a continuous rate infusion. Postoperatively fentanyl can be given by continuous rate infusion to provide analgesia, doses at the low end of the dose range should be used and respiratory function monitored. Its clearance is similar to morphine whilst its elimination half-life is longer, reflecting its higher lipid solubility and volume of distribution. There appear to be significant differences in pharmacokinetics between different avian species so doses should not be extrapolated across species.

Safety and handling: If animals are sent home with transdermal fentanyl patches the owners must be warned about the dangers of patch ingestion by humans or other animals.

Contraindications: No information available.

Adverse reactions: Intraoperative administration is likely to cause respiratory depression, therefore, respiration should be monitored and facilities must be available to provide positive pressure ventilation. Rapid i.v. injection can cause severe bradycardia, even asystole, therefore the drug should be given slowly. A reduction in heart rate is likely whenever fentanyl is given, atropine can be administered to counter bradycardia if necessary. Apart from the effects on heart rate, fentanyl has limited other effects on cardiovascular function when used at clinical dose rates.

Drug interactions: Fentanyl can be used to reduce the dose requirement for other anaesthetic drugs in patients with cardiovascular instability or systemic disease.

DOSES

Mammals: **Ferrets:** 10–30 µg (micrograms)/kg/h i.v. CRI, reduce to 1.25–5 µg/kg/h for postoperative analgesia; **Rabbits:** 7.5 µg/kg q20min; 30–100 µg/kg/h by CRI during anaesthesia; **Primates:** 5–10 µg/kg i.v. or 10–25 µg/kg/h CRI^a, 0.05–0.15 µg/kg i.m. as needed; **Hedgehogs:** 0.1 mg/kg i.m. in sedation combination.

Birds: **Red-tailed hawks:** 20 µg (micrograms) bolus i.v. followed by 0.2–0.5 µg/kg/min i.v. infusion reduced isoflurane requirement without effect on heart rate or blood pressure^b; **White (umbrella) cockatoos:** 0.02 mg/kg i.m. produced no analgesia but was well tolerated; 0.2 mg/kg

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produced some analgesia but signs of hyperactivity in some birds in first 15–30 min after administration^c. Anaesthetized Amazon parrots given fentanyl at 20 µg i.v. exhibited a dose-dependent isoflurane-sparing effect, but was accompanied by significant bradycardia and hypotension^d.

Reptiles: 12 µg (micrograms)/h transdermal fentanyl patch results in plasma concentrations above analgesic threshold in mammals, when tested in ball pythons^e and prehensile-tailed skinks^f. Analgesic efficacy not established, although anecdotally may be useful in snakes.

Amphibians, Fish: No information available.

References

- ^a Valverde CR, Mama KR, Kollias-Baker C, Steffey EP and Baggot JD (2000) Pharmacokinetics and cardiopulmonary effects of fentanyl in isoflurane-anaesthetized rhesus monkeys (*Macaca mulatta*). *American Journal of Veterinary Research* **61**(8), 931–934
- ^b Pavez JC, Hawkins MG, Pascoe PJ, Knych HKD and Kass PH (2011) Effect of fentanyl target-controlled infusions on isoflurane minimum anaesthetic concentration and cardiovascular function in red-tailed hawks (*Buteo jamaicensis*). *Veterinary Anaesthesia and Analgesia* **38**(4), 344–351
- ^c Hoppes S, Flammer K, Hoersch K, Papich M and Paul-Murphy J (2003) Disposition and analgesic effects of fentanyl in white cockatoos (*Cacatua alba*). *Journal of Avian Medicine and Surgery* **17**(3), 124–131
- ^d Hawkins MG, Pascoe PJ, Knych HKD *et al.* (2018) Effects of three fentanyl plasma concentrations on the minimum alveolar concentration of isoflurane in Hispaniolan Amazon parrots (*Amazona ventralis*). *American Journal of Veterinary Research* **79**(6), 600–605
- ^e Kharbush RJ, Gutwillig A, Hartzler KE *et al.* (2017) Antinociceptive and respiratory effects following application of transdermal fentanyl patches and assessment of brain µ-opioid receptor mRNA expression in ball pythons. *American Journal of Veterinary Research* **78**(7), 785–795
- ^f Gamble KC (2008) Plasma fentanyl concentrations achieved after transdermal fentanyl patch application in prehensile-tailed skinks, *Corucia zebrata*. *Journal of Herpetological Medicine and Surgery* **18**(3), 81–85

Fentanyl/Fluanisone

(Hypnorm) POM-V CD SCHEDULE 2

Formulations: Injectable: 0.2 mg/ml fentanyl with 10 mg/ml fluanisone.

Action: Fentanyl is a pure mu opioid agonist and fluanisone is a butyrophenone; the combination produces neuroleptanalgesia.

Use: Authorized for use in rabbits, guinea pigs, mice and rats for anaesthesia. Provides sedation and analgesia for restraint and to allow minor procedures to be carried out in these species and can also be used in primates. Combined with a benzodiazepine it can be used to produce anaesthesia with muscle relaxation for surgery. It can be used for premedication prior to induction of anaesthesia with propofol or alfaxalone in rabbits. Fentanyl/fluanisone produces a long duration of sedation and analgesia in small mammals (approximately 30–60 min), the duration of action is dependent on dose and varies between species. When used in combination with a benzodiazepine to produce anaesthesia, the lowest end of the dose range of fentanyl/fluanisone should be used. Hypnorm is miscible with midazolam at appropriate dilutions, enabling a single injection to produce anaesthesia. Duration of anaesthesia is approximately 20–40 min, although recovery can be very prolonged. Various dose regimens for different species have been published. Prolonged respiratory depression in the recovery period can

be avoided by the administration of a partial mu agonist (buprenorphine) at the end of the procedure. This will reverse respiratory depression induced by fentanyl and provide continued analgesia. Small mammals should be weighed before drug administration, accurate dosing is imperative to prevent overdose. Oxygen supplementation via a face mask is recommended during sedation and anaesthesia of all animals.

Safety and handling: Normal precautions should be observed.

Contraindications: Animals with pre-existing respiratory compromise.

Adverse reactions: Respiratory depression can occur when given in high doses, particularly during the recovery period. Administration of buprenorphine in the recovery period can ameliorate respiratory depression and sedation, and provide ongoing analgesia. More complete antagonism of the fentanyl portion may be achieved using naloxone, but this removes any analgesia that was provided by the fentanyl. Measures should be taken to maintain normothermia during the sedation/ anaesthesia and recovery period.

Drug interactions: No information available.

DOSES

When used for sedation is generally given as part of a combination.

See Appendix for sedation protocols in all species.

Mammals: Rabbits: 0.1–0.5 ml/kg i.m.; Guinea pigs: 0.5–1 ml/kg i.m.; Rats: 0.2–0.5 ml/kg i.m., i.p.; Mice: 0.1–0.5 ml/kg i.p.; Primates: 0.1–0.3 ml/kg i.m.^a. Doses at the highest 50% of these ranges are associated with extremely long recoveries and potentially hypothermia. Reversal with partial or complete opioid antagonists may be required in such cases. Doses for sick animals should start at the lowest end of the range.

Birds, Reptiles, Amphibians, Fish: No information available.

References

- ^a Bertrand HG, Sandersen CF and Flecknell PA (2017) Preliminary assessment of midazolam, fentanyl and fluanisone combination for the sedation of rhesus macaques (*Macaca mulatta*). *Laboratory Animals* **51**(4), 405–411

Finasteride

(Proscar*) **POM**

Formulations: Oral: 5 mg tablet.

Action: Competitively inhibits dihydrotestosterone (DHT) production within the prostate. DHT is the main hormonal stimulus for the development of benign prostatic hyperplasia.

Use: Treatment of benign prostatic hyperplasia associated with adrenal gland disease.

Safety and handling: People who are or may become pregnant should not handle this drug.

Contraindications: Do not use in breeding ferrets.

Adverse reactions: Secreted into semen and causes fetal anomalies.

Drug interactions: No information available.

DOSES

Mammals: Ferrets: 5 mg/kg p.o. q24h.

Birds, Reptiles, Amphibians, Fish: No information available.

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Fipronil

(Amflee Combo Spot-on Solution for Cats and Ferrets, Bob Martin Clear Plus Spot-on Solution for Cats and Ferrets, Chanonil Plus Spot-on Solution for Cats and Ferrets, Effipro, Eziflea Plus Spot-on Solution for Cats and Ferrets, FiprocLEAR Combo Spot-on Solution, Fipronil, Fiprosport, Fleanil Duo Spot-on Solution for Cats and Ferrets, Fleatix plus; Frontline Combo Spot On Cat, Frontline Plus Spot On Cat, Fyberix Combo Spot-on Solution for Cats and Ferrets, PestiGon Combo Spot-on Solution for Cats and Ferrets, Zeronil Plus; Zerotal Plus Spon-on Solution for Cats and Ferrets) **NFA-VPS, POM-V**

Formulations: Topical: 10% w/v fipronil spot-on pipettes in a wide range of sizes (Frontline and many others). Those formulations authorized for ferrets contain 50 mg S-methoprene with 60 mg fipronil (Frontline Combo/Plus and others); pyriproxyfen (Effipro Duo); permethrin (Frontect); eprinomectin, S-methoprene and praziquantel (Broadline). Also 0.25% w/v fipronil spray in alcohol base (Effipro and Frontline sprays) in a range of sizes.

Action: Fipronil interacts with ligand-gated (GABA) chloride channels, blocking pre- and postsynaptic transfer of chloride ions, resulting in death of parasites on contact.

Use: Authorized in ferrets for treatment of flea and tick infestations (Amflee Combo, Bob Martin Clear Plus, Chanonil Plus, Eziflea Plus, FiprocLEAR Combo, Fleanil Duo, Fleatix Plus, Frontline Combo, Frontline Plus, Fyberix Combo, PestiGon Combo, Zeronil Plus, Zerotal Plus). Can also be used for treatment of fleas, lice and ticks in other small mammals (except rabbits) and birds. Treatment of mites and ticks in reptiles. Has been used to treat *Hirstiella* mites in green iguanas, but still needs further safety evaluation in other species. Beware of use in juvenile reptiles and immediately after skin slough due to increased permeability of skin and associated toxicity. Treatment of the environment is also recommended. Care with overuse of alcohol-based spray in birds^a.

Safety and handling: Normal precautions should be observed.

Contraindications: **Do not use in rabbits.** Beware of use in debilitated reptiles, those which have recently shed their skin and in small species where overdosage and toxicity may occur. May be harmful to aquatic organisms.

Adverse reactions: Local pruritus or alopecia may occur at the site of application.

Drug interactions: A low dose of amitraz has been shown to have a synergistic effect on the speed of tick kill, thus reducing the risk of transmission for tick-borne pathogens.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Mammals: Ferrets: Spray 3–6 ml/kg (6–12 pumps/kg 100 ml application) q30–60d; One spot-on pipette of 0.5 ml/animal (50 mg fipronil, 60 mg S-methoprene) applied topically to back of neck; **Rodents:** 7.5 mg/kg

topically (15 pumps/kg 100 ml application) q30–60d. **Other mammals:** apply lightly q14d. Do not use in rabbits.

Birds: Use spray form q30–60d. Apply to cotton wool and dab behind head, under wings and at base of tail (raptors/parrots) or lightly under each wing (pigeons/passerines). 1 drop 10% spot on formulation applied to the skin of the neck of budgerigars on days 0, 7 and 14 effective against *Cnemidoptes pilae* infestation^b.

Reptiles: Spray on to cloth first then wipe over surface of reptile before washing off after 5 minutes. Apply q7–14d until negative for ectoparasites.

Amphibians, Fish: Not indicated.

References

- ^a Kitulagodage M, Astheimer LB and Buttemer WA (2008) Diacetone alcohol, a dispersant solvent, contributes to acute toxicity of a fipronil-based insecticide in a passerine bird. *Ecotoxicology and Environmental Safety* **71**(2), 597–600
- ^b Akhtar S, Durrani UF, Mahmood AK et al. (2021) Comparative efficacy of ivermectin and fipronil spot on against *Cnemidoptes pilae* in budgerigars. *Indian Journal of Animal Research* **55**(1), 105–108

Florfenicol

(Florocol, Nuflor) **POM-V**

Formulations: Injectable: 300 mg/ml, 450 mg/ml solution.

Oral: 500 mg/g powder.

Action: Bacteriostatic antimicrobial that acts by binding to the 50S ribosomal subunit of susceptible bacteria, preventing bacterial protein synthesis.

Use: Treatment of bacterial diseases in fish.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: No information available.

Drug interactions: No information available.

DOSES

Classified as category C (Caution) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Fish: 25–50 mg/kg i.m., intracoelomic q24h or 5–50 mg/kg p.o. q24h for 10 days^{a,b}.

Mammals, Birds, Reptiles, Amphibians: No information available.

References

- ^a Yanong RP, Curtis EW, Simmons R et al. (2005) Pharmacokinetic studies of florfenicol in koi carp and threespot gourami (*Trichogaster trichopterus*) after oral and intramuscular treatment. *Journal of Aquatic Animal Health* **17**(2), 129–137
- ^b Lewbart G, Papich MG and Whitt-Smith D (2005) Pharmacokinetics of florfenicol in the red pacu (*Piaractus brachipomum*) after single dose intramuscular injection. *Journal of Veterinary Pharmacology and Therapeutics* **28**, 317–319

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Flubendazole

(Anti Fluke & Wormer, Flimabend, Flubenvet, Flukasol Suspension, Fluke-M) **ESPA**

Formulations: **Immersion:** 1% powder (Flubenvet) <0.8% liquid (Flukasol Suspension), 1% liquid (Anti Fluke & Wormer), 5% powder (Fluke-M, Flubenvet); 100 mg/ml suspension (Flimabend).

Action: Benzimidazoles bind to a structural protein of the parasite microtubules, important organelles involved in major cell processes, causing paralysis, death and expulsion of the parasites.

Use: Treatment of monogenetic trematode ectoparasites (e.g. skin and gill flukes) and internal nematodes in fish. Used to kill snails and hydra in aquaria. Flubendazole has a low solubility but a high potency.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: No information available.

Drug interactions: No information available.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Fish: 2 mg/l by prolonged immersion once and change at least 30% of water after 6 days. Follow manufacturer's recommendations for proprietary formulations.

Mammals, Birds, Reptiles, Amphibians: No information available.

Flucloxacillin

(Floxapen*, Flucloxacillin*) **POM**

Formulations: **Injectable:** Flucloxacillin sodium: 250 mg, 500 mg, 1 g powders for reconstitution. **Oral:** Flucloxacillin sodium: 250 mg or 500 mg capsules; powder for reconstitution with water giving a final concentration of 125 mg/5 ml, 250 mg/5 ml. Formulations of flucloxacillin with ampicillin are available (Co-fluampicil, Magnapen).

Action: Beta-lactamase-resistant, narrow-spectrum beta-lactam antibiotic. It binds to penicillin-binding proteins, decreasing bacterial cell wall strength and rigidity, and affecting cell division, growth and septum formation. Flucloxacillin is bactericidal and works in a time-dependent fashion.

Use: Stable in gastric acid so can be given orally but food significantly reduces its bioavailability. It is less active than penicillin G or V against *Streptococcus* and obligate anaerobic bacteria, and is indicated for the treatment of infections caused by beta-lactamase-producing *Staphylococcus*. Patients with significant renal or hepatic dysfunction may need dosage adjustment. The amount of sodium in flucloxacillin sodium may be clinically important for patients on restricted sodium intakes. Although flucloxacillin is absorbed from the GI tract, food has a significant inhibitory effect on its bioavailability; doses must be given on an empty stomach therefore limiting potential use in rabbits and rodents. As flucloxacillin kills in a time-dependent fashion, dosing regimens should be designed to maintain tissue concentrations above

the MIC throughout the interdosing interval. Use with care in hepatic disease or hepatic impairment.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not administer to animals with a history of sensitivity to beta-lactam antimicrobials. Use of penicillins should be avoided in rabbits, other small herbivores, hamsters and gerbils especially via the oral route.

Adverse reactions: Nausea, diarrhoea and skin rashes are the commonest adverse effects. Cholestatic hepatitis has been reported in humans. Avoid oral administration in small herbivores (e.g. rabbits), hamsters and gerbils as penicillins can cause fatal enterotoxaemia.

Drug interactions: Avoid the concomitant use of bacteriostatic antibiotics. The aminoglycosides (e.g. gentamicin) may inactivate penicillins when mixed together in parenteral solutions. A synergistic effect is seen when beta-lactam and aminoglycoside antimicrobials are used concurrently.

DOSES

Classified as category D (Prudence) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Rats: 200 mg/kg s.c. q8h^a.

Birds, Reptiles, Amphibians, Fish: No information available.

References

- ^a Gisby J, Beale AS, Bryant JE, Toseland CD (1994) *Staphylococcal osteomyelitis* – a comparison of co-amoxiclav with clindamycin and flucloxacillin in an experimental rat model. *Journal of Antimicrobial Chemotherapy* **34**(5), 755–764

Fluconazole

(Diflucan*, Fluconazole*) **POM**

Formulations: Oral: 50 mg, 150 mg, 200 mg capsules; 40 mg/ml suspension. **Injectable:** 2 mg/ml solution.

Action: Inhibition of the synthesis of ergosterol in fungal cell membranes, thus causing increased cell wall permeability and allowing leakage of cellular contents.

Use: Effective against *Blastomyces*, *Candida*, *Cryptococcus*, *Coccidioides*, *Histoplasma* and *Microsporium canis* infections and variably effective against *Aspergillus* and *Penicillium* infections. Has been used for treatment of dermatophytosis in green iguanas. In amphibians, it prolonged survival in frogs with chytridiomycosis but did not prevent mortality. It attains therapeutic concentrations in the CNS and respiratory tract. It is excreted by the kidney, producing high concentrations in urine. Reduce dose in animals with renal impairment and liver disease. This drug should be used until clinical signs have resolved and the organism is no longer present; this may take up to 2 months in some cases.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use in pregnant/lactating animals.

Adverse reactions: Adverse effects may include nausea and diarrhoea. May be hepatotoxic. May cause vomiting in birds.

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Drug interactions: Fluconazole (due to inhibition of cytochrome P450-dependent liver enzymes) may increase plasma theophylline concentrations. In humans, fluconazole has led to terfenadine toxicity when the two drugs were administered together. Fluconazole increases ciclosporin blood levels.

DOSES

Mammals: Rabbits: 25–43 mg/kg i.v. infusion (slow) q12h; 37.5 mg/kg p.o. q12h for *Aspergillus* keratitis; 5 mg/kg p.o. q24h for cryptococcal meningitis.

Birds: 10–20 mg/kg p.o. q12–48h^{a,1}.

Reptiles: 5 mg/kg p.o. q24h (iguanas with dermatophytosis).

Amphibians: 60 mg/kg p.o. q24h. Baths at 25 mg/l prolonged survival in frogs with chytridiomycosis but did not prevent mortality^b.

Fish: No information available.

References

- ^a Flammer K and Papich M (2006) Pharmacokinetics of fluconazole after oral administration of single and multiple doses in African grey parrots. *American Journal of Veterinary Research* **67**(3), 417–422
- ^b Berger L, Speare R, Marantelli G and Skerratt LF (2009) A zoospore inhibition technique to evaluate the activity of antifungal compounds against *Batrachochytrium dendrobatidis* and unsuccessful treatment of experimentally infected green tree frogs (*Litoria caerulea*) by fluconazole and benzalkonium chloride. *Research in Veterinary Science* **87**(1), 106–110
- ¹ Donnelly KA, Wellehan JF and Quesenberry K (2019) Gastrointestinal disease associated with non-*albicans* *Candida* species in six birds. *Journal of Avian Medicine and Surgery* **33**(4), 413–418

Fludrocortisone

(Fludrocortisone acetate*) **POM**

Formulations: Oral: 0.1 mg tablets.

Action: Aldosterone analogue that increases potassium excretion and sodium retention but which also has some glucocorticoid properties.

Use: Treatment of adrenocortical insufficiency (post-adrenalectomy). Fludrocortisone is about 125 times more potent as a mineralocorticoid than is hydrocortisone but it is also about 12 times more potent as a glucocorticoid (and therefore about 3 times more potent than prednisolone). Monitor sodium and potassium concentrations separately (not just the ratio) 4–6 hours post-pill. Supplemental doses of prednisolone may be required at times of metabolic or physical stress.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Hypertension, oedema (including cerebral oedema) and hypokalaemia with overdoses. Long-term overdose may result in clinical signs of hypercortisolism.

Drug interactions: Hypokalaemia may develop if fludrocortisone is administered concomitantly with amphotericin B or potassium-depleting diuretics (furosemide, thiazides).

DOSES

Mammals: Ferrets: post-adrenalectomy: 0.05–0.1 mg/kg p.o. q24h.

Birds, Reptiles, Amphibians, Fish: No information available.

Flumazenil

(Flumazenil*) **POM**

Formulations: Injectable: 0.1 mg/ml.

Action: Benzodiazepine antagonist. Flumazenil binds to and rapidly displaces benzodiazepines from the benzodiazepine receptor, thereby reversing their sedative and anxiolytic effects within 1–2 minutes.

Use: For the complete or partial reversal of the central sedative effects of benzodiazepines. The duration of action of flumazenil is short (<1 hour in mammals) therefore dosing may need to be repeated.

Safety and handling: Normal precautions should be observed.

Contraindications: In birds, precise dosing required as cardiac dysrhythmias may occur with overdosage

Adverse reactions: In humans, palpitations have been reported following rapid injection.

Drug interactions: No information available.

DOSES

Mammals: Rabbits: 0.01–0.1 mg/kg i.m., i.v. (anecdotally starting at the middle of the dose range appears effective); **Chinchillas:** 0.1 mg/kg s.c.^a; **Primates:** 0.02 mg/kg i.v.; **Sugar gliders, Hedgehogs:** 0.05 mg/kg s.c., i.m.^b.

Birds: Most species: 0.02–0.1 mg/kg i.m., i.v.; **Amazon parrots:** 0.05 mg/kg intranasally^c; **Parakeets:** 0.13 mg/kg intranasally; **Canaries:** 0.25–0.31 mg/kg intranasally; **Zebra finches:** 0.3 mg/kg i.m.^d.

Reptiles: Most species: 0.05 mg/kg i.m., s.c., i.v.; **Ball pythons:** 0.08 mg/kg s.c. reversed effects of midazolam but re-sedation occurred ~3 hours later^e.

Amphibians: Blue poison dart frogs: 0.05 mg/kg s.c.^f.

Fish: No information available.

References

- ^a Henke J, Baumgartner C, Röltgen I, Eberspächer E and Erhardt W (2004) Anaesthesia with midazolam/medetomidine/fentanyl in chinchillas (*Chinchilla lanigera*) compared to anaesthesia with xylazine/ketamine and medetomidine/ketamine. *Journal of Veterinary Medicine Series A* **51(5)**, 259–264
- ^b Hausmann KE, Doss GA and Mans C (2021) Subcutaneous tiletamine-zolazepam immobilization and effect of flumazenil reversal in African pygmy hedgehogs (*Atelerix albiventris*). *Journal of Exotic Pet Medicine* **36**, 42–46
- ^c Mans C, Guzman DSM, Lahner LL, Paul-Murphy J and Sladky KK (2012) Sedation and physiologic response to manual restraint after intranasal administration of midazolam in Hispaniolan Amazon parrots (*Amazona ventralis*). *Journal of Avian Medicine and Surgery* **26(3)**, 130–140
- ^d Prather JF (2012) Rapid and reliable sedation induced by diazepam and antagonized by flumazenil in zebra finches (*Taeniopygia guttata*). *Journal of Avian Medicine and Surgery* **26(2)**, 76–85
- ^e Larouche CB, Beaufrère H, Mosley C, Nemeth NM and Dutton C (2019) Evaluation of the effects of midazolam and flumazenil in the ball python (*Python regius*). *Journal of Zoo and Wildlife Medicine* **50(3)**, 579–588
- ^f Yaw TJ, Mans C, Martinelli L and Sladky KK (2020) Comparison of subcutaneous administration of alfaxalone–midazolam–dexmedetomidine with ketamine–midazolam–dexmedetomidine for chemical restraint in juvenile blue poison dart frogs (*Dendrobates tinctorius azureus*). *Journal of Zoo and Wildlife Medicine* **50(4)**, 868–873

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Fluoxetine

(Reconcile, Prozac*) **POM-V, POM**

Formulations: Oral: 8 mg, 16 mg, 20 mg, 32 mg, 64 mg tablets; 4 mg/ml liquid.

Action: Fluoxetine and its primary metabolite norfluoxetine block serotonin re-uptake in the brain, resulting in antidepressive activity and a raising in motor activity thresholds. It also has minor noradrenaline re-uptake inhibition properties.

Use: Has been used to control feather picking and other compulsive type behaviours in psittacines but relapse following discontinuation of therapy is common. Should be used with a specifically constructed behaviour modification plan and with caution.

Safety and handling: Normal precautions should be observed.

Contraindications: Known sensitivity to fluoxetine or other SSRIs, history of seizures.

Adverse reactions: Common reactions include lethargy, decreased appetite and vomiting, which may result in minor weight loss. Trembling, restlessness and other GI disturbances may also occur and must be distinguished from a paradoxical increase in anxiety which has been reported in some cases. Owners should be warned of a potential increase in aggression in response to medication.

Drug interactions: Fluoxetine should not be used within 2 weeks of treatment with an MAOI (e.g. selegiline) and an MAOI should not be used within 6 weeks of treatment with fluoxetine. Fluoxetine, like other SSRIs, antagonizes the effects of anticonvulsants and so is not recommended for use with epileptic patients or in association with other agents which lower seizure threshold, (e.g. phenothiazines). Caution is warranted if fluoxetine is used concomitantly with aspirin or other anticoagulants since the risk of increased bleeding in the case of tissue trauma may be increased.

DOSES

Mammals: Rats: 1–1.5 mg/kg p.o. q24h; Primates: 0.45–2.0 mg/kg p.o. q24h^a; 10 mg/kg p.o.^b **Sugar gliders:** 0.5–1 mg/kg p.o. q24h^c, 2–5 mg/kg p.o. q12h^d;

Birds: 1–4 mg/kg p.o. q24h.

Reptiles, Amphibians, Fish: No information available.

References

- ^a Fontenot MB, Musso MW, McFatter RM and Anderson GM (2009) Dose finding study of fluoxetine and venlafaxine for the treatment of self-injurious and stereotypic behaviour in rhesus macaques (*Macaca mulatta*). *Journal of the American Association for Laboratory Animal Science* **48**(2), 176–184
- ^b Sawyer EK and Howell LL (2011) Pharmacokinetics of fluoxetine in rhesus macaques following multiple routes of administration. *Pharmacology* **88**(1–2), 44–49
- ^c Desmarchelier MR (2021) Clinical Psychopharmacology for the Exotic Animal Practitioner. *Veterinary Clinics: Exotic Animal Practice* **24**(1), 17–35
- ^d Johnson-Delaney CA (2006) Practical marsupial medicine. In *Proceeding of the American Exotic Mammals Veterinarian Conference*. Bonita Springs (FL) pp. 51–60

Fluralaner

(Bravecto, Bravecto spot-on, Bravecto plus spot-on solution for cats, Exzolt Oral Solution for Use in Drinking Water for Chickens) **POM-V**

Formulations: Oral: 112.5 mg, 250 mg, 500 mg, 1000 mg, 1400 mg chewable tablets; 10 mg/ml oral solution (Exzolt) **Topical spot-ons for dogs and cats:** 280 mg/ml fluralaner, 3 sizes cats, 5 sizes dogs. **Topical spot-on for cats:** 280 mg/ml fluralaner and 14 mg/ml moxidectin, 3 sizes.

Action: Fluralaner acts at ligand-gated chloride channels, in particular those gated by the neurotransmitter GABA, thereby blocking pre- and post-synaptic transfer of chloride ions across cell membranes.

Use:

- Treatment of fleas and ticks.
- Treatment of ear mite (*Otodectes cynotis*).
- Treatment of *Demodex* spp.
- Treatment of *Caparinia tripilis*.
- Treatment of *Sarcoptes scabiei*.

Onset of action: Fleas within 8 hours and ticks within 12 hours (*Ixodes ricinus*). Immediate and persistent tick killing activity for 12 weeks for *Ixodes ricinus*, *Dermacentor reticulatus* and *D. variabilis*, 8 weeks for *Rhipicephalus sanguineus*. For reduction of the risk of infection with *Babesia canis*. Parasites need to start feeding on the host to become exposed to fluralaner; therefore, the risk of the transmission of parasite-borne diseases cannot be excluded. Can be used in breeding, pregnant and lactating animals.

Safety and handling: Normal precautions should be observed.

Contraindications: Use with caution in animals with pre-existing epilepsy.

Adverse reactions: Unknown in exotic pets. In dogs, administration of the oral product has resulted in mild GI signs or rarely convulsions and lethargy. Administration of spot-on products may result in erythema and alopecia at application site. In cats, apathy, tremors, anorexia, vomiting and hypersalivation have been reported.

Drug interactions: Fluralaner is highly bound to plasma proteins and might compete with other highly bound active substances such as NSAIDs and the coumarin derivative warfarin.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Mammals: Rabbits, Hamsters: 25 mg/kg p.o. once^{a,b,c,1}. Primates: 30–35 mg/kg p.o. once^d. Hedgehogs: 15 mg/kg p.o. once².

Reptiles: Ball pythons: 40 mg/kg p.o. once^e.

Birds, Amphibians, Fish: No information available.

References

- Sheinberg G, Romero C, Heredi R *et al.* (2017) Use of oral fluralaner for the treatment of *Psoroptes cuniculi* in 15 naturally infested rabbits. *Veterinary Dermatology* **28**(4), 393–e91
- d'Ovidio D and Santoro D (2021) Efficacy of fluralaner in the treatment of sarcoptic mange (*Sarcoptes scabiei*) in 12 pet rabbits. *Topics in Companion Animal Medicine* **43**, 100528
- Singh SK, Jaiswa AK, Kumari S *et al.* (2022) Therapeutic effects of oral fluralaner in pet

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rabbits with severe sarcoptic mange (*Sarcoptes scabiei*). *Veterinary Parasitology* **304**, 109693

- ^d Churgin SM, Lee FK, Groenvold K *et al.* (2018) Successful treatment of generalized demodicosis in red-handed tamarins (*Saguinus midas*) using a single administration of oral fluralaner. *Journal of Zoo and Wildlife Medicine* **49(2)**, 470–474
- ^e Gobble J (2022) Oral fluralaner (Bravecto®) use in the control of mites in 20 ball pythons (*Python regius*). *Journal of Herpetological Medicine and Surgery* **32(2)**, 111–112
- ¹ Brosseau G (2020) Oral fluralaner as a treatment for *Demodex aurati* and *Demodex criceti* in a golden (Syrian) hamster (*Mesocricetus auratus*). *The Canadian Veterinary Journal* **61(2)**, 135
- ² Romero C, Sheinberg Waisburd G, Pineda J *et al.* (2017) Fluralaner as a single dose oral treatment for *Caparinia tripilis* in a pygmy African hedgehog. *Veterinary Dermatology* **28(6)**, 622–e152

Flurbiprofen

(Ocufer*) **POM**

Formulations: Ophthalmic: 0.03% solution in single-use vials.

Action: COX inhibitor that reduces the production of prostaglandins and therefore reduces inflammation.

Use: Before cataract surgery. It is also useful for anterior uveitis and ulcerative keratitis when topical corticosteroids are contraindicated. Topical NSAIDs have the potential to increase intraocular pressure and should be used with caution in animals predisposed to glaucoma.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: As with other topical NSAIDs, flurbiprofen may cause local irritation. Topical NSAIDs can be used in ulcerative keratitis but with caution as they can delay epithelial healing. Topical NSAIDs have been associated with an increased risk of corneal 'melting' (keratomalacia) in humans, although this has not been reported in the veterinary literature. Regular monitoring is advised.

Drug interactions: Ophthalmic NSAIDs may be used safely with other ophthalmic pharmaceuticals, although concurrent use of drugs which adversely affect the corneal epithelium (e.g. gentamicin) may lead to increased corneal penetration of the NSAID. The concurrent use of topical NSAIDs with topical corticosteroids has been identified as a risk factor in humans for precipitating corneal problems.

DOSES

Mammals, Birds, Reptiles: 1 drop per eye q6–12h depending on severity of inflammation. 1 drop q30min for 4 doses preoperatively (presurgical protocols vary widely).

Amphibians, Fish: No information available.

Fluticasone

(Flixotide*) **POM**

Formulations: Inhalational: 50 µg (micrograms), 125 µg, 250 µg metered inhalations (Evohaler).

Action: Binds to specific nuclear receptors and affects gene transcription such that many aspects of inflammation are suppressed.

Use: Used as an inhaled corticosteroid in the management of inflammatory airway disease. Administer via commercially available chambers and masks specifically designed for veterinary use. Not useful for acute bronchospasm (and cases of fluticasone-induced bronchospasm have been reported in humans).

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Inhaled steroids are known to suppress the hypothalamic-pituitary-adrenal axis, although they are considered generally safer than systemic steroids.

Drug interactions: No information available.

DOSES

Mammals: Rabbits: 50–250 µg (micrograms)/rabbit q12–24h via feline spacer chamber administration.

Birds, Reptiles, Amphibians, Fish: No information available.

Formaldehyde (Formalin, Formol, Methanediol, Methanol)

(FMG Mixture, Formaldehyde 30% Solution) **ESPA**

Formulations: **Immersion:** proprietary formulations are available (Formaldehyde 30% Solution); also available in combination with malachite green (FMG Mixture).

Action: A biocide causing cell death by cross-linking with proteins.

Use: Treatment of protozoan and some monogenetic ectoparasites of fish. It has some effect on fungal infections of fish eggs (water mould) but little activity against most bacteria. Formalin is an aqueous solution of 37–40% formaldehyde gas (= 100% formalin); diluted solutions are described according to the content of formalin, not formaldehyde. It is strongly recommended that a proprietary formulation is used initially to avoid problems related to purity and to enable accurate dosing. Dilute the measured dose 1:100 before adding to the fish tank or pond. Formalin can be irritating to the gills and chemically removes dissolved oxygen from the water, thus extra aeration must be provided. Do not use in water >27°C. It is usually supplied with 12–15% methanol to stabilize the solution and prevent the formation of paraformaldehyde, which appears as a white precipitate and is toxic to fish. Toxicity is increased at higher temperatures, low pH and low water hardness. Formaldehyde is toxic to some species of elasmobranchs (e.g. sharks, rays), scaleless fish, fry, invertebrates, plants and algae. Considered to have a synergistic effect with malachite green against protozoan ectoparasites. Has also been used for management of multiple protozoan ectoparasites in a research colony of axolotls, although did not fully eliminate all protozoa.

Safety and handling: Formalin must be handled with care; wear protective gloves and use a respirator. It is volatile and irritating, carcinogenic and may cause contact hypersensitivity to skin and airways. Use in well-ventilated areas. Should be stored in dark bottles above 4°C to prevent paraformaldehyde formation.

Contraindications: Do not mix with potassium permanganate since this may result in fire or explosion.

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Adverse reactions: Fish gasping at the surface in water with low dissolved oxygen.

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Drug interactions: No information available.

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DOSES

See Appendix for guidelines on responsible parasiticide use.

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Amphibians: Axolotls: An 8 h static immersion bath containing 0.025 ml/l 37% formaldehyde, followed by a second treatment of an 8 h immersion bath containing 0.05 ml/l 37% formaldehyde on 2 consecutive days^a.

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Fish: All doses based on full strength formalin (= 37% formaldehyde): 0.125–0.25 ml/l by immersion for 30–60 min q24h for 2–3 days or 0.015–0.025 ml/l by prolonged immersion q48h for 3 treatments; follow the manufacturer's recommendations for proprietary formulations.

G**H**

Mammals, Birds, Reptiles: No information available.

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References

^a Baker BB, Meyer DN, Llaniguez JT et al. (2019) Management of multiple protozoan ectoparasites in a research colony of axolotls (*Ambystoma mexicanum*). *Journal of the American Association for Laboratory Animal Science* **58**(4), 479–484

J**K**

Formalin see **Formaldehyde**
Formol see **Formaldehyde**

L**M**

Framycetin

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(Canaural) **POM-V**

O

Formulations: Topical: 5 mg/g suspension (Canaural also contains fusidic acid, nystatin and prednisolone).

P

Action: Aminoglycosides inhibit bacterial protein synthesis and require an oxygen-rich environment to be effective, thus they are ineffective in low-oxygen sites (abscesses, exudates), making all obligate anaerobic bacteria resistant. They are bactericidal and their mechanism of killing is concentration-dependent, leading to a marked post-antibiotic effect, allowing pulse-dosing regimens which may limit toxicity.

Q**R**

Use: Treatment of aural infections. Framycetin is particularly effective against Gram-negative bacteria, although the combination preparation Canaural has a broad spectrum of activity.

S**T**

Safety and handling: Normal precautions should be observed.

U

Contraindications: Do not use in animals with a perforated tympanum. Do not use in conjunction with other products known to be ototoxic. Corticosteroid-free preparations are preferred in pregnant animals and rabbits to avoid the systemic effects of corticosteroids.

V

Adverse reactions: Aminoglycosides are potentially ototoxic, and ataxia, deafness and nystagmus may be observed where drops have been administered with a perforated tympanum. Local irritation.

X

Drug interactions: No information available.

Y

DOSES

Z

Classified as category C (Caution) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Widely used anecdotally for the treatment of otitis externa in several species and for parasitic otitis (*Otodectes*) in the ferret at a dose of 2–10 drops per ear q12h.

Birds, Reptiles, Amphibians, Fish: No information available.

Fresh water (dechlorinated) (Tap water, Reverse osmosis water)

Formulations: Liquid.

Action: Enters the cell of marine protozoan ectoparasites causing them to expand rapidly by osmotic pressure until they rupture. Monogenean trematodes are thought to release their hold and drop off, but not all die and some will recover if not removed from the environment.

Use: Treatment of skin and gill disease due to infestation with marine protozoan ectoparasites and monogenean trematodes. Prepare the treatment bath by increasing the water temperature to match that of the aquarium. Add sodium bicarbonate to increase the pH to match that of the aquarium, using a digital pH meter to confirm the result. If necessary, add one part sea water from the aquarium to five parts fresh water to produce a hyposaline solution and reduce the physiological stress on sensitive fish species. Initially, start with shorter bath treatment times and increase as necessary.

Safety and handling: None.

Contraindications: Severely infested fish with marked clinical signs may react badly to the osmotic stress but may tolerate and respond better to hyposaline treatment (see above).

Adverse reactions: Carefully monitor the reactions of individual fish, which may vary differently to the immediate osmotic shock. Some species (e.g. angelfish) will lose orientation and lie on their sides temporarily, before recovering. If this initial loss of balance persists, then remove and return the fish to the aquarium.

Drug interactions: No information available.

DOSES

Fish: Immersion for 3–15 minutes, repeat weekly prn^{1,2}.

Mammals, Birds, Reptiles, Amphibians: No information available.

References

¹ Herwig N (1979) *Handbook of drugs and chemicals used in the treatment of fish diseases*. Charles C Thomas, Springfield, Illinois

² Noga EJ (2010) *Fish Disease – Diagnosis and Treatment, 2nd edn*. Wiley-Blackwell, Oxford

Furosemide (Frusemide)

(Dimazon, Frusedale, Libeo, Frusol*) **POM-V, POM**

Formulations: **Injectable:** 50 mg/ml solution. **Oral:** 20 mg, 40 mg, 1 g tablets; 40 mg/5 ml oral solution.

Action: Loop diuretic, inhibiting the Na⁺/K⁺/Cl⁻ cotransporter in the thick ascending limb of the Loop of Henle. The net effect is a loss of sodium, potassium, chloride and water in the urine. It also increases excretion of

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calcium, magnesium and hydrogen as well as renal blood flow and glomerular filtration rate. Transient venodilation may occur following i.v. administration and in some species, bronchodilation may occur; the exact mechanism for both is unclear. Although the majority of avian nephrons lack a Loop of Henle, furosemide still appears effective in birds; however, the mechanism of action is not well understood. The exact mechanism of action in reptiles is also unclear, although it does appear to have a diuretic effect.

Use: Management of congestive heart failure (acute and chronic), non-cardiogenic oedema, hypercalcaemic nephropathy, acute renal failure, hyperkalaemia and hypertension. Use with caution in patients with severe electrolyte depletion, hepatic failure and diabetes mellitus.

Safety and handling: Normal precautions should be observed.

Contraindications: Dehydration and anuria. Do not use in dehydrated or hyperuricaemic birds.

Adverse reactions: Hypokalaemia, hypochloraemia, hypocalcaemia, hypomagnesaemia and hyponatraemia; dehydration, polyuria/polydipsia and prerenal azotaemia occur readily. A marked reduction in cardiac output can occur in animals with severe pulmonary disease, low-output heart failure, hypertrophic cardiomyopathy, pericardial or myocardial disorders, cardiac tamponade and severe hypertension. Other adverse effects include ototoxicity, GI disturbances, leucopenia, anaemia, weakness and restlessness.

Drug interactions: Nephrotoxicity/ototoxicity associated with aminoglycosides may be potentiated when furosemide is also used. Furosemide may induce hypokalaemia, thereby increasing the risk of digoxin toxicity. Increased risk of hypokalaemia if furosemide given with acetazolamide, corticosteroids, thiazides and theophylline. Concurrent administration of NSAIDs with furosemide may decrease efficacy and may predispose to nephrotoxicity, particularly in patients with poor renal perfusion. Furosemide may inhibit the muscle relaxation qualities of tubocurarine, but increase the effects of suxamethonium.

DOSES

Mammals: Ferrets: 1–4 mg/kg i.v., i.m., p.o. q8–12h; Rabbits: 1–4 mg/kg i.v., i.m. q4–6h initially; maintenance doses are often 1–2 mg/kg p.o. q8–24h; Rodents: 1–4 mg/kg s.c., i.m. q4–6h or 5–10 mg/kg s.c., i.m. q12h; Primates: 1–4 mg/kg p.o., s.c. q12h; **Sugar gliders:** 1–5 mg/kg p.o., i.m., s.c. q6–12h; **Hedgehogs:** 2.5–5 mg/kg p.o., i.m., s.c. q8h.

Birds: 0.1–6.0 mg/kg i.m., s.c., i.v. q6–24h^{1,2}.

Reptiles: 2–5 mg/kg i.m. q24h^a.

Amphibians: Not indicated.

Fish: No information available.

References

- ^a Parkinson LA and Mans C (2018) Effects of furosemide administration to water-deprived inland bearded dragons (*Pogona vitticeps*). *American Journal of Veterinary Research* **79**(11), 1204–1208
- ¹ Watson M (2011) Furosemide. *Journal of Exotic Pet Medicine* **20**(1), 60–63
- ² Fitzgerald BC, Dias S and Martorell J (2018) Cardiovascular drugs in avian, small mammal, and reptile medicine. *Veterinary Clinics: Exotic Animal Practice* **21**(2), 399–442

Fusidic acid

(Canaural, Fuciderm, **Isathal**) **POM-V**

Formulations: Topical: 5 mg/g fusidate suspension (Canaural also contains framycetin, nystatin and prednisolone); 0.5% fusidic acid + 0.1% betamethasone cream (Fuciderm); 1% fusidic acid viscous solution (Isathal).

Action: Inhibits bacterial protein synthesis.

Use: Authorized in rabbits for the topical treatment of conjunctivitis associated with staphylococcal infections (Isathal). Active against Gram-positive bacteria, particularly *Staphylococcus pseudintermedius*. It is used topically in the management of staphylococcal infections of the conjunctiva, skin or ear. Fusidic acid is able to penetrate skin and penetrate the cornea gaining access to the anterior chamber of the eye. The carbomer gel vehicle in the ocular preparation may also be efficacious as a surface lubricant.

Safety and handling: Avoid contamination of the container on application.

Contraindications: Corticosteroid-free preparations are preferred in pregnant animals, birds and rabbits to avoid the systemic effects of corticosteroids.

Adverse reactions: No information available.

Drug interactions: No information available.

DOSES

Classified as category D (Prudence) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Rabbits: 1 drop per eye q12–24h (Isathal); Other mammals: 1 drop per eye q12–24h as necessary (unauthorized). Topical formulations containing corticosteroids should be used with caution in rabbits and rodents.

Birds: Skin: apply a thin layer q24h; Ophthalmic: 1 drop per affected eye q12–24h.

Reptiles, Amphibians, Fish: No information available.

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Gabapentin (Gabapentinum)(Neurontin*) **POM-V CD SCHEDULE 3**

For mulations: Oral: 100 mg, 300 mg, 400 mg capsules; 600 mg, 800 mg film-coated tablets; 50 mg/ml solution. Different formulations available as a veterinary special. **VSP**

Action: Gabapentin is an analogue of the neurotransmitter gamma-aminobutyric acid (GABA). The precise mechanism of action of gabapentin is unknown; however, it has been suggested that it mediates its anticonvulsive effect by increasing synaptic levels of GABA in the CNS, most likely through increased synthesis of GABA. Gabapentin has also been demonstrated to decrease the influx of calcium ions into neurons via a specific subunit of voltage-dependent calcium channels. It is believed that some of the therapeutic effect of gabapentin is mediated through binding to these channels. Gabapentin does not interact with sodium-dependent channels and demonstrates no affinity for the common neurotransmitter receptors, including benzodiazepine, glutamate, glycine and dopamine. The mode of action of the analgesic effect of gabapentin is unknown.

Use: Adjunctive therapy in the treatment of seizures refractory to treatment with conventional therapy. Also treatment of neuropathic pain, particularly if insensitive to opioid analgesics. Also shows some promise to reduce apparent stress in animals visiting a veterinary practice, given approximately 3 hours prior to transit. After multiple dosing, peak plasma concentrations of gabapentin are usually achieved within 2 hours of a dose and steady state achieved within 1–2 days. Gabapentin therapy should only be withdrawn slowly. Monitoring of serum levels in human patients does not appear useful. Use with caution in patients with renal impairment, behavioural abnormalities or severe hepatic disease.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: The most commonly reported adverse effect is mild sedation and ataxia. False-positive readings have been reported with some urinary protein tests in human patients taking gabapentin. Hepatic toxicity has been reported as a rare side effect in human patients.

Drug interactions: The absorption of gabapentin from the GI tract is reduced by antacids containing aluminium with magnesium; it is recommended that gabapentin is taken at least 2 hours after the administration of such an antacid. Cimetidine has been reported to reduce the renal clearance of gabapentin but the product information does not consider this to be of clinical importance.

DOSES

Mammals: Ferrets: 3–5 mg/kg p.o. q8h; Rabbits: 2–5 mg/kg p.o. q8h; Rats: 30 mg/kg p.o. q8h; Hamsters: 50 mg/kg p.o. q24h. Doses of 15–30 mg/kg 3 hours prior to transit are suggested for stress reduction in rabbits.

Birds: 10–11 mg/kg p.o. q8–24h^a; Amazon parrots: 15 mg/kg p.o. q8h^b.

Reptiles, Amphibians, Fish: No information available.

References

- ^a Yaw TJ, Zaffarano BA, Gall A *et al.* (2015) Pharmacokinetic properties of a single administration of oral gabapentin in the great horned owl (*Bubo virginianus*). *Journal of Zoo and Wildlife Medicine* **46**(3), 547–552.
- ^b Baine K, Jones MP, Cox S and Martín-Jiménez T (2015) Pharmacokinetics of compounded intravenous and oral gabapentin in Hispaniolan Amazon parrots (*Amazona ventralis*). *Journal of Avian Medicine and Surgery* **29**(3), 165–173

Gentamicin

(Clinigel Vet, EasOtic, Genta, Otomax, **Tiacil**, Genticin*)
POM-V, POM

Formulations: **Injectable:** 40 mg/ml solution for i.v., i.m., s.c. injection (human preparation), 100 mg/ml solution for i.v., i.m., s.c. injection (Genta). **Ophthalmic/aural solution:** 0.5% solution (Tiacil); 0.3% ophthalmic gel (Clinigel). Gentamicin is a component of some topical ear preparations.

Action: Aminoglycosides inhibit bacterial protein synthesis and require an oxygen-rich environment to be effective, thus they are ineffective in low-oxygen sites (abscesses, exudates), making all obligate anaerobic bacteria resistant. They are bactericidal and their mechanism of killing is concentration-dependent, leading to a marked post-antibiotic effect, allowing pulse-dosing regimens which may limit toxicity.

Use: Authorized in rabbits for the treatment of blepharitis, conjunctivitis, keratoconjunctivitis (including postoperative keratitis) and anterior uveitis (Tiacil). Active against Gram-negative bacteria, but some staphylococcal and streptococcal (*Streptococcus faecalis*) species are also sensitive. All obligate anaerobic bacteria and many haemolytic streptococci are resistant. Use in domestic animals is limited by nephrotoxicity and, more rarely, ototoxicity and neuromuscular blockade. Microbial resistance is a concern, although many bacteria resistant to gentamicin may be susceptible to amikacin. Consider specific glomerular filtration rate measurements to assess risk prior to initiating therapy. The trough serum level should be allowed to fall below 2 µg (micrograms)/ml. When used for 'blind' therapy of undiagnosed serious infections, gentamicin is usually given in conjunction with a penicillin and/or metronidazole. Aminoglycosides are more active in an alkaline environment. Geriatric animals or those with reduced renal function should only be given this drug systemically when absolutely necessary, although the move to dosing q24h should reduce the likelihood of nephrotoxicity. Use with caution in rabbits, birds and reptiles. Fluid therapy is essential during treatment of reptiles; monitor uric acid levels in birds and reptiles. Used for the treatment of bacterial disease in fish.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use the aural preparation if the tympanum is perforated. Do not use in conjunction with other drugs considered to be nephrotoxic.

Adverse reactions: Gentamicin delays epithelial healing of corneal ulcers and may cause local irritation. Nephrotoxicity and ototoxicity are potential side effects. Cellular casts in urine sediment are an early sign of impending nephrotoxicity; however, urine must be examined immediately to detect their presence and their absence is not a guarantee of safety. Serum creatinine levels rise later and fatal acute

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renal failure may be inevitable when they do. Gentamicin should not be used during pregnancy. Known nephrotoxicity in some aglomerular species of fish (e.g. toadfish) and toxicity to sensory hair cells of the fish ear.

Drug interactions: Avoid concurrent use of other nephrotoxic, ototoxic or neurotoxic agents (e.g. amphotericin B, furosemide). Increase monitoring and adjust dosages when these drugs must be used together. Aminoglycosides may be chemically inactivated by beta-lactam antibiotics (e.g. penicillins, cephalosporins) or heparin when mixed *in vitro*. The effect of non-depolarizing muscle relaxants (e.g. atracurium, pancuronium, vecuronium) may be enhanced by aminoglycosides. Synergism may occur when aminoglycosides are used with beta-lactam antimicrobials. Activity may be reduced if used in conjunction with bacteriostatic antimicrobials.

DOSES

Classified as category C (Caution) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Ophthalmic: 1 drop per eye q6–8h. Severe ocular infections may require dosing q1–2h. A fortified topical solution (100 mg gentamicin in 5 ml of 0.3% solution, making 14.3 mg/ml) can be used. Otic: 1–3 drops (depending on weight of the animal) in affected ear or apply ointment to affected area q12h. **Ferrets:** 2–5 mg/kg i.v. (over 30 min), i.m., s.c. q12–24h; topically q6–8h; **Rabbits:** 4–8 mg/kg s.c., i.m., i.v. q24h; can be incorporated into antibiotic-impregnated beads (1 g/20 g methylmethacrylate) in surgical sites (e.g. in rabbits following abscess debridement), beads may require surgical removal at a later date; **Guinea pigs:** 6 mg/kg s.c. q24h; **Rats, Mice:** 4–20 mg/kg i.m. q12–24h; **Other rodents:** 2–5 mg/kg s.c., i.m. q12–24h; **Primates:** 2–4 mg/kg i.m., i.v. q12h, 3 mg/kg i.m. q6–8h^a; **Sugar gliders:** 1.5–2.5 mg/kg s.c., i.m. q12h; **Hedgehogs:** 2 mg/kg s.c., i.m. q8h. Ensure adequate renal function and hydration status before use. **All species:** Nebulize 50 mg in 10 ml saline for 15 min q8–12h.

Birds: 2–10 mg/kg i.v., i.m. q6–12h; topically q6–8h; nebulize 50 mg in 10 ml saline for 15 min q8–12h^b.

Reptiles: **Chelonians:** 2–4 mg/kg i.m. q72h^c; may also be nebulized at dilution of 10–20 mg gentamicin in 15 ml saline for 15–20 min q8–12h for respiratory tract infections in chelonians and lizards; **Most snakes:** 2.5 mg/kg i.m. q72h; **Blood pythons:** 2.5 mg/kg i.m. once, then 1.5 mg/kg i.m. q96h^d.

Amphibians: 3 mg/kg i.m. q24h; 1 mg/ml solution as 8h bath q24–48h; 2 mg/ml dilution topically q6–8h for ocular disease¹.

Fish: 2.5–3 mg/kg i.m. q2–3d.

References

- ^a Watson JR, Stoskopf MK, Rozmiarek H *et al.* (1991) Kinetic study of serum gentamicin concentrations in baboons after single-dose administration. *American Journal of Veterinary Research* **52**(8), 1285–1287
- ^b Ramsey EC and Vulliet R (1993) Pharmacokinetic properties of gentamicin and amikacin in the cockatiel. *Avian Diseases* **37**(2), 628–634
- ^c Beck K, Loomis M, Lewbart GL *et al.* (1995) Preliminary comparison of plasma concentrations of gentamicin injected into the cranial and caudal limb musculature of the Eastern box turtle (*Terrapene carolina carolina*). *Journal of Zoo and Wildlife Medicine* **26**(2), 265–268
- ^d Hilf M, Swanson D, Wagner R and Yu VL (1991) A new dosing schedule for gentamicin in blood pythons (*Python curtus*): a pharmacokinetic study. *Research in Veterinary Science* **50**(2), 127–130

- ¹ Teare JA, Wallace RS and Bush M (1991) Pharmacology of gentamicin in the leopard frog (*Rana pipiens*). *Proceedings of the Annual Conference of the American Association of Zoo Veterinarians*, pp. 128–131

Glucose (Dextrose)

(Aqupharm, Vetivex, Glucose intravenous infusion*)

POM-V, POM-VPS, POM

Formulations: Injectable: Sodium chloride 0.9% w/v, glucose monohydrate 5.5% w/v (Aqupharm No. 3 and Vetivex 3), sodium chloride 0.18% w/v, glucose monohydrate 4.4% w/v (Aqupharm No. 18 and Vetivex 18), other electrolyte solutions with glucose for i.v. use: Glucose 40% and 50% w/v.

Action: Source of energy for cellular metabolism. Osmotic agent.

Use: Dilute glucose solutions are used for fluid replacement (primarily where intracellular and interstitial losses have occurred). Concentrated glucose solutions are used parenterally as an energy source or in the treatment of hypoglycaemia. Patients requiring parenteral nutritional support will require mixtures comprising combinations of amino acids, glucose solutions and fat. **See also Amino acid solutions** for use in parenteral nutrition. Solutions containing >5% glucose are hypertonic and irritant if given other than i.v. The 50% solutions contain 1.7 kcal/ml (8.4 kJ/ml) glucose and are extremely hypertonic (2525 mOsm/l). Use with caution in patients with insulin resistance and diabetes mellitus.

Safety and handling: Multi-use vials of 5% glucose or higher rapidly support bacterial growth and strict aseptic technique is required, single patient use is advised.

Contraindications: No information available.

Adverse reactions: 10–50% solutions are irritant and hyperosmolar; administer through a jugular catheter or dilute appropriately. Glucose infusions may produce severe hypophosphataemia in some patients with prolonged starvation. If glucose loading produces signs of hyperglycaemia, insulin may be added to correct it. See comments under **Amino acid solutions** for use in parenteral nutrition solutions.

Drug interactions: No information available.

DOSES

Mammals:

- Fluid therapy: Fluid requirements depend on the degree of dehydration and ongoing losses.
- Parenteral nutrition: The amount required will be governed by the animal's physiological status, the parenteral nutrition admixture and its ability to tolerate high blood glucose levels. Generally, glucose is used to supply 40–60% of the energy requirement. Seek specialist advice before giving parenteral nutrition. **See Amino acid solutions.**
- Hypoglycaemia: 0.1–2 ml 50% dextrose i.v. slowly over 10 min. Note that to meet minimum needs for maintenance 1 ml/kg/h of 50% glucose is needed.

Birds: Hypoglycaemia: 50–100 mg/kg i.v. as slow bolus.

Reptiles: 50 mg/kg dextrose administered as a 5–10% solution by dilution in crystalloids given by i.v. slowly.

Amphibians, Fish: No information available.

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Glutamine(Glutamine*) **GSL****Formulations:** Oral: 500 mg powder or tablet.**Action:** Conditionally essential amino acid required for energy synthesis in the enterocytes. Supplementation in patients with stress starvation is believed to have beneficial effects on intestinal cell proliferation and for prevention of mucosal atrophy.**Use:** GI protectant in stress starvation (i.e. when nutritional support by any route is indicated) and to enhance GI healing in patients with severe GI epithelium damage, such as that caused by parvovirus enteritis. Use with caution in cases with epilepsy or liver disease.**Safety and handling:** Normal precautions should be observed.**Contraindications:** Avoid in patients with acute hepatic encephalopathy as it is partially metabolized to ammonia and glutamate.**Adverse reactions:** May have CNS effects at high doses.**Drug interactions:** Glutamine may antagonize the effects of lactulose in patients with hepatic encephalopathy and could potentially affect the efficacy of antiseizure medications.**DOSES****Mammals:** Ferrets: 0.5 g/kg p.o. in divided doses daily.**Birds, Reptiles, Amphibians, Fish:** No information available.**Glyceryl trinitrate (Nitroglycerin(e))**(Deponit*, Glyceryl trinitrate*, Glytrin*, Minitran*, Nitrocine*, Nitronal*, Percutol*, Transiderm-Nitro*) **POM****Formulations:** **Topical:** 2% ointment to be applied to skin equating to a dose of up to 800 µg (micrograms)/kg/h per inch of skin treated; 5 mg/24 h, 10 mg/24 h, 15 mg/24 h transdermal patches (Deponit, Minitran, Nitro-Dur, Transderm-Nitro). **Oral:** 0.003 mg, 0.005 mg sub-lingual tablet (Glyceryl trinitrate). **Injectable:** 1 mg/ml solutions (Nitrocine, Nitronal).**Action:** Systemic vasodilator. Although a potent coronary vasodilator, its major benefit in small animals follows from a reduction in venous return as a consequence of venodilation. A decrease in venous return reduces left ventricular filling pressures.**Use:** Short-term management of cardiogenic oedema (particularly acute pulmonary oedema) in animals with congestive heart failure. It is normally only used for 1–2 days. Its efficacy is debatable. Rotate application sites; suggested sites include the thorax, groin and inside the ears. Rub ointment well into the skin.**Safety and handling:** Owners should be cautioned to avoid contact with areas where the ointment has been applied and to wear non-permeable gloves when applying.**Contraindications:** Hypotension, hypovolaemia, cerebral haemorrhage, head trauma.**Adverse reactions:** Hypotension (reduce dose), tachycardia and a rash

at the site of application. Tachyphylaxis can occur. Headaches are common in humans and may be an adverse effect in animals also.

Drug interactions: Concurrent use of ACE inhibitors, anaesthetics, beta-blockers, calcium-channel blockers, corticosteroids and diuretics may enhance the hypotensive effect. NSAIDs may antagonize its hypotensive effects.

DOSES

Mammals: Ferrets: 1–3 mm topically to the skin q12–24h; Rodents: 3 mm topically q6–12h.

Birds, Reptiles, Amphibians, Fish: No information available.

Glycopyrronium (Glycopyrrolate)

(Robinul*) **POM**

Formulations: Injectable: 200 µg (micrograms)/ml solution.

Action: Blocks the action of acetylcholine at muscarinic receptors at the terminal ends of the parasympathetic nervous system, reversing parasympathetic effects. Its quaternary structure prevents it from crossing the blood–brain barrier and so it is devoid of central effects.

Use: Potent antisialagogue agent and has been used preoperatively to decrease oral and bronchial secretions. It is also used to inhibit vagal efferent activity and manage bradycardias caused by the administration of potent opioid drugs. Glycopyrronium is used with long-acting anticholinesterase drugs (e.g. neostigmine, pyridostigmine) during antagonism of neuromuscular block. Glycopyrronium is longer acting than atropine. Routine administration of glycopyrronium prior to anaesthesia as part of premedication is no longer recommended. It causes a reduction in oral and bronchial secretions by decreasing the water content, therefore secretions become more sticky. The main use in rodents is in reducing the often profuse hypersalivation in guinea pigs and chinchillas, which can be inhaled in dorsal recumbency. Administration of potent opioids in the perioperative period can promote bradyarrhythmias but it is better to monitor heart rate and give glycopyrronium to manage a low heart rate if necessary. Administration of very low doses of glycopyrronium i.v. can cause exacerbation of bradyarrhythmias due to a vagal stimulatory effect; giving another dose i.v. will usually cause an increase in heart rate. Glycopyrronium is devoid of central effects and therefore does not cause mydriasis. Glycopyrronium is preferred over atropine in rabbits due to the variable and unpredictable effects of atropine in this species.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Tachycardias following overdose of glycopyrronium are usually transient and do not require management. Ventricular arrhythmias may be treated with lidocaine if severe. The incidence of adverse effects is lower than that seen with atropine.

Drug interactions: When mixed with alkaline drugs (e.g. barbiturates) a precipitate may form. Antimuscarinics may enhance the actions of sympathomimetics and thiazide diuretics. The following may enhance the activity of glycopyrronium: antihistamines, quinidine, pethidine,

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benzodiazepines and phenothiazines. Combining glycopyrronium and alpha-2 adrenergic agonists is not recommended.

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Mammals: Rabbits: 10–20 µg (micrograms)/kg i.v., i.m. once; may be used in place of the largely ineffective atropine in this species;

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Rodents: 10–20 µg/kg i.v., i.m. once; **Primates:** 5–10 µg/kg i.m. once; **Sugar gliders:** 10–20 µg/kg s.c., i.m., i.v. once¹; **Hedgehogs:** 10–20 µg/kg s.c. once.

E

Birds: Premedication: 10–30 µg (micrograms)/kg i.m., i.v. once².

F

Reptiles: 10 µg (micrograms)/kg i.m., i.v., s.c. once. May be ineffective in green iguanas^a.

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Amphibians, Fish: No information available.

References

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^a Pace L and Mader DR (2002) Atropine and glycopyrrolate, route of administration and response in the green iguana (*Iguana iguana*). *Proceedings of the Association of Reptilian and Amphibian Veterinarians* p. 79

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¹ Johnson-Delaney CA (2000) Therapeutics of companion exotic marsupials. *Veterinary Clinics of North America: Exotic Animal Practice* **3(1)**, 173–181.

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² Lierz M and Rüdiger K (2012) Anesthesia and analgesia in birds. *Journal of Exotic Pet Medicine* **21(1)**, 44–58

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Haloperidol

(Haldol*, Halkid*, Haloperidol*) **POM**

Formulations: Oral: 200 µg (micrograms)/ml, 1 mg/ml, 2 mg/ml solution. **Injectable:** 5 mg/ml.

Action: A potent central dopamine type 2 receptor antagonist. Also has antagonistic activity against alpha-1 adrenergic, histaminergic and cholinergic receptors.

Use: Used for the management of compulsive disorders such as feather plucking and self-mutilation in birds. Should be used in association with a behaviour modification plan. Has been used in combination with LHRHa to induce ovulation in koi.

Safety and handling: Normal precautions should be observed.

Contraindications: Avoid in patients with a known hypersensitivity to butyrophenones, those with hypokalaemia, cardiac disease or a history of seizures.

Adverse reactions: Potential extra-pyramidal side effects.

Drug interactions: Should not be used with monoamine oxidase inhibitors (e.g. selegiline), tramadol, beta-blockers, or drugs that prolong the QT interval. Concurrent administration with itraconazole may require a reduction in haloperidol dosage.

DOSES

Mammals: Primates: 0.03–0.05 mg/kg i.m. q12h.

Birds: 0.1–0.9 mg/kg p.o. q12–24h, 1–2 mg/kg i.m. q14–21d.

Reptiles: Boids: 0.5–10 mg/kg i.m. q7–14d.

Fish: 0.5 mg/kg i.m. (in combination with LHRHa to induce ovulation in koi)^a.

Amphibians: No information available.

References

- ^a Arabaci M, Çağırgan H and Sarı M (2004). Induction of ovulation in ornamental common carp (Koi, *Cyprinus carpio* L.) using LHRHa ([d-Ser (tBu) 6, Pro9-NEt]-LHRH) combined with haloperidol and carp pituitary extract. *Aquaculture Research* **35**(1), 10–14

Heparin (low molecular weight) (Dalteparin, Enoxaparin)

(Clexane (enoxaparin)*, Fragmin (dalteparin)*) **POM**

Formulations: **Injectable:** 2,500 IU/ml, 100,000 IU/ml ampoules (dalteparin); 25,000 IU/ml multidose vial plus various pre-filled syringes at concentrations of 12,500 IU/ml and 25,000 IU/ml (dalteparin); pre-filled syringes of 100 mg/ml (enoxaparin). 100 mg enoxaparin is equivalent to 10,000 IU of anti-Factor Xa activity.

Action: Low molecular weight heparin (LMWH) is an anticoagulant that inhibits factor Xa and thrombin. When compared with unfractionated heparin (UFH), LMWH has reduced anti-IIa activity relative to anti-Xa activity (ratio of anti-Xa to anti-IIa is 2–4:1 compared with UFH 1:1). Thus, at therapeutic doses, LMWH has minimal effect on APTT. Therapeutic monitoring of LMWH is by anti-Xa activity (but this may not be practical).

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A **Use:** Treatment of thromboembolic complications and hypercoagulable syndromes (e.g. pulmonary thromboembolism, disseminated intravascular coagulation (DIC)) in mammals, although there is no information available in exotic pets. LMWH is also used in the treatment of myocardial infarction, atrial fibrillation, deep vein thrombosis and pulmonary thromboembolism in humans. Its use in DIC is controversial as no beneficial effect has been shown in controlled clinical trials in humans. LMWH is no longer used to try to prevent DIC. May be helpful for treatment of PTFE toxicosis in birds.

E **Safety and handling:** Normal precautions should be observed.

F **Contraindications:** Bleeding disorders or severe renal dysfunction.

G **Adverse reactions:** If an overdosage occurs protamine can be used as an antidote. Heparin should not be administered i.m. as it may result in haematoma formation. Its use in DIC may worsen haemorrhage especially if the patient is thrombocytopenic. Heparin induced thrombocytopenia syndrome is a serious concern in human patients.

H **Drug interactions:** Use with caution with other drugs that can cause changes in coagulation status (e.g. aspirin, NSAIDs). Heparin may antagonize ACTH, corticosteroids or insulin. Heparin may increase plasma levels of diazepam. The actions of heparin may be partially counteracted by antihistamines, digoxin and tetracyclines. Do not mix other drugs in the same syringe as heparin.

L **DOSES**

M **Birds:** PTFE toxicosis: 300 IU/kg i.v. once.

N **Mammals, Reptiles, Amphibians, Fish:** No information available.

O **Heparin (unfractionated) (UFH)**

(Heparin*, Hepsal*) **POM**

P **Formulations:** Injectable: 1,000–25,000 IU/ml solutions; 10 IU/ml in saline, 100 IU/ml in saline.

Q **Action:** Heparin is an anticoagulant that exerts its effects primarily by enhancing the binding of antithrombin III (AT III) to factors IIa, IXa, Xa, XIa and XIIa; it is only effective if adequate AT III is present. The AT III/clotting factor complex is subsequently removed by the liver. Heparin inactivates thrombin and blocks the conversion of fibrinogen to fibrin. The inhibition of factor XII activation prevents the formation of stable fibrin clots. Heparin does not significantly change the concentrations of clotting factors, nor does it lyse pre-existing clots.

R **Use:** Treatment of DIC and thromboembolic disease (**see Heparin (low molecular weight)** (LMWH)). Therapy must be carefully monitored as the activity of UFH is somewhat less predictable than LMWH. In veterinary medicine it is mainly used to maintain the patency of catheters/cannulae.

S **Safety and handling:** Normal precautions should be observed.

T **Contraindications:** Major bleeding disorders, increased risk of haemorrhage, thrombocytopenia.

U **Adverse reactions:** If an overdosage occurs protamine can be used as an antidote. Heparin should not be administered i.m. as it may result in haematoma formation. Its use in DIC may worsen haemorrhage

especially if the patient is thrombocytopenic. Heparin induced thrombocytopenia syndrome is a serious concern in human patients.

Drug interactions: Use with caution with other drugs that can cause changes in coagulation status (e.g. aspirin, NSAIDs). Heparin may antagonize ACTH, corticosteroids and insulin. Heparin may increase plasma levels of diazepam. The actions of heparin may be partially counteracted by antihistamines, digoxin and tetracyclines. Do not mix other drugs in the same syringe as heparin.

DOSES

Mammals: Catheter maintenance: 1250 IU in 100 ml water for injection.

Birds, Reptiles, Amphibians, Fish: No information available.

Human chorionic gonadotrophin see Chorionic gonadotrophin

Hyaluronate

(Hyabak*, Hylo-Care*, Hylo-Forte*, Hylo-Tear*, Oxyal*, Vismed Multi*) **P**

Formulations: Ophthalmic: 0.1%, 0.15%, 0.18%, 0.3%, 0.4% solution in 10 ml bottle.

Action: Viscoelastic fluid with mucomimetic properties. Sodium hyaluronate is also available in different formulations as a viscoelastic for intraocular surgery.

Use: Used as a tear replacement and is beneficial for the management of quantitative (keratoconjunctivitis sicca (KCS) or dry eye) and qualitative tear film disorders. It has longer corneal contact time than the aqueous tear substitutes.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: It is tolerated well and ocular irritation is unusual.

Drug interactions: No information available.

DOSES

Mammals: 1 drop per eye q4–6h, although it can be used hourly if required.

Birds, Reptiles, Amphibians, Fish: No information available.

Hydralazine

(Apresoline*, Hydralazine*) **POM**

Formulations: Injectable: 20 mg powder for reconstitution and slow i.v. infusion; Oral: 25 mg, 50 mg tablets.

Action: Hydralazine acts chiefly on arteriolar smooth muscle causing vasodilation; it is able to decrease systemic vascular resistance to about 50% of the baseline value. The effects of hydralazine are to reduce afterload and increase heart rate, stroke volume and cardiac output.

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A **Use:** Afterload reducer as adjunctive therapy of congestive heart failure secondary to severe or refractory mitral value insufficiency. It can be used to treat systemic hypertension. Hospitalization with frequent monitoring of blood pressure is advised during its use. It is not typically used as a first-line drug in hypertension. As hydralazine may cause sodium and water retention, concomitant use of diuretic therapy is often necessary. Give with food if possible.

B **Safety and handling:** Normal precautions should be observed.

C **Contraindications:** Hypovolaemia, hypotension, renal impairment or cerebral bleeding.

D **Adverse reactions:** Reflex tachycardia, severe hypotension (monitor and adjust doses as necessary), anorexia and vomiting in susceptible species.

E **Drug interactions:** The hypotensive effects of hydralazine may be enhanced by ACE inhibitors (e.g. enalapril), anaesthetics, beta blockers (e.g. propranolol), calcium-channel blockers (e.g. diltiazem, verapamil), corticosteroids, diuretics and NSAIDs. Sympathomimetics (e.g. phenylpropanolamine) may cause tachycardia. The pressor response to adrenaline may be reduced.

F **DOSES**

G **Mammals:** Guinea pigs: 1 mg/kg i.v. prn.

H **Birds, Reptiles, Amphibians, Fish:** No information available.

I **Hydrochlorothiazide**

J (Co-amiloride*, Moduret*, Moduretic*) **POM**

K **Formulations:** Oral: 25 mg hydrochlorothiazide + 2.5 mg amiloride, 50 mg hydrochlorothiazide + 5 mg amiloride tablets.

L **Action:** Thiazide diuretic that inhibits reabsorption of sodium and chloride in the distal convoluted tubule, resulting in sodium, chloride and water loss in the urine. It also causes excretion of potassium, magnesium and bicarbonate. It is formulated with a potassium-sparing diuretic (amiloride).

M **Use:** Additional therapy for congestive heart failure when the clinical signs have become refractory to furosemide. However, furosemide therapy should still be continued when using hydrochlorothiazide. It may also be used in the prevention of calcium oxalate urolithiasis. Thiazides have antihypertensive effects, although the exact mechanism is unclear.

N **Safety and handling:** Normal precautions should be observed.

O **Contraindications:** Renal impairment, as it tends to reduce glomerular filtration rate.

P **Adverse reactions:** Hyperglycaemia, hypokalaemia, hyponatraemia, hypochloraemia and volume contraction. It enhances the effects of the renin-angiotensin-aldosterone system in heart failure.

Q **Drug interactions:** Increased possibility of hypokalaemia developing if thiazides are used concomitantly with corticosteroids or loop diuretics (furosemide). Thiazide-induced hypokalaemia may increase the risk of digoxin toxicity. Thus, concomitant use of potassium-sparing diuretics (e.g. spironolactone) or potassium supplementation may be necessary

during prolonged administration. The concurrent administration of vitamin D or calcium salts with thiazides may exacerbate hypercalcaemia.

DOSES

Reptiles: 1 mg/kg p.o. q24–72h.

Mammals, Birds, Amphibians, Fish: No information available.

Hydroxyzine

(Atarax*, Ucerax*) **POM**

Formulations: Oral: 10 mg, 25 mg tablets; 2 mg/ml syrup.

Action: Binds to H1 histamine receptors preventing histamine from binding. Hydroxyzine is metabolized to cetirizine.

Use: Used in feather plucking in birds. Use with caution in cases with urinary retention, angle-closure glaucoma and pyloroduodenal obstruction.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: May cause mild sedation. May reduce seizure threshold.

Drug interactions: No information available.

DOSES

Mammals: Ferrets, Rabbits: 2 mg/kg p.o. q8–12h.

Birds: 2.2 mg/kg p.o. q8h^a.

Reptiles, Amphibians, Fish: No information available.

References

- ^a Krinsley M (1993) Use of Derm Caps liquid and hydroxyzine HCl for the treatment of feather picking. *Journal of the Association of Avian Veterinarians* **7**, 221

Hyoscine see Butylscopolamine

Hypromellose

(Hypromellose*, Isopto*, Tears Naturale*) **P**

Formulations: Ophthalmic: 0.3%, 0.5%, 1% solutions in 10 ml dropper bottle; 0.32% (single-use vial). Large variety of other formulations also available.

Action: Cellulose based tear substitute (lacrimomimetic).

Use: Lubrication of dry eyes. In cases of keratoconjunctivitis sicca (KCS or dry eye) it will improve ocular surface lubrication, tear retention and patient comfort while lacrostimulation therapy (e.g. topical ciclosporin) is initiated. It may also be used as a vehicle base for compounding ophthalmic drugs. Patient compliance is poor if used >q4h, consider using a longer acting tear replacement.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: No information available.

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Drug interactions: No information available.

DOSES

Mammals: 1 drop per eye q1h during anaesthesia.

Birds, Reptiles, Amphibians, Fish: No information available.

Imepitoin

(Pexion) **POM-V**

Formulations: Oral: 100 mg, 400 mg tablets.

Action: Imepitoin inhibits seizures via potentiation of the GABA_A receptor-mediated inhibitory effects on the neurons. Imepitoin also has a weak calcium-channel blocking effect, which may contribute to its anticonvulsive properties.

Use: For the management of epileptic seizures due to idiopathic epilepsy in dogs. The choice of initial medication is guided by patient requirements: imepitoin has a more rapid onset of action than phenobarbital (a steady state does not need to be achieved), does not require the determination of serum concentrations and has a less severe adverse effect profile; however, phenobarbital is less expensive and more efficacious.

Safety and handling: Normal precautions should be observed.

Contraindications: The medication should not be used with severely impaired liver, kidney or heart function.

Adverse reactions: The most frequent adverse effect reported is sedation. Other adverse effects are generally mild and transient and include polyphagia, hyperactivity, polyuria, polydipsia, somnolence, hypersalivation, emesis, ataxia, apathy, diarrhoea, prolapsed nictitating membrane, decreased sight and sensitivity to sound.

Drug interactions: Imepitoin has been used in combination with phenobarbital in a small number of cases and no harmful clinical interactions were reported.

DOSES

Mammals: Mice: Experimental doses have been reported of 25–60 mg/kg p.o., i.p.

Birds, Reptiles, Amphibians, Fish: No information available.

Imidacloprid

(Advantage, Advantix, Advaprid, Advocate, Endectrid, Eziflea, Imidaflea, Imidamox, Imidaspot, Imoxat, Moxiclear, Prevensa, Prinocate, Prinovox, Ridaflea)
POM-V, AVM-GSL

Formulations: Topical: 100 mg/ml imidacloprid either as sole agent or in combination with moxidectin or permethrin (e.g. Advantix, Advocate, Prinovox and others) in spot-on pipettes of various sizes. Numerous GSL and non-authorized formulations.

Action: Binds to postsynaptic nicotinic receptors resulting in paralysis and death of fleas and their larvae.

Use: Authorized in ferrets for the treatment and prevention of flea infestation and the prevention of heartworm disease (Advocate, Endectrid, Imidamox, Imoxat, Moxiclear, Prevensa, Prinocate, Prinovox). Authorized in rabbits for the treatment of and prevention of flea infestations (Advantage, Advaprid, Eziflea, Imidaflea, Imidaspot, Ridaflea). For the treatment of flea infestations, the additional use of an approved

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A insect growth regulator is recommended and the product should be applied every 4 weeks to all in-contact rabbits. Has been used in lizards for the treatment of mites.

B **Safety and handling:** Many combinations contain products that are dangerous to aquatic organisms and birds.

C **Contraindications:** No information available.

D **Adverse reactions:** Transient pruritus and erythema at the site of application may occur.

E **Drug interactions:** No information available.

F **DOSES**

G **See Appendix for guidelines on responsible parasiticide use.**

H **Mammals:** Ferrets: 0.4 ml pipette q30d; Rabbits: 0.4 ml, 0.8 ml pipettes (use the smaller size in rabbits <4 kg) or 0.125 ml/kg imidacloprid/permethrin formulation for fur mites (*Leporacus* spp.); Rabbits, Guinea pigs: 0.1 ml/kg imidacloprid/permethrin formulation; Rodents: 20 mg/kg (equivalent to 0.2 ml/kg).

I **Reptiles:** Bearded dragons, Frillneck lizards: 0.2 ml/kg of imidacloprid (100 mg/ml) + moxidectin (25 mg/ml) formulation has been used topically q14d for 3 doses for treatment of mites and internal parasites^a.

J **Birds, Amphibians, Fish:** No information available.

K **References**

L ^a Mehlhorn H, Schmahl G and Mevissen I (2005) Efficacy of a combination of imidacloprid and moxidectin against parasites of reptiles and rodents. *Parasitology Research* **97**(1), S97–S101

M **Imidapril**

N (Prilium) **POM-V**

O **Formulations:** Oral: 75 mg, 150 mg powders for reconstitution.

P **Action:** ACE inhibitor. It inhibits conversion of angiotensin I to angiotensin II and inhibits the breakdown of bradykinin. Overall effect is a reduction in preload and afterload via venodilation and arteriodilation, decreased salt and water retention via reduced aldosterone production and inhibition of the angiotensin-aldosterone-mediated cardiac and vascular remodelling. Efferent arteriolar dilation in the kidney can reduce intraglomerular pressure and therefore glomerular filtration. This may decrease proteinuria.

T **Use:** Treatment of congestive heart failure caused by mitral regurgitation or dilated cardiomyopathy. Often used in conjunction with diuretics when heart failure is present as most effective when used in these cases. Can be used in combination with other drugs to treat heart failure (e.g. pimobendan, spironolactone, digoxin). May be beneficial in cases of chronic renal insufficiency, particularly protein-losing nephropathies. May reduce blood pressure in hypertension. ACE inhibitors are more likely to cause or exacerbate prerenal azotaemia in hypotensive animals and those with poor renal perfusion (e.g. acute, oliguric renal failure). Use cautiously if hypotension, hyponatraemia or outflow tract obstruction are present. Regular monitoring of blood pressure, serum creatinine, urea and electrolytes is strongly recommended with ACE inhibitor treatment. In poultry has been shown to have a beneficial effect in pulmonary hypertension models.

Clinically appears effective in companion birds at similar doses and is easy to dose due to the liquid formulation.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use in animals with acute renal failure, congenital heart disease, haemodynamically relevant stenoses (e.g. aortic stenosis), obstructive hypertrophic cardiomyopathy or hypovolaemia.

Adverse reactions: Potential adverse effects include hypotension, hyperkalaemia and azotaemia. Monitor blood pressure, serum creatinine and electrolytes when used in cases of heart failure. Dosage should be reduced if there are signs of hypotension (weakness, disorientation). Anorexia, vomiting and diarrhoea are rare.

Drug interactions: Concomitant use of potassium-sparing diuretics (e.g. spironolactone) or potassium supplements could result in hyperkalaemia. However, in practice, spironolactone and ACE inhibitors appear safe to use concurrently. There may be an increased risk of nephrotoxicity and decreased clinical efficacy when used with NSAIDs. There is a risk of hypotension with concomitant administration of diuretics, vasodilators (e.g. anaesthetic agents, antihypertensive agents) or negative inotropes (e.g. beta-blockers).

DOSES

Mammals: Anecdotally this product has been used in rats for protein-losing nephropathy at 0.25 mg/kg p.o. q24h. The liquid formulation makes it easier to accurately titrate doses.

Birds: 3 mg/kg p.o. q24h (based on dose in chickens)^a.

Reptiles, Amphibians, Fish: No information available.

References

- ^a Hao, Xue-Qin, Shou-Yan Zhang *et al.* (2013) Imidapril inhibits right ventricular remodeling induced by low ambient temperature in broiler chickens. *Poultry Science* **92**(6), 1492–1497

Insulin

(Caninsulin, Prozinc, Actrapid*, Humulin*, Hypurin*, Insulatard*, Lantus*) **POM-V, POM**

Formulations: **Injectable:** 40 IU/ml, 100 IU/ml suspensions (for s.c. injection) or 100 IU/ml solutions (for s.c., i.v. or i.m. injection). There are many preparations (including soluble) authorized for use in humans; however, veterinary authorized preparations (lente and PZI), when available, are preferential for both legal and medical reasons.

Action: Binds to specific receptors on the cell surface which then stimulate the formation of glycogen from glucose, lipid from free fatty acids, protein from amino acids and many other metabolic effects.

Use: Treatment of insulin-dependent diabetes mellitus (IDDM) and occasionally as adjunctive therapy in the management of hyperkalaemia associated with urinary tract obstruction. There are various formulations of insulins from various species (see table). Neutral (soluble) insulin is the normal crystalline form. Lente insulins rely on different concentrations of zinc and size of zinc-insulin crystals to provide different durations of activity. Glargine insulin (Lantus) is a pH sensitive, long-acting formulation that precipitates at the site of injection. Hyperkalaemia associated with

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hypoadrenocorticism is often associated with hypoglycaemia and insulin should be avoided in those cases.

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Trade name	Species of insulin	Types available
Actrapid*	Human	Neutral
Caninsulin	Porcine	Lente
Lantus*	Human	Glargine
Humulin*	Human	Neutral, Isophane (NPH)
Hypurin*	Bovine	Neutral, Isophane, Lente, PZI
	Porcine	Neutral, Isophane
Insulatard*	Human or porcine	Isophane
Prozinc	Human	PZI

* = Not authorized for veterinary use.

I

Safety and handling: Normal precautions should be observed.

J

Contraindications: Hypoglycaemia.

K

Adverse reactions: Overdosage results in hypoglycaemia and hypokalaemia.

L

Drug interactions: Corticosteroids, ciclosporin, thiazide diuretics and thyroid hormones may antagonize the hypoglycaemic effects of insulin. Anabolic steroids, beta-adrenergic blockers (e.g. propranolol), phenylbutazone, salicylates and tetracycline may increase insulin's effect. Administer with caution and monitor patients closely if digoxin is given concurrently. Beta-adrenergic agonists, such as terbutaline, may prevent or lessen insulin-induced hypoglycaemia in humans.

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DOSES

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Mammals: Ferrets: 0.5–1.0 IU/ferret s.c. q12h (lente) or 0.1 IU/ferret s.c. q24h (ultralente); Guinea pigs: 1–2 IU/kg s.c. q12h; Chinchillas: 1 IU/kg s.c. q12h; Rats: 1–3 IU/rat s.c. q12h; Hamsters, Gerbils: 2 IU/animal s.c. q12h; Marmosets: 0.1 IU/animal s.c. q12h (NPH or lente insulin).

Q

Reptiles: Chelonians, Snakes: 1–5 IU/kg i.m. q24–72h; Lizards: 5–10 IU/kg i.m. q24–72h; adjust doses according to serial glucose measurement.

R

Birds, Amphibians, Fish: No information available.

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Iodophor see **Povidone–iodine**

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Iron salts (Ferrous fumarate, Ferrous gluconate, Ferrous sulphate, Iron dextran)

W

(Gleptosil, CosmoFer*, Diafer*, Monofer*, Venofer*)

X

POM-VPS, POM, GSL

Y

Formulations: Injectable: 50 mg/ml iron dextran (CosmoFer), 50 mg/ml iron (III) isomaltoside (Monofer), 100 mg/ml iron (III) isomaltoside (Monofer), 200 mg/ml iron dextran (Gleptosil and several other formulations authorized for pigs), 20 mg/ml iron sucrose (Venofer).

Z

Oral: 200 mg FeSO₄ tablet (ferrous sulphate), ferrous gluconate and

other formulations in variable tablet and liquid preparations; ferrous fumarate preparations also typically contain folic acid.

Action: Essential for oxygen-binding in haemoglobin, electron transport chain and oxidative phosphorylation, and other oxidative reactions in metabolism.

Use: Treatment of iron deficiency anaemia and conditions where red blood cell synthesis is high and iron stores are depleted. The oral route should be used if possible. Iron absorption is complex and dependent in part on physiological demand, diet composition, current iron stores and dose. Valid reasons for administering iron parenterally are failure of oral therapy due to severe GI adverse effects, continuing severe blood loss, iron malabsorption, or a non-compliant patient. Modified-release preparations should be avoided as they are ineffective. In most species, iron is absorbed in the duodenum (but there is no specific information in exotic pets) and the release of iron from modified-release preparations occurs lower down the GI tract. Absorption is enhanced if administered 1 hour before or several hours after feeding. Reduce dosage if GI side effects occur.

Safety and handling: Normal precautions should be observed.

Contraindications: Severe infection or inflammation, intolerance to the oral preparation, any anaemia other than iron deficiency anaemia, presence of GI ulcers. Also contraindicated in patients with hepatic, renal (particularly pyelonephritis) or cardiac disease, and untreated urinary tract infections.

Adverse reactions: Parenteral iron may cause arrhythmias, anaphylaxis, shunting of iron to reticuloendothelial stores and iron overload. Oral iron may cause nausea, vomiting, constipation and diarrhoea. The faeces of animals treated with oral iron may be dark in appearance. High doses may be teratogenic and embryotoxic (injectable iron dextran).

Drug interactions: Chloramphenicol can delay the response to iron dextran and its concurrent use should be avoided. Oral preparations bind to tetracyclines and penicillamine causing a decrease in efficacy. Antacids, milk and eggs significantly decrease the bioavailability of oral iron.

DOSES

Mammals: 10 mg/kg i.m. of iron dextran once weekly or prn.

Birds: 10 mg/kg i.m. of iron dextran prn^a.

Reptiles, Amphibians, Fish: No information available.

References

- ^a Bos JH, Todd B, Tell LA, Ramsey EC and Fowler ME (1992) Treatment of anemic birds with iron dextran therapy: homologous and heterologous blood transfusions. *Tijdschrift voor diergeneeskunde* **117**(Suppl. 1), 22–23

Isoeugenol see **Eugenol**

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A Isoflurane**(Isocare, Isofane, IsoFlo, Isoflurane Vet, Isoflurin, Iso-vet, Vetflurane, VetOne Fluriso) POM-V****Formulations:** Inhalational: 250 ml bottle of liquid isoflurane.**Action:** Halogenated ethyl methyl ether and structural isomer of enflurane. The mechanism of action of volatile anaesthetic agents is not fully understood.

Use: Authorized in ferrets, guinea pigs, chinchillas, hamsters, gerbils, rats, mice, ornamental birds, reptiles (Isocare, IsoFlo, Isoflurin, Iso-vet, Vetflurane, VetOne Fluriso); rabbits, rats, mice, ornamental birds, reptiles (Isofane); ornamental birds (IsofluraneVet) for induction and maintenance of anaesthesia. Isoflurane is potent and highly volatile so should only be delivered from a suitable calibrated vaporizer. It is less soluble in blood than halothane but more soluble than sevoflurane, therefore induction and recovery from anaesthesia are quicker than halothane but slower than sevoflurane. The concentration of isoflurane required to maintain anaesthesia depends on the other drugs used in the anaesthesia protocol; the concentration should be adjusted according to clinical assessment of anaesthetic depth. MAC approximately 1.2–1.7% in most species. Isoflurane has a pungent smell and induction of anaesthesia using chambers or masks may be less well tolerated compared with sevoflurane. In amphibians, isoflurane can be delivered by bubbling the solution through water, intubation or by direct application to the skin. In fish, isoflurane is added direct to the water by spraying through a fine hypodermic needle positioned under the water surface but control of the anaesthetic concentration is difficult due to its relative insolubility and is therefore not recommended in fish.

Safety and handling: Measures should be adopted to prevent contamination of the environment with isoflurane during anaesthesia and when handling the agent.**Contraindications:** Avoid gaseous induction in rabbits and chelonians as they can breath-hold and develop serious complications.**Adverse reactions:** Isoflurane causes dose-dependent hypotension by causing vasodilation, particularly in skeletal muscle. This adverse effect does not wane with time. Isoflurane is a more potent respiratory depressant than halothane, respiratory depression is dose-dependent. Isoflurane does not sensitize the myocardium to catecholamines to the extent that halothane does, but can generate arrhythmias in certain conditions. Isoflurane is not metabolized by the liver (0.2%) and has less effect on liver blood flow compared with halothane. In ferrets, isoflurane can cause marked depression in haematological parameters (especially haematocrit, RBC count, Hb concentration) rapidly after induction, so care should be taken in interpreting blood results if isoflurane is used for restraint, or in anaemic or debilitated ferrets. In American green tree frogs, the use of compounded topical isoflurane jelly has been associated with fatalities, so use is to be avoided in this species^a.**Drug interactions:** Opioid agonists, benzodiazepines and N₂O reduce the concentration of isoflurane required to achieve surgical anaesthesia.

DOSES

Mammals: Ferrets, Rabbits, Primates, Sugar gliders, Hedgehogs: 3–5% isoflurane concentration is required to induce anaesthesia in unpremedicated patients, 1.5–3.0% is required for maintenance. The expired concentration required to maintain surgical anaesthesia in 50% of rabbits is 2.05%. **Rodents:** 2–5% for induction, 0.25–4.0% for maintenance.

Birds: 3–5% isoflurane concentration is required to induce anaesthesia in unpremedicated patients^{b,c}, 1.5–2.5% in 100% oxygen for maintenance. Administration of other anaesthetic agents and opioid analgesics reduces the dose requirement of isoflurane, therefore the dose should be adjusted according to individual requirement.

Reptiles: Lizards, Snakes: 3–5% in 100% oxygen for induction^d, 1.8–4% in 100% oxygen for maintenance^e.

Amphibians: 3–5% in 100% oxygen for induction, 1–2% in 100% oxygen for maintenance or 0.28 ml/100 ml water in bath (closed container), bubbled through water to effect or 0.007–0.015 ml/g topically (closed container), remove excess after induction or mixture of 3 ml isoflurane, 3.5 ml KY jelly and 1.5 ml water applied at 0.025–0.035 ml/g (closed container), remove excess after induction^f.

Fish: 0.4–0.75 ml/l for induction, 0.25–0.4 ml/l for maintenance (add to water).

References

- ^a Zec S, Clark-Price SC, Coleman DA and Mitchell MA (2014) Loss and return of righting reflex in American green tree frogs (*Hyla cinerea*) after topical application of compounded sevoflurane or isoflurane jelly: a pilot study. *Journal of Herpetological Medicine and Surgery* **24**(3), 72–76
- ^b Curro TG, Brunson DB and Paul-Murphy J (1994) Determination of the ED50 of isoflurane and evaluation of the isoflurane-sparing effect of butorphanol in cockatoos (*Cacatua* spp.). *Veterinary Surgery* **23**(5), 429–433
- ^c Ludders JW, Mitchell GS and Rhode J (1990) Minimal anesthetic concentration and cardiopulmonary dose response of isoflurane in ducks. *Veterinary Surgery* **19**(4), 304–307
- ^d Bertelsen MF, Mosley C, Crawshaw GJ, Dyson D and Smith DA (2005) Inhalation anesthesia in Dumeril's monitor (*Varanus dumerili*) with isoflurane, sevoflurane, and nitrous oxide: effects of inspired gases on induction and recovery. *Journal of Zoo and Wildlife Medicine* **36**(1), 62–69
- ^e Barter LS, Hawkins MG, Brosnan RJ, Antognini JF and Pypendop BH (2006) Median effective dose of isoflurane, sevoflurane, and desflurane in green iguanas. *American Journal of Veterinary Research* **67**(3), 392–397
- ^f Stetter MD, Rahael B, Indiviglio F *et al.* (1999) Isoflurane anaesthesia in amphibians: comparison of five application methods. *Proceedings of the Annual Conference of the American Association of Zoo Veterinarians*, pp. 255–257

Itraconazole

(Fungitraxx, Itrafungol, Sporanox*) **POM, POM-V**

Formulations: Oral: 100 mg capsule, 10 mg/ml oral solution; 50 mg tablets available as veterinary special. **VSP**

Action: Triazole antifungal agent that inhibits the cytochrome systems involved in the synthesis of ergosterol in fungal cell membranes, causing increased cell wall permeability and allowing leakage of cellular contents.

Use: Authorized for use in Psittaciformes, Falconiformes, Accipitriformes, Strigiformes and Anseriformes for treatment of aspergillosis and in Psittaciformes for treatment of candidiasis (Fungitraxx). Treatment of

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blastomycosis, coccidioidomycosis, cryptococcosis, sporotrichosis, histoplasmosis, a variety of dermatomycoses and Malassezia. It is widely distributed in the body, although low concentrations are found in tissues with low protein contents, e.g. CSF, ocular fluid and saliva. Itraconazole extends the activity of methylprednisolone. Treatment of chytridiomycosis in amphibians. Treatment of systemic mycosis in fish.

Safety and handling: Normal precautions should be observed.

Contraindications: Pregnancy. Avoid use if liver disease is present.

Adverse reactions: Vomiting, diarrhoea, anorexia, salivation, depression and apathy, abdominal pain, hepatic toxicosis, drug eruption, ulcerative dermatitis and limb oedema have been reported. It has a narrow safety margin in birds and should be discontinued if emesis or anorexia occurs. Avoid or use with great care in Grey Parrots as it is not well tolerated.

Drug interactions: In humans, antifungal imidazoles and triazoles inhibit the metabolism of antihistamines (particularly terfenadine), oral hypoglycaemics, antiepileptics and glucocorticoids. Antacids, omeprazole, H₂ antagonists and adsorbents may reduce the absorption of itraconazole. Plasma concentrations of ciclosporin, digoxin, benzodiazepines, glucocorticoids and vincristine may be increased by itraconazole.

DOSES

Mammals: Ferrets: dermatophytosis: 15 mg/kg p.o. q24h;

Rabbits: dermatophytosis: 10 mg/kg p.o. q24h for 15 days; pulmonary aspergillosis: 20–40 mg/kg p.o. q24h; **Rodents:** 2.5–10 mg/kg p.o. q24h;

Mice: blastomycosis: 50–150 mg/kg p.o. q24h; **Primates:** 10 mg/kg p.o. q24h (gastroenteritis), 5–10 mg/kg p.o. q12h (dermatophytosis); **Sugar gliders:** 5–10 mg/kg p.o. q24h; **Hedgehogs:** 5–10 mg/kg p.o. q24h; 10 mg/kg p.o. q12h^a.

Birds:

- Candidiasis: 10 mg/kg p.o. q24h for 14 days.
- Aspergillosis treatment: 5–10 mg/kg p.o. q12–24h^{b,c} for 8 weeks, repeat course if fungal elements still remain; **Raptors:** Doses of up to 20 mg/kg p.o. q12–24h may be used; **Grey Parrots:** Avoid or use with great care and only at lowest dose and longest interval (voriconazole at 12–18 mg/kg p.o. q12h or terbinafine more advisable).
- Aspergillosis prophylaxis: 10–20 mg/kg p.o. q24h in susceptible species (e.g. Gyrs and their hybrids, Goshawks, Snowy Owls, Golden Eagles) during stressful events (e.g. manning) where drug therapy should be started 5 days before the stressful event starts.

Reptiles: 5–10 mg/kg p.o. q24h (most species). Metabolically scaled doses have been suggested for the treatment of *Aspergillus* infections^d; **Spiny lizard:** 23.5 mg/kg p.o. q24h for 3 days (pharmacokinetic study maintained systemic itraconazole levels for a further 6 days)^e.

Amphibians: 0.0025% bath (2.50 ml aqueous itraconazole (1% Sporanox oral solution) to 1 l amphibian Ringer's solution); 5 minute bath daily for 6 days has been shown to be the lowest effective treatment concentration for chytridiomycosis. Up to 0.01% bath may be required for more severe infections. Do not use with larvae^f.

Fish: 1–5 mg/kg p.o. q24h for 1–7 days.

References

^a Bexton S and Nelson H (2016) Comparison of two systemic antifungal agents, itraconazole and terbinafine, for the treatment of dermatophytosis in European hedgehogs (*Erinaceus europaeus*). *Veterinary Dermatology* **27**(6), 500–504

- ^b Jones MP, Orosz SE, Cox SK and Frazier DL (2000) Pharmacokinetic disposition of itraconazole in red-tailed hawks (*Buteo jamaicensis*). *Journal of Avian Medicine and Surgery* **14**(1), 15–22
- ^c Orosz SE (1996) Pharmacokinetic properties of itraconazole in blue-fronted Amazon parrots (*Amazona aestiva aestiva*). *Journal of Avian Medicine and Surgery* **10**(3), 168–173
- ^d Girling SJ and Fraser MA (2009) Treatment of *Aspergillus* species infection in reptiles with itraconazole at metabolically scaled doses. *Veterinary Record* **165**(2), 52–54
- ^e Gamble KC, Alvarado TP and Bennett CL (1997) Itraconazole plasma and tissue concentrations in the spiny lizard (*Sceloporus* spp.) following once daily dosing. *Journal of Zoo and Wildlife Medicine* **28**(1), 89–93
- ^f Brannelly LA (2014) Reduced itraconazole concentration and durations are successful in treating *Batrachochytrium dendrobatidis* infection in amphibians. *Journal of Visualized Experiments* **85**, e51166. doi: 10.3791/51166

Ivermectin

(Animec, Anti-Parasite Spot On, Bimectin, Ivomec, Panomec, Qualimec, Virbamec, **Xeno**) **POM-V, POM-VPS, ESPA**

Formulations: **Injectable:** 1% w/v solution. **Topical:** 10 µg (micrograms), 25 µg, 50 µg, 150 µg, 450 µg spot-on tubes; 1 mg/ml, 10 mg/ml drops; 200 µg/ml spray.

Action: Interacts with GABA and glutamate-gated channels leading to flaccid paralysis of parasites.

Use: Prevention and treatment of internal and external parasites in mammals, birds and reptiles (except chelonians).

Safety and handling: Normal precautions should be observed.

Contraindications: Highly toxic to many species of fish and other aquatic organisms. Do not use in chelonians as can result in neurotoxicity and death. Indigo snakes and skinks have also been suggested to be sensitive to ivermectin so administration to these species is not recommended.

Adverse reactions: Neurotoxicity may be seen if it crosses mammalian blood–brain barrier.

Drug interactions: Dose adjustments may be required when administered concurrently with other therapeutic agents transported by P-glycoprotein.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Mammals: Most species: 0.2–0.5 mg/kg s.c., p.o. topically q7–14d; **Primates:** 0.2 mg/kg p.o., s.c., i.m., or topically, repeat in 14 days, 0.1 mg/kg p.o. for 3 alternate days (*Trichuris*)^a; **Sugar gliders,** **Hedgehogs:** 0.2–0.4 mg/kg s.c., p.o., repeat at 14, 28 days and 42 days (acariasis, nematodes).

Birds: 200 µg/kg i.m., s.c., p.o. q7–14d^b; **Raptors:** Capillariasis: 0.5–1 mg/kg i.m., p.o. q7–14d; *Serratospiculum*: 1 mg/kg p.o., i.m. q7–14d (moxidectin or doramectin may be given at same dose rates); **Passerines,** **Small psittacids:** Systemic dosing as above or 0.2 mg/kg applied topically to skin using 0.02% solution (in propylene glycol) q7–14d; **Pigeons:** 0.5 ml applied topically to bare skin using 0.02% solution q7–14d; **Ornamental birds:** Mites and lice: 1 drop of 0.1% solution/50 g bodyweight weekly for 3 weeks.

Reptiles: 0.2 mg/kg s.c., p.o. once, repeat in 10–14 days until negative

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for parasites; may use as environmental control for snake mites (*Ophionyssus natricis*) at dilution of 5 mg/l water sprayed in tank q7–10d (if pre-mix ivermectin with propylene glycol this facilitates mixing with water). Do not use in chelonians.

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Amphibians: 0.2–0.4 mg/kg p.o. q14d or 2 mg/kg topically, repeat in 14–21 days or 10 mg/l for 60 minute bath, repeat in 14 days.

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Fish: Do not use.

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References

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^a Moudgil AD and Singla LD (2018) Molecular confirmation and anthelmintic efficacy assessment against natural trichurid infections in zoo-housed non-human primates. *Journal of Medical Primatology* **47**(6), 388–392

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^b Lierz M (2001) Evaluation of the dosage of ivermectin in falcons. *Veterinary Record* **148**(19), 596–600

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Kaolin

(Kaogel VP, Prokolin) **AVM-GSL, general sale**

Formulations: Oral: Kaogel VP: aqueous suspension containing Kaolin Light. Combination products with pectin, magnesium trisilicate, aluminium hydroxide and phosphate, bismuth salts, calcium carbonate or tincture of morphine are also available.

Action: Adsorbent antidiarrhoeal agent with possible antisecretory effect.

Use: Treatment of diarrhoea of non-specific origin. Although stool consistency may improve, studies do not show that fluid balance is corrected or that the duration of morbidity is shortened.

Rarely indicated in birds; should always establish if loose droppings are true diarrhoea or polyuria. If diarrhoea, dietary indiscretion is unusual and will usually need to perform cytology and culture to establish presence of infectious agents.

Safety and handling: Normal precautions should be observed.

Contraindications: Intestinal obstruction or perforation.

Adverse reactions: No information available, but the use of kaolin is not advisable in rabbits, guinea pigs and chinchillas due to the risk of decreasing GI motility.

Drug interactions: May decrease the absorption of lincomycin, trimethoprim and sulphonamides.

DOSES

Mammals: Ferrets, Hamsters: 1–2 ml/kg kaolin/pectin p.o. q2–6h; Primates: 0.5–1 ml/kg kaolin/pectin p.o. q2–6h.

Birds: Kaolin/pectin mixture: 15 ml/kg p.o. once.

Reptiles, Amphibians, Fish: No information available.

Ketamine

(Anaestamine, Anesketin, Ketabel, Ketamidor, **Ketaset injection**, Ketavet, Ketexx, Narketan-10, Nimatek, Vetalar-V) **POM-V CD SCHEDULE 2**

Formulations: Injectable: 100 mg/ml solution.

Action: Antagonizes the excitatory neurotransmitter glutamate at N-methyl-D-aspartate (NMDA) receptors in the CNS. It interacts with opioid receptors in a complex fashion, antagonizing mu receptors, whilst showing agonist actions at delta and kappa receptors. It does not interact with GABA receptors.

Use: Authorized in rabbits, guinea pigs, hamsters, rats and mice (Anaestamine) and in primates (Ketaset, Vetalar) for sedation or anaesthesia. Provision of chemical restraint or dissociative anaesthesia. Ketamine may also provide profound visceral and somatic analgesia and inhibits central sensitization through NMDA receptor blockade and is used to provide perioperative analgesia as an adjunctive agent, although optimal doses to provide analgesia have not been elucidated. Dissociative anaesthesia is associated with mild stimulation of cardiac output and blood pressure, modest respiratory depression and the preservation of cranial nerve reflexes. For example, the eyes remain

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A open during anaesthesia and should be protected using a bland ophthalmic ointment. Used alone at doses adequate to provide general anaesthesia ketamine causes skeletal muscle hypertonicity and movement may occur that is unrelated to surgical stimulation. **B** These effects are normally controlled by the co-administration of **C** alpha-2 adrenergic agonists and/or benzodiazepines. When ketamine is combined with alpha-2 agonists (such as medetomidine **D** or dexmedetomidine) reversal of the alpha-2 agonist should be delayed until 45 min after ketamine administration. In fish, ketamine **E** does not induce a surgical plane of anaesthesia on its own and has been used in combination with medetomidine. Elasmobranchs **F** (sharks and rays) are more susceptible to the effects of ketamine than bony fish.

G **Safety and handling:** Normal precautions should be observed.

H **Contraindications:** Not recommended for animals whose eyes are at risk of perforation or who have raised intraocular pressure.

I **Adverse reactions:** Cardiovascular depression, rather than stimulation, and arrhythmias may arise in animals with a high sympathetic nervous system tone (e.g. animals in shock or severe cardiovascular disease). Tachycardia can also arise after administration of high doses i.v. **J** Respiratory depression may be marked in some animals. Ketamine may result in spacey, abnormal behaviour for 1–2 hours during recovery. **K** Prolonged administration of ketamine by infusion may result in drug accumulation and prolong recovery. Intramuscular injection may be painful.

L **Drug interactions:** No information available.

M **DOSES**

N **When used for sedation is generally given as part of a combination.**
O **See Appendix for sedation protocols in all species.**

P **Mammals:**

- Q** • **Ferrets:** 10–30 mg/kg i.m., s.c. alone gives immobilization and some analgesia but there is poor muscle relaxation and prolonged recovery. For perioperative/postoperative analgesia: 0.3–1.2 mg/kg/h CRI following 2–5 mg/kg i.v. loading dose. For postoperative analgesia: 0.1–0.4 mg/kg/h CRI.
- R** • **Rabbits:** 15–30 mg/kg i.m., s.c. alone gives moderate to heavy sedation with some analgesia but there is poor muscle relaxation and prolonged recovery. Intraoperatively: May be used as infusion of 10 µg (micrograms)/kg/min; postoperatively: 2–5 µg/kg/min; both preceded by a 250–500 µg/kg loading dose.
- S** • **Guinea pigs:** 10–50 mg/kg i.m., s.c. will provide immobilization, with little muscle relaxation and some analgesia, however it is suggested to use the lower end of the dose range to sedate first and then induce with isoflurane or sevoflurane, or give as part of a combination.
- T** • **Other rodents and small mammals:** 10–50 mg/kg i.m., s.c. will provide immobilization, however it is suggested to use the lower end of the dose range and to induce anaesthesia with isoflurane or sevoflurane, or use in combination with other agents.
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Birds: Largely superseded by gaseous anaesthesia.

Reptiles: **Chelonians:** 20–60 mg/kg i.m., i.v.^a; **Lizards:** 25–60 mg/kg i.m., i.v.; **Snakes:** 20–80 mg/kg i.m., i.v.; All doses for ketamine given alone,

but this is likely to result in variable sedation, poor muscle relaxation and prolonged recovery at higher dose rates. Usually used at lower doses combined with alpha-2 agonists and/or opioids/midazolam/alfaxalone to provide deep sedation/light anaesthesia. Lower doses are recommended in debilitated reptiles.

Amphibians: 50–150 mg/kg s.c., i.m. alone has long induction and recovery times. It is suggested to use 20–40 mg/kg i.m. in combination with diazepam at 0.2–0.4 mg/kg i.m.

Fish: 66–88 mg/kg i.m. alone or 1–2 mg/kg i.m. ketamine in combination with 0.05–0.1 mg/kg i.m. medetomidine, reversed with 0.2 mg/kg i.m. atipamezole.

References

- ^a Holz P and Holz RM (1994) Evaluation of ketamine, ketamine/xylazine and ketamine/midazolam anesthesia in red-eared sliders (*Trachemys scripta elegans*). *Journal of Zoo and Wildlife Medicine* **25**(4), 531–53

Ketoprofen

(Ketofen) **POM-V**

Formulations: Injectable: 1% solution. Oral: 5 mg, 20 mg tablets.

Action: COX-1 inhibition reduces the production of prostaglandins, while lipoxygenase enzyme inhibition has a potent effect on the vascular and cellular phases of inflammation. It has antipyretic, analgesic and anti-inflammatory effects.

Use: Relief of acute pain from musculoskeletal disorders and other painful disorders. Management of chronic pain. Ketoprofen is not COX-2 selective and is not recommended for preoperative administration. Do not administer perioperatively until the animal is fully recovered from anaesthesia and normotensive. Liver disease will prolong the metabolism of ketoprofen, leading to the potential for drug accumulation and overdose with repeated dosing. Administration of ketoprofen to animals with renal disease must be carefully evaluated. Ketoprofen has shown no significant analgesic effect in chain dogfish or koi, although it had some anti-inflammatory activity in koi that reduced muscle damage. It has also shown a significant reduction in the minimum anaesthetic concentration in goldfish.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not give to dehydrated, hypovolaemic or hypotensive patients or those with GI disease or blood clotting problems. Do not give to pregnant animals or animals <6 weeks of age. Do not use in vultures.

Adverse reactions: GI signs may occur in all animals after NSAID administration. Stop therapy if this persists beyond 1–2 days. Some animals develop signs with one NSAID and not another. A 1–2-week wash-out period should be allowed before starting another NSAID after cessation of therapy. Stop therapy immediately if GI bleeding is suspected. There is a small risk that NSAIDs may precipitate cardiac failure in humans and this risk in animals is unknown.

Drug interactions: Do not administer concurrently or within 24 hours of other NSAIDs and glucocorticoids. Do not administer with other potentially nephrotoxic agents (e.g. aminoglycosides).

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DOSES

Mammals: Ferrets: 1–3 mg/kg p.o., s.c., i.m. q24h; Rabbits: 1–3 mg/kg i.m., s.c. q24h; Hamsters, Gerbils, Rats: Up to 5 mg/kg p.o., i.m., s.c. q24h; **Other rodents:** 1–3 mg/kg s.c., i.m. q12–24h.

Birds: 1–5 mg/kg i.m. q8–24h^a. Do not use in vultures^b.

Reptiles: Green iguanas: 2 mg/kg i.v. no more frequently than q24h, longer dosing intervals have been suggested based on pharmacokinetic data; Bearded dragons: 2 mg/kg i.m. q24h^c.

Fish: 0.5–2 mg/kg i.m.^{d,e}.

Amphibians: No information available.

References

- ^a Graham JE, Kollias-Baker C, Craigmill AL, Thomasy SM and Tell LA (2005) Pharmacokinetics of ketoprofen in Japanese quail (*Coturnix japonica*). *Journal of Veterinary Pharmacology and Therapeutics* **28**(4), 399–402
- ^b Naidoo V, Wolter K, Cromarty D *et al.* (2010) Toxicity of non-steroidal anti-inflammatory drugs to Gyps vultures: a new threat from ketoprofen. *Biology Letters* **6**(3), 339–341
- ^c Vigneault A, Lair S, Gara-Boivin C, Beauchamp G and Vergneau-Grosset C (2022) Evaluation of the safety of multiple intramuscular doses of ketoprofen in bearded dragons (*Pogona vitticeps*). *Journal of Herpetological Medicine and Surgery* **32**(2), 123–129
- ^d Harms CA, Lewbart GA, Swanson CR, Kishimori JM and Boylan SM (2005) Behavioral and clinical pathology changes in koi carp (*Cyprinus carpio*) subjected to anesthesia and surgery with and without intra-operative analgesics. *Comparative Medicine* **55**(3), 221–226
- ^e Ward JL, McCartney SP, Chinnadurai SK and Posner LP (2012) Development of a minimum-anesthetic-concentration depression model to study the effects of various analgesics in goldfish (*Carassius auratus*). *Journal of Zoo and Wildlife Medicine* **43**(2), 214–222

Ketorolac

(Acular* Ketorolac trometamol) **POM**

Formulations: Ophthalmic: 0.5% drops in 5 ml bottle. Injectable: 10 mg/ml, 30 mg/ml solution.

Action: COX inhibitor that reduces the production of prostaglandins and therefore reduces inflammation.

Use: Treatment of anterior uveitis and ulcerative keratitis when topical corticosteroids are contraindicated. Topical NSAIDs have the potential to increase intraocular pressure and should be used with caution in animals predisposed to glaucoma.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: As with other topical NSAIDs, ketorolac trometamol may cause local irritation. Topical NSAIDs can be used in ulcerative keratitis but with caution as they can delay epithelial healing. Topical NSAIDs have been associated with an increased risk of corneal 'melting' (keratomalacia) in humans, although this has not been reported in the veterinary literature. Regular monitoring is advised.

Drug interactions: Ophthalmic NSAIDs may be used safely with other ophthalmic pharmaceuticals although concurrent use of drugs which adversely affect the corneal epithelium (e.g. gentamicin) may lead to increased corneal penetration of the NSAID. The concurrent use of topical NSAIDs with topical corticosteroids has been identified as a risk factor in humans for precipitating corneal problems.

DOSES

Mammals: 1 drop per eye q6–24h depending on severity of inflammation^a.

Birds: 1 drop per eye q12h.

Reptiles: Eastern box turtles: 0.25 mg/kg i.m. q24h^b.

Amphibians, Fish: No information available.

References

- ^a Waterbury L and Flach A (2006) Comparison of ketorolac tromethamine, diclofenac sodium and loteprednol etabonate in an animal model of ocular inflammation. *Journal of Ocular Pharmacology and Therapeutics* **22**, 155–159
- ^b Cerreta AJ, Masterson CA, Lewbart GA, Dise DR and Papich MG (2019) Pharmacokinetics of ketorolac in wild Eastern box turtles (*Terrapene carolina carolina*) after single intramuscular administration. *Journal of Veterinary Pharmacology and Therapeutics* **42(2)**, 154–159

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Lactulose(Duphalac*, Lactugal*, Lactulose*, Laevolac*) **P****Formulations:** Oral: 3.3 g/5 ml lactulose in a syrup base. Lactugal is equivalent to 62–74% w/v of lactulose.**Action:** Metabolized by colonic bacteria resulting in the formation of low molecular weight organic acids (lactic, formic, acetic acids). These acids increase osmotic pressure, causing a laxative effect, acidifying colonic contents and thereby trapping ammonia as ammonium ions, which are then expelled with the faeces.**Use:** Used to reduce blood ammonia levels in patients with hepatic encephalopathy and to treat constipation. Reduce the dose if diarrhoea develops. Some animals do not like the taste of lactulose. An alternative is lactitol (β -galactosidosorbitol) as a powder to add to food (500 mg/kg/day in 3 or 4 doses, adjusted to produce two or three soft stools per day), although its efficacy in the management of hepatic encephalopathy has not been extensively evaluated.**Safety and handling:** Normal precautions should be observed.**Contraindications:** Do not administer orally to severely encephalopathic animals at risk of inhalation.**Adverse reactions:** Excessive doses cause flatulence, diarrhoea, cramping and dehydration.**Drug interactions:** Synergy may occur when lactulose is used with oral antibiotics (e.g. neomycin). Do not use lactulose with other laxatives. Oral antacids may reduce the colonic acidification efficacy of lactulose. Lactulose syrup contains some free lactose and galactose, and so may alter insulin requirements in diabetic patients.**DOSES****Mammals:** Ferrets: 0.15–0.75 ml/kg p.o. q12h; Rodents: 0.5 ml/kg p.o. q12h.**Birds:** Appetite stimulant, hepatic encephalopathy: 0.2–1 ml/kg p.o. q8–12h.**Reptiles:** 0.5 ml/kg p.o. q24h.**Amphibians, Fish:** No information available.**Latanoprost**(Latanoprost*, Xalatan*) **POM****Formulations:** Ophthalmic: 50 μ g (micrograms)/ml (0.005%) solution in 2.5 ml bottle (or 0.2 ml vials).**Action:** Agonist for receptors specific for prostaglandin F. It reduces intraocular pressure by increasing outflow.**Use:** Management of primary glaucoma and is useful in the emergency management of acute primary glaucoma (superseding mannitol, acetazolamide and dichlorphenamide). Often used in conjunction with other topical antiglaucoma drugs such as carbonic anhydrase inhibitors. Latanoprost may be useful in the management of lens subluxation despite being contraindicated in anterior lens luxation. Latanoprost has comparable activity to travatoprost.

Safety and handling: Store in refrigerator until opened (then at room temperature).

Contraindications: Uveitis and anterior lens luxation.

Adverse reactions: Miosis, conjunctival hyperaemia and mild irritation may develop. Increased iridal pigmentation has been noted in humans.

Drug interactions: Do not use in conjunction with thiomersal-containing preparations.

DOSES

Mammals: Rabbits: 1 drop per eye q8–24h.

Birds, Reptiles, Amphibians, Fish: No information available.

Levamisole

(Chanaverm, Levacide, Levamole, Levasure) **POM-VPS, ESPA**

Formulations: Injectable: 7.5% solution. Oral: 7.5% oral solution.

Action: Imidazothiazoles act by interfering with parasite nerve transmission causing muscular spasm and rapid expulsion from the host.

Use: Treatment of nematodes, particularly lungworm, in hedgehogs. Not recommended in birds as low therapeutic index and severe toxicity reported in some species. Treatment of non-encysted GI nematodes in fish (e.g. *Camallanus*) and some immunostimulatory effects have been reported in some species. Limited efficacy against nematode eggs, therefore repeat treatment after 3 weeks.

Safety and handling: Normal precautions should be observed.

Contraindications: Avoid in debilitated animals

Adverse reactions: May cause sterility in zebrafish. May cause anorexia at higher doses.

Drug interactions: No information available.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Mammals: Primates: 7.5 mg/kg s.c., repeat after 14 days if necessary, 5–10 mg/kg p.o., repeat in 14 days; **Hedgehogs:** 10 mg/kg s.c., repeat in 48h, and then after 14 days if necessary; **Sugar gliders:** 10 mg/kg p.o.

Birds: Not recommended.

Reptiles: 5–10mg/kg s.c., i.m. once^a.

Amphibians: 6.5–13.5 mg/kg applied topically, repeat in 10 days^b.

Fish: 2.5–10 mg/kg p.o. (in feed) q24h for 7 days or 1–2 mg/l by immersion for 24 hours or 50 mg/l by immersion for 2 hours. Follow manufacturer's recommendations for proprietary formulations.

References

- ^a Corum O, Corum DD, Atik O *et al.* (2019) Pharmacokinetics of levamisole in the red-eared slider turtle (*Trachemys scripta elegans*). *Journal of Veterinary Pharmacology and Therapeutics* **42**(6), 654–659
- ^b Bianchi CM, Johnson CB, Howard LL and Crump P (2014) Efficacy of fenbendazole and levamisole treatments in captive Houston Toads (*Bufo [Anaxyrus] houstonensis*). *Journal of Zoo and Wildlife Medicine* **45**(3), 564–568

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A **Levetiracetam (S-Etiracetam, Levetirasetam)** (Desitrend*, Keppra*) **POM**

B Formulations: Oral: 250 mg, 500 mg, 750 mg and 1 g tablets;
C 100 mg/ml oral 300 ml solution; coated granules 250 mg/sachet.
D **Injectable:** 100 mg/ml intravenous solution (5 ml vial); formulated with
 sodium chloride for injection at 500 mg/100 ml, 1000 mg/100 ml and
 1500 mg/100 ml.

E **Action:** The mechanism of anticonvulsant action is unknown but has
 been shown to bind to the synaptic vesicle protein SV2A within the
F brain, which may protect against seizures.

G **Use:** As primary or adjunctive maintenance therapy for management of
 epileptic seizures refractory to conventional therapy. Used at a higher
 dose, in addition to conventional maintenance therapy, as pulse therapy
H for cluster seizures. Constant intravenous infusion used for emergency
 control of status epilepticus. Levetiracetam is rapidly absorbed from the GI
I tract with peak plasma concentrations reached in <2 hours of oral dosing.
 Steady state is rapidly achieved within 2 days. Plasma protein binding is
J minimal. The plasma half-life is short, being around 7 hours in human
 patients. Withdrawal of levetiracetam therapy or transition to or from
 another type of antiepileptic therapy should be done gradually. Use with
K caution and in reduced doses in patients with renal and severe hepatic
 impairment. Levetiracetam has also been used for the management of
L neuropathic pain (similar to gabapentin) in human patients.

M **Safety and handling:** Normal precautions should be observed.

N **Contraindications:** No information available.

O **Adverse reactions:** The most commonly reported adverse effects in
 humans are sedation, weakness and dizziness. Blood dyscrasias such as
 neutropenia, pancytopenia and thrombocytopenia may also develop.

P **Drug interactions:** Minimal, although there is some evidence to
 suggest that enzyme-inducing anticonvulsants (including phenobarbital
Q and phenytoin) may modestly reduce levetiracetam levels, but not by
 clinically relevant amounts.

R **DOSES**

S **Mammals:** Prairie dogs: 20 mg/kg p.o. q8h.

T **Birds:** 50 mg/kg p.o. q8h; 100–150 mg/kg p.o. q12h; individual variations
 seen so drug monitoring essential^{a,b}.

U **Reptiles, Amphibians, Fish:** No information available.

V **References**

- W** ^a Schnellbacher R, Beaufrière H, Arnold RD *et al.* (2014) Pharmacokinetics of
 levetiracetam in healthy Hispaniolan Amazon parrots (*Amazona ventralis*) after oral
 administration of a single dose. *Journal of Avian Medicine and Surgery* **28**(3), 193–200
X ^b Kabakchiev C, Lanieste D, James F *et al.* (2020) Diagnosis and long-term management
 of post-traumatic seizures in a white-crowned pionus (*Pionus senilis*). *Journal of the*
Y *American Veterinary Medical Association* **256**(10), 1145–1152
Z

Levothyroxine see **L-Thyroxine**

Lidocaine (Lignocaine)

(EMLA, Intubeaze, Lignadrin, Lignol, Locaine, Locovetic, Lidoderm*) **POM-V**

Formulations: Injectable: 1%, 2% solutions (some contain adrenaline).

Topical: 2% solution (Intubeaze), 4% solution (Xylocaine); 2.5% cream with prilocaine (EMLA); 5% transdermal patches (Lidoderm).

Action: Local anaesthetic action is dependent on reversible blockade of the sodium channel, preventing propagation of an action potential along the nerve fibre. Sensory nerve fibres are blocked before motor nerve fibres, allowing a selective sensory blockade at low doses. Lidocaine also has class 1b antiarrhythmic actions, decreasing the rate of ventricular firing, action potential duration and absolute refractory period, and increasing the relative refractory period. Lidocaine has a rapid onset of action and intermediate duration of action. Addition of adrenaline to lidocaine increases the duration of action by reducing the rate of systemic absorption.

Use: Provision of local or regional analgesia using perineural, infiltration, local i.v. or epidural techniques. It is generally recommended that adrenaline-free solutions be used for epidural administration. Also used to provide systemic analgesia when given i.v. by continuous rate infusion. First-line therapy for rapid or haemodynamically significant ventricular arrhythmias. May also be effective for some supraventricular arrhythmias, such as bypass-mediated supraventricular tachycardia, and for cardioversion of acute-onset or vagally-mediated atrial fibrillation. Widely used topically to desensitize mucous membranes (such as the larynx prior to intubation). EMLA cream is used to anaesthetize the skin before vascular cannulation. It must be placed on the skin for approximately 45–60 min to ensure adequate anaesthesia; covering the skin with an occlusive dressing promotes absorption. EMLA is very useful to facilitate venous catheter placement in the ears of conscious rabbits. Infusions of lidocaine reduce the inhaled concentrations of anaesthetic required to produce anaesthesia and prevent central sensitization to surgical noxious stimuli. Systemic lidocaine is best used in combination with other analgesic drugs to achieve balanced analgesia. Systemic lidocaine, as well as improving GI motility in painful rabbits by providing analgesia, also has a directly prokinetic effect. Lidocaine will accumulate after prolonged administration, leading to a delayed recovery.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not give lidocaine solutions containing adrenaline i.v. Do not use solutions containing adrenaline for complete ring block of an extremity because of the danger of ischaemic necrosis. Do not use preparations with adrenaline in birds. Use cautiously in small fish: do not overdose small fish or exceed 1–2 mg/kg total dose.

Adverse reactions: Depression, seizures, muscle fasciculations, vomiting, bradycardia and hypotension. Pronounced systemic effects including respiratory depression and sedative effects have been observed in amphibians at high dosages. If reactions are severe, decrease or discontinue administration. Seizures may be controlled with i.v. diazepam or pentobarbital. Monitor the ECG carefully during therapy.

Drug interactions: Cimetidine and propranolol may prolong serum lidocaine clearance if administered concurrently. Other antiarrhythmics may cause increased myocardial depression.

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DOSES

Note: 1 mg/kg is 0.05 ml/kg of a 2% solution.

Mammals: Local anaesthesia: Apply to the affected area with a small gauge needle to an appropriate volume. Total dose that should be injected should not exceed 4 mg/kg (ideally 1–3 mg/kg). **Rabbits:** 0.3 ml/kg for epidural anaesthesia using 2% solution or 1–2 mg/kg i.v. bolus for cardiac arrhythmias or 2–4 mg/kg intratracheal for cardiac arrhythmias. Intraoperative analgesia given by continuous rate infusion: 1–2 mg/kg loading dose (given slowly over 10–15 min) followed by 50–100 µg (micrograms)/kg/min. Postoperatively, similar dose rates can be used but should be adjusted according to pain assessment and be aware of the likelihood of accumulation, allowing an empirical reduction in dose rate over time^{a,b}; Topical: Apply thick layer of cream to the skin and cover with a bandage for 45–60 min prior to venepuncture.

Birds: <4 mg/kg as local infusion/nerve block. Do not use preparations with adrenaline^c.

Reptiles, Amphibians: 1–2 mg/kg as local infusion/nerve block; Neuraxial anaesthesia: 2 mg/kg in bearded dragons^d; 4 mg/kg in turtles^e. Do not exceed 5 mg/kg total dose per animal due to risk of side effects including cardiotoxicity, respiratory depression and sedation.

Fish: 1–2 mg/kg as local infusion/nerve block¹; Do not exceed 1–2 mg/kg total dose in small fish.

References

- ^a Schnellbacher RW, Carpenter JW, Mason DE *et al.* (2013) Effects of lidocaine administration via continuous rate infusion on the minimum alveolar concentration of isoflurane in New Zealand White rabbits (*Oryctolagus cuniculus*). *American Journal of Veterinary Research* **74**(11), 1377–1384
- ^b Schnellbacher RW, Divers SJ, Comolli JR *et al.* (2017) Effects of intravenous administration of lidocaine and buprenorphine on gastrointestinal tract motility and signs of pain in New Zealand White rabbits after ovariohysterectomy. *American Journal of Veterinary Research* **78**(12), 1359–1371
- ^c da Cunha AF, Strain GM, Rademacher N, Schellbacher R and Tully TM (2013) Palpation and ultrasound-guided brachial plexus blockade in Hispaniolan Amazon parrots (*Amazona ventralis*). *Veterinary Anaesthesia and Analgesia* **40**(1), 96–102
- ^d Ferreira TH and Mans C (2019) Evaluation of neuraxial anesthesia in bearded dragons (*Pogona vitticeps*). *Veterinary Anaesthesia and Analgesia* **46**(1), 126–134
- ^e Mans C (2014) Clinical technique: Intrathecal drug administration in turtles and tortoises. *Journal of Exotic Pet Medicine* **23**(1), 67–70
- ¹ Chatigny F, Kamunde C, Creighton CM and Stevens ED (2017) Uses and doses of local anesthetics in fish, amphibians, and reptiles. *Journal of the American Association for Laboratory Animal Science* **56**(3), 244–253

Lignocaine see **Lidocaine**

Lincomycin

(Lincocin, Lincoject, Linco-Sol, Linco-Spectin, Ocnil)

POM-V

Formulations: Injectable: 100 mg/ml solution for i.v. or i.m. use. Oral: powder for solution, some formulations combined with other drugs such as spectinomycin (Linco-Spectin).

Action: Lincosamide antibiotic that inhibits bacterial protein synthesis. It is bacteriostatic or bactericidal, depending on the organism and drug concentration. Being a weak base, it is ion-trapped in fluid that is more

acidic than plasma and therefore concentrates in prostatic fluid, milk and intracellular fluid.

Use: Active against Gram-positive cocci (including penicillin-resistant staphylococci) and many obligate anaerobes. The lincosamides (lincomycin and clindamycin) are particularly indicated for staphylococcal bone and joint infections. Clindamycin is more active than lincomycin, particularly against obligate anaerobes, and is better absorbed from the gut. Administer slowly if using i.v. route.

Safety and handling: Normal precautions should be observed.

Contraindications: Rapid i.v. administration should be avoided since this can result in collapse due to cardiac depression and peripheral neuromuscular blockade. Do not use oral lincosamides in rabbits, other small herbivores or hamsters, including placing them into ontogenic abscesses.

Adverse reactions: Human patients on lincomycin may develop colitis. Although not a major veterinary problem, patients developing diarrhoea (particularly if it is haemorrhagic) whilst taking the medication should be monitored carefully. Toxicity is a possibility in patients with liver disease; weigh the risk versus the potential benefits before use of this drug in such patients. Lincosamides can cause fatal enterotoxaemia in small herbivores and hamsters. May be nephrotoxic in reptiles.

Drug interactions: The action of neuromuscular blocking agents may be enhanced if given with lincomycin. The absorption of lincomycin may be reduced by kaolin. Lincosamide antimicrobials should not be used in combination with chloramphenicols or macrolides as these combinations are antagonistic.

DOSES

Classified as category C (Caution) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Ferrets: 11 mg/kg p.o. q8h; Sugar gliders: 30 mg/kg p.o., s.c., i.m. q24h.

Birds: 50–75 mg/kg p.o., i.m. q12h; **Pigeons:** lincomycin/spectinomycin preparation: 500–750 mg combined activity/l water.

Reptiles: 5 mg/kg i.m. q12–24h.

Amphibians, Fish: No information available.

Liquid paraffin see **Paraffin**

Lomustine (CCNU)

(Lomustine*) **POM**

Formulations: Oral: 40 mg capsule.

Action: Interferes with the synthesis and function of DNA, RNA and proteins. Antitumour activity correlates best with formation of interstrand crosslinking of DNA. Lomustine is highly lipid-soluble, allowing rapid transport across the blood–brain barrier.

Use: Treatment of primary and metastatic brain tumours in humans. Its use in animals is less well defined but has been reported to have some efficacy in the treatment of brain tumours, mast cell tumours, refractory

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A lymphoma, histiocytic sarcoma and epitheliotrophic lymphoma.
B S-Adenosylmethionine and silybin may be used to prevent or treat
 lomustine hepatotoxicity.

C **Safety and handling:** **Cytotoxic drug: see specialist texts for further
 advice on chemotherapeutic agents.**

D **Contraindications:** Bone marrow suppression. Pre-existing liver disease.

E **Adverse reactions:** Myelosuppression is the dose-limiting toxicity, with
 neutropenia developing 7 days after administration. Thrombocytopenia
 can also be seen, often with no other concurrent cytopenias. GI and
 cumulative dose-related and potentially irreversible hepatic toxicity have
 been reported in the dog, but not in the ferret.

F **Drug interactions:** Do not use with other myelosuppressive agents.
G Lomustine requires hepatic microsomal enzyme hydroxylation for the
 production of antineoplastic metabolites. In humans cimetidine
 enhances the toxicity of lomustine.

H DOSES

I **See Appendix for chemotherapy protocols in ferrets.**

J **Mammals:** Ferrets: Anecdotally used in lymphoma at doses extrapolated
 from cats (cat dose: 40–60 mg/m² p.o. q21d, but dosing intervals may
 need to be increased to 6 weeks).

K **Birds, Reptiles, Amphibians, Fish:** No information available.

L

M Loperamide

(Imodium*, Norimode*) **POM, P, GSL**

N **Formulations:** Oral: 2 mg capsule (Imodium, Norimode); 0.2 mg/ml
 syrup (Imodium).

O **Action:** Opioid agonist that alters GI motility by acting on receptors in
 the myenteric plexus. It normally has no central action.

P **Use:** Management of non-specific acute and chronic diarrhoea, and
 irritable bowel syndrome. May be used for symptomatic treatment of
 severe GI motility problems in rabbits, but for as short a period as
 possible to avoid GI stasis. Used in birds for cerebral oedema and anuric
 renal failure.

S **Safety and handling:** Normal precautions should be observed.

T **Contraindications:** Intestinal obstruction.

U **Adverse reactions:** Constipation will occur in some cases.

V **Drug interactions:** No information available.

W DOSES

X **Mammals:** Ferrets: 0.2 mg/kg p.o. q12h; Rabbits: 0.04–0.2 mg/kg p.o.
 q8–12h; Rodents: 0.1 mg/kg p.o. q8h; Primates: 0.04 mg/kg p.o. q8h.

Y **Birds, Reptiles, Amphibians, Fish:** No information available.

Z

Lufenuron

(Program, Program plus) **POM-V, AVM-GSL**

Formulations: Oral: 67.8 mg, 204.9 mg, 409.8 mg tablets (Program); 46 mg, 115 mg, 230 mg, 460 mg lufenuron with milbemycin (ratio of 20 mg lufenuron: 1 mg milbemycin) tablets (Program plus); 133 mg, 266 mg suspension (Program for cats). **Injectable:** 40 mg, 80 mg as 100 mg/ml suspension (Program).

Action: Inhibition of chitin synthetase leads to a failure of chitin production which means that flea eggs fail to hatch.

Use: Prevention of flea infestation (*Ctenocephalides felis*, *C. canis*). For treatment of flea infestations the additional use of an approved adulticide is recommended. All animals in the household should be treated. Lufenuron has an additional antifungal action but specific doses for the effective treatment of dermatophytosis are currently unknown. Tablets/suspension should be administered with food. Can be administered during pregnancy and lactation (Program). Used to treat crustacean ectoparasites (*Argulus* spp.) in freshwater fish and amphibians including tadpoles.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: No information available.

Drug interactions: No information available.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Mammals: Ferrets: 30–45 mg/kg p.o. q1month; Rabbits: 30 mg/kg p.o. q1month.

Amphibians: Axolotls: 0.1 mg/l by immersion as required (ideally use suspension formulation)^a.

Fish: 0.1 mg/l by immersion as required (ideally use suspension formulation)^b.

Birds, Reptiles: No information available.

References

^a MacHale JS and Hedley J (2021) Successful treatment of anchor worm (*Lernaea cyprinacea*) using lufenuron in the Mexican axolotl (*Ambystoma mexicanum*). *Journal of Herpetological Medicine and Surgery* **31**(2), 107–110

^b Mayer J, Hensel P, Mejia-Fava J, Brandão J and Divers S (2013) The use of lufenuron to treat fish lice (*Argulus* spp.) in koi (*Cyprinus carpio*). *Journal of Exotic Pet Medicine* **22**, 65–69

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A **Malachite green (Aniline green, Bright green)** (Malachite) **ESPA**

Formulations: Immersion: Proprietary formulations are available; also available in combination with formalin.

Action: A biocide (carbinol is the lipid-soluble form) which passes through biological membranes and acts as a cellular respiratory toxin.

Use: For the treatment of fungal infections (water mould) of fish and fish eggs. Some effect against protozoan ectoparasites and myxozoan parasites of fish. Some activity against Gram-positive bacteria in fish. It is strongly recommended that a proprietary formulation is used initially to avoid problems related to purity and enable accurate dosing. Malachite green is absorbed through the gills and cellular toxicity is irreversible.

Toxicity is increased at higher temperatures, low pH and low water hardness. It is toxic to some species of small fish and plants. It is inactivated by light, therefore UV and aquarium lights should be switched off during treatment. Manufactured as a dyeing agent; impurities in some forms of malachite green salts are toxic. Must not be used in fish food due to absorption and long persistence of residues in tissues.

Safety and handling: Normal precautions should be observed. Teratogen and suspected carcinogen. Will stain many objects, particularly plastics.

Contraindications: Do not use in tetras, small tropical fish or scaleless species.

Adverse reactions: Overdose, particularly in sensitive species, causes irreversible respiratory toxicity.

Drug interactions: Considered to have a synergistic effect with formalin (formaldehyde solution) against protozoan ectoparasites.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Fish: 50–60 mg/l for 10–30 seconds by immersion or 1–2 mg/l for 30–60 min by immersion or 0.1 mg/l by prolonged immersion or 100 mg/l applied topically to skin lesions; follow manufacturer's recommendations for proprietary formulations.

Mammals, Birds, Reptiles, Amphibians: No information available.

T **Mannitol (Cordycepic acid)** (Mannitol*) **POM**

Formulations: Injectable: 10%, 15%, 20% solutions.

Action: Mannitol is an inert sugar alcohol that acts as an osmotic diuretic.

Use: Reduction of intracranial pressure (most effective in acute elevations of intracranial pressure), treatment of acute glaucoma and may also be used in the treatment of oliguric renal failure. Reduction in intracranial and intraocular fluid pressure occurs within 15 minutes of the start of a mannitol infusion and lasts for 3–8 hours after the infusion is discontinued; diuresis occurs after 1–3 hours. A 5.07% solution in water is isosmotic with serum. It is recommended that an in-line filter be

used when infusing concentrated mannitol. There is some evidence that bolus administration (over 20–30 minutes) may be more effective for reduction of intracranial pressure than continuous administration. When used as treatment for raised intracranial pressure, hypovolaemia should be avoided to maintain cerebral perfusion pressure.

Safety and handling: Any crystals that have formed during storage should be dissolved by warming prior to use. The formation of crystals is a particular problem with the 20% formulations, which are supersaturated.

Contraindications: Prolonged administration may lead to the accumulation of mannitol in the brain and worsening of the cerebral oedema and raised intracranial pressure. Use with care in intracranial haemorrhage (except during intracranial surgery), take care to avoid volume overload in generalized oedema, severe congestive heart failure, pulmonary oedema or anuric renal failure (before rehydration).

Adverse reactions: The most common adverse reactions seen are fluid and electrolyte imbalances. Infusion of high doses may result in circulatory overload and acidosis. Thrombophlebitis may occur and extravasation of the solution may cause oedema and skin necrosis. Mannitol causes diarrhoea if given orally. Rarely mannitol may cause acute renal failure in human patients.

Drug interactions: Diuretic-induced hypokalaemia may occur when ACE inhibitors are used with potassium-depleting diuretics. Potassium-depleting diuretics should be used with care in conjunction with beta-blockers. Nephrotoxicity has been described with concurrent use of mannitol and ciclosporin in human patients. Mannitol may result in temporary impairment of the blood–brain barrier for up to 30 min after administration of high doses. Mannitol should never be added to whole blood for transfusion or given through the same set by which the blood is being infused. Do not add KCl or NaCl to concentrated mannitol solutions (20% or 25%) as a precipitate may form.

DOSES

Mammals: Ferrets: 0.5–1.0 g/kg i.v. infusion over 20 min;

Primates: 0.25–0.5 g/kg i.v. infusion over 5–10 min.

Birds: Acute renal failure and cerebral oedema: 0.2–2 mg/kg slow i.v.

Reptiles, Amphibians, Fish: No information available.

Marbofloxacin

(Aurizon, Efex, Marbocyl, Marboxidin, Marfloquin, Softiflox, Ubiflox) **POM-V**

Formulations: **Injectable:** 200 mg powder for reconstitution giving 10 mg/ml when reconstituted. **Oral:** 5 mg, 20 mg, 80 mg tablets.

Topical: Compound preparation containing 3 mg/ml of marbofloxacin along with clotrimazole and dexamethasone (aural use).

Action: Broad-spectrum bactericidal antibiotic inhibiting bacterial DNA gyrase. The bactericidal effect is concentration-dependent particularly against Gram-negative bacteria, meaning that pulse-dosing regimens may be effective. Low urinary pH may reduce the activity.

Use: Ideally fluoroquinolone use should be reserved for infections where culture and sensitivity testing predicts a clinical response and

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where first- and second-line antimicrobials would not be effective. Active against mycoplasmas and many Gram-positive and particularly Gram-negative organisms, including *Pasteurella*, *Staphylococcus*, *Pseudomonas aeruginosa*, *Klebsiella*, *Escherichia coli*, *Proteus* and *Salmonella*. Fluoroquinolones are effective against beta-lactamase-producing bacteria. Marbofloxacin is relatively ineffective in treating obligate anaerobic infections. Fluoroquinolones are highly lipophilic drugs that attain high concentrations within cells in many tissues and are particularly effective in the management of soft tissue, urogenital (including prostatitis) and skin infections. In rabbits, PU/PD has been noted when using the aural preparation due to uptake of corticosteroids and, ideally, a corticosteroid-free preparation should be compounded when an aural fluoroquinolone is required.

Safety and handling: Normal precautions should be observed.

Contraindications: Fluoroquinolones are relatively contraindicated in growing small mammals as cartilage abnormalities have been reported in young mice, rats, guinea pigs and rabbits after administration of other fluoroquinolones. Care may be needed using fluoroquinolones in breeding females.

Adverse reactions: Some animals show GI signs (nausea, vomiting). Use with caution in epileptics until further information is available, as fluoroquinolones potentiate CNS adverse effects when administered concurrently with NSAIDs in humans.

Drug interactions: Adsorbents and antacids containing cations (Mg^{2+} , Al^{3+}) may bind to fluoroquinolones, preventing their absorption from the GI tract. The absorption of fluoroquinolones may be inhibited by sucralfate and zinc salts; doses should be at least 2 hours apart. Enrofloxacin increases plasma theophylline concentrations. Preliminary data suggest this may not be clinically significant with marbofloxacin unless used in patients with renal insufficiency. Cimetidine may reduce the clearance of fluoroquinolones and should be used with caution in combination.

DOSES

Classified as category B (Restrict) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: 2–10 mg/kg p.o., s.c., i.m. q24h (usually started at ~5 mg/kg p.o. q24h)

Birds: 10 mg/kg p.o., i.m., i.v. q24h^a; **Macaws:** 2.5 mg/kg p.o., i.m. q24h^b.

Reptiles: Ball pythons: 10 mg/kg s.c., i.m., p.o. q48h^c. Chinese soft-shelled turtles: 10 mg/kg p.o., i.m.^d.

Amphibians, Fish: No information available.

References

- ^a García-Montijano M, González F, Waxman S *et al.* (2003) Pharmacokinetics of marbofloxacin after oral administration to Eurasian buzzards (*Buteo buteo*). *Journal of Avian Medicine and Surgery* **17**(4), 185–190
- ^b Carpenter JW, Hunter RP, Olsen JH *et al.* (2006) Pharmacokinetics of marbofloxacin in blue and gold macaws (*Ara ararauna*). *American Journal of Veterinary Research* **67**(6), 947–950
- ^c Coke RL, Isaza R, Koch DE, Pellerin MA and Hunter RP (2006) Preliminary single-dose pharmacokinetics of marbofloxacin in ball pythons (*Python regius*). *Journal of Zoo and Wildlife Medicine* **37**(1), 6–10
- ^d Shan Q, Zheng G, Liu S *et al.* (2015) Pharmacokinetic/pharmacodynamic relationship of marbofloxacin against *Aeromonas hydrophila* in Chinese soft-shelled turtles (*Trionyx sinensis*). *Journal of Veterinary Pharmacology and Therapeutics* **38**(6), 537–542

Maropitant

(Cerenia, Prevomax, Vetemex) **POM-V**

Formulations: Injectable: 10 mg/ml solution. Oral: 16 mg, 24 mg, 60 mg, 160 mg tablets.

Action: Selective NK-1 receptor antagonist, developed to block substance P from binding within the chemoreceptor trigger zone. Highly protein bound, with a long duration of activity (24 hours).

Use: Treatment and prevention of vomiting in susceptible species, including that caused by chemotherapy and motion sickness. In cases of frequent vomiting, treatment by injection is recommended. It should be used in combination with investigation into the cause of vomiting and with other supportive measures and specific treatments. Tablets best given with food; avoid prolonged fasting before administration. Has been suggested to affect visceral pain in rabbits, reducing the response to colorectal stimulation^a, in addition to increasing colonic peristalsis^b. Has been shown to reduce the severity of lesions associated with ulcerative dermatitis in susceptible mice, and may be of benefit in the treatment of non-parasitic, non-specific self-trauma in mice and possibly other rodents^c.

Safety and handling: Normal precautions should be observed. Do not attempt to remove the tablet by pushing through the blister packing as this will damage the tablet.

Contraindications: No specific contraindications but it would be sensible not to use maropitant where GI obstruction or perforation could be present or for longer than 48 hours without a definitive diagnosis. Metabolized by the liver so use with caution in patients with hepatic disease.

Adverse reactions: Transient pain reaction during injection is reported with associated hyperexcitability and tachypnoea, but no significant lasting adverse reactions. Pain on injection can be reduced by injecting the product at refrigerated temperatures.

Drug interactions: No compatibility studies exist, and therefore the injection should not be mixed with any other agent. Should not be used concurrently with calcium-channel antagonists as maropitant has an affinity to calcium-channels. Highly bound to plasma proteins and may compete with other highly bound drugs.

DOSES

Mammals: Ferrets, Rabbits: 1 mg/kg s.c. q24h^d; Mice: Ulcerative dermatitis: 1 mg/kg q24h^e; Hedgehogs: 2 mg/kg s.c. q12h^e; Sugar gliders: 0.2 mg/kg s.c. q24h. Primates: 1 mg/kg s.c. once^f.

Birds, Reptiles, Amphibians, Fish: No information available.

References

- ^a Okano S, Ikeura Y and Inatomi N (2002) Effects of tachykinin NK1 receptor antagonists on the viscerosensory response caused by colorectal distention in rabbits. *Journal of Pharmacology and Experimental Therapeutics* **300**, 925–931
- ^b Onori L, Agio A, Taddei G et al. (2003) Peristalsis regulation by tachykinin NK1 receptors in the rabbit isolated distal colon. *American Journal of Physiology: Gastrointestinal and Liver Physiology* **285**, G325–G331
- ^c Williams-Fritze MJ, Carlson Scholz JA, Zeiss C et al. (2011) Maropitant citrate for treatment of ulcerative dermatitis in mice with a C57BL/6 background. *Journal of the American Association of Laboratory Animal Science* **50**, 221–226

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- ^d Ozawa SM, Hawkins MG, Drzenovich TL, Kass PH and Knych HK (2019) Pharmacokinetics of maropitant citrate in New Zealand White rabbits (*Oryctolagus cuniculus*). *American Journal of Veterinary Research* **80**(10), 963–968
- ^e Martínez-Jiménez D, Garner B, Coutermarsh-Ott S *et al.* (2017) Eosinophilic leukemia in three African pygmy hedgehogs (*Atelerix albiventris*) and validation of Luna stain. *Journal of Veterinary Diagnostic Investigation* **29**(2), 217–223
- ^f Steinbach JR, MacGuire J, Chang S, Dierks E and Roble GS (2018) Assessment of pre-operative maropitant citrate use in macaque (*Macaca fascicularis* and *Macaca mulatta*) neurosurgical procedures. *Journal of Medical Primatology* **47**(3), 178–184

Mavacoxib

(Trocoxil) **POM-V**

Formulations: Oral: 6 mg, 20 mg, 30 mg, 75 mg, 95 mg chewable tablets.

Action: Selectively inhibits COX-2 enzyme, thereby limiting the production of prostaglandins involved in inflammation. The prolonged duration of action of mavacoxib in other species is seen in birds though intervals are shorter than in dogs. As such this is a drug that may be suitable for birds where owner compliance is difficult or where interventions may need to be minimized (e.g. zoo species).

Use: Treatment of pain and inflammation associated with degenerative joint disease. Pharmacokinetic data are available for some birds but there are large species differences in dosing interval. There is no data on efficacy, although mavacoxib did show an effect in an avian inflammatory model^a. Longer term toxicity has not been studied. Some dogs are poor metabolisers of mavacoxib. The situation is unknown in birds, so it is important to evaluate response to treatment.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not give to dehydrated, hypovolaemic or hypotensive patients or those with GI disease or blood clotting problems. Administration of mavacoxib to animals with renal disease must be carefully evaluated. Liver disease prolongs the metabolism of mavacoxib, leading to the potential for drug accumulation and overdose with repeated dosing, therefore, use is not recommended. In birds it is not known what the effects may be in young animals, so is best avoided in juveniles.

Adverse reactions: Should an animal require anaesthesia or develop any illness while receiving mavacoxib, then care must be taken to avoid dehydration, hypotension and hypovolaemia, and prompt intervention to manage these conditions should be implemented if they occur. Although the duration of action of mavacoxib is prolonged, symptomatic management of any side effects associated with drug administration is recommended only until the clinical signs resolve. There is a small risk that NSAIDs may precipitate cardiac failure in humans and this risk in birds is unknown, though other -coxib drugs (e.g. celecoxib) appear well tolerated.

Drug interactions: Do not administer concurrently with other NSAIDs and glucocorticoids. Do not administer another NSAID within 1 month of dosing with mavacoxib. Do not administer with other potentially nephrotoxic agents (e.g. aminoglycosides).

DOSES

Birds: Cockatiels, African grey parrots: 4 mg/kg p.o. q10–12d^{b,c}; Lovebirds: 4 mg/kg p.o. q5d.

Mammals, Reptiles, Amphibians, Fish: No information available.

References

- ^a Gasthuys E, Houben R, Haesendonck R *et al.* (2019) Development of an *in vivo* lipopolysaccharide inflammation model to study the pharmacodynamics of COX-2 inhibitors celecoxib, mavacoxib, and meloxicam in cockatiels (*Nymphicus hollandicus*). *Journal of Avian Medicine and Surgery* **33**(4), 349–36
- ^b Montesinos A, Ardiaca M, De baere S, Devreese M and Antonissen G (2019) Population pharmacokinetics of mavacoxib after oral administration of a single dose to African grey parrots. *Proceedings of the International Conference on Avian, Herpetological and Exotic Mammal Medicine*; London, p130
- ^c Dhondt L, Devreese M, Croubels S, De Baere S, Haesendonck R, Goessens T *et al.* (2017) Comparative population pharmacokinetics and absolute oral bioavailability of COX-2 selective inhibitors celecoxib, mavacoxib and meloxicam in cockatiels (*Nymphicus hollandicus*). *Scientific Reports* **7**(1), 1–12

Mebendazole/Closantel(Supaverm) **POM-VPS****Formulations:** Oral: 7.5% mebendazole and 5% closantel suspension.**Action:** Binds to a structural protein of the parasite microtubules causing paralysis, death and expulsion of the parasites. Closantel uncouples oxidative phosphorylation in the cell mitochondria, resulting in inhibition of ATP synthesis, leading to alterations in energy metabolism and death of the parasite.**Use:** For the treatment of external monogenean trematode infestation in koi carp.**Safety and handling:** Normal precautions should be observed.**Contraindications:** Do not administer to brood fish due to embryotoxic and teratogenic potential. Toxic to many other species.**Adverse reactions:** No information available.**Drug interactions:** No information available.**DOSES****See Appendix for guidelines on responsible parasiticide use.****Fish:** 1 ml/400 l by permanent immersion (equivalent to 0.187 mg/l mebendazole and 0.125 mg/l closantel) ^a.**Mammals, Birds, Reptiles, Amphibians:** No information available.**References**

- ^a Marshall CJ (1999) Use of Supaverm for the treatment of monogenean infestation in koi carp (*Cyprinus carpio*). *Fish Veterinary Journal* **4**, 33–39

Medetomidine(Domitor, Dorbene, Dormilan, Medetor, Sedastart, Sedator, Sededorm) **POM-V****Formulations:** Injectable: 1 mg/ml solution.**Action:** Agonist at peripheral and central alpha-2 adrenoreceptors producing dose-dependent sedation, muscle relaxation and analgesia.**Use:** Provides sedation and premedication when used alone or in combination with opioid analgesics. Medetomidine combined with ketamine is used to provide a short duration (20–30 min) of surgical anaesthesia. Specificity for the alpha-2 receptor is greater for medetomidine than for xylazine, but is lower than for dexmedetomidine.**A****B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A Medetomidine is a potent drug that causes marked changes in the cardiovascular system including an initial peripheral vasoconstriction that results in an increase in blood pressure and a compensatory bradycardia. **B** After 20–30 min vasoconstriction wanes, while blood pressure returns to normal values. Heart rate remains low due to the central sympatholytic effect of alpha-2 agonists. These cardiovascular changes result in a fall in cardiac output; central organ perfusion is well maintained at the expense of redistribution of blood flow away from the peripheral tissues. **C** Respiratory system function is well maintained; respiration rate may fall but is accompanied by an increased depth of respiration. Oxygen supplementation is advisable in all animals. The duration of analgesia from a 10 µg (micrograms)/kg dose is approximately 1 hour in dogs. Combining medetomidine with an opioid provides improved analgesia and sedation. **D** Lower doses of medetomidine should be used in combination with other drugs. Reversal of sedation or premedication with atipamezole shortens the recovery period, which may be advantageous. Analgesia should be provided with other classes of drugs before atipamezole. Using medetomidine in combination with other drugs in the lower dose range can provide good sedation and analgesia with minimal side effects. **E** Similarly to dexmedetomidine, medetomidine may be used in low doses to manage excitation during recovery from anaesthesia and to provide perioperative analgesia when administered by continuous rate infusion. **F** Intramuscular injection of medetomidine has been used in combination with ketamine in elasmobranchs (sharks and rays) and some bony fish. **G** **H** **I** **J** **K** **L**

Safety and handling: Normal precautions should be observed.

M **Contraindications:** Do not use in animals with cardiovascular or other systemic disease. Use in geriatric patients is not advisable. Do not use in pregnant animals. Do not use when vomiting is contraindicated. Not recommended in diabetic animals. **N**

O **Adverse reactions:** Causes diuresis by suppressing ADH secretion, a transient increase in blood glucose by decreasing endogenous insulin secretion, mydriasis and decreased intraocular pressure. Transient lens opacity may be seen in small mammals, especially rats, this rarely lasts more than 1–2 hours. Vomiting after i.m. administration is common, so medetomidine should be avoided when vomiting is contraindicated (e.g. foreign body, raised intraocular pressure). Due to effects on blood glucose, use in diabetic animals is not recommended. Spontaneous arousal from deep sedation following stimulation can occur with all alpha-2 agonists; aggressive animals sedated with medetomidine must still be managed with caution. **P** **Q** **R** **S** **T**

U **Drug interactions:** When used for premedication, medetomidine will significantly reduce the dose of all other anaesthetic agents required to maintain anaesthesia. Drugs for induction of anaesthesia should be given slowly and to effect to avoid inadvertent overdose, the dose of volatile agent required to maintain anaesthesia can be reduced by up to 70%. Do not use in patients likely to require or receiving sympathomimetic amines. **V** **W** **X**

DOSES

Y **When used for sedation is generally given as part of a combination. See Appendix for sedation protocols in all species.** **Z**

Mammals: Ferrets: 80–100 µg (micrograms)/kg i.m., s.c. in combination with an opioid and ketamine; **Rabbits:** 0.05–0.1 mg/kg i.v. or 0.1–0.3 mg/kg i.m., s.c. in combination with an opioid and ketamine;

Rodents, Other small mammals: Doses from 100–200 µg/kg i.p., i.m., s.c. in combination with ketamine and/or opioids or as premedication prior to induction with a volatile anaesthetic.

Birds: **See Appendix.**

Reptiles: 50–200 µg (micrograms)/kg i.m.; may be combined with ketamine and/or opioids/midazolam/alfaxalone to provide deep sedation/light anaesthesia (**see Appendix**); **Desert tortoises:** 150 µg/kg i.m.^a.

Fish: 0.05–0.1 mg/kg combined with 1–2 mg/kg ketamine i.m.^b;

Bonitos: 0.4 mg/kg combined with 4 mg/kg ketamine i.m.; **Mackerel:**

0.6–4.2 mg/kg combined with 53–228 mg/kg ketamine i.m.^c;

Sharks: 0.06–0.08 mg/kg combined with 5 mg/kg ketamine i.m.^d.

Reverse with atipamezole at five times the dose of medetomidine.

Amphibians: No information available.

References

- ^a Sleeman JM and Gaynor J (2000) Sedative and cardiopulmonary effects of medetomidine and reversal with atipamezole in Desert tortoises (*Gopherus agassizii*). *Journal of Zoo and Wildlife Medicine* **31**(1), 28–35
- ^b Williams TD, Christiansen J and Nygren S (1993) A comparison of intramuscular anesthetics in teleosts and elasmobranchs. *Proceedings of the International Association for Aquatic Animal Medicine annual conference, 1993*
- ^c Williams TD, Rollins M and Block BA (2004) Intramuscular anesthesia of bonito and Pacific mackerel with ketamine and medetomidine and reversal of anesthesia with atipamezole. *Journal of the American Veterinary Medical Association* **225**(3), 417–421
- ^d Stetter MD (2001) Fish and amphibian anesthesia. *Veterinary Clinics of North America: Exotic Animal Practice* **4**(1), 69–82

Medroxyprogesterone

(Depo-Provera*, Provera*) **POM**

Formulations: Injectable: 150 mg/ml suspension. **Oral:** 5 mg tablets.

Action: Alters the transcription of DNA leading to alterations in cellular metabolism which mimic progesterone.

Use: Used in primates as a contraceptive. Used in birds to manage feather plucking, persistent ovulation and sexual behavioural problems, but not recommended due to side effects.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use in pregnancy or diabetes mellitus.

Adverse reactions: Temperament changes (listlessness and depression), increased thirst or appetite, cystic endometrial hyperplasia/pyometra, diabetes mellitus, adrenocortical suppression, reduced libido (males), mammary enlargement/neoplasia and lactation. Subcutaneous injections may cause a permanent local alopecia, skin atrophy and depigmentation. There are many side effects in birds including liver damage, obesity and diabetes mellitus.

Drug interactions: No information available.

DOSES

Mammals: **Primates:** 5–10 mg/animal p.o. q24h for 5–10 days; 150 mg/animal i.m. q30d^a; **Lemurs:** 5 mg/kg i.m. q6wk.

Birds: Persistent ovulation, feather plucking, sexual behavioural problems: 5–50 mg/kg i.m. Use with great care due to side effects. Repeat in 4–6 weeks if necessary.

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A**Reptiles, Amphibians, Fish:** No information available.**B****References**

- ^a Cruzen CL, Baum ST and Colman RJ (2011) Glucoregulatory function in adult rhesus macaques (*Macaca mulatta*) undergoing treatment with medroxyprogesterone acetate for endometriosis. *Journal of the American Association for Laboratory Animal Science* **50**(6), 921–925

C**D****Melatonin****E**(Circadin*) **POM****F**

Formulations: Oral: 2 mg tablets. Melatonin is also available in many over-the-counter formulations of various sizes and often with other drugs added.

G**H**

Action: Hormone which is involved in the neuroendocrine control of seasonal hair loss.

I

Use: Palliative treatment of adrenocortical disease in ferrets.

J

Safety and handling: Normal precautions should be observed.

K

Contraindications: No information available.

L

Adverse reactions: No information available.

M

Drug interactions: No information available.

DOSES

Mammals: Ferrets: 0.5 mg p.o. q24h.

Birds, Reptiles, Amphibians, Fish: No information available.

N**O****Meloxicam****P**

(Inflacam, Loxicom, Meloxidolor, Meloxidyl, Meloxivet, **Metacam**, Rheumocam) **POM-V**

Q

Formulations: Oral: 0.5 mg/ml suspension for cats and guinea pigs, 1.5 mg/ml oral suspension for dogs; 1.0 mg, 2.5 mg tablets for dogs. **Injectable:** 2 mg/ml solution for cats, 5 mg/ml solution.

R

Action: Preferentially inhibits COX-2 enzyme thereby limiting the production of prostaglandins involved in inflammation.

S**T****U****V****W****X****Y****Z**

Use: Authorized in guinea pigs for alleviation of mild to moderate post-operative pain associated with soft tissue surgery (Metacam 0.5 mg/ml oral suspension). Alleviation of inflammation and pain in both acute and chronic musculoskeletal disorders and the reduction of postoperative pain and inflammation following orthopaedic and soft tissue surgery. All NSAIDs should be administered cautiously in the perioperative period as they may adversely affect renal perfusion during periods of hypotension. If hypotension during anaesthesia is anticipated, delay meloxicam administration until the animal is fully recovered from anaesthesia and is normotensive. Liver disease will prolong the metabolism of meloxicam leading to the potential for drug accumulation and overdose with repeated dosing. The oral dose (standard liquid preparation) may be administered directly into the mouth or mixed with food. Administration to animals with renal disease must be carefully evaluated. Be careful of species differences in effect in reptiles.

Safety and handling: After first opening a bottle of liquid oral suspension use contents within 6 months. Shake the bottle of the oral suspension well before dosing. The shelf-life of a broached bottle of injectable solution is 28 days.

Contraindications: Do not give to dehydrated, hypovolaemic or hypotensive patients or those with GI disease or blood clotting problems. Administration of meloxicam to animals with renal disease must be carefully evaluated and is not advisable in the perioperative period. Do not give to pregnant animals or animals <6 weeks of age.

Adverse reactions: GI signs may occur in all animals after NSAID administration. Stop therapy if this persists beyond 1–2 days. Some animals develop signs with one NSAID and not another. A 1–2 week wash-out period should be allowed before starting another NSAID after cessation of therapy. Stop therapy immediately if GI bleeding is suspected. There is a small risk that NSAIDs may precipitate cardiac failure in humans and this risk in animals is not known.

Drug interactions: Do not administer concurrently or within 24 hours of other NSAIDs and glucocorticoids. Do not administer with other potentially nephrotoxic agents (e.g. aminoglycosides).

DOSES

Mammals: Ferrets: 0.2 mg/kg p.o., s.c., i.m. q24h; Rabbits: 0.3–0.6 mg/kg s.c., p.o. q12–24h^a; studies have shown that rabbits may require a dose exceeding 0.3 mg/kg q24h to achieve optimal plasma levels of meloxicam over a 24-hour interval and doses of 1.5 mg/kg s.c., p.o. are well tolerated for 5 days; 1 mg/kg is tolerated for 29 days; Guinea pigs: Authorized at a dose of 0.2 mg/kg q24h on day 1 and 0.1 mg/kg subsequently, with doses up to 0.5 mg/kg on an individual patient basis where necessary; Rats: 1–2 mg/kg s.c., p.o. q24h; Mice: 2 mg/kg s.c., p.o. q24h; Primates: 0.1–0.2 mg/kg p.o. q24h, 0.2 mg/kg i.m. q24h^b, 0.1–0.2 mg/kg s.c. q24h; Sugar gliders: 0.2 mg/kg p.o., s.c. q24h; Hedgehogs: 0.2 mg/kg p.o., s.c. q24h, 0.5 mg/kg p.o. q12h.

Birds: Raptors: 0.5 mg/kg p.o. q12h^c; Zebra finches: 1–2 mg/kg i.m. q12h for <4 days; African grey parrots: 1 mg/kg i.m., p.o. q24h for <12 days; Amazon parrots: 1 mg/kg i.m. q12h^d, 1.5 mg/kg p.o. q12h; Cockatiels: 0.5–1 mg/kg i.m., p.o. q12–24h (based on experimental inflammatory model).

Reptiles: Doses of 0.1–0.5 mg/kg p.o., s.c., i.m. q24–48h have been suggested although analgesic effect is unproven in reptiles; Bearded dragons: 0.4 mg/kg i.m. q24h; Green iguanas: 0.2 mg/kg i.v., p.o. q24h^e; Red-eared sliders: 0.2 mg/kg i.m., i.v., intracoelomic^f.

Amphibians: 0.4 mg/kg p.o., s.c., intracoelomic q24h.

Fish: No information available.

References

- ^a Delk KW, Carpenter JW, KuKanich B, Nietfeld JC and Kohles M (2014) Pharmacokinetics of meloxicam administered orally to rabbits (*Oryctolagus cuniculus*) for 29 days. *American Journal of Veterinary Research* **75**(2), 195–199
- ^b Bauer C, Frost P and Kirschner S (2014) Pharmacokinetics of 3 formulations of meloxicam in cynomolgus macaques (*Macaca fascicularis*). *Journal of the American Association of Laboratory Animal Science* **53**, 502–511
- ^c Lacasse C, Gamble KC and Boothe DM (2013) Pharmacokinetics of a single dose of intravenous and oral meloxicam in Red-tailed Hawks (*Buteo jamaicensis*) and Great Horned Owls (*Bubo virginianus*). *Journal of Avian Medicine and Surgery* **27**(3), 204–210
- ^d Cole GA, Paul-Murphy J, Krugner-Higby L et al. (2009) Analgesic effects of intramuscular administration of meloxicam in Hispaniolan parrots (*Amazona ventralis*)

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with experimentally induced arthritis. *American Journal of Veterinary Research* **70**(12), 1471–1476

^e Divers SJ, Papich M, McBride M *et al.* (2010) Pharmacokinetics of meloxicam following intravenous and oral administration in green iguanas (*Iguana iguana*). *American Journal of Veterinary Research* **71**(11), 1277–1283

^f Uney K, Altan F, Aboubakr M, Cetin G and Dik B (2016) Pharmacokinetics of meloxicam in red-eared slider turtles (*Trachemys scripta elegans*) after single intravenous and intramuscular injections. *American Journal of Veterinary Research* **77**(5), 439–444

Mepivacaine

(Intra-epicaine) **POM-V**

Formulations: Injectable: 2% solution.

Action: Local anaesthetic action is dependent on reversible blockade of the sodium channel, preventing propagation of an action potential along the nerve fibre. Sensory nerve fibres are blocked before motor nerve fibres, allowing a selective sensory blockade at low doses.

Use: Blockade of sensory nerves to produce analgesia following perineural or local infiltration. Instillation into joints to provide intra-articular analgesia. Mepivacaine has less intrinsic vasodilator activity than lidocaine and is thought to be less irritant to tissues. It is of equivalent potency to lidocaine but has a slightly longer duration of action (100–120 minutes). It does not require addition of adrenaline to prolong its effect.

Safety and handling: Normal precautions should be observed.

Contraindications: Mepivacaine should not be injected i.v.

Adverse reactions: Inadvertent i.v. injection may cause convulsions and/or cardiac arrest.

Drug interactions: No information available.

DOSES

Mammals: Use the minimum volume required to achieve an effect. Toxic doses of mepivacaine have not been established in companion animals.

Birds, Reptiles, Amphibians, Fish: No information available.

Metacaine see **Tricaine mesylate**

Metamizole see **Butylscopolamine**

Methadone

(Comfortan, Synthadon) **POM-V CD SCHEDULE 2**

Formulations: Injectable: 10 mg/ml solution. Oral: 10 mg tablets.

Action: Analgesia mediated by the mu opioid receptor.

Use: Management of moderate to severe pain in the perioperative period. Incorporation into sedative and pre-anaesthetic medication protocols to provide improved sedation and analgesia. Methadone has similar pharmacological properties to morphine, and is useful in similar situations. It provides profound analgesia with a duration of action of 3–4 hours. Accumulation is likely to occur after prolonged repeated dosing which may allow the dose to be reduced or the dose interval to

be extended. Methadone can be given i.v. without causing histamine release and does not cause vomiting when given to animals preoperatively. Transient excitation may occur when methadone is given i.v. Methadone may be administered epidurally to provide analgesia. Respiratory function should be monitored when given i.v. to anaesthetized patients. The response to all opioids appears to vary between individual patients, therefore assessment of pain after administration is imperative. Methadone is metabolized in the liver, and some prolongation of effect may be seen with impaired liver function.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: In common with other mu agonists, methadone can cause respiratory depression, although this is unlikely when used at clinical doses in conscious animals. Respiratory depression may occur when given i.v. during general anaesthesia due to increased depth of anaesthesia. Methadone will cause constriction of GI sphincters (such as the pyloric sphincter) and may cause a reduction in GI motility when given over a long period. Methadone crosses the placenta and may exert sedative effects in neonates born to dams treated prior to parturition. Severe adverse effects can be treated with naloxone.

Drug interactions: Other CNS depressants (e.g. anaesthetics, antihistamines, barbiturates, phenothiazines, tranquillizers) may cause increased CNS or respiratory depression when used concurrently with narcotic analgesics.

DOSES

Mammals: Analgesia: Rabbits: 0.3–0.7 mg/kg slow i.v., i.m.

Primates: 1 mg/kg i.m.^a.

Birds, Reptiles, Amphibians, Fish: No information available.

References

- ^a Caramalac SM, Souza AFD, Caramalac SM *et al.* (2021) Comparative study between ketamine-S-dexmedetomidine and ketamine-S-midazolam-methadone in the anesthesia of capuchin monkeys (*Sapajus apella*). *Ciência Rural* **52**

Methanediol see **Formaldehyde**

Methanol see **Formaldehyde**

Methimazole see **Thiamazole**

Methoprene (S-Methoprene)

(Acclaim spray, **Amflee Combo**, **Bob Martin Clear Plus**, **Chanonil Plus**, **Eziflea Plus**, **FiprocLEAR Combo**, **Fleanil Duo**, **Fleatix Plus**, **Frontline Combo/Plus**, **Fyperix Combo**, **PestiGon Combo**, **R.I.P. flea spray**, **Zeronil Plus**, **Zerotol Plus**) **POM-V**, **NFA-VPS**, **AVM-GSL**

Formulations: Available in spot-on pipettes of various sizes always in combination with other agents. Those formulations authorized for ferrets contain 50 mg S-methoprene with 60 mg fipronil.

Environmental: S-methoprene with permethrin (Acclaim) or tetramethrine + permethrin (R.I.P. Fleas) household sprays.

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

A **Action:** Juvenile hormone analogue that inhibits larval development.

B **Use:** Authorized in ferrets for treatment of flea and tick infestations (Amflee Combo, Bob Martin Clear Plus, Chanonil Plus, Eziflea Plus, FiprocLEAR Combo, Fleanil Duo, Fleatix Plus, Frontline Combo, Frontline Plus, Fyperix Combo, PestiGon Combo, Zeronil Plus, Zerotal Plus). For treatment of flea infestations the topical products should be applied every 4 weeks to all in-contact animals. Bathing between 48 hours before and 24 hours after topical application is not recommended. Minimum treatment interval 4 weeks. Can be used in pregnant and lactating females. Treat infested household as directed with spray; keep away from birds and fish. Environmental sprays also have some efficacy against house dust mites *Dermatophagoides farinae* and *D. pteronyssinus*.

C **Safety and handling:** Normal precautions should be observed.

D **Contraindications:** Do not use on ferrets less than 6 months old.

E **Adverse reactions:** Local pruritus or alopecia may occur at the site of application. May be harmful to aquatic organisms.

F **Drug interactions:** No information available.

G **DOSES**

H **See Appendix for guidelines on responsible parasiticide use.**

I **Mammals:** Ferrets: One pipette of 0.5 ml/animal (50 mg fipronil, 60 mg S-methoprene) applied topically to back of neck.

J **Birds, Reptiles, Amphibians, Fish:** No information available.

M

N **Methotrexate**

O (Matrex*, Methotrexate*) **POM**

P **Formulations:** Oral: 2.5 mg, 10 mg tablets.

Q **Action:** An S-phase-specific antimetabolite antineoplastic agent; competitively inhibits folic acid reductase which is required for purine synthesis, DNA synthesis and cellular replication. This results in inhibition of DNA synthesis and function.

R **Use:** Treatment of lymphoma, although its use in animals is often limited by toxicity. In humans it is used to treat refractory rheumatoid arthritis; however, data are lacking with regards to its use in immune-mediated polyarthritides. Monitor haematological parameters regularly.

S **Safety and handling:** **Cytotoxic drug; see specialist texts for further advice on chemotherapeutic agents.**

T **Contraindications:** Pre-existing myelosuppression, severe hepatic or renal insufficiency, or hypersensitivity to the drug.

V **Adverse reactions:** GI ulceration, mucositis, hepatotoxicity, nephrotoxicity and haemopoietic toxicity may be seen, particularly with high doses. Low blood pressure and skin reaction are seen in humans.

W **Drug interactions:** Methotrexate is highly bound to serum albumin and thus may be displaced by phenylbutazone, phenytoin, salicylates, sulphonamides and tetracycline, resulting in increased blood levels and toxicity. Folic acid supplements may inhibit the response to methotrexate. Methotrexate increases the cytotoxicity of cytarabine. Cellular uptake is decreased by hydrocortisone, methylprednisolone and

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penicillins, and is increased by vincristine. Concurrent use of NSAIDs increases the risk of haematological, renal and hepatic toxicity.

DOSES

See Appendix for chemotherapy protocols in ferrets.

Mammals: Ferrets: 0.5 mg/kg i.v. once as part of a chemotherapy protocol.

Birds, Reptiles, Amphibians, Fish: No information available.

Methylthioninium chloride (Methylene blue)

(Methylthioninium chloride*) **POM, AVM-GSL**

Formulations: Injectable: 10 mg/ml (1% solution). **Immersion:** Several proprietary preparations are available.

Action: Acts as an electron donor to methaemoglobin reductase.

Use: Methaemoglobinaemia. Use an in-line filter if possible. Used for the treatment of protozoans and some monogenean ectoparasites, and external fungal infections in freshwater fish. It is strongly recommended that propriety formulations are used initially to avoid problems related to purity and to enable accurate dosing.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use unless adequate renal function is demonstrated.

Adverse reactions: May cause a Heinz body haemolytic anaemia and renal failure. Considered to have poor efficacy, stains many objects (especially plastics) and is toxic to some scaleless fish species, plants and nitrifying bacteria in biological filtration systems.

Drug interactions: No information available.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Fish: 1–3 mg/l by immersion; follow the manufacturer's recommendation for proprietary formulations.

Mammals, Birds, Reptiles, Amphibians: No information available.

Metoclopramide

(Emeprid, Metomotyl, Vomend, Maxolon*, Metoclopramide*)
POM-V, POM

Formulations: Injectable: 5 mg/ml solution in 10 ml multidose vials or clear glass ampoules. **Oral:** 10 mg tablets; 15 mg capsules; 1 mg/ml solution.

Action: Antiemetic and upper GI prokinetic stimulant; distal intestinal motility is not significantly affected. The antiemetic effect is a result of central dopamine (D2) receptor antagonism, and at higher doses 5HT₃ antagonism, at the chemoreceptor trigger zone. The gastric prokinetic effect is a result of local D2 antagonism and stimulation of muscarinic acetylcholine and 5HT₄ receptors leading to increases in oesophageal

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sphincter pressure, the tone and amplitude of gastric contractions and peristaltic activity in the duodenum and jejunum, and relaxing the pyloric sphincter by sensitizing tissues to acetylcholine. There is no effect on gastric, pancreatic or biliary secretions and nor does metoclopramide depend on an intact vagal innervation to affect motility.

Use: Vomiting of many causes can be reduced by this drug. The prokinetic effect may be beneficial in reflux oesophagitis and in emptying the stomach prior to induction of general anaesthesia. High doses are needed to abolish all reflux during anaesthesia. The prokinetic effect may also help to prevent postoperative ileus in rabbits; however, it is only effective in adult rabbits.

Safety and handling: Injection is light-sensitive. Obscure fluid bag if used in a continuous rate infusion.

Contraindications: Do not use where GI obstruction or perforation is present or highly suspected, or for >72 hours without a definitive diagnosis. Relatively contraindicated in epileptic patients.

Adverse reactions: Unusual, and probably relate to relative overdosing and individual variations in bioavailability. They include changes in mentation (depression, nervousness, restlessness) and behaviour. It may also cause sedation and extrapyramidal effects (movement disorders characterized as slow to rapid twisting movements involving the face, neck, trunk or limbs). Metoclopramide reduces renal blood flow, which may exacerbate pre-existing renal disease. Very rarely, allergic reactions may occur.

Drug interactions: The activity of metoclopramide may be inhibited by antimuscarinic drugs (e.g. atropine) and narcotic analgesics. The effects of metoclopramide may decrease (e.g. cimetidine, digoxin) or increase (e.g. oxytetracycline) drug absorption. The absorption of nutrients may be accelerated, thereby altering insulin requirements and/or timing of its effects in diabetics. Phenothiazines may potentiate the extrapyramidal effects of metoclopramide. The CNS effects of metoclopramide may be enhanced by narcotic analgesics or sedatives.

DOSES

Mammals: Ferrets, Rabbits, Guinea pigs: 0.5–1 mg/kg s.c., p.o. q6–12h; Primates: 0.2–0.5 mg/kg i.m. q8–24h, **Hedgehogs:** 0.2–0.5 mg/kg p.o., s.c.; **Sugar gliders:** 0.05–0.1 mg/kg i.m., s.c., p.o. q6–12h.

Birds: 0.3–2.0 mg/kg p.o., i.m. q8–24h^a.

Reptiles: 0.05–1 mg/kg p.o., i.m. q24h (anecdotal). Ineffective in desert tortoises at these doses^b.

Amphibians, Fish: No information available.

References

- ^a Beaufre H, Nevarez J, Taylor WM et al. (2010) Fluoroscopic study of the normal gastrointestinal motility and measurements in the Hispaniolan Amazon parrot (*Amazona ventralis*). *Veterinary Radiology & Ultrasound* **51**(4), 441–446
- ^b Tothill A, Johnson J, Branvold H, Paul C and Wimsatt J (2000) Effect of cisapride, erythromycin, and metoclopramide on gastrointestinal transit time in the desert tortoise, (*Gopherus agassizii*). *Journal of Herpetological Medicine and Surgery* **10**(1), 16–20

Metronidazole

(Eradia, Metrobactin, Stomorgyl, Flagyl*, Metrolyl*, Metronidazole*) **POM-V, POM**

Formulations: **Injectable:** 5 mg/ml i.v. infusion. **Oral:** 200 mg, 400 mg, 500 mg tablets; 25 mg metronidazole + 46.9 mg spiramycin tablets, 125 mg metronidazole + 234.4 mg spiramycin tablets, 250 mg metronidazole + 469 mg spiramycin tablets (Stomorgyl 2, 10 and 20, respectively); 40 mg/ml, 125 mg/ml oral solution.

Action: Synthetic nitroimidazole with antibacterial and antiprotozoal activity. Its mechanism of action on protozoans is unknown but in bacteria it appears to be reduced spontaneously under anaerobic conditions to compounds that bind to DNA and cause cell death. Spiramycin is a macrolide antibacterial that inhibits bacterial protein synthesis.

Use: Treatment of anaerobic infections, giardiasis and other protozoal infections, and in the management of hepatic encephalopathy. Metronidazole may have effects on the immune system by modulating cell-mediated immune responses. It is absorbed well from the GI tract and diffuses into many tissues including bone, CSF and abscesses. Spiramycin (a constituent of Stomorgyl) is active against Gram-positive aerobes including *Staphylococcus*, *Streptococcus*, *Bacillus* and *Actinomyces*. Metronidazole has been used as an appetite stimulant in reptiles. In ferrets it is used in combination with amoxicillin, and bismuth subsalicylate, ranitidine or omeprazole (triple therapy) for treatment of *Helicobacter mustelae*. In rabbits it is the treatment of choice for enterotoxaemia due to *Clostridium spiroforme*. Metronidazole is frequently used in combination with penicillin or aminoglycoside antimicrobials to improve anaerobic spectrum. Some texts recommend doses in excess of 25 mg/kg. There is a greater risk of adverse effects with rapid i.v. infusion or high total doses. May have a role in treatment of chinchillas with giardiasis (however, use with care in this species). Used for the treatment of some protozoan ectoparasites and internal flagellates (e.g. *Hexamita*, *Spironucleus*) in fish.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not administer to indigo or king snakes due to toxicity. Do not use in very small birds, such as zebra finches, due to toxicity.

Adverse reactions: Adverse effects in animals are uncommon and are generally limited to vomiting, CNS toxicity (nystagmus, ataxia, knuckling, head tilt and seizures), hepatotoxicity and haematuria. Prolonged therapy or the presence of pre-existing hepatic disease may predispose to CNS toxicity. Use with caution in the first trimester of pregnancy as it may be teratogenic. Use has been associated with reduced food intake^a, and anecdotally with hepatopathy in chinchillas. Has been associated with toxicity in indigo snakes, king snakes and rattle snakes at doses >40–100 mg/kg.

Drug interactions: Phenobarbital or phenytoin may enhance metabolism of metronidazole. Cimetidine may decrease the metabolism of metronidazole and increase the likelihood of dose-related adverse effects. Spiramycin should not be used concurrently with other

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antibiotics of the macrolide group as the combination may be antagonistic. Metronidazole is relatively insoluble in water and requires thorough mixing before adding to fish tanks.

DOSES

Classified as category D (Prudence) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

See Appendix for guidelines on responsible parasiticide use.

Mammals: Ferrets: 15–20 mg/kg p.o. q12h or 50–75 mg/kg p.o. q24h for 14 days with clarithromycin and omeprazole for *Helicobacter*; **Rabbits, Chinchillas, Guinea pigs:** 10–20 mg/kg p.o. q12h or 40 mg/kg p.o. q24h; 50 mg/kg p.o. q12h for 5 days may be required for giardiasis in chinchillas but use with caution; **Other rodents:** 20–40 mg/kg p.o. q24h; **Primates:** 25 mg/kg p.o. q12h; **Sugar gliders:** 25 mg/kg p.o. q12–24h for 7–10 days; **Hedgehogs:** 20 mg/kg p.o. q12h.

Birds: Raptors: 50 mg/kg p.o. q24h for 5 days; **Parrots:** 30 mg/kg p.o. q12h; **Pigeons:** 40–50 mg/kg p.o. q24h for 5–7 days or 100 mg/kg p.o. q48h for 3 doses or 200 mg/kg p.o. once; **Passerines:** 50 mg/kg p.o. q12h or 200 mg/l water daily for 7 days.

Reptiles:

- Anaerobic bacterial infections: **Green iguanas, Snakes (*Elaphe* spp.):** 20 mg/kg p.o. q24–48h^{b,c}.
- Protozoal infections: **Chelonians:** 100–125 mg/kg p.o., repeat after 14 days (use lower doses of 50 mg/kg p.o. q24h for 3–5 days for severe infections); **Chameleons:** 40–60 mg/kg p.o., repeat after 14 days; **Milksnakes:** 40 mg/kg p.o., repeat after 14 days; **Other snakes:** 100 mg/kg p.o., repeat after 14 days.

Amphibians: 50 mg/kg p.o. q24h for 3–5 days or 50 mg/l as a bath for up to 24 hours.

Fish: 25 mg/l by immersion q48h for 3 doses or 100 mg/kg p.o. q24h for 3 days.

References

- Mans C, Fink DM, Giammarco HE and Ciarrocchi C (2021) Effects of compounded metronidazole and metronidazole benzoate oral suspensions on food intake in healthy chinchillas (*Chinchilla lanigera*). *Journal of Exotic Pet Medicine* **36**, 75–79
- Kolmstetter CM, Frazier DL, Cox SK and Ramsay EC (2001) Pharmacokinetics of metronidazole in the green iguana (*Iguana iguana*). *Bulletin of the Association of Reptilian and Amphibian Veterinarians* **8**(3), 4–7
- Kolmstetter CM, Cox SK and Ramsay EC (2001) Pharmacokinetics of metronidazole in the yellow rat snake (*Elaphe obsoleta quadrivittata*). *Journal of Herpetological Medicine and Surgery* **11**(2), 4–8

Miconazole

(Daktarin, EasOtic, Malaseb, **Mycozole**, Surolan) **POM-V, ESPA**

Formulations: Topical: 2% cream/powder (Daktarin); 2% shampoo (Malaseb); 23 mg/ml suspension with prednisolone and polymyxin (Surolan), 10 mg/ml spray for topical administration (Mycozole).

Action: Inhibits cytochrome P450-dependent synthesis of ergosterol in fungal cells causing increased cell wall permeability and allowing leakage of cellular contents. Miconazole has activity against *Malassezia*, *Cryptococcus*, *Candida* and *Coccidioides*.

Use: Fungal skin and ear infections, including dermatophytosis and chytridiomycosis (amphibians).

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: No information available.

Drug interactions: No information available.

DOSES

Mammals: Rabbits: fungal otitis: 2–12 drops in affected ear q12–24h; dermatophytosis: apply a thin layer of cream topically to affected area twice daily, continue for 2 weeks after a clinical cure and negative fungal cultures. Bathe affected area daily (miconazole/chlorhexidine preparations). Avoid preparations containing steroids. **Chinchillas:** powder formulations may be added to dust baths for treatment of dermatophytosis at a rate of 1 tsp/bath.

Amphibians: Topical cream or 0.01% solution for chytridiomycosis¹.

Birds, Reptiles, Fish: No information available.

References

- ¹ Nichols DK and Lamirande EW (2001) Successful treatment of chytridiomycosis. *Froglog* **46**, 1

Midazolam

(Dormazolam Solution for Injection for Horses, Buccolam*, Epistatus*, Hypnovel* and several others) **POM CD Schedule 3**

Formulations: Injectable: 2 mg/ml, 5 mg/ml solutions. Oral: 10 mg/ml solution for buccal administration.

Action: Causes neural inhibition by increasing the effect of GABA on the GABA_A receptor, resulting in sedation, anxiolytic effects, hypnotic effects, amnesia, muscle relaxation and anticonvulsive effects. Compared with diazepam it is more potent, has a shorter onset and duration of action and is less irritant to tissues.

Use: Provides sedation with amnesia; as part of a premedication regime, as part of combined anaesthetic protocols, and in the emergency control of epileptic seizures (including status epilepticus). It provides unreliable sedation on its own, although it will sedate depressed animals. It is often used to offset muscle hypertonicity caused by ketamine. It is used with opioids and/or acepromazine for pre-anaesthetic medication in the critically ill. Midazolam can be diluted with saline, but avoid fluids containing calcium as this may result in precipitation of midazolam. Use with caution in severe hypotension, cardiac disease and respiratory disease. Antagonism with flumazenil is recommended if sedation persists longer than required, or hypothermia develops. Will have effects on gastrointestinal motility in birds, especially in combination with opioid. This sedation protocol, therefore, is not suitable for transit studies in birds.

Safety and handling: Normal precautions should be observed.

Contraindications: Avoid in myasthenia gravis and neonates.

Adverse reactions: In human patients, i.v. administration of midazolam has been associated with respiratory depression and severe hypotension. Excitement may occasionally develop.

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Drug interactions: Midazolam potentiates the effect of some anaesthetic agents, including propofol and some inhalation agents, reducing the dose required. Concurrent use of midazolam with NSAIDs (in particular diclofenac), antihistamines, barbiturates, opioid analgesics or CNS depressants may enhance the sedative effect. Opioid analgesics may increase the hypnotic and hypotensive effects of midazolam. Erythromycin inhibits the metabolism of midazolam.

DOSES

When used for sedation is generally given as part of a combination. See Appendix for sedation protocols in all species.

Mammals: Ferrets: 0.25–0.5 mg/kg i.v. or 0.3–1.0 mg/kg s.c., i.m.; Rabbits: 0.2–2 mg/kg i.v., i.m.; general anaesthesia: 0.25–1.0 mg/kg i.v. when used with fentanyl/fluanisone; **Chinchillas:** 1 mg/kg i.v. or 2 mg/kg i.m.; **Rodents:** 2–5 mg/kg i.v., i.m., i.p. but lower doses (e.g. 0.25–0.5 mg/kg) likely to be needed in sick individuals; **Primates:** 0.05–0.1 mg/kg i.v., i.m.; **Sugar gliders:** 0.1–0.5 mg/kg s.c., i.m., intranasal; **Hedgehogs:** 0.5–1 mg/kg i.m., 1–2 mg/kg intranasal, 1 mg/kg s.c.

Birds: 0.1–0.5 mg/kg i.m. or 0.05–0.15 mg/kg i.v. (premedicant) or 2–3 mg/kg intranasal^{a,b}. Can be combined with butorphanol for premedication; **Eurasian buzzards:** 2 mg/kg intranasal provides sedation^c; **Quaker parrots:** 1–2 mg/kg i.m. premedication has a significant isoflurane-sparing effect without effect on heart rate or body temperature^d.

Reptiles: 0.1–1 mg/kg i.m. for light sedation in snakes, lizards and chelonians^e; **Red-eared sliders:** 1.5 mg/kg i.m. provides variable levels of sedation^f; **Green iguanas:** 2 mg/kg i.m.; **Ball pythons:** 1–2 mg/kg i.m. Can be combined with ketamine and/or opioids/alpha-2 agonists/alfaxalone to provide deep sedation/light anaesthesia.

Amphibians: **Blue poison dart frogs:** 40 mg/kg s.c. in combination with dexmedetomidine and ketamine/alfaxalone for sedation.

Fish: No information available.

References

- ^a Mans C, Guzman DSM, Lahner LL, Paul-Murphy J and Sladky KK (2012) Sedation and physiologic response to manual restraint after intranasal administration of midazolam in Hispaniolan Amazon parrots (*Amazona ventralis*). *Journal of Avian Medicine and Surgery* **26**(3), 130–139
- ^b Doss GA, Fink DM and Mans C (2018) Assessment of sedation after intranasal administration of midazolam and midazolam-butorphanol in cockatiels (*Nymphicus hollandicus*). *American Journal of Veterinary Research* **79**(12), 1246–1252
- ^c Altundag Y, Kurt T, Özsoy S and Altundag İ (2021) Clinical evaluation of the intranasal administration of midazolam and reverse effects of flumazenil in Eurasian Buzzards (*Buteo buteo*). *Veterinarski Arhiv* **91**(6), 655–664
- ^d Zaheer OA, Sanchez A and Beaufrère H (2020) Minimum anesthetic concentration of isoflurane and sparing effect of midazolam in Quaker parrots (*Myiopsitta monachus*). *Veterinary Anaesthesia and Analgesia* **47**(3), 341–346
- ^e Arnett-Chinn ER, Hadfield CA and Clayton LA (2016) Review of intramuscular midazolam for sedation in reptiles at the National Aquarium, Baltimore. *Journal of Herpetological Medicine and Surgery* **26**(1–2), 59–63
- ^f Oppenheim YC and Moon PF (1995) Sedative effects of midazolam in red-eared slider turtles (*Trachemys scripta elegans*). *Journal of Zoo and Wildlife Medicine* **26**, 409–413

Milbemycin

(Milbactor, Milbemax, Milprazon, Milpro, Milquantel, Nexgard Spectra, Program plus) **POM-V**

Formulations: Oral: 2.5 mg/25 mg, 12.5 mg/125 mg milbemycin/praziquantel tablets (Milbactor, Milbemax, Milprazon, Milpro, Milquantel for dogs); 4 mg/10 mg, 16 mg/40 mg (Milbactor, Milbemax, Milprazon, Milpro, Milquantel for cats); 2.3 mg, 5.75 mg, 11.5 mg, 23 mg milbemycin with lufenuron (ratio 20 mg lufenuron: 1 mg milbemycin) tablets (Program plus). 1.875 mg, 3.75 mg, 7.5 mg, 15 mg, 30 mg milbemycin/afoxolaner (ratio 5 mg afoxolaner: 1 mg milbemycin) tablets (NexGard Spectra).

Action: Interacts with GABA and glutamate gated channels, leading to flaccid paralysis of parasites.

Use: Treatment of adult nematode infestation; roundworms (*Toxocara canis*, *T. cati*), hookworms (*Ancylostoma caninum*, *A. tubaeforme*) and whipworms (*Trichuris vulpis*). The addition of lufenuron provides flea control. For treatment of flea infestations the additional use of an authorized adulticide is recommended. The addition of praziquantel provides control of cestodes (*Dipylidium caninum*, *Taenia* spp., *Echinococcus*, *Mesocystoides*). It is also used for the prevention of heartworm disease (*Dirofilaria immitis*) in countries where this parasite is endemic. Can be used in pregnant and lactating females.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use in animals suspected of having heartworm disease. Not for use in any animal <0.5 kg.

Adverse reactions: No information available.

Drug interactions: No information available.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Mammals: Ferrets: 1.15–2.33 mg/kg p.o. q30d.

Birds, Reptiles, Amphibians, Fish: No information available.

Mineral oil see Paraffin

Mirtazapine

(Zispin) **POM**

Formulations: Oral: 15 mg tablets, 15 mg/ml solution.

Action: Tricyclic antidepressant that acts on central alpha-2 receptors which leads to increased noradrenaline levels within the brain. Also inhibits several serotonin receptors and histamine (H1) receptors.

Use: Appetite stimulation although effects not well-established in exotic pets. Can also be used as an antiemetic in conjunction with other drugs, but authorized preparations are preferred. Monitor animal carefully when using mirtazapine, particularly if there is also cardiac, hepatic or renal disease.

Safety and handling: Normal precautions should be observed.

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Contraindications: Do not use in patients with pre-existing haematological disease.

B

Adverse reactions: Sedation is common and can be profound. Can affect behaviour in many different ways. Has been associated with blood dyscrasias in humans.

C

Drug interactions: Several interactions known in humans, principally involving other behaviour-modifying drugs.

D

DOSES

E

Mammals: Rabbits: 3 mg/kg p.o. q24h for 2 days^a.

F

Birds, Reptiles, Amphibians, Fish: No information available.

G

References

^a Ozawa S, Thomson A and Petritz O (2022) Safety and efficacy of oral mirtazapine in New Zealand white rabbits (*Oryctolagus cuniculus*). *Journal of Exotic Pet Medicine* **40**, 16–20

H

Misoprostol

I

(Cytotec*) **POM**

J

Formulations: Oral: 200 µg (micrograms) tablet.

K

Action: Prostaglandin analogue which protects gastric mucosa by inhibiting gastric acid secretion and increases bicarbonate and mucus secretion, epithelial cell turnover and mucosal blood flow. It prevents, and promotes healing of, gastric and duodenal ulcers, particularly those associated with the use of NSAIDs. Some reports suggest it may not prevent gastric ulceration caused by methylprednisolone.

L**M****N**

Use: Protection against NSAID-induced gastric ulceration. In humans, doses of up to 20 µg/kg p.o. q6–12h are used to manage pre-existing NSAID-induced gastric ulceration, whilst doses of 2–5 µg/kg p.o. q6–8h are used prophylactically to prevent ulceration. Combinations with diclofenac are available for humans, but are not suitable for small animals because of different NSAID pharmacokinetics.

O**P**

Safety and handling: People who are or may become pregnant should not handle this drug.

Q**R**

Contraindications: Do not use in pregnant animals.

S

Adverse reactions: Diarrhoea, abdominal pain, nausea, vomiting and abortion.

T

Drug interactions: Use of misoprostol with gentamicin may exacerbate renal dysfunction.

U

DOSES

V

Mammals: Ferrets: 1–5 µg (micrograms)/kg p.o. q8h.

W

Birds, Reptiles, Amphibians, Fish: No information available.

X

Mitotane (o,p'-DDD)

Y

(Lysodren*) **POM**

Z

Formulations: Oral: 500 mg tablet or capsule.

Action: Necrosis of the adrenal cortex reducing the production of adrenal cortical hormones.

Use: Management of pituitary-dependent hyperadrenocorticism (HAC). However, other medications are authorized for this condition in dogs (**see Trilostane**). Has been used in the management of adrenal-dependent HAC, but with variable success. Mitotane is available from Europe for animals that have failed trilostane therapy. It should be given with food to improve its absorption from the intestinal tract. Following the initial 7–10 days therapy, an ACTH stimulation test should be performed to monitor the efficacy of therapy. In diabetic animals, the initial dose should be reduced by 30%. The addition of prednisolone is generally not recommended. If switching from trilostane to mitotane, then post-ACTH cortisol concentrations should be >200 nmol (nanomole)/l before starting mitotane. Not recommended to treat adrenal disease in ferrets due to their species-specific adrenal pathophysiology.

Safety and handling: Drug crosses skin and mucous membrane barriers. Wear gloves when handling this drug and avoid inhalation of dust.

Contraindications: No information available.

Adverse reactions: Anorexia, vomiting, diarrhoea and weakness, generally associated with too rapid a drop in plasma cortisol levels. They usually resolve with steroid supplementation. Acute-onset of neurological signs may be seen 2–3 weeks after initiation of therapy, possibly due to rapid growth of a pituitary tumour. Provide supplemental glucocorticoids during periods of stress.

Drug interactions: Barbiturates and corticosteroids increase the hepatic metabolism of mitotane. There may be enhanced CNS depression with concurrent use of CNS depressants. Spironolactone blocks the action of mitotane. Diabetic animals may have rapidly changing insulin requirements during the early stages of therapy.

DOSES

Mammals: Hamsters: 5 mg/animal p.o. q24h for 4 weeks.

Birds, Reptiles, Amphibians, Fish: No information available.

Morphine

(Morphine*, Oramorph*) **POM CD SCHEDULE 2**

Formulations: **Injectable:** 10 mg/ml, 15 mg/ml, 20 mg/ml, 30 mg/ml solution. **Oral:** 10 mg, 30 mg, 60 mg, 100 mg tablets. In addition there are suspensions, slow-release capsules, and granules in a wide range of strengths. **Rectal:** Suppositories are available in a wide range of strengths.

Action: Analgesia mediated by the mu opioid receptor.

Use: Management of moderate to severe pain in the perioperative period. Incorporation into sedative and pre-anaesthetic medication protocols to provide improved sedation and analgesia. Morphine is the reference opioid with which all others are compared. It provides profound analgesia and forms the mainstay of postoperative analgesic protocols in humans. In most species it has a short duration of action and continuous rate infusions can be used to overcome this limitation. The greater availability of data describing morphine by continuous rate infusion may justify its use over methadone for this method of administration. Accumulation is likely to occur after prolonged repeated

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dosing, which may allow the dose to be reduced or the dose interval to be extended. Morphine causes histamine release when given rapidly i.v., so it should be diluted and given slowly i.v. It commonly causes vomiting when given to animals preoperatively that are not in pain, therefore morphine should be avoided when vomiting is contraindicated (e.g. animals with raised intraocular pressure). Transient excitation may occur when morphine is given i.v. Preservative-free morphine can be administered into the epidural space where it will provide analgesia for up to 24 hours. Respiratory function should be monitored when morphine is given to anaesthetized patients. The response to all opioids appears to vary between individual patients, therefore assessment of pain after administration is imperative. Morphine is metabolized in the liver, some prolongation of effect may be seen with impaired liver function. Be careful of species differences in effect in reptiles and fish. There is marked interspecies variability in the response to morphine in fish and caution should be used when extrapolating from one species to another.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: In common with other mu agonists, morphine can cause respiratory depression, although this is unlikely when used at clinical doses. Respiratory depression may occur when given i.v. during general anaesthesia due to increased depth of anaesthesia. Vomiting is common after morphine administration and it causes constriction of GI sphincters (such as the pyloric sphincter) and may cause a reduction in GI motility when given over a long period. Morphine crosses the placenta and may exert sedative effects in neonates prior to parturition. Severe adverse effects can be treated with naloxone. A paradoxical increase in activity at lower dose rates has been reported in some fish species.

Drug interactions: Other CNS depressants (e.g. anaesthetics, antihistamines, barbiturates, phenothiazines, tranquilizers) may cause increased CNS or respiratory depression when used concurrently with narcotic analgesics.

DOSES

Mammals: Analgesia: **Ferrets:** 0.5–5 mg/kg i.m., s.c. q2–6h or 0.1 mg/kg epidural once^a; **Rabbits:** 0.1–0.4 mg/kg/h by CRI; **Rabbits, Rodents:** 0.5–2 mg/kg i.m., s.c. q2–4h; **Primates:** 1–2 mg/kg p.o., s.c., i.m., i.v. q4h. Whilst higher doses have been used in laboratory settings, these are on extremely healthy animals with no current pathology present, unlike clinical cases. Ceiling effects mean that increasing dose does not necessarily correspond to increasing analgesia; in practice, doses of 0.5–2.0 mg/kg are observed to have good analgesic effects.

Reptiles: Analgesia: 1–5 mg/kg i.m., higher doses have been suggested but may result in significant respiratory depression especially at higher doses in debilitated patients. Starting at the low end of the dose range is therefore recommended; **Red-eared sliders:** 1.5 mg/kg i.m.^b; **Tegus:** 5 mg/kg i.m.^c; **Bearded dragons:** 0.5 mg/kg via neuraxial injection for regional analgesia.

Amphibians: Analgesia: 38–42 mg/kg s.c.^d. May be used at 5 mg/100 ml combined with alfaxalone for immersion anaesthesia.

Fish: Analgesia: **Koi:** 5 mg/kg i.m.^e; **Goldfish:** 40 mg/kg intracoelomic^f.

Birds: No information available.

References

- ^a Sladky KK, Horne WA, Goodrowe KL *et al.* (2000) Evaluation of epidural morphine for postoperative analgesia in ferrets (*Mustela putorius furo*). *Contemporary Topics in Laboratory Animal Science* **39(6)**, 33–38
- ^b Sladky KK, Miletic V, Paul-Murphy J *et al.* (2007) Analgesic efficacy and respiratory effects of butorphanol and morphine in turtles. *Journal of the American Veterinary Medical Association* **230(9)**, 1356–1362
- ^c Leal WP, Carregaro AB, Bressan TF *et al.* (2017) Antinociceptive efficacy of intramuscular administration of morphine sulfate and butorphanol tartrate in tegus (*Salvator merianae*). *American Journal of Veterinary Research* **78(9)**, 1019–1024
- ^d Coble DJ, Taylor DK and Mook DM (2011) Analgesic effects of meloxicam, morphine sulphate, flunixin meglumine and xylazine hydrochloride in African clawed frogs (*Xenopus laevis*). *Journal of the American Association of Laboratory Animal Science* **50(3)**, 355–360
- ^e Baker TR, Baker BB, Johnson SM and Sladky KK (2013) Comparative analgesic efficacy of morphine sulfate and butorphanol tartrate in koi (*Cyprinus carpio*) undergoing unilateral gonadectomy. *Journal of the American Veterinary Medical Association* **243(6)**, 882–890
- ^f Newby NC, Wilkie MP and Stevens ED (2009) Morphine uptake, disposition, and analgesic efficacy in the common goldfish (*Carassius auratus*). *Canadian Journal of Zoology* **87(5)**, 388–399

Moxidectin

(Advocate, Bravecto Plus, Endectrid, Imidamox, Imoxat, Moxiclear, Prevensa, Prinocate, Prinovox) **POM-V**

Formulations: Topical: 10 mg/ml and 25 mg/ml moxidectin with imidacloprid in spot-on pipette; 14 mg/ml with fluralaner in spot-on pipette.

Action: Interacts with GABA and glutamate gated channels, leading to flaccid paralysis of parasites.

Use: Authorized in ferrets for the treatment and prevention of flea infestation and the prevention of heartworm disease (Advocate, Endectrid, Imidamox, Imoxat, Moxiclear, Prevensa, Prinocate, Prinovox). Used in rabbits for *Psoroptes cuniculi*. Used in birds for mite infestation. Used in lizards for the treatment of nematode infections and mites.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use larger pipette sizes in ferrets.

Adverse reactions: Transient pruritus and erythema at the site of application may occur. Highly toxic to aquatic organisms.

Drug interactions: No information available.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Mammals: Ferrets: 0.4 ml pipette (containing 4 mg moxidectin and 40 mg imidacloprid) monthly. If under heavy flea pressure can repeat once after 2 weeks; Rabbits: 0.2–0.3 mg/kg p.o., s.c., repeat in 10 days^a.

Birds: 0.2 mg/kg topically prn.

Reptiles: Bearded dragons, Frillneck lizards: 0.2 ml/kg of imidacloprid (100 mg/ml) + moxidectin (25 mg/ml) formulation has been used topically q14d for 3 treatments for treatment of mites and internal parasites^b.

Amphibians: 200 µg (micrograms)/kg s.c. q4months for nematodes.

Fish: No information available.

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References

- ^a Wagner R and Wendlberger U (2000) Field efficacy of moxidectin in dogs and rabbits naturally infested with *Sarcoptes* spp., *Demodex* spp. and *Psoroptes* spp. mites. *Veterinary Parasitology* **93**(2), 149–158
- ^b Mehlhorn H, Schmahl G and Mevissen I (2005) Efficacy of a combination of imidacloprid and moxidectin against parasites of reptiles and rodents. *Parasitology Research* **97**(1), S97–S101

Moxifloxacin(Moxivig*) **POM****Formulations:** Topical: 0.5% solution (5 mg/ml; 5 ml bottle).**Action:** Bactericidal antimicrobial which works by inhibiting the bacterial DNA gyrase enzyme, causing damage to the bacterial DNA. The fluoroquinolones work in a concentration-dependent manner.**Use:** Ideally, fluoroquinolone use should be reserved for infections where culture and sensitivity testing predicts a clinical response and where first- and second-line antimicrobials would not be effective. Fourth generation fluoroquinolone with broad-spectrum activity against a wide range of Gram-negative and some Gram-positive aerobes. Active against many ocular pathogens, including *Staphylococcus* spp. and *Pseudomonas aeruginosa*.**Safety and handling:** Normal precautions should be observed.**Contraindications:** No information available.**Adverse reactions:** May cause local irritation after application.**Drug interactions:** No information available.**DOSES****Classified as category B (Restrict) by the EMA.****See Appendix for guidelines on responsible antibiotic use.****Mammals, Birds, Reptiles:** 1 drop in affected eye q12h.**Amphibians, Fish:** No information available.

Naloxone

(Naloxone*, Narcan*) **POM**

Formulations: Injectable: 0.02 mg/ml, 0.4 mg/ml solutions.

Action: Competitive antagonist for opioid receptors, reversing the effects of opioid agonists.

Use: Treatment of opioid overdose to reverse the adverse effects of opioid agonists. Also used to identify persistent activity of opioid drugs. Onset of action i.v. is very rapid, but duration is short (30–40 min). Repeated doses or an infusion may be required to manage overdose of longer acting opioids such as morphine and methadone or high-dose fentanyl. Naloxone will also antagonize the effects of endogenous opioids, therefore it can cause antanalgesic effects in opioid naïve subjects. Administration to animals that could be in pain must therefore be considered carefully. Low dose naloxone i.v. will cause a transient elevation of unconsciousness when persistent opioid activity contributes to an unexpectedly long recovery from anaesthesia.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Indiscriminate use in animals that have undergone major surgery or trauma will expose the recipient to acute severe discomfort. In such cases the effects of opioid overdose (respiratory depression) should be managed by endotracheal intubation and artificial ventilation. Naloxone should be reserved for emergency situations when the effects of opioid overdose are severe.

Drug interactions: No information available.

DOSES

Mammals: Ferrets: 0.01–0.04 mg/kg i.v., i.m., s.c.; Rabbits: 0.01–0.1 mg/kg i.v., i.m., i.p.; Rodents: 0.01–0.1 mg/kg s.c., i.v., i.p.; Primates: 0.01–0.1 mg/kg s.c., i.m., i.v.; Hedgehogs: 0.1–0.16 mg/kg s.c., i.m.^a. Naloxone can be administered as a CRI at 0.02 mg/kg/h i.v. if a longer duration of opioid antagonism is required.

Birds: 2 mg/animal i.v.

Amphibians: 10 mg/kg s.c.

Reptiles, Fish: No information available.

References

- ^a Europaeus HE (1995) Chemical immobilization of free-ranging European hedgehogs (*Erinaceus europaeus*). *Journal of Zoo and Wildlife Medicine* **26**(2), 246–251

Nandrolone

(Laurabolin, Decadurabolin*) **POM-V, POM**

Formulations: Injectable: 25 mg/ml, 50 mg/ml (in oil).

Action: Binds to testosterone receptors and stimulates protein synthesis.

Use: For use wherever excessive tissue breakdown or extensive repair processes are taking place. Has also been advocated in the management of aplastic anaemia and anaemia associated with renal failure; however, may also have adverse effects on renal failure by increasing protein turnover. Monitor haematology to determine the efficacy of treatment and liver enzymes to monitor for hepatotoxicity.

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Safety and handling: Normal precautions should be observed.

Contraindications: Do not use in breeding animals, in pregnant animals or in those with diabetes mellitus.

Adverse reactions: Androgenic effects may develop. Use in immature animals may result in early closure of epiphyseal growth plates.

Drug interactions: The concurrent use of anabolic steroids with adrenal steroids may potentiate the development of oedema.

DOSES

Mammals: Ferrets: 1–5 mg/kg i.m. q7d; Rabbits: 2 mg/kg s.c., i.m. q7d.

Reptiles: 1 mg/kg i.m. q7–14d.

Birds, Amphibians, Fish: No information available.

Neomycin

(Neoiven, Neomay, Neopen, Maxitrol*, Nivemycin*)

POM-V, POM

Formulations: Oral: 500 mg tablets (Nivemycin). Parenteral: 100 mg/ml neomycin combined with 200 mg/ml penicillin G (Neopen). Topical:

Many dermatological, ophthalmic and otic preparations contain 0.25–0.5% neomycin.

Action: A bactericidal antimicrobial agent that inhibits bacterial protein synthesis once it has gained access to the bacterial cell via an oxygen-dependent carrier mechanism. As other aminoglycosides, neomycin operates a concentration-dependent cell killing mechanism, leading to a marked post-antibiotic effect.

Use: Active primarily against Gram-negative bacteria, although some *Staphylococcus* and *Enterococcus* species are sensitive. All obligate anaerobic bacteria and many haemolytic streptococci are resistant. Since parenteral neomycin is extremely nephrotoxic and ototoxic it is used topically for infections of the skin, ear or mucous membranes. It is also used orally to reduce the intestinal bacterial population in the management of hepatic encephalopathy. As with other aminoglycosides it is not absorbed after oral administration unless GI ulceration is present. This drug has been used (often combined with antimuscarinic agents) in the treatment of non-specific bacterial enteritides. However, other antibiotic drugs, if required at all, are better indicated for such use. Neomycin is more active in an alkaline environment.

Safety and handling: Normal precautions should be observed.

Contraindications: For systemic use, do not use in animals with pre-existing renal disease. Do not use ear preparations if the tympanum is ruptured.

Adverse reactions: Systemic toxicity, ototoxicity and nephrotoxicity may very occasionally occur following prolonged high-dose oral therapy or where there is severe GI ulceration/inflammatory bowel disease, as sufficient neomycin may be absorbed. Nephrotoxicity and ototoxicity are potential side effects associated with parenteral use. Some patients may develop a severe diarrhoea/malabsorption syndrome and bacterial or fungal superinfections. Topical ophthalmic preparation may cause local irritation.

Drug interactions: Absorption of digoxin, methotrexate, potassium and vitamin K may be decreased. Other ototoxic and nephrotoxic drugs, e.g. furosemide, should be used with caution in patients on oral neomycin therapy as the combinations are likely to be synergistic.

DOSES

Classified as category C (Caution) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Ferrets: 10–20 mg/kg p.o. q6h; Rabbits: 30 mg/kg p.o. q12h; Chinchillas, Guinea pigs: 15 mg/kg p.o. q12h; Rats, Mice: 25 mg/kg p.o. q12h; In-water medication: Gerbils, Rats, Mice: 2.6 mg/ml drinking water; Hamsters: 0.5 mg/ml drinking water; Ophthalmic: 1 drop per eye q6–8h; Otic: 2–12 drops per ear or apply liberally to skin q4–12h.

Birds: 10 mg/kg p.o. q8–12h.

Reptiles, Amphibians, Fish: No information available.

Nitenpyram

(Bob Martin Flea Tablets, Capstar, Johnson's 4 Fleas)

AVM-GSL

Formulations: Oral: 11.4 mg, 57 mg tablets.

Action: Postsynaptic binding to insect nicotinic receptors leads to insect paralysis and death. Kills fleas on animal within 30 minutes.

Use: Treatment of fleas. Should be used as part of a fully integrated flea control programme. All animals in the affected household should be treated. Treatment of flystrike in rabbits and rodents. Safe in pregnancy and lactation.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Transient increase in pruritus may be seen after administration due to fleas reacting to the product.

Drug interactions: No information available.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Mammals: 1 mg/kg p.o. once (minimum dose) or q24h.

Birds, Reptiles, Amphibians, Fish: No information available.

Nitroglycerin(e) see Glyceryl trinitrate

Nitrous oxide

(Entonox*, Nitrous oxide*) **POM**

Formulations: Inhalational: 100% nitrous oxide (N₂O) gas. Entonox is N₂O plus oxygen.

Action: Causes CNS depression.

Use: Used with oxygen to carry volatile anaesthetic agents such as isoflurane for the induction and maintenance of anaesthesia. N₂O reduces

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the concentration of inhalant agent required to maintain anaesthesia. Administration of N₂O at the beginning of volatile agent anaesthesia increases the speed of uptake of volatile agent from the alveoli (via the 2nd gas effect and concentration effect), hastening attainment of a stable plane of volatile agent anaesthesia. Oxygen must be supplemented for 5–10 min after N₂O is discontinued to prevent diffusion hypoxia. N₂O causes minimal respiratory and cardiovascular effects and is a useful addition to a balanced anaesthesia technique. A minimum oxygen concentration of 30% is required during anaesthesia. The inspired concentration of oxygen may fall to critically low levels when N₂O is used in rebreathing circuits during low flow rates. Do not use in such systems unless the inspired oxygen concentration can be measured on a breath-by-breath basis.

Safety and handling: Prolonged exposure can have serious adverse effects on human health. Scavenging is essential. N₂O is not absorbed by charcoal in passive scavenging systems utilizing activated charcoal.

Nitrous oxide is a potent greenhouse gas and is ozone-depleting. Nitrous oxide confers the largest carbon footprint of the anaesthetic gases within the acute sector, accounting for at least 80% of the total anaesthetic gas footprint of the NHS in 2019/2020. The negative environmental impact of nitrous oxide brings into question its continued use in veterinary medicine.

Contraindications: Do not give to patients with air-filled spaces within the body, e.g. pneumothorax or gastric dilatation. N₂O will cause a rapid expansion of any gas-filled space, increasing volume or pressure. It does not appear to cause problems in the normal caecum of hindgut fermenting species. Do not give to animals with marked respiratory compromise, due to the risks of hypoxia.

Adverse reactions: The cobalt ion present in vitamin B12 is oxidized by N₂O so that it is no longer able to act as the co-factor for methionine synthase. The result is reduced synthesis of methionine, thymidine, tetrahydrofolate and DNA. Exposure lasting only a few hours may lead to megaloblastic changes in bone marrow but more prolonged exposure (a few days) may result in agranulocytosis.

Drug interactions: No information available.

DOSES

Mammals, Birds, Reptiles: Inspired concentrations of 50–70%^a.

Amphibians, Fish: No information available.

References

- ^a Larouche CB, Mosley C, Beaufrère H, Dutton C (2019) Effects of midazolam and nitrous oxide on the minimum anesthetic concentration of isoflurane in the ball python (*Python regius*). *Veterinary Anaesthesia and Analgesia* **46**(6), 807–814

Nystatin

(Canaural, Nystan*, Nystatin*) **POM-V, POM**

Formulations: Oral: 100,000 IU/ml suspension. Topical: various products (Canaural also contains framycetin, fusidic acid and prednisolone).

Action: Binds to ergosterol, a major component of the fungal cell membrane, and forms pores in the membrane that lead to potassium leakage and death of the fungus.

Use: Antifungal agent with a broad spectrum of activity but noted for its activity against *Candida*, particularly *C. albicans*. Not absorbed from the GI tract. One study in birds showed a 78.5% sensitivity rate to nystatin in *C. albicans* so this is a good first use drug in candidiasis in birds^a.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Corticosteroid-free preparations are preferred in pregnant animals, birds and rabbits to avoid the systemic effects of corticosteroids.

Drug interactions: No information available.

DOSES

Mammals: All species: Topical: apply to affected areas q8–12h; **Rabbits:** 100,000 IU/kg p.o. q12h for GI tract yeast overgrowth; **Chinchillas:** 60,000–90,000 IU/kg p.o. q12h for 7–10 days.

Birds: Pigeons: 100,000 IU/kg p.o. q24h for 10 days; **Passerines:** 5000–300,000 IU/kg p.o. q12h for macrorhabdiosis (although there are doubts about its efficacy); **Other birds:** 300,000 IU/kg p.o. q12h¹.

Reptiles: 100,000 IU/kg p.o. q24h for 10 days.

Amphibians: Topical (1% cream) for cutaneous mycoses.

Fish: No information available.

References

^a Talazadeh, F, Ghorbanpoor, M and Shahriyari, A (2022) Candidiasis in Birds (Galliformes, Anseriformes, Psittaciformes, Passeriformes, and Columbiformes): A Focus on Antifungal Susceptibility Pattern of *Candida albicans* and Non-albicans Isolates in Avian Clinical Specimens. *Topics in Companion Animal Medicine* **46**, 100598.

¹ Tantaş A, Ak S and Özgür Y (1990) Aetiology, diagnostic criteria and therapeutic findings in candidiasis in parrots. *Veteriner Fakültesi Dergisi (İstanbul)* **16(1/2)**, 181–184

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Octreotide

(Sandostatin*, Sandostatin LAR*) **POM****Formulations:** Injectable: 50 µg (micrograms)/ml, 100 µg/ml, 200 µg/ml, 500 µg/ml solutions; depot preparation: 10 mg, 20 mg, 30 mg vials.**Action:** Somatostatin analogue that inhibits the release of several hormones.**Use:** May be useful in the management of gastric, enteric and pancreatic endocrine tumours (e.g. insulinoma, gastrinoma) and acromegaly. Variable responses have been reported in veterinary medicine. Most recent research suggests that it is not useful in most insulinomas. Tumours not expressing somatostatin receptors will not respond. There is limited information on the use of this drug in veterinary species. In humans doses up to 200 µg/person q8h are used. Similar doses of the aqueous preparation may be required in animals, but dosages for the depot preparation are not known.**Safety and handling:** Normal precautions should be observed.**Contraindications:** No information available.**Adverse reactions:** GI disturbances (anorexia, vomiting, abdominal pain, bloating, diarrhoea and steatorrhoea), hepatopathy and pain at injection sites have been recorded in humans.**Drug interactions:** No information available.**DOSES****Mammals:** Ferrets: 1–2 µg (micrograms)/kg s.c. q8–12h.**Birds, Reptiles, Amphibians, Fish:** No information available.**O**

Ofloxacin

(Exocin*) **POM****Formulations:** Topical: 0.3% solution.**Action:** Bactericidal antimicrobial which works by inhibiting the bacterial DNA gyrase enzyme, causing damage to the bacterial DNA. The fluoroquinolones work in a concentration-dependent manner.**Use:** Ideally fluoroquinolone use should be reserved for infections where culture and sensitivity testing predicts a clinical response and where first- and second-line antimicrobials would not be effective. For ophthalmic use when other antibiotic agents are ineffective. Active against many ocular pathogens, including *Staphylococcus* and *Pseudomonas aeruginosa*, although there is increasing resistance amongst some staphylococcal and streptococcal organisms. Better corneal penetration than ciprofloxacin.**Safety and handling:** Normal precautions should be observed.**Contraindications:** No information available.**Adverse reactions:** May cause local irritation after application.**Drug interactions:** No information available.**DOSES****Classified as category B (Restrict) by the EMA.****See Appendix for guidelines on responsible antibiotic use.**

Mammals: 1 drop to affected eye q6h; loading dose can be used: 1 drop to affected eye q15min for 4 doses.

Birds: 1 drop in affected eye q12h.

Reptiles: 1 drop in affected eye q8–12h.

Amphibians, Fish: No information available.

Oil of cloves see **Eugenol**

Omeprazole

(Gastrogard, Losec*, Mepradec*, Zanolprol*) **POM-V, POM**

Formulations: Oral: 10 mg, 20 mg, 40 mg capsules, gastro-resistant tablets, MUPS (multiple unit pellet system) tablets. **Injectable:** 40 mg vial for reconstitution for i.v. injection; discard remainder after use.

Action: Proton pump inhibitor. Ten times more potent than cimetidine in inhibiting gastric acid secretion and has a longer duration of activity (>24 hours).

Use: Management of gastric and duodenal ulcers, oesophagitis, and hypersecretory conditions secondary to gastrinoma (Zollinger–Ellison syndrome) or mast cell neoplasia. Gastrogard is authorized for use in equids, but the formulation (370 mg/g paste) makes accurate dosing of small animals impossible. Lansoprazole, rabeprazole and pantoprazole are similar drugs but have no known clinical advantage over omeprazole. Esomeprazole is a newer preparation containing only the active isomer of omeprazole.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Chronic suppression of acid secretion has caused hypergastrinaemia in laboratory animals, leading to mucosal cell hyperplasia, rugal hypertrophy and the development of carcinoids, and so treatment for a maximum of 8 weeks has been recommended. However, such problems have not been reported in companion animals. Adverse effects do include nausea, diarrhoea, constipation and skin rashes.

Drug interactions: Omeprazole may enhance the effects of phenytoin. There is a risk of interaction with tacrolimus, mycophenolate mofetil, clopidogrel, digoxin and itraconazole.

DOSES

Mammals: Ferrets: 0.7–4 mg/kg p.o. q24h for a maximum of 8 weeks;

Primates: 0.4 mg/kg p.o. q12h^a.

Birds, Reptiles, Amphibians, Fish: No information available.

References

- ^a Dubois A, Berg DE, Fiala N *et al.* (1998) Cure of *Helicobacter pylori* infection by omeprazole-clarithromycin-based therapy in non-human primates. *Journal of Gastroenterology* **33**(1), 18–22

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Ondansetron

(Zofran*) **POM**

Formulations: **Injectable:** 2 mg/ml solution in 2 ml and 4 ml ampoules. **Oral:** 4 mg, 8 mg tablets; 4 mg/5 ml syrup. **Rectal:** 16 mg suppositories.

Action: Potent antiemetic effects through action on the GI tract and the chemoreceptor trigger zone. It was developed for, and is particularly useful in, the control of emesis induced by chemotherapeutic drugs.

Use: Indicated for the management of nausea and vomiting in patients who are unable to tolerate, or whose signs are not controlled by, other drugs (e.g. maropitant, metoclopramide). Dolasetron, granisetron, palonosetron and tropisetron are similar drugs but have yet to be extensively used in companion animals.

Safety and handling: Normal precautions should be observed.

Contraindications: Intestinal obstruction.

Adverse reactions: In humans, constipation, headaches, occasional alterations in liver enzymes and, rarely, hypersensitivity reactions have been reported.

Drug interactions: Ondansetron may reduce the effectiveness of tramadol and so the dose of tramadol may need to be increased.

DOSES

Mammals: Ferrets: 1 mg/kg p.o. q12–24h.

Birds, Reptiles, Amphibians, Fish: No information available.

N

Oxantel see **Pyrantel**

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Oxfendazole

(Bovex, Ovidown) **POM-VPS**

Formulations: **Oral:** 2.265% suspension, 5% suspension in combination with selenium and cobalt.

Action: Benzimidazole. Inhibits fumarate reductase system of parasites, thereby blocking the citric acid cycle, and also reduces glucose absorption by the parasite.

Use: Treatment of GI parasites, specifically used for oxyurids in chelonians and iguanas. Also thought to have some efficacy against other endoparasites including ascarids, hookworms, whipworms, some cestodes, trematodes and *Giardia*. Avoid using formulations combined with selenium and cobalt.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: No information available, although other benzimidazoles have been associated with bone marrow suppression and feather abnormalities so these are potential risks.

Drug interactions: No information available.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Z

Birds: 10–40 mg/kg p.o. once.

Reptiles: Tortoises: 66 mg/kg p.o. once^a; Iguanas: 25 mg/kg p.o. once^b.

Amphibians: 5 mg/kg p.o. once.

Mammals, Fish: No information available.

References

- ^a Giannetto S, Brianti E, Poglayen G *et al.* (2007) Efficacy of oxfendazole and fenbendazole against tortoise (*Testudo hermanni*) oxyurids. *Parasitology Research* **100**(5), 1069–1073
- ^b Kehoe S, Divers S, Mayer J, Comoli J and Verocai GG (2020) Efficacy of single-dose oxfendazole to treat *Oxyurid* nematodiasis in the green iguana (*Iguana iguana*). *Journal of Herpetological Medicine and Surgery* **30**(3), 137–141

Oxytetracycline

(Aquatet, Engemycin, Oxyicare) **POM-V**

Formulations: Injectable: 100 mg/ml solution. Oral: 50 mg, 100 mg, 250 mg tablets. Feed supplement and soluble powders also available.

Action: Inhibits bacterial protein synthesis. The effect is bacteriostatic.

Use: Active against many Gram-positive and Gram-negative bacteria, *Rickettsiae*, mycoplasmas, spirochaetes and other microbes. One of the less lipid-soluble tetracyclines, it is excreted unchanged in urine and bile and undergoes enterohepatic recirculation. Has been used in combination with nicotinamide in the management of immune-mediated conditions, including discoid lupus erythematosus and lupoid onychodystrophy. Resistance to tetracyclines is widespread. Use with care in rabbits as narrow therapeutic range. Used for the treatment of bacterial disease in fish. High strength formulations (e.g. 100%) for immersion baths for fish should be used since bulking agents may have adverse effects on water quality.

Safety and handling: Normal precautions should be observed.

Contraindications: The concentrated injectable depot formulations used for cattle and sheep should never be given to small animals or fish. Avoid oral dosing in birds, other than for prophylaxis, as tetracyclines are poorly absorbed from the GI tract, rapidly lose potency in drinking water and put birds off drinking water due to their taste.

Adverse reactions: Include vomiting, diarrhoea, depression, hepatotoxicity (rare), fever, hypotension (following i.v. administration) and anorexia. Prolonged use may lead to development of superinfections. Although not well documented in veterinary medicine, tetracyclines induce dose-related functional changes in renal tubules in several species, which may be exacerbated by dehydration, haemoglobinuria, myoglobinuria or concomitant administration of other nephrotoxic drugs. Severe tubular damage has occurred following the use of outdated or improperly stored products and occurs due to the formation of a degradation product. Tetracyclines stain the teeth of children when used in the last 2–3 weeks of pregnancy or the first month of life. Although this phenomenon has not been well documented in animals, it is prudent to restrict the use of tetracyclines in all young animals. Injectable preparations in birds may cause toxicity or muscle necrosis. In rabbits and other small herbivores, higher doses (30 mg/kg) can be associated with enteritis. The carrier compound used in some injectable extended-release

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formulations may produce a sterile fluid-filled cavity at the site of injection in fish. Buffering with sodium bicarbonate may be required at high immersion dose rates to prevent solutions becoming acidic and harmful to fish.

Drug interactions: The bactericidal action of penicillins may be inhibited by oxytetracycline. Antacids containing divalent or trivalent cations (Mg^{2+} , Ca^{2+} , Al^{3+}), food or milk products bind tetracycline, reducing its absorption. Tetracyclines may increase the nephrotoxic effects of methoxyflurane. The GI effects of tetracyclines may be increased if administered concurrently with theophylline products. Efficacy may be reduced due to chelation when used as a bath treatment for fish in sea water or hard water with high levels of divalent metal cations (Ca^{2+} , Mg^{2+}). The drug has a very low bioavailability in some species of fish, including carp and goldfish. If the drug turns the water dark brown when decomposing in fish tanks, 50% of the water should be changed immediately.

DOSES

Classified as category D (Prudence) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Ferrets, Hamsters, Gerbils: 20–25 mg/kg i.m. q8–12h; Rabbits: 15 mg/kg i.m. q12h or 30 mg/kg of the 5–10% non-depot preparation s.c. q72h; Guinea pigs: 5 mg/kg i.m. q12h; Chinchillas: 15 mg/kg i.m. q12h or 50 mg/kg p.o. q12h; Rats: 20 mg/kg i.m. q8–12h or 10–20 mg/kg p.o. q8h; Mice: 100 mg/kg s.c. q12h or 10–20 mg/kg p.o. q8h; Primates: 10 mg/kg s.c., i.m. q24h, 25–50 mg/kg p.o.; Hedgehogs: 25–50 mg/kg p.o. q24h.

Birds: Raptors: 25–50 mg/kg p.o. q8h; Parrots: 50–100 mg/kg s.c. q2–3d (long-acting preparation); Pigeons: 50 mg/kg p.o. q6h or 80 mg/kg i.m. q48h (long-acting preparation) or 130–400 mg/l water^{a,b,c}; Passerines: 100 mg/kg p.o. q24h or 4–12 mg/l water for 7 days.

Reptiles: 6–10 mg/kg p.o., i.m., i.v. q24h.

Amphibians: 25–50 mg/kg p.o., s.c., i.m. q24h or 100 mg/l for a 1 hour bath.

Fish: 10–100 mg/l (freshwater fish) by prolonged immersion for 1–3 days, if poor response then change 50% of the water and repeat, or 55–83 mg/kg p.o. q24h for 10 days or 10–50 mg/kg i.m., intracoelomic q24h for 5–10 days^d.

References

- ^a Flammer K, Aucoin DP, Whitt DA and Styles DK (1990) Potential use of long-acting injectable oxytetracycline for treatment of chlamydiosis in Goffin's cockatoos. *Avian Diseases* **34**(1), 228–234
- ^b Osofsky A, Tell LA, Kass PH et al. (2005) Investigation of Japanese quail (*Coturnix japonica*) as a pharmacokinetic model of cockatiels (*Nymphicus hollandicus*) and Poicephalus parrots via comparison of the pharmacokinetics of a single intravenous injection of oxytetracycline hydrochloride. *Journal of Veterinary Pharmacology and Therapeutics* **28**(6), 505–513
- ^c Teare JA, Schwark WS, Shin SJ and Graham DL (1985) Pharmacokinetics of a long-acting oxytetracycline preparation in ring-necked pheasants, great horned owls and Amazon parrots. *American Journal of Veterinary Research* **46**(12), 2639
- ^d Vorbach BS, Chandasana H, Derendorf H and Yanong RP (2019) Pharmacokinetics of oxytetracycline in the Giant Danio (*Devario aequipinnatus*) following bath immersion. *Aquaculture* **498**, 12–16

Oxytocin

(Oxytocin S) **POM-V**

Formulations: Injectable: 10 IU/ml solution.

Action: Synthetic oxytocin.

Use: Induces parturition/egg laying when uterine inertia is present (as long as there is no uterine obstruction); evacuates uterine contents; decreases haemorrhage following parturition; and promotes milk 'let-down'. Before oxytocin is used in any species it is important to ensure that there is no evidence of obstructive dystocia and that blood calcium levels are adequate. Calcium supplementation should be administered 1 hour prior to oxytocin if ionized or total blood calcium levels are low. Can also be used i.v. (dose at 25% of i.m. dose, diluted in water for injection). Useful in chelonians, less effective in lizards and snakes. Ineffective in birds.

Safety and handling: Store in refrigerator.

Contraindications: Not recommended for use in egg retention in birds.

Adverse reactions: Overstimulation of the uterus can be hazardous to both mother and fetuses. In birds it can cause painful side effects due to stimulation of smooth muscle.

Drug interactions: Severe hypertension may develop if used with sympathomimetic pressor amines.

DOSES

Mammals: Ferrets: 0.2–3.0 IU/kg s.c., i.m.; Rabbits: 0.1–3.0 IU/kg s.c., i.m.; Rodents: 0.2–3.0 IU/kg s.c., i.m., i.v.; Mice: (milk let-down) 6.25 IU/kg s.c.; Primates (Common marmosets): 1–2 IU/animal i.v., i.m. q20min for 4 doses.

Birds: Do not use.

Reptiles: Egg retention: 2–10 IU/kg i.m. q90min for a maximum of 3 doses. Start at low end of dose range and better effect if calcium therapy used first. Red-eared sliders: 2 IU/kg i.m., i.v.^a.

Amphibians, Fish: No information available.

References

- ^a Di Ianni F, Parmigiani E, Pelizzone I et al. (2014) Comparison between intramuscular and intravenous administration of oxytocin in captive-bred red-eared sliders (*Trachemys scripta elegans*) and non-obstructive egg retention. *Journal of Exotic Pet Medicine* **23**, 79–84

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A Pancreatic enzyme supplements

(Lypex, Pancreatic Enzyme Supplement for Dogs and Cats, Panzym, Pro-Enzorb) **AVM-GSL, POM, P**

Formulations: Oral: The formulations vary in the amount and type of enzyme present. Readers are referred to individual products for further details. Many other formulations on human market.

Action: Exogenous replacement enzymes.

Use: Pancreatic enzymes (lipase, protease, amylase) are used to control signs of exocrine pancreatic insufficiency (EPI). Fresh raw, or fresh-frozen, pig pancreas (approximately 100 g per meal) is also an effective treatment (and is not a Specified Risk Material) but availability is limited and there is a risk of pathogen ingestion by this method. Non-enteric coated powders and enteric coated granules and tablets are available. Use the manufacturer's recommendations as the minimum required initially; the dose may be reduced empirically once a satisfactory response is achieved. Efficacy may be augmented by antibiotic control of secondary bacterial overgrowth and vitamin B12 therapy for any associated hypcobalaminaemia. Concomitant administration of acid blockers is not cost-effective and there is no requirement for pre-incubation with food. Follow dosing with food or water.

Safety and handling: Powder spilled on hands should be washed off or skin irritation may develop. Avoid inhaling powder as it causes mucous membrane irritation and may trigger asthma attacks in susceptible individuals. These risks are not associated with enteric coated pancreatic granules.

Contraindications: No information available.

Adverse reactions: Contact dermatitis of the lips is occasionally seen with powdered non-coated enzyme. Non-coated pancreatic enzymes may cause oral or oesophageal ulcers, and so dosing should be followed with food or water. High doses may cause diarrhoea and signs of GI cramping.

Drug interactions: The effectiveness may be diminished by antacids (magnesium hydroxide, calcium carbonate).

DOSES

Birds: Panzyme or Lypex: 1 capsule/kg bodyweight q24h mixed in food.

Mammals, Reptiles, Amphibians, Fish: No information available.

Paracetamol (Acetaminophen)

(Pardale V (paracetamol and codeine phosphate), Paracetamol*, Perfalgan*) **P, POM, POM-V**

Formulations: Oral: 500 mg tablet; 120 mg/5 ml, 250 mg/5 ml suspensions; 400 mg paracetamol and 9 mg codeine phosphate tablet (Pardale V). **Injectable:** 10 mg/ml solution.

Action: It has been proposed that its antipyretic actions are due to prostaglandin synthesis within the CNS; however its exact mechanism of action is unclear.

Use: Control of mild to moderate pain and as an antipyretic. Paracetamol has poor anti-inflammatory effects. It is believed to produce few GI side effects and therefore is commonly administered to patients with gastric ulceration, particularly if traditional NSAIDs are contraindicated; however, there are limited clinical data to support this practice. The authorized oral preparation of paracetamol contains codeine; however, due to a high first pass metabolism of opioids, this codeine is not bioavailable and therefore does not contribute to the analgesia.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use in snakes^a.

Adverse reactions: Overdose of paracetamol causes liver damage through the production of *N*-acetyl-*p*-aminobenzoquinonimine during metabolism, which causes hepatocyte cell death and centrilobular hepatic necrosis. Treatment of overdose with oral methionine or i.v. acetylcysteine is directed at replenishing hepatic glutathione.

Drug interactions: Metoclopramide enhances absorption of paracetamol, thereby enhancing its effects.

DOSES

Mammals: Rabbits: 200–500 mg/kg p.o. is often quoted, but clinical analgesic effects are noted at 10–20 mg/kg, and higher doses are not recommended; **Rodents:** 1–2 mg/ml of drinking water (use flavoured products); **Primates:** 5–10 mg/kg p.o. q6h.

Reptiles: Do not use in snakes^a.

Birds, Amphibians, Fish: No information available.

References

- ^a van den Hurk P and Kerckamp HM (2019) Phylogenetic origins for severe acetaminophen toxicity in snake species compared to other vertebrate taxa. *Comparative Biochemistry and Physiology Part C: Toxicology & Pharmacology* **215**, 18–24

Paraffin (Liquid paraffin, Mineral oil)

(Katalax, Lacri-Lube*, Liquid paraffin oral emulsion*, Simple Eye Ointment*) **AVM-GSL, P**

Formulations: Oral: White soft paraffin paste (Katalax); liquid paraffin (50/50 oil/water mix). **Topical:** 3.5 g, 4 g or 5 g ophthalmic ointment.

Action: Paraffin is a laxative; it softens stool by interfering with intestinal water resorption. It is also a lipid-based tear substitute that mimics the lipid portion of the tear film and helps prevent evaporation of tears.

Use: Paraffin is used to manage constipation. It is beneficial in the management of keratoconjunctivitis sicca, during general anaesthesia and for eyelid paresis. It is a long-acting ocular lubricant and is used when frequency of treatment is not easy. Liquid paraffin has been historically used as an aid to treating reduced GI motility in rabbits due to the concern that fur accumulation in the GI tract was a contributory factor. The use of such products may impair hydration of GI contents, as well as increase the risk of inhalation. Its use is therefore not recommended in this species.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not give orally in patients with a reduced gag reflex.

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A **Adverse reactions:** As paraffin is tasteless, normal swallowing may not be elicited if syringing orally; thus inhalation and subsequent lipoid pneumonia are a significant risk. Paraffin ointment may blur vision, although not often a problem in animals.

C **Drug interactions:** Reduced absorption of fat-soluble vitamins may follow prolonged use.

D **DOSES**

E **Mammals:** Ocular: Apply to eye at night or q6–12h prn. **Ferrets:** 10 mm Katalax paste p.o. q12–24h for constipation.

F **Birds, Reptiles, Amphibians, Fish:** No information available.

G **Paroxetine**

H (Paxil*, Seroxat*) **POM**

I **Formulations:** Oral: 20 mg, 30 mg tablets; 2 mg/ml liquid suspension.

J **Action:** Blocks serotonin re-uptake in the brain, resulting in antidepressive activity and raising motor activity thresholds.

K **Use:** Treatment of behavioural disorders in pet birds, such as feather plucking and self-mutilation.

L **Safety and handling:** Normal precautions should be observed.

M **Contraindications:** Known sensitivity to paroxetine or other SSRIs, history of seizures.

N **Adverse reactions:** Possible reactions include lethargy, decreased appetite and vomiting. Trembling, restlessness, GI disturbances and an apparent paradoxical increase in anxiety may occur in some cases. Owners should be warned of a potential increase in aggression in response to medication.

P **Drug interactions:** Paroxetine should not be used within 2 weeks of treatment with an MAOI (e.g. selegiline) and an MAOI should not be used within 6 weeks of treatment with paroxetine. Paroxetine, like other SSRIs, antagonizes the effects of anticonvulsants and so is not recommended for use with epileptic patients or in association with other agents which lower seizure threshold, e.g. phenothiazines. Caution is warranted if paroxetine is used concomitantly with aspirin or other anticoagulants, since the risk of increased bleeding in the case of tissue trauma may be increased.

T **DOSES**

U **Birds:** 1–2 mg/kg p.o. q12–24h^a.

V **Mammals, Reptiles, Amphibians, Fish:** No information available.

W **References**

- X** ^a van Zeeland YRA, Schoemaker NJ, Haritova A *et al.* (2013) Pharmacokinetics of paroxetine, a selective serotonin reuptake inhibitor in grey parrots (*Psittacus erithacus*): influence of pharmaceutical formulation and length of dosing. *Journal of Veterinary Pharmacology and Therapeutics* **36**(1), 51–58

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Penicillamine

(Penicillamine*) **POM**

Formulations: Oral: 125 mg, 250 mg tablets.

Action: Penicillamine is an orally administered chelating agent that binds copper, mercury and lead. It also binds to cystine.

Use: Oral treatment of lead poisoning. May be used in the management of lead toxicity, especially in birds although has a low therapeutic index in birds.

Safety and handling: Normal precautions should be observed.

Contraindications: Moderate to marked renal impairment and a history of penicillamine-related blood dyscrasias. Penicillamine can reduce GI absorption of dietary minerals, including zinc, iron, copper and calcium and, therefore, cause deficiencies with long-term use.

Adverse reactions: Serious adverse effects that have been described in humans given penicillamine include leucopenia, thrombocytopenia, fever, lymphadenopathy, skin hypersensitivity reactions and lupus-like reactions. May cause vomiting, hypoglycaemia and death in birds.

Drug interactions: The absorption of penicillamine is decreased if administered with antacids, food, or iron or zinc salts. An increase in the renal and haematological effects of penicillamine have been recorded in humans receiving it with cytotoxic drugs.

DOSES

Mammals: Ferrets: 10 mg/kg p.o. q24h; Rabbits: 30 mg/kg p.o. q12h.

Birds: Lead and zinc poisoning: 30–55 mg/kg p.o. q12h.

Reptiles, Amphibians, Fish: No information available.

Penicillin G (Benzyl penicillin)

(Depocillin, Duphaphen, Neopen, Norocillin) **POM-V**

Formulations: **Injectable:** comes in a variety of salts (sodium, procaine and benzathine) which affect solubility. Penicillin G sodium (highly soluble): 3 g powder for reconstitution for i.v. use; procaine penicillin (less soluble): 300 mg/ml suspension for s.c. use, slower release.

Action: Binds to penicillin-binding proteins involved in cell wall synthesis, decreasing bacterial cell wall strength and rigidity, and affecting cell division, growth and septum formation. As animal cells lack a cell wall the beta-lactam antibiotics are safe in most species. Kills bacteria in a time-dependent fashion.

Use: A beta-lactamase-susceptible antimicrobial. Narrow spectrum of activity and susceptible to acid degradation in the stomach. Used parenterally to treat infections caused by sensitive organisms (e.g. *Streptococcus*, *Clostridium*, *Borrelia burgdorferi*, *fusospirochaetes*). The sodium salt is absorbed well from s.c. or i.m. sites. Procaine penicillin is sparingly soluble, providing a 'depot' from which it is slowly released. When used for 'blind' therapy of undiagnosed infections, penicillins may be given in conjunction with an aminoglycoside such as gentamicin with or without metronidazole. As penicillin kills in a time-dependent fashion, it is important to maintain tissue concentrations above the MIC for the

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A organism throughout the interdosing interval. Patients with significant renal or hepatic dysfunction may need dosage adjustment.

B **Safety and handling:** After reconstitution penicillin G sodium is stable for 7 days if refrigerated, 24 hours if not.

C **Contraindications:** Use with caution in rabbits, and never orally, though it has been used in injectable forms long term for abscesses and osteomyelitis. Do not use in animals sensitive to beta-lactam antimicrobials. Do not administer penicillins to hamsters, gerbils, guinea pigs, chinchillas or degus. Do not give procaine penicillin to rats and mice.

D **Adverse reactions:** 600 mg of penicillin G sodium contains 1.7 mEq of Na⁺. This may be clinically important for patients on restricted sodium intakes. The i.m. administration of >600 mg/ml may cause discomfort. Oral doses of penicillins can cause fatal enterotoxaemia in rabbits and other small herbivores.

E **Drug interactions:** Avoid the concomitant use of bacteriostatic antibiotics. The aminoglycosides may inactivate penicillins when mixed in parenteral solutions *in vitro*, but they act synergistically when administered at the same time *in vivo*. Procaine can antagonize the action of sulphonamides and so procaine penicillin G should not be used with them. Avoid concurrent administration of aciclovir as co-administration associated with reduced elimination of acyclovir in red-eared sliders^a.

L DOSES

Classified as category D (Prudence) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

M **Mammals:** Ferrets: 20 mg/kg s.c., i.m. q12–24h; Rabbits: 40 mg/kg s.c. once every 7 days for 3 doses for *Treponema cuniculi*; other infections 40 mg/kg s.c. q12–24h^b; Rats (not procaine): 22 mg/kg s.c., i.m. q24h; Primates: 20–40 mg/kg s.c., i.m. q12h; Sugar gliders: 22–25 mg/kg s.c., i.m. q12–24h; Hedgehogs: 40 mg/kg s.c., i.m. q24h; Chinchillas, Guinea pigs, Hamsters, Gerbils, Degus: Do not use.

O **Birds, Reptiles, Amphibians, Fish:** No information available.

P References

- Q** ^a Corum D, Corum O, Atik OE et al. (2020) Effect of benzylpenicillin on intravenous pharmacokinetics of acyclovir in red-eared slider turtles (*Trachemys scripta elegans*). *Journal of Veterinary Pharmacology and Therapeutics* **43**(4), 319–324
- R** ^b Jekl V, Hauptman K, Minarikova A et al. (2016) Pharmacokinetic study of benzylpenicillin potassium after intramuscular administration in rabbits. *Veterinary Record* **179**(1), 18.

U Pentamidine isethionate

(Pentacarinat*) **POM**

Formulations: Injectable: 300 mg vials of powder for reconstitution.

Action: Kills protozoans by interacting with DNA. It is rapidly taken up by the parasites by a high-affinity energy-dependent carrier.

Use: Used for treatment of *Pneumocystis* infection. Pentamidine is a toxic drug and the potential to cause toxic damage to the kidney and liver in particular should be carefully considered prior to use.

Safety and handling: Care should be taken by staff handling this drug as it is a highly toxic agent. Similar precautions to those recommended

when handling cytotoxic agents used in cancer chemotherapy should be taken.

Contraindications: Impaired liver or kidney function. Never give by rapid i.v. injection due to cardiovascular effects.

Adverse reactions: Pain and necrosis at the injection site, hypotension, nausea, salivation, vomiting and diarrhoea. Hypoglycaemia and blood dyscrasias are also reported in humans.

Drug interactions: No information available.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Mammals: Ferrets: 3–4 mg/kg s.c. q48h for *Pneumocystis* pneumonia.

Birds, Reptiles, Amphibians, Fish: No information available.

Pentobarbital (Pentobarbitone)

(Dolethal, Euthanimal, **Euthasol**, **Euthatal**, **Euthoxin**, **Lethobarb**, **Pentobarbital for euthanasia**, **Pentoject**, **Release**, **Repose**) **POM-V CD SCHEDULE 3**

Formulations: Injectable: 200 mg/ml, 400 mg/ml, as either a blue, yellow or pink non-sterile aqueous solution.

Action: CNS depressant.

Use: Authorized for euthanasia in all animals (Lethobarb, Pentobarbital), rabbits and rodents (Euthasol, Repose), small mammals (Euthatal), polecats, rabbits, guinea pigs, hamsters, rats, mice, ornamental birds, pigeons, snakes, lizards, tortoises, frogs (Euthoxin, Release) and rodents (Pentoject). When it is predicted that euthanasia may be problematical (e.g. aggressive patients) it is recommended that premedication with an appropriate sedative is given¹. The animal should be restrained in order to forestall narcotic excitement until anaesthesia supervenes. The route of choice is i.v. if possible, but alternatives such as intraperitoneal, intracoelomic, intrarenal and intracardiac are possible when venepuncture is difficult to achieve. Intracardiac injection of pentobarbital should be performed under anaesthesia. There is no authorized pentobarbital product in the UK that is suitable for the emergency control of seizures. Fish may need to be immobilized, using an anaesthetic agent immersed in water, to enable accurate intravenous injection of pentobarbital.

Safety and handling: Normal precautions should be observed.

Contraindications: Should not be given i.m. as it is painful and slow to act. Do not use solutions intended for euthanasia to try to control seizures.

Adverse reactions: Narcotic excitement may be seen with agitated animals. Agonal gasping is sometimes seen. In fish, the heart can continue to contract even after brain death and removal from the body.

Drug interactions: Antihistamines and opioids increase the effect of pentobarbital.

DOSES

Mammals: Euthanasia: 150 mg/kg i.v., i.p. as rapidly as possible or to effect.

Birds: Euthanasia: 150 mg/kg i.v.

Reptiles: Euthanasia: 60–100 mg/kg i.v., intracoelomic or intracardiac.

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A **Amphibians:** Euthanasia: 60 mg/kg i.v., intracoelomic or via lymph sacs (e.g. frogs and toads).

B **Fish:** Euthanasia: 60–100 mg/kg i.v., intracoelomic, intracardiac.

References

¹ Leary S, Underwood W, Anthony R, et al. (2020) *AVMA Guidelines for the Euthanasia of Animals, 2020 edn*. American Veterinary Medical Association, Schaumburg, IL, USA

Pentobarbitone see **Pentobarbital**

Pentosan polysulphate

(Cartrophen, Osteopen) **POM-V**

Formulations: Injectable: 100 mg/ml solution.

Action: Semi-synthetic polymer of pentose carbohydrates with heparin-like properties that binds to damaged cartilage matrix comprising aggregated proteoglycans and stimulates the synthesis of new aggregated glycosaminoglycan molecules. Ability to inhibit a range of proteolytic enzymes may be of particular importance. Modulates cytokine action, stimulates hyaluronic acid secretion, preserves proteoglycan content and stimulates articular cartilage blood flow, resulting in analgesic and regenerative effects.

Use: Used as a disease modifying agent to reduce the pain and inflammation associated with osteoarthritis in mammals; however, there is no evidence to support its use in exotic species. Administered by aseptic s.c. injection, using an insulin syringe for accurate dosing. The manufacturer recommends monitoring haematocrit and total solids. Based on its use in cats with idiopathic cystitis (because cats suffering from this condition have been shown to have reduced concentrations of GAGs within the protective mucosal layer of the bladder), it has been suggested that pentosan polysulphate may be of benefit in the management of guinea pigs suffering from chronic idiopathic, non-obstructive, lower urinary tract disease. However, there is currently no evidence to support this contention. There is good evidence to suggest that this drug is of minimal value in acute cases.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use if septic arthritis is present or if renal or hepatic impairment exists. As it may induce spontaneous bleeding, do not use in animals with bleeding disorders.

Adverse reactions: Pain at the injection site has been reported. Because of its fibrinolytic action the possibility of bleeding from undiagnosed tumours or vascular abnormalities exists.

Drug interactions: The manufacturers state that pentosan polysulphate should not be used concurrently with steroids or non-steroidal anti-inflammatory drugs, including aspirin, or used concomitantly with coumarin-based anticoagulants or heparin. However, many animals suffering from osteoarthritis that might benefit from pentosan polysulphate treatment are concurrently receiving NSAID therapy. The risk of bleeding associated with concurrent administration of pentosan polysulphate and COX-2 preferential and COX-2 selective NSAIDs is probably low in animals with no history of blood clotting disorders.

DOSES

Mammals: Rabbits: 3 mg/kg s.c. q5–7d for 4 doses for osteoarthritis;
Guinea pigs: 3 mg/kg s.c. q5–7d for 4 doses for osteoarthritis and idiopathic cystitis.

Birds, Reptiles, Amphibians, Fish: No information available.

Permanganate of potash see **Potassium permanganate**

Permethrin see **Imidacloprid**

Pethidine (Meperidine)

(Pethidine, Demerol*, Meperidine*) **POM-V, POM CD**

SCHEDULE 2

Formulations: **Injectable:** 10–50 mg/ml solutions. 50 mg/ml solution is usually used in veterinary practice.

Action: Analgesia mediated by the mu opioid receptor.

Use: Management of mild to moderate pain. Incorporation into sedative and pre-anaesthetic medication protocols to provide improved sedation and analgesia. Pethidine has a fast onset (10–15 min) and short duration (45–60 min) of action. Frequent redosing is used for analgesia. The short duration of action may be desirable in some circumstances (e.g. when a rapid recovery is required or in animals with compromised liver function). It shares common opioid effects with morphine but also has anticholinergic effects, producing a dry mouth and sometimes an increase in heart rate. Be careful of species differences in effect in reptiles.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not give i.v. Not advisable to use in animals at risk from histamine release (e.g. some skin allergies, asthma, mast cell tumours).

Adverse reactions: Histamine released during i.v. injection causes hypotension, tachycardia and bronchoconstriction. Histamine-mediated reactions may also occur after i.m. injection, resulting in local urticaria. Pethidine crosses the placenta and may exert sedative effects in animals born to dams treated prior to parturition. Severe adverse effects can be treated with naloxone.

Drug interactions: Other CNS depressants (e.g. anaesthetics, antihistamines, barbiturates, phenothiazines, tranquilizers) may cause increased CNS or respiratory depression when used concurrently with narcotic analgesics. Pethidine may produce a serious interaction if administered with monoamine oxidase inhibitors (MAOIs). The mechanism of this interaction is not clear but effects include coma, convulsions and hyperpyrexia.

DOSES

Mammals: Analgesia: **Ferrets:** 5–10 mg/kg i.m., s.c. q2–3h;
Rabbits: 10 mg/kg i.m., s.c. q2–3h; **Rodents:** 10–20 mg/kg i.m. q2–3h;
Primates: 2–4 mg/kg i.v., i.m. q2–4h.

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Reptiles: Analgesia: Tortoises: 20 mg/kg i.m. q12–24h^a.

Birds, Amphibians, Fish: No information available.

References

- ^a Wambugu SN, Towett PK, Kiama SG, Abelson KS and Kanui TI (2010) Effects of opioids in the formalin test in the Speke's hinged tortoise (*Kinixys spekii*). *Journal of Veterinary Pharmacology and Therapeutics* **33**(4), 347–351

Phenobarbital (Phenobarbitone)

(Epiphen, Epityl, Phenobarbital (non-proprietary), Phenoleptil, Gardenal*) **POM-V, POM CD SCHEDULE 3**

Formulations: Oral: 12.5 mg, 15 mg, 25 mg, 30 mg, 50 mg, 60 mg, 100 mg tablets; 4% (40 mg/ml) solution. **Injectable:** 15 mg/ml, 30 mg/ml, 60 mg/ml, 200 mg/ml solutions (phenobarbital sodium BP).

Action: Thought to mediate its antiepileptic effect through affinity for the GABA_A receptor, resulting in a GABA-ergic effect; GABA being the major inhibitory mammalian neurotransmitter with prolonged opening of the chloride channel. Phenobarbital also blocks the AMPA receptor, inhibiting release of the excitatory neurotransmitter glutamate. This combined potentiation of GABA and inhibition of glutamate leads to reduced neuronal excitability.

Use: Phenobarbital and imepitoin are the initial medications of choice for the management of epileptic seizures due to idiopathic epilepsy. The choice of initial medication is guided by patient requirements:

phenobarbital is less expensive and more efficacious, whilst imepitoin has a more rapid onset of action than phenobarbital (does not need to achieve a steady state), does not require the determination of serum concentrations and has a less severe adverse effect profile. Doses are mainly anecdotal in exotic species and information is based on that for dogs. One pharmacokinetic study has been performed in birds.

Phenobarbital is rapidly absorbed after oral administration in dogs; maximal plasma concentrations reached within 4–8 hours. Wide range of elimination half-life (40–90 hours) in different dogs. Steady state serum concentrations are not reached until 7–10 days after treatment is initiated and the full clinical effect of a dose cannot be ascertained until this point. Serum concentrations should be determined after starting treatment or dose alterations, once a steady state has been reached. If <15 µg (micrograms)/ml the dose should be increased accordingly. If seizures are not adequately controlled dose may be increased up to a maximum serum concentration of 45 µg/ml. Plasma concentrations above this level are associated with increased hepatotoxicity. With chronic therapy, induction of the hepatic microsomal enzyme system results in a decreased half-life, particularly during the first 6 months of therapy. As a result, the dose may need to be increased. Phenobarbital levels should be assessed every 6–12 months. Any termination of phenobarbital therapy should be performed gradually with a recommended protocol of: reduce the dose by 25% of the original dose each month (month 1: 75% of the original phenobarbital dose; month 2: 50% of the original phenobarbital dose; month 3: 25% of the original phenobarbital dose).

Safety and handling: Normal precautions should be observed.

Contraindications: Do not administer to animals with impaired hepatic function. Not for use in pregnant and nursing animals, although the risk

associated with uncontrolled seizures may be greater than the risk associated with phenobarbital. Do not use to control seizures resulting from hepatic disease (e.g. portosystemic shunt), hypoglycaemia or toxic causes where the clinical signs are mediated through the GABA channels (ivermectin and moxidectin toxicity) as this may exacerbate the seizures. Do not administer high doses by i.v. or i.m. injection in animals with marked respiratory depression.

Adverse reactions: Sedation, ataxia, polyphagia and PU/PD. Polyphagia and PU/PD are likely to persist throughout therapy. Ataxia and sedation occur commonly following initiation of therapy but usually resolve within 1 week, although they may continue if high doses are used. Hepatic toxicity is rare, but may occur at high serum concentrations (or as a rare idiosyncratic reaction within 2 weeks of starting treatment). Hyperexcitability has been reported in dogs on subtherapeutic dose levels. Haematological abnormalities, including neutropenia, anaemia and thrombocytopenia, may occur. Long-term administration in the absence of hepatotoxicity is associated with: moderate increase in liver size on abdominal radiographs; no change in liver echogenicity or architecture on ultrasonography; no evidence of morphological liver damage on histology; significant increase in ALP and, to a lesser extent, ALT activity; transiently decreased albumin (up to 6 months after starting therapy) and increased GGT; and no changes in AST, bilirubin or fasting bile acids. Therefore, liver function should be assessed by other parameters, in particular a bile acid assay, persistent decrease in albumin levels, serum AST, bilirubin and ultrasonographic examination of the liver. Phenobarbital treatment does not affect adrenal function tests (ACTH stimulation test and low dose dexamethasone test) despite acceleration of dexamethasone metabolism. Phenobarbital significantly decreases total T4 and free T4, and cholesterol levels tend to increase towards the upper limits of the normal range.

Drug interactions: The effect of phenobarbital may be increased by other CNS depressants (antihistamines, narcotics, phenothiazines). Phenobarbital may enhance the metabolism of, and therefore decrease the effect of, corticosteroids, beta-blockers, metronidazole and theophylline. Barbiturates may enhance the effects of other antiepileptics. Cimetidine, itraconazole and chloramphenicol increase serum phenobarbital concentration through inhibition of the hepatic microsomal enzyme system.

DOSES

Mammals: Ferrets: 1–2 mg/kg p.o. q12–24h or 2–10 mg/kg/h i.v. CRI; Guinea pigs, Gerbils: 10–25 mg/kg i.v., i.p. q12–24h; Primates: 1–6 mg/kg p.o. q24h, 2 mg/kg i.v.

Birds: 3.5–7 mg/kg p.o. q12h^a.

Reptiles, Amphibians, Fish: No information available.

References

- ^a Powers LV and Papich MG (2011) Pharmacokinetics of orally administered phenobarbital in African grey parrots (*Psittacus erithacus erithacus*). *Journal of Veterinary Pharmacology and Therapeutics* **34**(6), 615–617

Phenobarbitone see **Phenobarbital**
Phenoxetol see **Phenoxyethanol**

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Phenoxybenzamine

(Dibenyline*) **POM****Formulations:** Oral: 10 mg capsule. Injectable: 50 mg/ml solution.**Action:** An alpha-adrenergic blocker that irreversibly blocks presynaptic and postsynaptic receptors, producing a so-called chemical sympathectomy.**Use:** Reflex dyssynergia/urethral spasm. If concurrent beta-blockers are also used (for severe tachycardia/arrhythmias), only start these once alpha blockade is in place (to avoid a hypertensive crisis). Use with extreme caution in animals with pre-existing cardiovascular disease.**Safety and handling:** Normal precautions should be observed.**Contraindications:** No information available.**Adverse reactions:** Adverse effects associated with alpha-adrenergic blockade include hypotension, miosis, tachycardia and nasal congestion.**Drug interactions:** There is an increased risk of a first dose hypotensive effect if administered with beta-blockers or diuretics. Phenoxybenzamine will antagonize effects of alpha-adrenergic sympathomimetic agents (e.g. phenylephrine).**K**

DOSES

Mammals: Ferrets: 0.5–1 mg/kg p.o. q12h.**Birds, Reptiles, Amphibians, Fish:** No information available.**N**

Phenoxyethanol (Ethylene glycol monophenyl ether, Phenoxetol)

(Aqua-sed, Koi Sedate) **ESPA****Formulations:** Immersion: 100% liquid.**Action:** Precise mechanism of action is unknown but may involve an effect on nerve cell membranes and suppress activity in the CNS.**Use:** For the sedation, immobilization, anaesthesia and euthanasia of fish¹. The drug is fairly soluble but requires vigorous whisking into water to improve solubility. It does not accumulate in the fish tissues after induction of anaesthesia and therefore can be used for prolonged periods. It is considered safer and more potent at lower temperatures. It has some antibacterial properties and has been included at low concentrations in some proprietary formulations.**Safety and handling:** Irritant. Normal precautions should be observed.**Contraindications:** No information available.**Adverse reactions:** There can be a long induction time, and hyperactivity during induction and recovery. There is a narrow safety margin in some species and there may be involuntary muscle activity during anaesthesia.**Drug interactions:** No information available.**Y**

DOSES

Fish: Anaesthesia: 0.1–0.5 ml/l by immersion, follow manufacturer's recommendations for proprietary formulations; Euthanasia: 2.0 ml/l by immersion^{2,3}.

Mammals, Birds, Reptiles, Amphibians: No information available.

References

- ¹ Underwood W and Anthony R (2020) *AVMA guidelines for the euthanasia of animals: 2020 edition*. American Veterinary Medical Association, Illinois
- ² Ross LG and Ross B (1999) *Anaesthesia and Sedative Techniques for Aquatic Animals*. Blackwell Science, Oxford
- ³ Sneddon LU (2012) Clinical anaesthesia and analgesia in fish. *Journal of Exotic Pet Medicine* **21**, 32–43

Phenylephrine

(Phenylephrine hydrochloride*) **POM**

Formulations: Injectable: 1% (10 mg/ml) solution. Ophthalmic: 2.5%, 10% solution (single-dose vials).

Action: Directly stimulates the alpha-adrenergic receptors in the iris dilator musculature.

Use: When applied topically to the eye causes vasoconstriction and mydriasis (pupil dilation). Ophthalmic uses include mydriasis prior to intraocular surgery (often in conjunction with atropine) and differentiation of involvement of superficial conjunctival vasculature from deep episcleral vasculature (by vasoconstriction). It is also used in the diagnosis of Horner's syndrome (HS) (denervation hypersensitivity) by determining the time to pupillary dilation, following administration of 1% phenylephrine topically to both eyes. Essentially, the shorter the time to pupillary dilation, the closer the lesion to the iris: <20 minutes suggests third-order HS; 20–45 min suggests second-order HS; 60–90 min suggests first-order HS or no sympathetic denervation of the eye. If 10% phenylephrine is used, mydriasis occurs in 5–8 minutes in post-ganglionic (third-order neuron) lesions. Vasoconstrictors should be used with care. Although they raise blood pressure, they do so at the expense of perfusion of vital organs (e.g. kidney). In many patients with shock, peripheral resistance is already high and to raise it further is unhelpful.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: These include hypertension, tachycardia and reflex bradycardia.

Drug interactions: There is a risk of arrhythmias if phenylephrine is used in patients receiving digoxin or with volatile anaesthetic agents. When used concurrently with oxytocic agents the pressor effects may be enhanced, leading to severe hypertension.

DOSES

Mammals: Ophthalmic use: 1 drop approximately 2 hours before intraocular surgery. 1 drop as a single dose for vasoconstriction. 1 drop to both eyes for diagnosis of Horner's syndrome.

Birds, Reptiles, Amphibians, Fish: No information available.

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A Phenylpropanolamine (Diphenylpyraline)(Proin, Propalin, Urilin) **POM-V****B Formulations:** Oral: 40 mg/ml phenylpropanolamine syrup (equivalent to 50 mg/ml phenylpropanolamine hydrochloride); 15 mg and 50 mg phenylpropanolamine hydrochloride tablets (equivalent to 12 mg and 40 mg phenylpropanolamine, respectively).**C Action:** Increases urethral outflow resistance and has some peripheral vasoconstrictive effects.**E Use:** Treatment of incontinence secondary to urinary sphincter incompetence. May also be useful in the management of nasal congestion. Incontinence may recur if doses are delayed or missed. The onset of action may take several days.**G Safety and handling:** Normal precautions should be observed.**H Contraindications:** No information available.**I Adverse reactions:** May include restlessness, aggressiveness, irritability and hypertension. Cardiotoxicity has been reported.**J Drug interactions:** No information available.**K DOSES****L Mammals:** Rabbits: 5–10 mg/rabbit p.o. q12h, reducing to lowest effective dose gradually.**Birds, Reptiles, Amphibians, Fish:** No information available.**M Phytomenadione see Vitamin K1****N Pimobendan**(Cardisure, Fortekor-Plus, Pimocard, Vetmedin) **POM-V****Q Formulations:** Injectable: 0.75 mg/ml solution (5 ml vial, Vetmedin). Oral: 5 mg hard capsules or 1.25 mg, 5 mg or 10 mg chewable tablets (Vetmedin); 1.25 mg, 2.5 mg, 5 mg, 10 mg flavoured tablets (Cardisure, Pimocard). Available in compound preparations with benazepril (1.25 mg pimobendan/2.5 mg benazepril; 5 mg pimobendan/10 mg benazepril) (Fortekor-Plus).**S Action:** Inodilator producing both positive inotropic and vasodilatory effects. Inotropic effects are mediated via sensitization of the myocardial contractile apparatus to intracellular calcium and by phosphodiesterase (PDE) III inhibition. Calcium sensitization allows for a positive inotropic effect without an increase in myocardial oxygen demand. Vasodilation is mediated by PDE III and V inhibition, resulting in arterio- and venodilation.**W Use:** Use in exotic species is anecdotal and extrapolated from use in dogs. Indicated for use with concurrent congestive heart failure therapy (e.g. furosemide, ACE inhibitors). The presence of food may reduce bioavailability.**X Safety and handling:** Normal precautions should be observed.**Y Contraindications:** Do not use in hypertrophic cardiomyopathy and in cases where augmentation of cardiac output via increased contractility is not possible (e.g. aortic stenosis).**Z**

Adverse reactions: A moderate positive chronotropic effect and vomiting may occur in some cases, which may be avoided by dose reduction.

Drug interactions: The positive inotropic effects are attenuated by drugs such as beta-blockers and calcium-channel blockers (especially verapamil). No interaction with digitalis glycosides has been noted.

DOSES

Mammals: Ferrets: 0.5 mg/kg p.o. q12h; Rabbits: 0.1–0.3 mg/kg p.o. q12–24h; Rodents: 0.2–0.4 mg/kg p.o. q12h; Primates: 0.2 mg/kg p.o. q24h; Sugar gliders: 0.3–0.5 mg/kg p.o. q12h; Hedgehogs: 0.3 mg/kg p.o. q12h.

Birds: 0.25 mg/kg p.o. q12h^{1,2} has been used for clinical cases, with doses of 10 mg/kg showing no side effects. Pharmacokinetic studies in Amazon parrots indicate that doses up to 10 mg/kg p.o. q12h may be required for optimal effect³.

Reptiles, Amphibians, Fish: No information available.

References

- ¹ Brandao J, Reynolds CA, Beaufre H et al. (2016) Cardiomyopathy in a Harris hawk (*Parabuteo unicinctus*). *Journal of the American Veterinary Association* **249**(2), 221–227
- ² Sedacca CD, Campbell TW, Bright JM, Webb BT and Aboellail TA (2009) Chronic cor pulmonale secondary to pulmonary atherosclerosis in an African grey parrot. *Journal of the American Veterinary Medical Association* **234**(8), 1055–1059
- ³ Guzman DS, Beaufre H, KuKanich B et al. (2014) Pharmacokinetics of single oral dose of pimobendan in Hispaniolan Amazon parrots (*Amazona ventralis*). *Journal of Avian Medicine and Surgery* **28**(2), 95–101

Piperacillin

(Tazocin*) **POM**

Formulations: Injectable: 2.25 g, 4.5 g powder (2 g or 4 g piperacillin sodium + 0.25 g or 0.5 g tazobactam (Tazocin)).

Action: Beta-lactam antibiotics bind penicillin-binding proteins involved in cell wall synthesis, decreasing bacterial cell wall strength and rigidity, and affecting cell division, growth and septum formation. As animal cells lack a cell wall the beta-lactam antibiotics are extremely safe in most species. The effect is bactericidal and killing occurs in a time-dependent fashion.

Use: Piperacillin is a ureidopenicillin. It is reserved for life-threatening infections (e.g. endocarditis or septicaemia) where culture and sensitivity testing predict a clinical response. These include infections caused by *Pseudomonas aeruginosa* and *Bacteroides fragilis* in neutropenic patients, although it has activity against other Gram-negative bacilli including *Proteus*. For pseudomonal septicaemias, antipseudomonal penicillins should be given with an aminoglycoside (e.g. gentamicin) as there is a synergistic effect. Piperacillin should usually be combined with a beta-lactamase inhibitor and therefore is co-formulated with tazobactam. Experience in veterinary species is limited, and doses are largely empirical.

Safety and handling: Normal precautions should be observed.

Contraindications: Avoid use in animals with reported sensitivity to penicillins. Do not administer to rabbits, guinea pigs, chinchillas, hamsters, gerbils or degus.

Adverse reactions: Nausea, diarrhoea and skin rashes are the commonest adverse effects in humans. Painful if given by i.m. injection.

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A The sodium content of each formulation may be clinically important for patients on restricted sodium intakes. Avoid use in animals with reported sensitivity to penicillins. Avoid administration in small herbivores (e.g. rabbits), hamsters and gerbils as penicillins can cause fatal enterotoxaemia.

B

C **Drug interactions:** Piperacillin enhances the effects of non-depolarizing muscle relaxants. Gentamicin inactivates piperacillin if mixed in the same syringe. Clinical experience with this drug is limited. There is synergism between the beta-lactams and the aminoglycosides.

D **DOSES**

E **Classified as category A (Avoid) by the EMA.**

F **See Appendix for guidelines on responsible antibiotic use.**

G **Mammals:** Hedgehogs: 10 mg/kg s.c. q8–12h; Rabbits, Chinchillas, Guinea pigs, Hamsters, Gerbils, Degus: Do not use.

H **Birds:** 100 mg/kg i.m., i.v. q12h^a.

I **Reptiles:** Chelonians, Lizards: 50–100 mg/kg i.m. q24h for 7–14 days; may be nebulized diluted 100 mg piperacillin/10 ml saline for 15–20 min q8–12h for lower respiratory tract infections; **Blood pythons:** <100 mg/kg i.m. q48h^b.

J **Amphibians:** 100 mg/kg s.c., i.m. q24h.

K **Fish:** No information available.

L **References**

- M** ^a Robbins PK, Tell LA, Needham ML and Craigmaill AL (2000) Pharmacokinetics of piperacillin after intramuscular injection in red-tailed hawks (*Buteo jamaicensis*) and great horned owls (*Bubo virginianus*). *Journal of Zoo and Wildlife Medicine* **31**(1), 47–51
- N** ^b Hilf M, Swanson D, Wagner R and Yu VL (1991) Pharmacokinetics of piperacillin in blood pythons (*Python curtus*) and *in vitro* evaluation of efficacy against aerobic Gram-negative bacteria. *Journal of Zoo and Wildlife Medicine* **22**(2), 199–203

P **Piperazine**

Q (Biozine, Easy Round Wormer, Piperazine Citrate Worm Tablets, Puppy Easy Worm Syrup, Roundworm, **Soluverm**)

R **AVM-GSL**

S **Formulations:** Oral: 100 mg, 105 mg, 416 mg, 500 mg tablets; 500 mg/g (50% w/w), 510 mg/g (51% w/w) powder; 58 mg/ml syrup. There are a variety of different formulations.

T **Action:** An anti-ascardial anthelmintic that blocks acetylcholine, thus affecting neurotransmission and paralysing the adult worm; it has no larvicidal activity.

U **Use:** Active against *Ascaris* spp. Ineffective against tapeworms and lung worms. Piperazine may be used in pregnant animals. Used for the treatment of non-encysted GI nematodes in fish.

W **Safety and handling:** Normal precautions should be observed.

X **Contraindications:** No information available.

Y **Adverse reactions:** Uncommon but occasionally vomiting or muscle tremors and ataxia have been reported.

Z **Drug interactions:** Piperazine and pyrantel have antagonistic mechanisms of action; do not use together.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Mammals: Ferrets: 50–100 mg/kg p.o., repeat in 2–3 weeks; Rabbits: 200 mg/kg p.o., repeat in 2–3 weeks; Chinchillas: 100 mg/kg p.o. q24h for 2 doses; Primates: 65 mg/kg p.o. q24h for 10 days; Sugar gliders: 50 mg/kg p.o. q24h. In-water medication (7 days on, 7 days off, 7 days on) at the following doses, is also possible: Guinea pigs, Hamsters: 10 mg/ml; Rats, Mice: 4–5 mg/ml.

Birds: Pigeons: 1.9 g/l water; Passerines: 3.7 g/l water for 12h. Repeat in 2–3 weeks.

Amphibians: 50 mg/kg p.o., repeat in 14 days.

Fish: 10 mg/kg in feed q24h for 3 doses or 110 mg/kg in feed once.

Reptiles: No information available.

Piroxicam

(Brexidol*, Feldene*, Piroxicam*) **POM**

Formulations: Oral: 10 mg, 20 mg capsules; 20 mg dissolving tablet. Injectable: 20 mg/ml solution.

Action: Inhibition of COX enzymes limits the production of prostaglandins involved in inflammation. Also limits tumour growth but the mechanism is still to be determined.

Use: In veterinary medicine, piroxicam has been used to treat certain tumours expressing COX receptors, e.g. transitional cell carcinoma of the bladder, prostatic carcinoma and colonic-rectal carcinoma and polyps. Piroxicam suppositories are available in the human field and may be useful in the management of colorectal polyps/neoplasia. Other NSAIDs are authorized for veterinary use in various inflammatory conditions but there is no information on the effect of these drugs in neoplastic conditions.

Safety and handling: Normal precautions should be observed.

Contraindications: Gastric ulceration, renal disease, concurrent use of corticosteroids.

Adverse reactions: As a non-specific COX inhibitor it may cause general adverse effects associated with NSAIDs, including GI toxicity, gastric ulceration and renal papillary necrosis (particularly if patient is dehydrated). There is a small risk that NSAIDs may precipitate cardiac failure in humans and this risk in animals is unknown.

Drug interactions: Do not use with corticosteroids or other NSAIDs (increased risk of gastric ulceration). Concurrent use with diuretics or aminoglycosides may increase risk of nephrotoxicity. Piroxicam is highly protein-bound and may displace other protein-bound drugs. The clinical significance of this is not well established.

DOSES

Mammals: Rabbits: 0.2 mg/kg p.o. q8h.

Birds: 0.5–0.8 mg/kg p.o. q12h.

Reptiles, Amphibians, Fish: No information available.

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A**PMSG** see **Serum gonadotrophin****B****C****Polysulphated glycosaminoglycan**(Adequan) **POM-V****D****Formulations:** Injectable: 100 mg/ml solution for i.m. injection.**E****Action:** Precursor to mucopolysaccharides, enzyme inhibitor, stimulates chondrocytes and synovial cells. Binds to damaged cartilage matrix consisting of aggregated proteoglycans and stimulates the synthesis of new glycosaminoglycan molecules and inhibits proteolytic enzymes.**F****G****Use:** Acts as a chondroprotective agent in the adjunctive management of non-infectious and non-immune-mediated arthritides. For intra-articular use surgical preparation of the joint is necessary and for both preparations aseptic technique is mandatory. Not a replacement, but rather an adjunct for other therapies.**H****I****Safety and handling:** Normal precautions should be observed.**J****Contraindications:** Not for use in patients where arthrotomy is anticipated because of a possible increase in bleeding. Not for use when infection is present or suspected.**K****L****Adverse reactions:** Not recommended for use in birds as fatal coagulopathies have been reported in one coraciiforme, two raptors, and one psittacine following intra-muscular injection^a. Intra-articular injection may cause pain and inflammation. Although rare, joint sepsis is possible.**M****N****Drug interactions:** None described to date.**O****DOSES****Mammals:** Rabbits: 2.2 mg/kg s.c., i.m. q3d for 21–28 days, then q14d as needed; Primates: 2 mg/kg i.m. q3–5d for 2–3 months.**P****Birds:** Do not use.**Q****Reptiles, Amphibians, Fish:** No information available.**R****References**^a Anderson K, Garner MM, Reed HH *et al.* (2013) Hemorrhagic diathesis in avian species following intramuscular administration of polysulfated glycosaminoglycan. *Journal of Zoo and Wildlife Medicine* **44**(1), 93–99**S****T****Polyvinyl alcohol**(Liquifilm Tears*, Sno Tears*) **P****U****V****Formulations:** Ophthalmic: 1.4%, 10 ml, 15 ml.**W****Action:** Polyvinyl alcohol is a synthetic resin tear substitute (lacromimetic).**X****Y****Use:** It is used for lubrication of dry eyes. In cases of keratoconjunctivitis sicca (KCS or dry eye) it will improve ocular surface lubrication, tear retention and patient comfort while lacrostimulation therapy (e.g. topical ciclosporin) is initiated. It is more adherent and less viscous than hypromellose. Patient compliance is poor if administered >q4h; consider using a longer acting tear replacement.**Z****Safety and handling:** Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: No information available.

Drug interactions: No information available.

DOSES

Mammals: 1 drop per eye q1h.

Birds, Reptiles, Amphibians, Fish: No information available.

Polyvinylpyrrolidone–iodine complex see **Povidone–iodine**

Potassium bromide

(Epilease, Libromide, Vetbromide, Potassium bromide solution) **POM-V, AVM-GSL**

Formulations: Oral: 325 mg, 600 mg tablets; 250 mg capsules; 250 mg/ml solution.

Action: Within the CNS it competes with transmembrane chloride transport and inhibits sodium, resulting in membrane hyperpolarization and elevation of the seizure threshold. Bromide competes with chloride in postsynaptic anion channels following activation by inhibitory neurotransmitters and therefore potentiates the effect of GABA. Acts synergistically with other therapeutic agents that have GABA-ergic effects (such as phenobarbital).

Use: Control of seizures in which the seizures are refractory to treatment with phenobarbital or where the use of phenobarbital or imepitoin is contraindicated. KBr is usually used in conjunction with phenobarbital. Although not authorized for this use, KBr has been used in conjunction with imepitoin. Bromide has a long half-life (>20 days) and steady state plasma concentrations may not be achieved for 3–4 months. Monitoring of serum drug concentrations should be performed and dose levels adjusted accordingly. The serum KBr concentration should reach 0.8–1.5 mg/ml to be therapeutic. The slow rise of plasma bromide levels after enteral administration limits its usefulness in status epilepticus. Bromide is well absorbed from the GI tract and eliminated slowly by the kidney in competition with chloride. High levels of dietary salt increase renal elimination of bromide. Consequently, it is important that the diet be kept constant once bromide therapy has started. Bromide will be measured in assays for chloride and will therefore produce falsely high 'chloride' results. Use with caution in renal disease.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Ataxia, sedation and somnolence are seen with overdosage. Skin reactions have been reported in animals with pre-existing skin diseases, e.g. flea bite dermatitis. Vomiting may occur after oral administration, particularly if high concentrations (>250 mg/ml) are used. Polyphagia, polydipsia and pancreatitis have also been reported. In the case of acute bromide toxicity, 0.9% NaCl i.v. is the treatment of choice. Less commonly behavioural changes, including irritability or restlessness, may be evident.

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Drug interactions: Bromide competes with chloride for renal reabsorption. Increased dietary salt, administration of fluids or drugs containing chloride, and use of loop diuretics (e.g. furosemide) may result in increased bromide excretion and decreased serum bromide concentrations.

B**C****DOSES****D**

Mammals: Ferrets: 22–30 mg/kg p.o. q24h in combination with phenobarbital; 70–80 mg/kg p.o. q24h if used alone.

E

Birds, Reptiles, Amphibians, Fish: No information available.

F**Potassium citrate****G**

(Cystopurin*, Potassium citrate BP*) **AVM-GSL**

H

Formulations: Oral: 30% solution. Various preparations are available.

I

Action: Enhances renal tubular resorption of calcium and alkalinizes urine.

J

Use: Management of calcium oxalate and urate urolithiasis, and fungal urinary tract infections. May be used to treat hypokalaemia, although potassium chloride or gluconate is preferred. Used to treat some forms of metabolic acidosis.

K**L**

Safety and handling: Normal precautions should be observed.

M

Contraindications: Renal impairment or cardiac disease.

N

Adverse reactions: Rare, but may include GI signs and hyperkalaemia.

O

Drug interactions: No information available.

P**DOSES**

Mammals: Rabbits: 33 mg/kg p.o. q8h; Guinea pigs: 10–30 mg/kg p.o. q12h.

Birds, Reptiles, Amphibians, Fish: No information available.

Q**Potassium permanganate (Permanganate of potash)****R**

(Permanganate dip) **ESPA**

S

Formulations: Immersion: Crystals for dissolution in water; solution (proprietary formulations are available).

T

Action: Oxidization of organic matter.

U

Use: Treatment of external bacterial infections and ectoparasites in freshwater fish. Used to disinfect aquarium plants and kill algae. Best used as a separate 30–60 min bath treatment. Effective treatment in ponds requires 2 mg/l of active chemical to be maintained throughout the treatment period. The permanganate ion (pink colour in solution) reacts with organic matter and is reduced to inactive manganese dioxide (clear or light brown colour). If the colour change takes place within 12 hours, more potassium permanganate should be added in 2 mg/l increments with care until the light pink colour is restored. Toxic at high pH because manganese dioxide may precipitate on to the gills. Stains many materials. Used in birds for cautery of broken bleeding nails/talons.

V**W****X****Y****Z**

Safety and handling: Normal precautions may be observed.

Contraindications: Do not use in sea water. Not recommended for use in aquaria since many species (e.g. zebrafish, *Danio* spp.) are sensitive to this chemical^a.

Adverse reactions: No information available.

Drug interactions: Do not use with formalin.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Birds: Effective by direct application for cautery of broken bleeding nails/talons.

Fish: External bacteria and parasites: 5 mg/l by immersion for 30–60 min or 2 mg/l by prolonged immersion (follow manufacturer's recommendations for proprietary formulations); Plant disinfection: 10 mg/l by immersion for 5–10 min.

Mammals, Reptiles, Amphibians: No information available.

References

- ^a Intorre L, Meucci V, Di Bello D *et al.* (2007) Tolerance of benzalkonium chloride, formalin, malachite green, and potassium permanganate in goldfish and zebrafish. *Journal of the American Veterinary Medical Association* **231**(4), 590–595

Potentiated sulphonamides see
Trimethoprim/Sulphonamide

Povidone–iodine (Iodophor, Polyvinylpyrrolidone–iodine complex, PVP)

(Tamodine, Betadine*, Vetasept*) **ESPA**

Formulations: Topical: 7.5% w/w povidone–iodine (0.75% w/w available iodine) solution.

Action: Biocidal effect on a broad range of organisms (bacteria, fungi, viruses, protozoa) by killing cells through iodination of lipids and oxidation of cytoplasmic and membrane compounds.

Use: Clean and treat wounds.

Safety and handling: Normal precautions should be observed.

Contraindications: Care should be taken when using with iodine-sensitive patients since potential for iodine to be absorbed by some animals (e.g. use in axolotls may cause metamorphosis).

Adverse reactions: No information available.

Drug interactions: No information available.

DOSES

Mammals, Birds, Reptiles, Amphibians, Fish: Topical: Apply undiluted with a cotton bud or swab to wet surface and rinse with fresh water.

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Pradofloxacin(Veraflox) **POM-V****Formulations:** Oral: 15 mg, 60 mg, 120 mg tablets; 25 mg/ml solution.**Action:** Broad-spectrum, concentration-dependent antibiotic that inhibits DNA gyrase and topoisomerase IV, blocking bacterial replication.**Use:** Fluoroquinolone use should be reserved for infections where culture and sensitivity testing predicts a clinical response and where first- and second-line antimicrobials would not be effective. Active against many Gram-negative and Gram-positive organisms. Much improved activity *versus* anaerobes compared with earlier generation fluoroquinolones. Like other members of the class, pradofloxacin has exceptional lipid solubility, attaining high concentrations especially in the urogenital tract including the prostate gland. Specific indications include superficial and deep pyodermas and wound infections associated with susceptible organisms such as *Staphylococcus pseudintermedius*, urinary tract infections associated with susceptible organisms such as *Escherichia coli* and *S. pseudintermedius*, severe periodontal disease associated with anaerobes such as *Porphyromonas* spp. and *Prevotella* spp. and acute severe upper respiratory tract infections associated with organisms such as *E. coli*, *Staphylococcus* spp. and *Pasteurella multocida*.**Safety and handling:** Normal precautions should be observed.**Contraindications:** Do not use in pregnant or lactating animals. Do not use in animals that are not skeletally mature due to potential adverse effects on cartilage. Do not use in animals with persistent cartilage lesions. Do not use in animals with neurological disease especially epilepsy.**Adverse reactions:** Potentially fatal GI dysbiosis.**Drug interactions:** Absorbents and antacids containing cations (Mg^{2+} , Al^{3+}) may bind fluoroquinolones, preventing their absorption from the GI tract. Their absorption may also be inhibited by sucralfate and zinc salts; separate dosing by at least 2 hours. Fluoroquinolones increase plasma theophylline concentrations. Cimetidine may reduce the clearance of fluoroquinolones and so should be used with caution with these drugs. Some fluoroquinolones may decrease the metabolism and increase nephrotoxicity of ciclosporin and tacrolimus in humans and therefore concurrent use in animals is best avoided until more research has been performed. May increase the action of orally administered anticoagulants.**DOSES****Classified as category B (Restrict) by the EMA.****See Appendix for guidelines on responsible antibiotic use.****Mammals:** Rabbits, Hindgut fermenting rodents: Do not use, as the spectra covered by this antimicrobial will change faecal microbiome and can lead to fatal caecal dysbiosis^{a,b}. **Rats:** Diarrhoea and dehydration has been noted. This drug is therefore not recommended in exotic pet mammals, in the absence of safety data.**Birds:** Not recommended for use in birds.**Reptiles:** No side effects seen in study in Eastern long-neck turtles^c but 7.5 mg/kg p.o. q24h was suggested to be ineffective against most bacterial pathogens.

Amphibians, Fish: No information available.

References

- ^a Xie S, Trott DJ, Saputra S *et al.* (2022) Pharmacokinetic profile and effect on the faecal microbiome of a single dose of pradofloxacin oral suspension in the rabbit (*Oryctolagus cuniculus*). *Journal of Veterinary Pharmacology and Therapeutics* **45(2)**, 203–212
- ^b Veraflox EPAR assessment report: https://www.ema.europa.eu/en/documents/assessment-report/veraflox-epar-public-assessment-report_en-0.pdf
- ^c Taylor E, Trott DJ, Kimble B, Xie S, Govendir M and McLelland DJ (2021) Pharmacokinetic profile of a single dose of an oral pradofloxacin suspension administered to eastern long-necked turtles (*Chelodina longicollis*). *Journal of Veterinary Pharmacology and Therapeutics* **44(4)**, 503–509

Praziquantel

(Anthelmin, Broadline Spot-on, Cazitel, Cestem, Dolpac, Droncit, Droncit Spot-on, Dronspot Spot-on, Drontal, Endoguard, **Fluke-Solve**, Milbactor, Milbemax, MilbeVet, Milprazon, Milpro, NexGard, Prazitel, Profender Spot-on, Veloxa, WormCat, WORMclear, Various other authorized proprietary preparations) **POM-V, NFA-VPS, AVM-GSL**

Formulations: Available in variety of tablets and spot-on pipettes either alone or in combination with other agents (e.g. febantel, pyrantel, fipronil, (S)-methoprene, oxantel, emodepside, tigolaner, milbemycin, esafloxolaner, eprinomectin). **For immersion:** 10 g, 100 g sachets (Fluke-Solve); 10 g, 50 g, 100 g sachets of powder containing praziquantel, nitroscanate and emamectin benzoate (Fluke-Solve Plus).

Action: Cestocide that increases cell membrane permeability of susceptible worms, resulting in loss of intracellular calcium and paralysis. This allows the parasites to be phagocytosed or digested.

Use: Used for the treatment of trematodes and cestodes including skin and gill trematode infections and tapeworms in ornamental fish. Switch off protein skimmers during treatment.

Safety and handling: Normal precautions should be observed. People who are or may become pregnant should not handle solutions containing emodepside.

Contraindications: Do not use at the same time as any other fish or pond treatments. Turn off UV and carbon filters for 24 hours after adding product to water. Do not use (Fluke-Solve Plus) in ponds containing orfe (*Leuciscus* spp.) or terrapins: use only in ponds containing koi carp exclusively.

Adverse reactions: Oral administration can occasionally result in anorexia, vomiting, lethargy and diarrhoea.

Drug interactions: No information available.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Mammals: Ferrets: 5–10 mg/kg p.o., repeat in 10–14 days; Rabbits: 5–10 mg/kg p.o., repeat in 10 days; **Gerbils, Rats, Mice:** 30 mg/kg p.o. q14d (for 3 treatments); **Primates:** 20 mg/kg p.o., i.m. once (cestodes) or 15–20 mg/kg p.o., i.m. once (trematodes) but might need repeating at 10 day intervals or 40 mg/kg i.m., p.o. once; **Hedgehogs:** 7 mg/kg p.o., repeat in 14 days; **Sugar gliders:** 5–10 mg/kg p.o., s.c., repeat in 10–14 days

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Birds: 10 mg/kg i.m.; 10–20 mg/kg p.o., repeat in 10–14 days.

Reptiles: 5–8 mg/kg p.o., repeat in 2 weeks in most species. Profender has been used topically in a variety of reptile species at doses up to 1.12 ml/kg, corresponding to 24 mg emodepside and 96 mg praziquantel/kg, once^{a,b}.

Amphibians: 8–24 mg/kg p.o. q7–21d or 10 mg/l bath for 3 hours repeat q7–21d.

Fish: 2 mg/l by immersion, repeat every 3 weeks for 3 doses; follow manufacturer's recommendations for proprietary formulations.

References

- ^a Schilliger L, Betremieux O, Rochet J, Krebber R and Schaper R (2009) Absorption and efficacy of a spot-on combination containing emodepside plus praziquantel in reptiles. *Revue de Médecine Vétérinaire* **160**(12), 557–561
- ^b Tang PK, Pellett S, Blake D and Hedley J (2017) Efficacy of a topical formulation containing emodepside and praziquantel (Profender®, Bayer) against nematodes in captive tortoises. *Journal of Herpetological Medicine and Surgery* **27**(3), 116–122

Prazosin

(Hypovase*, Prazosin*) **POM**

Formulations: Oral: 0.5 mg, 1 mg, 2 mg, 5 mg tablets.

Action: Prazosin is a postsynaptic alpha-1 blocking agent causing arterial and venous vasodilation. This leads to reduction in blood pressure and systemic vascular resistance.

Use: Adjunctive therapy of congestive heart failure secondary to mitral regurgitation in cases that are refractory to standard therapy. May be useful in promoting urine flow in patients with functional urethral obstruction and in the management of systemic or pulmonary hypertension. Efficacy may decline over time. Not often used.

Safety and handling: Normal precautions should be observed.

Contraindications: Hypotension, renal failure.

Adverse reactions: Hypotension, syncope, drowsiness, weakness, GI upsets.

Drug interactions: Concomitant use of beta-blockers (e.g. propranolol) or diuretics (e.g. furosemide) may increase the risk of a first dose hypotensive effect. Calcium-channel blockers may cause additive hypotension. Prazosin is highly protein-bound and so may be displaced by, or displace, other highly protein-bound drugs (e.g. sulphonamide) from plasma proteins.

DOSES

Mammals: Ferrets: 0.05–0.1 mg/kg p.o. q8h for urethral smooth muscle relaxation.

Birds, Reptiles, Amphibians, Fish: No information available.

Prednisolone

(PLT, Prednicare, Prednidale, Pred-forte*) **POM-V, POM**

Formulations: Ophthalmic: Prednisolone acetate 0.5%, 1% suspensions in 5 ml, 10 ml bottles (Pred-forte). **Topical:** Prednisolone is a component of many topical dermatological, otic and ophthalmic preparations.

Injectable: Prednisolone sodium succinate 10 mg/ml solution; 7.5 mg/ml suspension plus 2.5 mg/ml dexamethasone. **Oral:** 1 mg, 5 mg, 25 mg tablets. PLT is a compound preparation containing cinchophen.

Action: Binds to specific cytoplasmic receptors which then enter the nucleus and alter the transcription of DNA, leading to alterations in cellular metabolism which result in anti-inflammatory, immunosuppressive and antifibrotic effects. Also has glucocorticoid activity.

Use: Management of chronic allergic/inflammatory conditions (e.g. atopy, inflammatory bowel disease), immune-mediated conditions, hypoadrenocorticism, and lymphoproliferative and other neoplasms. In combination with cinchophen (PLT) it is used in the management of osteoarthritis. Prednisolone has approximately 4 times the anti-inflammatory potency and half the relative mineralocorticoid potency of hydrocortisone. It, like methylprednisolone, is considered to have an intermediate duration of activity and is suitable for alternate-day use. Animals on chronic therapy should be tapered off their steroids when discontinuing the drug. There are no studies comparing protocols for tapering immunosuppressive or anti-inflammatory therapy; it is appropriate to adjust the therapy according to laboratory or clinical parameters. For example, cases with immune-mediated haemolytic anaemia should have their therapy adjusted following monitoring of their haematocrit. There is no evidence that long-term low doses of glucocorticoids do, or do not, prevent relapse of immune-mediated conditions. Impaired wound healing and delayed recovery from infections may be seen. Use glucocorticoids with care in rabbits as they are sensitive to these drugs. If using in birds must make sure bird is genuinely pruritic and underlying infectious disease, e.g. aspergillosis, chlamydiosis, has been excluded before use. The use of steroids in most cases of shock and spinal cord injury is of no benefit and may be detrimental. Has been used successfully as part of the chemotherapy protocol in bearded dragons and in one green iguana.

Safety and handling: Shake suspension before use.

Contraindications: Do not use in pregnant animals. Systemic corticosteroids are generally contraindicated in patients with renal disease and diabetes mellitus. Topical corticosteroids are contraindicated in ulcerative keratitis.

Adverse reactions: Prolonged use of glucocorticoids suppresses the hypothalamic–pituitary axis (HPA), causing adrenal atrophy, and may cause significant proteinuria and glomerular changes. Catabolic effects of glucocorticoids lead to weight loss and cutaneous atrophy. Iatrogenic hyperadrenocorticism may develop with chronic use. Vomiting, diarrhoea and GI ulceration may develop; the latter may be more severe when corticosteroids are used in animals with neurological injury. Hyperglycaemia and decreased serum T4 values may be seen in patients receiving prednisolone. Corticosteroids should be used with care in birds as there is a high risk of immunosuppression and side effects, such as hepatopathy and a diabetes mellitus-like syndrome. In rabbits, even small single doses can potentially cause severe adverse reactions. Ferrets are particularly susceptible to GI ulceration, and concurrent gastric protectants may be advisable, especially in stressed animals.

Drug interactions: There is an increased risk of GI ulceration if used concurrently with NSAIDs. Hypokalaemia may develop if acetazolamide,

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amphotericin B or potassium-depleting diuretics (e.g. furosemide, thiazides) are administered concomitantly with corticosteroids. Glucocorticoids may antagonize the effect of insulin. The metabolism of corticosteroids may be enhanced by phenytoin or phenobarbital and decreased by antifungals (e.g. itraconazole).

DOSES

See Appendix for chemotherapy protocols in ferrets.

Mammals: Ferrets: Lymphoma; anti-inflammatory: 1–2 mg/kg p.o. q24h; postoperative management of adrenalectomy: 0.25–0.5 mg/kg p.o. q12h, taper to q48h; Rabbits: Anti-inflammatory: 0.25–0.5 mg/kg p.o. q12h for 3 days, then q24h for 3 days, then q48h; Primates: Anti-inflammatory: 0.5 mg/kg p.o., s.c., i.m. q24h; allergy, autoimmune disease: up to 2 mg/kg p.o., s.c., i.m. q24h; Sugar gliders: Anti-inflammatory: 0.1–0.2 mg/kg p.o., s.c., i.m. q24h; Hedgehogs: Allergy: 2.5 mg/kg p.o., s.c., i.m. q12h; Others: Anti-inflammatory: 1.25–2.5 mg/kg p.o. q24h. Ophthalmic: Dosage frequency and duration of therapy is dependent upon the type of lesion and response to therapy. Usually 1 drop in the affected eye(s) q4–24h, tapering in response to therapy. Care should be exercised in rabbits, as they are highly sensitive to the effects of steroids, even in topical preparations.

Birds: Pruritus: 1 mg/kg p.o. q12h, reduced to minimum effective dose as quickly as possible.

Reptiles: Anti-inflammatory: 2–5 mg/kg p.o. q24–48h; as part of chemotherapy protocol, used at 2 mg/kg p.o. q24h for 2 weeks, then 1 mg/kg p.o. q24h in a green iguana with lymphoma¹ and at 1 mg/kg p.o. q24h in bearded dragons with lymphoid leukaemia².

Amphibians, Fish: No information available.

References

- ¹ Folland DW, Johnston MS, Thamm DH and Reavill D (2011) Diagnosis and management of lymphoma in a green iguana (*Iguana iguana*). *Journal of the American Veterinary Medical Association* **239**(7), 985–991
- ² Hepps Keeney CM, Intile JL, Sims CS and Harrison TM. (2021) Lymphoid leukemia in five bearded dragons (*Pogona vitticeps*). *Journal of the American Veterinary Medical Association* **258**(7), 748–757

Prochlorperazine

(Buccastem*, Prochlorperazine*, Stemetil*) **POM**

Formulations: Injectable: 12.5 mg/ml solution in 1 ml ampoule. Oral: 3 mg, 5 mg tablets; 5 mg/5 ml syrup.

Action: Blocks dopamine, muscarinic acetylcholine and 5-HT₃ receptors in chemoreceptor trigger zone and vomiting centre.

Use: Predominantly to control motion sickness and emesis associated with vestibular disease.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Sedation, depression, hypotension and extrapyramidal reactions (rigidity, tremors, weakness, restlessness, etc.).

Drug interactions: CNS depressant agents (e.g. anaesthetics, narcotic analgesics) may cause additive CNS depression if used with prochlorperazine. Antacids or antidiarrhoeal preparations (e.g. bismuth

subsalicylate or kaolin/pectin mixtures) may reduce GI absorption of oral phenothiazines. Increased blood levels of both drugs may result if propranolol is administered with phenothiazines. Phenothiazines block alpha-adrenergic receptors, which may lead to unopposed beta activity causing vasodilation and increased cardiac rate if adrenaline is given.

DOSES

Mammals: Rabbits: 0.2–0.5 mg/kg p.o. q8h for reduction of possible motion sickness with torticollis.

Birds, Reptiles, Amphibians, Fish: No information available.

Proligestone

(Delvosteron) **POM-V**

Formulations: Injectable: 100 mg/ml suspension.

Action: Alters the transcription of DNA, leading to alterations in cellular metabolism which mimic those caused by progesterone.

Use: Authorized in ferrets to prevent the problems, such as vaginal and uterine infection and bone marrow depression, associated with prolonged oestrus and persistently high oestrogen levels. Can also be used for the treatment and prevention of false pregnancy in the rabbit. As coat colour changes may occasionally occur, injection into the medial side of the flank fold is recommended for thin-skinned or show animals.

Safety and handling: Normal precautions should be observed.

Contraindications: Best avoided in diabetic animals, as insulin requirements are likely to change unpredictably. Do not use in birds.

Adverse reactions: Proligestone does not appear to be associated with as many or as serious adverse effects as other progestogens (e.g. megestrol acetate, medroxyprogesterone acetate). However, adverse effects associated with long-term progestogen use, e.g. temperament changes (listlessness and depression), increased thirst or appetite, cystic endometrial hyperplasia/pyometra, diabetes mellitus, acromegaly, adrenocortical suppression, mammary enlargement/neoplasia and lactation, may be expected. Irritation at the site of injection may occur and calcinosis circumscripta at the injection site has been reported.

Drug interactions: No information available.

DOSES

Mammals: Ferrets: 50 mg/ferret s.c., if no response give 25 mg/ferret after 7 days, repeat at further 7 days; **Rabbits:** 30 mg/kg for pseudopregnancy.

Birds: Do not use.

Reptiles, Amphibians, Fish: No information available.

Propentofylline

(Canergy, Vitofyllin, Vivitonin) **POM-V**

Formulations: Oral: 50 mg, 100 mg tablets.

Action: Propentofylline is a xanthine derivative that increases blood flow to the heart, muscle and CNS via inhibition of phosphodiesterase. It also

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A has an antiarrhythmic action, bronchodilator effects, positive inotropic and chronotropic effects on the heart, inhibitory effects on platelet aggregation and reduces peripheral vascular resistance.

B **Use:** Anecdotally used in raptors for wing tip oedema and in parrots for egg yolk peritonitis. May also be utilised in treatment of bumblefoot in raptors. Also described in treatment of age-related cognitive dysfunction on raptors.

C **Safety and handling:** Normal precautions should be observed.

D **Contraindications:** No information available.

E **Adverse reactions:** May increase myocardial oxygen demand.

F **Drug interactions:** No information available.

G **DOSES**

H **Birds:** 5 mg/kg p.o. q12h^{1,2}.

I **Mammals, Reptiles, Amphibians, Fish:** No information available.

J **References**

¹ Chitty J (2020) Care of the geriatric raptor. *Veterinary Clinics: Exotic Animal Practice* **23**(3), 503–523

² Samour J, Wernick M and Zsivanovits P (2021) Therapeutic management of pododermatitis in falcon medicine: historical and modern perspective. *Archives of Veterinary and Animal Sciences* **3**(1), 1–5

L **Propofol**

M (Inductofol, Propodine, PropoFlo Plus, PropoFol-Lipuro
N Vet, Propomitor, Proposure, Vetofol) **POM-V**

O **Formulations:** Injectable: 10 mg/ml solution: lipid emulsion available both without a preservative or antibacterial or with a preservative (benzyl alcohol), 20 ml, 50 ml and 100 ml glass bottles. The solution containing a preservative can be used for up to 28 days after the vial is first broached.

P **Action:** The mechanism of action is not fully understood but it is thought to involve modulation of the inhibitory activity of GABA at GABA receptors.

Q **Use:** Induction of anaesthesia and maintenance of anaesthesia using intermittent boluses or a continuous rate infusion. The solution containing benzyl alcohol preservative should not be used for maintenance of anaesthesia by continuous rate infusion due to the risk of toxicity caused by prolonged administration. Injection i.v. produces a rapid loss of consciousness as the CNS takes up the highly lipophilic drug. Over the next few minutes propofol distributes to peripheral tissues and the concentration in the CNS falls such that, in the absence of further doses, the patient wakes up. Propofol does not have analgesic properties, therefore it is better used in combination with other drugs to maintain anaesthesia; for example, a continuous rate infusion of a potent opioid. Considerable care must be taken with administration in hypovolaemic animals and those with diminished cardiopulmonary, hepatic and renal reserves. Propofol has been used by immersion to induce general anaesthesia in a few species of fish (koi, goldfish, trout and Nile tilapia).

S **Safety and handling:** Shake the lipid emulsion well before use and do not mix with other therapeutic agents or therapeutic fluids prior to administration. If using a preparation that contains no bacteriostat,

opened bottles should be stored in a refrigerator and used within 8 hours or discarded. Once broached, the lipid preparation with a preservative has a shelf-life of 28 days.

Contraindications: No information available.

Adverse reactions: The rapid injection of large doses causes apnoea, cyanosis, bradycardia and severe hypotension. Problems are less likely when injection is made over 30–60 seconds. Propofol is not irritant to tissues but a pain reaction is commonly evident during i.v. injection; the underlying mechanism causing pain is unknown.

Drug interactions: No information available.

DOSES

Mammals: Ferrets: 2–8 mg/kg i.v.; Rabbits: Unpremedicated: 7.5–15 mg/kg i.v.; premedicated: 2–6 mg/kg i.v.; Rats: 7.5–10 mg/kg i.v.; Mice: 12–26 mg/kg i.v.; Primates: 2.5–5 mg/kg i.v. followed by 0.3–0.5 mg/kg/min CRI.

Birds: 10 mg/kg i.v. by slow infusion to effect: supplemental doses up to 3 mg/kg^a; 0.5 mg/kg/min i.v. infusion as CRI.

Reptiles: 5–10 mg/kg i.v., intraosseous; Red-eared sliders: 10–20 mg/kg i.v. (higher doses required for successful intubation)^b; Bearded dragons: 10 mg/kg i.v.^c; Green iguanas: 5–10 mg/kg intraosseous; Brown tree snakes: 5 mg/kg i.v.

Amphibians: Leopard frogs: 10 mg/kg injected in the sublingual plexus area (achieves sedation/light anaesthesia); Tiger salamanders: 10–35 mg/kg intracoelomic gives induction in 30 minutes and recovery in 24 hours^d. Euthanasia: 60–100 mg/kg intracoelomic or 100–400 mg/kg topically, remove and rinse after induction.

Fish: Anaesthesia: 2.5–10 mg/l by immersion^{e,f,g}.

References

- ^a Langlois I, Harvey RC, Jones MP and Schumacher J (2003) Cardiopulmonary and anesthetic effects of isoflurane and propofol in Hispaniolan Amazon parrots (*Amazona ventralis*). *Journal of Avian Medicine and Surgery* **17**(1), 4–10
- ^b Ziolo MS and Bertelsen MF (2009) Effects of propofol administered via the supravertebral sinus in red-eared sliders. *Journal of the American Veterinary Medical Association* **234**(3), 390–393
- ^c Perrin KL and Bertelsen MF (2017). Intravenous Alfaxalone and propofol anesthesia in the bearded dragon (*Pogona vitticeps*). *Journal of Herpetological Medicine and Surgery* **27**(3), 123–126
- ^d Mitchell MA, Riggs SM, Singleton CB et al. (2009) Evaluating the clinical and cardiopulmonary effects of clove oil and propofol in tiger salamanders (*Ambystoma tigrinum*). *Journal of Exotic Pet Medicine* **18**, 50–56
- ^e Oda A, Bailey KM, Lewbart GA, Griffith EH and Posner LP (2014) Physiologic and biochemical assessments of koi (*Cyprinus carpio*) following immersion in propofol. *Journal of the American Veterinary Medical Association* **245**(11), 1286–1291
- ^f Balko JA, Wilson SK, Lewbart GA, Gaines BR and Posner LP (2017) Propofol as an immersion anesthetic and in a minimum anesthetic concentration (MAC) reduction model in goldfish (*Carassius auratus*). *Journal of Zoo and Wildlife Medicine* **48**(1), 48–54
- ^g Oda A, Messenger KM, Carbajal L et al. (2018) Pharmacokinetics and pharmacodynamic effects in koi carp (*Cyprinus carpio*) following immersion in propofol. *Veterinary Anaesthesia and Analgesia* **45**(4), 529–38

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Propranolol

(Inderal*, Propranolol*) **POM**

Formulations: Injectable: 1 mg/ml solution. Oral: 10 mg, 40 mg, 80 mg, 160 mg tablets.

Action: Non-selective beta-blocker. Blocks the chronotropic and inotropic effects of beta-1 adrenergic stimulation on the heart, thereby reducing myocardial oxygen demand. Blocks the dilatory effects of beta-2 adrenergic stimulation on the vasculature and bronchial smooth muscle. The antihypertensive effects are mediated through reducing cardiac output, altering the baroreceptor reflex sensitivity and blocking peripheral adrenoceptors.

Use: Management of cardiac arrhythmias (sinus tachycardia, atrial fibrillation or flutter, supraventricular tachycardia, ventricular arrhythmias), hypertrophic cardiomyopathy or obstructive heart disease. Potential efficacy as an additional antihypertensive drug and can be used in pheochromocytoma if combined with an alpha-blocker. Used to reverse some of the clinical features of thyrotoxicosis prior to surgery in patients with hyperthyroidism. May be used in behavioural therapy to reduce somatic signs of anxiety and is therefore useful in the management of situational anxieties and behavioural problems where contextual anxiety is a component. Some authors suggest using propranolol in combination with phenobarbital for the management of fear- and phobia-related behaviour problems. There is a significant difference between i.v. and oral doses. This is a consequence of propranolol's lower bioavailability when administered orally as a result of decreased absorption and a high first pass effect. Wean off slowly when using chronic therapy.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use in patients with bradyarrhythmias, acute or decompensated congestive heart failure. Relatively contraindicated in animals with medically controlled congestive heart failure as is poorly tolerated. Do not administer concurrently with alpha-adrenergic agonists (e.g. adrenaline).

Adverse reactions: Bradycardia, AV block, myocardial depression, heart failure, syncope, hypotension, hypoglycaemia, bronchospasm, diarrhoea and peripheral vasoconstriction. Depression and lethargy are occasionally seen as a result of CNS penetration. Propranolol may exacerbate any pre-existing renal impairment. Sudden withdrawal of propranolol may result in exacerbation of arrhythmias or the development of hypertension.

Drug interactions: The hypotensive effect of propranolol is enhanced by many agents that depress myocardial activity including anaesthetic agents, phenothiazines, antihypertensive drugs, diuretics and diazepam. There is an increased risk of bradycardia, severe hypotension, heart failure and AV block if propranolol is used concurrently with calcium-channel blockers. Concurrent digoxin administration potentiates bradycardia. The metabolism of propranolol is accelerated by thyroid hormones, thus reducing its effect. The dose of propranolol may need to be decreased when initiating carbimazole therapy. Oral aluminium hydroxide preparations reduce propranolol absorption. Cimetidine may decrease the metabolism of propranolol, thereby increasing its levels in the blood. Propranolol enhances the effects of muscle relaxants (e.g.

suxamethonium, tubocurarine). Hepatic enzyme induction by phenobarbital or phenytoin may increase the rate of metabolism of propranolol. There is an increased risk of lidocaine toxicity if administered with propranolol due to a reduction in lidocaine clearance. The bronchodilatory effects of theophylline may be blocked by propranolol. Although the use of propranolol is not contraindicated in patients with diabetes mellitus, insulin requirements should be monitored as propranolol may enhance the hypoglycaemic effect of insulin.

DOSES

Mammals: Ferrets: 0.2–2 mg/kg p.o., s.c. q12–24h for hypertrophic cardiomyopathy.

Birds: Supraventricular tachycardia: 0.04 mg/kg slow i.v.

Reptiles, Amphibians, Fish: No information available.

Prostaglandin E2 see **Dinoprostone**
Prostaglandin F2 see **Dinoprost**
tromethamine

Proxymetacaine (Proparacaine) (Proxymetacaine*) POM

Formulations: Ophthalmic: 0.5% ($\pm$ fluorescein 0.25%), 1.0% solution (single-use vials).

Action: Local anaesthetic action is dependent on reversible blockade of the sodium channel, preventing propagation of an action potential along the nerve fibre. Sensory nerve fibres are blocked before motor nerve fibres, allowing a selective sensory blockade at low doses.

Use: Proxymetacaine is used on the ocular surface (cornea and conjunctival sac), the external auditory meatus and the nares. It acts rapidly (within 10 seconds) and provides anaesthesia for 25–55 minutes in the conjunctival sac depending on the species. Serial application increases duration and depth of anaesthesia. Topical anaesthetics block reflex tear production and should not be applied before a Schirmer tear test.

Safety and handling: Store in refrigerator and in the dark; reduced efficacy if stored at room temperature for >2 weeks.

Contraindications: Do not use for therapeutic purposes.

Adverse reactions: Conjunctival hyperaemia is common; local irritation manifested by chemosis may occasionally occur for several hours after administration (less likely than with tetracaine). All topical anaesthetics are toxic to the corneal epithelium and delay healing of ulcers.

Drug interactions: No information available.

DOSES

Mammals: Ophthalmic: 1–2 drops per eye; duration 1 hour in rabbits.

Birds, Reptiles, Amphibians, Fish: No information available.

PVP see **Povidone–iodine**

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A Pyrantel

(Anthelmin, Cazitel, Cestem, Dolpac, Drontal, Endoguard, Prazitel, Veloxa, WORMclear, Various authorized proprietary products) **POM-V, ESPA**

Formulations: Available in variety of formulations in combination with other agents (e.g. febantel, praziquantel). **Oral:** 10 mg, 50 mg, 80 mg, 87 mg, 120 mg, 125 mg, 175 mg tablets; 5 mg/ml liquid. NB: Some formulations and doses give content of pyrantel in terms of pyrantel embonate/pamotate (50 mg pyrantel is equivalent to 144 mg pyrantel embonate/pamotate).

Action: A cholinergic agonist which interferes with neuronal transmission in parasites and thereby kills them. Febantel and oxantel are derivatives of pyrantel with increased activity against whipworms.

Use: Control of *Toxocara canis*, *Toxascaris leonina*, *Trichuris vulpis*, *Uncinaria stenocephala*, *Ancylostoma caninum* and *A. braziliensis*.

Safety and handling: Normal precautions should be observed.

Contraindications: Safety has not been established in pregnant or lactating animals and therefore its use is not recommended.

Adverse reactions: Vomiting and diarrhoea may be observed.

Drug interactions: The addition of febantel or oxantel has a synergistic effect. Do not use with levamisole, piperazine or cholinesterase inhibitors.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Mammals: Ferrets: 4.4 mg/kg (pyrantel embonate) p.o., repeat in 14 days; Rabbits: 5–10 mg/kg (pyrantel embonate/pamoate) p.o., repeat in 10–21 days; Rodents: 50 mg/kg (pyrantel embonate/pamoate) p.o., repeat as required; **Primates:** 11 mg/kg p.o., repeat in 14 days^a.

Reptiles: 5 mg/kg p.o., repeat in 14 days.

Amphibians: 5 mg/kg p.o. q14d.

Birds, Fish: No information available.

References

- ^a Bentzel DE and Bacon DJ (2007) Comparison of various anthelmintic therapies for the treatment of *Trypanoxyuris microon* infection in owl monkeys (*Aotus nancymae*). *Comparative Medicine* **2**, 206–209

Pyrimethamine

(Daraprim*, Fansidar*) **P**

Formulations: Oral: 25 mg tablet. Fansidar also contains sulfadoxine.

Action: Interference with folate metabolism of the parasite and thereby prevents purine synthesis (and therefore DNA synthesis).

Use: Infections caused by *Toxoplasma gondii*, *Neospora caninum* and *Plasmodium* spp. Should not be used in pregnant or lactating animals without adequate folate supplementation. Used in birds to treat toxoplasmosis, sarcocystosis and leucocytozoonosis.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Depression, anorexia and reversible bone marrow suppression (within 6 days of the start of therapy). Folate supplementation (5 mg/day) may prevent bone marrow suppression.

Drug interactions: Increased antifolate effect if given with phenytoin or sulphonamides. Folate supplementation for the host will reduce the efficacy of the drug if given concomitantly and should thus be given a few hours before pyrimethamine.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Mammals: Ferrets: 0.5 mg/kg p.o. q12h; Primates: Toxoplasmosis: 2 mg/kg p.o. q24h for 3 days, then 1 mg/kg p.o. q24h for 28 days; *Plasmodium*: 10 mg/kg p.o. q24h; *Encephalitozoon cuniculi*: 0.5 mg/kg p.o. q12h. Monitor for folate deficiency and supplement with folate.

Birds: 0.5–1 mg/kg p.o. q12h for 28 days¹.

Reptiles, Amphibians, Fish: No information available.

References

- ¹ Doneley RJ (2009) Bacterial and parasitic diseases of parrots. *Veterinary Clinics of North America: Exotic Animal Practice* **12**(3), 417–432

Pyriproxyfen

(Indorex household spray) **AVM-GSL**

Formulations: Environmental spray.

Action: Juvenile hormone analogue. Arrests the development of flea larvae in the environment.

Use: Use as part of a comprehensive flea control programme in ferrets and rabbits (anecdotal) in conjunction with on-animal adulticide products. Anecdotally has been used in aviaries to control tick infestations.

Safety and handling: Normal precautions should be observed. The product should not enter watercourses as this may be dangerous for fish and other organisms.

Contraindications: Do not use directly on animals.

Adverse reactions: None reported.

Drug interactions: None reported.

DOSES

See Appendix for guidelines on responsible parasiticide use.

Use in the environment as directed.

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Ranitidine

(Ranitidine*, Zantac*) **POM**

Formulations: Injectable: 25 mg/ml solution. Oral: 75 mg, 150 mg, 300 mg tablets; 15 mg/ml syrup. Currently, nearly all preparations of ranitidine are unavailable due to carcinogenic qualities of the human preparations. For veterinary use, specials manufacturers should be contacted. **VSP**

Action: Ranitidine is a histamine (H₂) receptor antagonist blocking histamine-induced gastric acid secretion. It is more potent than cimetidine but has lower bioavailability (50%) and undergoes hepatic metabolism. It also has a prokinetic effect through stimulation of local muscarinic acetylcholine receptors, which may be of benefit when GI motility is impaired.

Use: Management of gastric and duodenal ulcers, idiopathic, uraemic or drug-related erosive gastritis, oesophagitis, and hypersecretory conditions secondary to gastrinoma or mast cell neoplasia. Often used prophylactically in ferrets to prevent stress-associated gastric ulceration. If used for the treatment of ulceration, then treatment should continue for 2 weeks after remission of clinical signs which means typically a 1 month course. Absorption is not clinically significantly affected by food intake, anticholinergic agents or antacids. Used for its prokinetic effect in rabbits, chinchillas and guinea pigs. Currently cimetidine is the only antiulcer drug with a veterinary market authorization. However, the use of ranitidine is justified under the cascade when predominantly used for its prokinetic effect.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Rarely reported but include cardiac arrhythmias and hypotension, particularly if administered rapidly i.v.

Drug interactions: It is advisable, though not essential, that sucralfate is administered 2 hours before H₂ blockers. Stagger oral doses of ranitidine when used with other antacids, digoxin or metoclopramide by 2 hours as it may reduce their absorption or effect.

DOSES

Mammals: Ferrets: 3.5 mg/kg p.o. q12h; Rabbits: 4–6 mg/kg p.o., s.c. q8–24h; Guinea pigs, Chinchillas: 5 mg/kg p.o. q12h as a prokinetic; Primates: 0.5 mg/kg p.o. q12h (anti-ulcer); Hedgehogs: 2mg/kg s.c. q12h¹.

Birds, Reptiles, Amphibians, Fish: No information available.

References

¹ Lee SY and Park HM (2012) Gastroesophageal intussusception with megaesophagus in a hedgehog (*Atelerix albiventris*). *Journal of Exotic Pet Medicine* **21**(2), 168–171

Retinol see **Vitamin A**

Rifampin (Rifampicin)

(Rifadin*, Rifampicin*, Rimactane*) **POM**

Formulations: Oral: 150 mg, 300 mg capsules; 20 mg/ml syrup.

Action: Bactericidal drug binding to the beta subunit of RNA polymerase and causing abortive initiation of RNA synthesis.

Use: Wide spectrum of antimicrobial activity including bacteria (particularly Gram-positive), *Chlamydia*, *Rickettsia*, some protozoans and poxviruses. Very active against *Staphylococcus aureus* and *Mycobacterium tuberculosis*. Gram-negative aerobic bacteria are usually innately resistant. Obligate anaerobes (Gram-positive or -negative) are usually susceptible. Exact indications for small animal veterinary practice remain to be fully established. It has been suggested as part of the combination of treatments for mycobacterial infections in primates in combination with ethambutol and isoniazid. It may also have a place in the management of chlamydiosis, ehrlichiosis and bartonellosis. Chromosomal mutations readily lead to resistance, therefore rifampin should be used in combination with other antimicrobial drugs to prevent the emergence of resistant organisms. Various combinations of clarithromycin, enrofloxacin, clofazimine and doxycycline have been used with rifampin in the management of mycobacteriosis, including in birds. Until controlled studies are conducted to investigate the value of rifampin in these infections, recommendations remain empirical.

Safety and handling: People who are or may become pregnant should not handle crushed or broken tablets or the syrup without the use of gloves.

Contraindications: Rifampin may be teratogenic at high doses and should not be administered to pregnant animals. It should not be administered to animals with liver disease.

Adverse reactions: Rifampin metabolites may colour urine, saliva and faeces orange-red.

Drug interactions: Rifampin is a potent hepatic enzyme inducer and increases the rate of metabolism of other drugs in humans, including barbiturates, theophylline and itraconazole. Increased dosages of these drugs may be required if used in combination with rifampin.

DOSES

Classified as category A (Avoid) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Rabbits: 40 mg/kg p.o. q12h rifampin in conjunction with 50 mg/kg p.o. q24h azithromycin or 80 mg/kg p.o. q12h clarithromycin for *Staphylococcus osteomyelitis*^a.

Birds: 10–20 mg/kg p.o. q12–24h.

Reptiles, Amphibians, Fish: No information available.

References

- ^a Shirtliff ME, Mader JT and Cathoun J (1999) Oral rifampin plus azithromycin or clarithromycin to treat osteomyelitis in rabbits. *Clinical Orthopaedics and Related Research* (1976–2007) **359**, 229–236

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Robenacoxib(Onsior) **POM-V****Formulations:** Oral: 5 mg, 6 mg, 10 mg, 20 mg, 40 mg flavoured tablets. **Injectable:** 20 mg/ml solution.**Action:** Selectively inhibits COX-2 enzyme, thereby limiting the production of prostaglandins involved in inflammation. Robenacoxib is tissue selective, defined as being preferentially distributed and concentrated at sites of inflammation combined with having a short half-life in plasma (approximately 2 hours). This may confer advantages in terms of reducing exposure of target side effect organs (e.g. liver and kidneys) to robenacoxib in a 24-hour dose interval, although there are currently no clinical data to support this contention, particularly in exotic species and birds.**Use:**

- Alleviation of inflammation and pain in both acute and chronic musculoskeletal disorders.
- Treatment of proventricular dilatation disease (PDD) caused by avian bornavirus infection) in psittacids^{a,1}.
- The oral dose may be administered directly into the mouth or mixed with a small amount of food. Dose rates and indications are anecdotal though widely reported.

Safety and handling: Normal precautions should be observed.**Contraindications:** Do not give to dehydrated, hypovolaemic or hypotensive patients, or those with GI disease or blood clotting problems. Administration of robenacoxib to animals with renal disease must be carefully evaluated and is not advisable in the perioperative period. If hypotension during anaesthesia is anticipated, delay robenacoxib administration until the animal is fully recovered and normotensive.**Adverse reactions:** None reported, though can be expected in line with other NSAIDs.**Drug interactions:** Different NSAIDs should not be administered within 24 hours of each other. The optimal wash-out period between NSAID administration and glucocorticoid administration is unknown – allowing 3–5 days between the two classes of drugs is recommended. Do not administer with other potentially nephrotoxic agents, e.g. aminoglycosides. As this drug is highly protein-bound, there may be interactions with other protein-bound drugs (e.g. sulphonamides) reducing binding of robenacoxib thus reducing duration and increasing likelihood of adverse reactions. Similarly, efficacy of concurrently administered ACE inhibitors and furosemide may be reduced.**DOSES****Birds:** 2–10 mg/kg i.m. q7d for 30 days, then monthly for PDD.**Mammals, Reptiles, Amphibians, Fish:** No information available.**References**

^a Hoppes SM and Shivaprasad HL (2020) Update on avian bornavirus and proventricular dilatation disease: diagnostics, pathology, prevalence, and control. *Veterinary Clinics: Exotic Animal Practice* **23**(2), 337–351.

¹ Romagnano A (2014) Proventricular Dilatation Disease. In: *Proceedings of BSAVA Congress*

Rocuronium

(Esmeron*) **POM**

Formulations: Injectable: 10 mg/ml solution.

Action: Inhibits the actions of acetylcholine at the neuromuscular junction by binding competitively to the nicotinic acetylcholine receptor on the post-junctional membrane.

Use: Provision of neuromuscular blockade during anaesthesia. This may be to improve surgical access through muscle relaxation, facilitate positive pressure ventilation or intraocular surgery. Rocuronium is very similar to vecuronium but it has a more rapid onset of action and shorter duration to spontaneous recovery. Its availability in aqueous solution and longer shelf-life increase convenience. Monitoring (using a nerve stimulator) and reversal of the neuromuscular blockade is recommended to ensure complete recovery before the end of anaesthesia. The neuromuscular blockade caused by rocuronium can be rapidly reversed using sugammadex (a cyclodextrin developed to reverse aminosteroidal neuromuscular blocking agents) at a dose of 8 mg/kg i.v. in dogs. Hypothermia, acidosis and hypokalaemia will prolong the duration of action of neuromuscular blockade. Hepatic disease may prolong duration of action of rocuronium; atracurium is preferred in this group of patients. The effects of renal disease on duration of action of rocuronium require further investigation.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not administer unless the animal is adequately anaesthetized and facilities to provide positive pressure ventilation are available.

Adverse reactions: Causes an increase in heart rate and a mild hypertension when used at high doses. May cause transient paralysis of lower eyelid in birds.

Drug interactions: Neuromuscular blockade is more prolonged when rocuronium is given in combination with volatile anaesthetics, aminoglycosides, clindamycin and lincomycin.

DOSES

Birds: Tawny owls: 0.35 mg/eye^a; Little owls: 0.2 mg/eye topically^b; **Scops owls:** 0.15 mg/eye topically^c; **Kestrels:** 0.12 mg/eye topically^d; **Amazon parrots:** 20 µl (microlitre)/eye topically. Doses can be repeated doses for ocular surgery without significant side effects^e.

Reptiles: 0.25–0.5 mg/kg i.m. Not recommended as a substitute for analgesia or general anaesthesia.

Mammals, Amphibians, Fish: No information available.

References

- ^a Barsotti G, Briganti A, Spratte JR, Ceccherelli R and Breggi G (2010) Mydriatic effect of topically applied rocuronium bromide in tawny owls (*Strix aluco*): comparison between two protocols. *Veterinary Ophthalmology* **13**(S1), 9–13
- ^b Barsotti G, Asti M, Giani E, Ceccherelli R, and Briganti A (2019) Effect of topical ophthalmic instillation of rocuronium bromide on the intraocular pressure of kestrels (*Falco tinnunculus*) and little owls (*Athene noctuae*). *Journal of the American Veterinary Medical Association*, **255**(12), 1359–1364
- ^c Cantero F, Ortillés Á, Peña MT et al. (2021) Safety and efficacy of unilateral topical application of rocuronium bromide in healthy scops owls (*Otus scops*). *Veterinary Ophthalmology*, **24**(2), 169–176

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^d Barsotti G, Briganti A, Spratte JR, Ceccherelli R and Breggi G (2012) Safety and efficacy of bilateral topical application of rocuronium bromide for mydriasis in European kestrels (*Falco tinnunculus*). *Journal of Avian Medicine and Surgery* **26**(1), 1–5

^e Dongo PS, Pinto DG, Guimarães MB *et al.* (2021). Effects of bilateral topical administration of repeated doses of rocuronium bromide as a mydriatic agent in Amazon parrots (*Amazona aestiva* and *Amazona amazonica*). *Journal of Exotic Pet Medicine* **39**, 11–16

S-Adenosylmethionine (SAmE)

(Denamarin, Denosyl, Doxion, Hepaticare, Hepatosyl Plus, MaxxiSAmE, Samylin, Zentonil Advanced, various authorized proprietary products) **GSL**

Formulations: Oral: 90 mg, 100 mg, 200 mg, 225 mg, 400 mg, 425 mg tablets; 50 mg, 100 mg, 200 mg capsules; 75 mg, 300 mg, 400 mg powder.

Action: S-Adenosylmethionine (SAmE) is an endogenous molecule synthesized by cells throughout the body and is a component of several biochemical pathways. SAmE is especially important in hepatocytes because of their central role in metabolism. In humans, antidepressant effects of SAmE are also documented.

Use: Adjunctive treatment for liver disease, especially for acute hepatotoxin-induced liver disease. SAmE has been shown to increase hepatic glutathione levels; a potent antioxidant which protects hepatocytes from toxins and death. Can also be used in patients on long-term therapy with potentially hepatotoxic drugs. The use as an antidepressant therapy in animals has yet to be established. The safety of exogenous SAmE has not been proven in pregnancy; therefore, it should be used with caution.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: None reported. GI signs (nausea, vomiting, diarrhoea), dry mouth, headache, sweating and dizziness are occasionally reported in humans.

Drug interactions: Concurrent use of SAmE with tramadol, meperidine, pentazocine, MAOIs including selegiline, SSRIs such as fluoxetine, or other antidepressants (e.g. amitriptyline) could cause additive serotonergic effects. SAmE may increase the clearance of drugs that undergo hepatic glucuronidation, including paracetamol, diazepam and morphine.

DOSES

Mammals: Ferrets, Rodents: 20–100 mg/kg p.o. daily for liver support.

Reptiles: 30 mg/kg p.o. daily for liver support.

Birds, Amphibians, Fish: No information available.

Salbutamol

(Ventolin*) **POM**

Formulations: Inhalational: 100 µg (micrograms) per metered inhalation (Evohaler). Injectable: 0.5 mg/ml.

Action: Selective beta-2 stimulation causes smooth muscle relaxation and bronchodilation.

Use: Treatment of bronchospasm in inflammatory airway disease and irritation. Used in rats for mycoplasmosis-related chronic obstructive pulmonary disease (COPD).

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

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A **Adverse reactions:** In humans side effects of the beta-2 agonists include headache, muscle cramps and palpitation. Other side effects include tachycardia, arrhythmias, peripheral vasodilation, and disturbances of sleep and behaviour.

C **Drug interactions:** In humans there is an increased risk of side effects if salbutamol is used by patients also taking diuretics, digoxin, theophylline or corticosteroids.

D **DOSES**

E **Mammals:** Rats: 100 µg (micrograms)/rat q4–6h (use a small chamber) or as needed for relief of bronchospasm.

F **Birds:** 50 µg (micrograms) s.c., i.m., i.v. q8h.

G **Reptiles, Amphibians, Fish:** No information available.

H **Selamectin**

I (Evicto, Selehold, Stronghold, Stronghold Plus) **POM-V**

J **Formulations:** Topical: Spot-on pipettes of various sizes containing 6% or 12% selamectin (Evicto, Selehold, Stronghold). Combined with sarolaner (Stronghold Plus).

K **Action:** Interacts with GABA and glutamate-gated channels leading to flaccid paralysis of the parasite.

L **Use:** Treatment and prevention of flea and ear mite infestations, *Trixacaris*, *Cheyletiella*, sarcoptic acariasis, biting lice, hookworms (*Ancylostoma tubaeforme*), adult roundworms (*Toxocara canis*, *T. cati*) and prevention of heartworm disease (*Dirofilaria immitis*). Frequent shampooing may reduce the efficacy of the product. Can be used in lactation and pregnancy.

O **Safety and handling:** Highly toxic to aquatic organisms; therefore, take care with disposal.

P **Contraindications:** None.

Q **Adverse reactions:** Transient pruritus and erythema at the site of application may occur. Do not use in chelonians as could potentially cause neurotoxicity and death similar to ivermectin.

R **Drug interactions:** No information available.

S **DOSES**

See Appendix for guidelines on responsible parasiticide use.

T **Mammals:** Ferrets, Rabbits, Rodents: 6–15 mg/kg monthly; Sugar gliders, Hedgehogs: 6–18 mg/kg monthly; **Primates:** 6 mg/kg^a.

U **Birds:** 20 mg/kg single dose¹.

V **Reptiles:** Do not use in chelonians.

W **Amphibians:** Bullfrogs: 6 mg/kg single-dose^b.

X **Fish:** No information available.

Y **References**

^a Wang T, Yang GY, Yan HJ *et al.* (2008) Comparison of efficacy of selamectin, ivermectin and mebendazole for the control of gastrointestinal nematodes in rhesus macaques, China. *Veterinary Parasitology* **153**(1–2), 121–125. D'Agostino JJ, West G, Boothe DM, *et al.* (2007) Plasma pharmacokinetics of selamectin after a single topical administration in the American bullfrog (*Rana catesbeiana*). *Journal of Zoo and Wildlife Medicine* **38**(1), 51–55

¹ DiGeronimo PM (2016) Therapeutic review: selamectin. *Journal of Exotic Pet Medicine* **25**, 80–83

Serum gonadotrophin (PMSG, Equine chorionic gonadotrophin)

(PMSG-Intervet) **POM-V**

Formulations: Injectable: 5000 IU freeze-dried plug.

Action: Mimics action of FSH (follicle-stimulating hormone).

Use: Induction of oestrus by stimulation of ovarian follicle development. Increases spermatogenesis, though with low efficiency. Induction of ovulation in amphibians.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Anaphylactoid reactions may occur rarely.

Drug interactions: No information available.

DOSES

Mammals: Guinea pigs: 1000 IU/animal i.m., repeat in 7–10 days.

Amphibians: Induction of ovulation: 50–200 IU s.c., i.m. followed by 600 IU hCG after 72 h; **Barred frogs:** 2 doses of PMSG (50 IU and 25 IU at 6 and 4 days, respectively), prior to 2 doses of 100 IU hCG 24 h apart^a.

Birds, Reptiles, Fish: No information available.

References

- ^a Clulow J, Clulow S, Guo J, *et al.* (2012) Optimisation of an oviposition protocol employing human chorionic and pregnant mare serum gonadotropins in the barred frog (*Mixophyes fasciolatus*) (Myobatrachidae). *Reproductive Biology and Endocrinology* **10**, 60

Sevoflurane

(SevoFlo, Sevohale, Sevotek) **POM-V**

Formulations: Inhalational: 250 ml bottle.

Action: The mechanism of action of volatile anaesthetic agents is not fully understood.

Use: Induction and maintenance of anaesthesia. Sevoflurane is potent and highly volatile so should only be delivered from a suitable calibrated vaporizer. It is less soluble in blood than isoflurane; therefore, induction and recovery from anaesthesia is quicker. Sevoflurane has a less pungent smell than isoflurane and induction of anaesthesia using chambers or masks is usually well tolerated in small animals. The concentration of sevoflurane required to maintain anaesthesia depends on the other drugs used in the anaesthesia protocol; the concentration should be adjusted according to clinical assessment of anaesthetic depth. The cessation of administration results in rapid recovery, which may occasionally be associated with signs of agitation. Sevoflurane does not sensitize the myocardium to catecholamines to the extent that halothane does.

Safety and handling: Measures should be adopted to prevent contamination of the environment.

Contraindications: No information available.

Adverse reactions: Causes a dose-dependent hypotension that does not wane with time. The effects of sevoflurane on respiration are

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dose-dependent and comparable with isoflurane. Sevoflurane crosses the placental barrier and will affect neonates delivered by caesarean section. Sevoflurane is degraded by soda lime to compounds that are nephrotoxic in rats (principally Compound A). Conditions accelerating degradation (i.e. low gas flows, high absorbent temperatures and high sevoflurane concentrations) should be avoided in long operations.

Drug interactions: Opioid agonists, benzodiazepines and nitrous oxide reduce the concentration of sevoflurane required to achieve surgical anaesthesia. The effects of sevoflurane on the duration of action of non-depolarizing neuromuscular blocking agents are similar to those of isoflurane, i.e. greater potentiation compared with halothane.

DOSES

Mammals, Birds: The expired concentration required to maintain surgical anaesthesia in 50% of recipients is about 2.5% in most animals (minimum alveolar concentration, MAC)^{a,b}. The MAC in rabbits is 3.7%. Administration of other anaesthetic agents and opioid analgesics reduces the dose requirement of sevoflurane; therefore, the dose should be adjusted according to individual requirement. 6–8% sevoflurane concentration is required to induce anaesthesia in unpremedicated patients.

Reptiles: Induction: 6–8% in 100% oxygen; **Maintenance:** 3–5% in 100% oxygen^{c,d}.

Amphibians: Induction: 37.5 µl (microlitres)/g of a topical mixture (3 parts liquid sevoflurane, 3.5 parts KY jelly, 1.5 parts distilled water) in a closed chamber^e; Gaseous chamber induction: In cane toads, 1.75% sevoflurane resulted in loss of righting reflex within 15 minutes in 50% of toads (*Rhinella marina*)^f.

Fish: No information available.

References

- ^a Joyner PH, Jones MP, Ward D *et al.* (2008) Induction and recovery characteristics and cardiopulmonary effects of sevoflurane and isoflurane in bald eagles. *American Journal of Veterinary Research* **69**(1), 13–22
- ^b Phair KA, Larsen RS, Wack RF, Shilo-Benjamini Y and Pypendop BH (2012) Determination of the minimum anesthetic concentration of sevoflurane in thick-billed parrots (*Rhynchopsitta pachyrhyncha*). *American Journal of Veterinary Research* **73**(9), 1350–1355
- ^c Barter LS, Hawkins MG, Brosnan RJ, Antognini JF and Pypendop BH (2006) Median effective dose of isoflurane, sevoflurane, and desflurane in green iguanas. *American Journal of Veterinary Research* **67**(3), 392–397
- ^d Bertelsen MF, Mosley C, Crawshaw GJ, Dyson D and Smith DA (2005) Inhalation anesthesia in Dumeril's monitor (*Varanus dumerili*) with isoflurane, sevoflurane, and nitrous oxide: effects of inspired gases on induction and recovery. *Journal of Zoo and Wildlife Medicine* **36**(1), 62–69
- ^e Zec S, Clark-Price SC, Coleman DA and Mitchell MA (2014) Loss and return of righting reflex in American green tree frogs (*Hyla cinerea*) after topical application of compounded sevoflurane or isoflurane jelly: a pilot study. *Journal of Herpetological Medicine and Surgery* **24**(3), 72–76
- ^f Morrison KE, Strahl-Heldreth D and Clark-Price SC (2016) Isoflurane, sevoflurane and desflurane use in cane toads (*Rhinella marina*). *Veterinary Record Open* **3**(1), 185

Sildenafil

(Revatio*, Sildenafil*, Viagra*) **POM**

Formulations: Oral: 20 mg, 25 mg, 50 mg tablets, 10 mg/ml oral suspension (Revatio). **Injectable:** 0.08 mg/ml (Revatio IV). Special reformulations available: 6.25 mg, 10 mg and 12.5 mg tablets.

Action: Pulmonary vasculature vasodilation, due to an increase in vascular levels of cGMP caused by inhibition of cGMP-specific phosphodiesterase type V. Reduction and prevention of nephrotoxicity from aminoglycoside administration, due to its selective phosphodiesterase-5 inhibition effects.

Use: Indicated for the treatment of pulmonary arterial hypertension and prevention of renal damage when treated with nephrotoxic drugs. Sildenafil decreases pathological changes in alveoli, bronchioles, interstitial tissue, and arterioles of rats with COPD.

Safety and handling: Normal precautions should be observed.

Contraindications: Systemic hypotension, significant hepatic or renal impairment or bleeding disorders.

Adverse reactions: Vomiting, dizziness and raised intraocular pressure.

Drug interactions: Cimetidine, erythromycin, itraconazole and phenobarbital increase plasma sildenafil concentration. Sildenafil treatment should not be started at the same time as an alpha-adrenergic blocker. Avoid concomitant use of nitrates, which significantly enhance its hypotensive effect.

DOSES

Mammals: Rats: 5 mg/kg q24h for nephroprotective effects, 2–4.5 mg/kg q24h for respiratory effects^{a,b}.

Birds, Reptiles, Amphibians and Fish: No information available.

References

- ^a Morsy MA, Ibrahim SA, Amin EF *et al.* (2014) Sildenafil ameliorates gentamicin-induced nephrotoxicity in rats: role of iNOS and eNOS. *Journal of Toxicology*, doi: 10.1155/2014/89382
- ^b Ren Z, Li J, Shen J *et al.* (2020) Therapeutic sildenafil inhibits pulmonary damage induced by cigarette smoke exposure and bacterial inhalation in rats. *Pharmaceutical Biology* **58**(1), 116–123

Silver sulfadiazine

(Flamazine*) **POM**

Formulations: Topical: 1% cream (water-soluble).

Action: Slowly releases silver in concentrations that are toxic to bacteria and yeasts. The sulfadiazine component also has anti-infective qualities.

Use: Topical antibacterial and antifungal drug particularly active against Gram-negative organisms such as *Pseudomonas aeruginosa*. Used in the management of second- and third-degree burns, and flystrike. Up to 10% may be absorbed, depending on the size of area treated. Used for the postoperative treatment of skin wounds and localized external bacterial infections in fish however, it can delay or worsen the wound healing process in common carp^a.

Safety and handling: Use gloves.

Contraindications: Do not use in neonates or pregnant animals.

Adverse reactions: Patients hypersensitive to sulphonamides may react to silver sulfadiazine. It may accumulate in patients with impaired hepatic or renal function.

Drug interactions: No information available.

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DOSES**Classified as category D (Prudence) by the EMA.****See Appendix for guidelines on responsible antibiotic use.**

Mammals: Apply sparingly to wounds q12–24h. Otitis (resistant *Pseudomonas/refractory Malassezia*): Dilute 1:1 with sterile water and apply topically.

Birds: Burns/skin infection: Apply antiseptically to the affected area to a thickness of approximately 1.5 mm. Initially, apply as often as necessary to keep wound covered, then reduce as healing occurs to once a day applications. Dressings may be applied if necessary. Keep the affected area clean. Otitis (resistant *Pseudomonas/refractory Malassezia*): Dilute 1:1 with sterile water and apply topically.

Reptiles, Amphibians: Apply topically to wounds q24–72h^b.

Fish: Apply topically to wounds keeping site out of water for 30–60 seconds.

References:

- ^a Ang J, Pierezan F, Kim S et al. (2021) Use of topical treatments and effects of water temperature on wound healing in common carp (*Cyprinus carpio*). *Journal of Zoo and Wildlife Medicine* **52**(1), 103–116
- ^b Archibald KE, Harrison T, Troan B, Smith D and Minter LJ (2020) Effect of multiradiance low-level laser therapy and topical silver sulfadiazine on healing characteristics of dermal wounds in marine toads (*Rhinella marina*). *Veterinary Medicine International* doi:10.1155/2020/8888328

Silybin (Milk thistle, Silibinin, Silymarin)

(Denamarin, Doxion, Hepaticare, Hepato Support, Hepatosyl Plus, Samylin, Zentonil Advanced, Various authorized proprietary products) **AVM-GSL**

Formulations: Oral: 9 mg, 24 mg, 25 mg, 35 mg, 40 mg, 50 mg, 70 mg, 100 mg tablets; 10 mg, 40 mg, 53 mg powder.

Action: Silybin is the active component of milk thistle or silymarin. It acts as an antioxidant and free radical scavenger, promotes hepatocyte protein synthesis, increases the level of glutathione, and stimulates biliary flow and the production of hepatoprotective bile acids. It also inhibits leucotriene production so reducing the inflammatory response.

Use: Adjunctive treatment for liver disease, especially for acute hepatotoxin-induced liver disease. Silybin has been shown to increase hepatic glutathione levels; a potent antioxidant which protects hepatocytes from toxic damage. Can also be used in patients on long-term therapy with potentially hepatotoxic drugs.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: GI signs, pruritus and headaches have been recognized in primates.

Drug interactions: Silybin may inhibit microsomal cytochrome P450 isoenzyme 2C9 (CYP2C9). May increase plasma levels of beta-blockers (e.g. propranolol), calcium-channel blockers (e.g. verapamil), diazepam, lidocaine, metronidazole, pethidine and theophylline. Silymarin may increase the clearance of drugs that undergo hepatic glucuronidation, including paracetamol, diazepam and morphine. Clinical significance

has not been determined for this interaction and the usefulness of silymarin for treating paracetamol toxicity has not been determined.

DOSES

Mammals: Ferrets: 50–250 mg/kg p.o. q24h. (Note: rabbits and herbivorous rodents may denature active principles in the stomach.)

Birds: 50–75 mg/kg p.o. q12h.

Reptiles, Amphibians, Fish: No information available.

Sodium bicarbonate

(Sodium bicarbonate*) **POM**

Formulations: **Injectable:** 1.26%, 4.2%, 8.4% solutions for i.v. infusion (8.4% solution = 1 mmol/ml). **Oral:** 300 mg, 500 mg, 600 mg tablets; 100% BP powder.

Action: Provision of bicarbonate ions.

Use: Management of severe metabolic acidosis, to alkalinize urine, and as an adjunctive therapy in the treatment of hypercalcaemic or hyperkalaemic crisis. Active correction of acid–base imbalance requires blood gas analysis. Do not attempt specific therapy unless this facility is immediately available. 1 g of sodium bicarbonate provides 11.9 mEq of Na⁺ and 11.9 mEq of bicarbonate. In hypocalcaemic patients use sodium bicarbonate cautiously and administer slowly. As oral sodium bicarbonate (especially at higher doses) may contribute significant amounts of sodium, use with caution in patients on salt-restricted intakes, e.g. those with congestive heart failure. Used for buffering water in fish tanks (i.e. increasing alkalinity) and for raising pH during the use of some acidic anaesthetic agents by immersion (e.g. tricaine).

Safety and handling: Normal precautions should be observed.

Contraindications: Should not be used in animals that are unable to effectively expel carbon dioxide (e.g. hypoventilating, hypercapnoeic patients).

Adverse reactions: Excessive use of sodium bicarbonate i.v. can lead to metabolic alkalosis, hypernatraemia, congestive heart failure, a shift in the oxygen dissociation curve causing decreased tissue oxygenation, and paradoxical CNS acidosis leading to respiratory arrest.

Drug interactions: Sodium bicarbonate is incompatible with many drugs and calcium salts: do not mix unless checked beforehand. Alkalinization of the urine by sodium bicarbonate decreases the excretion of quinidine and sympathomimetic drugs, and increases the excretion of aspirin, phenobarbital and tetracyclines (especially doxycycline).

DOSES

Mammals:

- Severe metabolic acidosis: mmol NaHCO₃ required = base deficit × 0.5 × bodyweight (kg) (0.3 is recommended instead of 0.5 in some references). Give half the dose slowly i.v. over 3–4 hours, recheck blood gases and clinically re-evaluate the patient. Avoid over-alkalinization.
- Acutely critical situations (e.g. cardiac arrest): 1 mmol/kg i.v. over 1–2 min followed by 0.5 mmol/kg at intervals of 10 min during the arrest.
- Adjunctive therapy of hypercalcaemia: 0.5–1 mmol/kg i.v. over 30 min.

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- Adjunctive therapy of hyperkalaemia: 2–3 mmol/kg i.v. over 30 min.
- Metabolic acidosis secondary to renal failure or to alkalinize the urine: Initial dose 8–12 mg/kg p.o q8h and then adjust dose to maintain total CO₂ concentrations at 18–24 mEq/l. The dose may be increased to 50 mg/kg to adjust urine pH in patients with normal renal, hepatic and cardiac function.

Fish: Add powder as required to the aquarium water and monitor pH using a digital pH meter.

Birds, Reptiles, Amphibians: No information available.

Sodium chloride

(Aquapharm, Hypertonic saline, Sodium chloride, Vetivex)
POM-V

Formulations: **Injectable:** 0.45% to 7% NaCl solutions; 0.18% NaCl with 4% glucose and 0.9% NaCl with 5% glucose solutions. **Oral:** 300 mg, 600 mg tablets. **Ophthalmic:** 5% ointment (compounded by an ocular pharmacy). **Immersion:** Crystals for dissolution in water.

Action: Expands plasma volume and replaces lost extracellular fluid. In freshwater aquaria and ponds, nitrite toxicity resulting from poor water quality causes oxidation of haemoglobin to methaemoglobin. Chloride ions inhibit nitrite absorption through the gill epithelium and reduce the toxic effect in fish. It also reduces the osmotic pressure between the freshwater environment and the fish tissues.

Use: When used for fluid replacement NaCl (0.45% and 0.9%) will expand the plasma volume compartment. Compared with colloids, 2.5 to 3.0 times as much fluid must be given because the crystalloid is distributed to other sites. Normal saline is also the treatment of choice for patients with hypercalcaemia or hyperchloraemic alkalosis. Sodium chloride solutions are often used as a drug diluent. Hypertonic saline is used to expand the circulating blood volume rapidly in animals with shock, particularly during the preoperative period. The hypertonic ophthalmic ointment is used in the management of corneal oedema. Hypertonic saline solutions have very high sodium concentrations and it is important to monitor serum sodium concentrations before and after their administration; maintenance with an isotonic crystalloid is usually required after administration to correct electrolyte and fluid disturbances created by the administration of the hypertonic solution. Oral sodium supplementation is recommended by some authors in the long-term management of hypoadrenocorticism. Used for the treatment of external protozoans in amphibians and fish. Used as a short-term bath to control external bacterial and parasitic infections and help remove excess mucus from gills. Reduces stress associated with transportation in fish. Used for the reduction of toxicity due to poor water quality and to aid the healing of wounds in freshwater fish. Inadequate biological filtration or water changes results in increased nitrite levels in the environment, which adversely affects fish health and causes methaemoglobinaemia. Although salt will reduce the toxic effects of poor water quality, it is essential that the underlying causes are addressed. Salt has been used as an adjunctive treatment of carp infected with carp edema virus (koi sleepy disease). Do not use salt with yellow prussate of soda (YPS, sodium ferrocyanide) added as an anti-caking agent or iodized salt. Use lower

dose rates initially for prolonged immersion, since small fish and some species (e.g. catfish) are sensitive to salt. In some tropical fish species, neurological clinical signs such as abnormal swimming behaviour may occur within a few days at concentrations of up to 1.5 g/l and mortality may occur at 3 g/l. Avoid use with fish that navigate by electric field (e.g. elephant nose fish, knife fish). Fish must be observed during high salt concentration baths and should be removed if exhibiting signs of distress (i.e. loss of activity and balance). Remove plants that may not tolerate salt before prolonged immersion treatments.

Safety and handling: Hypertonic saline solutions should be regarded as drugs and not as intravenous fluids and should be stored separately to prevent confusion.

Contraindications: Do not use if zeolite is used in the filtration system as this will cause the release of absorbed ammonia into the environment.

Adverse reactions: Peripheral oedema is more likely to occur after crystalloids because muscle and subcutaneous capillaries are less permeable to protein. Normal saline contains higher amounts of chloride than plasma, which will increase the risk of acidosis. The degree of acidosis is not likely to be a problem in a healthy patient but acidosis may be exacerbated in a compromised patient. Hypertonic saline administered at fluid rates >1 ml/kg/min can cause a vagally mediated bradycardia, therefore the rate of fluid administration must be carefully controlled. The ophthalmic ointment may cause a stinging sensation. In some tropical fish species, neurological clinical signs such as abnormal swimming behaviour may occur within a few days at concentrations of up to 1.5 g/l and mortality may occur at 3 g/l. Avoid use with fish that navigate by electric field (e.g. elephant nose fish, knife fish).

Drug interactions: No information available.

DOSES

Fluid therapy: Fluid requirements depend on the degree of dehydration and ongoing losses.

Mammals:

- Corneal oedema: Apply a small amount of ointment q4–24h.
- Hypotension/shock: 5 ml/kg of 7.5% solution over 5–10 minutes. Solutions of this concentration are hypertonic, therefore they should be used with caution and with other appropriate fluid replacement strategies. Hypertonic NaCl may be combined with colloid solutions to stabilize the increase in vascular volume provided by the hypertonic solution.

Amphibians: External protozoans: 6 g/l as a 24 h bath for 3–5 days.

Fish:

- External protozoans: 6 g/l as a 24 h bath for 3–5 days.
- Ectoparasite control: 30 g/l as a bath for 5–10 min weekly or as needed.
- Osmoregulation: 1–5 g/l by prolonged immersion.

Birds, Reptiles: No information available.

Spiramycin see **Metronidazole**

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Streptomycin

(Devomycin) **POM-V**

Formulations: **Injectable:** Streptomycin (250 mg/ml); Streptomycin (150 mg/ml) with dihydrostreptomycin (150 mg/ml) (Devomycin D).

Action: Inhibits bacterial protein synthesis, resulting in a bactericidal effect that is concentration-dependent.

Use: Active against a range of Gram-negative and some Gram-positive pathogens although resistance is quite widespread and it is less active than other aminoglycosides. It is specifically indicated in the treatment of infections caused by *Leptospira* and *Mycobacterium tuberculosis* (in combination with other drugs). Aminoglycosides require an oxygen-rich environment to be effective, thus they are ineffective in sites of low oxygen tension (abscesses, exudates) and all obligate anaerobes are resistant. Use of streptomycin is limited and if an aminoglycoside is indicated other members of the family are more commonly employed (e.g. gentamicin). There is a marked post-antibiotic effect, allowing the use of pulse-dosing regimens which may limit toxicity. Dosing 2–3 times a week is used to treat mycobacteriosis in humans.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use in guinea pigs, hamsters, gerbils, rats and mice. Do not use in raptors and use with caution in other bird breeds.

Adverse reactions: Streptomycin is one of the more ototoxic aminoglycosides, interfering with balance and hearing, which can be permanent. Nephrotoxicosis may be a problem but is less likely than with other aminoglycosides. Toxic to certain rodents and birds, especially raptors (neurotoxic). Fatal reaction to streptomycin and penicillin reported in parrots. Oral doses can cause fatal enterotoxaemia in rabbits and potentially other small herbivores.

Drug interactions: Increased risk of nephrotoxicity when used with cephalosporins (notably cefalotin) and cytotoxic drugs. Ototoxicity is increased with loop diuretics. The effects of neostigmine and pyridostigmine may be antagonized by aminoglycosides. The effect of non-depolarizing muscle relaxants (e.g. pancuronium) may be enhanced. Penicillin and streptomycin act synergistically. Aminoglycosides may be chemically inactivated by beta-lactam antibiotics (e.g. penicillins, cephalosporins) or heparin when mixed *in vitro*.

DOSES

Classified as category C (Caution) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Mammals: Rabbits: 50 mg/kg streptomycin in combination with 40 mg/kg penicillin s.c. q24h. Do not use in guinea pigs, hamsters, gerbils, rats and mice.

Birds: Not recommended^a.

Reptiles, Amphibians, Fish: No information available.

References

^a Hauser H (1960) Fatal reaction to streptomycin and penicillin in parrots with pulmonary mycosis. *Monatshefte für Veterinärmedizin* **15**, 632–634

Sucralfate

(Antepsin*, Antepsin suspension*, Carafate*) **POM**

Formulations: Oral: 1 g tablet; 0.2 g/ml suspension.

Action: In an acidic medium an aluminium ion detaches from the compound, leaving a very polar, relatively non-absorbable ion. This ion then binds to proteinaceous exudates in the upper GI tract, forming a chemical diffusion barrier over ulcer sites, preventing further erosion from acid, pepsin and bile salts. However, its major action appears to relate to stimulation of mucosal defences and repair mechanisms (stimulation of bicarbonate and PGE production and binding of epidermal growth factor). These effects are seen at neutral pH.

Use: Treatment of oesophageal, gastric and duodenal ulceration, used with an H₂ receptor antagonist or proton pump inhibitor but given separately. The efficacy of sucralfate as a phosphate binder in renal failure is uncertain.

Safety and handling: Normal precautions should be observed.

Contraindications: Perforated ulcer.

Adverse reactions: Minimal; constipation is the main problem in humans. Bezoar formation and hypophosphataemia are also reported in humans.

Drug interactions: Sucralfate may decrease the bioavailability of H₂ antagonists, phenytoin and tetracycline. Although there is little evidence to suggest that this is of clinical importance, it may be a wise precaution to administer sucralfate at least 2 hours before these drugs. Sucralfate interferes significantly with the absorption of fluoroquinolones and digoxin.

DOSES

Mammals: Ferrets: 25–125 mg/kg p.o. q6–12h; Rabbits: 25 mg/kg p.o. q8–12h; Rodents: 25–50 mg/kg p.o. q6–8h; Primates: 500 mg/animal p.o. q6–12h; Hedgehogs: 10 mg/kg p.o. q8–12h.

Birds: 25 mg/kg p.o. q8h (anecdotal); 100 mg/kg p.o. q24h has been used for treatment following ventriculotomy in a pigeon¹.

Reptiles: 500–1000 mg/kg p.o. q6–8h (anecdotal); 200 mg/kg p.o. q24h has been used for treatment following intestinal surgery in a green iguana².

Amphibians, Fish: No information available.

References

- ¹ Laniesse D, Beaufrère H, Mackenzie S *et al.* (2018). Perforating foreign body in the ventriculus of a pet pigeon (*Columba livia domestica*). *Journal of the American Veterinary Medical Association* **253**(12), 1610–1616
- ² Wills S, Beaufrère H, Watrous G, Oblak ML and Smith DA (2016). Proximal duodenoileal anastomosis for treatment of small intestinal obstruction and volvulus in a green iguana (*Iguana iguana*). *Journal of the American Veterinary Medical Association* **249**(9), 1061–1066

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A Sulfadimethoxine(Coxi Plus) **POM-V****Formulations:** Oral: 1000 mg/4 g sachet.**Action:** Competitively inhibits bacterial and protozoal synthesis of folic acid.**Use:** Coccidiosis in ferrets, hedgehogs, primates, rabbits, birds and reptiles; toxoplasmosis in passerine birds. Use with care in reptiles with reduced renal function, renal failure or dehydration.**Safety and handling:** Normal precautions should be observed.**Contraindications:** No information available.**Adverse reactions:** No information available.**Drug interactions:** No information available.**H DOSES****See Appendix for guidelines on responsible parasiticide use.****Mammals:** Ferrets: 50 mg/kg p.o. once, then 25 mg/kg p.o. q24h for 5–10 days; **Rabbits, Rodents:** 50 mg/kg p.o. once, then 25 mg/kg p.o. q24h for 10–20 days; **Primates:** 50 mg/kg p.o. once, then 25 mg/kg p.o. q24h; **Hedgehogs:** 2–20 mg/kg p.o. q24h; **Other small mammals:** 10–15 mg/kg p.o. q12h.**Birds:** 1 g/l of drinking water daily for 2 days, then 3 days off and 2 days on.**Reptiles:** 50 mg/kg p.o. q24h for 21 days for treatment of coccidiosis in bearded dragons^a.**Amphibians, Fish:** No information available.**N References**^a Walden M and Mitchell MA (2012) Evaluation of three treatment modalities against *Isospora amphiboluri* in inland bearded dragons (*Pogona vitticeps*). *Journal of Exotic Pet Medicine* **21**(3), 213–218**S Sulfasalazine**(Salazopyrin*, Sulphasalazine*) **POM****Formulations:** Oral: 500 mg tablet; 250 mg/ml suspension.**Action:** Sulfasalazine is a pro-drug: a diazo bond binding sulfapyridine to 5-ASA is cleaved by colonic bacteria to release free 5-ASA, which acts locally in high concentrations in the colon as an anti-inflammatory.**Use:** Used in the management of colitis. There is a significant risk of keratoconjunctivitis sicca (KCS) and periodic Schirmer tear tests should be performed.**Safety and handling:** Normal precautions should be observed.**Contraindications:** No information available.**Adverse reactions:** Uncommon but include KCS, vomiting, allergic dermatitis and cholestatic jaundice. Owners should be made aware of the seriousness of KCS and what signs to monitor. The cause of the KCS is not clear. Historically sulfapyridine has been blamed. Olsalazine has been recommended as the incidence of KCS is less with its use, though not completely abolished. It is possible that 5-ASA may sometimes be responsible.

Drug interactions: The absorption of digoxin may be inhibited by sulfasalazine, and the measurement of serum folate concentration may be interfered with. Sulfasalazine may cause a reduction in serum thyroxine concentrations.

DOSES

Mammals: Ferrets: 10–20 mg/kg q8–24h¹.

Birds, Reptiles, Amphibians, Fish: No information available.

References

- ¹ Williams BH (2000) Therapeutics in ferrets. *Veterinary Clinics of North America: Exotic Animal Practice* **3**(1), 131-153

Sulphonamide see **Trimethoprim/sulphonamide**

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A **T4** see **L-Thyroxine****B**
C
Telmisartan**(Semintra) POM-V****D**
Formulations: Oral: 4 mg/ml solution, 10 mg/ml solution.**E**
Action: Angiotensin II receptor (type AT1) antagonist which acts to inhibit the effects of angiotensin (i.e. vasoconstriction; increased aldosterone synthesis; sodium and water retention; and renal, vascular and cardiac remodelling). In the kidney, angiotensin II may result in glomerular capillary hypertension and increased protein in the glomerular filtrate, which could trigger or potentiate interstitial fibrosis.**G**
Use: Suggested for the reduction of proteinuria associated with chronic kidney disease in rats and rabbits. Telmisartan is authorized for use in cats, in which it has been shown to delay deterioration in proteinuria over 6 months in cats with chronic kidney disease (IRIS stages IIa to IV, urine specific gravity <1.035 and no co-morbidities). It is used, off licence, to treat protein-losing nephropathy in rats, and chronic kidney disease in rabbits. Monitoring of blood pressure is recommended in animals that develop clinical signs referable to hypotension or those undergoing general anaesthesia. There is some evidence of efficacy in reducing hypertension in rabbits^{a,b}. Rats and rabbits have been shown to be the most sensitive species to telmisartan-related GI damage, and so if any signs of GI disturbance are noted treatment should be discontinued^c.**H**
I
Safety and handling: Normal precautions should be observed.**J**
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Contraindications: The safety of telmisartan has not been established in breeding, pregnant, lactating or skeletally immature animals.**L**
M
Adverse reactions: Mild and transient GI signs (inappetence, diarrhoea, and regurgitation and vomiting in those species in which this is possible). Rats and rabbits are considered most sensitive to GI inflammation and ulceration. Healthy cats administered 5 times the recommended dose for 6 months experienced decreases in blood pressure and RBC count and increases in blood urea nitrogen, this may be a consideration in other species.**N**
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Drug interactions: Avoid concurrent use of ACE inhibitors.**P**
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DOSES**U**
Mammals: Rabbits, Rats: 1 mg/kg p.o. q24h.**V**
Birds, Reptiles, Amphibians, Fish: No information available.**W**
References

- X**
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- ^a Maeda S, Nishizaki M, Yamawake N *et al.* (2010) Effect of high-dose telmisartan on the prevention of recurrent atrial fibrillation in hypertensive patients. *Journal of Atrial Fibrillation* **3**(3), 289
 - ^b Hu ZP, Fang XL, Qian HY, *et al.* (2014) Telmisartan prevents angiotensin II-induced endothelial dysfunction in rabbit aorta via activating HGF/Met system and PPAR γ pathway. *Fundamental & Clinical Pharmacology* **28**(5), 501–511
 - ^c European Medicines Agency (2005) Micardis: EPAR–Scientific Discussion. Available from: <https://www.ema.europa.eu/en/medicines/human/EPAR/micardis>

Terbinafine

(Lamisil[®]) **POM**

Formulations: Oral: 250 mg tablets. **Topical:** 1% cream.

Action: Inhibits ergosterol synthesis by inhibiting squalene epoxidase, an enzyme that is part of the fungal cell wall synthesis pathway.

Use: Management of dermatophytosis, subcutaneous and systemic fungal infections including aspergillosis in birds and chytridiomycosis in amphibians. Optimal therapeutic regimes are still under investigation. Pre-treatment and monitoring CBC, renal and liver function tests are advised.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Vomiting, diarrhoea, increased liver enzymes and pruritus.

Drug interactions: No information available.

DOSES

Mammals: Rodents: 10–30 mg/kg p.o. q24h for 4–6 weeks; **Primates:** 8.25 mg/kg p.o. q24h; **Hedgehogs:** Doses of 100 mg/kg p.o. q12h are reported^a, although 20–30 mg/kg appears clinically effective in resolving dermatophytosis.

Birds: 10–15 mg/kg p.o. q12h or nebulization of 1 mg/ml for 20 min q8h^{b,c,d}.

Reptiles: Red-eared sliders: 15 mg/kg p.o. q12–24h^e; **Bearded dragons:** 20 mg/kg p.o. q24–48h used for treatment of *Nannizziopsis guarroi*^f.

Amphibians, Fish: No information available.

References

- ^a Bexton S and Nelson H (2016) Comparison of two systemic antifungal agents, itraconazole and terbinafine, for the treatment of dermatophytosis in European hedgehogs (*Erinaceus europaeus*). *Veterinary Dermatology* **27**, 500–e133
- ^b Bechert U, Christensen JM, Poppenga R, Fahmy SA and Redig P (2010) Pharmacokinetics of terbinafine after single oral dose administration in red-tailed hawks (*Buteo jamaicensis*). *Journal of Avian Medicine and Surgery* **24**(2), 122–130
- ^c Emery LC, Cox SK and Souza MJ (2012) Pharmacokinetics of nebulized terbinafine in Hispaniolan Amazon parrots (*Amazona ventralis*). *Journal of Avian Medicine and Surgery* **26**(3), 161–166
- ^d Evans EE, Emery LC, Cox SK and Souza MJ (2013) Pharmacokinetics of terbinafine after oral administration of a single dose to Hispaniolan Amazon parrots (*Amazona ventralis*). *American Journal of Veterinary Research* **74**(6), 835–838
- ^e Eshar D, KuKanich B, Avni-Magen N and Joo H (2021) Terbinafine pharmacokinetics following single-dose oral administration in red-eared slider turtles (*Trachemys scripta elegans*): a pilot study. *Journal of Zoo and Wildlife Medicine* **52**(2), 520–528
- ^f McEntire MS, Reinhart JM, Cox SK and Keller KA (2022) Single-dose pharmacokinetics of orally administered terbinafine in bearded dragons (*Pogona vitticeps*) and the antifungal susceptibility patterns of *Nannizziopsis guarroi*. *American Journal of Veterinary Research* **83**(3), 256–263

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A Terbutaline(Bricanyl*, Monovent*) **POM****Formulations:** Injectable: 0.5 mg/ml solution. Oral: 5 mg tablets; 1.5 mg/5 ml syrup.**Action:** Selective beta-2 adrenergic agonist that directly stimulates bronchodilation.**Use:** Bronchodilation. Use with caution in patients with diabetes mellitus, hyperthyroidism, hypertension or seizure disorders.**Safety and handling:** Normal precautions should be observed.**Contraindications:** No information available.**Adverse reactions:** Fine tremor, tachycardia, hypokalaemia, hypotension and hypersensitivity reactions. Administration i.m. may be painful.**Drug interactions:** There is an increased risk of hypokalaemia if theophylline or high doses of corticosteroids are given with high doses of terbutaline. Use with digitalis glycosides or inhalational anaesthetics may increase the risk of cardiac arrhythmias. Beta-blockers may antagonize its effects. Other sympathomimetic amines may increase the risk of adverse cardiovascular effects.**K DOSES****Mammals:** Rabbits: 0.312–1.25 mg/animal p.o. q8h; Rats: 5 mg/kg p.o. q12h.**Birds:** 0.1 mg/kg i.m. in an emergency.**Reptiles, Amphibians, Fish:** No information available.**O Tetracaine (Amethocaine)**(Amethocaine hydrochloride*) **POM****Formulations:** Ophthalmic: 0.5%, 1% solution (single-use vials).**Action:** Local anaesthetic action is dependent on reversible blockade of the sodium channel, preventing propagation of an action potential along the nerve fibre. Sensory nerve fibres are blocked before motor nerve fibres, allowing a selective sensory blockade at low doses.**Use:** Local anaesthesia of the ocular surface (cornea and conjunctival sac). Although effective, it is rarely used in veterinary practice. An alternative topical ophthalmic anaesthetic such as proxymetacaine is advised. Duration of action has not been reported in companion animal species. Topical anaesthetics block reflex tear production and should not be applied before a Schirmer tear test.**Safety and handling:** Store in refrigerator.**Contraindications:** Do not use for therapeutic purposes.**Adverse reactions:** Tetracaine often causes marked conjunctival irritation, chemosis and pain on application. All topical anaesthetics are toxic to the corneal epithelium and delay healing of ulcers.**Drug interactions:** No information available.**Y DOSES****Mammals, Birds:** 1 drop per eye, single application.**Reptiles, Amphibians, Fish:** No information available.

Tetracosactide (Tetracosactrin, ACTH)

(Synacthen*) **POM**

Formulations: Injectable: 0.25 mg/ml solution for intravenous use. In the event of availability problems, an alternative lyophilized formulation (0.25 mg/ml) for i.m. use may be imported on a named patient basis.

Action: ACTH analogue that binds to specific receptors on the cell membrane of adrenocortical cells and induces the production of steroids from cholesterol.

Use: To stimulate cortisol production in the diagnosis of hypo- (but not hyper-) adrenocorticism (Addison's disease) in ferrets. Availability problems at time of writing. It is recommended to use lower doses than previously published and reserve for diagnosis of hypoadrenocorticism.

Safety and handling: Normal precautions should be observed. Small aliquots of the intravenous and intramuscular preparations may be frozen and thawed once without undue loss of activity.

Contraindications: No information available.

Adverse reactions: None reported.

Drug interactions: None reported.

DOSES

Mammals: Ferrets: 1 µg (microgram)/kg i.m. for diagnosis of hypoadrenocorticism.

Birds, Reptiles, Amphibians, Fish: No information available.

Theophylline

(Corvental-D) **POM-V**

Formulations: Oral: 100 mg, 200 mg, 500 mg sustained-release capsules.

Action: Causes inhibition of phosphodiesterase, alteration of intracellular calcium, release of catecholamine, and antagonism of adenosine and prostaglandin, leading to bronchodilation and other effects.

Use: Spasmolytic agent and has a mild diuretic action. It has been used in the treatment of small airway disease. Beneficial effects include bronchodilation, enhanced mucociliary clearance, stimulation of the respiratory centre, increased sensitivity to P_aCO_2 , increased diaphragmatic contractility, stabilization of mast cells and a mild inotropic effect. Theophylline has a low therapeutic index and should be dosed on a lean bodyweight basis. Administer with caution in patients with severe cardiac disease, gastric ulcers, hyperthyroidism, renal or hepatic disease, severe hypoxia or severe hypertension. Therapeutic plasma theophylline values are 5–20 µg (micrograms)/ml.

Safety and handling: Normal precautions should be observed.

Contraindications: Patients with a known history of arrhythmias or seizures.

Adverse reactions: Vomiting, diarrhoea, polydipsia, polyuria, reduced appetite, tachycardia, arrhythmias, nausea, twitching, restlessness, agitation, excitement and convulsions. Most adverse effects are related

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A to the serum level and may be symptomatic of toxic serum concentrations. The severity of these effects may be decreased using modified-release preparations. They are more likely to be seen with more frequent administration.

B

C **Drug interactions:** Agents that may increase the serum levels of theophylline include cimetidine, diltiazem, erythromycin, fluoro-quinolones and allopurinol. Phenobarbital may decrease the serum concentration of theophylline. Theophylline may decrease the effects of pancuronium. Theophylline and beta-adrenergic blockers (e.g. propranolol) may antagonize each other's effects. Theophylline administration with halothane may cause an increased incidence of cardiac dysrhythmias and with ketamine an increased incidence of seizures.

D **DOSES**

E **Mammals:** Ferrets: 4.25–10 mg/kg p.o. q8–12h; Guinea pigs, Rats: 10–20 mg/kg p.o. q8–12h.

F **Birds:** 10 mg/kg p.o. q12h.

G **Reptiles, Amphibians, Fish:** No information available.

H **Thiamazole (Methimazole)**

I (Felimazole, Thiafeline, Thyronorm) **POM-V**

J **Formulations:** Oral: 1.25 mg, 2.5 mg, 5 mg tablets; 5 mg/ml solution. Also available as a transdermal formulation on a named patient basis.

K **Action:** Interferes with the synthesis of thyroid hormones by inhibiting peroxidase-catalysed reactions (blocks oxidation of iodide), the iodination of tyrosyl residues in thyroglobulin, and the coupling of mono- or di-iodotyrosines to form T3 and T4. There is no effect on iodine uptake and it does not inhibit peripheral de-iodination of T4 to T3.

L **Use:** Control of thyroid hormone levels in animals with hyperthyroidism. Monitor therapy on the basis of serum thyroxine concentrations (4–6 hours after dosing) and adjust dose accordingly for long-term medical management. Assess haematology, biochemistry and serum total T4 regularly, adjusting dosage as necessary. Transdermal thiamazole gels can also be used, particularly in those that develop GI side effects from the oral formulations. However, this route is not as reliable as oral medication or as safe for humans who apply the gel.

M **Safety and handling:** Normal precautions should be observed.

N **Contraindications:** Do not use in pregnant or lactating animals.

O **Adverse reactions:** Vomiting (in target species capable of this) and inappetence/anorexia may be seen but are often transient. Jaundice, cytopenias, immune-mediated diseases and dermatological changes (pruritus, alopecia and self-induced trauma) are reported but rarely seen. Treatment of hyperthyroidism can decrease glomerular filtration rate, thereby raising serum urea and creatinine values, and can occasionally unmask occult renal failure. Animals that have an adverse reaction to carbimazole are likely also to have an adverse reaction to thiamazole. Laboratory studies in rats and mice have shown evidence of teratogenic and embryotoxic effects. Following long-term treatment with thiamazole in rodents, an increased risk of neoplasia in the thyroid gland has been shown to occur.

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Drug interactions: Phenobarbital may reduce clinical efficacy. Benzimidazole drugs reduce hepatic oxidation and may lead to increased circulating drug concentrations. Thiamazole should be discontinued before iodine-131 treatment.

DOSES

Mammals: Guinea pigs: 0.5–2.0 mg/kg p.o. q24h.

Reptiles: Snakes: 1 mg/kg p.o. q24h for 30 days.

Birds, Amphibians, Fish: No information available.

Thiamine see Vitamin B1

L-Thyroxine (T4, Levothyroxine)

(Leventa, Soloxine, Thyforon) **POM-V**

Formulations: Oral: 0.1 mg, 0.2 mg, 0.3 mg, 0.5 mg, 0.8 mg tablets; 1 mg/ml solution.

Action: Binds to specific intracellular receptors and alters gene expression.

Use: Treatment of hypothyroidism. No studies have been performed on exotic species and doses are anecdotal. Avian plasma total T4 levels are extremely low and reptile total T4 levels vary widely depending on species and external factors. Currently no European laboratories have tests validated for these levels. Thus, diagnosis of true hypothyroidism in birds and reptiles is controversial and there are few confirmed cases. L-Thyroxine has been used to induce moult in birds; however, this is not recommended. Cases of pre-existing cardiac disorders require lower doses initially.

Safety and handling: Normal precautions should be observed.

Contraindications: Uncorrected adrenal insufficiency.

Adverse reactions: Clinical signs of overdosage include tachycardia, excitability, nervousness and excessive panting. May cause feather dystrophy if moult induced too quickly in birds.

Drug interactions: The actions of catecholamines and sympathomimetics are enhanced by thyroxine. Diabetic patients receiving thyroid hormones may have altered insulin requirements; monitor carefully during the initiation of therapy. Oestrogens may increase thyroid requirements by increasing thyroxine-binding globulin. The therapeutic effect of ciclosporin, digoxin and digitoxin may be reduced by thyroid hormones. Tachycardia and hypertension may develop when ketamine is given to patients receiving thyroid hormones. In addition many drugs may affect thyroid function tests and therefore monitoring of therapy.

DOSES

Mammals: Rodents: 5 µg (micrograms)/kg p.o. q12h.

Birds: 0.02 mg/kg p.o. q12–24h. Dissolve 1 mg in 28.4 ml water and give 0.4–0.5 ml/kg q12–24h.

Reptiles: Tortoises: 0.02 mg/kg p.o. q24–48h has been reported for the management of hypothyroidism in a Galapagos tortoise and an African spurred tortoise^{1,2}.

Amphibians, Fish: No information available.

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References

- ¹ Franco KH and Hoover JP (2009) Levothyroxine as a treatment for presumed hypothyroidism in an adult male African spurred tortoise. *Journal of Herpetological Medicine and Surgery* **19**(2), 47–49
- ² Norton TM, Jacobson ER, Caligiuru R and Kollias GV (1989) Medical management of a Galapagos tortoise (*Geochelone elephantopus*) with hypothyroidism. *Journal of Zoo and Wildlife Medicine* **20**, 212–216

Timolol maleate(Azarga*, CoSopt*, Timolol*, Timoptol*) **POM**

Formulations: Ophthalmic: 0.25%, 0.5% solutions (5 ml bottle, single-use vials; 0.5% solution most commonly used); 1% brinzolamide with 0.5% timolol (Azarga); 2% dorzolamide with 0.5% timolol (CoSopt) (5 ml bottle, single-use vials).

Action: A topical non-selective beta-blocker that decreases aqueous humour production via beta-adrenoreceptor blockade in the ciliary body.

Use: Management of glaucoma. It can be used alone or in combination with other topical glaucoma drugs, such as a topical carbonic anhydrase inhibitor. Dorzolamide/timolol or brinzolamide/timolol may be more effective than either drug alone. The combination causes miosis and is therefore not the drug of choice in uveitis or anterior lens luxation. Although no specific guidelines are available, it may be appropriate to use 0.25% solution in smaller patients to reduce the risks of systemic absorption.

Safety and handling: Normal precautions should be observed.

Contraindications: Avoid in uncontrolled heart failure and asthma.

Adverse reactions: Ocular adverse effects include miosis, conjunctival hyperaemia and local irritation. Systemic absorption may occur following topical application causing bradycardia and reduced blood pressure.

Drug interactions: Additive adverse effects may develop if given concurrently with oral beta-blockers. Concomitant administration of timolol with verapamil may cause a bradycardia and asystole. Prolonged AV conduction times may result if used with calcium antagonists or digoxin.

DOSES

Mammals: Rabbits: 1 drop per eye q12h.

Birds, Reptiles, Amphibians, Fish: No information available.

TMS see **Tricaine mesilate****Tobramycin**(Nebcin*, Tobramycin*) **POM**

Formulations: Injectable: 40 mg/ml solution.

Action: Aminoglycosides inhibit bacterial protein synthesis. They are bactericidal and their mechanism of killing is concentration-dependent, leading to a marked post-antibiotic effect, allowing pulse-dosing regimens which may limit toxicity.

Use: Treatment of Gram-negative infections. It is less active against most Gram-negative organisms than gentamicin but appears to be more active against *Pseudomonas aeruginosa*. Aminoglycosides are ineffective at sites of low oxygen tension (e.g. abscesses) and all obligate anaerobic bacteria are resistant. The doses below are for general guidance only, and should be assessed according to the clinical response. Cellular casts found in the urine sediment are an early sign of nephrotoxicity. Monitor renal function during use. If giving i.v., administer slowly. Geriatric animals or those with decreased renal function should only be given this drug systemically when absolutely necessary and then q12h or less frequently.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use ophthalmic product where corneal ulceration is present. Aminoglycosides may be contraindicated in small herbivores.

Adverse reactions: Tobramycin is considered to be less nephrotoxic than gentamicin.

Drug interactions: Avoid concurrent use of other nephrotoxic, ototoxic or neurotoxic agents (e.g. amphotericin B, furosemide). Increase monitoring and adjust dosages when these drugs must be used together. Aminoglycosides may be chemically inactivated by beta-lactam antibiotics (e.g. penicillins, cephalosporins) or heparin when mixed *in vitro*. The effect of non-depolarizing muscle relaxants (e.g. pancuronium) may be enhanced by aminoglycosides. Synergism may occur when aminoglycosides are used with penicillins or cephalosporins.

DOSES

Classified as category D (Prudence) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Birds: 2.5–5 mg/kg i.m. q8–12h.

Reptiles: 2.5 mg/kg i.m. q24–72h. Concurrent fluid therapy advised, especially if hydration status poor or uncertain.

Mammals, Amphibians, Fish: No information available.

Tolfenamic acid

(Tolfedine) **POM-V**

Formulations: Injectable: 40 mg/ml solution. Oral: 6 mg, 20 mg, 60 mg tablets.

Action: Inhibition of cyclo-oxygenase (COX) but uncertain if preferentially inhibits COX-2 over COX-1. COX inhibition limits the production of prostaglandins involved in inflammation. Also reported to have a direct antagonistic action on prostaglandin receptors.

Use: Alleviation of acute and chronic inflammation and pain. Liver disease will prolong the metabolism of tolfenamic acid leading to the potential for drug accumulation and overdose with repeated dosing. Use with caution in renal diseases and in the perioperative period, as may adversely affect renal perfusion during periods of hypotension. There is emerging evidence, using *in vitro* models and dog tumour cell lines, that tolfenamic acid may have anticancer activity against some tumour types.

Safety and handling: Normal precautions should be observed.

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A **Contraindications:** Do not give to dehydrated, hypovolaemic or hypotensive patients or those with GI disease or blood clotting problems. Do not give to pregnant animals or animals <6 weeks old.

B **Adverse reactions:** GI signs may occur in all animals after NSAID administration. Stop therapy if this persists beyond 1–2 days. Some animals develop signs with one NSAID drug and not another. A 1–2-week wash-out period should be allowed before starting another NSAID after cessation of therapy. Stop therapy immediately if GI bleeding is suspected. There is a small risk that NSAIDs may precipitate cardiac failure in humans and this risk in animals is unknown.

C **Drug interactions:** Do not administer concurrently with, or within 24 hours of, other NSAIDs and glucocorticoids. Do not administer with other potentially nephrotoxic agents, e.g. aminoglycosides.

D **DOSES**

E **Mammals:** Guinea pigs: 2 mg/kg s.c. q24h.

F **Birds, Reptiles, Amphibians, Fish:** No information available.

G **Toltrazuril (Triazinone)** **(Baycox, Zorabel) POM-V**

H **Formulations:** Oral: 25 mg/ml, 50 mg/ml solution.

I **Action:** Damages all intracellular development stages of *Eimeria* spp. Interferes with the division of the nucleus and with the activity of the mitochondria, which are responsible for the respiratory metabolism of coccidia. In the magrometes, toltrazuril damages the so-called wall-forming bodies. In all intracellular developmental stages, severe vacuolization occurs due to inflation of the endoplasmic reticulum.

J **Use:** For the treatment of coccidiosis in rabbits and microsporidiosis (e.g. *Glugae*) in fish. May be used in combination with clindamycin for treatment of *Choleoeimeria*. The highly alkaline solution is unpalatable and irritant and should be diluted with at least an equal volume of water immediately prior to oral administration in rabbits. It has been used experimentally and effectively against a number of ciliate, myxozoan and microsporidian parasites in fish.

K **Safety and handling:** Normal precautions should be observed.

L **Contraindications:** No information available.

M **Adverse reactions:** In rabbits, oral irritation when concentrated solutions are used; higher doses have resulted in transient inappetence.

N **Drug interactions:** No information available.

O **DOSES**

P **See Appendix for guidelines on responsible parasiticide use.**

Q **Mammals:** Rabbits: 2.5–10 mg/kg p.o. q24h for 2–3 days, repeat in 7–14 days^a.

R **Birds:** Raptors: Coccidiosis: 10 mg/kg p.o. q24h for 2 doses; repeat weekly for 3 weeks; **Blue-crowned laughing thrushes:** 10 mg/kg p.o. q24h for 2 doses; repeat weekly as needed^b.

S **Reptiles:** 5–15 mg/kg p.o. q24h. **Rankin's dragons:** 10 mg/kg p.o. q24h for 3 days repeated once in 7 days in combination with clindamycin for the treatment of *Choleoeimeria*^c.

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Fish: 5–20 mg/l by immersion for 1–4 hours q48h, repeated 3 times.

Amphibians: No information available.

References

- ^a Redrobe SP, Gakos G, Elliot SC *et al.* (2010) Comparison of toltazuril and sulphadimethoxine in the treatment of intestinal coccidiosis in pet rabbits. *Veterinary Record* **167**(8), 287–290
- ^b Jamriška J, Lavilla LA, Thomasson A *et al.* (2013) Treatment of toxoplasmosis in the blue-crowned laughing thrush (*Dryonastes courtoisi*). *Avian Pathology* **42**(6), 569–571
- ^c Stöhr AC, Globokar-Vrhovec M and Pantchev N (2020) *Choleoecimeria* spp. prevalence in captive reptiles in Germany and a new treatment option in a Lawson's Dragon (*Pogona henrylawsoni*). *Journal of Herpetological Medicine and Surgery* **30**(4), 261–269

Tramadol

(Tralieve, Tramadol ER*, Ultracet*, Ultram*, Zamadol*)

POM-V, POM CD SCHEDULE 3

Formulations: Oral: 50 mg tablets; 100 mg, 200 mg, 300 mg extended release tablets; 80 mg chewable tablets. Smaller tablet sizes (10 mg, 25 mg) are available from some veterinary wholesalers; 5 mg/ml, 100 mg/ml oral liquid. **Injectable:** 50 mg/ml solution (may be difficult to source in the UK).

Action: Some metabolites of tramadol are agonists at all opioid receptors, particularly mu receptors. The parent compound also inhibits the re-uptake of noradrenaline and 5-HT, and stimulates presynaptic 5-HT release, which provides an alternative pathway for analgesia involving the descending inhibitory pathways within the spinal cord. In humans, good and poor metabolizers of tramadol are described, with good metabolizers developing more opioid-like effects following drug administration and improved analgesia. Whether similar individual differences in metabolism of tramadol occur in other species is currently unknown.

Use: Management of mild to moderate acute pain and as an adjunctive analgesic in the management of chronic pain resulting from osteoarthritis or neoplasia. The recommended dose range is currently largely empirical due to a lack of combined PK/PD studies. Perioperatively, injectable tramadol is used instead of opioids to provide analgesia for acute pain, although the injectable preparation can be difficult to obtain in the UK. Tramadol has similar actions to morphine but causes less respiratory depression, sedation and GI side effects. It is attractive as an adjunct to manage chronic pain because it can be given orally; however, a larger body of evidence to support dose recommendations is needed. For example, although used in rabbits, 4.4 mg/kg i.v. had no clinically significant impact on the MAC of isoflurane. Oral tramadol at 11 mg/kg in rabbits did not reach a plasma concentration of tramadol or O-desmethyltramadol that would provide sufficient analgesia in humans for clinically acceptable periods.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Contraindicated in humans with epilepsy. Owners should be informed that there may be a slightly increased risk of seizures in treated animals. Sedation can occur in some animals and dosages should be adjusted for the individual.

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Drug interactions: Tramadol can be given in combination with other classes of analgesic drugs such as NSAIDs, amantadine and gabapentin. It has the potential to interact with drugs that inhibit central 5-HT and noradrenaline re-uptake such as tricyclic antidepressants (e.g. amitriptyline), monoamine oxidase inhibitors (e.g. selegiline), selective serotonin re-uptake inhibitors and some opioids (e.g. fentanyl, pethidine and buprenorphine), causing serotonin syndrome that can result in seizures and death. Should signs of serotonin syndrome develop (manifest in mild form as hyperthermia, elevated blood pressure and CNS disturbances such as hypervigilance and excitation) these must be managed symptomatically and contributing drug treatments stopped.

DOSES

Mammals: Ferrets: 5 mg/kg p.o., s.c. q12–24h; Rabbits: 3–10 mg/kg p.o. q8–12h (note: analgesic dose not determined)^{a,b}; Rats: 10–20 mg/kg p.o., s.c. q8–12h; Mice: 10–40 mg/kg s.c. q12h; Primates: 3 mg/kg p.o. (analgesic effect not determined) or 1.5 mg/kg i.v. q24^c; Hedgehogs: 2–4 mg/kg p.o. q12h; Sugar gliders: 2 mg/kg p.o. q12h.

Birds: Bald eagles: 5 mg/kg p.o. q12h; Common buzzards: 4 mg/kg i.m. (reduced thermal withdrawal response for 3 hours post-dosing); Red-tailed hawks: 15 mg/kg p.o. q12h to achieve human therapeutic levels; Hispaniolan Amazon parrots: 30 mg/kg p.o. q6h^{d,e} (achieved human therapeutic levels and reduced thermal withdrawal response for 6 hours post-dosing).

Reptiles: Chelonians: 5–10 mg/kg i.m., p.o. q24–48h^f; Bearded dragons: 11 mg/kg p.o.

Amphibians, Fish: No information available.

References

- ^a Egger CM, Souza MJ, Greenacre CB, Cox SK and Rohrbach BW (2009) Effect of intravenous administration of tramadol hydrochloride on the minimum alveolar concentration of isoflurane in rabbits. *American Journal of Veterinary Research* **70**(8), 945–949
- ^b Souza MJ, Greenacre CB and Cox SK (2008) Pharmacokinetics of orally administered tramadol in domestic rabbits (*Oryctolagus cuniculus*). *American Journal of Veterinary Research* **69**(8), 979–982
- ^c Kelly KR, Pypendop BH and Christe KL (2015) Pharmacokinetics of tramadol following intravenous and oral administration in male rhesus macaques (*Macaca mulatta*). *Journal of Veterinary Pharmacology and Therapeutics* **38**, 375–382
- ^d Geelen S, Guzman DSM, Souza MJ et al. (2013) Antinociceptive effects of tramadol hydrochloride after intravenous administration to Hispaniolan Amazon parrots (*Amazona ventralis*). *American Journal of Veterinary Research* **74**(2), 201–206
- ^e Souza MJ, Gerhardt L and Cox S (2013) Pharmacokinetics of repeated oral administration of tramadol hydrochloride in Hispaniolan Amazon parrots (*Amazona ventralis*). *American Journal of Veterinary Research* **74**(7), 957–962
- ^f Baker BB, Sladky KK and Johnson SM (2011) Evaluation of the analgesic effects of oral and subcutaneous tramadol administration in red-eared slider turtles. *Journal of the American Veterinary Medical Association* **238**(2), 220–227

Travoprost

(Travatan*) **POM**

Formulations: Ophthalmic: 40 µg (micrograms)/ml (0.004%) solution in 2.5 ml bottle.

Action: Agonist for receptors specific for prostaglandin F. It reduces intraocular pressure by increasing uveoscleral outflow.

Use: Its main indication is in the management of primary glaucoma and it is useful in the emergency management of acute primary glaucoma (superseding mannitol and acetazolamide). Often used in conjunction with other topical antiglaucoma drugs such as carbonic anhydrase inhibitors. It may be useful in the management of lens subluxation despite being contraindicated in anterior lens luxation. Travoprost has comparable activity to latanoprost.

Safety and handling: Normal precautions should be observed.

Contraindications: Uveitis and anterior lens luxation.

Adverse reactions: Miosis, conjunctival hyperaemia and mild irritation may develop. Increased iridial pigmentation has been noted in humans.

Drug interactions: Do not use in conjunction with thiomersal-containing preparations.

DOSES

Mammals: Rabbits: 1 drop per eye q12h.

Birds, Reptiles, Amphibians, Fish: No information available.

Trazodone

(Desyrel*, Molipaxin*, Oleptro*, Trazodone*) **POM**

Formulations: Oral: 50 mg, 100 mg, 150 mg tablets/capsules; 10 mg/ml liquid.

Action: Trazodone is an atypical antidepressant with mixed serotonergic agonistic (5HT_{1A}) and antagonistic actions (other 5HT receptor sites).

Use: Chronic anxiety-related problems that are unresponsive to other pharmacological interventions. Should be used in conjunction with a behaviour modification plan. Use with caution and carefully monitor patients with renal disease. Less risk in patients with cardiac disease than tricyclic antidepressants such as amitriptyline. Reduction of anxiety associated with hospitalization, including postoperative confinement in rabbits and may be used for the management of short-term anxiety-related problems, e.g. travel or examination anxiety.

Safety and handling: Normal precautions should be observed.

Contraindications: Glaucoma, history of seizures or urinary retention and severe liver disease.

Adverse reactions: Sedation, vomiting, excitability and dry mouth. Third eyelid protrusion described in some cats but not reported in exotic pets.

Drug interactions: Should not be used with monoamine oxidase inhibitors or drugs metabolized by cytochrome P450 2D6 (e.g. chlorphenamine, cimetidine). There is a risk of serotonin syndrome if combined with other serotonergic substances, but adjunctive therapy is sometimes used (see below). Ketoconazole will inhibit the breakdown of trazodone, leading to increased blood levels, while carbamazepine will have the opposite effect; itraconazole may be similar, although there is no clinical evidence of this.

DOSES

Mammals: Rabbits: For the management of short-term anxiety associated with hospitalization: 40 mg/kg p.o. 3 hours prior to the visit^a.

Birds, Reptiles, Amphibians, Fish: No information available.

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References

- ^a Rickertl K, Reed J and Brundage C (2020) Physiological effect of trazodone hydrochloride use for anxiety and sedation in rabbits (*Oryctolagus cuniculus*). *The FASEB Journal* **34**, 1–1.

Tretinoin see **Vitamin A****Triazinone** see **Toltrazuril****Tricaine mesilate (Metacaine, TMS, Tricaine mesylate, Tricaine methane sulphonate)**(Finquel, MS-222, Nytox, Tricaine, **Tricaine PHARMAQ**)**POM-VPS****Formulations:** Immersion: 100% powder for dissolution in water.**Action:** Tricaine is highly lipid soluble and rapidly absorbed across the gills, resulting in anaesthesia by impeding peripheral nerve signal transmission to the CNS.

Use: Authorized for use in an immersion bath for sedation, immobilisation and anaesthesia of fish. Can also be used for sedation, immobilisation and anaesthesia of amphibians and for euthanasia of fish and amphibians. Ideally, the drug should be dissolved in water from the tank or pond of origin to minimize problems due to changes in water chemistry. The dry powder is very soluble in both fresh and marine water, and may be added directly or made into a stock solution (e.g. 100 mg/ml) for more accurate dosing but should be used within 12 hours of preparation as advised by some manufacturers. Before use, the pH of the anaesthetic solution should be buffered with sodium bicarbonate to the same pH of the water of origin. The anaesthetic solution should be used on the day of preparation and be well aerated during use. Food should be withheld for 12–24 h before anaesthesia to reduce the risk of regurgitation. The stage of anaesthesia reached is determined by the concentration used and the duration of exposure, since absorption continues throughout the period of immersion. Potency and toxicity increases with increasing temperature. Salinity and water hardness reduce toxicity and increase the dose required. Fully aquatic amphibians may be anaesthetized by complete immersion with access to air but partial immersion is recommended for all other species. Small fish are more sensitive to the drug than large fish. Obligate and facultative air-breathing fish species (e.g. bettas, arowana, some catfish and gouramis) will require higher dosages. Different species vary in their response and may require different concentrations. It is recommended to use the lower dose rates to test the selected drug concentration and exposure time with a small group before medicating large numbers. Anaesthetized fish should be returned to clean water from their normal environment to allow recovery. For euthanasia, 10 times the normal anaesthetic dose should be used and the fish kept in the solution for at least 60 minutes after respiration ceases.

Safety and handling: The powder should be stored dry and stock solutions stored in sealed dark bottles. A case of reversible retinal toxicity has been reported in a human with long history of skin exposure to tricaine. Normal precautions should be observed.

Contraindications: Should not be used in the following tropical fish species: *Apistogramma (Mikrogeophagus) ramirezi*, *Baltantiocheilus melanopterus*, *Epiplatys suratusensis*, *Melanotaenia maccullochi*, *Mono-dactylus argenteus*, *Phenacogrammus interruptus* and *Scatophagus argus*.

Adverse reactions: No information available.

Drug interactions: No information available.

DOSES

Amphibians: Axolotls: Immersion in 2 g/l solution for 20 minutes for 15–30 minute anaesthesia^a; **African clawed frogs:** Immersion in 1–2 g/l for 30–60 minute anaesthesia^b; **White's tree frogs:** 0.5–2 g/l (lower dose for mild sedation, higher dose for surgical anaesthesia)^c.

Fish: 50–250 mg/l by immersion for induction of anaesthesia; 25–100 mg/l by immersion for maintenance of anaesthesia¹.

Mammals, Birds, Reptiles: No information available.

References

- ^a Zullian C, Dodelet-Devillers A, Roy S and Vachon P (2016) Evaluation of the anesthetic effects of MS222 in the adult Mexican axolotl (*Ambystoma mexicanum*). *Veterinary Medicine* **7**, 1–7
- ^b Lalonde-Robert V, Beaudry F and Vachon P (2012) Pharmacologic parameters of MS222 and physiologic changes in frogs (*Xenopus laevis*) after immersion at anesthetic doses. *Journal of the American Association of Laboratory Animal Science* **51**, 464–468
- ^c Krisp AR, Hausmann JC, Sladky KK *et al.* (2020) Anesthetic efficacy of MS-222 in White's tree frogs (*Litoria caerulea*). *Journal of Herpetological Medicine and Surgery* **30**, 38–41
- ¹ Sneddon LU (2012) Clinical anesthesia and analgesia in fish. *Journal of Exotic Pet Medicine* **21**, 32–43

Trilostane

(Veteryl) **POM-V**

Formulations: Oral: 10 mg, 30 mg, 60 mg, 120 mg capsules.

Action: Blocks adrenal synthesis of glucocorticoids. Effects on mineralocorticoids are relatively minor.

Use: Treatment of pituitary- and adrenal-dependent hyperadrenocorticism. It is not considered to be useful in treating ferrets with hyperadrenocorticism. Perform ACTH stimulation tests (start test 3–5h post-dosing) at 10 days, 4 weeks, 12 weeks and then every 3 months. The aim is for a post-ACTH cortisol of 40–120 nmol/l. In cases where clinical signs persist or polydipsia appears within the 24-hour period, ACTH stimulation tests performed later in the day and/or sequential cortisol determinations may be needed for dose adjustment (either mg/kg or frequency). Dosage adjustments may be necessary even after prolonged periods of stability.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use in patients with renal or hepatic insufficiency.

Adverse reactions: In humans, idiosyncratic reactions include diarrhoea, colic, muscle pain, nausea, hypersalivation and rare cases of skin changes (rash or pigmentation). Clinical hypoadrenocorticism can be seen. Adrenal necrosis has been reported. Adrenal hyperplasia has been noted with prolonged treatment but the effects of this are unknown. Prolonged adrenal suppression after drug withdrawal has been noted in some cases.

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Drug interactions: Trilostane should not be administered concurrently with other drugs that suppress adrenal function (e.g. mitotane, itraconazole).

DOSES

Mammals: Rodents: 2–4 mg/kg p.o. q24h.

Birds, Reptiles, Amphibians, Fish: No information available.

Trimethoprim/Sulphonamide (Potentiated sulphonamides)

(Co-Trimazine, Diatrim, Duphatrim, Metaxol, Norodine, **Sulfatrim**, Tribriessen, Trimacare, Trimediazine, Trimedoxine, Trinacol, Septrin*) **POM-V**

Formulations: Trimethoprim and sulphonamide are formulated in a ratio of 1:5.

- **Injectable:** trimethoprim 40 mg/ml and sulfadiazine 200 mg/ml (240 mg/ml total) suspension.
- **Oral:** 16 mg/ml trimethoprim and 80 mg/ml sulfamethoxazole (Sulfatrim), 200 mg sulfamethoxazole and 40 mg trimethoprim per 5 ml = 48 mg/ml suspension (Septrin paediatric suspension).
Trimethoprim and sulfadiazine are also available in a variety of tablet sizes designated by the amount of trimethoprim (e.g. 20 mg, 80 mg).
Note that concentration is listed in terms of total combined quantity of both components in 5 ml of liquid.

Action: Trimethoprim and sulphonamides block sequential steps in the synthesis of tetrahydrofolate, a cofactor required for the synthesis of many molecules, including nucleic acids. Sulphonamides block the synthesis of dihydropteroic acid by competing with para-aminobenzoic acid, and trimethoprim inhibits the enzyme dihydrofolate reductase, preventing the reduction of dihydrofolic acid to tetrahydrofolic acid. This two-step mechanism ensures that bacterial resistance develops more slowly than to either agent alone. In addition, the effect of the combination tends to be bactericidal as against a bacteriostatic effect of either agent alone.

Use: Authorized in rabbits, pigeons and bearded dragons for treatment of coccidiosis (Sulfatrim). Many organisms are susceptible, including *Nocardia*, *Brucella*, Gram-negative bacilli, some Gram-positive organisms (*Streptococcus*), plus *Pneumocystis carinii*, *Toxoplasma gondii* and *Plasmodium*. *Pseudomonas* and *Leptospira* are usually resistant. Trimethoprim/sulphonamide is useful in the management of urinary, respiratory tract and prostatic infections, but ineffective in the presence of necrotic tissue. Trimethoprim alone may be used for urinary, prostatic, systemic salmonellosis and respiratory tract infections. Fewer adverse effects are seen with trimethoprim alone. Trimethoprim is a weak base which becomes ion-trapped in fluids that are more acidic than plasma (e.g. prostatic fluid and milk). Monitor tear production particularly during long-term use and in animals susceptible to keratoconjunctivitis sicca. Ensure patients receiving sulphonamides are well hydrated and are not receiving urinary acidifying agents. Used for the treatment of bacterial infections and intestinal coccidiosis in fish and reptiles.

Safety and handling: Normal precautions should be observed.

Contraindications: Avoid use in animals with keratoconjunctivitis sicca (KCS) or previous history of adverse reaction to sulphonamides (e.g. KCS or polyarthritis). Beware use in reptiles where renal disease is suspected.

Adverse reactions: Acute hypersensitivity reactions are possible with sulphonamide products; they may manifest as a type III hypersensitivity reaction. Sulphonamides may reversibly suppress thyroid function. Dermatological reactions (e.g. toxic epidermal necrolysis) have been associated with the use of sulphonamides in some animals. Sulphonamide crystal formation can occur in the urinary tract, particularly in animals producing very concentrated and/or acidic urine.

Drug interactions: Antacids may decrease the bioavailability of sulphonamides if administered concomitantly. Urinary acidifying agents will increase the tendency for sulphonamide crystals to form within the urinary tract. Concomitant use of drugs containing procaine may inhibit the action of sulphonamides since procaine is a precursor for para-amino benzoic acid. When using the Jaffe alkaline picrate reaction assay for creatinine determination, trimethoprim/sulphonamide may cause an overestimation of approximately 10%.

DOSES

Classified as category D (Prudence) by the EMA.

See Appendix for guidelines on responsible antibiotic use.

Doses (mg) of total product (trimethoprim + sulphonamide).

Mammals: Ferrets: 15–30 mg/kg p.o., s.c. q12h; Rabbits: 20–30 mg/kg p.o., s.c. q12h (for 10–14 days for coccidiosis); Guinea pigs, Chinchillas, Hamsters: 15–30 mg/kg p.o., i.m., s.c. q12–24h; Rats, Mice, Gerbils: 50–100 mg/kg p.o., s.c. q24h; Primates: 15 mg/kg p.o. q12h or 30 mg/kg s.c., i.m. q24h; Prosimians: 25 mg/kg p.o. q12h; Marsupials: 10–20 mg/kg p.o., i.m. q12–24h; Sugar gliders: 10–20 mg/kg p.o. q12–24h; Hedgehogs: 30 mg/kg p.o., s.c., i.m. q12h.

Birds: 8–30 mg/kg i.m. q12h or 20–100 mg/kg p.o. q12h; Pigeons: 475–970 mg/l drinking water or 25–50 mg/kg p.o. q12h for 10–14 days for coccidiosis (Sulfatrim).

Reptiles: 15–30 mg/kg p.o. q24–48h; Bearded dragons: 15–20 mg/kg p.o. q24h for 7–14 days for coccidiosis.

Amphibians: 3 mg/kg p.o., s.c. q24h.

Fish: 30 mg/kg i.m. q48h for 10 days or 30 mg/kg in feed q24h for 10 days.

Tropicamide

(Mydriacyl*, Tropicamide*) **POM**

Formulations: Ophthalmic: 0.5%, 1% solution, single-use vials, 5 ml bottle.

Action: Inhibits acetylcholine at the iris sphincter and ciliary body muscles, causing mydriasis (pupil dilation) and cycloplegia (paralysis of the ciliary muscle). Ineffective in birds and reptiles because of the complex arrangement of musculature in the iris and ciliary body.

Use: Synthetic, short-acting antimuscarinic used for mydriasis and cycloplegia. It is the mydriatic of choice for intraocular examination due to its rapid onset (20–30 min) and short duration of action (4–12 h).

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A Tropicamide is more effective as a mydriatic than as a cycloplegic and is therefore less effective than atropine in relieving ciliary body muscle spasm associated with uveitis. Use with care in patients with lens luxation.

B **Safety and handling:** Normal precautions should be observed.

C **Contraindications:** Avoid in glaucoma.

D **Adverse reactions:** No information available.

E **Drug interactions:** No information available.

F **DOSES**

F **Mammals:** Rabbits, Chinchillas, Hamsters, Rats: 1 drop per eye, repeat after 20–30 min if necessary.

G **Birds, Reptiles:** Ineffective because of the complex arrangement of musculature in the iris and ciliary body.

H **Amphibians, Fish:** No information available.

I **Tylosin**

J (Bilosin, Pharmasin, Tylan, Tylovvet) **POM-V**

K **Formulations:** Injectable: 200 mg/ml solutions (Bilosin, Tylan, Tylovvet). Oral: 100 g/bottle soluble powder (Tylan).

L **Action:** A bacteriostatic macrolide antibiotic that binds to the 50S ribosomal subunit, suppressing bacterial protein synthesis.

M **Use:** Tylosin has good activity against mycoplasmas and has the same antibacterial spectrum of activity as erythromycin but is generally less active against bacteria.

N **Safety and handling:** Normal precautions should be observed.

O **Contraindications:** Do not give other than via nebulization to rabbits, guinea pigs, chinchillas, hamsters or degus.

P **Adverse reactions:** GI disturbances. The activity of tylosin is enhanced in an alkaline pH. Tylosin can cause pain at the site of injection.

Q **Drug interactions:** Not well documented in small animals. It does not appear to inhibit the same hepatic enzymes as erythromycin.

R **DOSES**

S **Classified as category C (Caution) by the EMA.**

T **See Appendix for guidelines on responsible antibiotic use.**

U **Mammals:** Ferrets, Rats, Mice: 10 mg/kg p.o., i.m., s.c. q12h; Not recommended for rabbits, guinea pigs, chinchillas, hamsters or degus other than by nebulization, as for birds.

V **Birds:** 20–40 mg/kg i.m. q8–12h (not chickens) or by nebulization of 100 mg diluted in 5 ml DMSO and 10 ml saline; **Pigeons:** 50 mg/kg p.o. q24h, 25 mg/kg i.m. q6–8h or 800 mg/l drinking water; **Passerines:** 1 g/l drinking water for 7–10 days.

X **Reptiles:** 5 mg/kg i.m. q24h every 10–60 days.

Y **Amphibians, Fish:** No information available.

Z

Ursodeoxycholic acid (UDCA)

(Destolit*, Ursodeoxycholic acid*, Ursofalk*, Ursogal*)

POM

Formulations: Oral: 150 mg, 300 mg, 500 mg tablets; 250 mg capsule; 50 mg/ml suspension.

Action: A relatively hydrophilic bile acid with cytoprotective effects in the biliary system. It inhibits ileal absorption of hydrophobic bile acids, thereby reducing their concentration in the body pool; hydrophobic bile acids are toxic to hepatobiliary cell membranes and may potentiate cholestasis. It also has an immunomodulatory effect and may modify apoptosis of hepatocytes.

Use: An adjunctive therapy for patients with liver disease, particularly where cholestasis is present. Anecdotal use only in ferrets.

Safety and handling: Normal precautions should be observed.

Contraindications: No information available.

Adverse reactions: Vomiting, in those species which can do so, is a rare effect. Serious hepatotoxicity has been recognized in rabbits and non-human primates. Some human patients have an inability to sulphate lithocholic acid (a naturally occurring metabolite of UDCA), which is a known hepatotoxin; the veterinary significance of this is unclear.

Drug interactions: Aluminium-containing antacids may bind to UDCA, thereby reducing its efficacy.

DOSES

Mammals: Ferrets: 10–15 mg/kg p.o. q12–24h.

Birds, Reptiles, Amphibians, Fish: No information available.

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Vecuronium(Norcuron*) **POM****Formulations:** Injectable: 10 mg powder for reconstitution.**Action:** Inhibits the actions of acetylcholine at the neuromuscular junction by binding competitively to the alpha subunit of the nicotinic acetylcholine receptor on the post-junctional membrane.**Use:** Used topically as a mydriatic in birds; ensure solution does not contain surface-penetrating agent. Provision of neuromuscular blockade during anaesthesia. This may be to improve surgical access through muscle relaxation, to facilitate positive pressure ventilation or for intraocular surgery. Intermediate dose-dependent duration of action of approximately 20 min. Has no cardiovascular effects and does not cause histamine release. Monitoring (using a nerve stimulator) and reversal of the neuromuscular blockade are recommended to ensure complete recovery before the end of anaesthesia. Hypothermia, acidosis and hypokalaemia will prolong the duration of neuromuscular blockade. In healthy animals repeated doses are relatively non-cumulative and it can be given by infusion i.v. to maintain neuromuscular blockade. It is metabolized by the liver; therefore in animals with liver dysfunction atracurium is advised rather than vecuronium.**Safety and handling:** Unstable in solution and so is presented as a freeze-dried powder. The prepared solution can be diluted further if required.**Contraindications:** Do not administer systemically unless the animal is adequately anaesthetized and facilities to provide positive pressure ventilation are available.**Adverse reactions:** No information available.**Drug interactions:** Neuromuscular blockade is more prolonged when vecuronium is given in combination with volatile anaesthetics, aminoglycosides, clindamycin and lincomycin.**DOSES****Birds:** 1 drop of 0.8 mg/ml solution in 0.9% saline applied topically to eye; repeat after 2 minutes ^{a,b}.**Reptiles:** 1 drop of 0.4% solution applied topically to eye q15–20min ^c.**Mammals, Amphibians, Fish:** No information available.**References**

- ^a Hendrix DV and Sims MH (2004) Electroretinography in the Hispaniolan Amazon parrot (*Amazona ventralis*). *Journal of Avian Medicine and Surgery* **18**(2), 89–94
- ^b Ramer JC, Paul-Murphy J, Brunson D and Murphy CJ (1996) Effects of mydriatic agents in cockatoos, African gray parrots and Blue-fronted Amazon parrots. *Journal of the American Veterinary Medical Association* **208**(2), 227–230
- ^c Somma AT, Lima L, Lange RR, Turner-Giannico A and Montiani-Ferreira F (2015) The eye of the red-eared slider turtle: morphologic observations and reference values for selected ophthalmic diagnostic tests. *Veterinary Ophthalmology* **18**, 61–70

Verapamil

(Cordilox*, Securon*, Verapamil*) **POM**

Formulations: Injectable: 2.5 mg/ml solution. Oral: 40 mg, 80 mg, 120 mg, 160 mg tablets.

Action: Inhibits inward movement of calcium ions through slow (L-type) calcium channels in myocardial cells, cardiac conduction tissue and vascular smooth muscle. Verapamil causes a reduction in myocardial contractility (negative inotrope), depressed electrical activity (slows AV conduction) and vasodilation (cardiac vessels and peripheral arteries and arterioles).

Use: In rabbits may be used perioperatively to minimize formation of surgical adhesions. Patients with severe hepatic disease may have a reduced ability to metabolize the drug; reduce the dose by 70%.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use in patients with 2nd or 3rd degree AV block, hypotension, sick sinus syndrome, left ventricular dysfunction or heart failure.

Adverse reactions: Can cause hypotension, bradycardia, dizziness, precipitation or exacerbation of congestive heart failure, nausea, constipation and fatigue in humans. The likelihood of adverse reactions is decreased by dilution and slow administration of the product where given i.v. but even with these precautions, adverse cardiac reactions may be seen in some cases, caution is strongly advised.

Drug interactions: Do not use concurrently with beta-blockers. Both drugs have a negative inotropic and chronotropic effect and the combined effect can be profound. Co-administration with sodium channel blockers may also lead to cardiovascular depression and hypotension. Verapamil activity may be adversely affected by vitamin D or calcium salts. Cimetidine may increase the effects of verapamil. Verapamil may increase the blood levels of digoxin, digitoxin or theophylline, leading to potentially toxic effects from these drugs. Calcium-channel blockers may increase intracellular vincristine. The neuromuscular blocking effects of non-depolarizing muscle relaxants may be enhanced by verapamil.

DOSES

Mammals: Rabbits: 0.2 mg/kg very slow i.v., s.c., p.o. after surgery and repeated q8h for 9 doses or 2.5–25 µg (micrograms)/kg/h i.p., s.c.; rapid i.v. administration can cause cardiac failure and sudden death, and s.c. or p.o. routes are advised instead; **Hamsters:** 0.25–0.5 mg/kg s.c. q12h.

Birds, Reptiles, Amphibians, Fish: No information available.

Vincristine

(Oncovin*, Vincristine*) **POM**

Formulations: Injectable: 1 mg, 2 mg, 5 mg vials.

Action: Interferes with microtubule assembly, causing metaphase arrest and ultimately resulting in cell death.

Use: With other neoplastic agents in the treatment of ferret neoplastic diseases, particularly lymphoma. Has been used to treat lymphoma in a green iguana. Use with caution in patients with hepatic disease, leucopenia,

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A infection, or pre-existing neuromuscular disease. Solution is locally irritant and must be administered i.v. through a carefully pre-placed catheter.

B **Safety and handling:** **Cytotoxic drug; see specialist texts for further advice on chemotherapeutic agents.** Store under refrigeration.

C **Contraindications:** No information available.

D **Adverse reactions:** Include peripheral neuropathy, ileus, GI tract toxicity/constipation and severe local irritation if administered perivascularly. Potentially myelosuppressive.

E **Drug interactions:** Concurrent administration of vincristine with drugs that inhibit cytochromes of the CYP3A family may result in decreased metabolism of vincristine and increased toxicity. If vincristine is used in combination with asparaginase it should be given 12–24 hours before the enzyme. Administration of asparaginase with or before vincristine may reduce clearance of vincristine and increase toxicity.

H DOSES

I **See Appendix for chemotherapy protocols in ferrets.**

J **Mammals:** Ferrets: 0.12–0.2 mg/kg i.v. as part of chemotherapy protocol for lymphoma.

K **Birds:** 0.1 mg/kg i.v. q7–14d¹.

L **Reptiles:** 0.008 mg/kg i.v. as part of chemotherapy protocol for lymphoma (one case report in the green iguana)².

M **Amphibians, Fish:** No information available.

N References

- ¹ Rivera S, McClearen JR and Reavill DR (2009) Treatment of nonepitheliotropic cutaneous B-cell lymphoma in an umbrella cockatoo (*Cacatua alba*). *Journal of Avian Medicine and Surgery* **23**(4), 294–303
- ² Folland DW, Johnston MS, Thamm DH and Reavill D (2011) Diagnosis and management of lymphoma in a green iguana (*Iguana iguana*). *Journal of the American Veterinary Medical Association* **239**(7), 985–991

P Virkon S

Q (Virkon Aquatic tablets*)

R **Formulations:** Marketed as a water sanitizer and comprises a mixture of oxyzones, fruit acid salts and potassium monopersulphonate.

S **Immersion:** 5 g tablet for dissolution in water.

T **Action:** No information available.

U **Use:** For the reduction of viruses, bacteria and some parasites in koi carp ponds. This is a modified version of Virkon S with the amount of detergent reduced since this is toxic to aquatic animals, and the fragrance removed. **Do not** add directly to the pond: the tablets must be dissolved in water before adding slowly to the pond at the point where the water enters from the filtration system, to aid dispersal. An initial dose of 8 mg/l is recommended. If further weekly dosing at 2 mg/l is required, then 10% of the pond water should be changed prior to dosing. The product is deactivated by UV light. Repeated dosing can result in a build-up of the surfactant, resulting in some foaming, which can be reduced by partial water changes.

V **Safety and handling:** Normal precautions should be observed. The dust is irritating to skin and the respiratory system and may cause serious eye damage.

Contraindications: No information available.

Adverse reactions: No information available.

Drug interactions: No information available.

DOSES

Fish: 2–8 mg/l by prolonged immersion initially, then 2 mg/l weekly as required.

Mammals, Birds, Reptiles, Amphibians: No information available.

Vitamin A (Retinol, Isotretinoin, Tretinoin)

(Roaccutane*) **POM**

Formulations: **Injectable:** Vitamin A (retinol) 50,000 IU/ml (only available on special order from France and the USA at time of writing).

Oral: 10 mg, 20 mg isotretinoin capsules (Roaccutane), component of various multivitamin powder and liquid supplements. **Topical:** 0.025% tretinoin cream; 0.01% tretinoin gel.

Action: Nutritional fat-soluble hormone that regulates gene expression. An essential nutrient for many metabolic processes including normal reproduction, immune function, vision and epithelial tissues. Tretinoin (all-trans retinoic acid) is the acid form of vitamin A and isotretinoin (13-cis retinoic acid) is an isomer of tretinoin.

Use: Treatment of hypovitaminosis A. Also used in conjunction with other appropriate therapies for sebaceous adenitis in rabbits. Animals receiving oral dosing should be monitored for vitamin A toxicity. Avoid concurrent use of oral and topical preparations because of toxicity. Avoid using formulations of vitamins A, D3 and E that are authorized for farm animals as they are too concentrated for small animal use. Carnivorous and insectivorous reptiles may not be able to convert plant based carotenoids to vitamin A so an animal source of vitamin A (retinyl palmitate) is recommended for these species.

Safety and handling: Vitamin A is teratogenic; gloves should be worn when applying topical preparations. Avoid contact with eyes, mouth or mucous membranes. Minimize exposure of the drug to sunlight.

Contraindications: Do not use in pregnant animals.

Adverse reactions: Many adverse effects are reported in humans following the use of oral isotretinoin, predominantly involving the skin, haematological parameters, hepatotoxicity, nervous system and bone changes. Depression, anorexia and skin sloughing are reported in reptiles following overdose. Teratogenic if administered in the first trimester or at high doses. Redness and skin pigmentation may be seen after several days. It changes the lipid content of tears, which can result in keratoconjunctivitis sicca. It may also cause hyperlipidaemia and can be hepatotoxic at high doses. Prolonged use of vitamin A can promote loss of calcium from bone and lead to hypercalcaemia. Do not use topical preparations simultaneously with other topical drugs.

Drug interactions: Numerous, depending on preparation and route given. Consult specialist tests before using with another drug. Oral vitamin A may alter ciclosporin levels, which should therefore be monitored closely.

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DOSES

Mammals: Rabbits: 500–1000 IU/kg i.m. once; Guinea pigs, Sugar gliders: 50–500 IU/kg i.m. or 10 mg beta carotene/kg food; Hamsters: 50–500 IU/kg i.m. or 2 µg (micrograms) vitamin A palmitate/g food; Hedgehogs: 400 IU/kg p.o., i.m. q24–48h; only give parenterally for up to 2 doses, thereafter orally.

Birds: 1000–20,000 IU/bird i.m. once or p.o. q12h.

Reptiles: 1000–2000 IU/kg p.o. q7–14d.

Amphibians: 1000 IU/kg p.o. daily for 14 days, then q7d.

Fish: No information available.

Vitamin B complex

(Anivit 4BC, Duphafral, Duphafral Extravite, Multivitamin injection, Vitamin B tablets) **POM-VPS, general sale**

Formulations: Various preparations containing varying quantities of vitamins are available, authorized for farm animals only. Most are for parenteral use and all those are POM-VPS.

Action: Cofactors for enzymes of intermediary metabolism and biosynthesis.

Use: Multiple deficiencies of B vitamins may occur in patients with renal or hepatic disease or significant anorexia. Dosages and routes vary with individual products. Check manufacturer's recommendations prior to use. Most products are intended for large animal use and some may contain vitamin C and other vitamins or minerals.

Safety and handling: All B vitamins are photosensitive and must be protected from light. Multidose vials require aseptic technique for repeated use.

Contraindications: No information available.

Adverse reactions: Anaphylaxis may be seen when used i.v. and products should be given slowly and/or diluted with i.v. fluids. Use of large animal products which also contain fat-soluble vitamins (A, D, E, K) may lead to toxicity.

Drug interactions: None reported.

DOSES

Mammals: Rabbits: 0.02–0.4 ml/rabbit s.c., i.m. q24h or as required; Rodents: 0.02–0.2 ml/kg s.c., i.m. q24h or as required; **Marsupials:** 0.01–0.02 ml/kg i.m. q24h or as required; **Sugar gliders:** 0.01–0.2 ml/kg s.c. i.m.; Hedgehogs: 1 ml/kg s.c., i.m. q24h or as required.

Birds: 1–3 mg/kg i.m. once. Should aim to achieve 10–30 mg/kg thiamine. Care must be taken not to exceed 3 mg/kg pyridoxine HCl (vitamin B6) as toxicity (acute death in 24–48 hours) recorded in raptors and pigeons^a.

Reptiles, Amphibians, Fish: No information available.

References

- ^a Samour J (2013) Acute toxicity after administration of high doses of vitamin B6 (pyridoxine) in falcons. *Proceedings of the 1st International Conference on Avian, Herpetological and Exotic Mammal Medicine*, Weisbaden, Germany

Vitamin B1 (Thiamine)

(Vitamin B1) **POM-V, general sale**

Formulations: **Injectable:** 100 mg/ml solution (authorized for veterinary use, though only in farm animals). **Oral:** various.

Action: Cofactor for enzymes in carbohydrate metabolism, it forms a compound with ATP to form thiamine diphosphate/thiamine pyrophosphate employed in carbohydrate metabolism. It does not affect blood glucose.

Use: Thiamine supplementation is required in deficient animals. Although uncommon this may occur in animals fed raw fish diets (containing thiaminase) or uncooked soy products. Thiamine may be beneficial in alleviating signs of lead poisoning and ethylene glycol intoxication. It is used in birds, reptiles and amphibians for the treatment of cerebral cortical necrosis.

Safety and handling: Protect from air and light; multidose vials require aseptic technique for repeated use.

Contraindications: Do not use in pregnant animals unless absolutely necessary.

Adverse reactions: Anaphylaxis can be seen with i.v. use; dilute with fluids and/or give slowly if using i.v. Adverse effects in pregnant animals are documented.

Drug interactions: There are no specific clinical interactions reported, although thiamine may enhance the activity of neuromuscular blocking agents.

DOSES

Mammals: Ferrets, Rabbits: 1–2 mg/kg s.c., i.m. q12–24h for several days until signs resolve.

Birds: 10–30 mg/kg i.m. q24h^a.

Reptiles: Thiamine deficiency: 25–35 mg/kg p.o., i.m., s.c. q24h.

Amphibians: Thiamine deficiency: 25 mg/kg of feed fish.

Fish: No information available.

References

- ^a Carnarius M, Hafez HM, Henning A, Henning HJ and Lierz M (2008) Clinical signs and diagnosis of thiamine deficiency in juvenile goshawks (*Accipiter gentilis*). *Veterinary Record* **163**(7), 215–17

Vitamin B12 (Cyanocobalamin, Hydroxocobalamin)

(Anivit B12 250 and 1000, Cobalaplex, Neo-Cytamen (hydroxocobalamin), Vitbee 250 and 1000) **POM-VPS**

Formulations: **Injectable:** 0.25 mg/ml, 1 mg/ml solutions.

Action: Essential cofactor for enzymes involved in DNA and RNA synthesis and in carbohydrate metabolism.

Use: Cyanocobalamin is used to treat vitamin B12 deficiency. Such a deficiency may develop in patients with significant disease of the distal ileum, small intestinal bacterial overgrowth and exocrine pancreatic insufficiency. In humans, hydroxocobalamin has almost completely replaced cyanocobalamin in the treatment of vitamin B12 deficiency.

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A**Safety and handling:** Must be protected from light.**B****Contraindications:** Do not give i.v.**C****Adverse reactions:** Hypersensitivity to the phenol preservative in the injectable solutions can occur; patients should be monitored after injections for rash, fever and urticaria.**D****Drug interactions:** None reported.**E****DOSES****F****Mammals:** Rabbits: 20–50 µg (micrograms)/kg monthly to continue as long as deficiency is present.**G****Birds:** 0.25–5 mg/kg i.m. q7d.**H****Reptiles:** 0.05 mg/kg i.m., s.c.**I****Amphibians, Fish:** No information available.**J****Vitamin C (Ascorbic acid)****K**(Numerous trade names) **POM, general sale****L****Formulations:** Injectable: 100 mg/ml. Oral: Various strength tablets, capsules, powders and liquids.**M****N****Action:** Water-soluble antioxidant, also critical for crosslinking collagen precursors (growth and repair of tissue) and is involved in protein, lipid and carbohydrate metabolism.**O****Use:** Vitamin C is used to reduce methaemoglobinaemia associated with paracetamol toxicity. Supplemental vitamin C may be required in conditions of increased oxidative stress, in cachexic patients and in those requiring nutritional support. Vitamin C is not endogenously produced by guinea pigs, certain other South American rodents not typically kept as pets, and great apes, so must be present in their diet.**P****Safety and handling:** Normal precautions should be observed.**Q****Contraindications:** Avoid use in patients with liver disease.**R****Adverse reactions:** May cause anaphylaxis if given i.v. Vitamin C supplementation may increase liver damage by increasing iron accumulation. Prolonged use at excessive doses can increase the risk of urate, oxalate and cystine crystalluria and stone formation.**S****Drug interactions:** Large doses (oral or injectable) will acidify the urine and may increase the renal excretion of some drugs (e.g. mexiletine) and reduce the effect of some antibacterial drugs in the genitourinary system (e.g. aminoglycosides).**T****U****DOSES****V****Mammals:** Rabbits: 50–100 mg/kg p.o., s.c. q12h; Guinea pigs: Maintenance: 10–30 mg/kg/day p.o., s.c. or 200–400 mg/l drinking water; Hypovitaminosis C: 100–200 mg/kg p.o. q24h; Primates: 25 mg/kg p.o. q12h or 30–100 mg/kg p.o., s.c. q24h^{1,2}; Hedgehogs: 50–200 mg/kg p.o., s.c. q24h.**W****Reptiles:** 10–20 mg/kg i.m. q7d.**X****Birds, Amphibians, Fish:** No information available.**Y****References****Z**¹ Eisele PH, Morgan JP, Line AS and Anderson JH (1992) Skeletal lesions and anemia associated with ascorbic acid deficiency in juvenile rhesus macaques. *Laboratory Animal Science* **42**(3), 245–249

- ² Ratterree MS, Didier PJ, Blanchard JL, Clarke MR and Schaeffer D (1990) Vitamin C deficiency in captive non-human primates fed a commercial primate diet. *Laboratory Animal Science* **40**(2), 165–168

Vitamin D (1,25-dihydroxycolecalfiferol (active vitamin D3), colecalfiferol (vitamin D3))

(Zolcal D, Alfacalcidol*, Calcijex*, Calcitriol*, One-alpha*, Rocaltrol*) **POM-VPS, POM, ESPA**

Formulations: Oral: Alfacalcidol 2 µg (micrograms)/ml solution (One-alpha), 0.25–1 µg capsules (Alfacalcidol; One-alpha), Calcitriol 0.25 µg capsules (Calcitriol; Rocaltrol*). **Injectable:** Calcitriol 1 µg/ml solution (Calcijex).

Action: Vitamin D is a general term used to describe a range of hormones that influence calcium and phosphorus metabolism. They include vitamin D2 (ergocalciferol or calciferol), vitamin D3 (colecalfiferol), dihydrotachysterol, alfacalcidol and calcitriol (1,25-dihydroxycolecalfiferol, the active form of vitamin D3). These different drugs have differing rates of onset and durations of action. In conjunction with other hormones (calcitonin and parathormone) regulates calcium homeostasis through numerous complex mechanisms, including accretion of calcium to bone stores, absorption of calcium from dietary sources.

Use: Chronic management of hypocalcaemia when associated with low parathyroid hormone concentrations which are most commonly associated with iatrogenic hypoparathyroidism following thyroidectomy and immune-mediated hypoparathyroidism. Calcitriol has also been used in the management of renal secondary hyperparathyroidism; in this circumstance it reduces serum parathyroid hormone concentrations. Dihydrotachysterol has an onset of action within 24 hours and raises serum calcium within 1–7 days, with a discontinuation time of 1–3 weeks for serum calcium levels to normalize. Calcitriol and alfacalcidol (1-alpha-hydroxycolecalfiferol) have a rapid onset of action (1–2 days) and a short half-life (<1 day); they are the preferred forms for use. Vitamin D requires two hydroxylations (one in the liver and the other in the kidney) to become active. Thus, only the active form (calcitriol) should be used in patients with renal failure. Vitamin D3 is used in birds and reptiles for the treatment of nutritional secondary hyperparathyroidism in combination with calcium and ultraviolet light. Vitamin D has a very narrow therapeutic index and toxic doses are easily achieved resulting in soft tissue calcification. Serum calcium and preferably ionized calcium concentrations need to be monitored closely and frequently. Avoid using formulations of vitamins A, D3 and E that are authorized for farm animals as they are too concentrated for small animal use.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use in patients with hyperphosphataemia or malabsorption syndromes. Do not use in pregnant animals.

Adverse reactions: Hypercalcaemia and hyperphosphataemia.

Drug interactions: Corticosteroids may negate the effect of vitamin D preparations. Sucralfate decreases absorption of vitamin D. Drugs that induce hepatic enzyme systems (e.g. barbiturates) will increase the metabolism of vitamin D and lower its effective dose. Magnesium- or

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calcium-containing antacids may cause hypermagnesaemia or hypercalcaemia when used with vitamin D. Thiazide diuretics may also cause hypercalcaemia with concurrent use. Hypercalcaemia may potentiate the toxic effects of verapamil or digoxin; monitor carefully.

DOSES

Mammals: Primates: 110 IU/100 g p.o. q24h (marmosets) or 2000 IU/kg.

Birds: Hypovitaminosis D: 3300–6600 IU/kg i.m. once (calcitriol)¹.

Reptiles: 200–1000 IU/kg p.o., i.m. q7d. Oral administration may be safer than i.m.

Amphibians: 2–3 IU/ml in a continuous bath with 2.3% calcium gluconate; 100–400 IU/kg p.o. q24h.

Fish: No information available.

References

- ¹ de Matos R (2008) Calcium metabolism in birds. *Veterinary Clinics of North America: Exotic Animal Practice* **11**(1), 59–82

Vitamin E (Alpha tocopheryl acetate)

(Multivitamin injection, Vitamin E suspension) **POM-VPS**

Formulations: Oral: 20 mg/ml, 100 mg/ml suspension.

Injectable: component of multivitamin injectable preparations.

Action: Lipid-soluble antioxidant also regulates gene expression and is involved in cellular metabolism of sulphur compounds.

Use: Vitamin E supplementation is very rarely required in small animals. Patients with exocrine pancreatic insufficiency and other severe malabsorptive diseases may be at risk of developing deficiency. Its use has been suggested for numerous conditions, including discoid lupus, demodicosis and hepatic diseases including fibrosis. These are, however, only anecdotal suggestions and there may be some significant risks. Avoid using formulations of vitamins A, D3 and E that are authorized for farm animals as they are too concentrated for small animal use.

Safety and handling: Normal precautions should be observed.

Contraindications: Do not use in patients at high risk for thrombosis. Do not use in neonates.

Adverse reactions: Thrombosis. Anaphylactoid reactions have been reported.

Drug interactions: Vitamin E may enhance vitamin A absorption, utilization and storage. Vitamin E may alter ciclosporin pharmacokinetics and, if used concurrently, ciclosporin therapy should be monitored by checking levels.

DOSES

Mammals: Primates: 3.75 IU/kg i.m. q3d with selenium at 1.15 mg/kg i.m. for 30 days (myopathy); **Marsupials:** 25–100 IU/animal i.m. q24h or 400 IU/animal i.m. post-capture restraint (macropods).

Birds: 0.06 mg/kg i.m. q7d in psittacids for hypovitaminosis E; once only in raptors to prevent/treat capture myopathy^a.

Reptiles: 1 IU/kg i.m. q24h for 7–14 days.

Amphibians: 1 mg/kg p.o., i.m. q7d or 200 IU/kg of feed.

Fish: No information available.

References

- ^a Schink B, Hafez HM and Lierz M (2008) Alpha-tocopherol in captive falcons: reference values and dietary impact. *Journal of Avian Medicine and Surgery* **22(2)**, 99–102

Vitamin K1 (Phytomenadione)

(Vitamin K1 Laboratoire TVM, Konaktion*) **POM-V, NFA-VPS**

Formulations: Injectable: 10 mg/ml. Oral: 50 mg tablets.

Action: Involved in the formation of active coagulation factors II, VII, IX and X by the liver.

Use: Toxicity due to coumarin and its derivatives. Before performing liver biopsy in patients with prolonged coagulation times. Deficient states may also occur in prolonged significant anorexia. Although vitamin K is a fat-soluble vitamin its biological behaviour is like that of a water-soluble vitamin; it has a relatively short half-life and there are no significant storage pools. It may still require 6–12 hours for effect. One-stage prothrombin time is the best method of monitoring therapy. Use a small gauge needle when injecting s.c. or i.m. in a patient with bleeding tendencies.

Safety and handling: Normal precautions should be observed.

Contraindications: Avoid giving i.v. if possible.

Adverse reactions: Anaphylactic reactions have been reported following i.v. administration. Safety not documented in pregnant animals.

Drug interactions: Many drugs will antagonize the effects of vitamin K, including aspirin, chloramphenicol, allopurinol, diazoxide, cimetidine, metronidazole, erythromycin, itraconazole, propranolol and thyroid drugs as well as coumarin-based anticoagulants. If the patient is on other long-term medications it is advisable to check specific literature. The absorption of oral vitamin K is reduced by mineral oil.

DOSES**Mammals:**

- Ferrets:
 - Known 1st generation coumarin toxicity or vitamin K1 deficiency: Initially 2.5 mg/kg s.c. in several sites, then 1–2.5 mg/kg in divided doses p.o. q8–12h for 5–7 days.
 - Known 2nd generation coumarin (brodifacoum) toxicity: Initially 5 mg/kg s.c. in several sites, then 2.5 mg/kg p.o. q12h for 3 weeks, then re-evaluate coagulation status. The patient's activity should be restricted for 1 week following treatment. Evaluate the coagulation status 3 weeks after cessation of treatment.
 - Known inandione (diphacinone) or unknown anticoagulant toxicity: Initially 2.5–5 mg/kg s.c. over several sites, then 2.5 mg/kg p.o. divided q8–12h for 3–4 weeks. Re-evaluate coagulation status 2 days after stopping therapy. If the PT time is elevated, continue therapy for 2 additional weeks. If not elevated, repeat PT time in 2 days. If normal, the animal should be rested for 1 week, if abnormal then continue therapy for an additional week and re-check PT times as above.

A**B****C****D****E****F****G****H****I****J****K****L****M****N****O****P****Q****R****S****T****U****V****W****X****Y****Z**

- **Rabbits, Rodents:** 1–10 mg/kg i.m. as needed depending upon clinical signs/clotting times.
- **Primates:** 1–5 mg/animal i.m. q24h.
- **Sugar gliders:** 2 mg/kg s.c. q24–72h.

Birds: 0.2–2.5 mg/kg i.m., p.o. q6–12h until stable, then q24h¹.

Reptiles: 0.5 mg/kg i.m. q24h.

Amphibians, Fish: No information available.

References

- ¹ Murray M and Tseng F (2008) Diagnosis and treatment of secondary anticoagulant rodenticide toxicosis in a red-tailed hawk (*Buteo jamaicensis*). *Journal of Avian Medicine and Surgery* **22**(1), 41–46

Voriconazole

(Vfend*, Voriconazole*) **POM**

Formulations: Oral: 50 mg, 200 mg tablets; 40 mg/ml oral suspension. Injectable: 200 mg powder for reconstitution.

Action: Triazole antifungal agent that inhibits the cytochrome systems involved in the synthesis of ergosterol in fungal cell membranes, causing increased cell wall permeability and allowing leakage of cellular contents.

Use: Treatment of aspergillosis, candidiasis, blastomycosis, coccidioidomycosis, cryptococcosis, sporotrichosis, histoplasmosis, a variety of dermatomycoses and *Malassezia*. It is widely distributed in the body, although low concentrations are found in tissues with low protein contents (e.g. CSF, ocular fluid and saliva). Treatment of chytridiomycosis in amphibians. Reduced drug costs when using generic forms have made this more accessible for veterinary use. First choice agent in Grey parrots for aspergillosis. In acute disease or when spinal forms of aspergillosis are suspected, the intramuscular route should be utilised initially.

Safety and handling: Normal precautions should be observed.

Contraindications: Pregnancy. Avoid use if liver disease is present.

Adverse reactions: Vomiting, diarrhoea, anorexia, salivation, depression and apathy, abdominal pain, hepatic toxicosis, drug eruption, ulcerative dermatitis and limb oedema could potentially occur. Macaws and some raptor species (eg old world vultures) also appear sensitive to side effects. Neurological signs and death within 48 hours of injection have been reported in certain snake species (cottonmouths and Dumeril's ground boas) so do not use in these species.

Drug interactions: In humans, antifungal imidazoles and triazoles inhibit the metabolism of antihistamines (particularly terfenadine), oral hypoglycaemics, and antiepileptics and glucocorticoids. Plasma concentrations of ciclosporin, benzodiazepines and vincristine may be increased by voriconazole.

DOSES

Mammals: Rabbits: 1% solution applied topically for fungal keratitis.

Birds: Raptors: 12.5 mg/kg p.o. q12h (some recommend 12.5 mg/kg q12h, then q24h after 3 days)^a; 12.5 mg/kg i.m. q24h; **Grey parrots:** 12–18 mg/kg p.o. q12h^b; **Amazon parrots:** 18 mg/kg p.o. q8h^c; Similar dose rates may be used via the intramuscular or intravenous route in

urgent cases or in spinal cases^d. Macaws appear more sensitive to side effects – it is worth starting doses at 10 mg/kg p.o. q12h and reducing if vomiting, biliverdinuria or anorexia seen.

Reptiles: Bearded dragons: 10 mg/kg p.o. q24h^e. Do not use in cottonmouth snakes or Dumeril's ground boas.

Amphibians: Poison dart frogs: 1.25 µg (micrograms)/ml q24h topically via spray for 7 days^f.

Fish: No information available.

References

- ^a Di Somma A, Bailey T, Silvanose C and Garcia-Martinez C (2007) The use of voriconazole for the treatment of aspergillosis in falcons (*Falco* species). *Journal of Avian Medicine and Surgery* **21**(4), 307–317
- ^b Flammer K, Nettifee Osborne JA, Webb DJ et al. (2008) Pharmacokinetics of voriconazole after oral administration of single and multiple doses in African grey parrots (*Psittacus erithacus timneh*). *American Journal of Veterinary Research* **69**(1), 114–121
- ^c Sanchez-Migallon Guzman D, Flammer K, Papich MG et al. (2010) Pharmacokinetics of voriconazole after oral administration of single and multiple doses in Hispaniolan Amazon parrots (*Amazona ventralis*). *American Journal of Veterinary Research* **71**(4), 460–467
- ^d Azmanis P, Pappalardo L, Sara ZA, Silvanose C, and Naidoo V (2020) Pharmacokinetics of voriconazole after a single intramuscular injection in large falcons (*Falco* spp.). *Medical Mycology* **58**(5), 661–666
- ^e Van Waeyenberghe L, Baert K, Pasmans F et al. (2010) Voriconazole, a safe alternative for treating infections caused by the *Chrysosporium* anamorph of *Nannizziopsis vriesii* in bearded dragons (*Pogona vitticeps*). *Medical Mycology* **48**(6), 880–885
- ^f Martel A, Van Rooij P, Vercauteren G et al. (2011) Developing a safe antifungal treatment protocol to eliminate *Batrachochytrium dendrobatidis* from amphibians. *Medical Mycology* **49**(2), 143–149

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Xylazine

B (Chanazine, Nerfasin, Rompun, Sedaxylan, Virbaxyl, Xylacare, Xylapan) **POM-V****C** **Formulations:** Injectable: 20 mg/ml solution.**D** **Action:** Agonist at peripheral and central alpha-2 adrenoreceptors, producing dose-dependent sedation, muscle relaxation and analgesia.**E** **Use:** Has been almost totally superseded by medetomidine or dexmedetomidine in smaller patients and is no longer recommended. **F** Used to provide sedation and premedication when used alone or in combination with opioid analgesics. Xylazine combined with **G** ketamine is used to provide a short duration (20–30 minutes) of surgical anaesthesia. Xylazine is less specific for the alpha-2 adrenoreceptor than **H** are medetomidine and dexmedetomidine and causes significant alpha-1 adrenoreceptor effects. This lack of specificity is likely to be associated with the poorer safety profile of xylazine compared with medetomidine **I** and dexmedetomidine. Xylazine also sensitizes the myocardium to catecholamine arrhythmias, which increases the risk of cardiovascular complications. Xylazine is a potent drug that causes marked changes **J** in the cardiovascular system. It should not be used in animals with cardiovascular or systemic disease affecting cardiovascular **K** performance. Atipamezole is not authorized as a reversal agent for xylazine, but it is effective and can be used to reverse the effects of **L** xylazine if an overdose is given. Spontaneous arousal from deep sedation following stimulation can occur with all alpha-2 agonists; **M** aggressive animals sedated with xylazine must still be managed with caution. Xylazine stimulates growth hormone production and may be **N** used to assess the pituitary gland's ability to produce this hormone (xylazine stimulation test). **O****P** **Safety and handling:** Normal precautions should be observed.**Q** **Contraindications:** Do not use in animals with cardiovascular or other systemic disease. Use of xylazine in geriatric patients is also not **R** advisable. It causes increased uterine motility and should not be used in pregnant animals, nor in animals likely to require or receiving **S** sympathomimetic amines. Due to effects on blood glucose, use in diabetic animals is not recommended. Avoid when vomiting is contraindicated (e.g. foreign body, raised intraocular pressure).**T** **Adverse reactions:** Xylazine has diverse effects on many organ systems as well as the cardiovascular system. It causes a diuresis by **U** suppressing ADH secretion, a transient increase in blood glucose by decreasing endogenous insulin secretion, mydriasis and decreased **V** intraocular pressure.**W** **Drug interactions:** When used for premedication, xylazine will significantly reduce the dose of all other anaesthetic agents required to maintain anaesthesia.**X**

DOSES

Y **When used for sedation is generally given as part of a combination. See Appendix for sedation protocols in all species.****Z** **Mammals:** Primates: 0.5–6 mg/kg i.m. for sedation or 0.5–1 mg/kg in combination with ketamine for anaesthesia; **Sugar gliders:** 5 mg/kg in combination with ketamine; **Hedgehogs:** 0.5–1.0 mg/kg i.m.

Birds, Reptiles: No longer recommended as superseded by medetomidine and dexmedetomidine.

Amphibians: 10 mg/kg intracoelomic q12–24h appears to have an analgesic effect in laboratory studies in *Xenopus* frogs^a.

Fish: No information available.

References

- ^a Terril-Robb L, Suckow M and Grigdesby C (1996) Evaluation of the analgesic effects of butorphanol tartarate, xylazine hydrochloride and flunixin meglumine in leopard frogs (*Rana pipiens*). *Contemporary Topics in Laboratory Animal Science* **35**, 54–56

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Zinc salts(Numerous trade names) **GSL****Formulations:** Oral: various zinc sulphate, zinc gluconate, zinc acetate and chelated zinc preparations.**Action:** Primarily involved in DNA and RNA synthesis, although also involved in essential fatty acid synthesis, WBC function and numerous reactions in intermediary metabolism. When administered orally can reduce GI absorption and hepatic uptake of copper.**Use:** Zinc-responsive dermatoses. Proposed benefits also exist in chronic liver disease and hepatic encephalopathy. Bioavailability of elemental zinc varies depending on formulation: zinc acetate and chelated forms: highest; gluconate: intermediate; sulphate: lowest. Higher bioavailability is also associated with improved tolerance.**Safety and handling:** Normal precautions should be observed.**Contraindications:** Patients with copper deficiency.**Adverse reactions:** Nausea, vomiting and occasional diarrhoea. Haemolysis may occur with large doses or serum levels >10 mg/ml particularly if a coexistent copper deficiency exists.**Drug interactions:** Significant interactions with other divalent heavy metals such as iron and copper can occur and long-term administration of zinc may lead to decreased hepatic copper or iron stores and functional deficiency. Penicillamine and ursodeoxycholic acid may potentially inhibit zinc absorption; the clinical significance is unclear. Zinc salts may chelate oral tetracycline and reduce its absorption; separate doses by at least 2 hours. Zinc salts may reduce the absorption of fluoroquinolone antibiotics.**DOSES****Mammals:** Primates: 2.5 µg (micrograms)/animal p.o. q24h for 3 days.**Birds, Reptiles, Amphibians, Fish:** No information available.

Appendix I: general information

Abbreviations

In general abbreviations should not be used in prescription writing. However, it is recognized that at present some Latin abbreviations are used when prescribing. These should be limited to those listed here.

Abbreviations used in prescription writing

a.c.	Before meals
ad. lib.	At pleasure
amp.	Ampoule
b.i.d.	Twice a day
cap.	Capsule
g	Gram
h	Hour(s)
i.c.	Intracoelomic
i.m.	Intramuscular
i.p.	Intraperitoneal
i.v.	Intravenous
m ²	Square metre
mg	Milligram
ml	Millilitre
o.m.	In the morning
o.n.	At night
p.c.	After meals
p.o.	By mouth, orally
prn	As required
q	Every, e.g. q8h = every 8 hours
q.i.d./q.d.s	Four times a day
q.s.	A sufficient quantity
s.c.	Subcutaneous
s.i.d.	Once a day
Sig:	Directions/label
stat.	Immediately
susp.	Suspension
tab.	Tablet
t.i.d./t.d.s.	Three times a day

Other abbreviations used in this Formulary

ACE	Angiotensin converting enzyme
ACTH	Adrenocorticotrophic hormone
ADH	Antidiuretic hormone
AV	Atrioventricular
CBC	Complete blood count
CHF	Congestive heart failure
CNS	Central nervous system
COX	Cyclo-oxygenase
CRI	Continuous rate infusion
CSF	Cerebrospinal fluid
d	Day(s)
DIC	Disseminated intravascular coagulation
DNA	Deoxyribonucleic acid
ECG	Electrocardiogram
EMA	European Medicines Agency
GI	Gastrointestinal
Hb	Haemoglobin
MAOI	Monoamine oxidase inhibitor
MIC	Minimum inhibitory concentration
min	Minute
NSAID	Non-steroidal anti-inflammatory drug
PT	Prothrombin
PU/PD	Polyuria/polydipsia
RBC	Red blood cell
SSRI	Selective serotonin reuptake inhibitor
STC	Special Treatment Certificate
WBC	White blood cell
wk	Week(s)

Writing a prescription

A 'veterinary prescription' is defined by EU law as 'any prescription for a veterinary medicinal product issued by a professional person qualified to do so in accordance with applicable national law'. The word 'veterinary' takes its normal meaning 'of or for animals'. In the UK there are two classes of medicines available only on veterinary prescription, POM-V and POM-VPS, described in the **Introduction**. Only in the case of POM-V medicines does the veterinary prescription have to be issued by a veterinary surgeon. The act of prescribing is taken to mean the decision made by the prescriber as to which product should be supplied, taking into account the circumstances of the animals being treated, the available authorized veterinary medicinal products and the need for responsible use of medicines.

A written prescription for a veterinary medicine must include the following and be printed or written legibly in ink or otherwise so as to be indelible:

- Name, address, and telephone number of the person prescribing the product
- Qualifications of the person prescribing the product (it is good practice to include their RCVS or SQP number)
- Name and address of the owner or keeper of the animal
- Identification (including the species) of the animal or group of animals to be treated
- Premises at which the animal(s) are kept, if this is different from the address of the owner or keeper
- Date
- Signature (or other authentication) of the person prescribing the product - sign in ink with your normal signature
- Name and amount of the product prescribed
 - Use product or approved generic name for drugs in capital letters – do not abbreviate. Ensure the full name is stated, to include the pharmaceutical form and strength
 - Write out microgram/nanogram – do not abbreviate
 - Always put a 0 before an initial decimal point (e.g. 0.5 mg), but avoid the unnecessary use of a decimal point (e.g. 3 mg not 3.0 mg)
 - State duration of treatment where known and the total quantity to be supplied
 - Give precise instructions concerning route/dose/formulation. Directions should preferably be in English without abbreviation. It is recognized that some Latin abbreviations are used (see p. 289)
- Necessary warnings
- Withdrawal period, if relevant
- The directions that the prescriber wishes to appear on the labelled product. It is good practice to include the words 'For animal treatment only'
- A declaration that 'This prescription is for an animal under my care' or words to that effect
- If it is prescribed under the cascade, a statement to that effect. If drugs that are not authorized for veterinary use are going to be used when there is an alternative that is 'higher' in the prescribing cascade, there should be a clear clinical justification made on an individual basis and recorded in the clinical notes or on the prescription.

Written prescriptions for POM-V or POM-VPS medicines:

- Are valid for 6 months unless the prescriber states a shorter period
- May only be used once unless the prescriber specifies that it is repeatable
- If repeatable, the number of repeat supplies that may be made must be specified (if the prescription is not repeatable, it is considered good practice for this to be stated).

In addition to the general prescription requirements above, a written prescription for a Schedule 2 or 3 Controlled Drug (CD) should state an exact dose in words as well as in figures (e.g. not 'as directed'), and it must include the RCVS number of the veterinary surgeon prescribing the drug. A written prescription for Schedule 2 or 3 CDs can only be dispensed once and only within 28 days. Single prescriptions with multiple dispenses (repeatable prescriptions) are not allowed for Schedule 2 and 3 CDs. It is good practice to mark the prescription 'no repeats'.

For all prescriptions, any alterations invalidate the prescription – it must be rewritten.

The following is a standard form of prescription used:

From: <i>Address of practice</i>	<i>Date</i>
<i>Telephone No.</i>	
<i>Animal's name and identification (species, breed, age and sex)</i>	<i>Owner's name Owner's address</i>
	<i>Premises where animal(s) are kept if different from above</i>
Rx • <i>Print name, strength and formulation of drug</i> • <i>Total quantity to be supplied</i> • <i>Amount to be administered</i> • <i>Frequency of administration</i> • <i>Duration of treatment</i> • <i>Any warnings</i> • <i>If not a POM-V and prescribed under the 'Cascade', this must be stated</i> • <i>For animal treatment only</i> • <i>For an animal under my care</i> • <i>Withdrawal period, if relevant</i> Non-repeat/repeat X 1, 2 or 3 Name, qualifications, RCVS number and signature of veterinary surgeon OR SQP number and signature	

Dosing small and exotic animals

Veterinary surgeons who are unfamiliar with the actions of a particular drug in a given exotic species are advised to consult the drug monograph and more complete references.

The size of some species of exotic pets makes dosing difficult and care must be taken when calculating small doses. Some points to bear in mind when dosing are:

- Where powders are to be dissolved in water, sterile water for injection should be used
- Most solutions may be diluted with water for injection or 0.9% saline
- Dilution will be necessary when volumes <0.1 ml are to be administered
- Suspensions cannot be diluted
- Use 1 ml syringes for greatest accuracy
- Specialist laboratories with a Veterinary Specials Authorization should be contacted to reformulate drugs.

Doses provided in this Formulary are intended for exotic pet species. They should not be used for commercial production species of fish or poultry, or large mammals. Established poultry texts should be consulted for treating individual chickens.

Composition of intravenous fluids

Fluid	Na ⁺ (mmol/l)	K ⁺ (mmol/l)	Ca ²⁺ (mmol/l)	Cl ⁻ (mmol/l)	HCO ₃ ⁻ (mmol/l)	Dext. (g/l)	Osmol. (mosl/l)
0.45% NaCl	77			77			155
0.9% NaCl	154			154			308
5% NaCl	856			856			1722
Ringer's	147	4	2	155			310
Lactated Ringer's (Hartmann's)	131	5	2	111	29 *		280
Darrow's	121	35		103	53 *		312
0.9% NaCl + 5.5% Dext.	154			154		50	560
0.18% NaCl + 4% Dext.	31			31		40	264
Duphalyte **		2.6	1.0	3.6		454	Unknown

Dext. = Dextrose; Osmol. = Osmolality. * Bicarbonate is present as lactate.

** Also contains a mixture of vitamins and small quantities of amino acids and 1.2 mmol/l of MgSO₄.

Percentage solutions

The concentration of a solution may be expressed on the basis of weight per unit volume (w/v) or volume per unit volume (v/v).

% w/v = number of grams of a substance in 100 ml of a liquid

% v/v = number of ml of a substance in 100 ml of liquid

% Solution	g or ml/100 ml	mg/ml	Solution strength
100	100	1000	1:1
10	10	100	1:10
1	1	10	1:100
0.1	0.1	1	1:1000
0.01	0.01	0.1	1:10,000

Guidelines for responsible antibiotic use

While antibiotics are essential medicines for treating bacterial infections, it is important to remember that their use leads to selection of resistant strains of bacteria. Resistance may be intrinsic (chromosomal DNA changes) or acquired (by plasmid transfer).

It is important that the veterinary profession uses antibiotics responsibly in order to:

- Minimize the selection of resistant veterinary pathogens (and therefore safeguard animal health)
- Minimize possible resistance transfer to human pathogens
- Retain the right to prescribe certain antibiotics.

Following these guidelines will help to maximize the therapeutic success of antibiotic agents while at the same time minimizing the development of antibiotic resistance, thereby safeguarding antimicrobials for future veterinary and human use. These guidelines should be read in conjunction with the individual drug monographs, updated *BSAVA Guide to the Use of Veterinary Medicines* and the **PROTECT ME** guidance (www.bsavalibrary.com).

Following the **PROTECT ME** guidance can help reduce resistance:

Prescribe only when necessary

Consider non-bacterial disease (e.g. viral infection), nutritional imbalance or metabolic disorders where antibiotic therapy would be redundant. Remember also that some bacterial disease will self-resolve without antibiotics. Offer a non-prescription form to support a decision not to prescribe antibiotic therapy.

Reduce prophylaxis

Antibiotics are not a substitute for surgical asepsis and the need for prophylactic antibiotics in surgery should be carefully considered. Prophylactic antibiotics are only appropriate in a few medical cases (e.g. immunocompromised patients).

Offer other options

Consider therapeutic alternatives (e.g. lavage and debridement of infected material, analgesia, cough suppressants, fluid therapy or

nutritional modification). Use topical preparations as these reduce the selection pressure on resident intestinal flora (the microbiome). Use effective hygiene techniques and antiseptics to prevent infections.

Treat effectively

Before prescribing antibiotics, consider which bacteria are likely to be involved and how effectively the chosen drug will penetrate the target site. Use the shortest effective course and avoid underdosing. Ensure compliance with appropriate formulation and clear instructions.

Employ narrow spectrum

Unnecessarily broad-spectrum antibiotics could promote antibiotic resistance; selecting narrow-spectrum antibiotics limits the effects on commensal bacteria. Use culture results to support de-escalation (switching to a narrower spectrum antibiotic) whenever possible.

Culture appropriately

A sample for culture should be collected **before** starting antibiotic therapy. Culture is essential when prolonged (>1 week) treatment courses are anticipated, when resistance is likely (e.g. hospital-acquired infections) and with life-threatening infections. If first-line treatment fails, do not use another antibiotic without culture and susceptibility results (**avoid cycling antibiotics**).

Tailor practice policy to patients

A customized practice policy can guide antibiotic selection to address the bacterial infections and resistance patterns that you encounter, minimizing inappropriate use.

Monitor culture results

Track and record culture profiles and update your practice policy accordingly. Monitor for preventable infections (e.g. postoperative) and alter practices if needed. Audit your own antibiotic use, particularly of critically important antibiotics (e.g. fluoroquinolones and ceftiofur).

Educate others

Share this important message to reduce the threat from multi-resistant strains of bacteria and improve the health of pets and people.

European Medicines Agency categorization of antibiotics

In 2019, the EMA launched a public consultation on its updated scientific advice on the categorization of antibiotics. The scientific advice ranks antibiotics by considering both the risk that their use in animals causes to public health through the possible development of antimicrobial resistance and the need to use them in veterinary medicine. It now addresses all classes of antibiotics, including those classified as critically important antimicrobials for human health by the World Health Organization (**see Useful websites**).

Readers are encouraged to check and consider the EMA's updated scientific advice on the categorization of antibiotics when prescribing these medicines for animals in their care. The categorization can also be used as a tool for the preparation of treatment guidelines (**see Useful websites**). The classification comprises four categories:

- **Category A (Avoid)** includes antibiotics that are currently not authorized in veterinary medicine in the European Union (EU). These

medicines may not be used in food-producing animals and may be given to individual companion animals only under exceptional circumstances

- **Category B (Restrict)** refers to quinolones, 3rd- and 4th-generation cephalosporins and polymyxins. Antibiotics in this category are critically important in human medicine and their use in animals should be restricted to mitigate the risk to public health
- **Category C (Caution)** covers antibiotics for which alternatives in human medicine generally exist in the EU, but only few alternatives are available in certain veterinary indications. These antibiotics should only be used when there are no antibiotic substances in Category D that would be clinically effective
- **Category D (Prudence)** includes antibiotics that should be used as first line treatments, whenever possible. These antibiotics can be used in animals in a prudent manner. This means that unnecessary use and long treatment periods should be avoided, and group treatment should be restricted to situations where individual treatment is not feasible.

Antibiotics in small mammals

Antibiotic therapy in several small mammal species poses a greater risk when compared with other species due to the suppression of normal bacterial flora, resulting in overgrowth of other species, notably *Clostridium*, resulting in enterotoxaemia and death. Mice, rats, ferrets and usually gerbils are fairly resistant, whereas hamsters, guinea pigs, chinchillas, degus and rabbits are more susceptible. The risk of enterotoxaemia is related to several factors, including the drug selected, the dose, the route of administration and the animal's nutritional status and general health. Antibiotics that have been associated with this problem when given orally include penicillins, lincosamides, cephalosporins and erythromycin. Some species appear more sensitive to certain drugs than others, while some medications appear well-tolerated by alternative routes (e.g. injectable benzylpenicillin in rabbits). See individual monographs for more details.

References

Allerton F and Nuttall T (2021) Antimicrobial use: importance of bacterial culture and susceptibility testing. *In Practice* **43**, 500–510
Sykes J and Greene J (2011) *Infectious Diseases of the Dog and Cat*, 4th edn. Saunders, Philadelphia

Antimicrobial susceptibility

		Gram positive				
		Cocci				
		MRSA/MRSP	Staphylococcus	Enterococcus	Streptococcus	
Aminoglycosides	Amikacin	●	●	●	●	
	Gentamicin	●	●	●	●	
Beta-lactams	Amoxicillin	●	●	●	●	
	Ampicillin	●	●	●	●	
	Amoxicillin-clavulanate	●	●	●	●	
Cephalosporins	Cefalexin	●	●	●	●	
	Cefotaxime (3rd gen)	●	●	●	●	
	Cefovecin (3rd gen)	●	●	●	●	
	Ceftazidime (3rd gen)	●	●	●	●	
	Ceftiofur (3rd gen)	●	●	●	●	
	Cefuroxime (2nd gen)	●	●	●	●	
Fluoroquinolones	Enrofloxacin	●	●	●	●	
	Marbofloxacin	●	●	●	●	
	Pradofloxacin	●	●	●	●	
Lincosamides	Clindamycin	●	●	●	●	
	Lincomycin	●	●	●	●	
Macrolides	Azithromycin	●	●	●	●	
	Erythromycin	●	●	●	●	
Nitroimidazoles	Metronidazole	●	●	●	●	
Sulphonamides	Trimethoprim sulfamethoxazole	●	●	●	●	
Tetracyclines	Doxycycline	●	●	●	●	
	Oxytetracycline	●	●	●	●	

Antimicrobial susceptibility chart listing the most likely susceptibility and resistance patterns to commonly available antibiotics. This chart is a summary and does not predict the susceptibility or resistance of an individual isolate; each case must be verified using up-to-date literature published by drug manufacturers. * = Extended-spectrum beta-lactamases (ESBLs) are normally inhibited by clavulanic acid; however, AmpC is resistant to clavulanic acid.

[illegible]

Reproduced from Allerton and Nuttall (2021), with permission from the publisher. Data from this Formulary, Sykes and Greene (2011), and published Veterinary Committee on Antimicrobial Susceptibility Testing and Clinical and Laboratory Standards Institute breakpoints.

- Susceptible
- Caution with interpretation and dosing
- Intrinsic resistance
- Acquired resistance

Guidelines for responsible parasiticide use

Parasiticides are essential medicines for the treatment of animal ectoparasites (e.g. ticks, fleas, mites) and endoparasites (e.g. tapeworms, roundworms, protozoa), and for prophylaxis both in the UK and in travelling pets. These parasites can have direct animal health and welfare impacts, indirect impacts through the transmission of disease and pose zoonotic risks. In some cases, parasiticide treatment is legally mandated (e.g. tapeworm treatment of travelling pets).

Overuse of parasiticides in farm animal and equine practice has led to widespread resistance, particularly within endoparasites. Whilst resistance is not currently a significant concern in UK small animal practice, overuse of these agents may lead to increased prevalence of resistant organisms and reduced efficacy of parasiticide treatments.

Parasiticides are harmful to a wide range of invertebrates. In addition, there is increasing recognition of environmental contamination from products used to treat companion animals, with potentially significant detrimental impact on wildlife ecosystems. Significant declines in terrestrial, including bee, and aquatic invertebrate populations are likely worsened by widespread environmental pesticide contamination, including from companion animal parasiticides. Environmental contamination can occur from topical products (such as seen with bathing after treatment), urine and faecal excretion, wastewater from homes and from incorrect disposal of products. Although significant uncertainty surrounds the proportional impact of companion animal parasiticides, veterinary surgeons should be mindful of these concerns and reduce use where possible.

Recommendations for responsible parasiticide use include:

- Prescribing parasiticide treatments based on risk rather than providing blanket treatment. This should take account of health risks (animal, human and environmental), lifestyle and environmental factors, season, geographical prevalence and the results of parasite testing (e.g. faecal examination)
- Ensuring correct dosing and application, as well as avoiding spillage
- Disposing of unused products, packaging and faeces from treated animals responsibly
- Avoiding bathing or swimming after topical application. Avoiding topical products in animals likely to contaminate water
- Treating parasites as needed for the individual animal, including using narrow spectrum products where appropriate
- Encouraging targeted treatment through laboratory testing (e.g. faecal examination) and owner recognition of parasites (e.g. via regular examination for fleas, and for ticks after walks in high prevalence areas). Early treatment of parasites with a significant domestic environmental lifecycle (e.g. fleas) is likely to result in the use of less parasiticide than if severe infestations are allowed to develop
- Considering non-pharmacological interventions to reduce treatment requirements (such as vacuuming and cleaning of bedding, regular bathing of untreated animals, removing faeces from the environment, and avoiding scavenging, hunting or raw feeding)
- Educating owners about parasites and responsible parasiticide use to ensure adherence to guidance and early recognition of parasites.

Safety and handling of chemotherapeutic agents

Chemotherapy agents are genotoxic, mutagenic and teratogenic therefore, unsurprisingly, their use is considered an occupational health hazard. The use of these agents appears to be increasing due to demand from clients wishing to treat cancer in their pets, therefore, the number of events carrying an exposure risk is on the rise. In addition to the increased use of 'conventional' chemotherapy, innovative dosing strategies such as metronomic chemotherapy reduce the direct control that practices can exert over safe use (as clients are the primary administrators in this context). There is also increasing availability of novel agents such as small molecular c-kit inhibitors, which are not chemotherapeutic in the conventional sense but may carry similar risks especially in relation to pregnancy and should, therefore, be handled as cytotoxic drugs. Considering these risks together, great care must be taken to provide appropriate education to staff and clients in order that exposure to these drugs can be avoided.

Appropriate indication for the use of chemotherapy

Given the potential risks associated with chemotherapy exposure, both to the patient and those involved in preparing and handling chemotherapy agents, they should only be prescribed when absolutely indicated (i.e. for histologically-confirmed diseases when the likelihood of response is considered high). Investigational use should be confined to controlled clinical trials.

Routes to exposure

Exposure can occur during the preparation, handling and administration of the agents, as a result of cleaning spills, and through contact with bodily fluids or excreta of patients treated with chemotherapy. The environment in which the patient is treated and housed after chemotherapy is at risk of contamination (both within the hospital and in the home). Therefore, practice staff, owners and subsequent users of administration spaces are all at risk of exposure in the event of contamination. Exposure may occur via direct skin contact, inhalation of aerosolized drug particles, ingestion or needle stick injuries.

Approaches to mitigate the risk of exposure

Practices should assess the suitability of their environment and staffing (both level and training) for chemotherapy administration. In some cases, the facilities or staffing levels or training may be unsuitable. In these instances, patients should be referred. Similarly, veterinary surgeons should consider the suitability of patients and owners prior to starting treatment. This assessment is particularly relevant for oral medications as administration is not under the direct control of the veterinary surgeon and owners may become too relaxed about safety precautions.

The following guidance is provided to help minimize the risks associated with chemotherapy use.

Personnel and owners:

- The preparation and administration of cytotoxic drugs should only be undertaken by trained staff.
- Practices should develop standard operating procedures (SOPs) for preparing, handling and administering chemotherapeutic agents, for managing potentially contaminated spaces after the administration of chemotherapy or following spills, and for housing treated

patients. Staff should be trained in these SOPs with regular refreshers sessions held. Records of staff training should be kept.

- Owners and staff (including cleaners, animal carers and veterinary professionals) involved in the care of animals being treated with cytotoxic drugs must be informed of:
 - The risks of working with cytotoxic agents (to them as well as the patient)
 - The potential methods for preventing aerosol formation and the spread of contamination
 - The proper working practices for a safety cabinet (if available)
 - The instructions to clean contaminated spaces and spills
 - The principles of good personal protection and hygiene practice.
- Pregnant and immunocompromised personnel should not be involved in the process of preparing and/or administering cytotoxic agents, caring for animals that have been treated with cytotoxic drugs or cleaning the areas with which these animals have come into contact. Employees should be made aware that the risk to a fetus may be highest in the first trimester of pregnancy; consequently, employees planning to conceive and/or are in the early stages of pregnancy should be assigned to other duties. It is the responsibility of the employee to warn their employers if they are pregnant, likely to become pregnant or are immunocompromised.

Equipment and facilities:

- All areas where cytotoxic agents are prepared and/or administered, or where animals who have received cytotoxic drugs are being cared for, should be identified by a clear warning sign.
- Access to these areas should be restricted, ideally at all times, but at a minimum during chemotherapy preparation and administration.
- No rest activities or food or drink consumption should be permitted in the preparation, administration or post treatment kennelling areas at any time.
- There must be adequate materials for cleaning spilled cytotoxic agents (cytotoxic spill kit), clearly labelled and with associated use instructions, within the preparation and administration area(s).
- Ideally, a negative pressure pharmaceutical isolator with externally ducted exhaust filters, which has been properly serviced and checked, should be used. If such an isolator is not available, then a suitably modified Class 2B Biological Safety Cabinet (BSC) may be used.
- Regardless of whether a BSC is available, use of a needle-free closed or semi-closed system is mandatory to prevent aerosol formation, reduce the risk of stick injury and control exposure to carcinogenic compounds. Available systems include (but are not limited to): ChemoClave/Chemolock, Equashield, Onguard, PhaSeal and Spiros.

Preparation of cytotoxic drugs:

- Printed copies of preparation SOPs and drug data sheets should be available in the room in which the agents are to be prepared.
- Where available, the ventilated cabinet should be used for all chemotherapy preparation.
- If a BSC is not available, veterinary surgeons should consider acquiring pre-prepared patient specific doses of chemotherapy from a commercial company.
- During preparation and administration, disposable chemoprotective personal protective equipment (PPE) should be worn by all involved

staff. The PPE should include:

- A chemotherapy impermeable full-length gown. The gown should have elasticated cuffs
- Two pairs of chemotherapy impermeable (often nitrile) gloves. One seated underneath the cuff and one covering it entirely
- Eye protection in the form of goggles or a face shield
- Fitted respiratory protection is ideal and should be considered mandatory if chemotherapy is prepared outside of a BSC.
- The dosage of chemotherapy should be calculated, and the calculation checked by at least one colleague.
- All essential supplies should be kept inside the BSC. Absorbent padding should be placed in the BSC under the chemotherapy draw area.
- All administration consumables (tubing, fluid bags) should be prepared and primed outside the BSC prior to adding the chemotherapeutic agents (which should be done within the BSC). Heparinized saline should be avoided as it can interact with some chemotherapeutic drugs (e.g. doxorubicin/epirubicin).
- Chemotherapy injection spikes rather than needles should be used for withdrawing liquid chemotherapy.
- Manual handling of oral or topical medicines containing cytotoxic drugs should be avoided. Oral chemotherapy drugs should never be crushed or split. Several companies (e.g. BOVA, Chemopet, NOVA) supply compounded chemotherapy agents in various sizes suitable for different patients. This may be useful for drugs such as piroxicam, hydroxycarbamide, cyclophosphamide and lomustine. If reformulation is not possible, then adjusting the dosage regimen is often sufficient.
- When drug preparation is complete, the final product should be sealed in a plastic bag or other container for transport before it is taken out of the BSC. It should be clearly labelled with the drug name, patient name and dose.
- All potentially contaminated materials should be discarded in special cytotoxic waste disposal containers, which can be opened without direct contact with hands/gloves (e.g. a foot pedal). Local regulations for the disposal of this waste should be followed.
- The chemotherapy preparation space should be decontaminated and disinfected, and the outer pair of gloves discarded.
- Once drug preparation is complete, the inner pair of gloves can be discarded.
- There should be a clear procedure regarding how to handle cytotoxic drugs following an injection accident.

Administration of cytotoxic drugs:

- Prior to prescribing chemotherapy, the veterinary surgeon should make an assessment of the suitability of the patient to receive treatment (and be hospitalized should there be a significant adverse event).
- Drug administration should take place in a quiet and calm environment with restricted entry during the process. The external side of the door should be labelled to warn other staff. Personnel administering chemotherapy should not be disturbed.
- At the time of administration, the veterinary surgeon should again assess whether the patient is adequately calm and cooperative for safe administration. Anxious patients may benefit from the administration of mild anxiolytics prior to arrival at the clinic (e.g.

trazodone, gabapentin). Fractious or exuberant animals should be sedated (with the owner's consent).

- Two suitably trained members of staff should be present with the patient throughout the administration process. This is especially important when using vesicant drugs such as vinca alkaloids (vincristine, vinblastine) or anthracyclines (doxorubicin/epirubicin).
- A pre-administration checklist should be developed to confirm:
 - The correct patient identity
 - The blood results have been checked and are adequate for chemotherapy
 - The correct volume of drug has been accurately prepared
 - The catheter has been placed and is functioning properly
 - The patient is calm and can comfortably be restrained for the expected duration of administration
 - All staff are wearing adequate PPE (as described in the preparation section, although a single pair of gloves is acceptable).
- Drugs should be administered safely using a needle-free system (such as those described above).
- Drug administration should occur over a disposable absorbent pad (e.g. incontinence pad).
- The tubing should never be removed from a fluid bag containing a hazardous drug, nor should it be disconnected at any other point in the system. The intravenous catheter, tubing and bag should be removed from the patient and discarded intact as a single unit when possible. A cytotoxic waste bin should be used.
- The patient should be clearly labelled as a chemotherapy recipient (e.g. different coloured name tag). The chemotherapy injection site should be wrapped in a coloured dressing that identifies this procedure has occurred.
- Hands should be washed with soap and water before leaving the drug administration area.
- Procedures should be in place for dealing with any spillages that occur and for the safe disposal of waste. In the event of contact with skin or eyes, the affected area should be washed with copious amounts of water or normal saline. Medical advice should be sought if the eyes are affected.

Procedures for nursing patients receiving chemotherapy:

- Special wards or designated kennels with clear identification that the patients are being treated with cytotoxic agents is required. As well as kennel notices, different coloured bedding can be a helpful indicator.
- Excreta (saliva, urine, vomit, faeces) are all potentially hazardous after the animal has been treated with cytotoxic drugs and should be handled and disposed of accordingly.
- For hospitalized patients, the period of risk (i.e. the period during which the patient is expected to be excreting chemotherapeutic drugs or metabolites) should be clearly marked on the kennel and chemotherapy impervious PPE (such as disposable gloves and protective clothing) should be worn when carrying out nursing procedures (see the ACVIM small animal consensus statement (Smith *et al.*, 2018) for more information on excretion periods).
- All materials that have come into contact with the animal during the period of risk should be considered as potentially contaminated.

- After the animal has left the ward, the kennel should be cleaned according to the chemotherapy specific cleaning protocol.

Guidelines for owners:

- All owners should be given written information about:
 - The potential hazards of cytotoxic drugs (to humans and treated pets)
 - The excretion period relevant to the drug(s) administered
 - How to deal with the patient's excreta (saliva, urine, vomit, faeces).
- If owners are to administer tablets themselves, then they should receive written information stating:
 - The PPE required
 - That oral chemotherapeutic agents should not be crushed or split.
- Medicine containers should be clearly labelled with 'cytotoxic contents' warning tape.
- Veterinary surgeons should:
 - Offer to provide suitable PPE
 - Confirm owners are using the provided PPE during subsequent consultations.

Cost and chemotherapy safety

Undoubtedly some of the mitigating measures described above will increase the cost of chemotherapy and, unfortunately, in some cases may make chemotherapy unaffordable. However, practices have legal responsibilities to safeguard their employees and the public, and veterinary professionals have a duty of care to their patients. Therefore, cost is not a reason for not following suitable safety procedures.

Further information

The guidance provided here is a summary of current best practice, but is subject to continual re-evaluation and, therefore, practices should review their approach from time to time. For further information, readers are encouraged to review the *ACVIM small animal consensus statement on safe use of cytotoxic chemotherapeutics in veterinary practice* (Smith *et al.*, 2018), *Safe handling of cytotoxic drugs in the workplace* (hse.gov.uk/healthservices/safe-use-cytotoxic-drugs.htm) and the NIOSH guidance on *Safe handling of hazardous drugs for veterinary healthcare workers* (cdc.gov/niosh).

Reference

Smith AN, Klahn S, Phillips B *et al.* (2018) ACVIM small animal consensus statement on safe use of cytotoxic chemotherapeutics in veterinary practice. *Journal of Veterinary Internal Medicine* **32**(3), 904–913

Drugs usage in renal and hepatic insufficiency

With failure of liver or kidney, the excretion of some drugs may be impaired, leading to increased serum concentrations. Note: this information is based on mammals and may not be directly applicable to birds, reptiles, amphibians and fish.

Renal failure

- a. Double the dosing interval or halve the dosage in patients with severe renal insufficiency. Use for drugs that are relatively non-toxic.
- b. Increase dosing interval 2-fold when creatinine clearance (Ccr) is 0.5–1.0 ml/min/kg, 3-fold when Ccr is 0.3–0.5 ml/min/kg and 4-fold when Ccr is <0.3 ml/min/kg.
- c. Precise dose modification is required for some toxic drugs that are excreted solely by glomerular filtration (e.g. aminoglycosides). This is determined by using the dose fraction K_r to amend the drug dose or dosing interval according to the following equations:

Modified dose reduction = normal dose $\times K_r$
Modified dose interval = normal dose interval/ K_r
where K_r = patient Ccr/normal Ccr

Where Ccr is unavailable, Ccr may be estimated at 88.4/serum creatinine (μmol (micromoles)/l) (where serum creatinine is <350 $\mu\text{mol/l}$). K_r may be estimated at 0.33 if urine is isosthenuric or 0.25 if the patient is azotaemic.

Drug	Nephrotoxic	Dose adjustment in renal failure
Amikacin	Yes	c
Amoxicillin	No	a
Amphotericin B	Yes	c
Ampicillin	No	a
Cefalexin	No	b
Chloramphenicol	No	N, A
Digoxin	No	c
Gentamicin	Yes	c
Oxytetracycline	Yes	CI
Penicillin	No	a
Streptomycin	Yes	b
Tobramycin	Yes	c
Trimethoprim/sulphonamide	Yes	b, A

a, b, c = Refer to section above on dose adjustment; A = Avoid in severe renal failure; CI = Contraindicated; N = normal dose.

Hepatic insufficiency

Drug clearance by the liver is affected by many factors and thus it is not possible to apply a simple formula to drug dosing. The table below is adapted from information in the human literature.

Drug	DI	CI
Aspirin		✓
Azathioprine		✓
Cefotaxime	✓	
Chloramphenicol		✓
Clindamycin		✓
Cyclophosphamide	✓	
Diazepam		✓
Doxorubicin	✓	
Doxycycline	✓	
Furosemide	✓	
Hydralazine	✓	
Lidocaine	✓	
Metronidazole	✓	
Morphine	✓	
NSAIDs	✓	
Oxytetracycline		✓
Pentobarbital	✓	
Phenobarbital	✓	
Propranolol	✓	
Theophylline	✓	
Vincristine	✓	

CI = Contraindicated; avoid use if at all possible. DI = A change in dose or dosing interval may be required.

Further reading

British National Formulary No. 84 (2022) British Medical Association and the Royal Pharmaceutical Society of Great Britain

Carpenter JW (2018) *Exotic Animal Formulary*, 5th edn. Elsevier, St Louis

Compendium of Data Sheets for Animal Medicines (2022) National Office of Animal Health, Enfield, Middlesex

Giguère S, Prescott JF, Baggot JD and Walker RD (2013) *Antimicrobial Therapy in Veterinary Medicine*, 5th edn. Wiley Blackwell, Iowa

Monthly Index of Medical Specialties (2022) Haymarket Medical Publications, London

Papich MG (2015) *Saunders Handbook of Veterinary Drugs, 4th edn.* Saunders Elsevier, St Louis
 Plumb DC (2018) *Plumb's Veterinary Drug Handbook, 9th edn.* Wiley Blackwell, New Jersey (also available from www.plumbsveterinarydrugs.com)

Useful websites

www.bnf.org

British National Formulary – registration required through academic institutions to use BNF online but can order paper copy of BNF from this site.

www.bsava.com

British Small Animal Veterinary Association – links to position statements on responsible use of antibacterials and parasite control.

www.bsavalibrary.com

British Small Animal Veterinary Association online Library – links to *Journal of Small Animal Practice*, and contains the *BSAVA Guide to the Use of Veterinary Medicines* and searchable online Formularies (free access for BSAVA members).

www.bva.co.uk

British Veterinary Association.

www.bvzs.co.uk

British Veterinary Zoological Society.

cdc.gov/niosh

Centers for Disease Control and Prevention – NIOSH guidance on the safe handling of hazardous drugs.

www.ema.europa.eu/en

European Medicines Agency.

www.gov.uk/government/organisations/veterinary-medicines-directorate

Veterinary Medicines Directorate.

www.hse.gov.uk/healthservices/safe-use-cytotoxic-drugs.htm

Health and Safety Executive – safe handling of cytotoxic drugs in the workplace.

www.medicines.org.uk/emc

Electronic Medicines Compendium.

www.ncbi.nlm.nih.gov/pubmed

PubMed is a widely used free service of the U.S. National Library of Medicine and the National Institutes of Health that allows users to search abstracts in the medical literature. All major veterinary publications covered.

www.noahcompendium.co.uk

NOAH compendium site.

www.rcvs.org.uk

Royal College of Veterinary Surgeons.

www.who.int

World Health Organization.

www.wiley.com

Journal of Small Animal Practice – free for BSAVA members, free abstracts and pay per article for others.

Websites relating to the manufacture of extemporaneous products

www.gov.uk/government/publications/human-and-veterinary-medicines-register-of-licensed-manufacturing-sites

Human and veterinary medicines – links to the Register of licensed manufacturing sites – manufacturer specials, human (MS) and the Register of licensed manufacturing sites – manufacturer specials authorisation, veterinary (MANSA).

www.bova.co.uk

Site for a company that will reformulate many drugs into conveniently sized tablets, liquids, pastes and transdermal gels.

www.chemopet.co.uk

Site for a company that will reformulate a wide range of injectable and oral chemotherapy drugs.

<https://healthcare.pccarx.co.uk/>

Site for a company that will reformulate many drugs into conveniently sized tablets.

www.novalabs.co.uk/

Site for a company that will reformulate many drugs into conveniently sized tablets or liquids.

www.summitvetpharma.co.uk

Site for a company that will reformulate many drugs into conveniently sized tablets, liquids and transdermal gels.

Appendix II: protocols

Chemotherapy protocols

The metabolism of small mammals, birds and reptiles is markedly different from dogs, cats and humans. Therefore, extrapolation of doses is risky to the patient and extrapolation of safety data (such as the length of time that an animal may excrete a cytotoxic drug) is risky to the owners and staff handling the animals. The lack of any proper evidence base for the use of cytotoxic agents in these species means that any attempt to use them cannot be shown to be safe, either to the animal or to the people who come into contact with that animal. For this reason, only limited information on doses of these agents is generally provided in the Formulary, other than for ferrets where established protocols have been published^a (see below). The use of these agents in other species should be undertaken with caution until such time as better monitoring facilities are available in veterinary practice and there are sufficient cases to undertake effective clinical trials.

Reference

^a Webb JK, Graham JE, Burgess KE and Antinoff N (2019) Presentation and survival time of domestic ferrets (*Mustela putorius furo*) with lymphoma treated with single-and multi-agent protocols: 44 cases (1998–2016). *Journal of Exotic Pet Medicine* **31**, 64–67

Chemotherapy protocols for lymphoma: ferrets

Note: the suggested protocols are based on clinical application and case studies and extrapolated from other species; there are no primary controlled trials in ferrets. A CBC should be performed prior to the administration of any chemotherapeutic agents.

Option 1

Week	Day	Drug	Dose and route
1	1	Prednisolone	1–2 mg/kg p.o. q12h continued throughout course of therapy
		Vincristine	0.025 mg/kg i.v.
	3	Cyclophosphamide	10 mg/kg p.o., s.c.
2	8	Vincristine	0.025 mg/kg i.v.
3	15	Vincristine	0.025 mg/kg i.v.
4	22	Vincristine	0.025 mg/kg i.v.
	24	Cyclophosphamide	10 mg/kg p.o., s.c.
7	46	Cyclophosphamide	10 mg/kg p.o., s.c.
9	63	Prednisolone	Taper dose down gradually to 0 over the next 4wks

Option 2

Week	Drug	Dose and route
1	Vincristine	0.025 mg/kg i.v.
	Asparaginase ^a	400 IU/kg i.p.
	Prednisolone	1 mg/kg p.o. q24h continued throughout course of therapy
2	Cyclophosphamide	10 mg/kg p.o., s.c.
3	Doxorubicin	1 mg/kg i.v.
4	Vincristine	0.025 mg/kg i.v.
5	Cyclophosphamide	10 mg/kg p.o., s.c.
6	Doxorubicin	1 mg/kg i.v.
8	Vincristine	0.025 mg/kg i.v.
10	Cyclophosphamide	10 mg/kg p.o., s.c.
12	Vincristine	0.025 mg/kg i.v.
14	Methotrexate	0.5 mg/kg i.v.

^a Premedicate with diphenhydramine at 1–2 mg/kg i.m. 30 minutes prior to administration to prevent an anaphylactic response.

Option 3

Week	Drug	Dose and route
–3	Asparaginase ^a	400 IU/kg s.c.
1	Vincristine	0.12 mg/kg i.v.
	Prednisolone	1 mg/kg p.o. q24h continued throughout course of therapy
	Cyclophosphamide	10 mg/kg p.o.
2	Vincristine	0.12 mg/kg i.v.
3	Vincristine	0.12 mg/kg i.v.
4, 7, 10, 13, etc. (q3wks)	Vincristine	0.12 mg/kg i.v.
	Cyclophosphamide	10 mg/kg p.o.
Rescue	Doxorubicin ^a	1–2 mg/kg i.v. slow infusion

^a Premedicate with diphenhydramine at 1–2 mg/kg i.m., s.c. 30 minutes prior to administration to prevent an anaphylactic response.

Option 4 (avoiding i.v. route)

Week	Drug	Dose and route
1	Asparaginase ^a	10,000 IU/m ² s.c.
	Cyclophosphamide	250 mg/m ² p.o., s.c. (in 50 ml/kg NaCl s.c.)
	Prednisolone	2 mg/kg p.o. q24h for 7 days then q48h throughout course of therapy
2	Asparaginase ^a	10,000 IU/m ² s.c.
3	Asparaginase ^a	10,000 IU/m ² s.c.
	Cytosine arabinoside	300 mg/m ² s.c. q24h for 2 days (dilute 100 mg in 1 ml sterile water)
4	Asparaginase ^a	10,000 IU/m ² s.c.
5	Cyclophosphamide	250 mg/m ² p.o., s.c. (in 50 ml/kg NaCl)
7	Methotrexate	0.8 mg/kg i.m.
9	Cyclophosphamide	250 mg/m ² p.o., s.c. (in 50 ml/kg NaCl)
11	Cytosine arabinoside	300 mg/m ² s.c. q24h for 2 days (dilute 100 mg in 1 ml sterile water)
	Chlorambucil	2 mg tablet/ferret p.o.
13	Cyclophosphamide	250 mg/m ² p.o., s.c. (in 50 ml/kg NaCl)
15	Procarbazine	50 mg/m ² p.o. q24h for 14 days
18	Cyclophosphamide	250 mg/m ² p.o., s.c. (in 50 ml/kg NaCl)
20	Cytosine arabinoside	300 mg/m ² s.c. q24h for 2 days (dilute 100 mg in 1 ml sterile water)
23	Cyclophosphamide	250 mg/m ² p.o., s.c. (in 50 ml/kg NaCl)
26	Procarbazine	50 mg/m ² p.o. q24h for 14 days
28	Repeat weeks 20–26 if not in remission for 3 cycles	

Note: perform a CBC at weeks 2, 4, 7, 8, 12, 16, 17 and 27; if severe myelosuppression, reduce dose by 25% for subsequent treatments of the previously used drug.

^aPremedicate with diphenhydramine at 1–2 mg/kg i.m., s.c. 30 minutes prior to administration to prevent an anaphylactic response.

Sedation/immobilization protocols

Sedative combinations for ferrets

- Ketamine (5–8 mg/kg i.m.) plus medetomidine (80–100 µg (micrograms)/kg i.m.) or dexmedetomidine (40–50 µg (micrograms)/kg i.m.) to which can be added butorphanol (0.1–0.2 mg/kg i.m.) or buprenorphine (0.02 mg/kg i.m.) based on studies in healthy ferrets^{a,b}.
- Ketamine (5–20 mg/kg i.m.) plus midazolam (0.25–0.5 mg/kg i.m.) or diazepam (0.25–0.5 mg/kg i.m.) will provide immobilization or, at the higher doses, a short period of anaesthesia.
- Ketamine (7–10 mg/kg i.m., s.c.) plus medetomidine (20 µg (micrograms)/kg i.m., s.c.) plus midazolam (0.5 mg/kg i.m., s.c.) will provide anaesthesia; concurrent oxygenation is recommended^b.
- Midazolam (0.2–0.3 mg/kg i.m.) plus butorphanol (0.2–0.3 mg/kg i.m.) provides moderate sedation for non-invasive procedures^c.

References

- ^a Ko JC, Heaton-Jones TG and Nicklin CF (1997) Evaluation of the sedative and cardiorespiratory effects of medetomidine, medetomidine-butorphanol, medetomidine-ketamine and medetomidine-butorphanol-ketamine in ferrets. *Journal of the American Animal Hospital Association* **33**(5), 438–448
- ^b Scherthaner A, Lendl CE, Hartmann K et al. (2011) Medetomidine/midazolam/ketamine anaesthesia in ferrets: effects on cardiorespiratory parameters and evaluation of plasma drug concentrations. *Veterinary Anaesthesia and Analgesia* **38**(5), 439–450
- ^c Kapaldo N and Eshar D (2022) Ferret Sedation and Anesthesia. *Veterinary Clinics of North America: Exotic Animal Practice* **25**(1), 273–296.

Sedative combinations for rabbits

- Ketamine (3–5 mg/kg i.v. or 5–10 mg/kg i.m., s.c.) in combination with medetomidine (0.05–0.1 mg/kg i.v. or 0.1–0.3 mg/kg s.c., i.m.) or dexmedetomidine (0.025–0.05 mg/kg i.v. or 0.05–0.15 mg/kg s.c., i.m.) and butorphanol (0.05–0.1 mg/kg i.m., i.v., s.c.) or buprenorphine (0.02–0.05 mg/kg i.m., i.v., s.c.).
- Fentanyl/fluanisone (0.1–0.3 ml/kg i.m.) plus diazepam (0.5–1 mg/kg i.v., i.m. or 2.5–5.0 mg/kg i.p.) or midazolam (0.25–1.0 mg/kg i.v., i.m., i.p.).

The combinations above will provide immobilization/light anaesthesia, usually sufficient to allow intubation for maintenance with a volatile agent.

- Ketamine (15 mg/kg i.m.) in combination with medetomidine (0.25 mg/kg i.m.) and buprenorphine (0.03 mg/kg i.m.) will provide general anaesthesia based on studies in healthy animals^{a,b,c}, but use of lower doses of medetomidine and ketamine followed by intubation and use of a volatile agent is recommended in practice
- Midazolam (0.5–2 mg/kg i.m.) plus butorphanol (0.5–2 mg/kg i.m.) provides moderate sedation for non-invasive procedures^d. Can also be given i.v. starting at lower end of dose range.

The doses above are for use in clinically healthy animals. As prey species are good at hiding signs of disease and because subclinical disease, especially respiratory tract disease, is common, careful preanaesthetic assessment of such patients and titration of doses to effect by i.v. administration, where possible, is advised.

References

- ^a Grint NJ and Murison PJ (2008) A comparison of ketamine-midazolam and ketamine-medetomidine combinations for induction of anaesthesia in rabbits. *Veterinary Anaesthesia and Analgesia* **35**(2), 113–121

- ^b Murphy KL, Roughan JV, Baxter MG and Flecknell PA (2010) Anaesthesia with a combination of ketamine and medetomidine in the rabbit: effect of premedication with buprenorphine. *Veterinary Anaesthesia and Analgesia* **37(3)**, 222–229
- ^c Orr HE, Roughan JV and Flecknell PA (2005) Assessment of ketamine and medetomidine anaesthesia in the domestic rabbit. *Veterinary Anaesthesia and Analgesia* **32(5)**, 271–279
- ^d Gardhouse S and Sanchez A (2022) Rabbit sedation and anesthesia. *Veterinary Clinics of North America: Exotic Animal Practice* **25(1)**, 181–210

Sedative combinations for other small mammals

- Medetomidine (50 µg (micrograms)/kg i.m.) or dexmedetomidine (25 µg (micrograms)/kg i.m.) plus, if needed, ketamine (2–4 mg/kg i.m.)^{a,b,c,d,e,f}.
- **Sugar gliders:** Buprenorphine (0.01 mg/kg i.m.) in combination with midazolam^g.
- Other combinations as for rabbits: Combinations can also be administered i.p. in small rodents.

References

- ^a Bakker J, Uilenreef JJ, Pelt ER *et al.* (2013) Comparison of three different sedative-anaesthetic protocols (ketamine, ketamine-medetomidine and alfaxalone) in common marmosets (*Callithrix jacchus*). *BMC Veterinary Research* **9(1)**, 113
- ^b Buchanan KC, Burge RR and Ruble GR (1998) Evaluation of injectable anaesthetics for major surgical procedures in guinea pigs. *Contemporary Topics in Laboratory Animal Science* **37(4)**, 58–63
- ^c Dang V, Bao S, Ault A *et al.* (2008) Efficacy and safety of five injectable anaesthetic regimens for chronic blood collection from the anterior vena cava of guinea pigs. *Journal of American Association of Laboratory Animal Science* **47(6)**, 56–60
- ^d Hedenqvist P, Roughan JV and Flecknell PA (2000) Effects of repeated anaesthesia with ketamine/medetomidine and of pre-anaesthetic administration of buprenorphine in rats. *Laboratory Animals* **34(2)**, 207–211
- ^e Jang HS, Choi HS, Lee SH, Jang KH and Lee MG (2009) Evaluation of the anaesthetic effects of medetomidine and ketamine in rats and their reversal with atipamezole. *Veterinary Anaesthesia and Analgesia* **36(4)**, 319–327
- ^f Nevalainen T, Pyhälä L, Voipio HM and Virtanen R (1989) Evaluation of anaesthetic potency of medetomidine-ketamine combination in rats, guinea pigs and rabbits. *Acta Veterinaria Scandinavica* **85**, 139–143
- ^g Doss G and de Miguel Garcia C (2022) African pygmy hedgehog (*Atelerix albiventris*) and sugar glider (*Petaurus breviceps*) sedation and anesthesia. *Veterinary Clinics of North America: Exotic Animal Practice* **25(1)**, 257–272

Note: Reduce doses if animal is debilitated. For all small mammals, for deeper anaesthesia, intubation (if possible) and use of a volatile agent is recommended, rather than using higher doses of injectable agents.

Sedative combinations for birds

Injectable anaesthesia has largely been avoided in birds since the advent of 'safer' gaseous agents such as isoflurane and sevoflurane. However, systemic agents are becoming increasingly used either as induction agents (e.g. as part of a total intravenous anaesthetic (TIVA) protocol) or as premedication prior to gaseous anaesthesia. The latter is primarily used to reduce the dose of gaseous anaesthesia and therefore reduce stress and hypotensive effects. Parasympathetic agents (such as atropine) are rarely used as their effect is to make respiratory excretions more viscous, thus increasing the risk of tube blockage. TIVA is rarely used except in specialized situations, whilst intravenous induction tends to be used in field situations or in large birds (e.g. ratites) and those with a dive response (e.g. waterfowl, penguins) that may be hard to induce with gaseous agents. In some high-risk cases, sedation may be used in pet birds for non-invasive diagnostic procedures (e.g. blood sampling, radiography) or prior

to intravenous euthanasia. In these cases, the intranasal route is much less painful for the bird and is easily used, with midazolam/butorphanol mixtures being effective. For stress relief (e.g. in dyspnoeic birds), midazolam alone may be used. Flumazenil (0.05 mg/kg) is effective in reversing midazolam via the intramuscular or intranasal routes.

- Propofol: 10 mg i.v. by slow infusion to effect; supplemental doses up to 3 mg/kg^{a,b}; 0.5 mg/kg/min i.v. as CRI.
 - For large birds and those with a dive response: 2–4 mg/kg i.v. to effect.
 - **Bengalese finches**: 10–50 mg/kg s.c. with or without midazolam and/or butorphanol. Increasing dose linked with increasing length of sedation rather than depth
 - **Budgerigars**: 15 mg/kg s.c.; 10 mg/kg suitable for shorter non-invasive investigations
- Ketamine/diazepam combinations can be used for induction and muscle relaxation. Ketamine (30–40 mg/kg) plus diazepam (1.0–1.5 mg/kg) are given slowly i.v. to effect. May also be given i.m. but this produces different effects in different species and specific literature or specialist advice should be consulted. Ketamine (10 mg/kg i.m.) and diazepam (0.2–0.5 mg/kg i.m.) can be used as premedication/sedation (pigeons, Amazon parrots) prior to the administration of sevoflurane/isoflurane^{c,d}.
- Sedation: Midazolam:
 - 0.1–0.5 mg/kg i.m. or 0.05–0.15 mg/kg i.v. (premedicant) or 2–3 mg/kg intranasal. Can be combined with butorphanol (1–3 mg/kg intranasal) for premedication/sedation.
 - 2 mg/kg intranasal provides sedation in **Eurasian buzzards** (reversible with 0.05 mg/kg intranasal flumazenil).
 - 1–2 mg/kg i.m. premedication had a significant isoflurane-sparing effect on **Quaker parrots** without effecting the heart rate or body temperature.
- Sedation: Butorphanol (3 mg/kg intranasal) plus midazolam (3 mg/kg) may prolong anaesthetic recovery^e.
- **Raptors**: Ketamine (2–5 mg/kg i.m.) plus medetomidine (25–100 µg (micrograms)/kg) (lower dose rate i.v.; higher rate i.m.). This combination can be reversed with atipamezole at 65 µg (micrograms)/kg i.m. Ketamine should be avoided in vultures.

References

- ^a Fitzgerald G and Cooper JE (1990) Preliminary studies on the use of propofol in the domestic pigeon (*Columba livia*). *Research in Veterinary Science* **49**(3), 334–338
- ^b Hawkins MG, Wright BD, Pascoe PJ *et al.* (2003) Pharmacokinetics and anesthetic and cardiopulmonary effects of propofol in red-tailed hawks (*Buteo virginianus*) and great horned owls (*Bubo virginianus*). *American Journal of Veterinary Research* **64**(6), 677–683
- ^c Azizpour and Hassani Y (2012) Clinical evaluation of general anaesthesia in pigeons using a combination of ketamine and diazepam. *Journal of the South African Veterinary Association* **83**(1), 12
- ^d Paula VV, Otsuki DA, Auler Júnior JO *et al.* (2013) The effect of premedication with ketamine, alone or with diazepam, on anaesthesia with sevoflurane in parrots (*Amazona aestivalis*). *BMC Veterinary Research* **9**, 142
- ^e Doss GA, Fink DM and Mans C (2018) Assessment of sedation after intranasal administration of midazolam and midazolam/butorphanol in cockatiels (*Nymphicus hollandicus*). *American Journal of Veterinary Research* **79**(12), 1246–1252

Sedative combinations for reptiles

- Alfaxalone or propofol (**see monographs**) can be used to provide deep sedation or induce anaesthesia in reptiles. Additional analgesia should be provided as part of the pre-anaesthetic sedation (see below) for any potentially painful procedure and maintenance with a volatile inhalation agent will be required for prolonged anaesthesia.
- Ketamine as sole agent: Ketamine alone may result in variable sedation, poor muscle relaxation and prolonged recovery at higher dose rates. Usually combined with alpha-2 agonists and/or opioids/midazolam to provide deep sedation/light anaesthesia (see below).
- Alpha-2 agonists as sole agents: Although single agent use has been reported, it is generally preferable to use medetomidine or dexmedetomidine in combination with opioids and/or ketamine and/or midazolam for more reliable sedation (see below). Should be reversed with atipamezole (**see monograph**).
- Benzodiazepines as sole agents: Although single agent used has been reported, it is generally preferable to use midazolam in combination with opioids and/or ketamine and/or alpha-2 agonists for more reliable sedation (see below). Can be reversed with flumazenil if required (**see monograph**).
- Ketamine/alpha-2 agonist/opioid/benzodiazepine mixtures:
 - Most species for pre-anaesthetic sedation: Midazolam (1–2 mg/kg i.m.) plus butorphanol (0.5–2 mg/kg i.m.) or morphine (1–1.5 mg/kg i.m.) if required may be administered pre-anaesthetic for sedation or in combination with alfaxalone. Choice of opioid will depend on whether just sedation (butorphanol can be used) or sedation plus analgesia (morphine is preferred) is required
 - Chelonians:
 - Ketamine (5–10 mg/kg i.m., i.v.) plus medetomidine (100–200 µg (micrograms)/kg i.m., i.v.) (deep sedation to light anaesthesia in **Gopher tortoises** and **Red-eared sliders**)^{a,b}
 - Ketamine (10 mg/kg intranasal) plus dexmedetomidine (200 µg (micrograms)/kg intranasal) (deep sedation to light anaesthesia in **Red-eared sliders**)^c
 - Ketamine (2 mg/kg s.c.) plus dexmedetomidine (100 µg (micrograms)/kg s.c.) and midazolam (1 mg/kg s.c.) (moderate sedation in **Red-eared sliders**)^d
 - Ketamine (10 mg/kg i.m.) plus dexmedetomidine (100 µg (micrograms)/kg i.m.) and midazolam (1 mg/kg i.m.) (deep sedation to light anaesthesia in **Red-footed tortoises** and **Ornate box turtles**)^{e,f}
 - Lizards:
 - Midazolam (0.5–2 mg/kg i.m., s.c.) plus dexmedetomidine (50–100 µg (micrograms)/kg i.m., s.c.) plus ketamine if required (1–5 mg/kg i.m., s.c.) (sedation in variety of lizards)^b
 - Midazolam (1 mg/kg i.m.) plus dexmedetomidine (200 µg (micrograms)/kg i.m.) (sedation in **Tegus**)^h
 - Midazolam (1 mg/kg i.m.) plus dexmedetomidine (100 µg (micrograms)/kg i.m.) (light to moderate sedation in **Leopard geckos**)ⁱ
 - Snakes:
 - Midazolam (0.5 mg/kg i.m.) plus dexmedetomidine (50 µg (micrograms)/kg i.m.) (moderate sedation in **Royal pythons**)^j

- Butorphanol (10 mg/kg i.m.) followed by midazolam (0.5 mg/kg i.m.), dexmedetomidine (50 µg (micrograms)/kg i.m.) and ketamine (5 mg/kg i.m.) (light anaesthesia in Bullsnares)^k.

References

- ^a Dennis C and Heard DJ (2002) Cardiopulmonary effects of a medetomidine-ketamine combination administered intravenously in gopher tortoises. *Journal of the American Veterinary Medical Association* **220(10)**, 1516–1519
- ^b Greer LL, Jenne KJ and Diggs HE (2001) Medetomidine-ketamine anesthesia in red-eared slider turtles (*Trachemys scripta elegans*). *Journal of the American Association for Laboratory Animal Science* **40(3)**, 8–11
- ^c Cermakova E, Cepelchova V and Knotek Z (2018) Efficacy of two methods of intranasal administration of anaesthetic drugs in red-eared terrapins (*Trachemys scripta elegans*). *Veterinárni Medicina* **63(2)**, 87–93
- ^d Mans C, Drees R, Sladky KK, Hatt JM and Kircher PR (2013) Effects of body position and extension of the neck and extremities on lung volume measure via computed tomography in red-eared slider turtles (*Trachemys scripta elegans*). *Journal of the American Veterinary Medical Association* **243(8)**, 1190–1196
- ^e Eshar D, Rooney T A, Gardhouse S and Beaufrère H (2021) Evaluation of the effects of a dexmedetomidine-midazolam-ketamine combination administered intramuscularly to captive red-footed tortoises (*Chelonoidis carbonaria*). *American Journal of Veterinary Research* **82(11)**, 858–864
- ^f Rooney TA, Eshar D, Gardhouse S and Beaufrère H (2021) Evaluation of a dexmedetomidine-midazolam-ketamine combination administered intramuscularly in captive ornate box turtles (*Terrapene ornata ornata*). *Veterinary Anaesthesia and Analgesia* **48(6)**, 914–921
- ^g Budden L, Doss GA, Clyde VL and Mans C (2018) Retrospective evaluation of sedation in 16 lizard species with dexmedetomidine-midazolam with or without ketamine. *Journal of Herpetological Medicine and Surgery* **28(1–2)**, 47–50
- ^h Bisetto SP, Melo CF and Carregaro AB (2018) Evaluation of sedative and antinociceptive effects of dexmedetomidine, midazolam and dexmedetomidine-midazolam in tegus (*Salvator merianae*). *Veterinary Anaesthesia and Analgesia* **45(3)**, 320–328
- ⁱ Doss GA, Fink DM, Sladky KK and Mans C (2017) Comparison of subcutaneous dexmedetomidine-midazolam versus alfaxalone-midazolam sedation in leopard geckos (*Eublepharis macularius*). *Veterinary Anaesthesia and Analgesia* **44(5)**, 1175–1183
- ^j Yaw TJ, Mans C, Johnson S et al. (2020) Evaluation of subcutaneous administration of alfaxalone-midazolam and dexmedetomidine-midazolam for sedation of ball pythons (*Python regius*). *Journal of the American Veterinary Medical Association* **256(5)**, 573–579
- ^k Sadar MJ and Ambros B (2018) Use of alfaxalone or midazolam–dexmedetomidine–ketamine for implantation of radiotransmitters in bullsnares (*Pituophis catenifer sayi*). *Journal of Herpetological Medicine and Surgery* **28(3–4)**, 93–98

Sedative combinations for fish

The dose rates below are for induction of anaesthesia. A lower dose (approximately 20–25% of the induction dose rate) is required for maintenance of anaesthesia in a separate container or using a recirculating system. All anaesthetics are administered by immersion and the stage of anaesthesia reached is determined by the concentration used and the duration of exposure, since absorption continues throughout the period of immersion. There are significant species differences in the response to drugs and it is advised that lower dose rates are used for unfamiliar species, marine and tropical fish. Many products and stock solutions should be kept in a dark bottle and protected from light. Ideally, the anaesthetic solution should be made up using water from the tank or pond of origin to minimize problems due to changes in water chemistry. It should be used on the day of preparation and well aerated during use. Food should be withheld for 12–24 hours before anaesthesia to reduce the risk of regurgitation, which may cause

damage to gill tissues. Monitoring heart rate during prolonged procedures using a Doppler probe or ultrasound scanner is advisable since this is a direct reflection of the level of anaesthesia. Following the procedure, anaesthetized fish should be returned to clean water from their normal environment to allow recovery.

- Tricaine mesilate (MS-222) (50–250 mg/l by immersion) produces an acidic solution and should be buffered with sodium bicarbonate to maintain the same pH as the original environmental water conditions. The dry powder is very soluble in water and can be added directly or a stock solution can be made up to facilitate accurate dosing.
- Benzocaine (Optomease Vet) (25–200 mg/l by immersion) is insoluble in water and must be dissolved in acetone or ethanol (e.g. a stock solution of 100 g benzocaine/l of ethanol produces 100 mg/ml to facilitate accurate dosing).
- 2-Phenoxyethanol (Aqua-sed) (0.1–0.5 ml/l by immersion) must be whisked vigorously into the water to improve solubility.
- Alfaxalone (Alfaxan) (5–10 mg/l by immersion) for induction ^a.
- Propofol (Norofol) (2.5–10 mg/l by immersion) produces a cloudy solution and reduces visibility of the patient, therefore, care is required while monitoring vital signs during anaesthesia ^b.

References

- Minter LJ, Bailey KM, Harms CA, Lewbart GA and Posner LP (2014) The efficacy of alfaxalone for immersion anaesthesia in koi carp (*Cyprinus carpio*). *Veterinary Anaesthesia and Analgesia* **41**(4), 398–405
- Oda A, Bailey KM, Lewbart GA, Griffith EH and Posner LP (2014) Physiologic and biochemical assessments of koi (*Cyprinus carpio*) following immersion in propofol. *Journal of the American Veterinary Medical Association* **245**(11), 1286–1291

Euthanasia of fish

It is recommended that dose rates of approximately 10 times the induction dose of the drugs listed above are used by immersion for euthanasia. Fish should remain in the solution for at least 60 minutes after respiration (opercular movement) has ceased and the absence of a heartbeat confirmed using a Doppler probe or by ultrasound scanner. In cases where tissue samples are required for histopathological examination, euthanasia can be completed when opercular movement has stopped by performing a secondary procedure (e.g. cervical section and pithing, or intravenous injection of pentobarbital (60–100 mg/kg)). Oil of clove (clove oil, eugenol) (500–1000 mg/ml by immersion or undiluted 10 drops/l by immersion) is available without prescription at most pharmacies and can be used by owners for emergency euthanasia. The oil must be whisked vigorously into the water to improve its solubility. It is at the veterinary surgeon's discretion as to whether to provide direction to owners regarding emergency euthanasia at home.

Anaesthetics, analgesics and NSAIDs

Adrenoreceptor antagonists

Atipamezole 24

Antiarthritis drugs

Pentosan polysulphate 212

Polysulphated
glycosaminoglycan 222

Inhalational anaesthetics

Isoflurane 152

Nitrous oxide 197

Sevoflurane 245

Injectable anaesthetics

Alfaxalone 7

Fentanyl/Fluanisone 118

Ketamine 157

Propofol 232

(see also *Sedation/immobilization
protocols in the Appendix II*)

Local anaesthetics

Benzocaine 33

Bupivacaine 38

Eugenol 112

Lidocaine 165

Mepivacaine 180

Phenoxyethanol 216

Tricaine mesilate 268

Muscle relaxants

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Mydriatics

Atracurium 25

Non-opioid analgesics

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Carprofen 49

Celecoxib 58

Dimethylsulfoxide 94

Ketoprofen 159

Mavacoxib 174

Meloxicam 178

Paracetamol 206

Piroxicam 221

Robenacoxib 240

Tolfenamic acid 263

Opioid analgesics

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Butorphanol 42

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Morphine 191

Naloxone 195

Pethidine 213

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Sedatives

Acepromazine 1

Dexmedetomidine 86

Diazepam 88

Medetomidine 175

Midazolam 187

Xylazine 286

(see also *Sedation/immobilization
protocols in the Appendix*)

Sympathomimetics

Adrenaline 5

Anti-infectives

(see also *Guidelines for responsible
antibiotic use in the Appendix*)

Anthelmintics

Emodepside 106

Fenbendazole 115

Levamisole 163

Mebendazole/Closantel 175

Oxfendazole 202

Piperazine 220

Praziquantel 227

Pyrantel 236

Antibiotics

Aminoglycosides

Amikacin 12

Framycetin 130

Gentamicin 135

Neomycin 196

Streptomycin 252

Tobramycin 262

Beta-lactams

Amoxicillin 16

Ampicillin 19

Aztreonam 30

Cloxacillin 75

Co-amoxiclav 76

Flucloxacillin 122

Penicillin G 209

Piperacillin 219

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 - Cefotaxime 52
 - Cefovecin 53
 - Ceftazidime 55
 - Ceftiofur 56
 - Cefuroxime 57
- Fluoroquinolones
 - Ciprofloxacin 69
 - Enrofloxacin 109
 - Marbofloxacin 171
 - Moxifloxacin 194
 - Ofloxacin 200
 - Pradofloxacin 226
- Lincosamides
 - Clindamycin 73
 - Lincomycin 166
- Macrolides
 - Azithromycin 28
 - Clarithromycin 71
 - Erythromycin 111
 - Tylosin 272
- Nitromidazoles
 - Metronidazole 185
- Sulphonamides
 - Silver sulfadiazine 247
 - Trimethoprim/Sulphonamide 270
- Tetracyclines
 - Doxycycline 102
 - Oxytetracycline 203
- Others
 - Benzalkonium chloride 32
 - Chloramine-T 61
 - Chloramphenicol 62
 - Copper sulphate 78
 - Florfenicol 121
 - Fusidic acid 133
 - Potassium permanganate 224
 - Rifampin 239
 - Trimethoprim/Sulphonamide 270

Antifungals

- Amphotericin B 18
- Bronopol 36
- Clotrimazole 75
- Copper sulphate 78
- Enilconazole 108
- Flubendazole 122
- Fluconazole 123
- Formaldehyde 129
- Itraconazole 153
- Malachite green 170
- Miconazole 186
- Nystatin 198
- Silver sulfadiazine 247
- Voriconazole 284

Antiprotozoals

- Amprolium 20
- Carnidazole 48
- Chloroquine 64
- Clazuril 72
- Fresh water (dechlorinated) 131
- Metronidazole 185
- Pentamidine isethionate 210
- Pyrimethamine 236
- Sulfadimethoxine 254
- Toltrazuril 264

Antivirals

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- Amantadine 11
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Anti-neoplastic

(see also *Safety and handling of chemotherapeutic agents and Chemotherapy protocols in the Appendix*)

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Anthracycline glycosides

- Doxorubicin 101

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- Lomustine 167
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- Diphenhydramine 96
- Doxepin 100
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Endocrine

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NOTES

Drug-related Adverse Event reporting

What are Adverse Events?

Adverse Events (AEs) are defined as 'any untoward medical occurrence that may present during treatment with a pharmaceutical product regardless of perceived causation'. AEs may include effects on the patient, humans handling the products or patient, or the environment. Lack of efficacy events are also AEs.

Who manages Adverse Event reports and who can report them?

The Veterinary Medicines Directorate (VMD) and Marketing Authorization Holders (MAHs) work together to collate information about, identify and mitigate risks associated with veterinary pharmaceuticals. These processes rely on spontaneous reporting of AEs. Anyone can report an AE, including veterinary professionals, owners and other interested parties. Under the RCVS Code of Professional Conduct veterinary surgeons and veterinary nurses are required to report adverse events.

Why is reporting Adverse Events important?

Prior to receiving a marketing authorization (licence), veterinary medicines undergo a significant amount of testing and evaluation. During this phase many of the more common AEs associated with a medicine will be identified. However, pre-marketing testing cannot hope to identify all possible AEs that will occur once a product is in widespread clinical use in a diverse population of animals or for 'off-label' use. In addition, products without an authorization do not undergo such testing.

Not all AEs are true adverse drug reactions (ADRs). AE reports are collated and analysed to identify statistically associated drug–event combinations. Reported cases are also reviewed to determine whether a causal relationship is likely. As the identification of true ADRs relies on statistical analysis, higher reporting rates allow the VMD to identify and mitigate any potential harm caused by a particular drug more quickly. A further benefit is that reporting both known and unknown events makes ADRs more prevalent in particular species, breeds, diseases and combinations of drugs more likely to be identified.

How to report an Adverse Event?

Adverse events can be reported to the VMD or the MAH (for authorized products); these organizations share AE information so there is no need to report to both. Reports to the VMD can be made via a dedicated webpage (<https://www.gov.uk/report-veterinary-medicine-problem>). A paper form can be downloaded if required. Most practices participating in the Small Animal Veterinary Surveillance Network (SAVSNET) can also report via their practice management systems.

The VMD pharmacovigilance team can be contacted on email at **adverse.events@vmd.gov.uk** or by telephone on **01932 336911**.

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